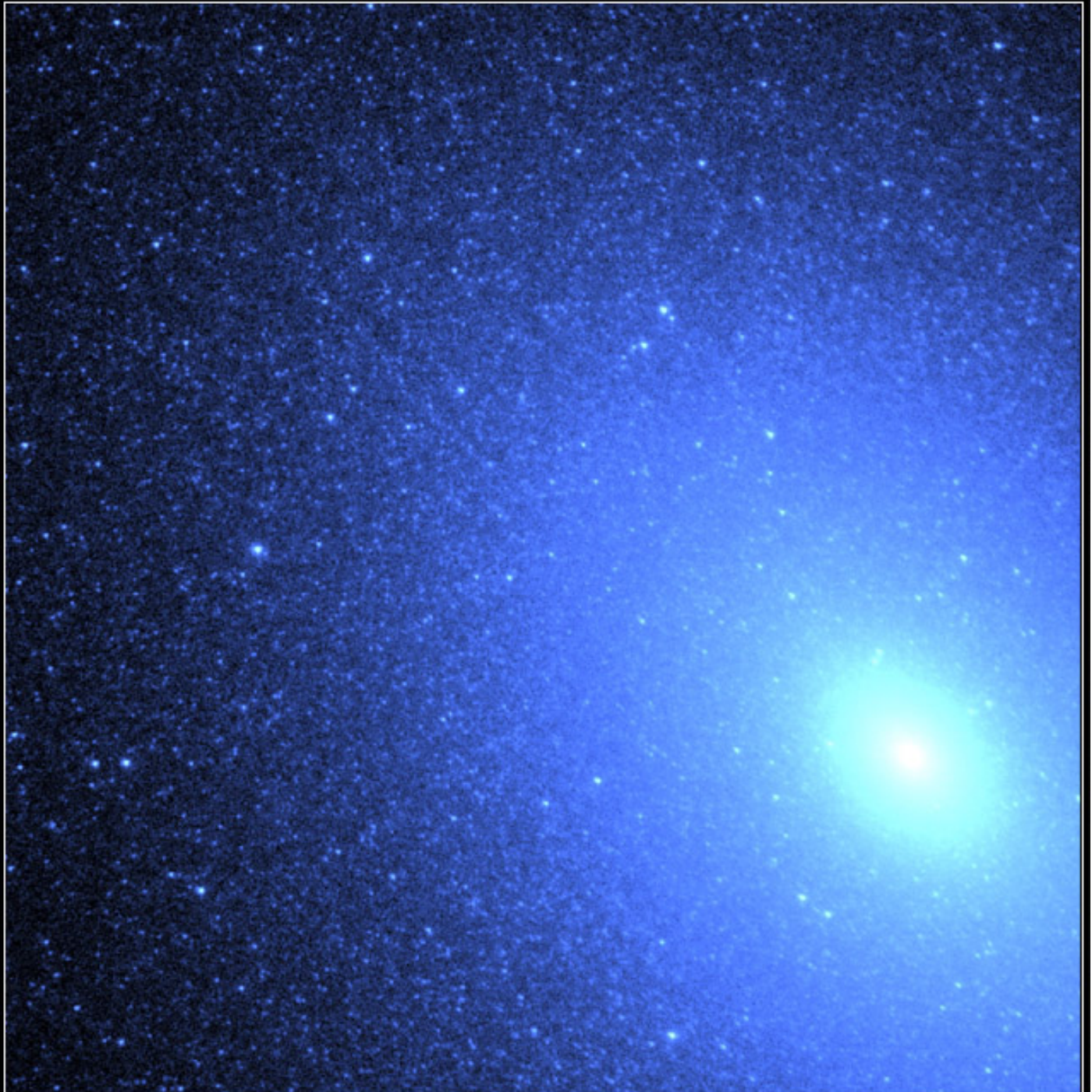


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Dwarf Elliptical Galaxy M32 **HST • STIS**
NASA and T. Brown (NASA Goddard Space Flight Center)
STScI-PRC99-40

Hubble Space Telescope's exquisite resolution has allowed astronomers to resolve, for the first time, hot blue stars deep inside an elliptical galaxy. The swarm of nearly 8,000 blue stars resembles a blizzard of snowflakes near the core (lower right) of the neighboring galaxy M32, located 2.5 million light-years away in the constellation Andromeda.

Hubble confirms that the ultraviolet light comes from a population of extremely hot helium-burning stars at a late stage in their lives. Unlike the Sun, which burns hydrogen into helium, these old stars exhausted their central hydrogen long ago, and now burn helium into heavier elements.

The observations, taken in October 1998, were made with the camera mode of the Space Telescope Imaging Spectrograph (STIS) in ultraviolet light. The STIS field of view is only a small portion of the entire galaxy, which is 20 times wider on the sky. For reference, the full moon is 70 times wider than the STIS field-of-view. The bright center of the galaxy was placed on the right side of the image, allowing fainter stars to be seen on the left side of the image.

Thirty years ago, the first ultraviolet observations of elliptical galaxies showed that they were surprisingly bright when viewed in ultraviolet light. Before those pioneering UV observations, old groups of stars were assumed to be relatively cool and thus extremely faint in the ultraviolet. Over the years since the initial discovery of this unexpected ultraviolet light, indirect evidence has accumulated that it originates in a population of old, but hot, helium-burning stars. Now Hubble provides the first direct visual evidence.

Nearby elliptical galaxies are thought to be relatively simple galaxies comprised of old stars. Because they are among the brightest objects in the Universe, this simplicity makes them useful for tracing the evolution of stars and galaxies.

Early Cosmology:

Cosmology is the study of the Universe and its components, how it formed, how it has evolved and what is its future. Modern cosmology grew from ideas before recorded history. Ancient man asked questions such as "What's going on around me?" which then developed into "How does the Universe work?", the key question that cosmology asks.

Many of the earliest recorded scientific observations were about cosmology, and the pursuit of understanding has continued for over 5000 years. Cosmology has exploded in the last 10 years with radically new information about the structure, origin and evolution of the Universe obtained through recent technological advances in telescopes and space observatories and basically has become a search for the understanding of not only what makes up the Universe (the objects within it) but also its overall architecture.



Modern cosmology is on the borderland between science and philosophy, close to philosophy because it asks fundamental questions about the Universe, close to science since it looks for answers in the form of empirical understanding by observation and rational explanation. Thus, theories about cosmology operate with a tension between a philosophical urge for simplicity and a wish to include all the Universe's features versus the sheer complexity of it all.

Very early cosmology, from Neolithic times of 20,000 to 100,000 years ago, was extremely local. The Universe was what you immediately interacted with. Things outside your daily experience appeared supernatural, and so we call this time the Magic Cosmology.



Later in history, 5,000 to 20,000 years ago, humankind begins to organize themselves and develop what we now call culture. A greater sense of permanence in your daily existences leads to the development of myths, particularly creation myths to explain the origin of the Universe. We call this the Mythical Cosmology.



The third stage, what makes up the core of modern cosmology grew out of ancient Greek, later Christian, views. The underlying theme here is the use of observation and experimentation to search for simple, universal laws. We call this the Geometric Cosmology.

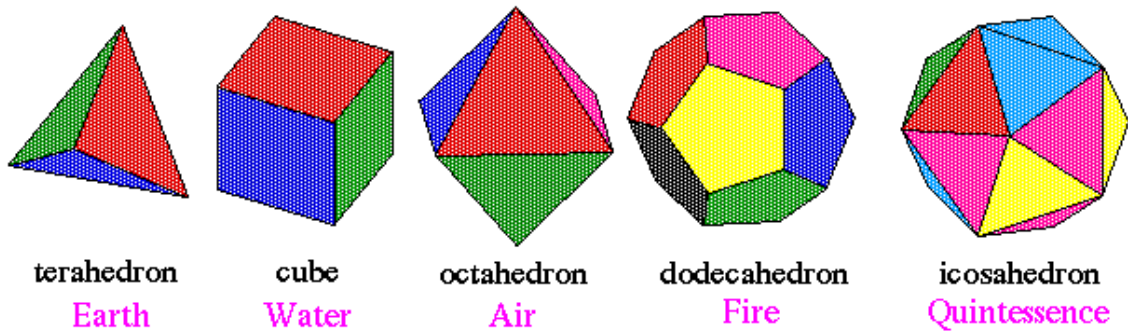


The earliest beginnings of science was to note that there exist patterns of cause and effect that are manifestations of the Universe's rational order. We mostly develop this idea as small children (touch hot stove = burn/pain). But the extrapolation of a rational order to cosmology requires a leap of faith in the beginning years of science, later supported by observation and experimentation.

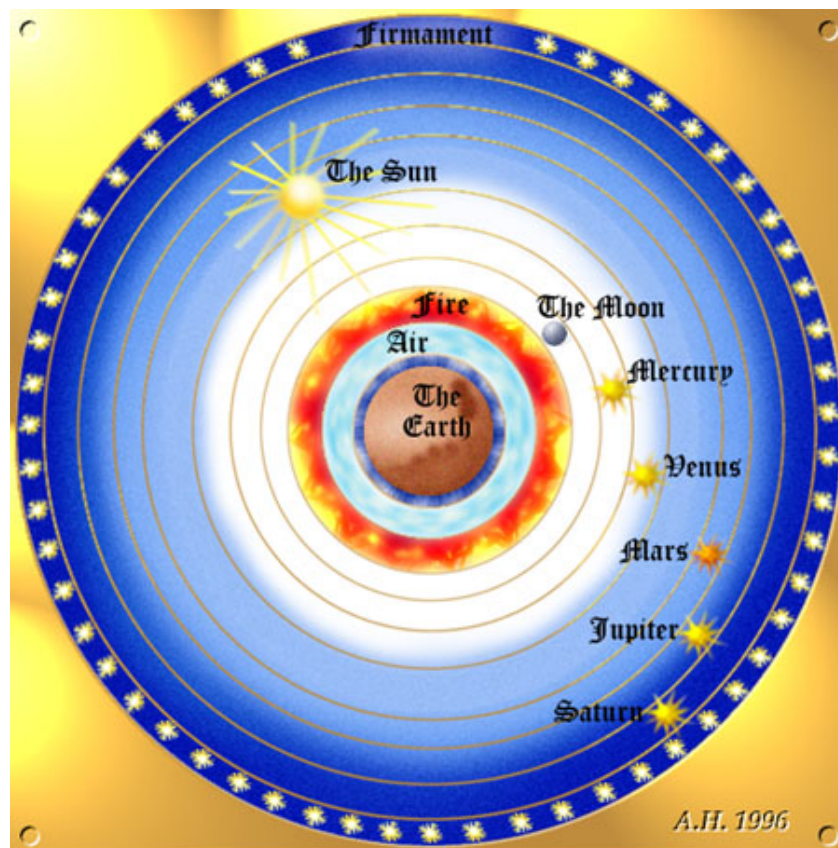
Greek Cosmology

The earliest cosmology was an extrapolation of the Greek system of four elements in the Universe (earth, water, fire, air) and that everything in the Universe is made up of some combination of these four primary elements. In a seemingly unrelated discovery, [Euclid](#), a Greek mathematician, proofed that there are only five solid shapes that can be made from simple polygons (the triangle, square and hexagon). [Plato](#), strongly influenced by this pure mathematical discovery, revised the four element theory with the proposition that there were five elements to the Universe (earth, water, air, fire and quintessence) in correspondence with the five regular solids.

The five Platonic solids



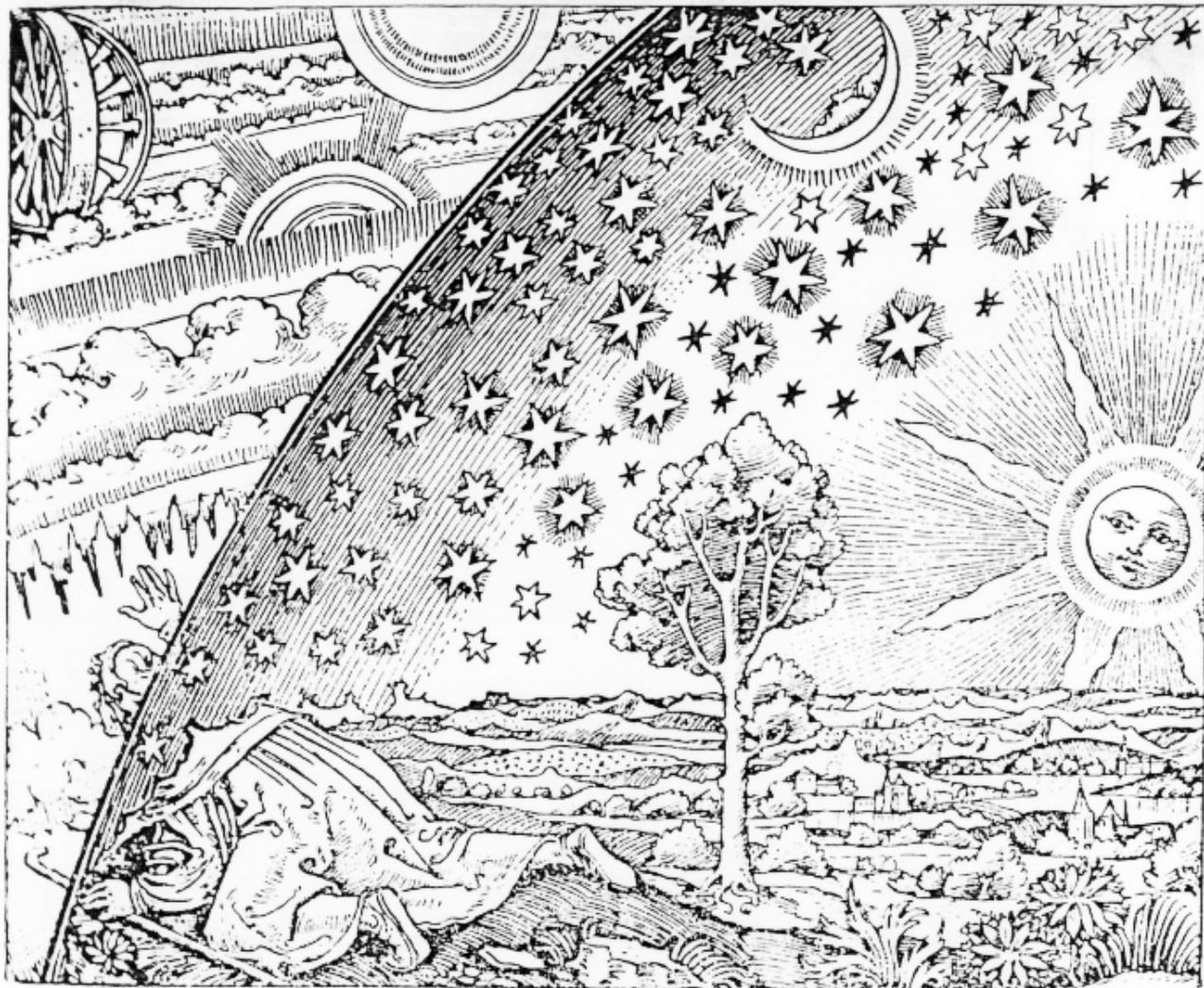
Each of these five elements occupied a unique place in the heavens (earth elements were heavy and, therefore, low; fire elements were light and located up high). Thus, Plato's system also became one of the first cosmological [models](#) and looked something like the following diagram:



Like any good scientific model, this one offers explanations and various predictions. For example, hot air rises to reach the sphere of Fire, so heated balloons go up. Note that this model also predicts some incorrect things, such as all the planets revolve around the Earth, called the [geocentric theory](#).

Middle Ages

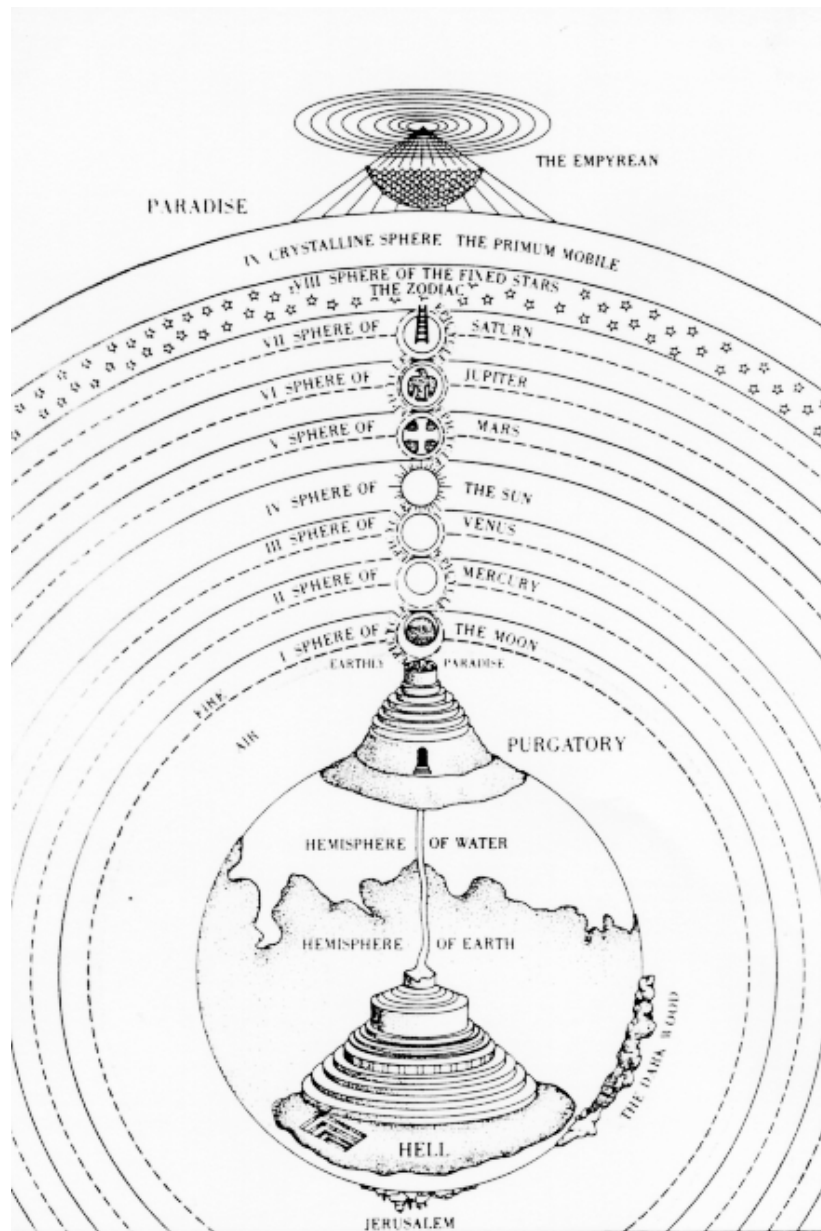
The distinction between what makes up matter (the primary elements) and its form became a medieval Christian preoccupation, with the sinfulness of the material world opposed to the holiness of the heavenly realm. The medieval Christian cosmology placed the heavens in a realm of perfection, derived from Plato's [Theory of Forms](#)



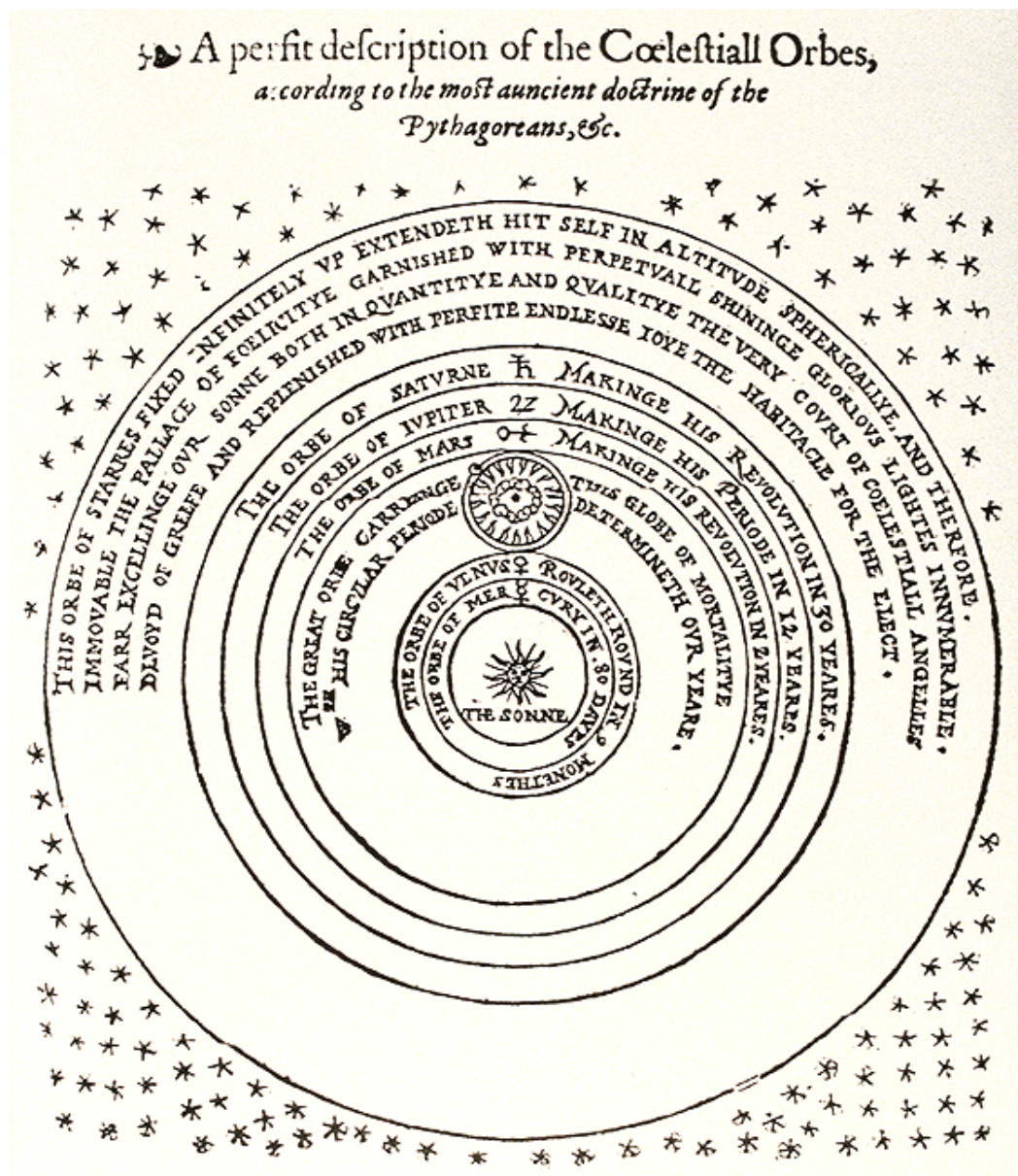
SPACE WAS FINITE and had a definite edge, according to the Aristotelian cosmology accepted during medieval times. Here a man is shown looking beyond the edge of space to the Empyrean abode of God beyond. The illustration is often said to be a 16th-century German woodcut; according to Owen Gingerich of Harvard University, it is more likely a piece of art nouveau that was apparently published for the first time in 1907 in *Weltall und Menschheit*, edited by

Hans Kraemer. In either case the picture clearly demonstrates a dilemma posed by Immanuel Kant known as Kant's antinomy of space. Kant believed that the universe had to be finite in extent and homogeneous in composition, and that space had to obey the laws of Euclidean geometry. Actually, however, all those assumptions cannot be true at once. Newton, Leibniz and Einstein had different ways of resolving the dilemma, shown in illustrations on next two pages.

Before the scientific method was fully developed, many cosmological models were drawn from religious or inspirational sources. One such was the following scheme taken from [Dante's](#) 'The Divine Comedy'.



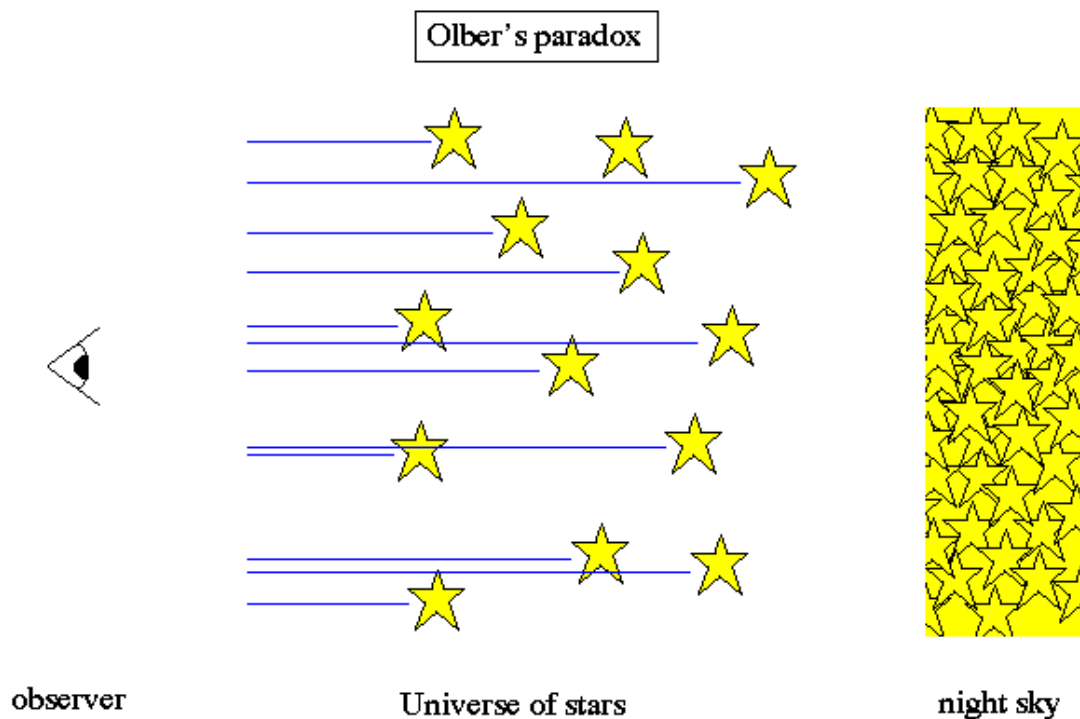
The political and intellectual authority of the medieval church declined with time, leading to the creative anarchy of the Renaissance. This produced a scientific and philosophical revolution including the birth of modern physics. Foremost to this new style of thinking was a strong connection between ideas and facts (the scientific method).



Since cosmology involves observations of objects very far away (therefore, very faint) advancement in our understanding of the cosmos has been very slow due to limits in our technology. This has changed dramatically in the last few years with the construction of large telescopes and the launch of space-based observatories.

Olber's Paradox:

The oldest cosmological [paradox](#) concerns the fact that the night sky should appear as bright as the surface of the Sun in a very large (or infinite), ageless Universe.



observer

Universe of stars

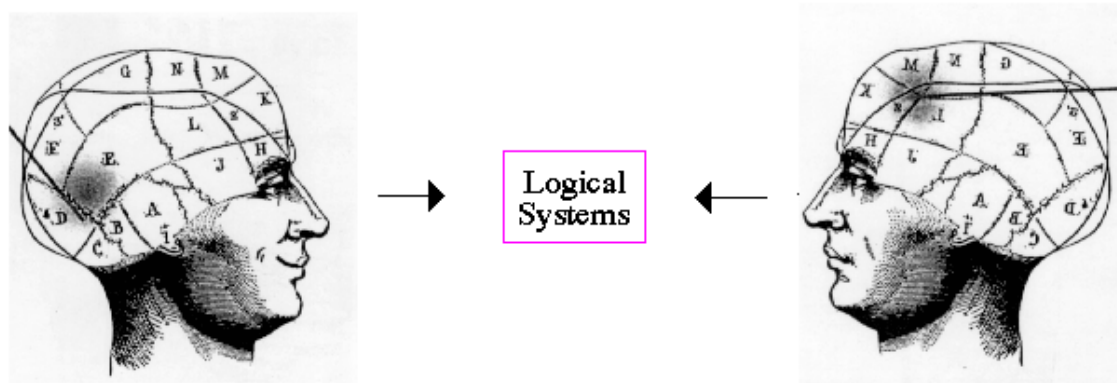
night sky

Note that the paradox cannot be resolved by assuming that parts of the Universe are filled with absorbing dust or dark matter, because eventually that material would heat up and emit its own light.

The resolution of Olber's paradox is found in the combined observation that 1) the speed of light is finite (although a very high velocity) and 2) the Universe has a finite age, i.e. we only see the light from parts of the Universe less than 15 billion light years away.

Rationalism:

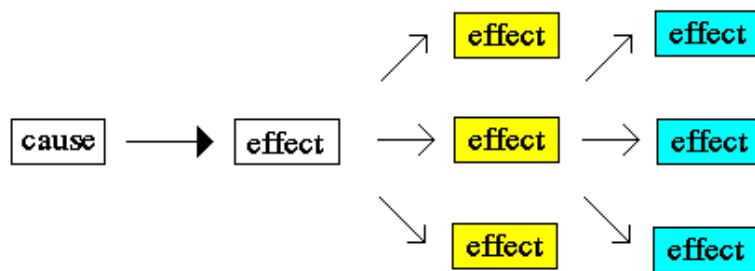
The main purpose of science is to trace, within the chaos and flux of phenomena, a consistent structure with order and meaning. This is called the philosophy of [rationalism](#). The purpose of scientific understanding is to coordinate our experiences and bring them into a [logical system](#).



Throughout history, intellectual efforts are directed towards the discovery of pattern, system and structure, with a special emphasis on order. Why? control of the unpredictable, fear of the unknown, and a person who seeks to understand and discover is called a [scientist](#).

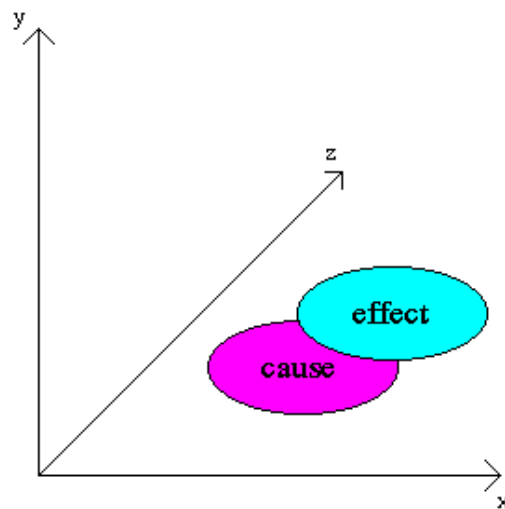
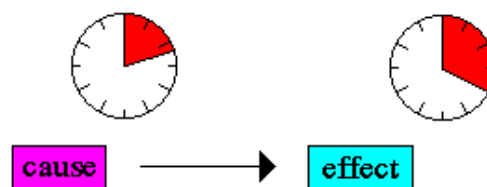
Cause+Effect:

The foundation for rationalism rests squarely on the principle of locality, the idea that correlated events are related by a chain of causation.



There are three components to cause and effect:

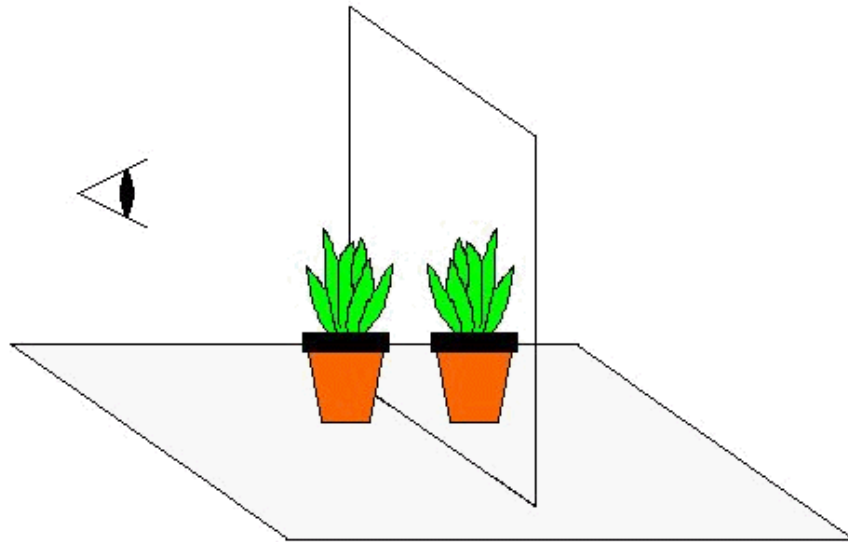
- contiguity in space
- temporal priority of the cause (i.e. its first)
- necessary connection

spatial connection**temporal connection****energy connection**



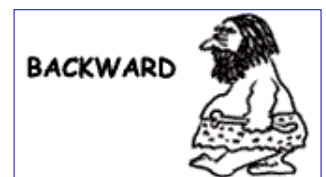
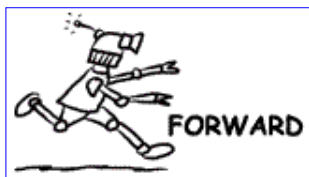
The necessary connection in cause and effect events is the exchange of [energy](#), which is the foundation of information theory => knowledge is power (energy).

Also key to cause and effect is the concept that an object's existence and properties are independent of the observation or experiment and rooted in [reality](#).



objects in a mirror are 'unreal' in the sense of not having existence in the material world

Causal links build an existence of patterns that are a manifestation of the Universe's rational order. Does the chain of cause and effect ever end? Is there an 'Initial Cause'?

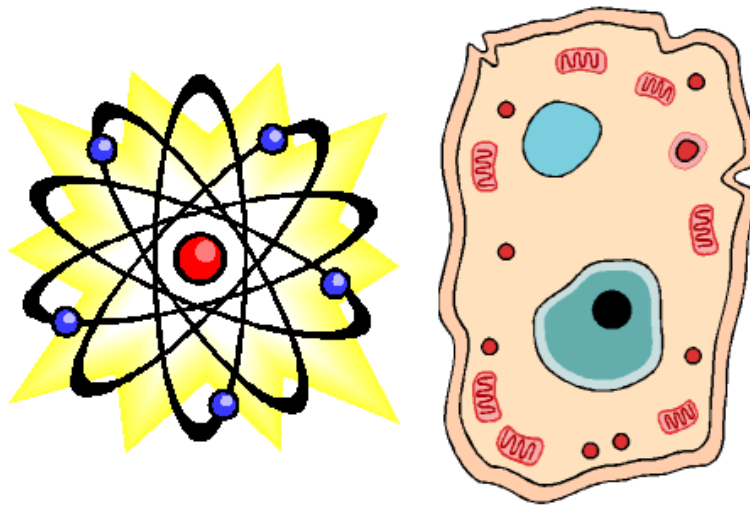


Science:

The tool of the philosophy of rationalism is called science. [Science](#) is any system of knowledge that is concerned with the physical world and its phenomena and entails unbiased observations and/or systematic experimentation. In general, a science involves a pursuit of knowledge covering general truths or the operations of fundamental [laws of nature](#).



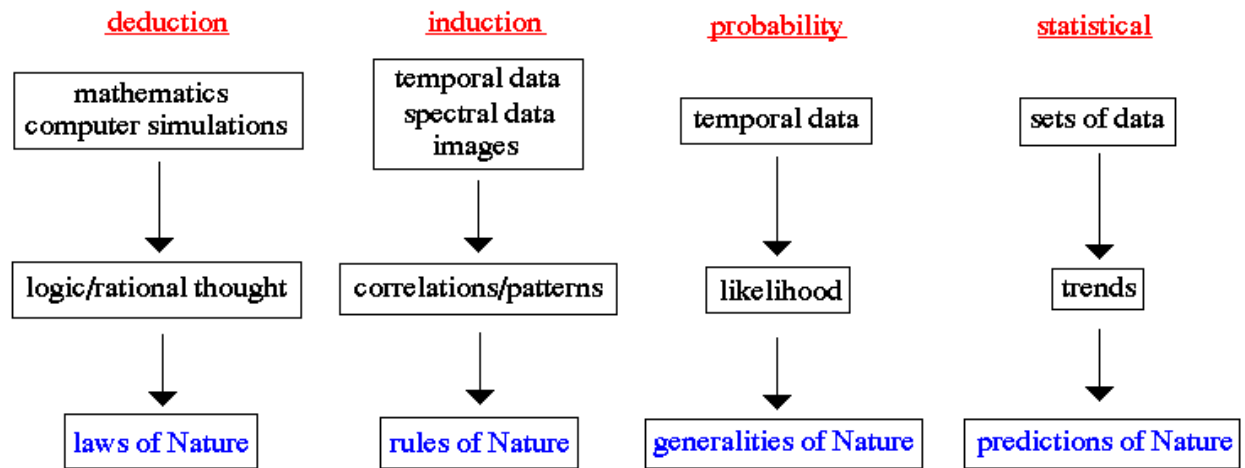
Science is far from a perfect instrument of knowledge, but it provides something that other philosophies often fail to provide, concrete results. Science is a "candle in the dark" to illuminate [irrational beliefs or superstitions](#).



Science does not, by itself, advocate courses of human action, but it can certainly illuminate the possible consequences of alternative courses. In this regard, science is both imaginative and disciplined, which is central to its power of prediction.

The keystone to science is proof or evidence/data, which is not to be confused with certainty. Except in pure mathematics, nothing is known for certain (although much is certainly false). Central to the scientific method is a system of [logic](#).

Scientific arguments of logic basically take on four possible forms; 1) the pure method of deduction, where some conclusion is drawn from a set of propositions (i.e. pure logic), 2) the method of induction, where one draws general conclusions from particular facts that appear to serve as evidence, 3) by probability, which passes from frequencies within a known domain to conclusions of stated likelihood, and 4) by statistical reasoning, which concludes that, on the average, a certain percentage of a set of entities will satisfy the stated conditions. To support these methods, a scientist also uses a large amount of [skepticism](#) to search for any [fallacies](#) in arguments.



The fact that scientific reasoning is so often successful is a remarkable property of the Universe, the dependability of Nature.

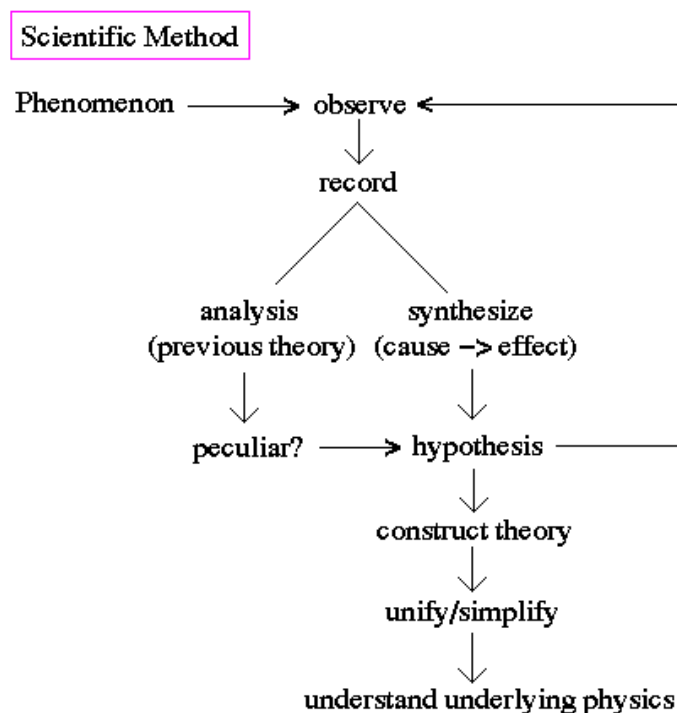
Scientific Method:

Of course, the main occupation of a scientist is [problem solving](#) with the goal of understanding the Universe. To achieve this goal, a scientist applies the scientific method. The scientific method is the rigorous standard of procedure and discussion that sets reason over irrational belief. The process has four steps:

- observation/experimentation
- deduction
- hypothesis
- falsification

Note the special emphasis on falsification, not verification. A powerful hypothesis is one that is actually highly vulnerable to falsification and that can be tested in many ways.

The underlying purpose of the scientific method is the construction of simplifying ideas, models and theories, all with the final goal of understanding.

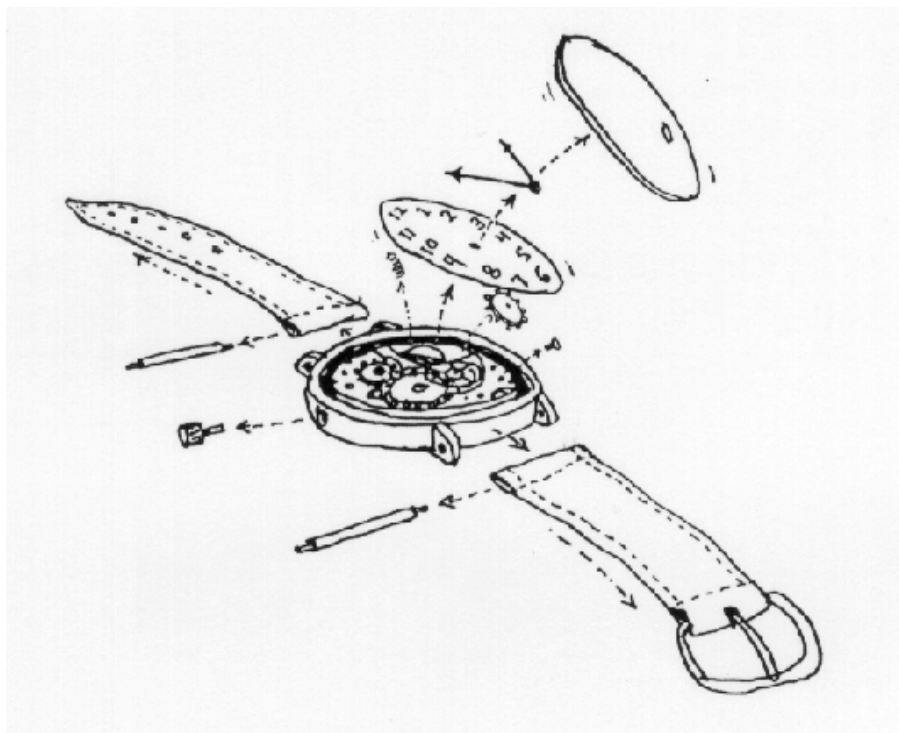


The only justification for our concepts of `electron', `mass', `energy', or `time' is that they serve to represent the complexity of our experiences. It is an ancient debate on whether humankind invents or discovers physical laws. Whether natural laws exist independent of our culture or whether we impose these laws on Nature as crude approximations.

Science can be separated from pseudo-science by the [principle of falsifiability](#), the concept that ideas must be capable of being proven false in order to be scientifically valid.

Reductionism:

[Reductionism](#) is the belief that any complex set of phenomena can be defined or explained in terms of a relatively few simple or primitive ones.



For example, atomism is a form of reductionism in that it holds that everything in the Universe can be broken down into a few simple entities (elementary particles) and laws to describe the interactions between them. This idea became modern chemistry which reduces all chemical properties to ninety or so basic elements (kinds of atoms) and their rules of combination.

Reductionism is very similar to, and has its roots from, [Occam's Razor](#), which states that between competing ideas, the simplest theory that fits the facts of a problem is the one that should be selected.

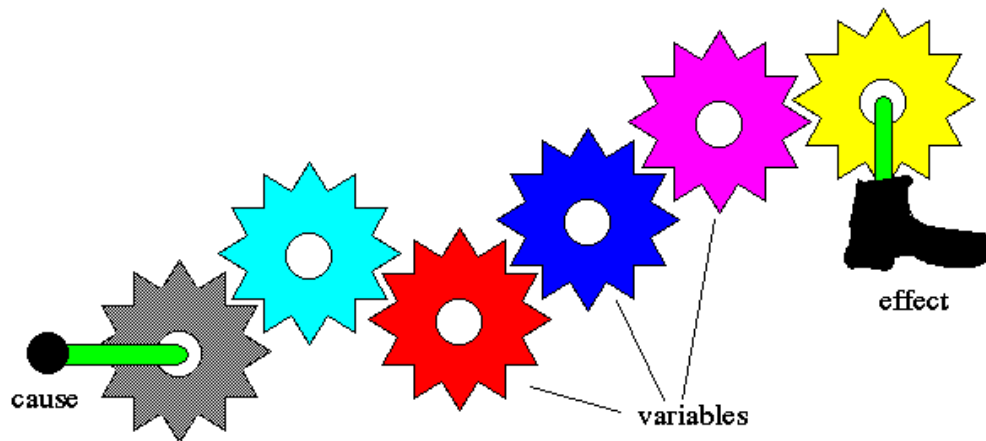
Reductionism was widely accepted due to its power in prediction and formulation. It is, at least, a good approximation of the macroscopic world (although it is completely wrong for the microscope world, see quantum physics).

Too much success is a dangerous thing since the reductionist philosophy led to a wider [paradigm](#), the methodology of [scientism](#), the view that everything can and should be reduced to the properties of matter ([materialism](#)) such that emotion, aesthetics and religious experience can be reduced to biological instinct, chemical imbalances in the brain, etc. The 20th century reaction against reductionism is [relativism](#). Modern science is somewhere in between.

Determinism:

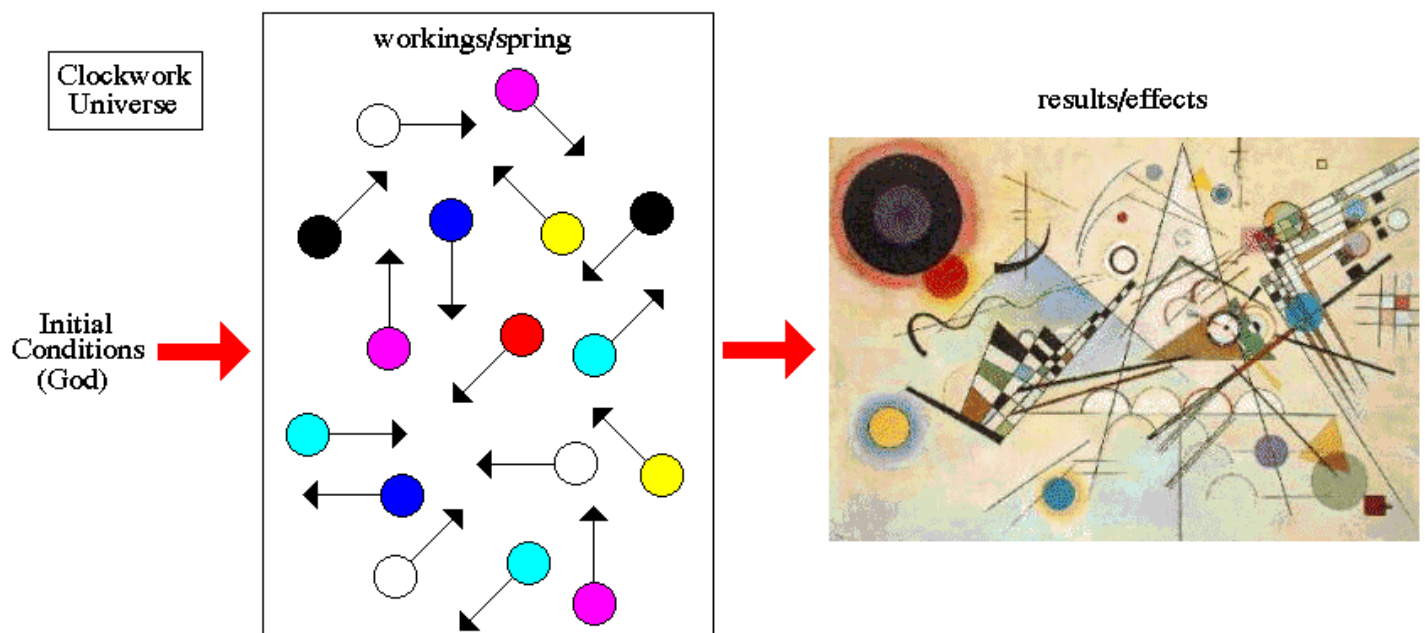
Closely associated with reductionism is [determinism](#), the philosophy that everything has a cause, and that a particular cause leads to a unique effect. Another way of stating this is that for everything that happens there are conditions such that, given them, nothing else could happen, the outcome is determined.

Determinism



Determinism implies that everything is predictable given enough information.

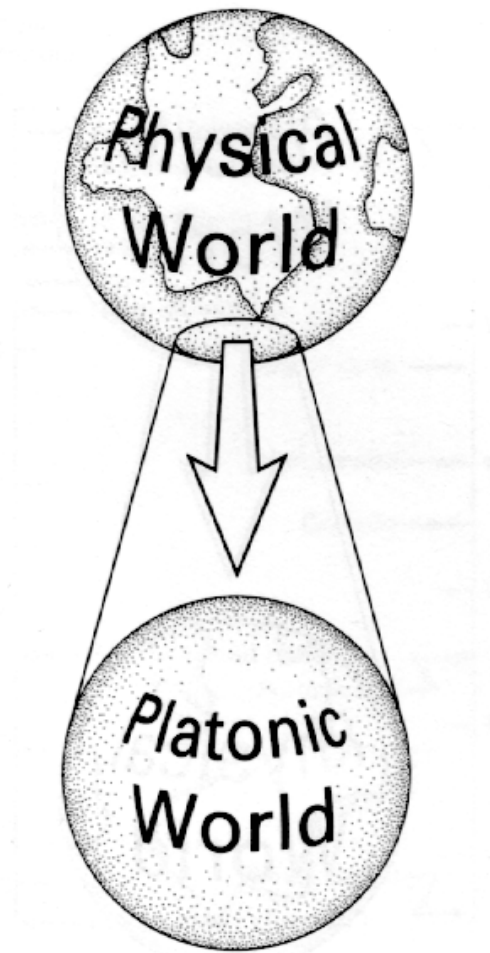
[Newtonian or classical physics](#) is rigidly determinist, both in the predictions of its equations and its foundations, there is no room for chance, surprise or creativity. Everything is as it has to be, which gave rise to the concept of a [clockwork Universe](#).



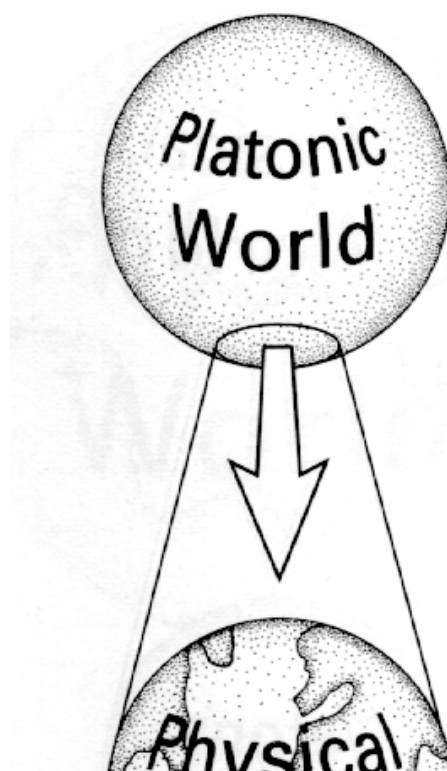
Mathematics and Science:

The belief that the underlying order of the Universe can be expressed in mathematical form lies at the heart of science and is rarely questioned. But whether mathematics a human invention or if it has an independent existence is a question for [metaphysics](#).

There exists two schools of thought. One that mathematical concepts are mere idealizations of our physical world. The world of absolutes, what is called the Platonic world, has existence only through the physical world. In this case, the mathematical world would be thought of as emerging from the world of physical objects.



The other school is attributed to Plato, and finds that Nature is a structure that is precisely governed by timeless mathematical laws. According to Platonists we do not invent mathematical truths, we discover them. The Platonic world exists and physical world is a shadow of the truths in the Platonic world. This reasoning comes about when we realize (through thought and experimentation) how the behavior of Nature follows mathematics to an extremely high degree of accuracy. The deeper we probe the laws of Nature, the more the physical world disappears and becomes a world of pure math.





Mathematics transcends the physical reality that confronts our senses. The fact that mathematical theorems are discovered by several investigators indicates some objective element to mathematical systems. Since our brains have evolved to reflect the properties of the physical world, it is of no surprise that we discover mathematical relationships in Nature.

Galileo's Laws of Motion:

Galileo Galilei stressed the importance of obtaining knowledge through precise and quantitative experiment and observation. Man and Nature are considered distinct and experiment was seen as a sort of dialogue with Nature. Nature's rational order, which itself is derived from God, was manifested in definite laws.



Aside from his numerous inventions, [Galileo](#) also laid down the first accurate laws of motion for masses. Galileo realized that all bodies accelerate at the same rate regardless of their size or mass. Everyday experience tells you differently because a feather falls slower than a cannonball. Galileo's genius lay in spotting that the differences that occur in the everyday world are in incidental complication (in this case, air friction) and are irrelevant to the real underlying properties (that is, gravity) which is pure mathematical in its form. He was able to abstract from the complexity of real-life situations the simplicity of an idealized law of gravity.

Key among his investigations are:

- developed the concept of motion in terms of [velocity](#) (speed and direction) through the use of inclined planes.
- developed the idea of [force](#), as a cause for motion.
- determined that the natural state of an object is rest or uniform motion, i.e. objects always have a velocity, sometimes that velocity has a magnitude of zero = rest.
- objects resist change in motion, which is called [inertia](#).

Galileo also showed that objects fall with the same speed regardless of their mass. The fact that a feather falls slowly than a steel ball is due to amount of air resistance that a feather experiences (alot) versus the steel ball (very little).

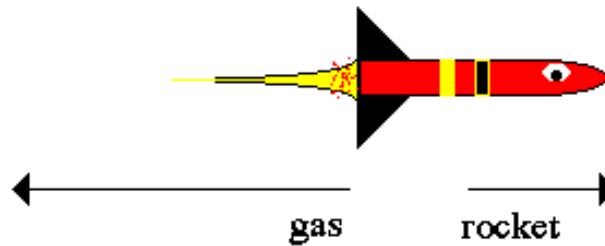


Newtonian Physics:

Newtonian or classical physics is reductionist, holding that all physical reality can be reduced to a few particles and the laws and forces acting among them. Newtonian physics is free of spiritual or psychological forces = emphasis on [objectivity](#).

[Newton](#) expanded on the work of Galileo to better define the relationship between energy and motion. In particular, he developed the following concepts:

- the change in velocity of an object is called acceleration, and is caused by a force

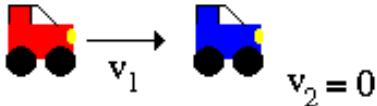


- The resistance an object has to changes in velocity is called inertia and is proportional to its mass
- Momentum is a quantity of motion (kinetic) energy and is equal to mass times velocity

$$\text{momentum} = \text{mass} \times \text{velocity}$$

if the mass ↑ then the momentum ↑
 if the velocity ↑ then the momentum ↑

Key to the clockwork universe concept are the conservation laws in Newtonian physics. Specifically, the idea that the total momentum of an interaction is conserved (i.e. it is the same before and after).

Before  $mv_1 + m(0) = mv_1$

After  $m(0) + mv_\gamma = mv_\gamma$
therefore, $v_\gamma = v_1$

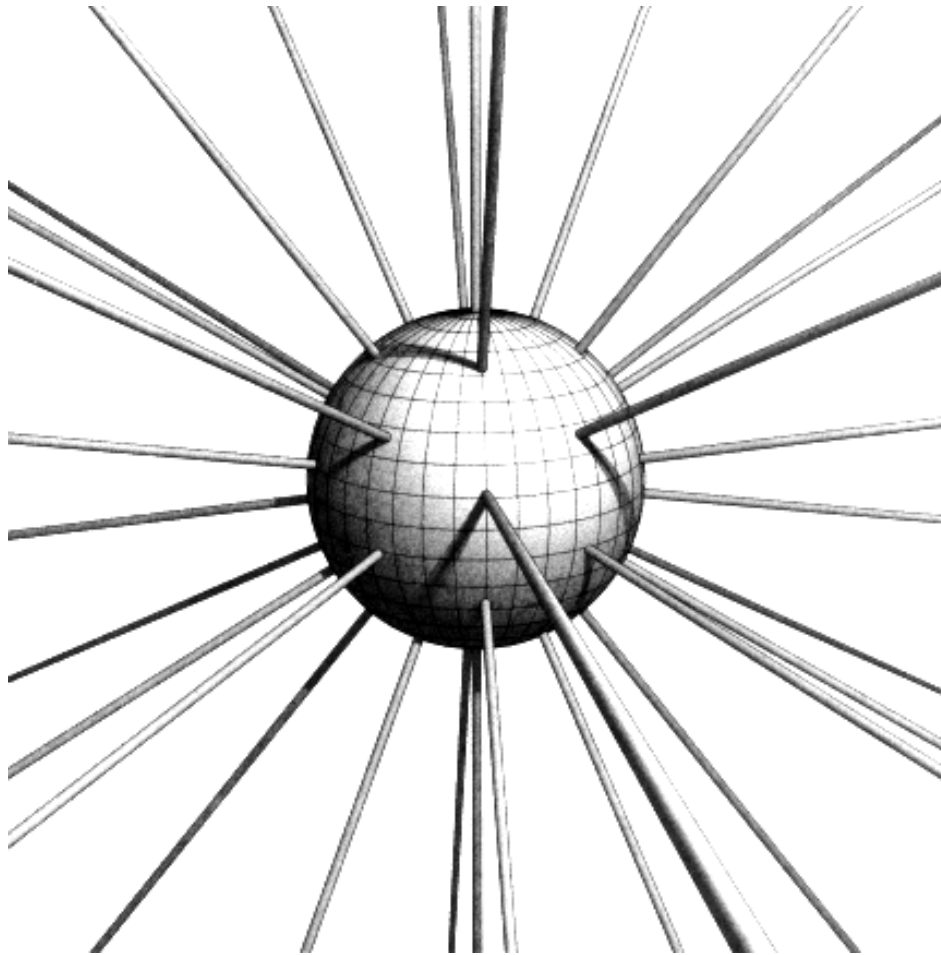
Before  $mv_1 + 5m(0) = mv_1$

After  $m(0) + 5mv_\gamma = 5mv_\gamma$
therefore, $v_\gamma = \frac{v_1}{5}$

Conservation laws allow detailed predictions from initial conditions, a highly deterministic science.

Newton's Law of Universal Gravitation:

Galileo was the first to notice that objects are ``pulled" towards the center of the Earth, but Newton showed that this same force (gravity) was responsible for the orbits of the planets in the Solar System.



Objects in the Universe attract each other with a force that varies directly as the product of their masses and inversely as the square of their distances

Newton's Law of Universal Gravitation

The force of gravity, F_g , is given by

$$F_g = \frac{G m_1 m_2}{R^2}$$

where,

G = gravitational constant = $6.668 \times 10^{-8} \text{ dynes cm}^3 \text{ g}^{-2}$

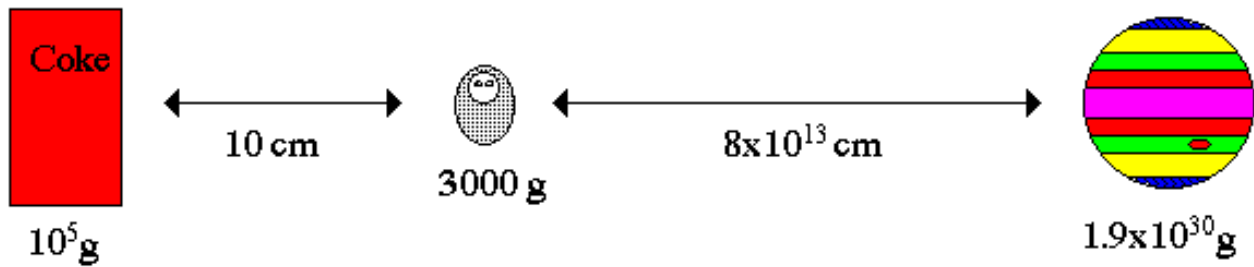
m_1 = mass of object #1

m_2 = mass of object #2

R = distance between the objects

All masses, regardless of size, attract other masses with gravity. You don't notice the force from nearby objects because their mass is so small compared to the mass of the Earth. Consider the following example:

Example: what is the force of gravity from Jupiter on you at birth compared to the force of gravity from a nearby Coke machine?



$$F_{\text{Coke}} = \frac{G(10^5 \text{ g})(3000 \text{ g})}{(10 \text{ cm})^2}$$

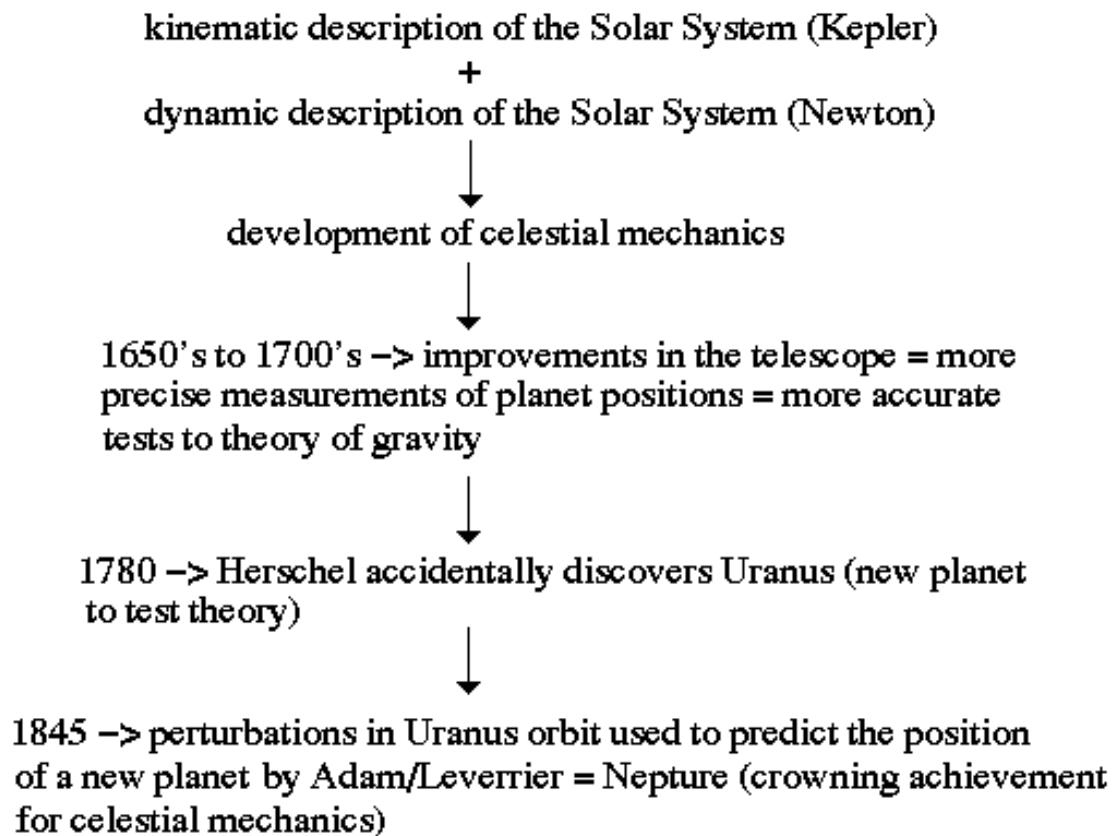
$$= 2 \times 10^{-1} \text{ dynes}$$

$$F_{\text{Jupiter}} = \frac{G(1.9 \times 10^{30} \text{ g})(3000 \text{ g})}{(8 \times 10^{13} \text{ cm})^2}$$

$$= 5.9 \times 10^{-2} \text{ dynes}$$

therefore, the force of gravity from the Coke machine is about 3 times more than the force of gravity from Jupiter.

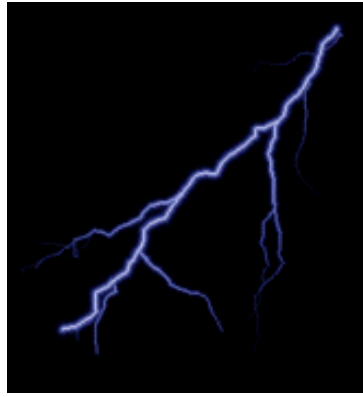
Newton's development of the underlying cause of planetary motion, gravity, completed the [solar system model](#) begun by the Babylonians and early Greeks. The mathematical formulation of Newton's dynamic model of the solar system became the science of celestial mechanics, the greatest of the deterministic sciences.



Although Newtonian mechanics was the grand achievement of the 1700's, it was by no means the final answer. For example, the equations of orbits could be solved for two bodies, but could *not* be solved for three or more bodies. The [three body problem](#) puzzled astronomers for years until it was learned that some mathematical problems suffer from [deterministic chaos](#), where dynamical systems have apparently random or unpredictable behavior.

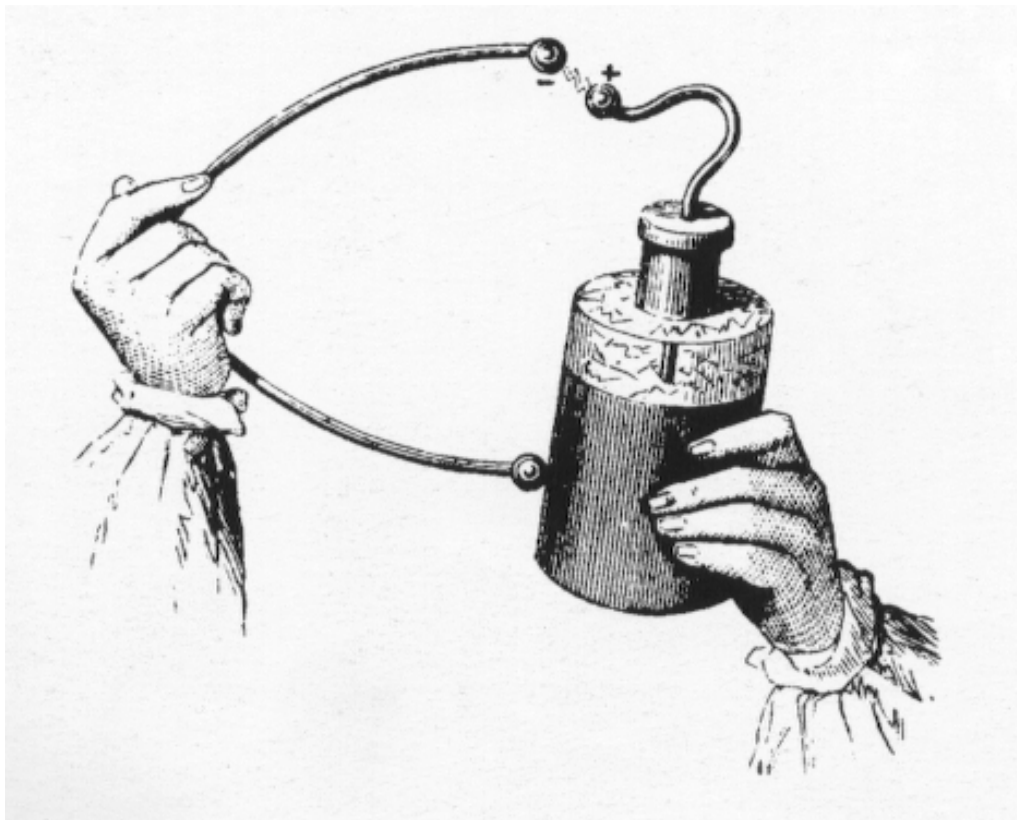
Electricity:

The existence of [electricity](#), the phenomenon associated with stationary or moving electric charges, has been known since the Greeks discovered that amber, rubbed with fur, attracted light objects such as feathers. Ben Franklin proved the electrical nature of lightning (the famous key experiment) and also established the conventional use of negative and positive types of charges.



By the 18th century, physicist Charles Coulomb defined the quantity of electricity (electric charge) later known as a coulomb, and determined the force law between electric charges, known as Coulomb's law. Coulomb's law is similar to the law of gravity in that the electrical force is inversely proportional to the distance of the charges squared, and proportional to the product of the charges.

By the end of the 18th century, we had determined that electric charge could be stored in a conducting body if it is insulated from its surroundings. The first of these devices was the Leyden jar, consisted of a glass vial, partly filled with sheets of metal foil, the top of which was closed by a cork pierced with a wire or nail. To charge the jar, the exposed end of the wire is brought in contact with a friction device.

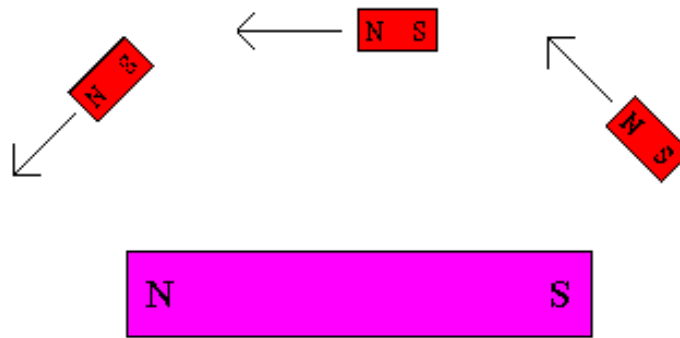


Modern atomic theory explains this as the ability of atoms to either lose or gain an outer electron and thus exhibit a net positive charge or gain a net negative charge (since the electron is negative). Today we know that the basic quantity of electric charge is the electron, and one coulomb is about 6.24×10^{18} electrons.

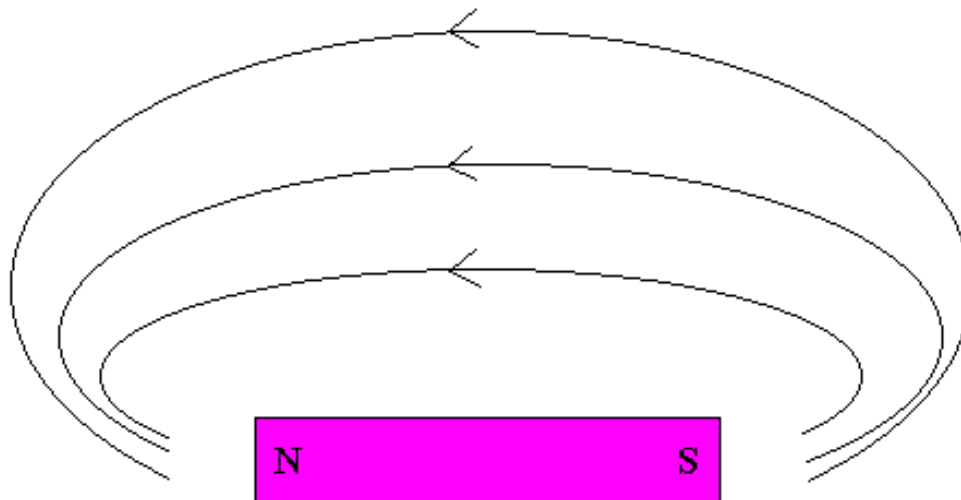
The battery was invented in the 19th century, and electric current and static electricity were shown to be manifestations of the same phenomenon, i.e. current is the motion of electric charge. Once a laboratory curiosity, electricity becomes the focus of industrial concerns when it is shown that electrical power can be transmitted efficiently from place to place and with the invention of the incandescent lamp.

to map a field,

use test particles to map the forces, then

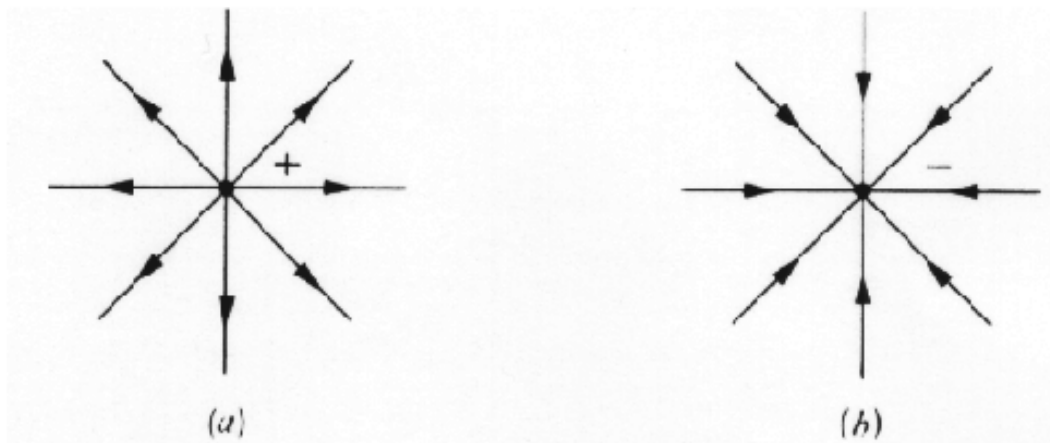


connect the dots

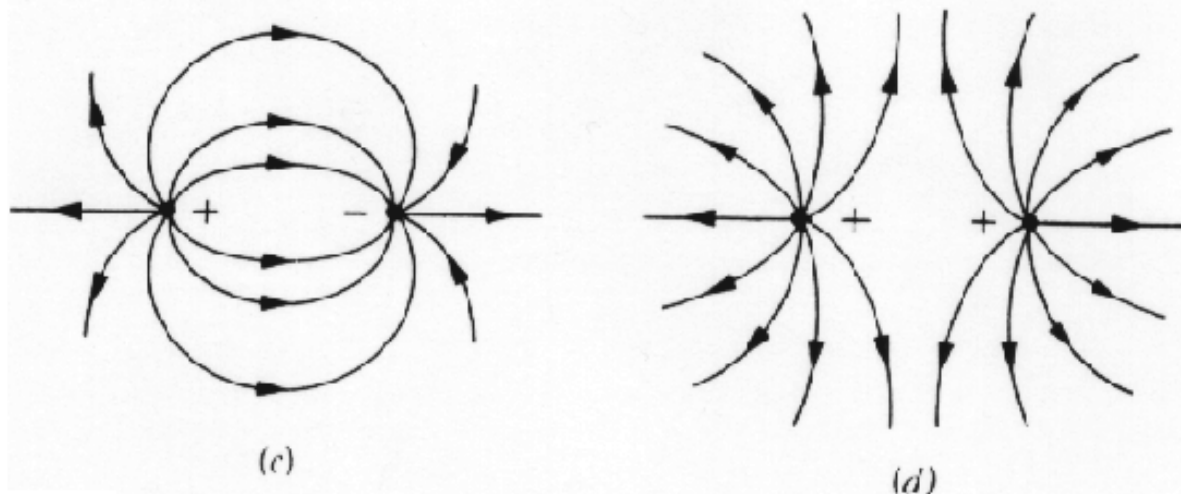


The discovery of Coulomb's law, and the behavior or motion of charged particles near other charged particles led to the development of the electric field concept. A field can be considered a type of energy in space, or energy with position. A field is usually visualized as a set of lines surrounding the body, however these lines do not exist, they are strictly a mathematical construct to describe motion. Fields are used in electricity, magnetism, gravity and almost all aspects of modern physics.

A single, isolated charge acts as a source of an electric field (a) or a sink (b)



The field of two charges has a complicated shape, each charge disturbs the field of the other



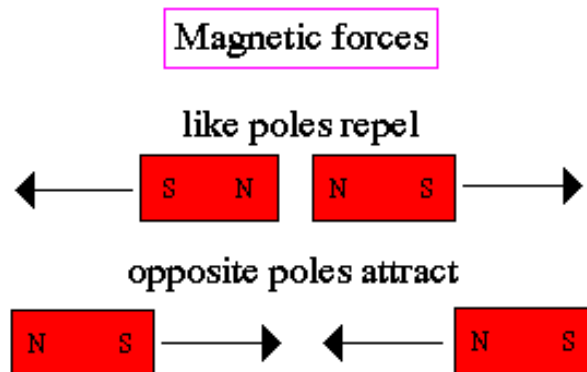
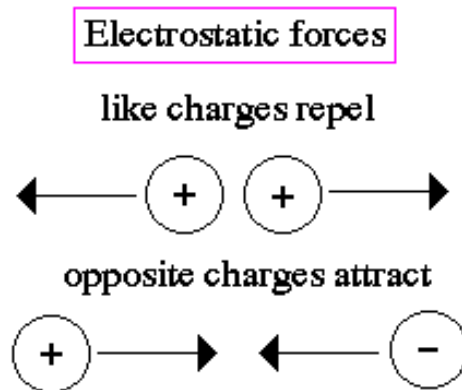
Opposite charges attract reflected by the field lines which link them together (c). Like charges repel, no field lines connect them (d).

An electric field is the region around an electric charge in which an electric force is exerted on another charge. Instead of considering the electric force as a direct interaction of two electric charges at a distance from each other, one charge is considered the source of an electric field that extends outward into the surrounding space, and the force exerted on a second charge in this space is considered as a direct interaction between the electric field and the second charge.

Magnetism:

Magnetism is the phenomenon associated with the motion of electric charges,

although the study of magnets was very confused before the 19th century because of the existence of ferromagnets, substances such as iron bar magnets which maintain a magnetic field where no obvious electric current is present (see below). Basic magnetism is the existence of magnetic fields which deflect moving charges or other magnets. Similar to electric force in strength and direction, magnetic objects are said to have 'poles' (north and south, instead of positive and negative charge). However, magnetic objects are always found in pairs, there do not exist isolated poles in Nature.



The most common source of a magnetic field is an electric current loop. The motion of electric charges in a pattern produces a magnetic field and its associated magnetic force. Similarly, spinning objects, like the Earth, produce magnetic fields, sufficient to deflect compass needles.

Today we know that permanent magnets are due to dipole charges inside the magnet at the atomic level. A dipole charge occurs from the spin of the electron around the nucleus of the atom. Materials (such as metals) which have incomplete electron shells will have a net magnetic moment. If the material has a highly ordered crystalline pattern (such as iron or nickel), then the local magnetic fields of the atoms become coupled and the material displays a large scale bar magnet behavior.

Electromagnetism:

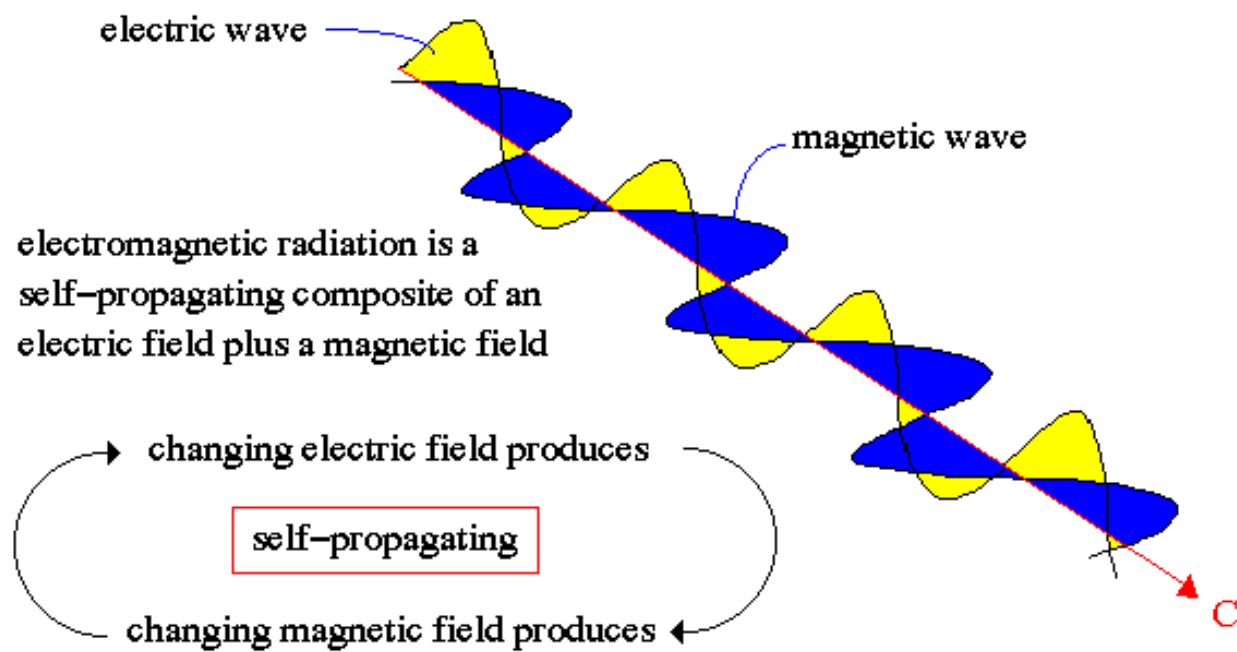
Although conceived of as distinct phenomena until the 19th century, [electricity](#)

and [magnetism](#) are now known to be components of the unified theory of [electromagnetism](#).

A connection between electricity and magnetism had long been suspected, and in 1820 the Danish physicist Hans Christian Orsted showed that an electric current flowing in a wire produces its own magnetic field. Andre-Marie Ampere of France immediately repeated Orsted's experiments and within weeks was able to express the magnetic forces between current-carrying conductors in a simple and elegant mathematical form. He also demonstrated that a current flowing in a loop of wire produces a magnetic dipole indistinguishable at a distance from that produced by a small permanent magnet; this led Ampere to suggest that magnetism is caused by currents circulating on a molecular scale, an idea remarkably near the modern understanding.

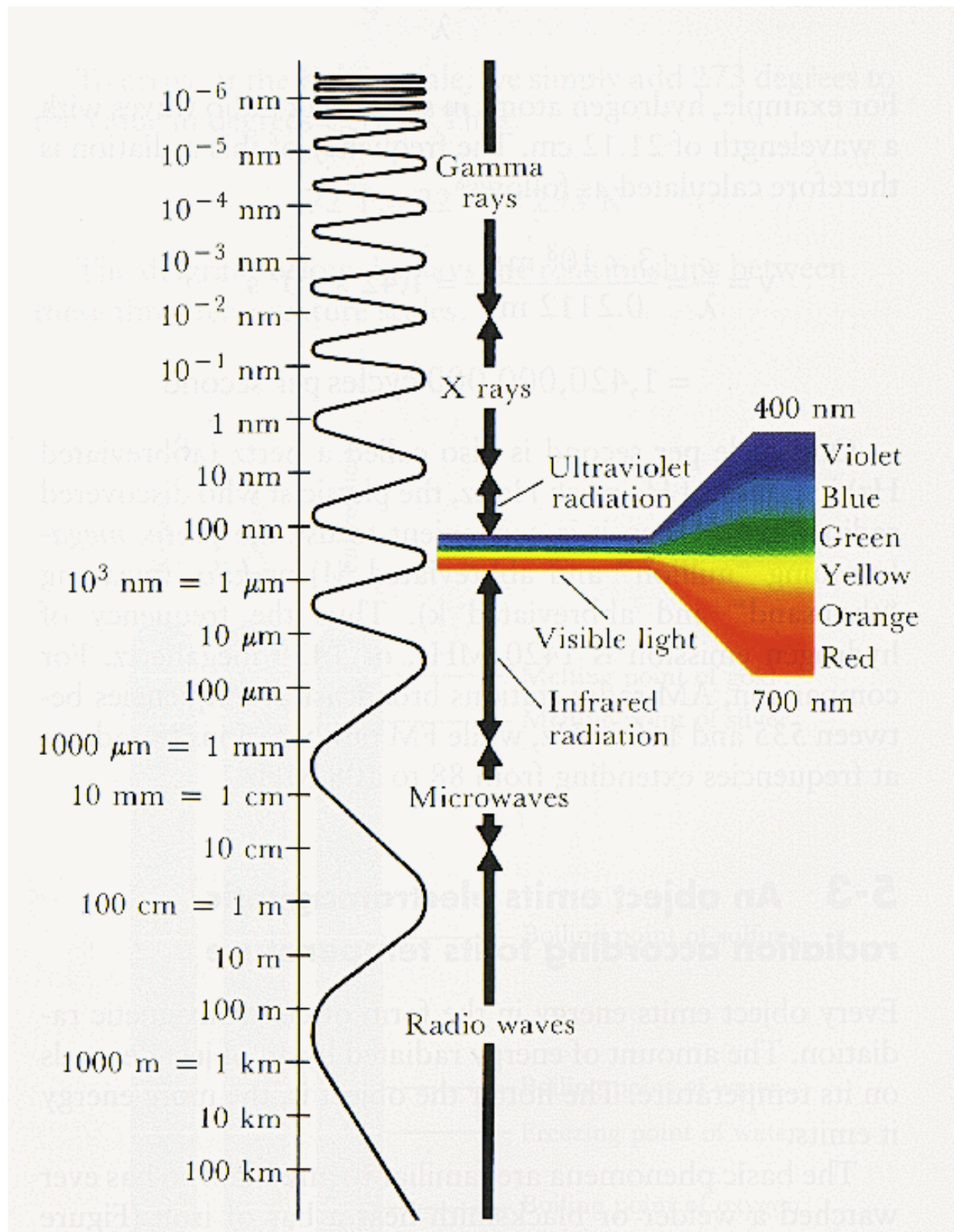
[Faraday](#), in the early 1800's, showed that a changing electric field produces a magnetic field, and that vice-versus, a changing magnetic field produces an electric current. An electromagnet is an iron core which enhances the magnetic field generated by a current flowing through a coil, was invented by William Sturgeon in England during the mid-1820s. It later became a vital component of both motors and generators.

The unification of electric and magnetic phenomena in a complete mathematical theory was the achievement of the Scottish physicist [Maxwell](#) (1850's). In a set of four elegant equations, Maxwell formalized the relationship between electric and magnetic fields. In addition, he showed that a linear magnetic and electric field can be self-reinforcing and must move at a particular velocity, the speed of light. Thus, he concluded that light is energy carried in the form of opposite but supporting electric and magnetic fields in the shape of waves, i.e. self-propagating electromagnetic waves.

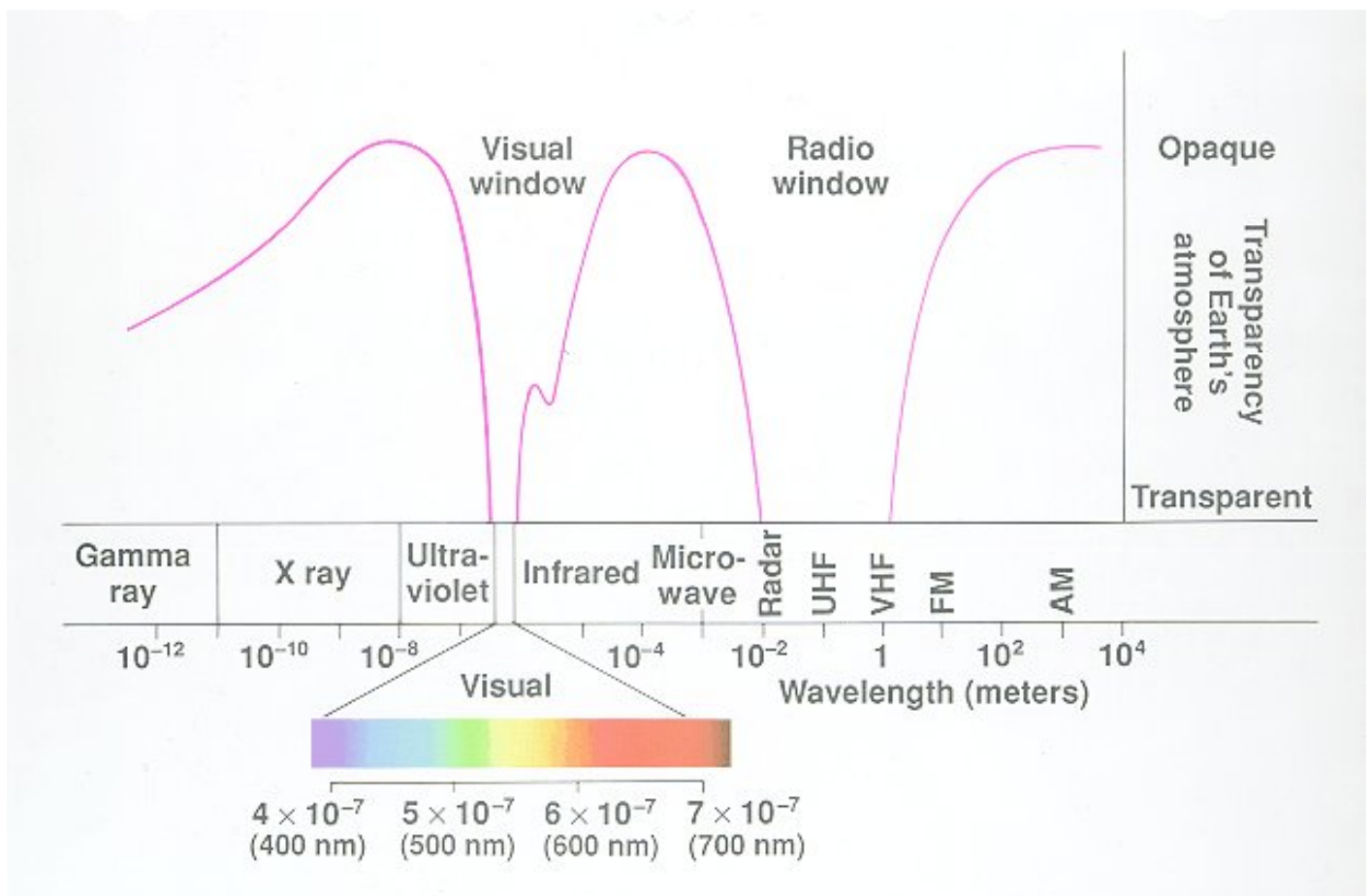
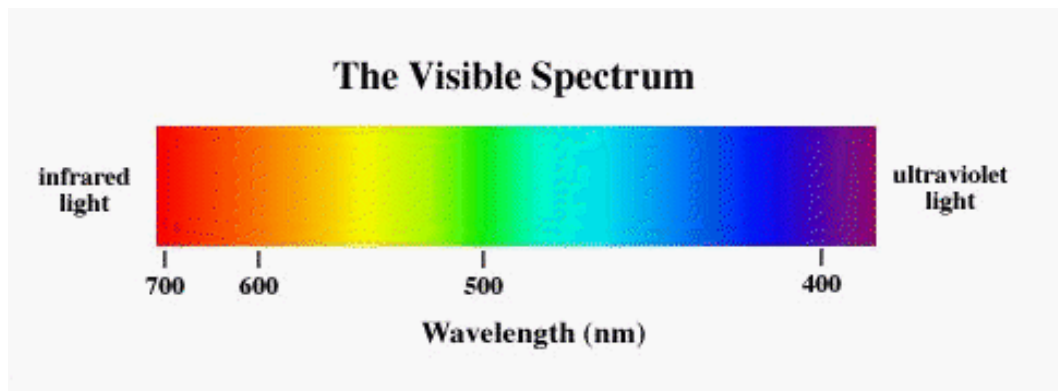


Electromagnetic Radiation (a.k.a. Light):

The wavelength of the light determines its characteristics. For example, short wavelengths are high energy gamma-rays and x-rays, long wavelengths are radio waves. The whole range of wavelengths is called the [electromagnetic spectrum](#).



Our eyes only see over the following range of wavelengths:



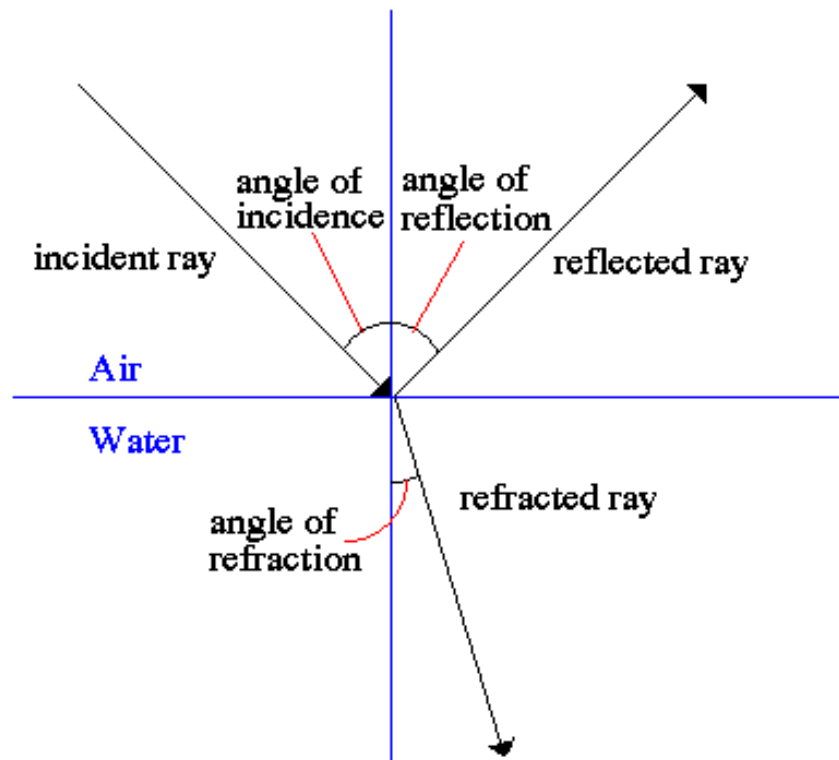
Wave Properties:

Due to its wave-like nature, light has three properties when encountering a medium:

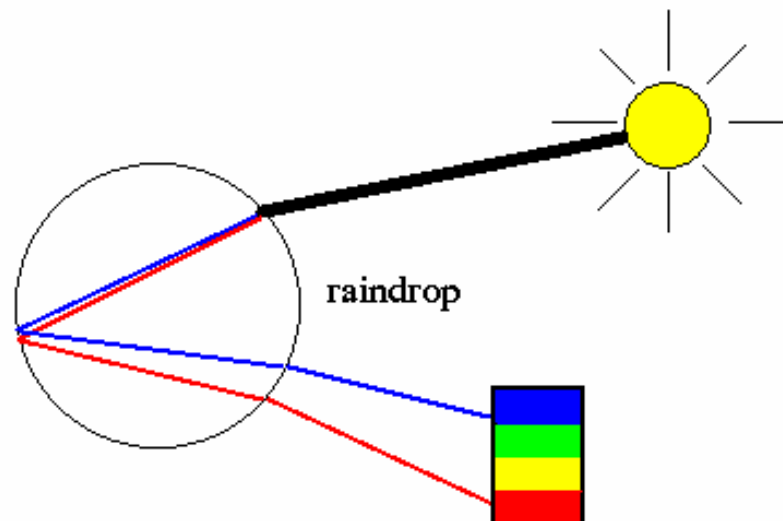
- 1) [reflection](#)
- 2) [refraction](#)
- 3) [diffraction](#)

When a light ray strikes a medium, such as oil or water, the ray is both refracted and reflected as shown below:

Reflection and Refraction

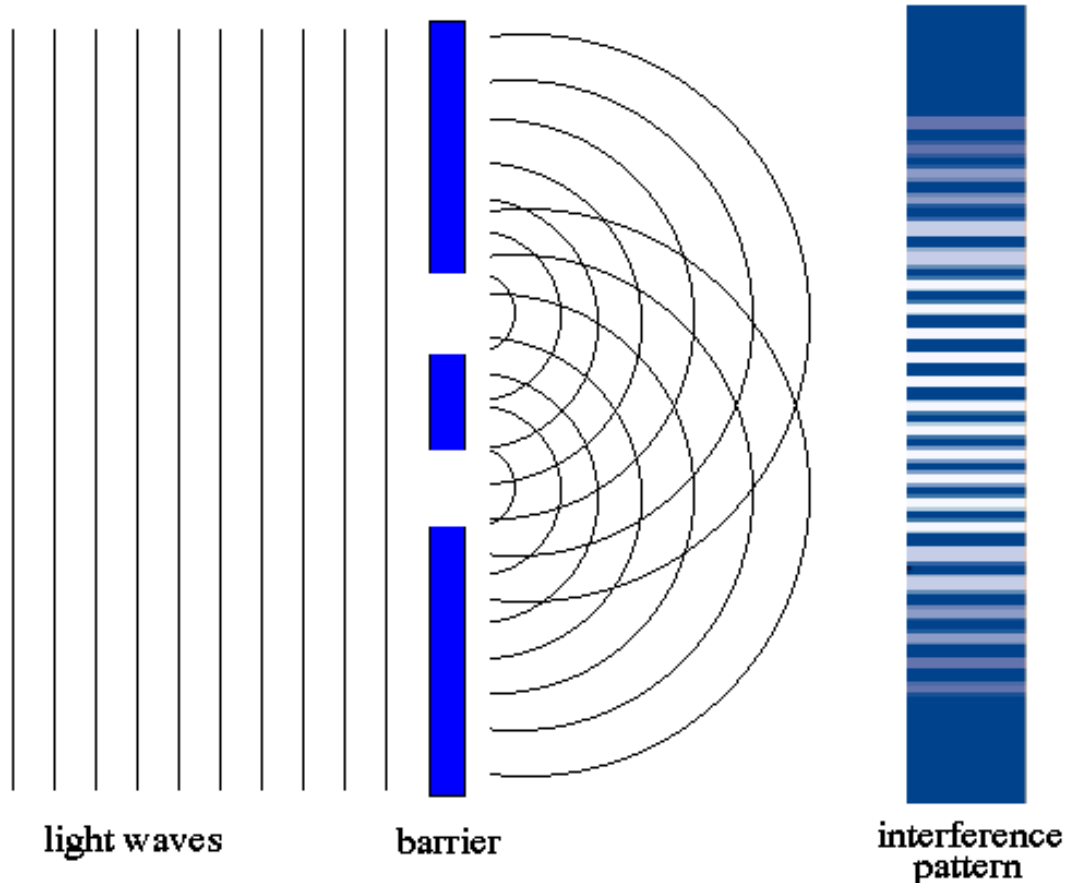


The angle of refraction is greater for a denser medium and is also a function of wavelength (i.e. blue light is more refracted compared to red and this is the origin to rainbows from drops of water)



Diffraction is the constructive and destructive interference of two beams of light that results in a wave-like pattern

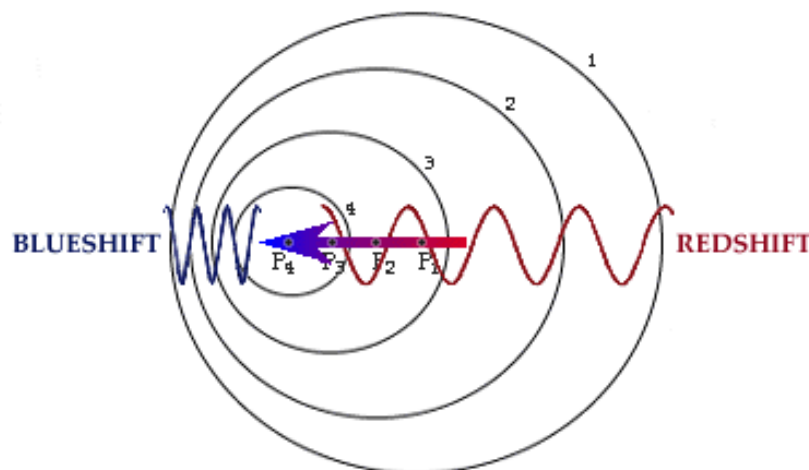
Interference



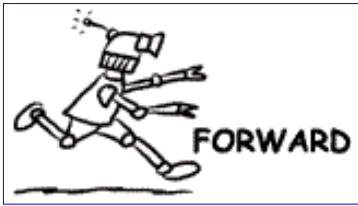
[click here to see interference movie](#)

Doppler effect:

The Doppler effect occurs when an object that is emitting light is in motion with respect to the observer. The speed of light does not change, only the wavelength. If the object is moving towards the observer the light is "compressed" or blueshifted. If the object is moving away from the observer the light is "expanded" or redshifted.



We can use the Doppler effect to measure the orbital velocity of planets and the rotation of the planets.



Atomic Theory:

The ancient philosopher, Heraclitus, maintained that everything is in a state of flux. Nothing escapes change of some sort (it is impossible to step into the same river). On the other hand, Parmenides argued that everything is what it is, so that it cannot become what is not (change is impossible because a substance would have to transition through nothing to become something else, which is a logical contradiction). Thus, change is incompatible with being so that only the permanent aspects of the Universe could be considered real.

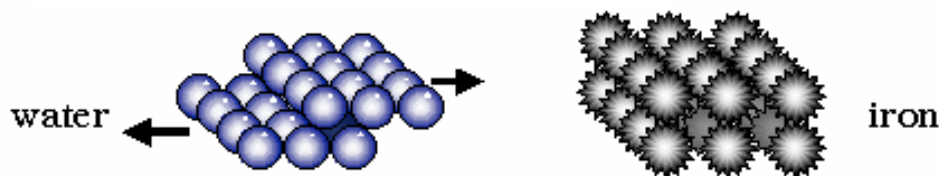
An ingenious escape was proposed in the fifth century B.C. by Democritus. He hypothesized that all matter is composed of tiny indestructible units, called atoms. The atoms themselves remain unchanged, but move about in space to combine in various ways to form all macroscopic objects. Early atomic theory stated that the characteristics of an object are determined by the shape of its atoms. So, for example, sweet things are made of smooth atoms, bitter things are made of sharp atoms.

In this manner permanence and flux are reconciled and the field of atomic physics was born. Although Democritus' ideas were to solve a philosophical dilemma, the fact that there is some underlying, elemental substance to the Universe is a primary driver in modern physics, the search for the ultimate subatomic particle.

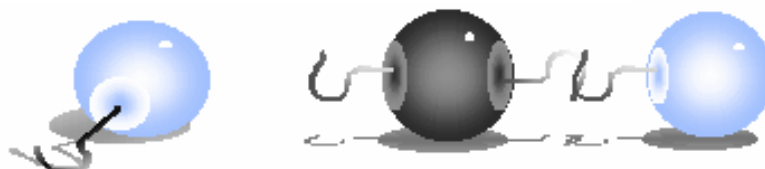
ELEMENTS			
	<u>Wt.</u>		<u>Wt.</u>
⊙ Hydrogen	1	⊙ Copper	56
① Azote	5	Ⓛ Lead	90
● Carbon	6	Ⓢ Silver	190
○ Oxygen	7	ⓖ Gold	190
Ⓟ Phosphorus	9	Ⓟ Platina	190
⊕ Sulfur	13	☿ Mercury	167

It was John Dalton, in the early 1800's, who determined that each chemical element is composed of a unique type of atom, and that the atoms differed by their masses. He devised a system of chemical symbols and, having ascertained the relative weights of atoms, arranged them into a table. In addition, he formulated the theory that a chemical combination of different elements occurs in simple numerical ratios by weight, which led to the development of the laws of definite and multiple proportions.

Greeks: atoms determine properties



Dalton: atoms determine composition

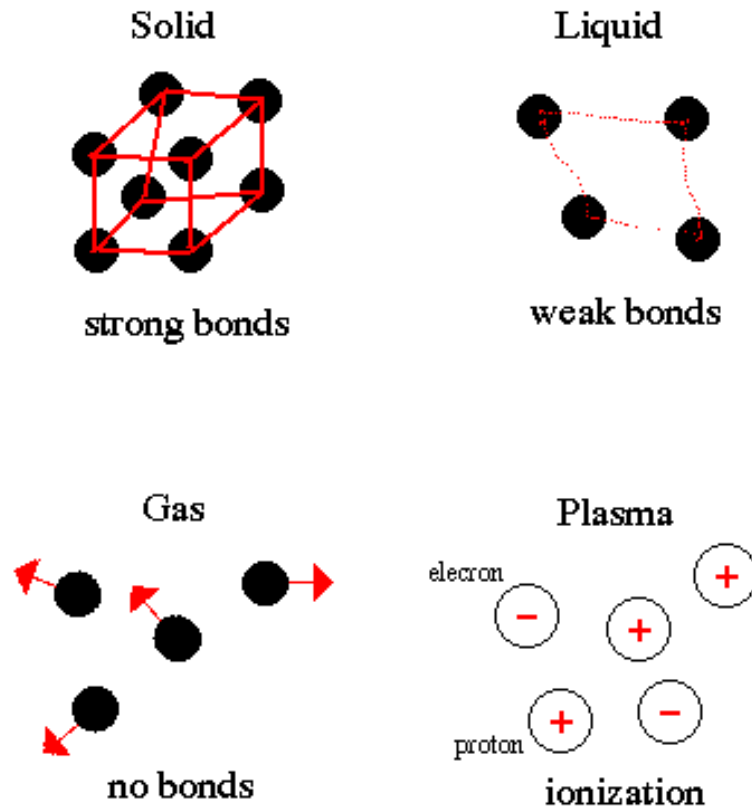


He then determined that compounds are made of molecules, and that molecules are composed of atoms in definite proportions. Thus, atoms determine the composition of matter, and compounds can be broken down into their individual elements.

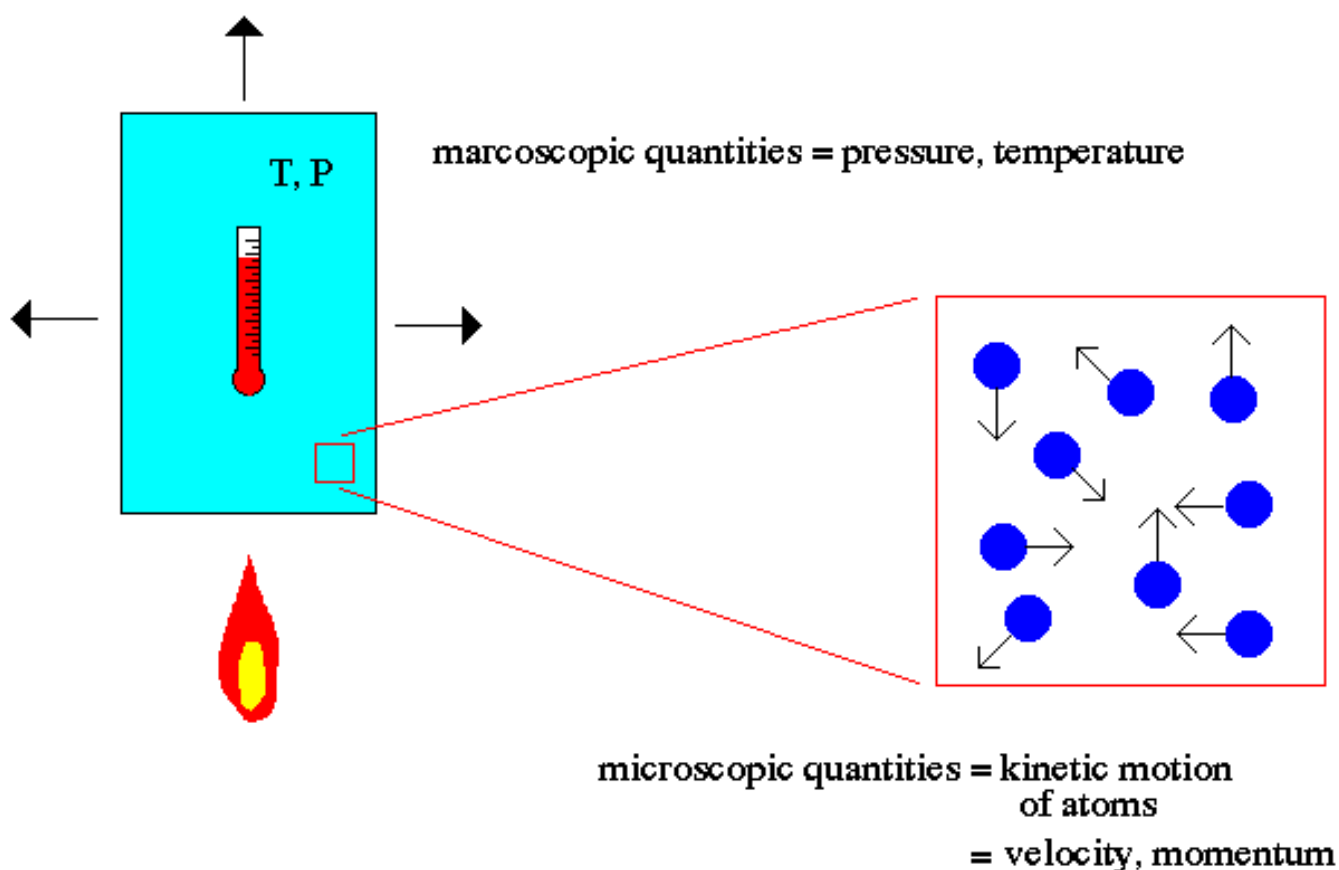
The first estimates for the sizes of atoms and the number of atoms per unit volume were made by Joseph Loschmidt in 1865. Using the ideas of kinetic theory, the idea that the properties of a gas are due to the motion of the atoms that compose it, Loschmidt calculated the mean free path of an atom based on diffusion rates. His result was that there are 6.022×10^{23} atoms per 12 grams of carbon. And that the typical diameters of an atom is 10^{-8} centimeters.

Matter:

Matter exists in four states: solid, liquid, gas and plasma. Plasmas are only found in the coronae and cores of stars. The state of matter is determined by the strength of the bonds between the atoms that makes up matter. Thus, is proportional to the temperature or the amount of energy contained by the matter.



The change from one state of matter to another is called a phase transition. For example, ice (solid water) converts (melts) into liquid water as energy is added. Continue adding energy and the water boils to steam (gaseous water) then, at several million degrees, breaks down into its component atoms.



The key point to note about atomic theory is the relationship between the macroscopic world (us) and the microscopic world of atoms. For example, the macroscopic world deals with concepts such as temperature and pressure to describe matter. The microscopic world of atomic theory deals with the kinetic motion of atoms to explain macroscopic quantities.

Temperature is explained in atomic theory as the motion of the atoms (faster = hotter). Pressure is explained as the momentum transfer of those moving atoms on the walls of the container (faster atoms = higher temperature = more momentum/hits = higher pressure).

Ideal Gas Law:

Macroscopic properties of matter are governed by the [Ideal Gas Law](#) of chemistry.

An ideal gas is a gas that conforms, in physical behavior, to a particular, idealized relation between pressure, volume, and temperature. The ideal gas law states that for a specified quantity of gas, the product of the volume, V , and pressure, P , is proportional to the absolute temperature T ; i.e., in equation form, $PV = kT$, in which k is a constant. Such a relation for a substance is called its equation of state and is sufficient to describe its gross behavior.

Although no gas is perfectly described by the above law, the behavior of real gases is

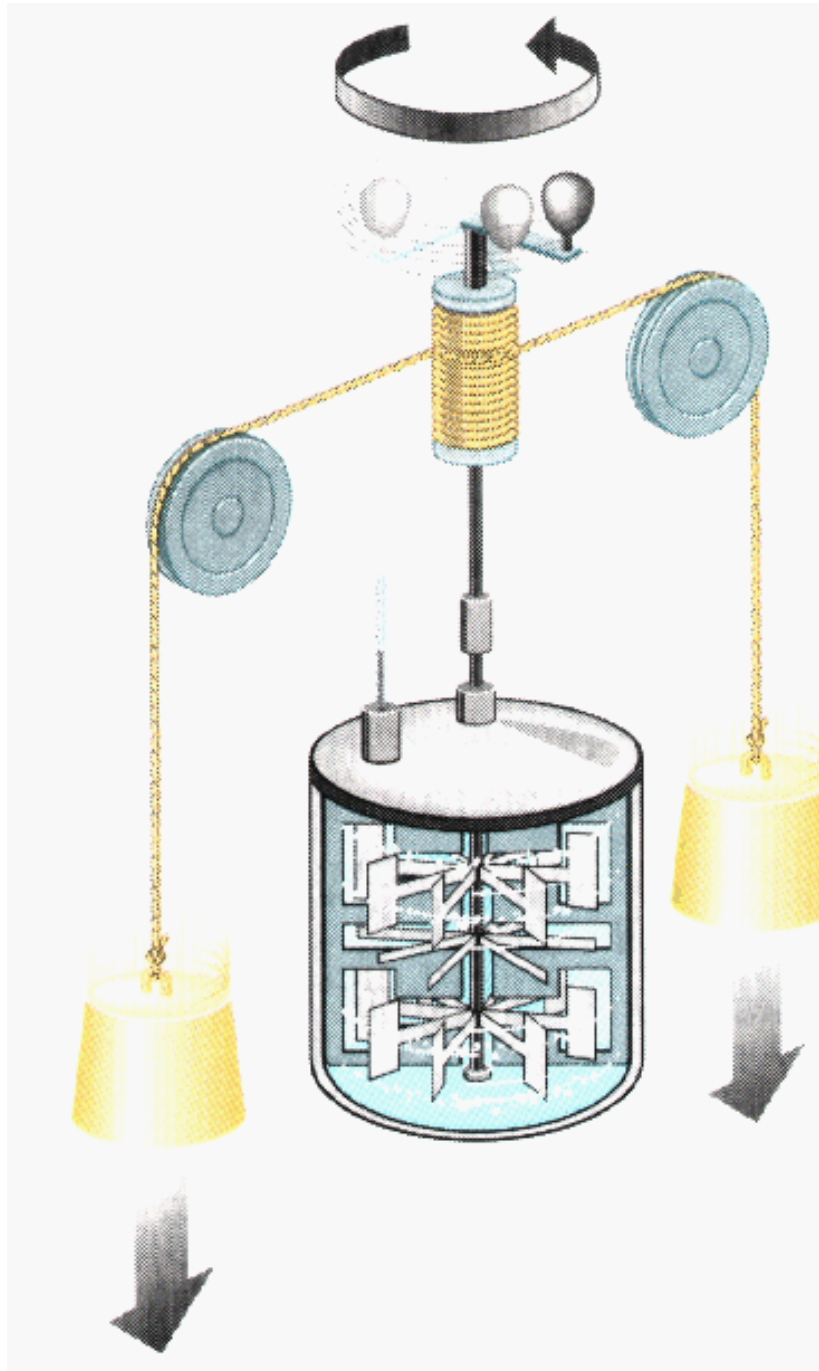
described quite closely by the ideal gas law at sufficiently high temperatures and low pressures (such as air pressure at sea level), when relatively large distances between molecules and their high speeds overcome any interaction. A gas does not obey the equation when conditions are such that the gas, or any of the component gases in a mixture, is near its triple point.

The ideal gas law can be derived from the kinetic theory of gases and relies on the assumptions that (1) the gas consists of a large number of molecules, which are in random motion and obey Newton's deterministic laws of motion; (2) the volume of the molecules is negligibly small compared to the volume occupied by the gas; and (3) no forces act on the molecules except during elastic collisions of negligible duration.

Thermodynamics:

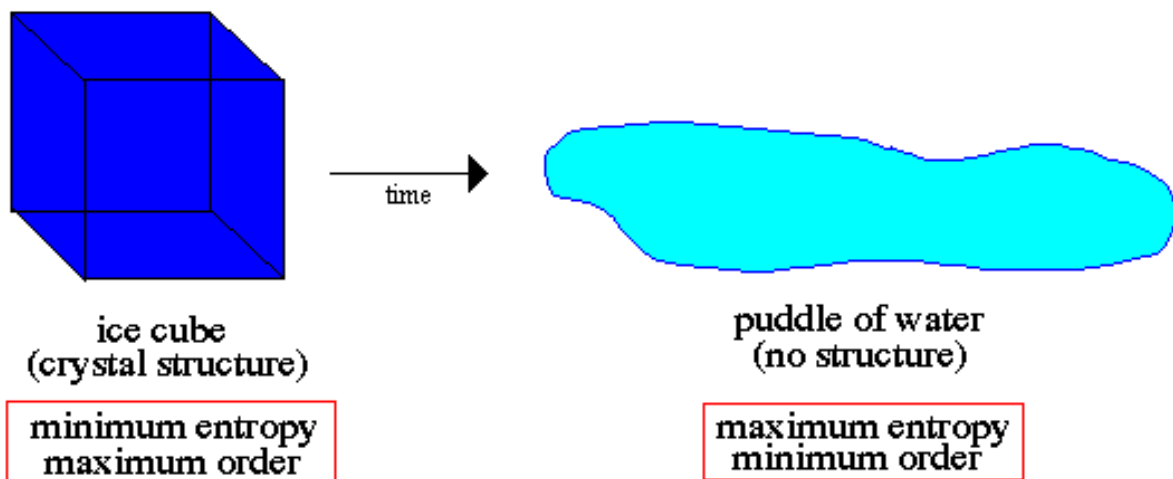
The study of the relationship between heat, work, temperature, and energy, encompassing the general behavior of physical system is called [thermodynamics](#).

The first law of thermodynamics is often called the law of the conservation of energy (actually mass-energy) because it says, in effect, that when a system undergoes a process, the sum of all the energy transferred across the system boundary--either as heat or as work--is equal to the net change in the energy of the system. For example, if you perform physical work on a system (e.g. stir some water), some of the energy goes into motion, the rest goes into raising the temperature of the system.



The second law of thermodynamics states that, in a closed system, the [entropy](#) increases. Cars rust, dead trees decay, buildings collapse; all these things are examples of entropy in action, the spontaneous movement from order to disorder.

Entropy



Classical or Newtonian physics is incomplete because it does not include irreversible processes associated with the increase of entropy. The entropy of the whole Universe always increased with time. We are simply a local spot of low entropy and our destiny is linked to the unstoppable increase of disorder in our world => stars will burn out, civilizations will die from lack of power.

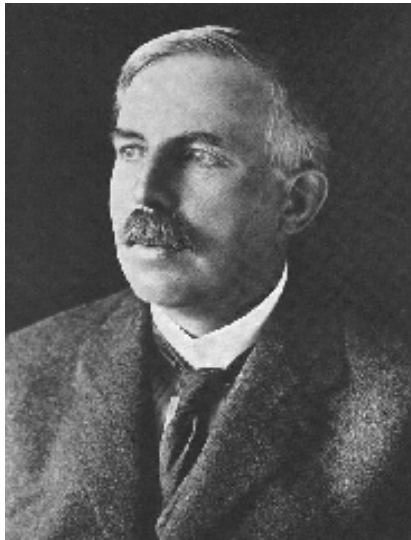
The approach to equilibrium is therefore an irreversible process. The tendency toward equilibrium is so fundamental to physics that the second law is probably the most universal regulator of natural activity known to science.

The concept of temperature enters into thermodynamics as a precise mathematical quantity that relates heat to entropy. The interplay of these three quantities is further constrained by the third law of thermodynamics, which deals with the absolute zero of temperature and its theoretical unattainability.

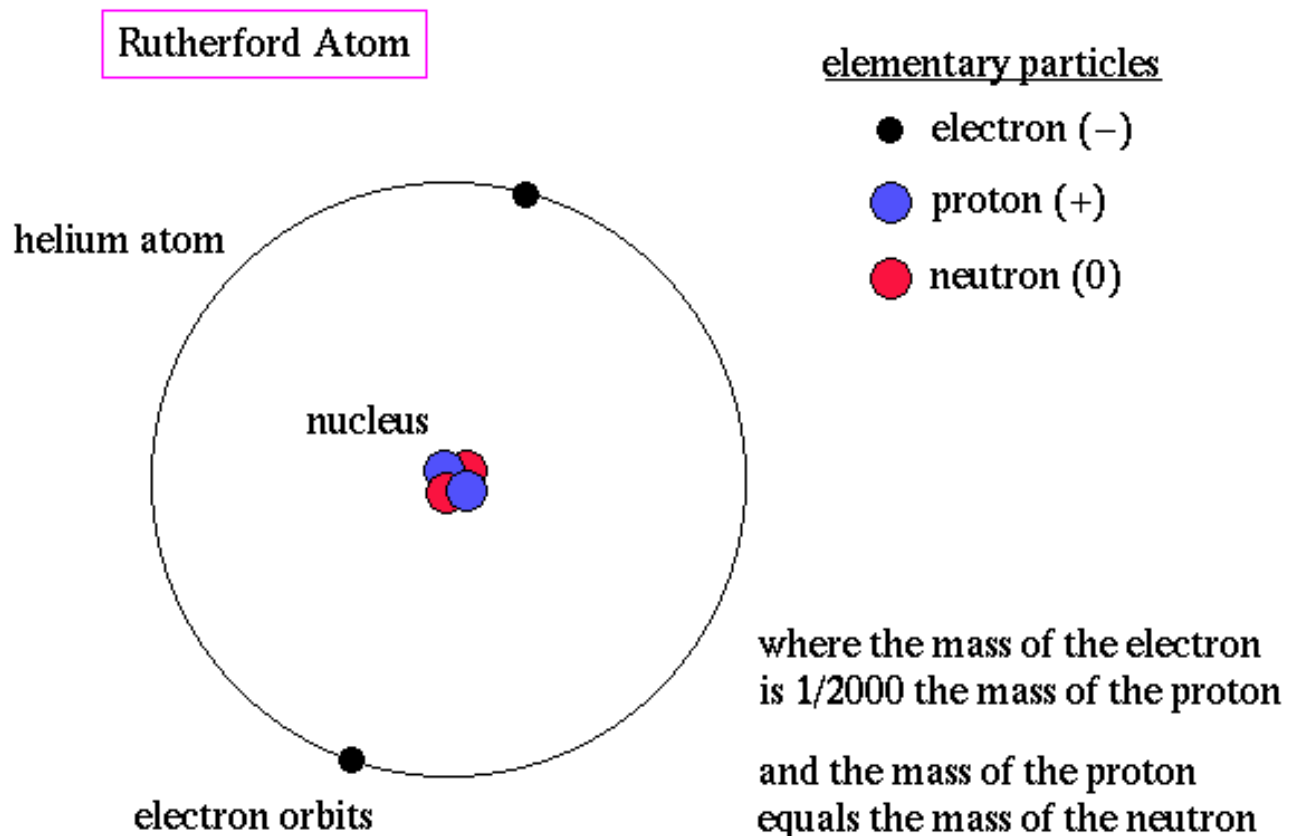
Absolute zero (approximately -273 C) would correspond to a condition in which a system had achieved its lowest energy state. The third law states that, as this minimum temperature is approached, the further extraction of energy becomes more and more difficult.

Rutherford Atom:

Ernest Rutherford is considered the father of nuclear physics. Indeed, it could be said that Rutherford invented the very language to describe the theoretical concepts of the atom and the phenomenon of radioactivity. Particles named and characterized by him include the alpha particle, beta particle and proton. Rutherford overturned Thomson's atom model in 1911 with his well-known gold foil experiment in which he demonstrated that the atom has a tiny, massive nucleus.



His results can best explained by a model for the atom as a tiny, dense, positively charged core called a nucleus, in which nearly all the mass is concentrated, around which the light, negative constituents, called electrons, circulate at some distance, much like planets revolving around the Sun.



The Rutherford atomic model has been alternatively called the nuclear atom, or the planetary model of the atom.



BACKWARD

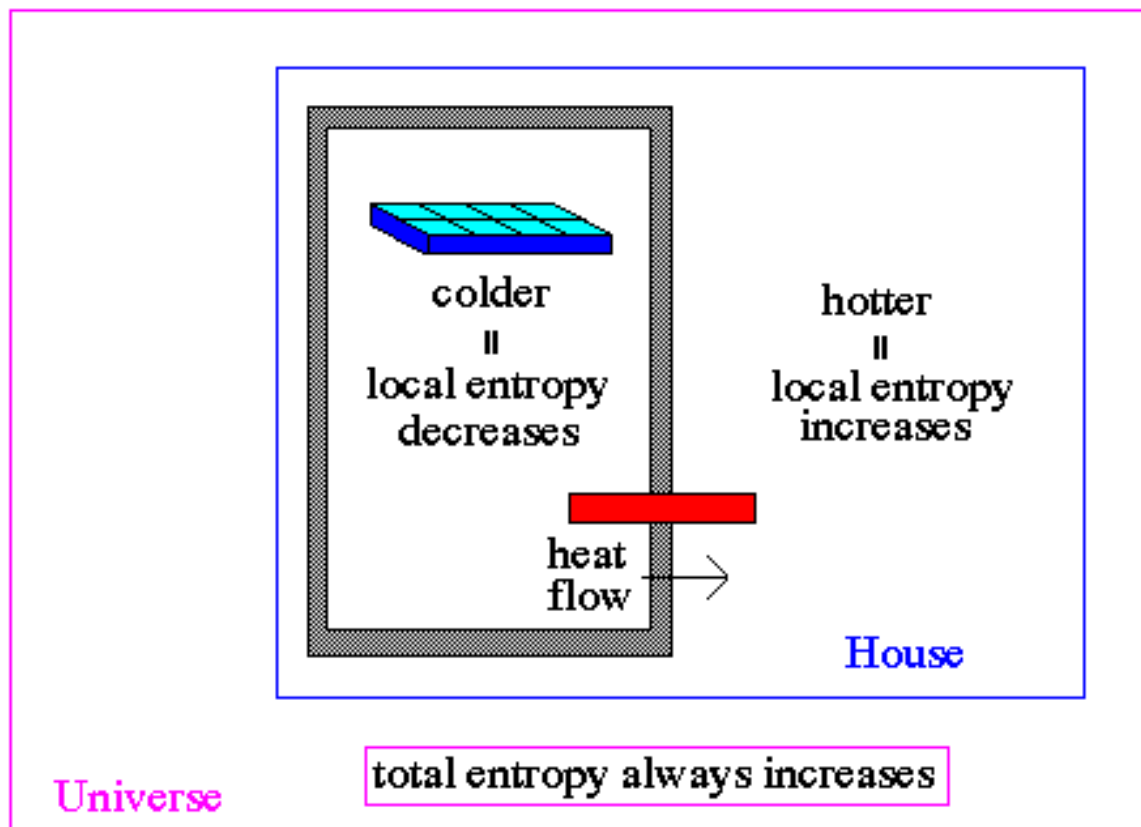




Entropy:

Cars rust, dead trees decay, buildings collapse; all these things are examples of [entropy](#) in action, the spontaneous and continuous movement from order to [disorder](#).

The measure of entropy must be global. For example, you can pump heat out of a refrigerator (to make ice cubes), but the heat is placed in the house and the entropy of the house increases, even though the local entropy of the ice cube tray decreases. So the sum of the entropy in the house and refrigerator increases.



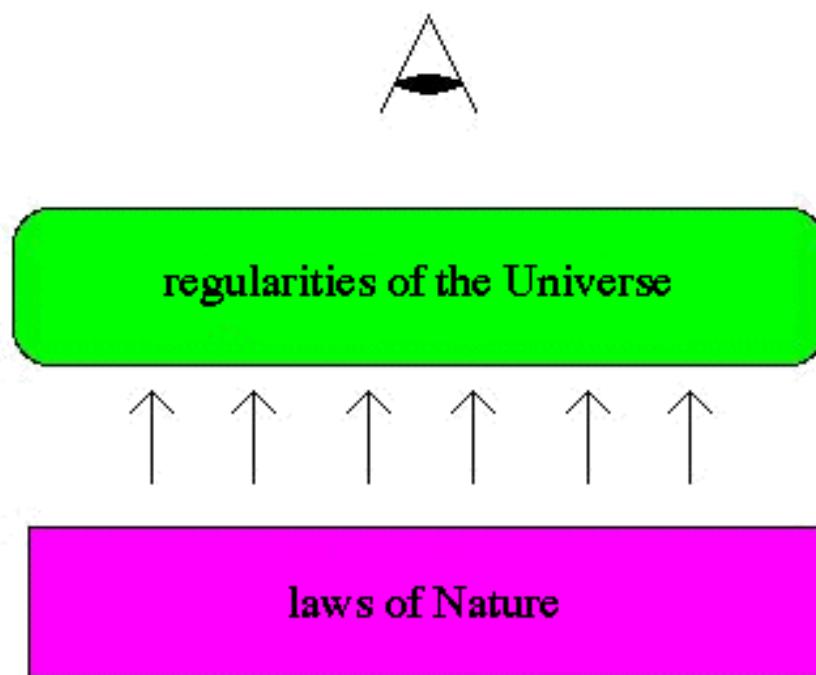
The concept of entropy applies to many other physical systems other than heat. For example, information flow suffers from entropy. A signal is always [degraded by random noise](#).

The entropy of the whole Universe always increased with time. We are simply a local spot of low entropy and our destiny is linked to the unstoppable increase of disorder in our world => stars will burn out, civilizations will die from lack of power.

Irreversibility:

Classical physics is a science upon which our belief in a deterministic, time-reversible description of Nature is based. Classical physics does not include any distinction between the past and the future. The Universe is ruled by deterministic laws, yet the macroscopic world is not reversible. This is known as Epicurus' clinamen, the dilemma of being and becoming, the idea that some element of chance is needed to account for the deviation of material motion from rigid predetermined evolution.

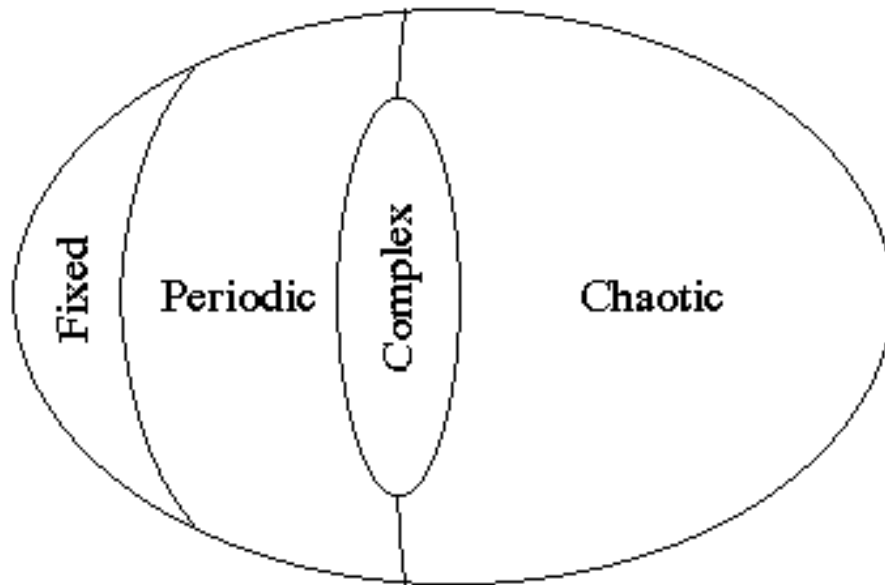
The astonishing success of simple physical principles and mathematical rules in explaining large parts of Nature is not something obvious from our everyday experience. On casual inspection, Nature seems extremely complex and random. There are few natural phenomenon which display the precise sort of regularity that might hint of an underlying order. Where trends and rhythms are apparent, they are usually of an approximate and qualitative form. How are we to reconcile these seemingly random acts with the supposed underlying lawfulness of the Universe?



For example, consider falling objects. Galileo realized that all bodies accelerate at the same rate regardless of their size or mass. Everyday experience tells you differently because a feather falls slower than a cannonball. Galileo's genius lay in spotting that the differences that occur in the everyday world are in incidental complication (in this case, air friction) and are irrelevant to the real underlying properties (that is, gravity). He was able to abstract from the complexity of real-life situations the simplicity of an idealized law of gravity. Reversible processes appear to be idealizations of real processes in Nature.

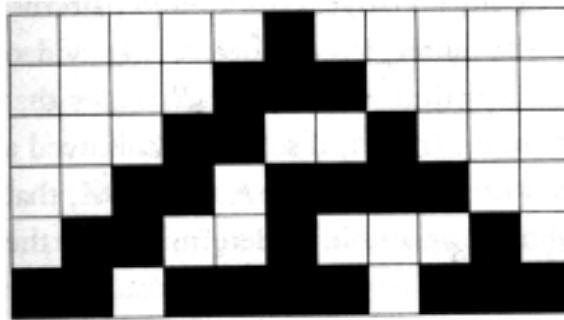
Probability-based interpretations make the macroscopic character of our observations responsible for the irreversibility that we observe. If we could follow an individual molecule we would see a time reversible system in which the each molecule follows the laws of Newtonian physics. Because we can only describe the number of molecules in each compartment, we conclude that the system evolves towards equilibrium. Is irreversibility merely a consequence of the approximate macroscopic character of our observations? Is it due to our own ignorance of all the positions and velocities?

Irreversibility leads to both order and disorder. Nonequilibrium leads to concepts such as self-organization and dissipative structures (Spatiotemporal structures that appear in far-from-equilibrium conditions, such as oscillating chemical reactions or regular spatial structures, like snowflakes). Objects far from equilibrium are highly organized thanks to temporal, irreversible, nonequilibrium processes (like a [pendulum](#)).



Information movement in complex dynamical systems is such that the regime at the left is where information is frozen, to the right is a flexible regime where behavior such as crystal growth is found. To the far right, information moves so freely that its structure cannot be maintained; the regime is too chaotic. Only in the center region can information be stable enough to support message structure.

The behavior of complex systems is not truly random, it is just that the final state is so sensitive to the initial conditions that it is impossible to predict the future behavior without infinite knowledge of all the motions and energy (i.e. a butterfly in South America influences storms in the North Atlantic).



5 generations

The growth of a two-state (on or off) one-dimensional cellular automaton. The row of eight boxes on top shows the rule set: for each combination of three cells in generation 0, there is a determined result for the next-generation cell below the triplet. Beginning from a single seed, these rules are applied consistently, each generation represented by a horizontal row of cells. The first image shows five generations of growth, the second image shows fifty, and the third shows 100 generations, by which time it is clear that simple rules have generated considerable complexity.

50 generations

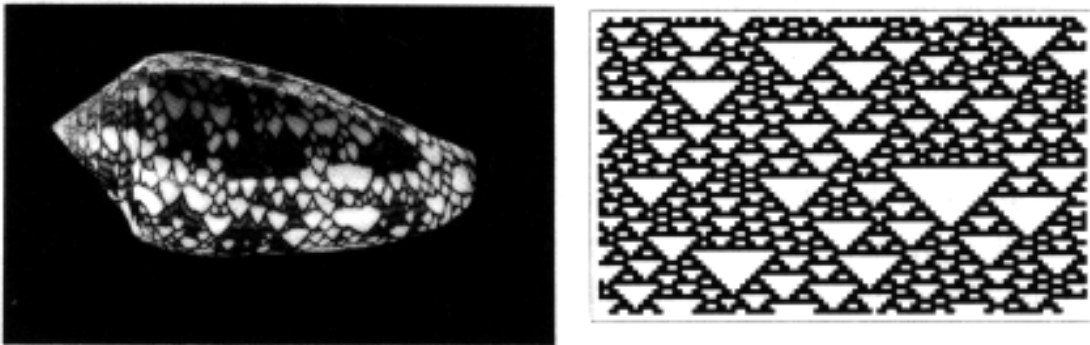


100 generations



Although this is 'just' a mathematical game, there are many examples of the same shape and complex behavior occurring in Nature.

A comparison between the natural pattern on a mollusk shell and the pattern of a simple one-dimensional cellular automaton.



Individual descriptions are called trajectories, statistical descriptions of groups are called ensembles. Individual particles are highly deterministic, trajectories are fixed. Yet ensembles of particles follow probable patterns and are uncertain. Does this come from ignorance of all the trajectories or something deeper in the laws of Nature? Any predictive computation will necessarily contain some input errors because we cannot measure physical quantities to unlimited precision.

Note that relative probabilities evolve in a deterministic manner. A statistical theory can remain deterministic. However, macroscopic irreversibility is the manifestation of the randomness of probabilistic processes on a microscopic scale. Success of reductionism was based on the fact that most simple physical systems are linear, the whole is the sum of the parts. Complexity arrives in nonlinear systems.

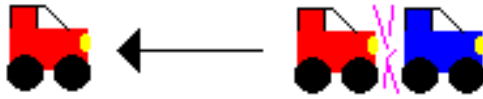
Arrow of Time:

Why do we perceive [time](#) as always moving forward? Why are our memories always of the past and never of the future? All the fundamental Newtonian laws are time reversible. Collisions look the same forwards or backwards. A box of gas molecules obeying Newton's laws perfectly does not have an inbuilt arrow of time. However, it is possible to show that the continual random molecular motions will cause the entire ensemble to visit and revisit every possible state of the box, much like the continual shuffling of a deck of cards will eventually reproduce any sequence.

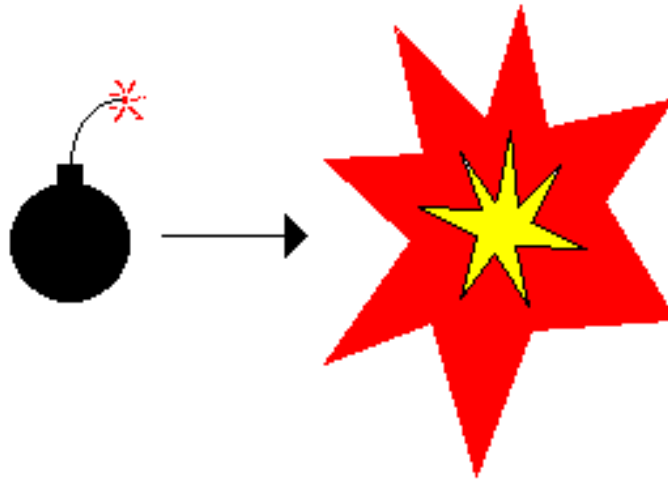
Arrow of Time



time reversible



not time reversible

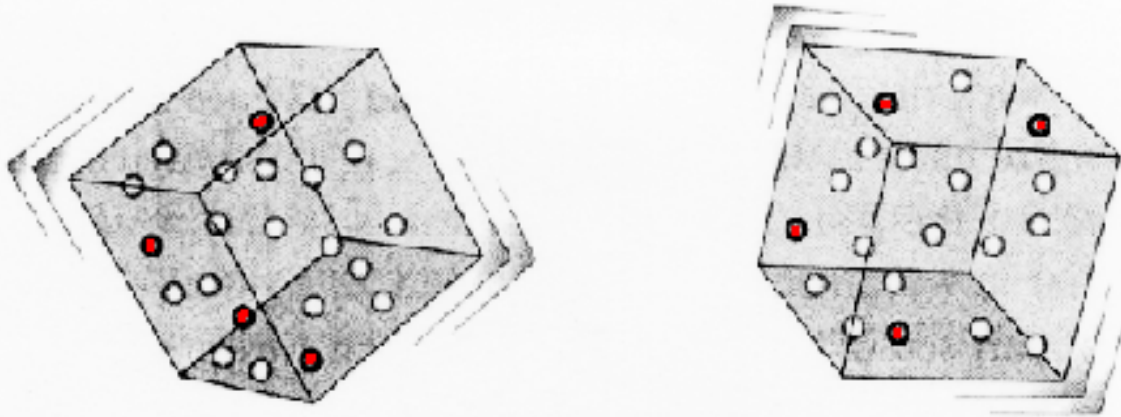


there is a temporal direction
in the macroscopic world

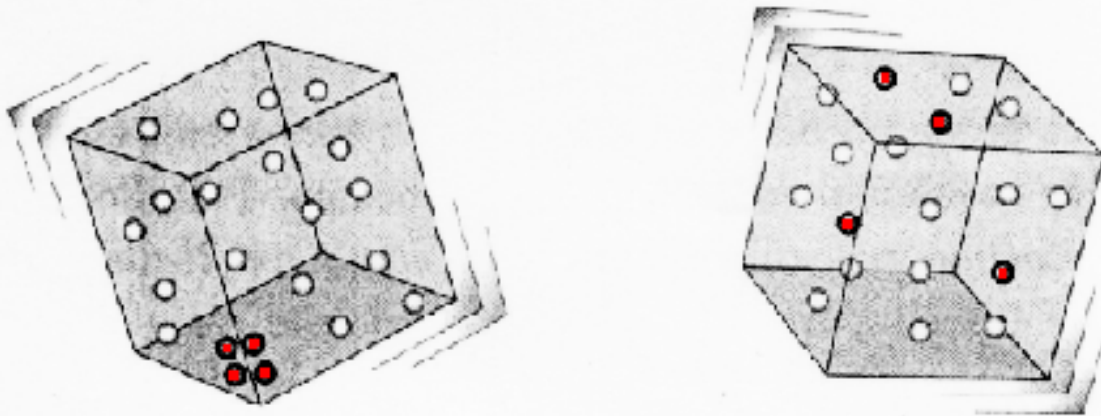
This ability of Nature to be divided into a multitude of states makes it easier to understand why thermodynamical systems move toward equilibrium, known as Poincare's theorem. If a box of gas is in a low entropy state at one moment, it will very probably soon be in a less ordered state since given the large number of states for it to evolve to, most of those states are of higher entropy. So just by the laws of chance, the box has a higher probability of becoming a higher entropy state rather than a lower one since there are so many more possible high entropy states.

Poincare's theorem claims that if every individual state has the same chance of being visited, then obviously mixed-up states are going to turn up much more often than the less mixed-up or perfectly ordered states, simply because there are many more of them.

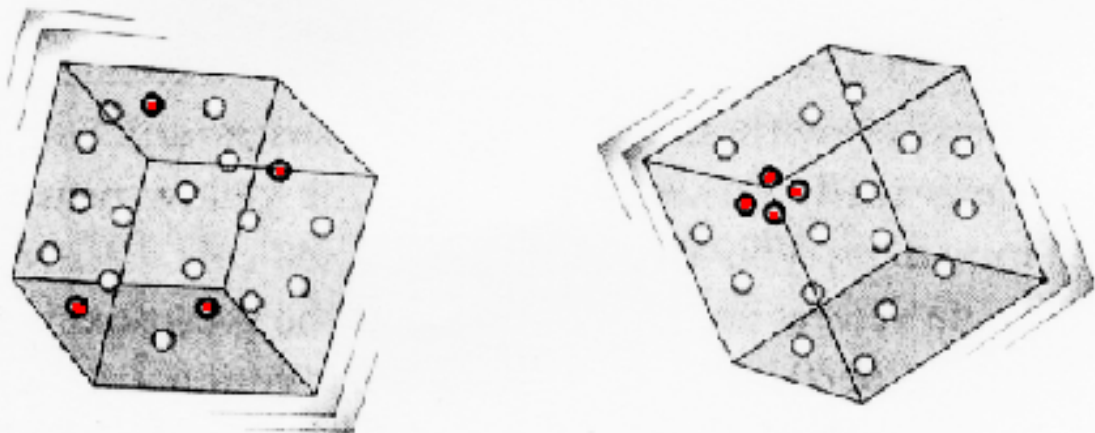
This happens . . .



This happens . . .



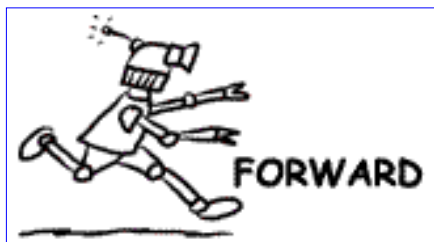
This hardly ever happens . . .



Thermodynamical events, such as a growing tree, are not reversible. Cracked eggs do not repair themselves. Defined by these events, time

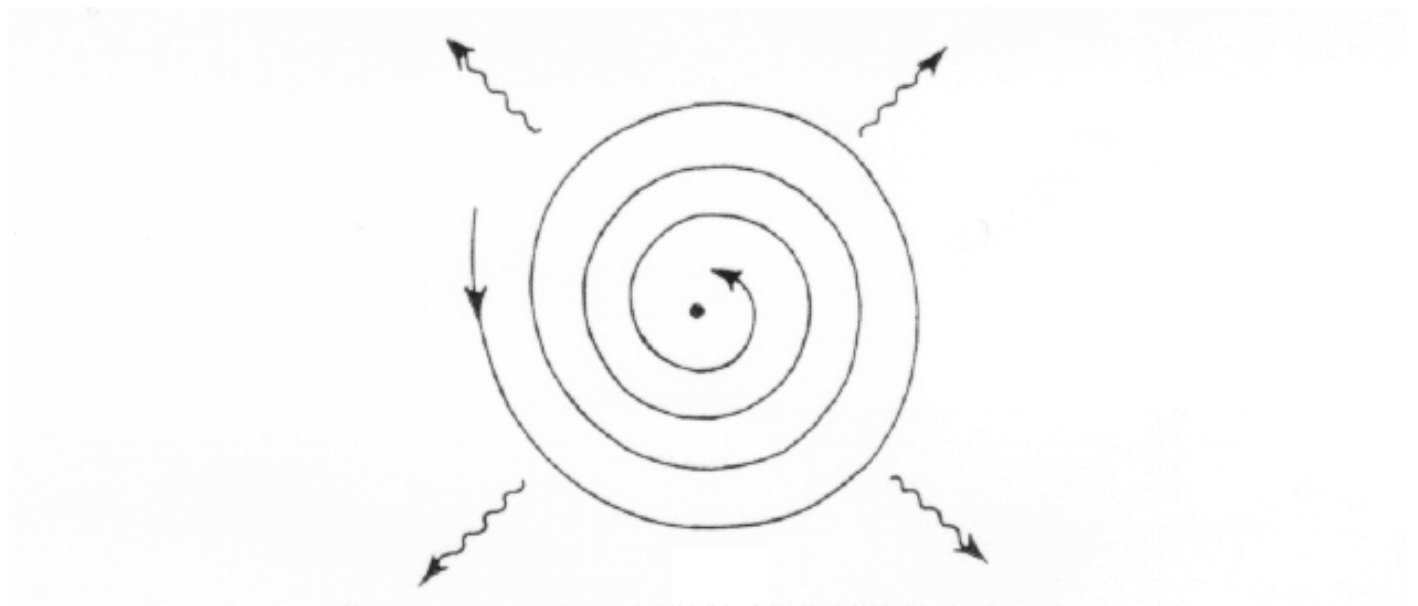
has an arrow, a preferred direction.

Entropy and the arrow of time are strongly linked. Increasing entropy is in the direction of positive time. However, a study of the components to systems shows that the parts are describable in terms of time-symmetric laws. In other words, the microscopic world is ruled by time-symmetric laws, but the macroscopic world has a particular direction.



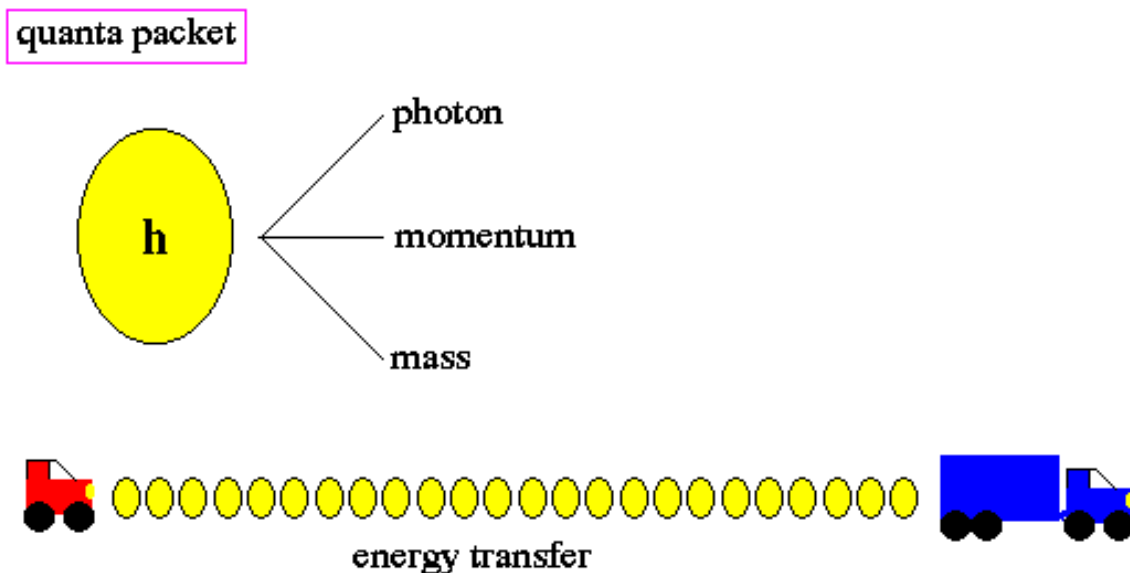
Planck's constant:

In the early 1900's, German physicist E. Planck noticed fatal flaw in our physics by demonstrating that the electron in orbit around the nucleus accelerates. Acceleration means a changing electric field (the electron has charge), when means photons should be emitted. But, then the electron would lose energy and fall into the nucleus. Therefore, atoms shouldn't exist!



According to classical physics, an electron in orbit around an atomic nucleus should emit electromagnetic radiation (photons) continuously, because it is continually accelerating in a curved path. The resulting loss of energy implies that the electron should spiral into the nucleus in a very short time (i.e. atoms can not exist).

To resolve this problem, Planck made a wild assumption that energy, at the sub-atomic level, can only be transferred in small units, called quanta. Due to his insight, we call this unit Planck's constant (h). The word [quantum](#) derives from quantity and refers to a small packet of action or process, the smallest unit of either that can be associated with a single event in the microscopic world.



Changes of energy, such as the transition of an electron from one orbit to another around the nucleus of an atom, is done in discrete quanta. Quanta are not divisible and the term quantum leap refers to the abrupt movement from one discrete energy level to another, with no smooth transition. There is no "inbetween".

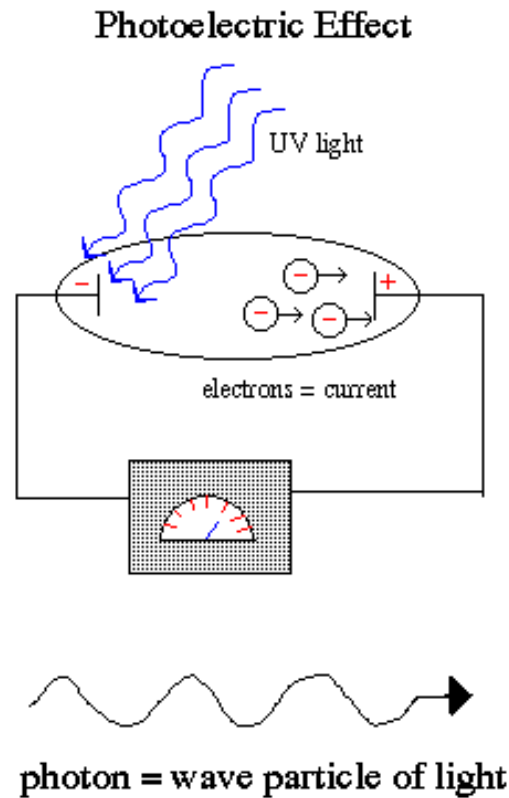
The quantization, or "jumpiness" of action as depicted in quantum physics differs sharply from classical physics which represented motion as smooth, continuous change. Quantization limits the energy to be transferred to photons and resolves the UV catastrophe problem.

Wave-Particle Dualism:

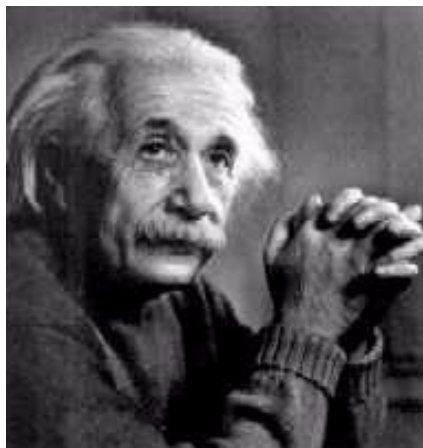
The wave-like nature of light explains most of its properties:

- reflection/refraction
- diffraction/interference
- Doppler effect

But, the results from [stellar spectroscopy](#) (emission and absorption spectra) can only be explained if light has a particle nature as shown by [Bohr's atom](#) and the photon description of light.



This dualism to the nature of light is best demonstrated by the [photoelectric effect](#), where a weak UV light produces a current flow (releases electrons) but a strong red light does not release electrons no matter how intense the red light.



[Einstein](#) explained the photoelectric effect by assuming that light exists in a particle-like state, packets of energy (quanta) called photons. There is no current flow for red light because the packets of energy carried by each individual red photons are too weak to knock the electrons off the atoms no matter how many red photons you beamed onto the cathode. But the individual UV photons were each strong enough to release the electron and cause a current flow.

It is one of the strange, but fundamental, concepts in modern physics that light has both a wave and particle state (but not at the same time), called [wave-particle dualism](#).

de Broglie Matter Waves:

Perhaps one of the key questions when Einstein offered his photon description of light is, does an electron have wave-like properties? The response to this question arrived from the Ph.D. thesis of Louis de Broglie in 1923. de Broglie argued that since light can display wave and particle properties, then matter can also be a particle and a wave too.



One way of thinking of a matter wave (or a photon) is to think of a wave packet. Normal waves look with this:



having no beginning and no end. A composition of several waves of different wavelength can produce a wave packet that looks like this:



So a photon, or a free moving electron, can be thought of as a wave packet, having both wave-like properties and also the single position and size we associate with a particle. There are some slight problems, such as the wave packet doesn't really stop at a finite distance from its peak, it also goes on for every and every. Does this mean an electron exists at all places in its trajectory?

de Broglie also produced a simple formula that the wavelength of a matter particle is related to the momentum of the particle. So energy is also connected to the wave

property of matter.

While de Broglie waves were difficult to accept after centuries of thinking of particles as solid things with definite size and positions, electron waves were confirmed in the laboratory by running electron beams through slits and demonstrating that interference patterns formed.

How does the de Broglie idea fit into the macroscopic world? The length of the wave diminishes in proportion to the momentum of the object. So the greater the mass of the object involved, the shorter the waves. The wavelength of a person, for example, is only one millionth of a centimeter, much too short to be measured. This is why people don't 'tunnel' through chairs when they sit down.

Uncertainty Principle:

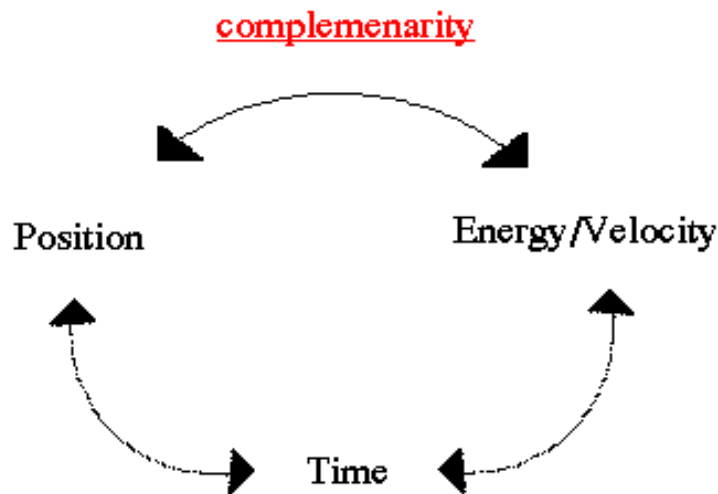
Classical physics was on loose footing with problems of [wave/particle duality](#), but was caught completely off-guard with the discovery of the [uncertainty principle](#).



The uncertainty principle, developed by W. Heisenberg, is a statement of the effects of wave-particle duality on the properties of subatomic objects. Consider the concept of momentum in the wave-like microscopic world. The momentum of wave is given by its wavelength. A wave packet like a photon or electron is a composite of many waves. Therefore, it must be made of many momentums. But how can an object have many momentums?

Of course, once a measurement of the particle is made, a single momentum is observed. But, like fuzzy position, momentum before the observation is intrinsically uncertain. This is what is known as the uncertainty principle, that certain quantities, such as position, energy and time, are unknown, except by probabilities. In its purest form, the uncertainty principle states that accurate knowledge of [complementarity](#) pairs is

impossible. For example, you can measure the location of an electron, but not its momentum (energy) at the same time.



Mathematically we describe the uncertainty principle as the following, where 'x' is position and 'p' is momentum:

$$\Delta x \Delta p > \frac{h}{2\pi}$$

This is perhaps the most famous equation next to $E=mc^2$ in physics. It basically says that the combination of the error in position times the error in momentum must always be greater than Planck's constant. So, you can measure the position of an electron to some accuracy, but then its momentum will be inside a very large range of values. Likewise, you can measure the momentum precisely, but then its position is unknown.

Also notice that the uncertainty principle is unimportant to macroscopic objects since Planck's constant, h , is so small (10^{-34}). For example, the uncertainty in position of a thrown baseball is 10^{-30} millimeters.

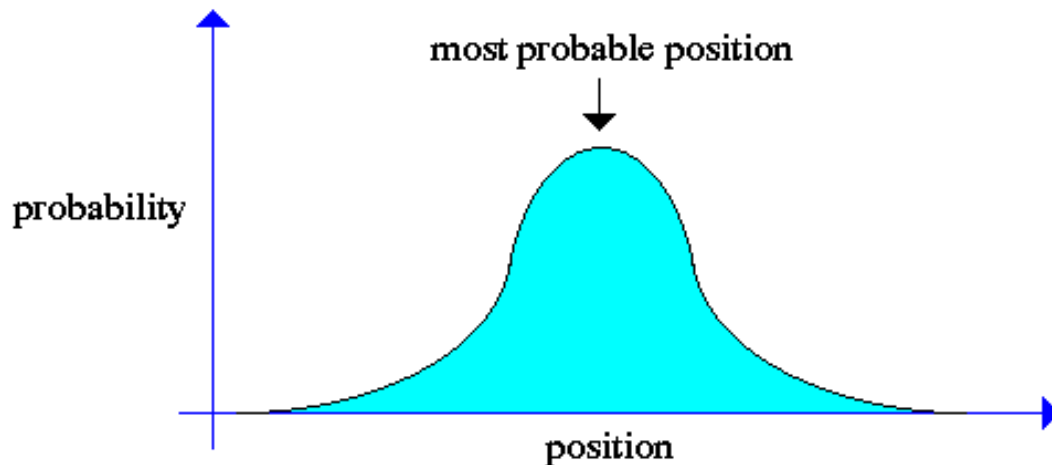
The depth of the uncertainty principle is realized when we ask the question; is our knowledge of reality unlimited? The answer is no, because the uncertainty principle states that there is a built-in uncertainty, indeterminacy, unpredictability to Nature.

Quantum Wave Function:

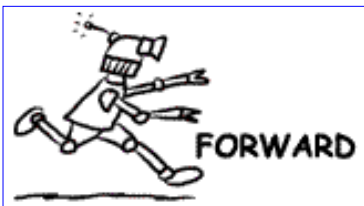
The wave nature of the microscopic world makes the concept of 'position' difficult for subatomic particles. Even a wave packet has some 'fuzziness' associated with it. An electron in orbit has no position to speak of, other than it is somewhere in its orbit.

To deal with this problem, quantum physics developed the tool of the quantum [wave function](#) as a mathematical description of the superpositions associated with a quantum entity at any particular moment.

Quantum Wave Function



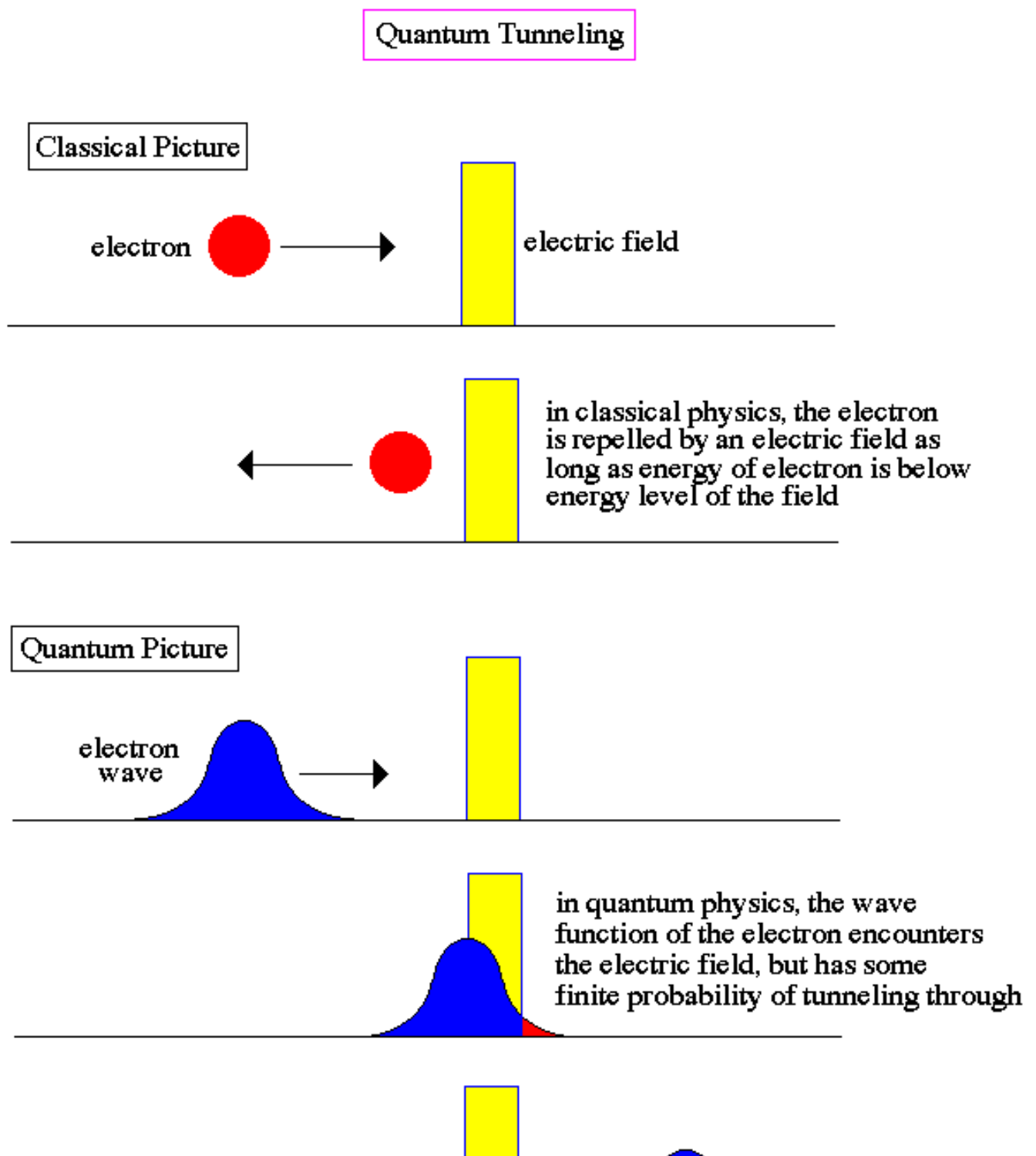
The key point to the wave function is that the position of a particle is only expressed as a likelihood or probability *until* a measurement is made. For example, striking an electron with a photon results in a position measurement and we say that the wave function has 'collapsed' (i.e. the wave nature of the electron converted to a particle nature).



Superposition:

The fact that quantum systems, such as electrons and protons, have indeterminate aspects means they exist as possibilities rather than actualities. This gives them the property of being things that might be or might happen, rather than things that are. This is in sharp contrast to Newtonian physics where things are or are not, there is no uncertainty except those imposed by poor data or limitations of the data gathering equipment.

The superposition of possible positions for an electron can be demonstrated by the observed phenomenon called [quantum tunneling](#).



this is the basis for transistors



Notice that the only explanation for quantum tunneling is if the position of the electron is truly spread out, not just hidden or unmeasured. It raw uncertainty allows for the wave function to penetrate the barrier. This is genuine indeterminism, not simply an unknown quantity until someone measures it.

It is important to note that the superposition of possibilities only occurs before the entity is observed. Once an observation is made (a position is measured, a mass is determined, a velocity is detected) then the superposition converts to an actual. Or, in quantum language, we say the wave function has collapsed.

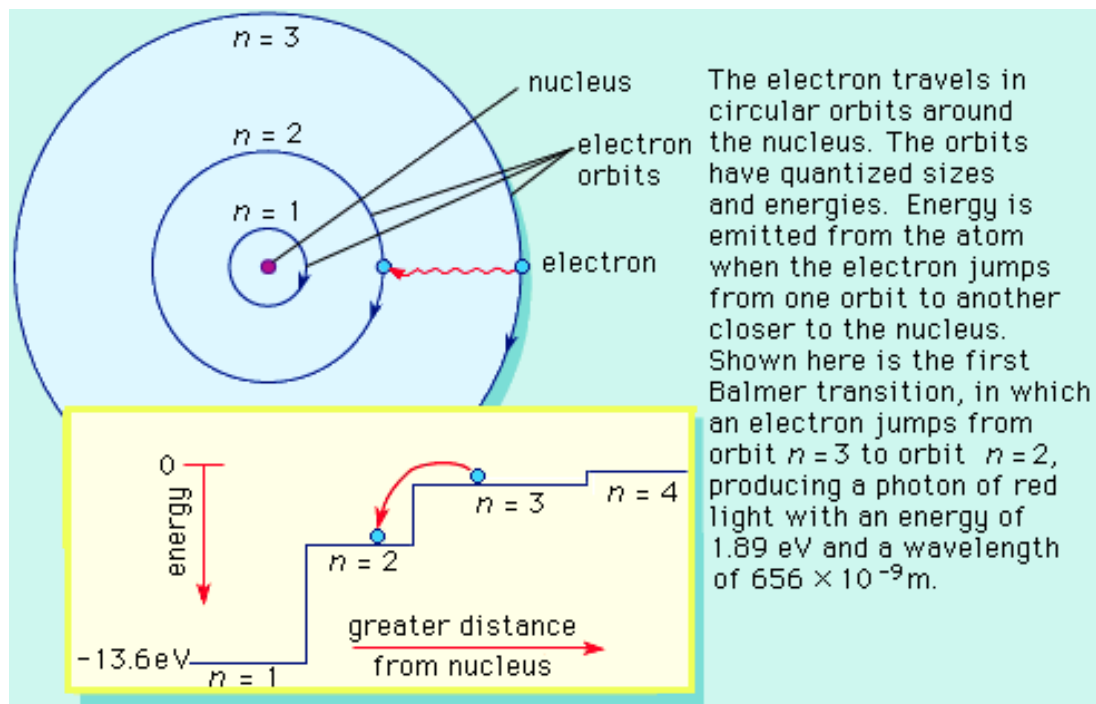
The collapse of the wave function by observation is a transition from the many to the one, from possibility to actuality. The identity and existence of a quantum entities are bound up with its overall environment (this is called contextualism). Like homonyms, words that depend on the context in which they are used, quantum reality shifts its nature according to its surroundings.

Bohr Atom:

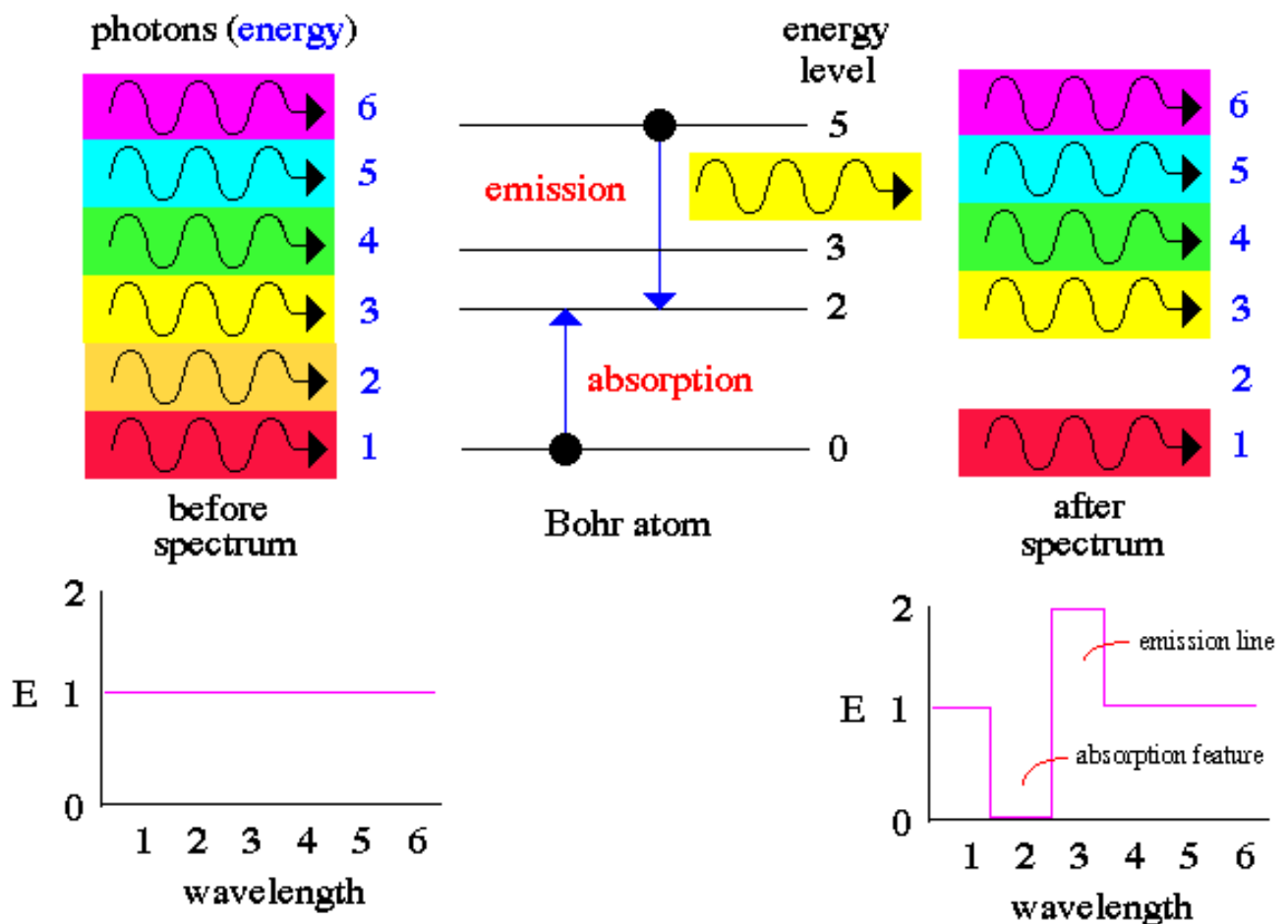
Perhaps the foremost scientists of the 20th century was Niels Bohr, the first to apply Planck's quantum idea to problems in atomic physics. In the early 1900's, Bohr proposed a quantum mechanical description of the atom to replace the early model of Rutherford.



The [Bohr model](#) basically assigned discrete orbits for the electron, multiples of Planck's constant, rather than allowing a continuum of energies as allowed by classical physics.



The power in the Bohr model was its ability to predict the [spectra](#) of light emitted by atoms. In particular, its ability to explain the spectral lines of atoms as the absorption and emission of photons by the electrons in quantized orbits.



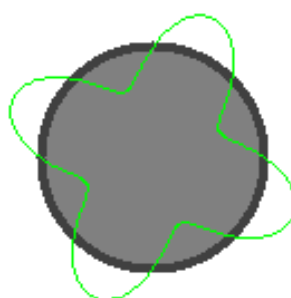
In principle, all of atomic and molecular physics, including the structure of atoms and

their dynamics, the periodic table of elements and their chemical behavior, as well as the spectroscopic, electrical, and other physical properties of atoms and molecules, can be accounted for by quantum mechanics => fundamental science.

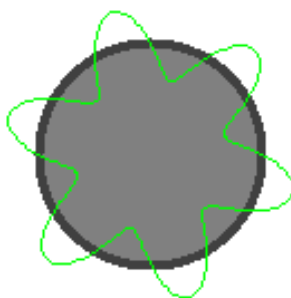
Quantum Mechanics:

The field of [quantum mechanics](#) concerns the description of phenomenon on small scales where classical physics breaks down. The biggest difference between the classical and microscopic realm, is that the quantum world can be not be perceived directly, but rather through the use of instruments. And a key assumption to quantum physics is that quantum mechanical principles must reduce to Newtonian principles at the macroscopic level (there is a continuity between quantum and Newtonian mechanics).

Quantum mechanics uses the philosophical problem of wave/particle duality to provide an elegant explanation to quantized orbits around the atom. Consider what a wave looks like around an orbit, as shown below.



n=4 orbit



n=6 orbit

Only certain wavelengths of an electron matter wave will 'fit' into an orbit. If the wavelength is longer or shorter, then the ends do not connect. Thus, de Broglie matter waves explain the Bohr atom such that on certain orbits can exist to match the natural wavelength of the electron. If an electron is in some sense a wave, then in order to fit into an orbit around a nucleus, the size of the orbit must correspond to a whole number of wavelengths.

Notice also that this means the electron does not exist at one single spot in its orbit, it has a wave nature and exists at all places in the allowed orbit (the uncertainty

principle). Thus, a physicist speaks of allowed orbits and allowed transitions to produce particular photons (that make up the fingerprint pattern of spectral lines).

Quantum mechanics was capable of bringing order to the uncertainty of the microscopic world by treatment of the wave function with new mathematics. Key to this idea was the fact that relative probabilities of different possible states are still determined by laws. Thus, there is a difference between the role of chance in quantum mechanics and the unrestricted chaos of a lawless Universe.

The quantum description of reality is objective (weak form) in the sense that everyone armed with a quantum physics education can do the same experiments and come to the same conclusions. Strong objectivity, as in classical physics, requires that the picture of the world yielded by the sum total of all experimental results to be not just a picture or model, but identical with the objective world, something that exists outside of us and prior to any measurement we might have of it. Quantum physics does not have this characteristic due to its built-in indeterminacy.

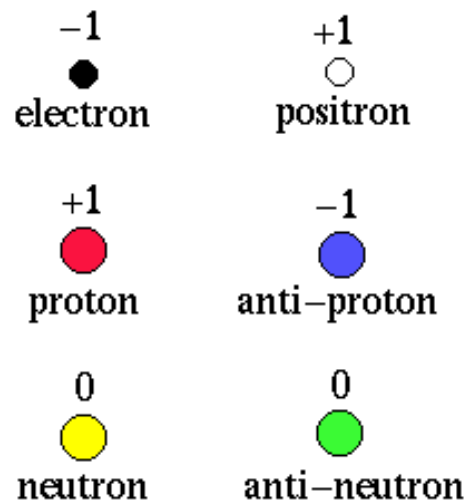
For centuries, scientists have gotten used to the idea that something like strong objectivity is the foundation of knowledge. So much so that we have come to believe that it is an essential part of the scientific method and that without this most solid kind of objectivity science would be pointless and arbitrary. However, quantum physics denies that there is any such thing as a true and unambiguous reality at the bottom of everything. Reality is what you measure it to be, and no more. No matter how uncomfortable science is with this viewpoint, quantum physics is extremely accurate and is the foundation of modern physics (perhaps then an objective view of reality is not essential to the conduct of physics). And concepts, such as cause and effect, survive only as a consequence of the collective behavior of large quantum systems.

Antimatter:

A combination of quantum mechanics and relativity allows us to examine subatomic processes in a new light. Symmetry is very important to physical theories. For example, conservation of momentum is required for symmetry in time. Thus, the existence of a type of 'opposite' matter was hypothesized soon after the development of quantum physics. 'Opposite' matter is called antimatter. Particles of antimatter has the same mass and characteristics of regular matter, but opposite in charge. When matter and antimatter come in contact they are both instantaneously converted into pure energy, in the form of photons.

Antimatter

All elementary particles have antimatter counterparts. Antimatter counterparts have the same mass, but opposite electric charge.



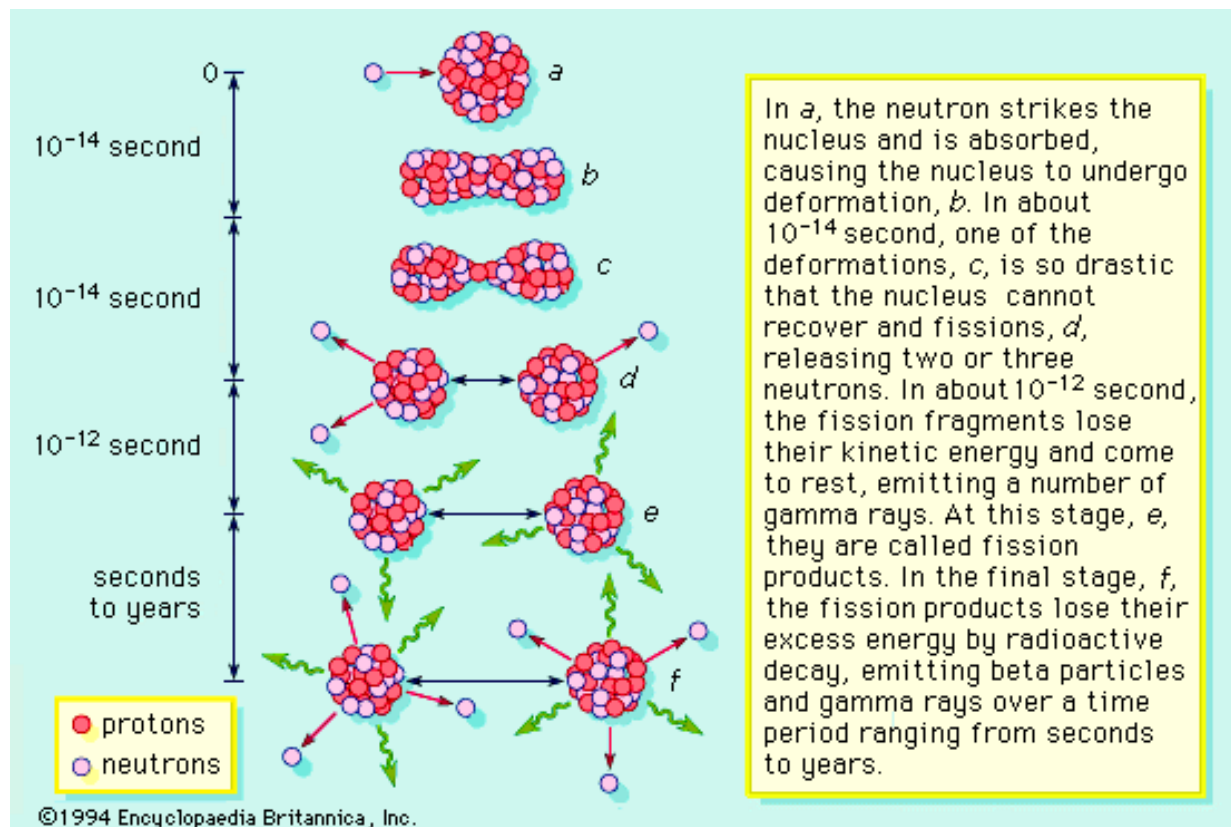
When matter and antimatter come in contact they are instantly converted into energy (annihilation).

Antimatter is produced all the time by the collision of high energy photons, a process called pair production, where an electron and its antimatter twin (the positron) are created from energy ($E=mc^2$).

Fission/Fusion:

One of the surprising results of quantum physics is that if a physical event is not specifically forbidden by a quantum rule, then it can and will happen. While this may seem strange, it is a direct result of the uncertainty principle. Things that are strict laws in the macroscopic world, such as the conservation of mass and energy, can be broken in the quantum world with the caveat that they can only be broken for very small intervals of time (less than a Planck time). The violation of conservation laws led to one of the greatest breakthroughs of the early 20th century, the understanding of radioactivity decay (fission) and the source of the power in stars (fusion).

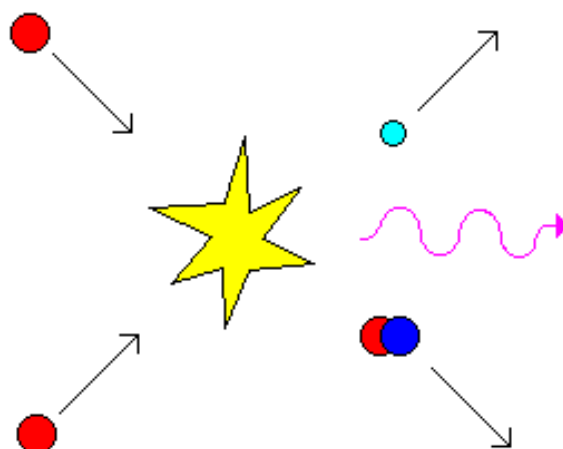
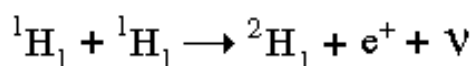
Nuclear fission is the breakdown of large atomic nuclei into smaller elements. This can happen spontaneously (radioactive decay) or induced by the collision with a free neutron. Spontaneous fission is due to the fact that the wave function of a large nucleus is 'fuzzier' than the wave function of a small particle like the alpha particle. The uncertainty principle states that, sometimes, an alpha particle (2 protons and 2 neutrons) can tunnel outside the nucleus and escape.



Induced fission occurs when a free neutron strikes a nucleus and deforms it. Under classical physics, the nucleus would just reform. However, under quantum physics there is a finite probability that the deformed nucleus will tunnel into two new nuclei and release some neutrons in the process, to produce a chain reaction.

Fusion is the production of heavier elements by the fusing of lighter elements. The process requires high temperatures in order to produce sufficiently high velocities for the two light elements to overcome each others electrostatic barriers.

Fusion



the 1st stage of the proton–proton chain is the fusion of two protons to produce deuterium, a positron and a neutrino – this is the energy source in the core of the Sun

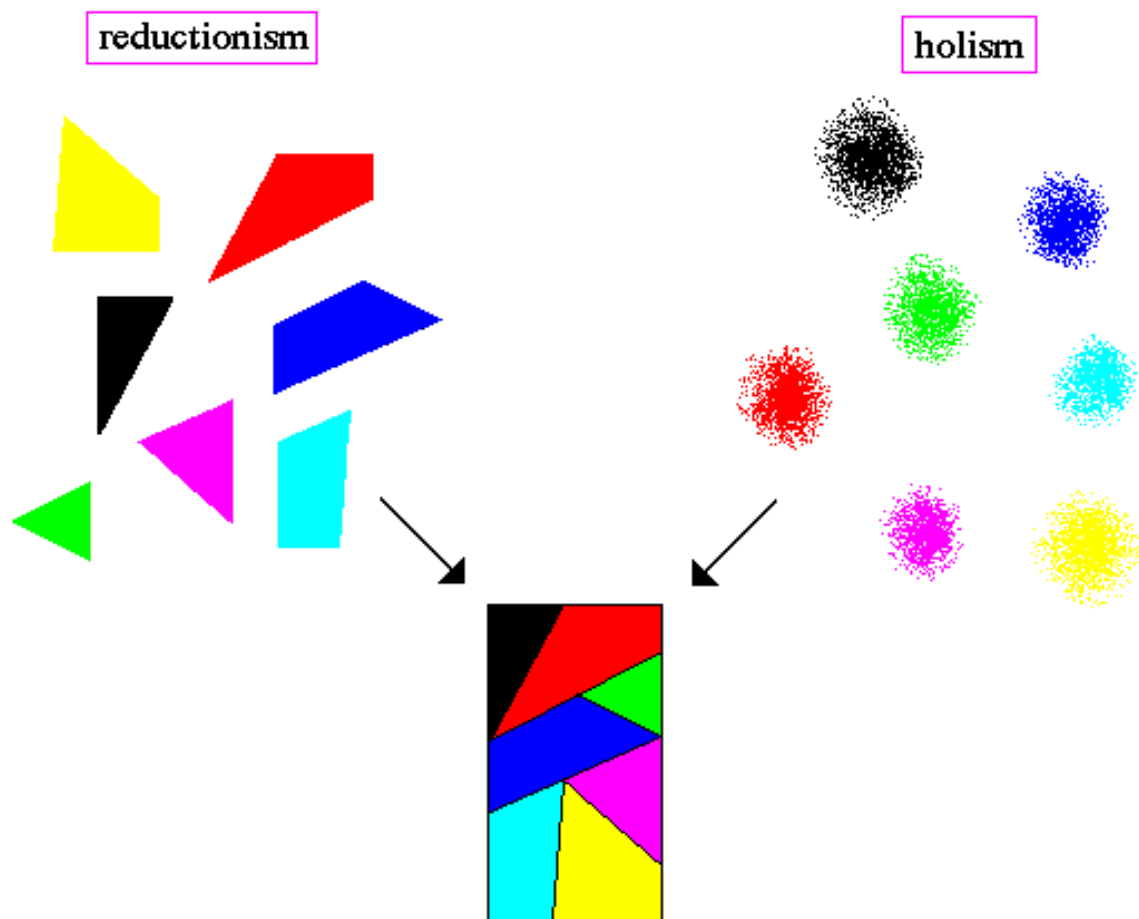
Even for the high temperatures in the center of a star, fusion requires the quantum tunneling of a neutron or proton to overcome the repulsive electrostatic forces of an atomic nuclei. Notice that both fission and fusion release energy by converting some of the nuclear mass into gamma-rays, this is the famous formulation by Einstein that $E=mc^2$.

Although it deals with probabilities and uncertainties, the quantum mechanics has been spectacularly successful in explaining otherwise inaccessible atomic phenomena and in meeting every experimental test. Its predictions are the most precise and the best checked of any in physics; some of them have been tested and found accurate to better than one part per billion.

Holism:

This is the holistic nature of the quantum world, with the behavior of individual particles being shaped into a pattern by something that cannot be explained in terms of the Newtonian reductionist paradigm. Newtonian physics is reductionistic, quantum physics is [holistic](#).

Where a reductionist believes that any whole can be broken down or analyzed into its separate parts and the relationships between them, the holist maintains that the whole is primary and often greater than the sum of its parts. Nothing can be wholly reduced to the sum of its parts.

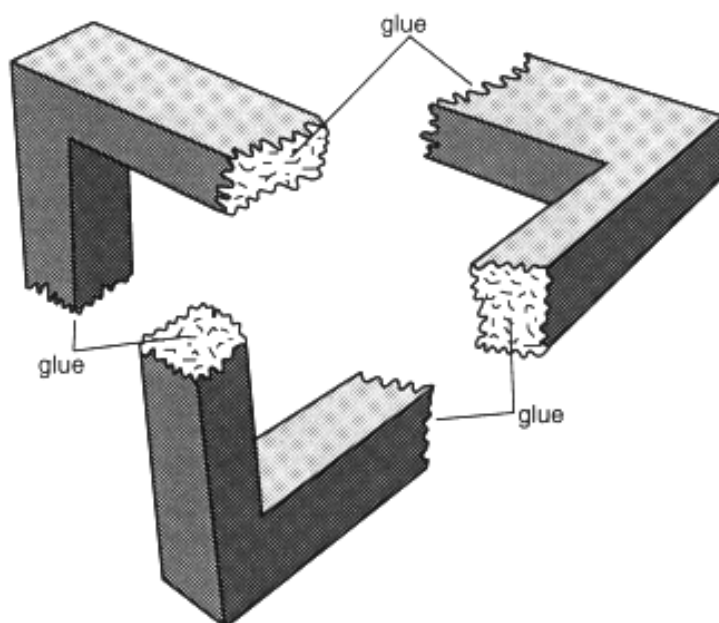


The atom theory of the Greeks viewed the Universe as consists of indestructible atoms. Change is a rearrangement of these atoms. An earlier holism of [Parmenides](#) argued that at some primary level the world is a changeless unity, indivisible and [wholly continuous](#).

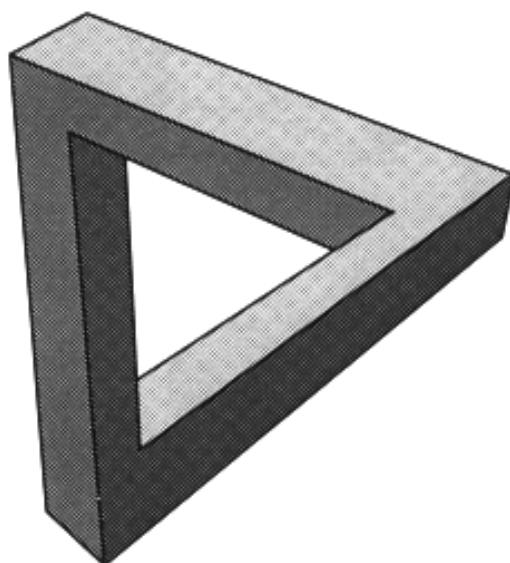
The highest development of quantum theory returns to the philosophy of Parmenides by describing all of existence as an excitation of the underlying [quantum vacuum](#), like ripples on a universal pond. The substratum of all is the quantum vacuum, similar to Buddhist idea of permanent identity.

Holism

as an example of the emergent property of holism consider the following three pieces of a triangle



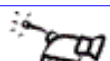
and now consider what those pieces look like when assembled (if possible)



an impossible triangle

Quantum reality is a bizarre world of both/and, whereas macroscopic world is ruled by either/or. The most outstanding problem in modern physics is to explain how the both/and is converted to either/or during the act of observation.

Note that since there are most probable positions and energy associated with the wave function, then there is some reductionism available for the observer. The truth is somewhere between Newton and Parmenides.



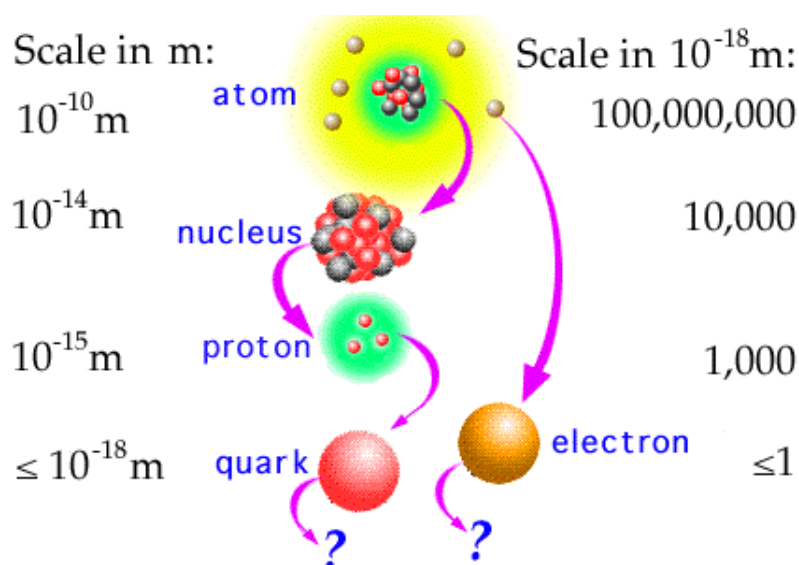


Elementary Particles :

One of the primary goals in modern physics is to answer the question "What is the Universe made of?" Often that question reduces to "What is matter and what holds it together?" This continues the line of investigation started by Democritus, Dalton and Rutherford.

Modern physics speaks of fundamental building blocks of Nature, where fundamental takes on a reductionist meaning of simple and structureless. Many of the particles we have discussed so far appear simple in their properties. All electrons have the exact same characteristics (mass, charge, etc.), so we call an electron fundamental because they are all non-unique.

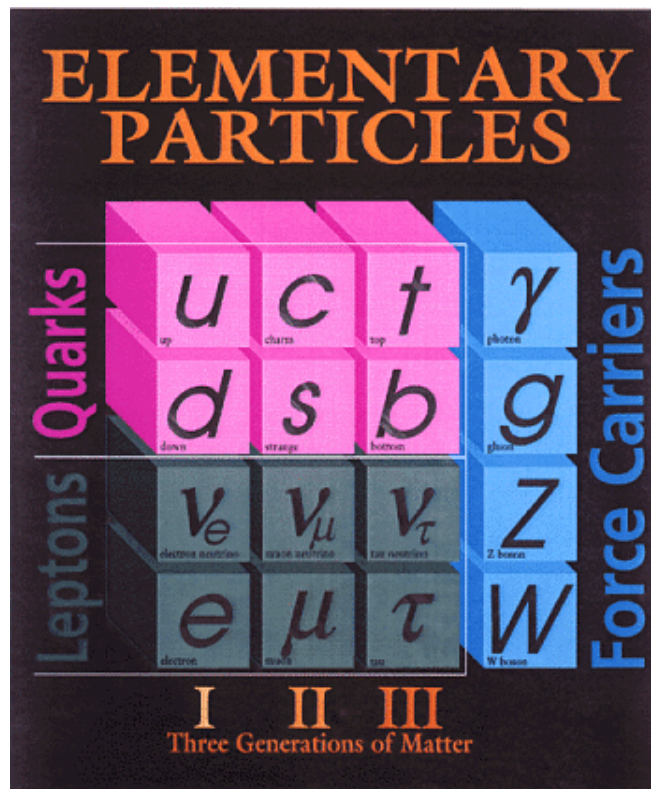
The search for the origin of matter means the understanding of [elementary particles](#). And with the advent of holism, the understanding of elementary particles requires an understanding of not only their characteristics, but how they interact and relate to other particles and forces of Nature, the field of physics called [particle physics](#).



The study of particles is also a story of advanced technology begins with the search for the primary constituent. More than 200 subatomic particles have been discovered so far, all detected in sophisticated particle accelerators. However, most are not fundamental, most are composed of other, simpler particles. For example, Rutherford showed that the atom was composed of a nucleus and orbiting electrons. Later physicists showed that the nucleus was composed of neutrons and protons. More recent work has shown that protons and neutrons are composed of quarks.

Quarks and Leptons:

The two most fundamental types of particles are [quarks](#) and [leptons](#). The quarks and leptons are divided into 6 flavors corresponding to three generations of matter. Quarks (and antiquarks) have electric charges in units of $1/3$ or $2/3$'s. Leptons have charges in units of 1 or 0.



Normal, everyday matter is of the first generation, so we can concentrate our investigation to up and down quarks, the electron [neutrino](#) (often just called the neutrino) and [electrons](#).

Quarks

- u up quark — $2/3$ charge
- d down quark — $-1/3$ charge

Leptons

- e electron — -1 charge, lightest particle
- ν neutrino — low mass, no charge, weak interacting

Note that for every quark or lepton there is a corresponding antiparticle. For example, there is an up antiquark, an anti-electron (called a positron) and an anti-neutrino. Bosons do not have antiparticles since they are force carriers (see fundamental forces).

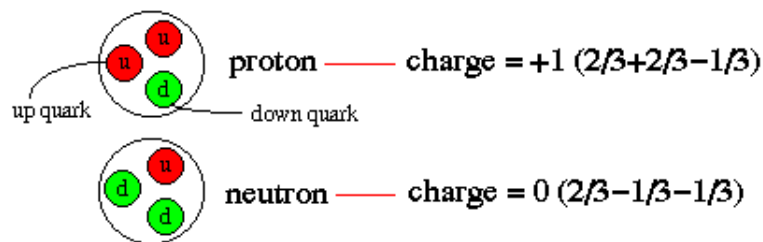
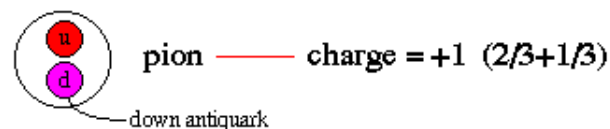
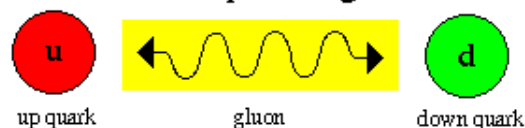
Bosons (force carriers)

 photon	massless, no charge, electromagnetic force carrier
 gluons	high mass, color charge (blue, green, red), strong force carrier
	high mass, weak force carrier
 graviton	massless, no charge, gravity force carrier

massless = moves at speed of light, long range
 high mass = moves at less than speed of light, short range

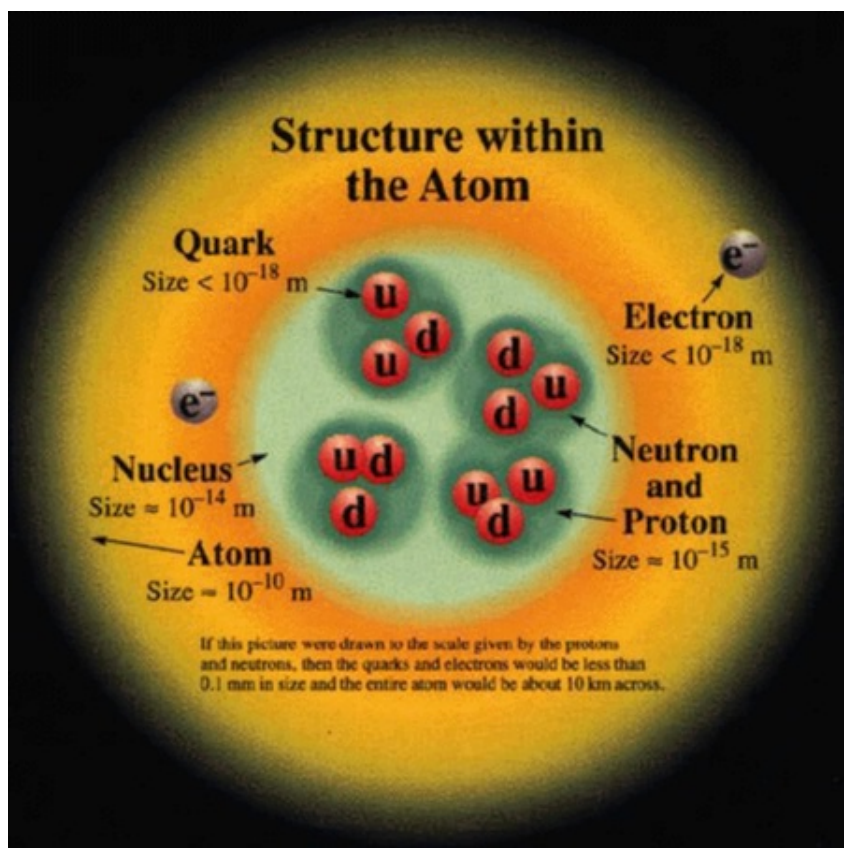
Baryons and Mesons:

Quarks combine to form the basic building blocks of matter, baryons and mesons. Baryons are made of three quarks to form the protons and neutrons of atomic nuclei (and also anti-protons and anti-neutrons). Mesons, made of quark pairs, are usually found in cosmic rays. Notice that the quarks all combine to make charges of -1, 0, or +1.

Atomic Nuclei = Combinations of QuarksBaryons = particles made of 3 quarksMesons = particles made of 2 quarksWhat binds quarks together?

the strong force carried by gluons

Thus, our current understanding of the structure of the atom is shown below, the atom contains a nucleus surrounded by a cloud of negatively charged electrons. The nucleus is composed of neutral neutrons and positively charged protons. The opposite charge of the electron and proton binds the atom together with electromagnetic forces.



The protons and neutrons are composed of up and down quarks whose fractional charges ($2/3$ and $-1/3$) combine to produce the 0 or +1 charge of the proton and neutron. The nucleus is bound together by the nuclear strong force (that overcomes the electromagnetic repulsion of like-charged protons)

Color Charge:

Quarks in baryons and mesons are bound together by the strong force in the form of the exchange of gluons. Much like how the electromagnetic force strength is determined by the amount of electric charge, the strong force strength is determined by a new quantity called color charge.

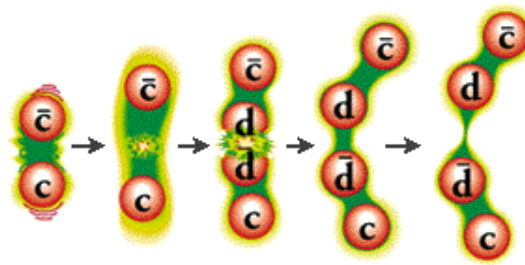
Quarks come in three colors, red, blue and green (they are not actually colored, we just describe their color charge in these terms). So, unlike electromagnetic charges which come in two flavors (positive and negative or north and south poles), color charge in quarks comes in three types. And, just to be more confusing, color charge also has its anti-particle nature. So there is anti-red, anti-blue and anti-green.

Gluons serve the function of carrying color when they interact with quarks. Baryons and mesons must have a mix of colors such that the result is white. For example, red, blue and green make white. Also red and anti-red make white.

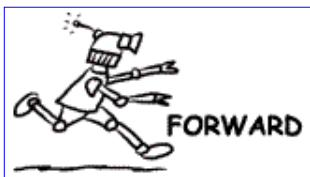
Quark Confinement:

There can exist no free quarks, i.e. quarks by themselves. All quarks must be bound to another quark or antiquark by the exchange of gluons. This is called quark confinement. The exchange of gluons produces a color force field, referring to the assignment of color charge to quarks, similar to electric charge.

The color force field is unusual in that separating the quarks makes the force field stronger (unlike electromagnetic or gravity forces which weaken with distance). Energy is needed to overcome the color force field. That energy increases until a new quark or antiquark is formed (energy equals mass, $E=mc^2$).



Two new quarks form and bind to the old quarks to make two new mesons. Thus, none of the quarks were at anytime in isolation. Quarks always travel in pairs or triplets.



Fundamental Forces :

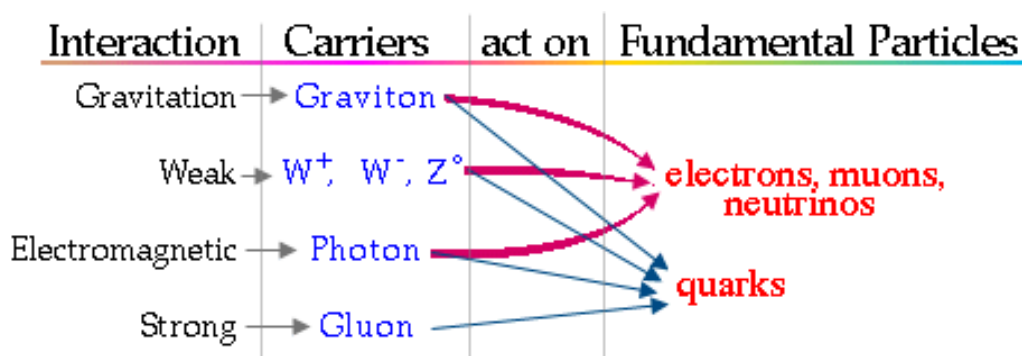
Matter is effected by forces or interactions (the terms are interchangeable). There are four fundamental forces in the Universe:

1. gravitation (between particles with mass)
2. electromagnetic (between particles with charge/magnetism)
3. strong nuclear force (between quarks)
4. weak nuclear force (operates between neutrinos and electrons)

The first two you are familiar with, gravity is the attractive force between all matter, electromagnetic force describes the interaction of charged particles and magnetics. Light (photons) is explained by the interaction of electric and magnetic fields.

The strong force binds quarks into protons, neutrons and mesons, and holds the nucleus of the atom together despite the repulsive electromagnetic force between protons. The weak force controls the radioactive decay of atomic nuclei and the reactions between leptons (electrons and neutrinos).

Current physics (called quantum field theory) explains the exchange of energy in interactions by the use of force carriers, called bosons. The long range forces have zero mass force carriers, the graviton and the photon. These operate on scales larger than the solar system. Short range forces have very massive force carriers, the W^+ , W^- and Z^0 for the weak force, the gluon for the strong force. These operate on scales the size of atomic nuclei.



So, although the strong force has the greatest strength, it also has the shortest range.

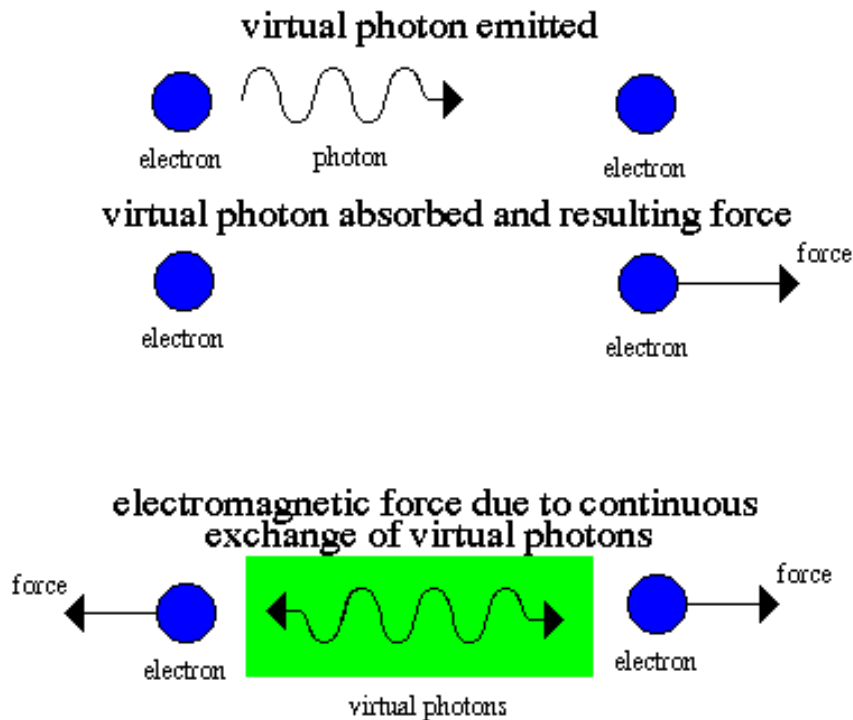
Quantum Electrodynamics :

The subfield of physics that explains the interaction of charged particles and light is called [quantum electrodynamics](#). Quantum electrodynamics (QED) extends quantum theory to fields of force, starting with electromagnetic fields.

Under QED, charged particles interact by the exchange of virtual photons, photons that do not exist outside of the interaction and only serve as carriers of

momentum/force.

Virtual Photons



Notice the elimination of action at a distance, the interaction is due to direct contact of the photons.

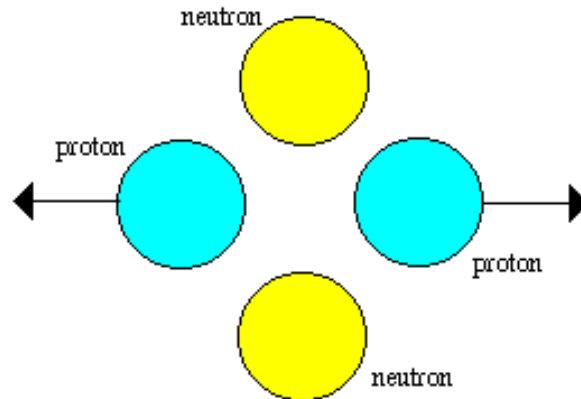
In the 1960's, a formulation of QED led to the unification of the theories of weak and electromagnetic interactions. This new force, called electroweak, occurs at extremely high temperatures such as those found in the early Universe and reproduced in particle accelerators. Unification means that the weak and electromagnetic forces become symmetric at this point, they behave as if they were one force.

Electroweak unification gave rise to the belief that the weak, electromagnetic and strong forces can be unified into what is called the Standard Model of matter.

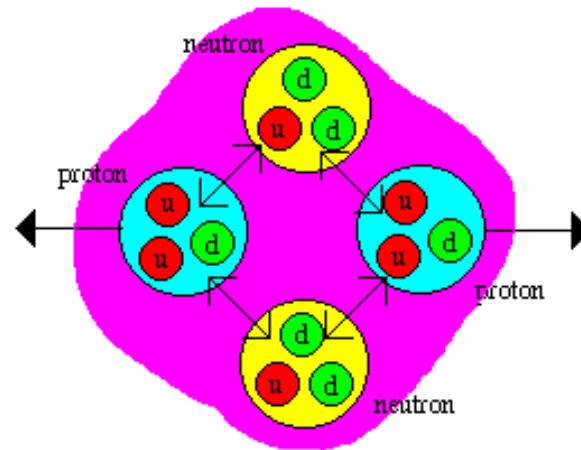
Quantum Chromodynamics:

[Quantum chromodynamics](#) is the subfield of physics that describes the strong or "color" force that binds quarks together to form baryons and mesons, and results in the complicated force that binds atomic nuclei together.

Color Force



an early dilemma for quantum mechanics was ‘‘how does the nucleus of an atom hold together with the repulsive electrostatic forces of the protons pulling it apart’’



the answer is that the color force between quarks in the proton and neutron produce the strong force, which overcomes the electrostatic force

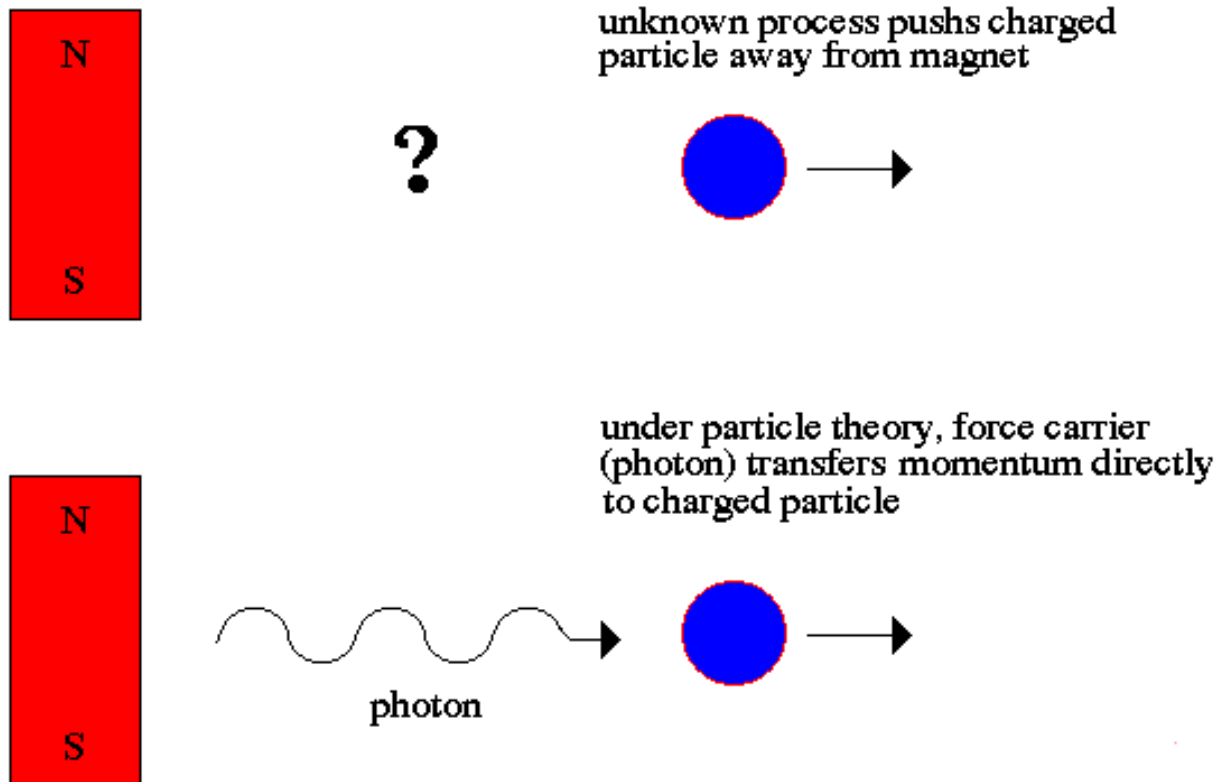
The strong force overcomes the electromagnetic or gravitational forces only on very short range. Outside the nucleus the effect of the strong force is non-existent.

Action at a Distance:

Newtonian physics assumes a direct connection between cause and effect. Electric and magnetic forces pose a dilemma for this interpretation since there is no direct contact between the two charges, rather there is an [action at a distance](#).

To resolve this dilemma it was postulated that there is an exchange of force carriers between charged particles. These force carriers were later identified with particles of light ([photons](#)). These particles served to transfer momentum by contact between charged particles, much like colliding cars and trucks.

Action at a Distance



However, this attempt to resolve the action at a distance paradox uses a particle nature to light, when observation of interference patterns clearly shows that light has a wave-like nature. It was this dual nature to light, of both particle and wave (see [wave/particle duality](#)), that led to the revolution known as [quantum physics](#).

Theory of Everything:

Is that it? Are quarks and leptons the fundamental building blocks? Answer = maybe. We are still looking to fill some holes in what is known as the [Standard Model](#).

The Standard Model is a way of making sense of the multiplicity of elementary particles and forces within a single scheme. The Standard Model is the combination of two schemes; the electroweak force (unification of electromagnetism and weak force) plus quantum chromodynamics. Although the Standard Model has brought a considerable amount of order to elementary particles and has led to important predictions, the model is not without some serious difficulties.

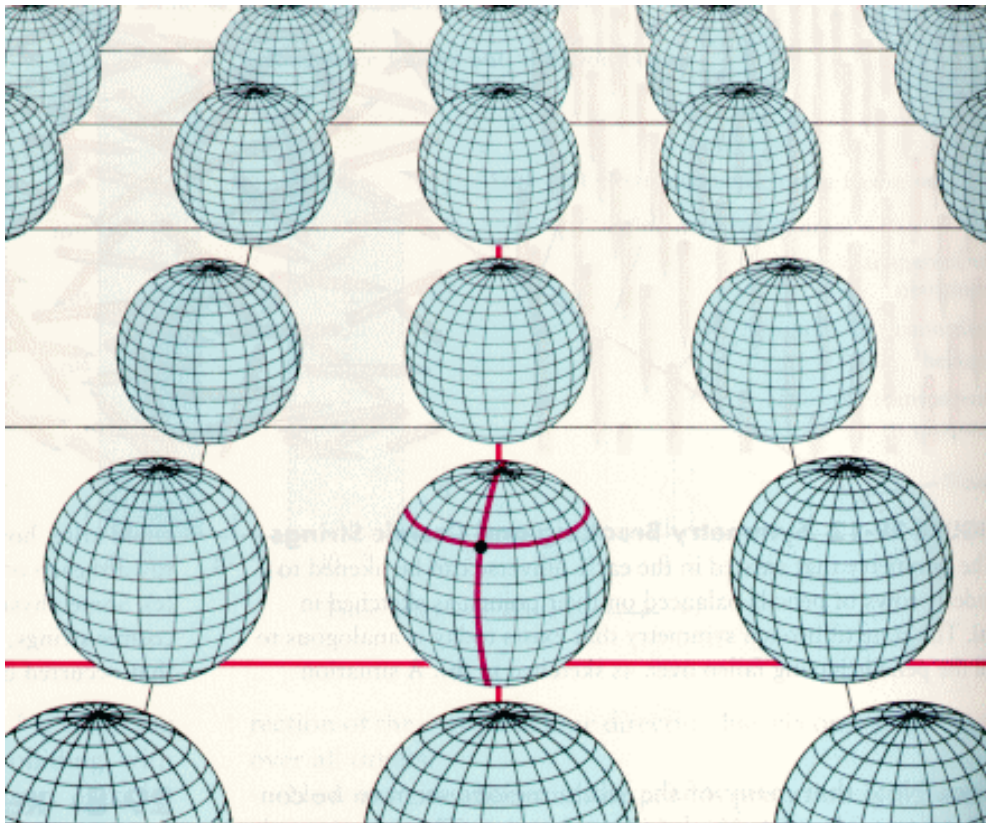
For example, the Standard Model contains a large number of arbitrary constants. Good choice of the constants leads to exact matches with experimental results. However, a good fundamental theory should be one where the constants are self-evident.

The Standard Model does not include the unification of all forces and, therefore, is incomplete. There is a strong expectation that there exists a [Grand Unified Field Theory \(GUTS\)](#) that will provide a deeper meaning to the Standard Model and explain the missing elements.

Supergravity:

Even a GUTS is incomplete because it would not include spacetime and therefore gravity. It is hypothesized that a ``Theory of Everything" (TOE) will bring together all the fundamental forces, matter and curved spacetime under one unifying picture. For cosmology, this will be the single force that controlled the Universe at the time of formation. The current approach to the search for a TOE is to attempt to uncover some fundamental symmetry, perhaps a symmetry of symmetries. There should be predictions from a TOE, such as the existence of the [Higgs particle](#), the origin of mass in the Universe.

One example of an attempt to formula a TOE is [supergravity](#), a quantum theory that unities particle types through the use of ten dimensional spacetime (see diagram below). Spacetime (4D construct) was successful at explaining gravity. What if the subatomic world is also a geometric phenomenon.



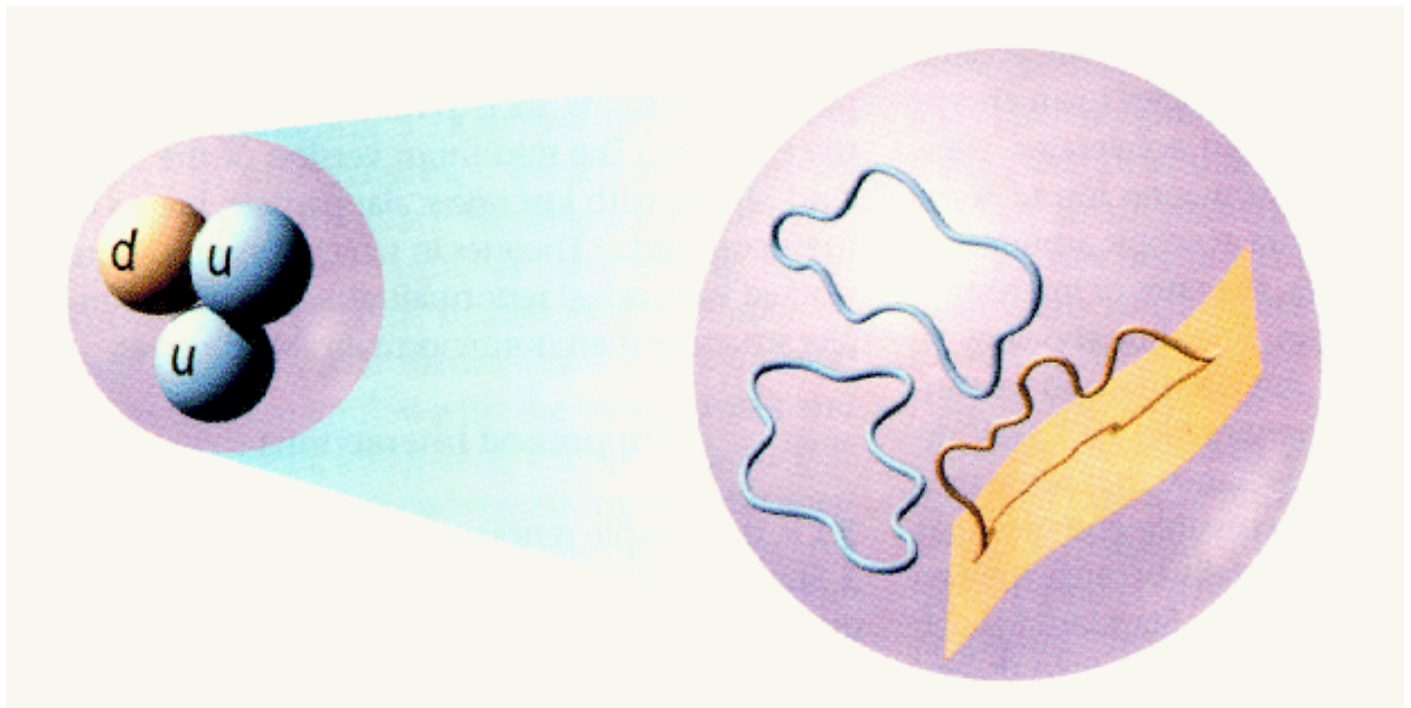
Many more dimensions of time and space could lie buried at the quantum level, outside our normal experience, only having an impact on the microscopic world of elementary particles.

It is entirely possible that beneath the quantum domain is a world of pure chaos,

without any fixed laws or symmetries. One thing is obvious, that the more our efforts reach into the realm of fundamental laws, the more removed from experience are the results.

String Theory:

Another recent attempt to form a TOE is through M (for membrane) or string theory. String theory is actually a high order theory where other models, such as supergravity and quantum gravity, appear as approximations. The basic premise to string theory is that subatomic entities, such as quarks and forces, are actually tiny loops, strings and membranes that behave as particles at high energies.

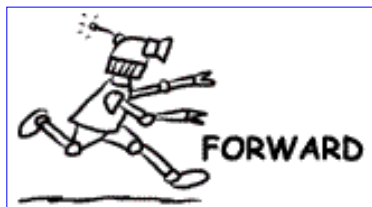


One of the problems in particle physics is the bewildering number of elementary particles (muons and pions and mesons etc). String theory answers this problem by proposing that small loops, about 100 billion billion times smaller than the proton, are vibrating below the subatomic level and each mode of vibration represents a distinct resonance which corresponds to a particular particle. Thus, if we could magnify a quantum particle we would see a tiny vibrating string or loop.

The fantastic aspect to string theory, that makes it such an attractive candidate for a TOE, is that it not only explains the nature of quantum particles but it also explains spacetime as well. Strings can break into smaller strings or combine to form larger strings. This complicated set of motions must obey self-consistent rules and the the constraint caused by these rules results in the same relations described by relativity theory.

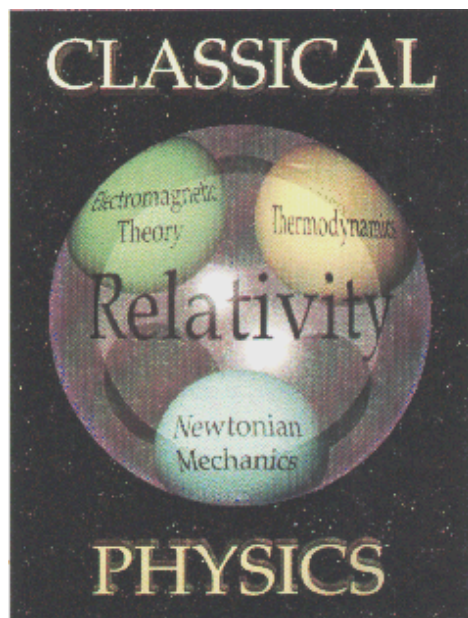
Another aspect of string theory that differs from other TOE candidates is its high aesthetic beauty. For string theory is a geometric theory, one that, like general relativity, describes objects and interactions through the use of geometry and does not

suffer from infinities or what is called normalization problems such as quantum mechanics. It may be impossible to test the predictions of string theory since it would require temperature and energies similar to those at the beginning of the Universe. Thus, we resort to judging the merit of this theory on its elegance and internal consistence rather than experiment data.



Relativity:

Einstein's theory of relativity deals with Newtonian physics when energies or velocities are near the speed of light. Relativity is usually thought of as modern physics since it was developed at the start of the 20th century and could only be tested in the realm available to scientists by high technology. However, relativity primarily completes the revolution that Newton started and is also highly deterministic as is much of classical physics.



In the holistic [viewpoint](#) of relativity theory, concepts such as length, mass and time take on a much more nebulous aspect than they do in the apparently rigid reality of our everyday world. However, what relativity takes away with one hand, it gives back in the form of new and truly fundamental constants and concepts.

The [theory of relativity](#) is traditionally broken into two parts, special and general relativity. Special relativity provides a framework for translating physical events and laws into forms appropriate for any inertial frame of reference. General relativity addresses the problem of accelerated motion and gravity.

Special Theory of Relativity:

By the late 1800's, it was becoming obvious that there were some serious problems for Newtonian physics concerning the need for absolute space and time when referring to events or interactions (frames of reference). In particular, the newly formulated theory of electromagnetic waves required that light propagation occur in a medium (the waves had to be waves on something).

In a Newtonian Universe, there should be no difference in space or time regardless of where you are or how fast you are moving. In all places, a meter is a meter and a second is a second. And you should be able to travel as fast as you want, with enough acceleration (i.e. force).

In the 1890's, two physicists (Michelson and Morley) were attempting to measure the Earth's velocity around the Sun with respect to Newtonian Absolute space and time. This would also test how light waves propagated since all waves must move through a medium. For light, this hypothetical medium was called the aether.

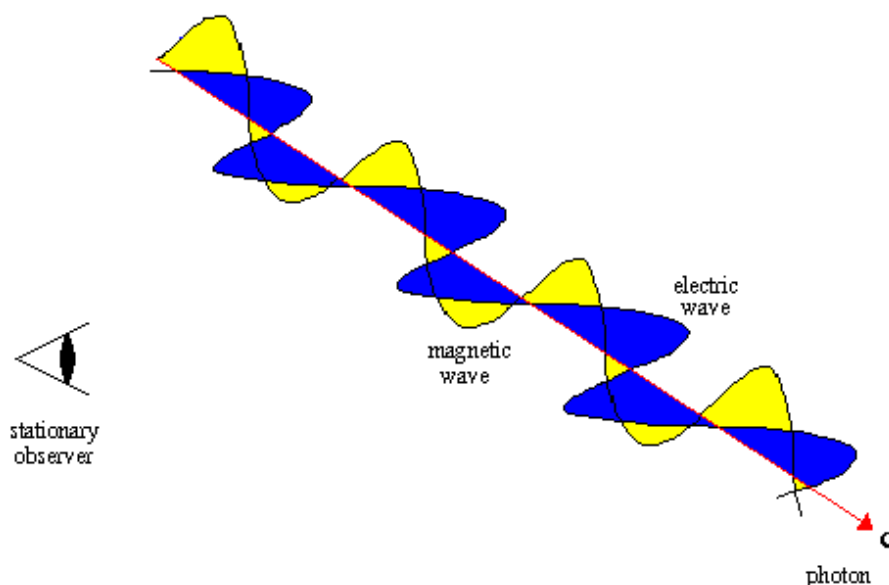
The results of the [Michelson-Morley experiment](#) was that the velocity of light was constant regardless of how the experiment was tilted with respect to the Earth's motion. This implied that

there was no aether and, thus, no absolute space. Thus, objects, or coordinate systems, moving with constant velocity (called inertial frames) were relative only to themselves.

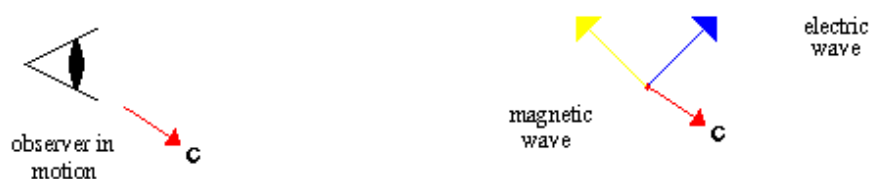
In Newtonian mechanics, quantities such as speed and distance may be transformed from one frame of reference to another, provided that the frames are in uniform motion (i.e. not accelerating).

Speed of Light

Einstein noticed there was a logical flaw in our notion of time and space (frames of reference) and light as electromagnetic waves. Consider a stationary observer who sees a passing photon as a self-propagating electric and magnetic wave.



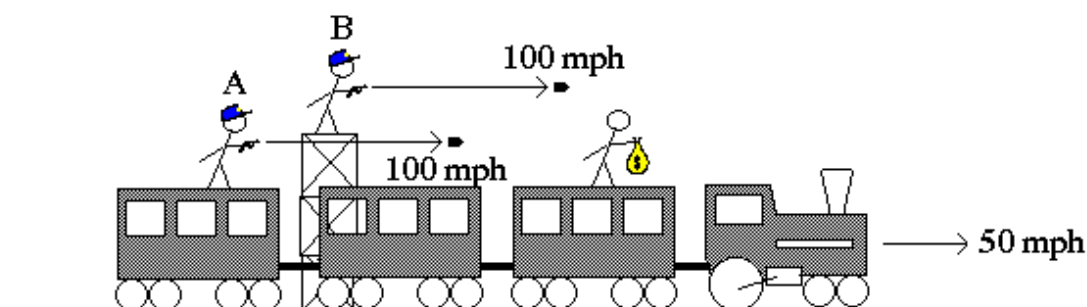
Now consider an observer who is moving at the speed of light. Since the observer is moving at the same speed as the waves, there is no wave motion and the electric and magnetic fields can not support themselves, i.e. the photon can not exist.



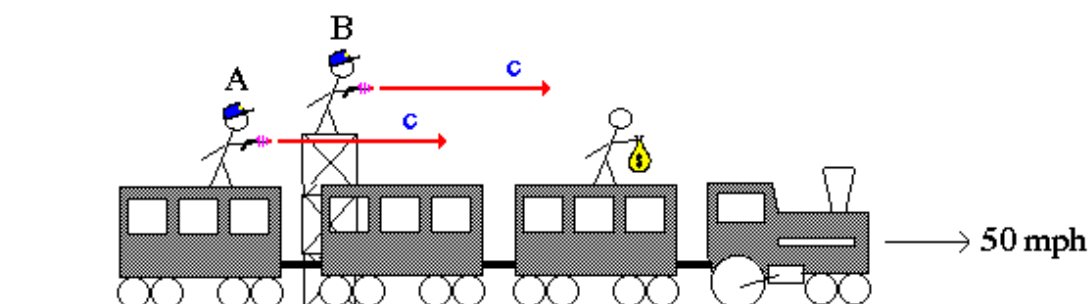
Einstein concluded that the speed of light is an upper limit to motion, that observers must always move slower than the speed of light, only massless particles (like photons) can move at the speed of light. This was the foundation for special relativity.

Considering the results of the Michelson-Morley experiment led Einstein to develop the [theory of special relativity](#). The key premise to special relativity is that the speed of light (called $c = 186,000$ miles per sec) is constant in all frames of reference, regardless of their motion. What this means can be best demonstrated by the following scenario:

Relativistic Train Robbery



bullet from A strikes train robber at 100 mph (due to inertia), bullet from B strikes train robber at $100 - 50 = 50$ mph



relativity states that the speed of light is constant in all frames, thus the light rays from both A and B strike the train robber at the same time, and at the speed of light, c .

This eliminates the paradox with respect to Newtonian physics and electromagnetism of what does a light ray 'look like' when the observer is moving at the speed of light. The solution is that only massless photons can move at the speed of light, and that matter must remain below the speed of light regardless of how much acceleration is applied.

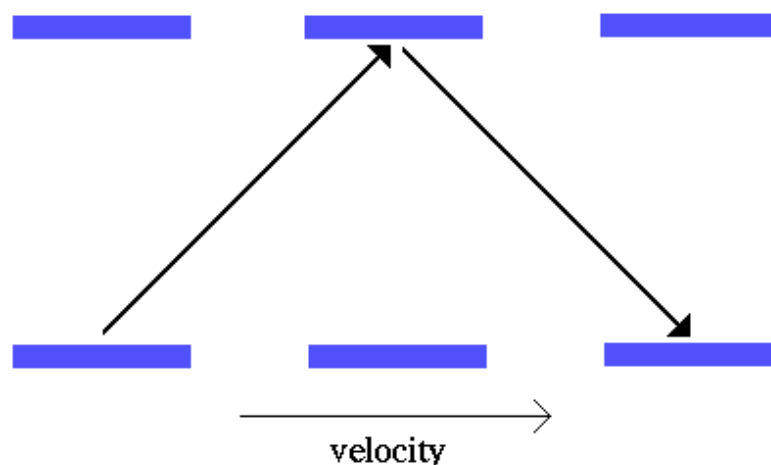
In special relativity, there is a natural upper limit to velocity, the speed of light. And the speed of light the same in all directions with respect to any frame. A surprising result to the speed of light limit is that clocks can run at different rates, simply when they are traveling a different velocities.

Photon Clock



A photon clock is a device that consists of two mirrors and a photon passing between them. Each time the photon strikes a mirror corresponds to one 'tick' of the clock.

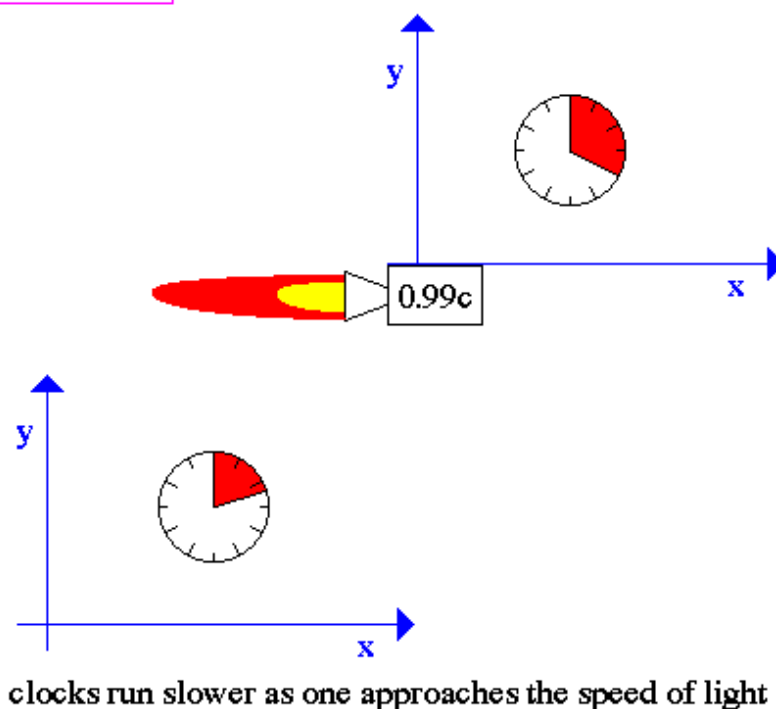
To understand time dilation, consider a moving photon clock. Since special relativity says that a photon must travel at a constant speed (c), then the path of the photon for a moving clock is longer than a clock at rest.



Therefore, the time between 'ticks' is longer for a moving clock, as seen from a rest frame.

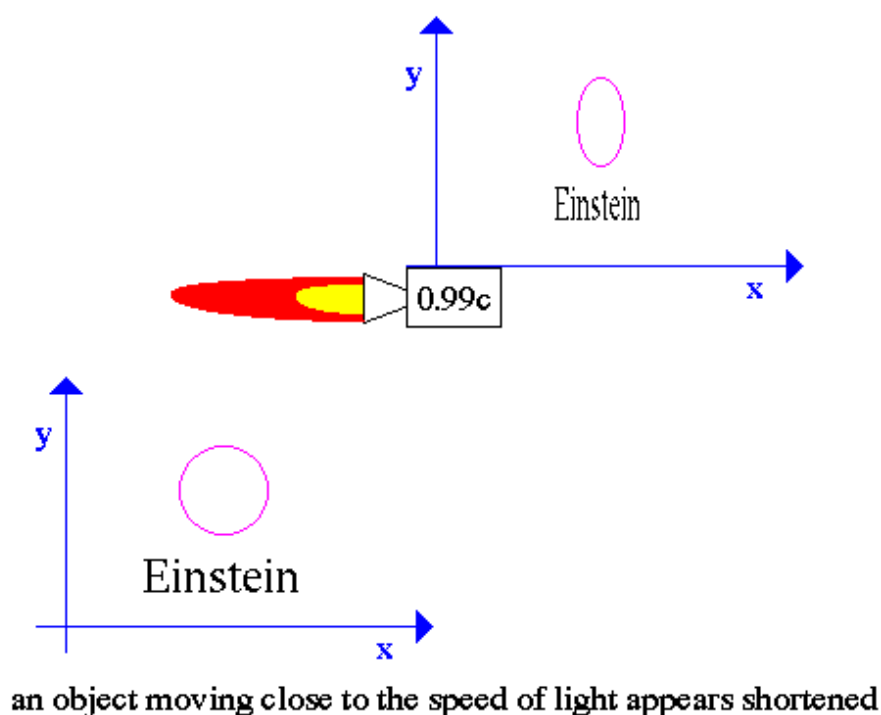
This means that time (and space) vary for frames of reference moving at different velocities with respect to each other. The change in time is called time dilation, where frames moving near the speed of light have slow clocks.

Time Dilation



Likewise, space is shortened in in high velocity frames, which is called Lorentz contraction.

Lorentz Contraction



Space-Time Lab

Time dilation leads to the famous [Twins Paradox](#), which is not a paradox but rather a simple fact of special relativity. Since clocks run slower in frames of reference at high velocity, then one can imagine a scenario were twins age at different rates when separated at birth due to a

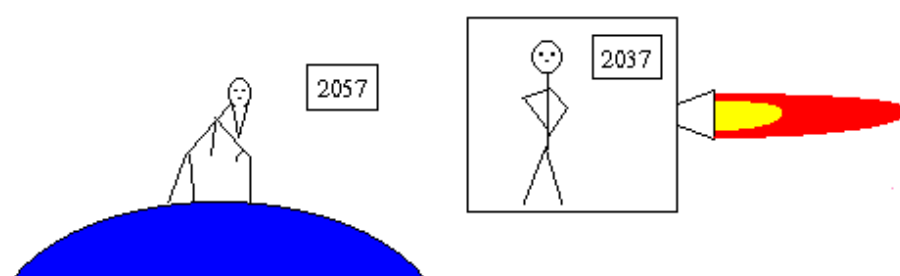
trip to the stars.

Twin Paradox

one set of twins leaves the Earth in a rocketship bound for the stars



60 years later the rocket returns to with the astronaut only 40 years old due to time dilation



It is important to note that all the predictions of special relativity, length contraction, time dilation and the twin paradox, have been confirmed by direct experiments, mostly using sub-atomic particles in high energy accelerators. The effects of relativity are dramatic, but only when speeds approach the speed of light. At normal velocities, the changes to clocks and rulers are too small to be measured.

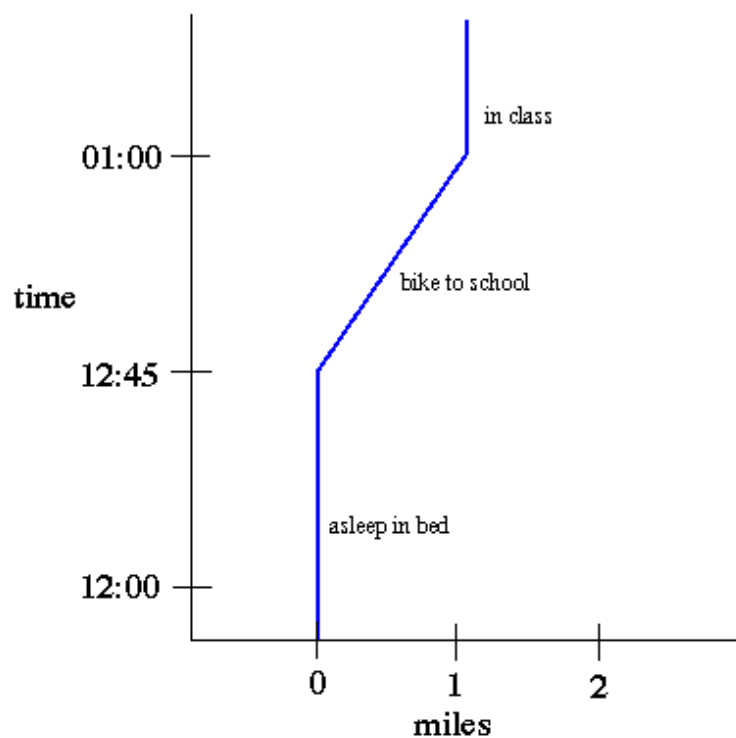
Spacetime:

Special relativity demonstrated that there is a relationship between spatial coordinates and temporal coordinates. That we can no longer reference where without some reference to when. Although time remains physically distinct from space, time and the three dimensional space coordinates are so intimately bound together in their properties that it only makes sense to describe them jointly as a four dimensional continuum.

Einstein introduced a new concept, that there is an inherent connection between geometry of the Universe and its temporal properties. The result is a four dimensional (three of space, one of time) continuum called [spacetime](#) which can best be demonstrated through the use of Minkowski diagrams and world lines.

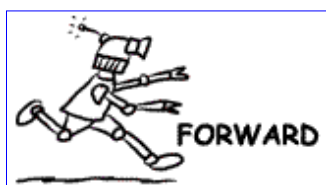
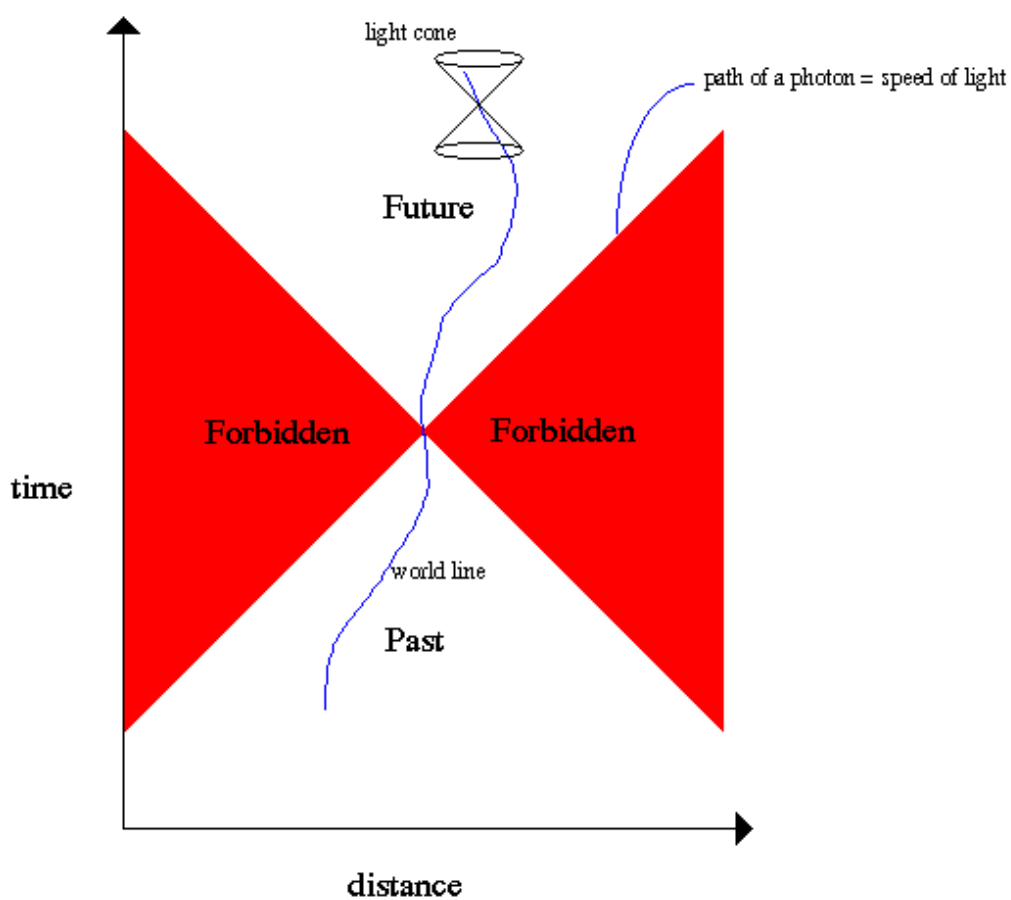
World Lines

A world line is the plot of an event or object in spacetime. Typically only one of the 3 spatial coordinates are shown in order to make a flat plot



Spacetime makes sense from special relativity since it was shown that spatial coordinates (Lorentz contraction) and temporal coordinates (time dilation) vary between frames of reference. Notice that under spacetime, time does not 'happen' as perceived by humans, but rather all time exists, stretched out like space in its entirety. Time is simply 'there'.

Light Cones



Mass-Energy Equivalence:

Since special relativity demonstrates that space and time are variable concepts from different frames of reference, then velocity (which is space divided by time) becomes a variable as well. If velocity changes from reference frame to reference frame, then concepts that involve velocity must also be relative. One such concept is momentum, motion energy.

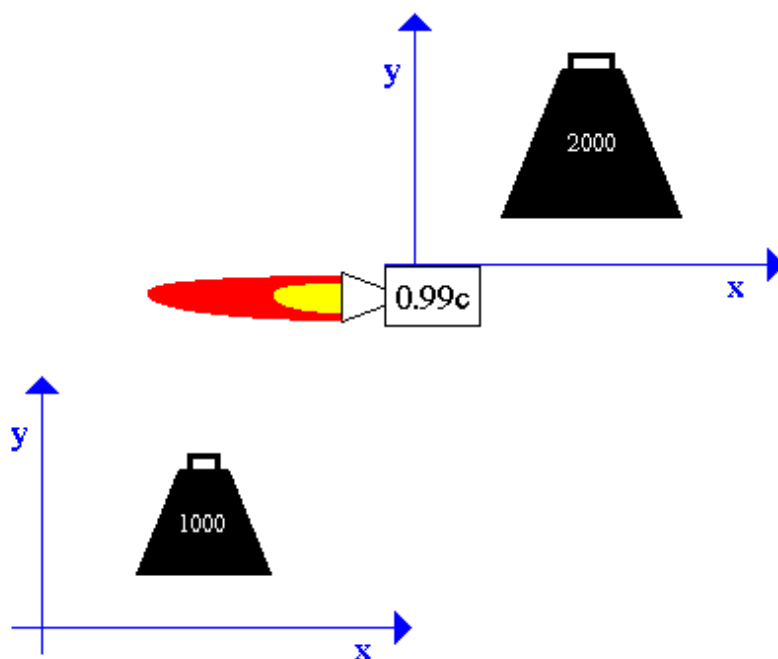
Momentum, as defined by Newtonian, can not be conserved from frame to frame under special relativity. A new parameter had to be defined, called relativistic momentum, which is conserved, but only if the mass of the object is added to the momentum equation.

This has a big impact on classical physics because it means there is an equivalence between mass and energy, summarized by the famous Einstein equation:

$$E = mc^2$$

The implications of this was not realized for many years. For example, the production of energy in nuclear reactions (i.e. fission and fusion) was shown to be the conversion of a small amount of atomic mass into energy. This led to the development of nuclear power and weapons.

As an object is accelerated close to the speed of light, relativistic effects begin to dominate. In particular, adding more energy to an object will not make it go faster since the speed of light is the limit. The energy has to go somewhere, so it is added to the mass of the object, as observed from the rest frame. Thus, we say that the observed mass of the object goes up with increased velocity. So a spaceship would appear to gain the mass of a city, then a planet, than a star, as its velocity increased.

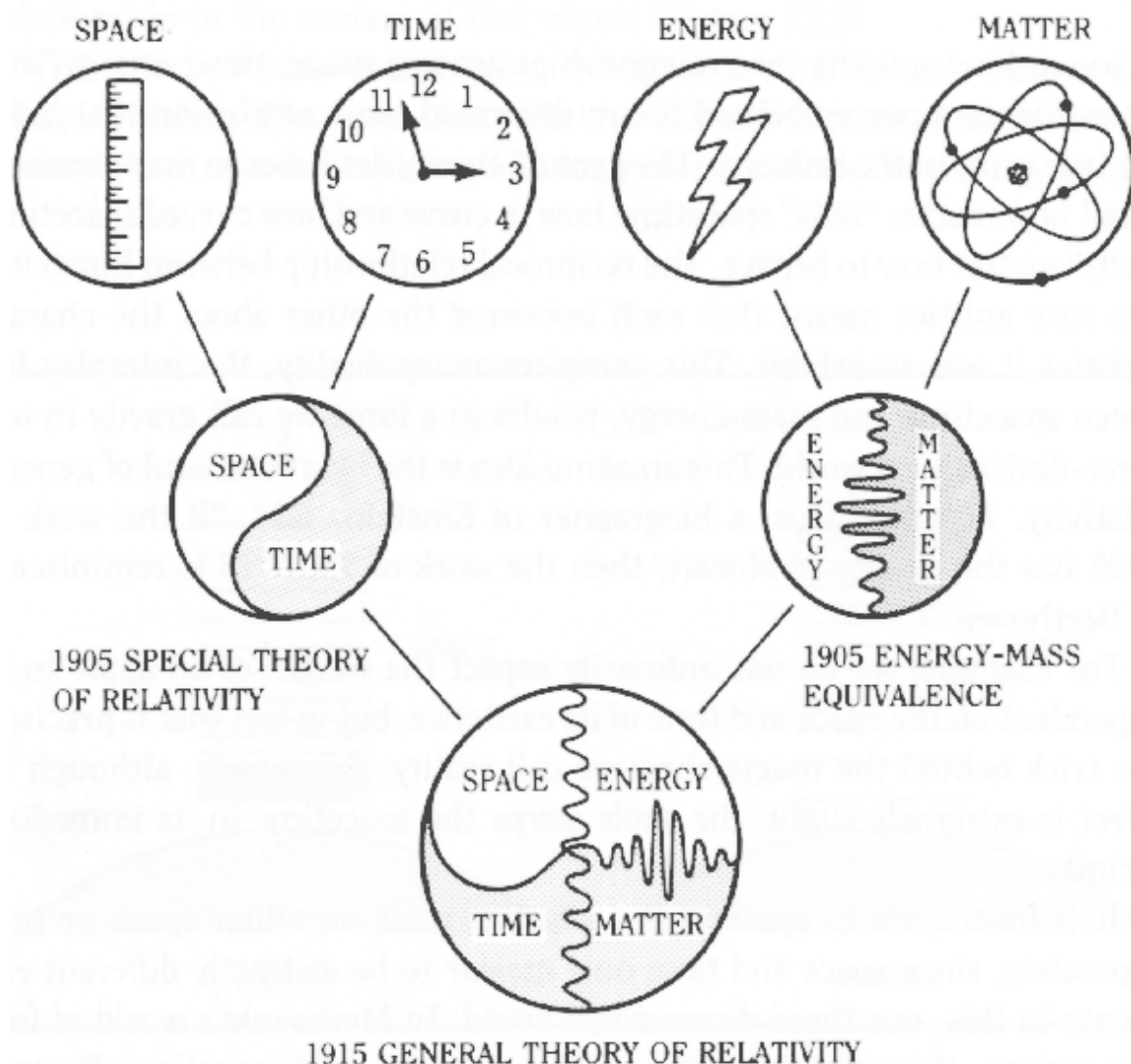


an objects mass increases as it approaches the speed of light

Likewise, the equivalence of mass and energy allowed Einstein to predict that the photon has momentum, even though its mass is zero. This allows the development of light sails and photoelectric detectors.

Spacetime and Energy:

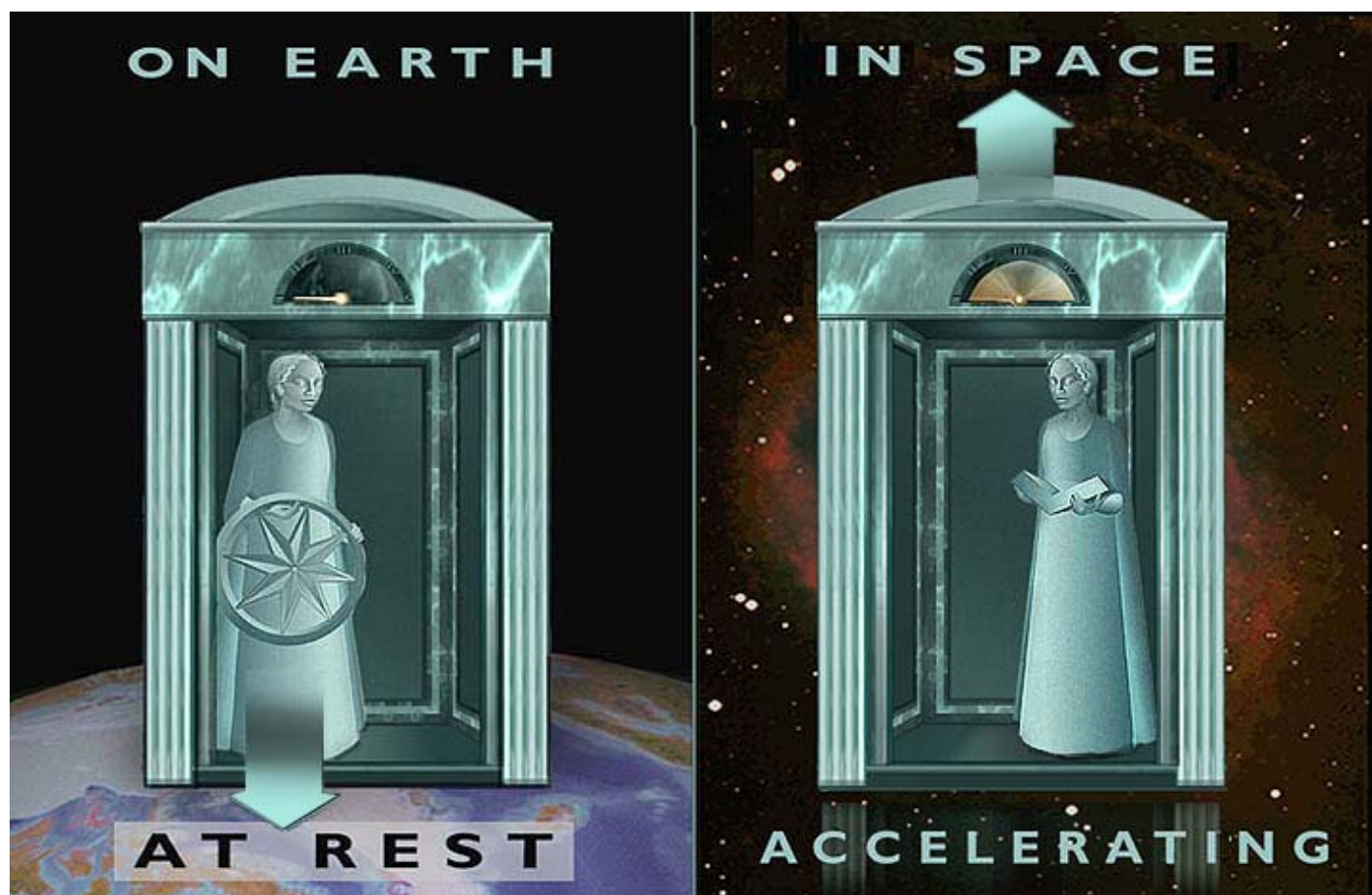
Special relativity and $E=mc^2$ led to the most powerful unification of physical concepts since the time of Newton. The previously separate ideas of space, time, energy and mass were linked by special relativity, although without a clear understanding of how they were linked.



The how and why remained to the domain of what is called general relativity, a complete theory of gravity using the geometry of spacetime. The origin of general relativity lies in Einstein's attempt to apply special relativity in accelerated frames of reference. Remember that the conclusions of relativity were founded for inertial frames, i.e. ones that move only at a uniform velocity. Adding acceleration was a complication that took Einstein 10 years to formulate.

Equivalence Principle:

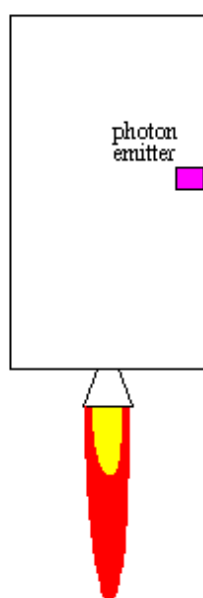
The [equivalence principle](#) was Einstein's 'Newton's apple' insight to gravitation. His thought experiment was the following, imagine two elevators, one at rest of the Earth's surface, one accelerating in space. To an observer inside the elevator (no windows) there is no physical experiment that he/she could perform to differentiate between the two scenarios.



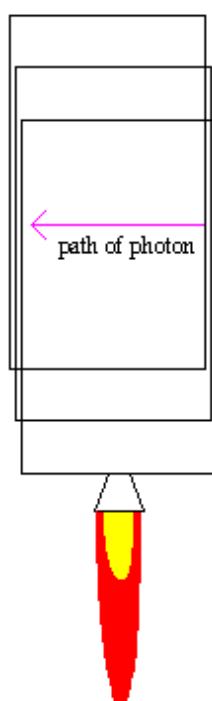
An immediate consequence of the equivalence principle is that [gravity bends light](#). To visualize why this is true imagine a photon crossing the elevator accelerating into space. As the photon crosses the elevator, the floor is accelerated upward and the photon appears to fall downward. The same must be true in a gravitational field by the equivalence principle.

Gravity Bends Light

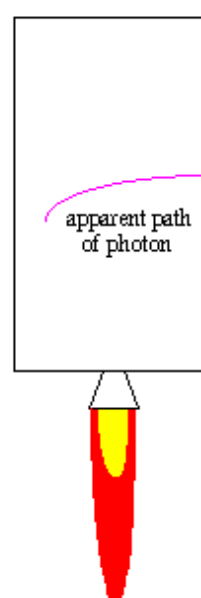
accelerating frame



view from outside



view from inside



by the equivalence principle, a photon will also “fall” in a gravitational field

The principle of equivalence renders the gravitational field fundamentally different from all other force fields encountered in nature. The new theory of gravitation, the general theory of relativity, adopts this characteristic of the gravitational field as its foundation.

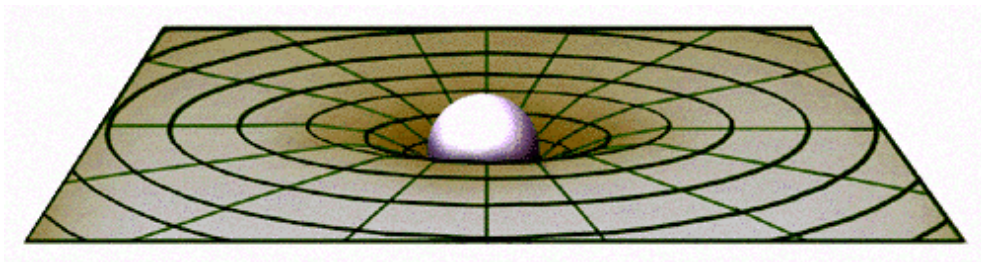
General Relativity :

The second part of relativity is the [theory of general relativity](#) and lies on two empirical findings that he elevated to the status of basic postulates. The first postulate is the relativity principle: local physics is governed by the theory of special relativity. The second postulate is the equivalence principle: there is no way for an observer to distinguish locally between gravity and acceleration.



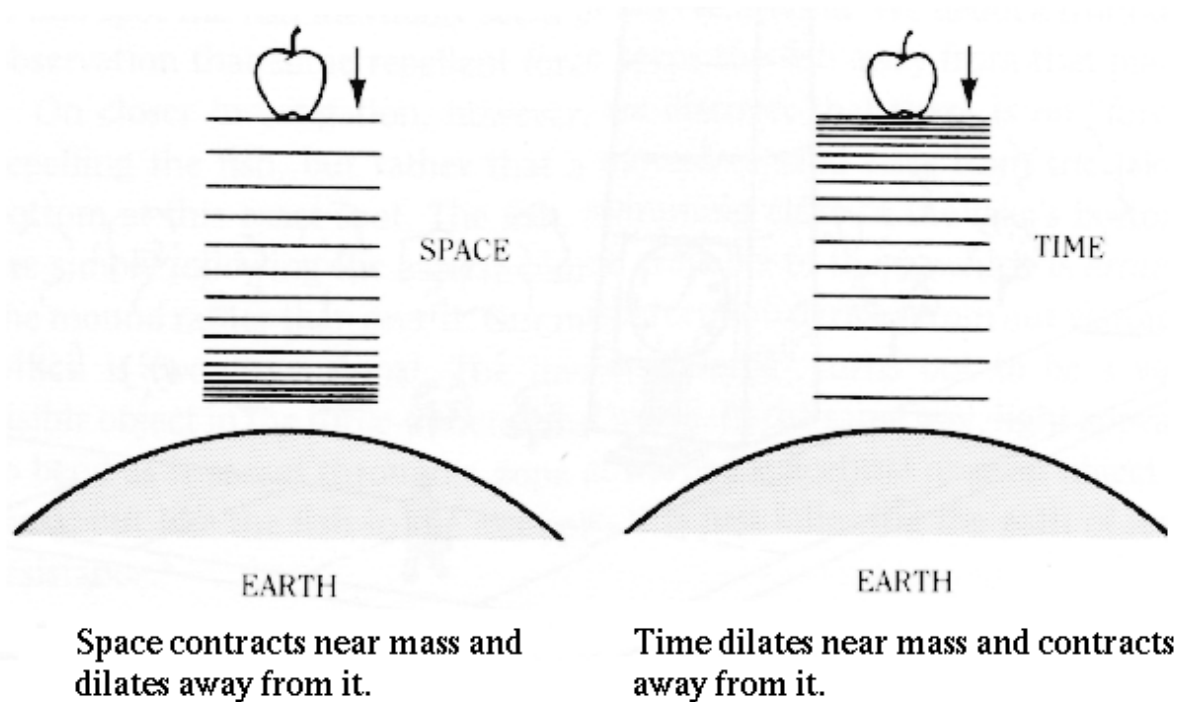
Einstein managed to fit a square peg into a round hole by modifying both the peg and the hole! His general theory of relativity resolved conflicts between Newton's theory of gravity and the special theory of relativity.

Einstein discovered that there is a relationship between mass, gravity and spacetime. Mass distorts spacetime, causing it to curve.



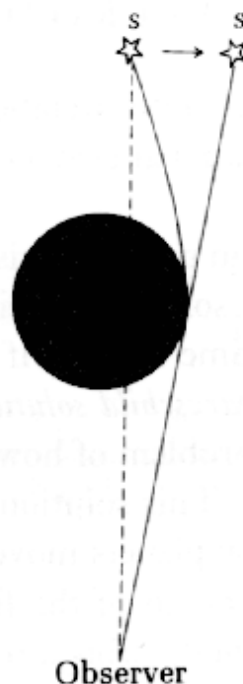
Gravity can be described as motion caused in [curved spacetime](#).

Thus, the primary result from general relativity is that gravitation is a purely geometric consequence of the properties of spacetime. Special relativity destroyed classical physics view of absolute space and time, general relativity dismantles the idea that spacetime is described by Euclidean or plane geometry. In this sense, general relativity is a field theory, relating Newton's law of gravity to the field nature of spacetime, which can be curved.



Gravity in general relativity is described in terms of curved spacetime. The idea that spacetime is distorted by motion, as in special relativity, is extended to gravity by the equivalence principle. Gravity comes from matter, so the presence of matter causes distortions or warps in spacetime. Matter tells spacetime how to curve, and spacetime tells matter how to move (orbits).

There were two classical test of general relativity, the first was that light should be [deflected](#) by passing close to a massive body. The first opportunity occurred during a total eclipse of the Sun in 1919.



Measurements of stellar positions near the darkened solar limb proved Einstein was right. Direct confirmation of [gravitational lensing](#) was obtained by the Hubble Space Telescope last year.

The second test is that general relativity predicts a time dilation in a gravitational field, so that,

relative to someone outside of the field, clocks (or atomic processes) go slowly. This was confirmed with atomic clocks flying airplanes in the mid-1970's.

The general theory of relativity is constructed so that its results are approximately the same as those of Newton's theories as long as the velocities of all bodies interacting with each other gravitationally are small compared with the speed of light--i.e., as long as the gravitational fields involved are weak. The latter requirement may be stated roughly in terms of the [escape velocity](#). A gravitational field is considered strong if the escape velocity approaches the speed of light, weak if it is much smaller. All gravitational fields encountered in the solar system are weak in this sense.

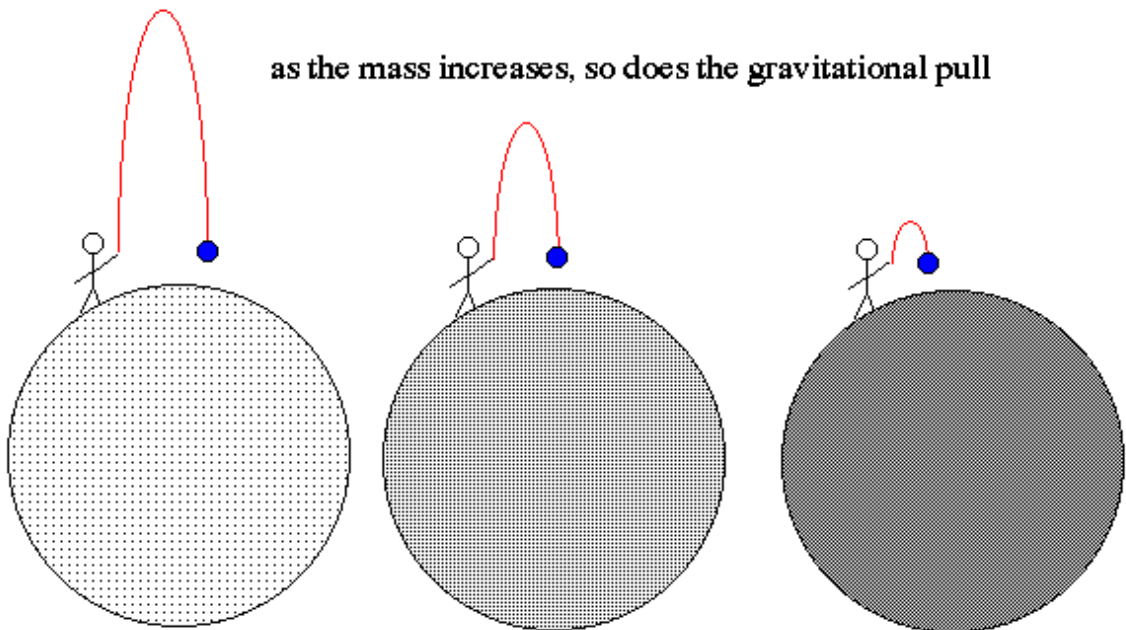
Notice that at low speeds and weak gravitational fields, general and special relativity reduce to Newtonian physics, i.e. everyday experience.

Black Holes:

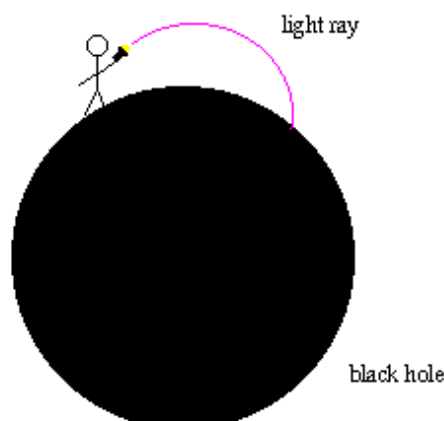
The fact that light is bent by a gravitational field brings up the following thought experiment. Imagine adding mass to a body. As the mass increases, so does the gravitational pull and objects require more energy to reach escape velocity. When the mass is sufficiently high enough that the velocity needed to escape is greater than the speed of light we say that a [black hole](#) has been created.

Black Hole

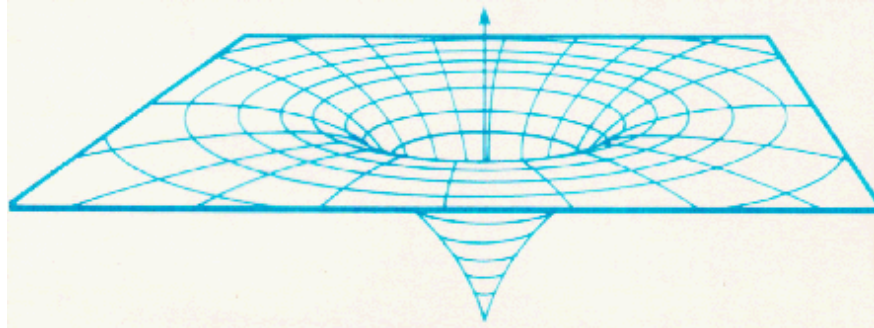
as the mass increases, so does the gravitational pull



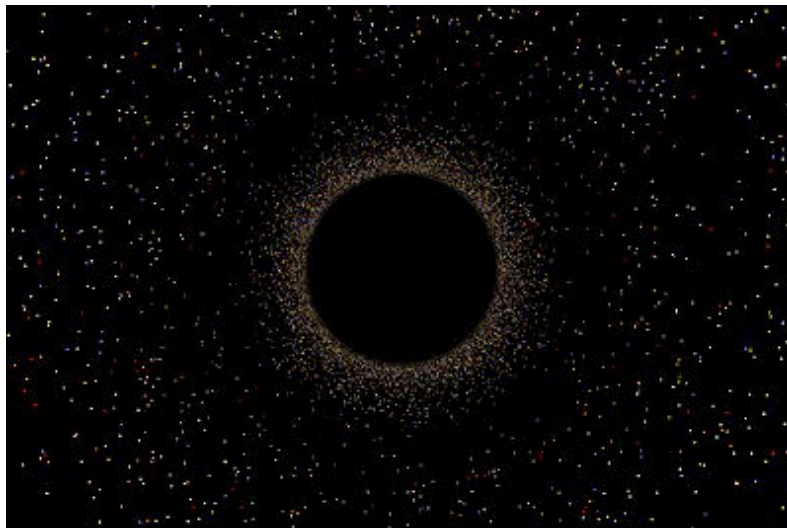
if the gravitational pull is such that even light cannot escape, then a black hole forms



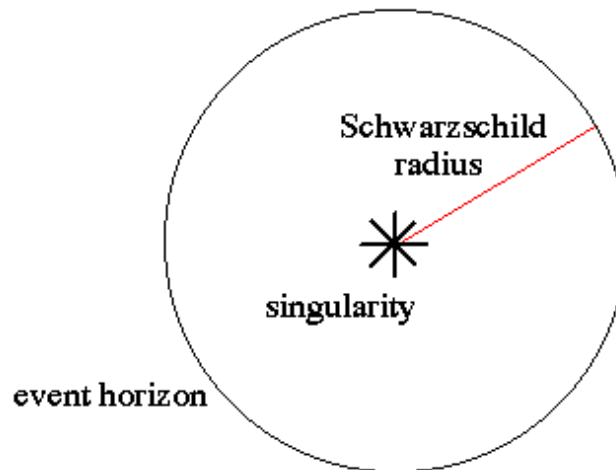
Another way of defining a black hole is that for a given mass, there is a radius where if all the mass is compressed within this radius the curvature of spacetime becomes infinite and the object is surrounded by an event horizon. This radius is called the [Schwarzschild radius](#) and varies with the mass of the object (large mass objects have large Schwarzschild radii, small mass objects have small Schwarzschild radii).



The Schwarzschild radius marks the point where the event horizon forms, below this radius no light escapes. The visual image of a black hole is one of a dark spot in space with no radiation emitted. Any radiation falling on the black hole is not reflected but rather absorbed, and starlight from behind the black hole is lensed.



Even though a black hole is invisible, it has properties and structure. The boundary surrounding the black hole at the Schwarzschild radius is called the event horizon, events below this limit are not observed. Since the forces of matter can not overcome the force of gravity, all the mass of a black hole compresses to infinity at the very center, called the singularity.

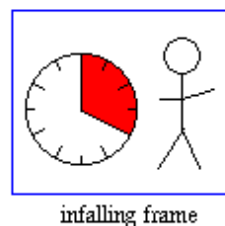
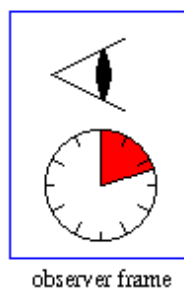
Black Hole Interior

A black hole can come in any size. Stellar mass black holes are thought to form from supernova events, and have radii of 5 km. Galactic black hole in the cores of some galaxies, millions of solar masses and the radius of a solar system, are built up over time by cannibalizing stars. Mini black holes formed in the early Universe (due to tremendous pressures) down to masses of asteroids with radii the size of a grain of sand.

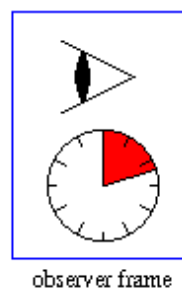


Note that a black hole is the ultimate entropy sink since all information or objects that enter a black hole never return. If an observer entered a black hole to look for the missing information, he/she would be unable to communicate their findings outside the event horizon.

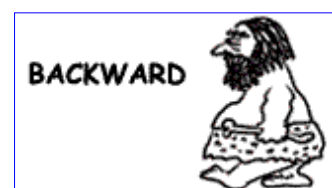
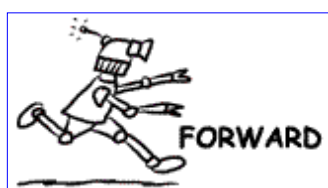
Falling into a Black Hole



the outside observer watches the infalling's frame clock slow until it freezes just above the event horizon



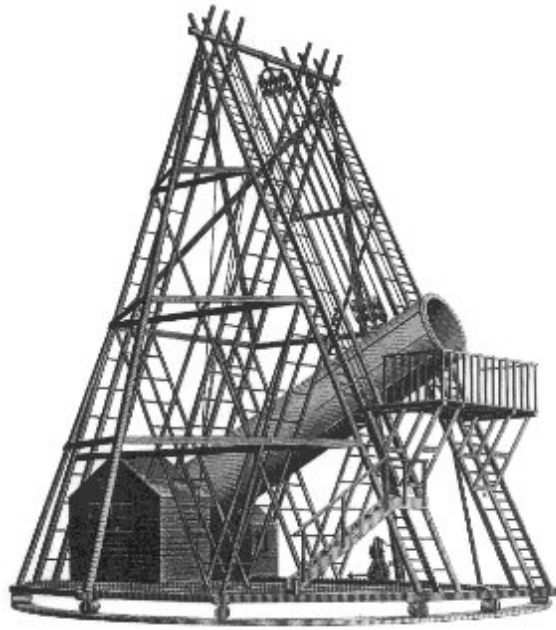
on the other hand, the infalling observer sees the rest of the Universe speeding up, watching the end of time just before falling into the event horizon



Galaxies:

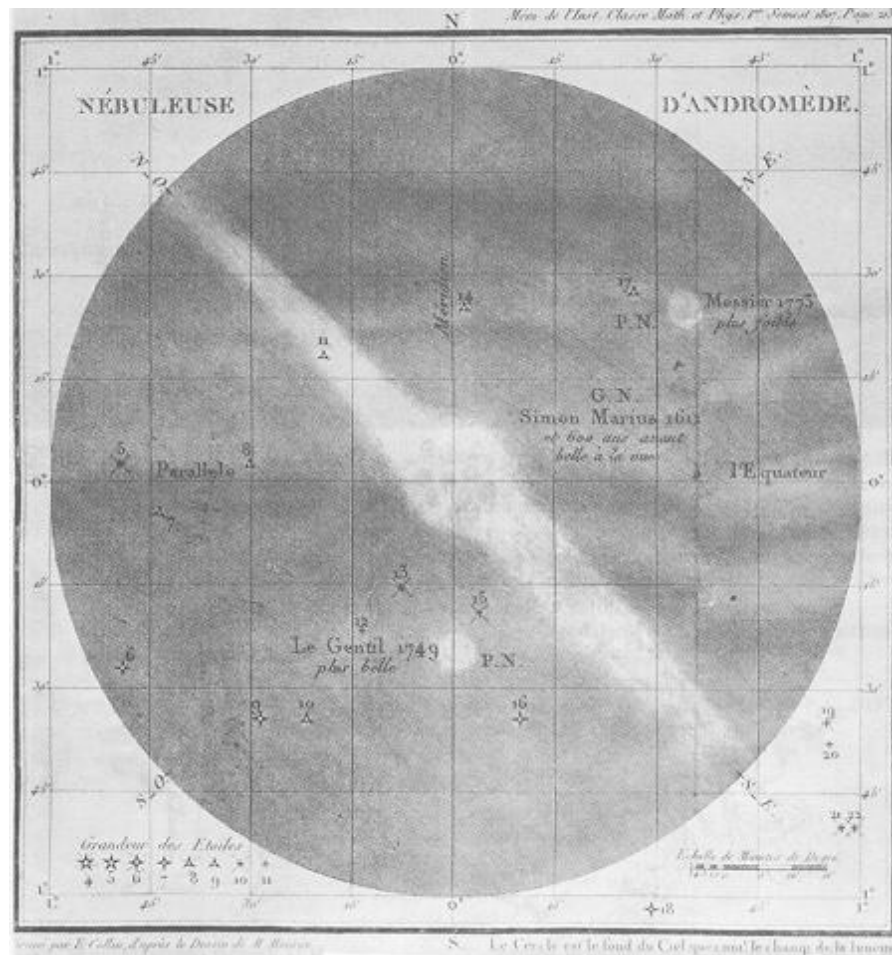
A [galaxy](#) is a collect of stars, gas and dust bound together by their common gravitational pull. Galaxies range from 10,000 to 200,000 [light-years](#) in size and between 10^9 and 10^{14} solar luminosities in brightness.

The discovery of `nebula', fuzzy objects in the sky that were not planets, comets or stars, is attributed to Charles Messier in the late 1700's. His collection of 103 objects is the first galaxy catalog. Herschel (1792-1871) used a large reflecting telescope to produce the first General Catalog of galaxies.

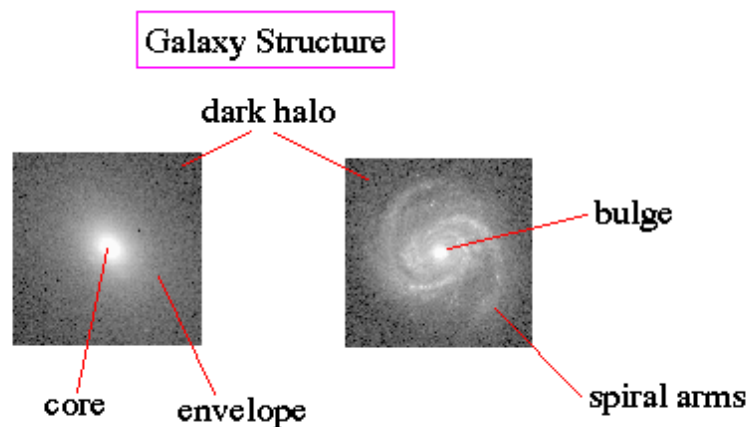


Before photographic plates, galaxies were drawn by hand by the astronomer.

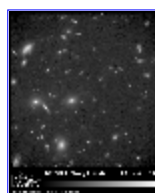




Galaxies have certain features in common. Gravity holds the billions of stars together, and the densest region is in the center, called a core or bulge. Some galaxies have spiral or pinwheel arms. All galaxies have a faint outer region or envelope and a mysterious dark matter halo.

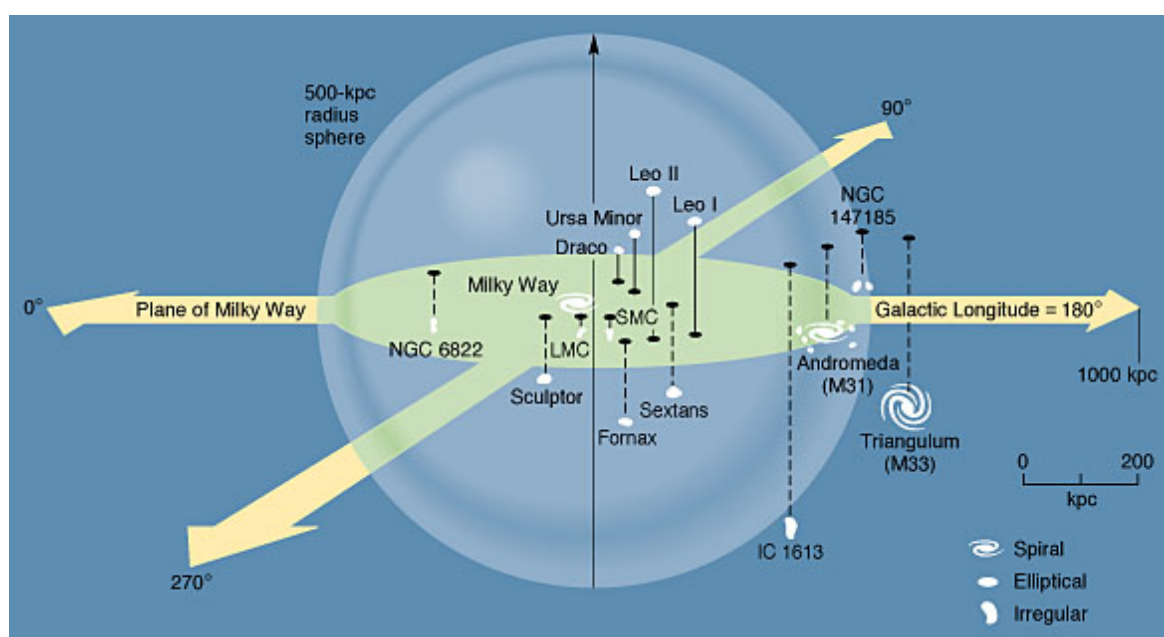


The contents of galaxies vary from galaxy type to galaxy type, and with [time](#).



Almost all galaxy types can be found in [groups](#) or [clusters](#). Many clusters of galaxies have a large,

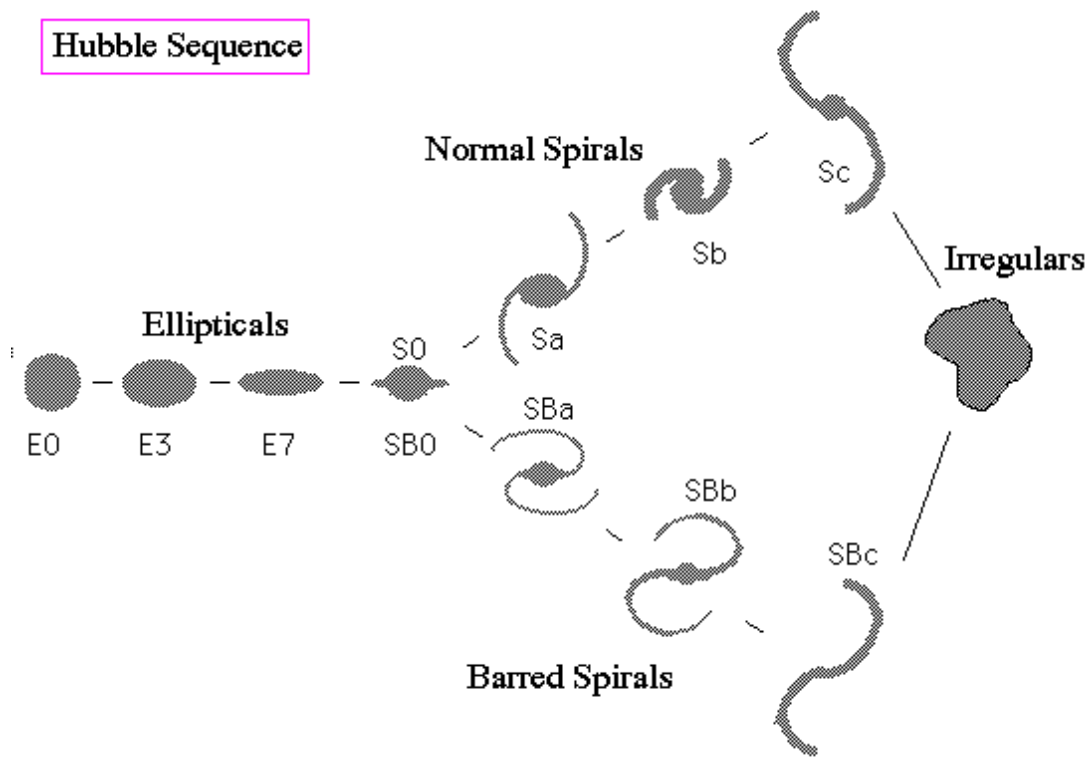
[supergiant galaxy](#) at its center which has grown by cannibalizing its neighbors. Our solar system is located in outer regions of a spiral galaxy we call the Milky Way. The nearest neighbor galaxy is [Andromeda Galaxy \(M31\)](#).



Above is a 3D plot of most of the Local Group of galaxies, the population of galaxies within 1000 kpc of the Milky Way. Clustering of dwarf satellite galaxies around the great Milky Way and Andromeda spirals can be seen.

Hubble sequence :

Almost all current systems of galaxy classification are outgrowths of the initial scheme proposed by American astronomer Edwin Hubble in 1926. In Hubble's scheme, which is based on the optical appearance of galaxy images on photographic plates, galaxies are divided into three general classes: ellipticals, spirals, and irregulars.



Elliptical galaxies :

Galaxies of this class have smoothly varying brightnesses, steadily decreasing outward from the center. They appear elliptical in shape, with lines of equal brightness made up of concentric and similar ellipses. These galaxies are nearly all of the same color: they are somewhat redder than the Sun. Ellipticals are also devoid of gas or dust and contain just old stars.



NGC 4881

All ellipticals look alike, NGC 4881 is a good example (NGC stands for New General Catalog). Notice how smooth and red NGC 4881 looks compared to the blue spirals to the right.



M32

A few ellipticals are close enough to us that we can resolve the individual stars within them, such as M32, a companion to the Andromeda Galaxy.

Spiral galaxies:

These galaxies are conspicuous for their spiral-shaped arms, which emanate from or near the nucleus and gradually wind outward to the edge. There are usually two opposing arms arranged symmetrically around the center. The nucleus of a spiral galaxy is a sharp-peaked area of smooth texture, which can be quite small or, in some cases, can make up the bulk of the galaxy. The arms are embedded in a thin disk of stars. Both the arms and the disk of a spiral system are blue in color, whereas its central areas are red like an elliptical galaxy.



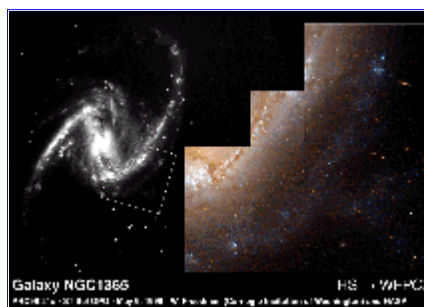
M100

Notice in the above picture of M100 from HST, that the center of the spiral is red/yellow and the arms are blue. Hotter, younger stars are blue, older, cooler stars are red. Thus, the center of a spiral is made of old stars, with young stars in the arms formed recently out of gas and dust.



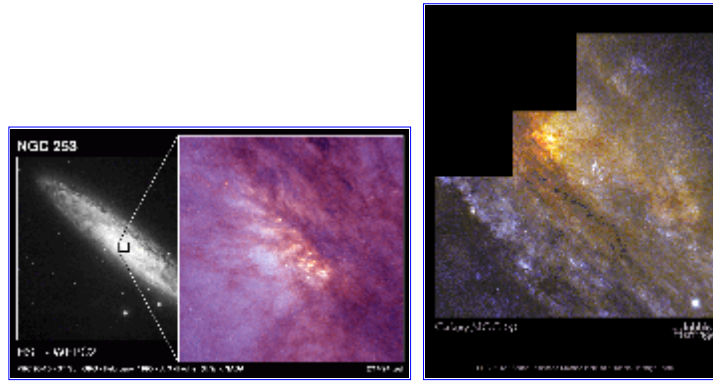
NGC 4639

The bulge of NGC 4639 is quite distinct from the younger, bluer disk regions.



NGC 1365

NGC 1365 is a barred spiral galaxy. Note the distinct dark lanes of obscuring dust in the bar pointing towards the bulge. A close-up of the spiral arms shows blue nebula, sites of current star formation.

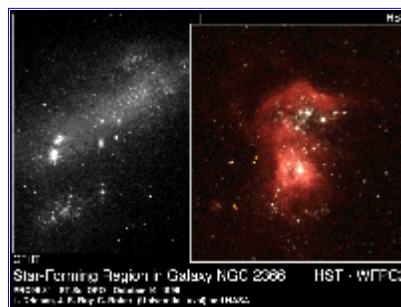


NGC 253 core and outer disk

NGC 253 is a typical Sa type galaxy with very tight spiral arms. As spiral galaxies are seen edge-on the large amount of gas and dust is visible as dark lanes and filaments crossing in front of the bulge regions.

Irregular galaxies :

Most representatives of this class consist of grainy, highly irregular assemblages of luminous areas. They have no noticeable symmetry nor obvious central nucleus, and they are generally bluer in color than are the arms and disks of spiral galaxies.



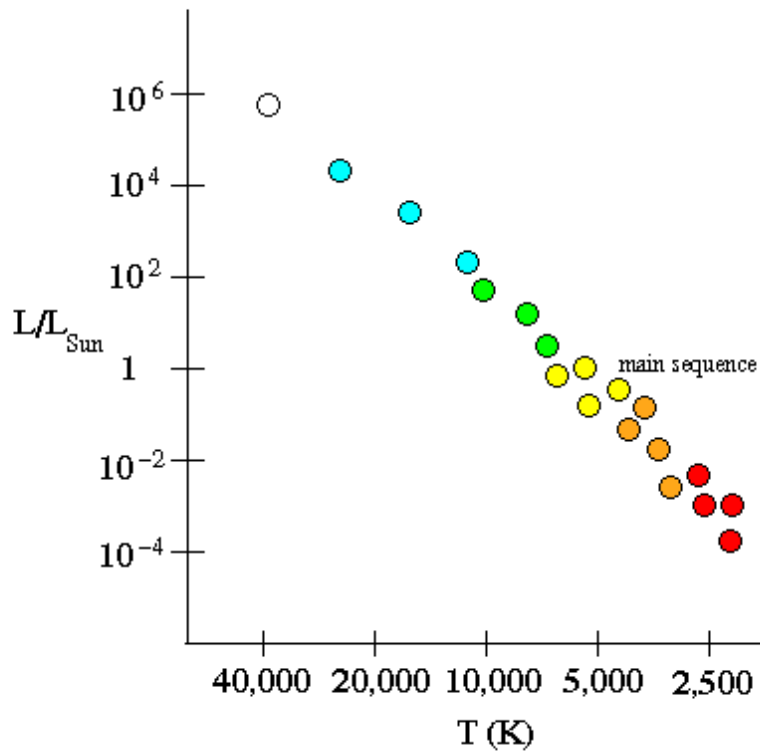
NGC 2363

NGC 2363 is an example of a nearby irregular galaxy. There is no well defined shape to the galaxy, nor are there spiral arms. A close-up of the bright region on the east side shows a cluster of new stars embedded in the red glow of ionized hydrogen gas.

Galaxy Colors:

The various colors in a galaxy (red bulge, blue disks) is due to the types of stars found in those galaxy regions, called its stellar population. Big, massive stars burn their hydrogen fuel, by thermonuclear fusion, extremely fast. Thus, they are bright and hot = blue. Low mass stars, although more numerous, are cool in surface temperature (= red) and much fainter. All this is displayed in a [Hertzsprung-Russell Diagram](#) of the young star cluster.

Hertzsprung–Russell Diagram

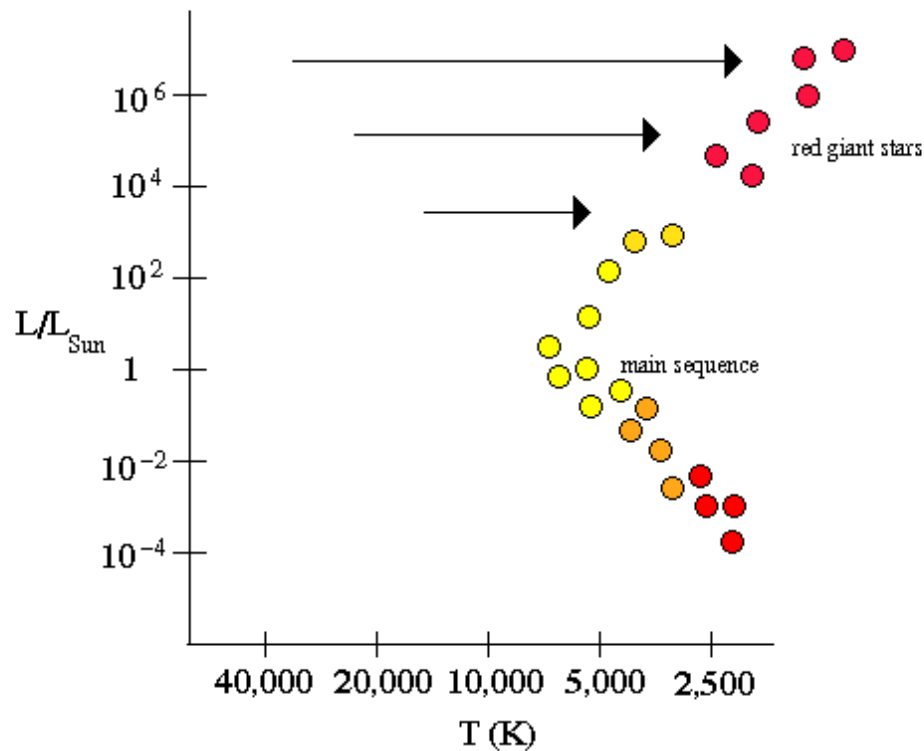


the HR diagram is a plot of stellar luminosity (with respect to the Sun) versus stellar temperature (color). Various types of stars occupy certain parts of the diagram.

A new cluster of stars will be composed of a range of stellar masses, which means a range of stellar colors $\rightarrow$ brightnesses

The hot blue stars use their core fuel much faster than the fainter, cooler red stars. Therefore, a young stellar population has a mean color that is blue (the sum of the light from all the stars in the stellar population) since most of the light is coming from the hot stars. An old stellar population is red, since all the hot stars have died off (turned into red giant stars) leaving the faint cool stars.

Old Stellar Population

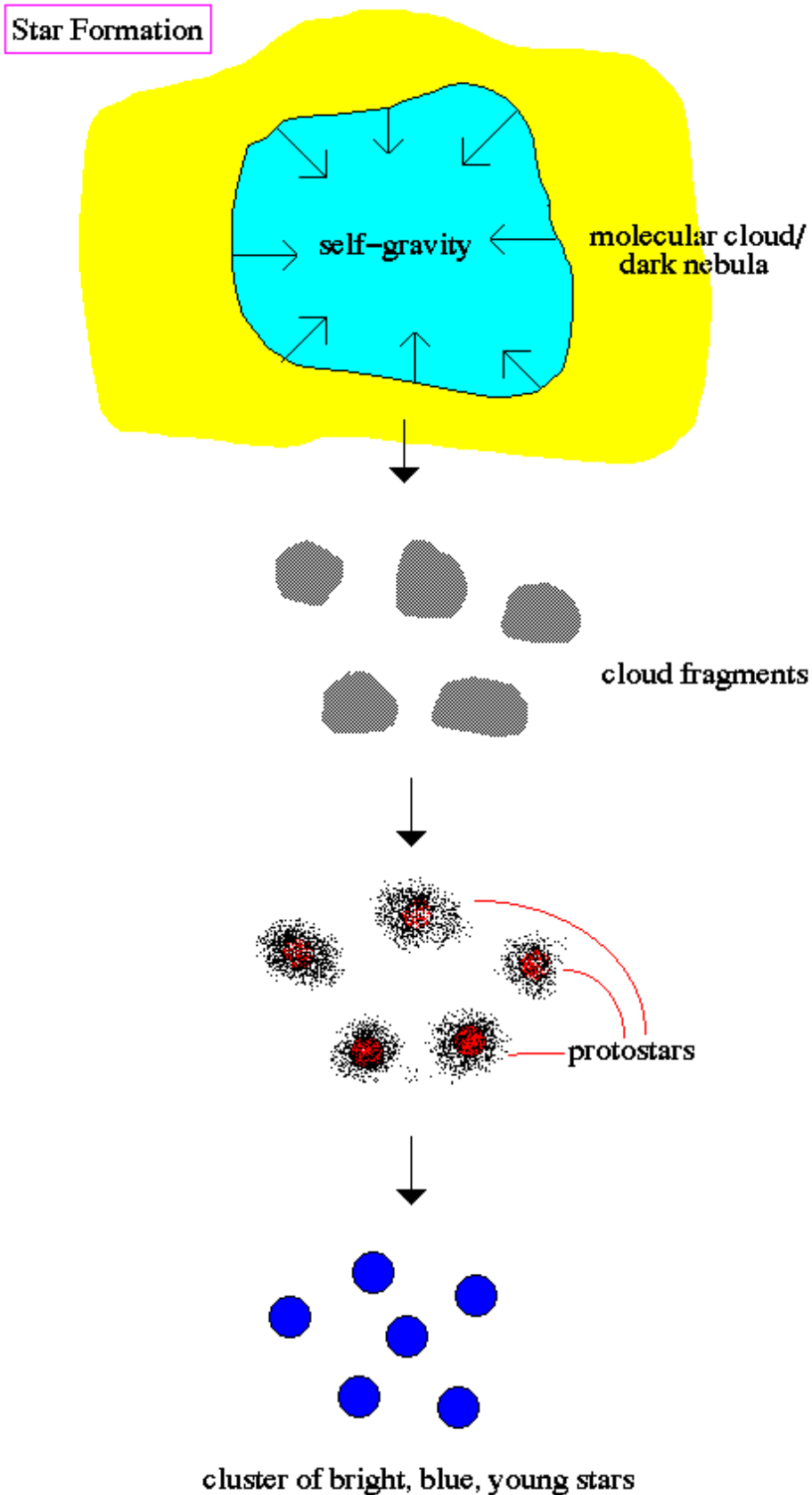


Over billions of years, bright blue stars evolve across the HR diagram to become red giant stars. The integrated color of the once blue young stellar population, becomes red as the red giants stars dominate.

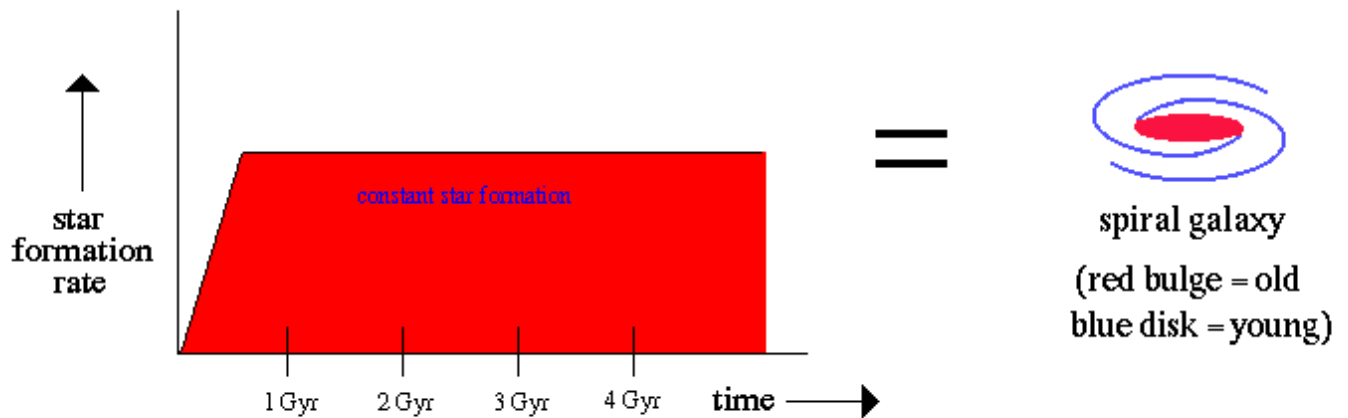
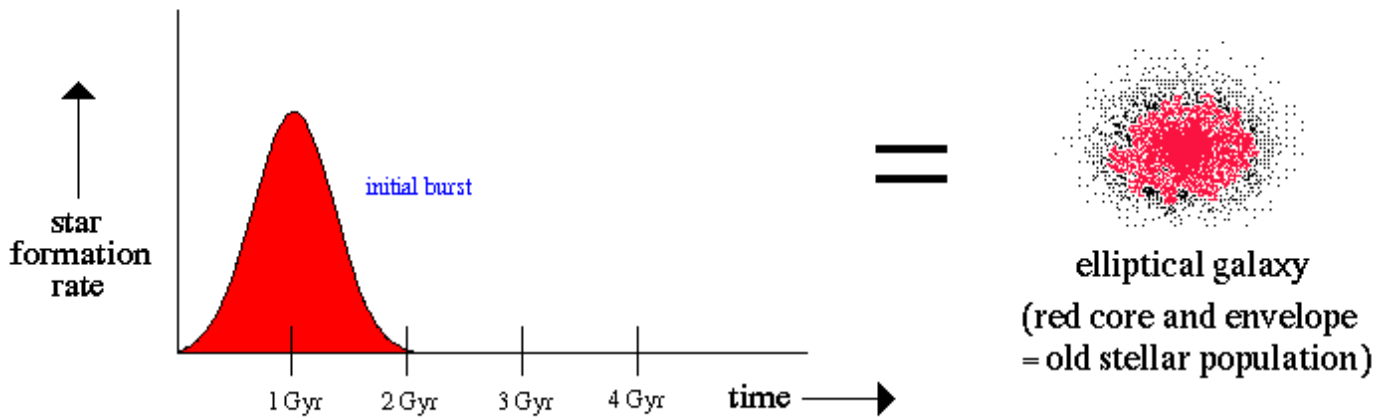
The bottom line is that the red regions of a galaxy are old, with no hot stars. The blue portions of a galaxy are young, meaning the stellar population that dominates this region is newly formed.

Star Formation :

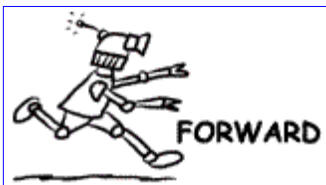
The one feature that correlates with the shape, appearance and color of a galaxy is the amount of current star formation. Stars form when giant clouds of hydrogen gas and dust collapse under their own gravity. As the cloud collapses it fragments into many smaller pieces, each section continues to collapse until thermonuclear fusion begins.



Initial conditions for a galaxy determines its rate of star formation. For example, elliptical galaxies collapse early and form stars quickly. The gas is used up in its early years and today has the appearance of a smooth, red object with no current star formation.



Spirals, on the other hand, form slower, with lower rates of star formation. The gas that 'fuels' star formation is used slower and, thus, there is plenty around today to continue to form stars within the spiral arms.

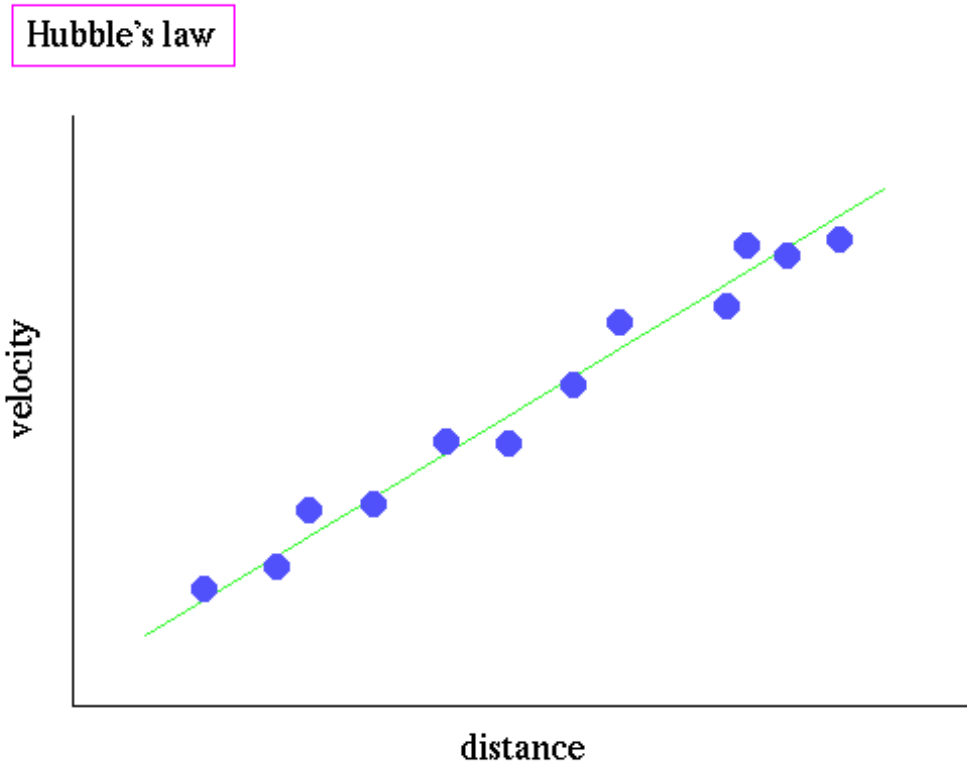


[Chapter 24 of Hartman and Impey](#)



Hubble's law:

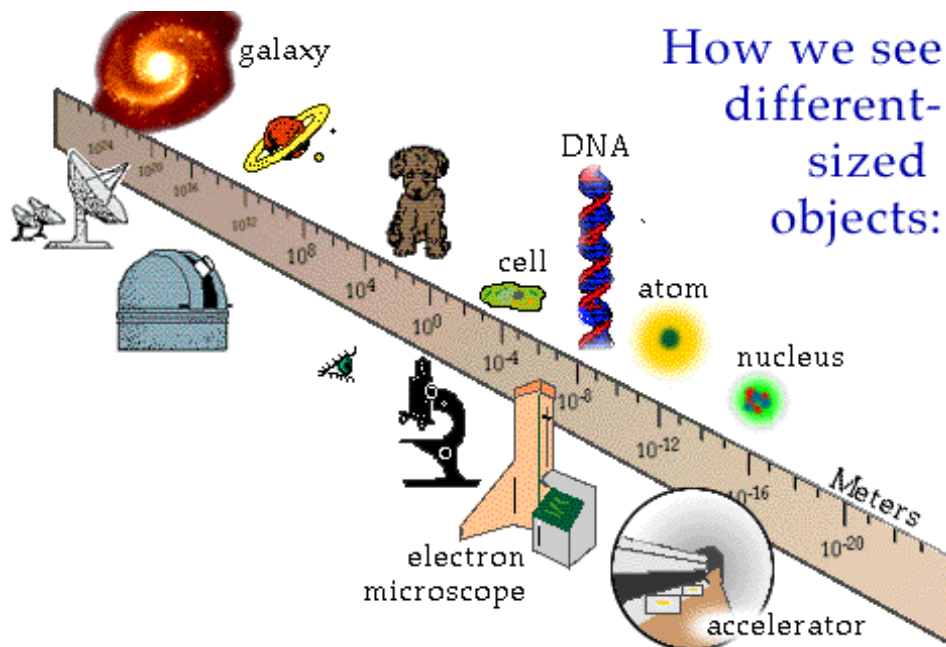
In the 1930's, Edwin Hubble discovered that all galaxies have a positive [redshift](#). In other words, all galaxies were receding from the Milky Way. By the Copernican principle (we are not at a special place in the Universe), we deduce that all galaxies are receding from each other, or we live in a dynamic, [expanding Universe](#).



The expansion of the Universe is described by a very simple equation called Hubble's law; the velocity of the recession of a galaxy is equal to a constant times its distance ($v=Hd$). Where the constant is called [Hubble's constant](#) and relates distance to velocity in units of [light-years](#).

Distance Scale:

The most important value for an astronomical object is its [distance](#) from the Earth. Since cosmology deals with objects larger and brighter than our Sun or solar system, it is impossible to have the correct frame of reference with respect to their size and luminosity as there is nothing to compare extragalactic objects with.



Before the 1920's, it was thought that galaxies were in fact objects within our own Galaxy, possibly regions forming individual stars. They were given the name "[nebula](#)", which we now use to denote regions of gas and dust within [galaxies](#).

At the turn of the century [Cepheid variable stars](#), a special class of [pulsating stars](#) that exhibit a particular period-luminosity relation, were discovered. In other words, it was found that their intrinsic brightness was proportional to their period of variation and, hence, could be used for measuring the distances to nearby galaxies.

In the late 1920's, Hubble discovered similar Cepheid stars in [neighboring galaxies](#) as was found in our own Galaxy. Since they followed the same period-luminosity relation, and they were very faint, then this implied that the neighboring galaxies were very far away. This proved that spiral 'nebula' were, in fact, external to our own Galaxy and sudden the Universe was vast in space and time.

Although Hubble showed that spiral nebula were external to our Galaxy, his estimate of their distances was off by a factor of 6. This was due to the fact that the calibration to Cepheids was poor at the time, combined with the primitive telescopes Hubble used.

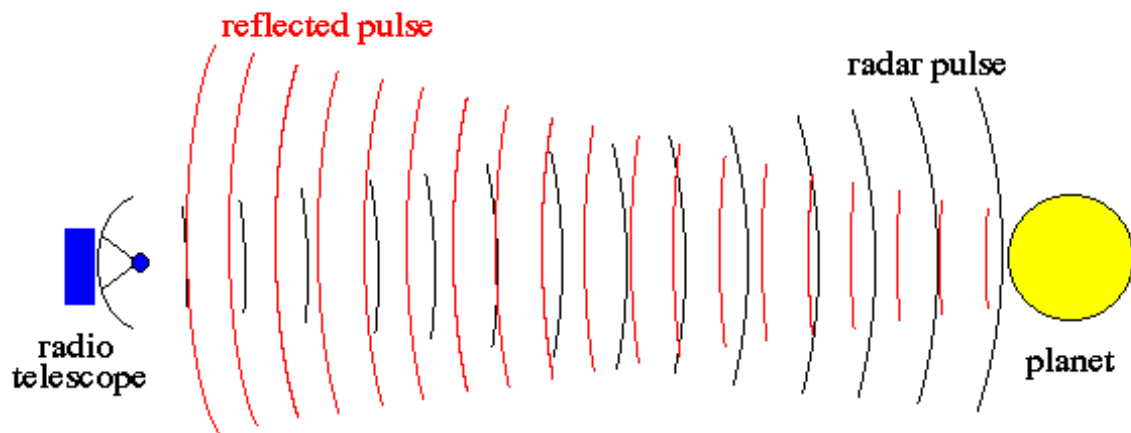
Modern efforts to obtain an estimate of Hubble's constant, the expansion rate of the Universe, find it necessary to determine the distance and the velocities of a large sample of galaxies. The hardest step in this process is the construct of the distance scale for galaxies, a method of determining the true distance to a particular galaxy using some property or characteristic that is visible over a range of galaxies types and distance.

The determination of the distance scale begins with the construction of ladder of primary, secondary and tertiary calibrators in the search for a standard candle.

Primary Calibrators:

The construction of the distance scale ladder is a process of building of a chain of objects with well determined distance. The bottom of this chain is the determination of the scale of objects in the Solar System. This is done through radar ranging, where a radio pulse is reflected off of the various planets in the Solar System.

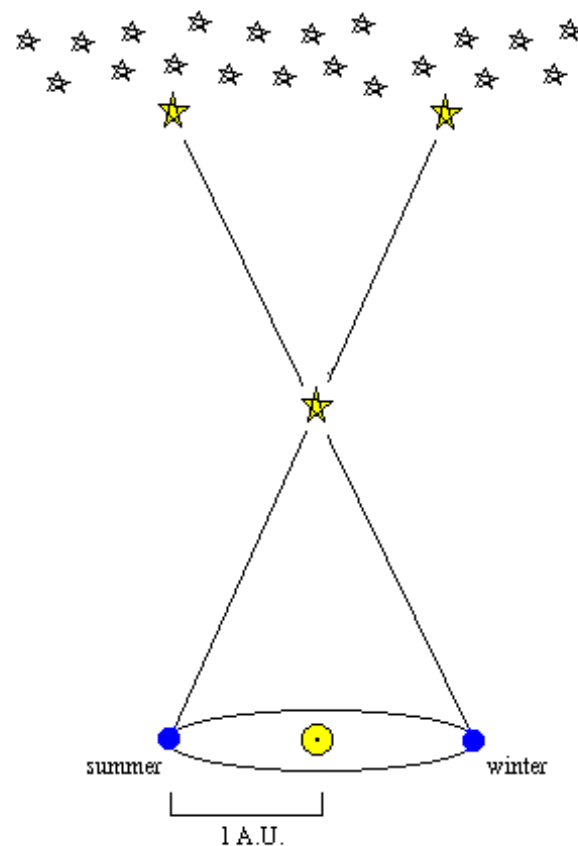
Radar Astrometry



a radio pulse is beamed to the planet in question, and reflected pulse is detected and timed, the time of reflect times the speed of light equals the distance to the planet

The most important value from solar system radar ranging is the exact distance of the Earth from the Sun, determined by triangular measurement of the Earth and terrestrial worlds. This allows an accurate value for what is called the Astronomical Unit (A.U.), i.e. the mean Earth-Sun distance. The A.U. is the ``yardstick" for measuring the distance to nearby stars by [parallax](#).

Stellar Parallax



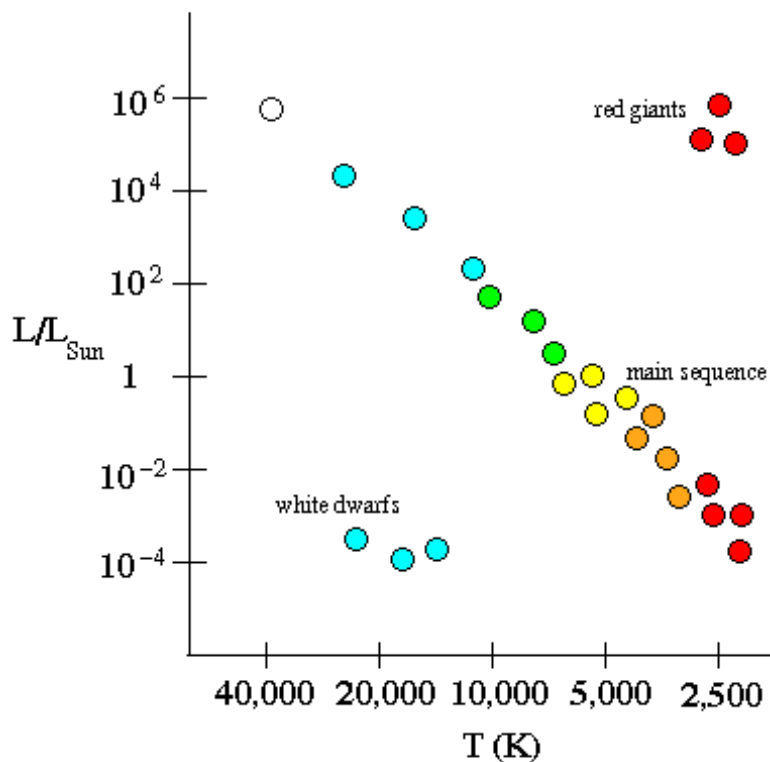
parallax is the apparent change of position of a nearby star with respect to background stars due to the motion of the Earth around the Sun

The parallax system is only good for stars within 300 light-years of the Earth due to limitations of measuring small changes in stellar position. Fortunately, there are hundreds of stars within this volume of space, which become the calibrators for secondary distance indicators.

Secondary Calibrators:

Secondary calibrators of the distance scale depend on statistical measures of stellar properties, such as the mean brightness of a class of stars. It has been known since the 1800's that stars follow a particular color-luminosity relation known as the [Hertzsprung-Russell Diagram](#).

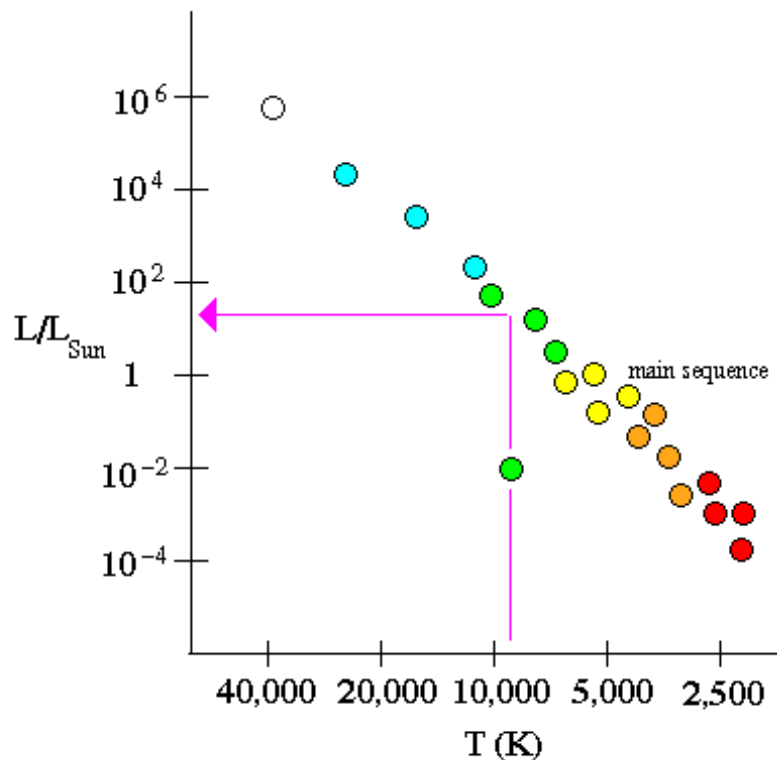
Hertzsprung–Russell Diagram



the HR diagram is a plot of stellar luminosity (with respect to the Sun) versus stellar temperature (color). Various types of stars occupy certain parts of the diagram. Stars spend most of their lifetimes on the main sequence

The existence of the main sequence for stars, a relationship between luminosity and color due to the stable, hydrogen-burning part of a star's life, allows for the use of spectroscopic parallax. A star's temperature is determined by its spectrum (some elements become ions at certain temperatures). With a known temperature, then an absolute luminosity can be read off the HR diagram.

Spectroscopic Parallax

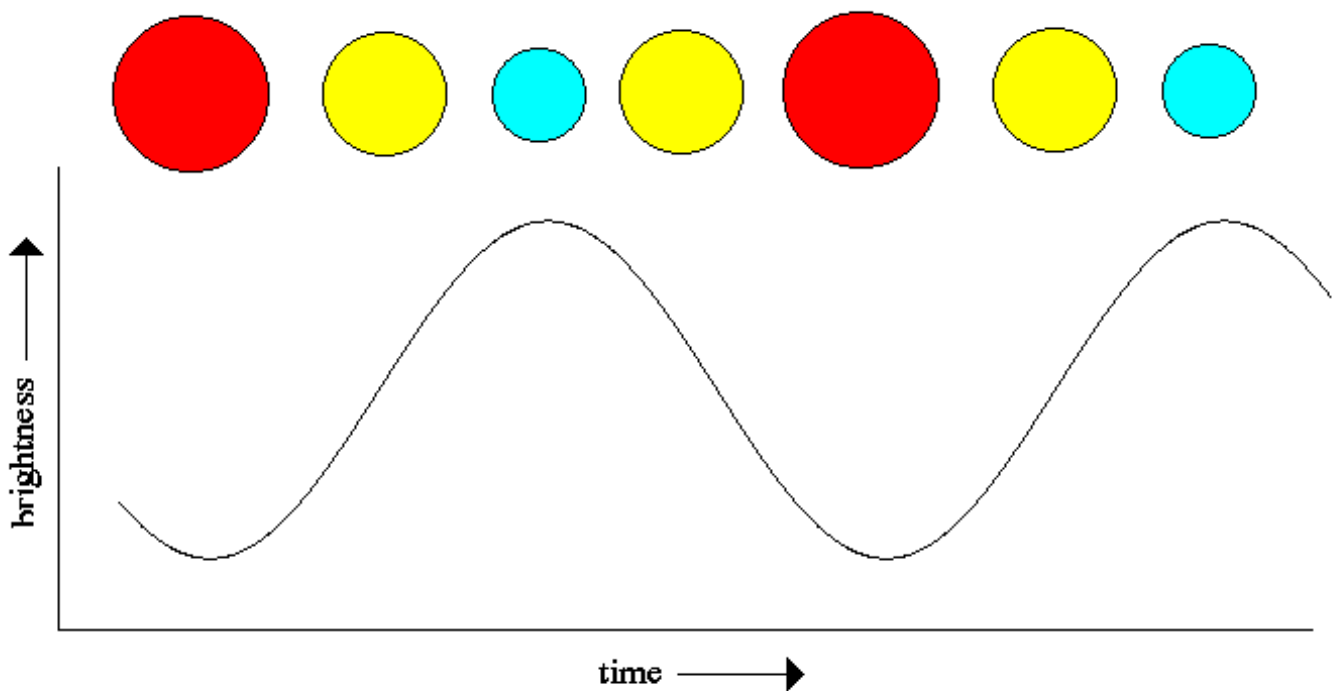


the true brightness of a star can be found if the color is known by matching the star to the main sequence. Knowledge of the observed brightness plus the true brightness derives the distance to the star.

The distance to a star is simply the ratio of its apparent brightness and its true brightness (imagine car headlights at a distance). The method allows us to measure the distances to thousands of local stars and, in particular, to nearby [star clusters](#) which harbor [variable stars](#).

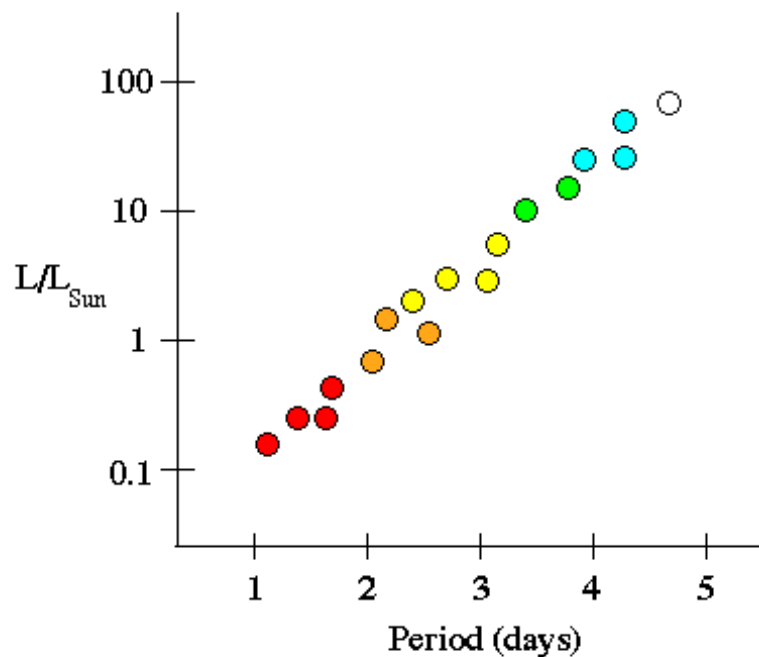
A variable star is a star where the brightness of the star changes over time (usually a small amount). This is traced by a [light curve](#), a plot of brightness and time.

Variable Star



Particular variable stars, such as Cepheids, have a period-luminosity relationship. Meaning that for a particular period of oscillation, they have a unique absolute brightness.

Period-Luminosity Diagram



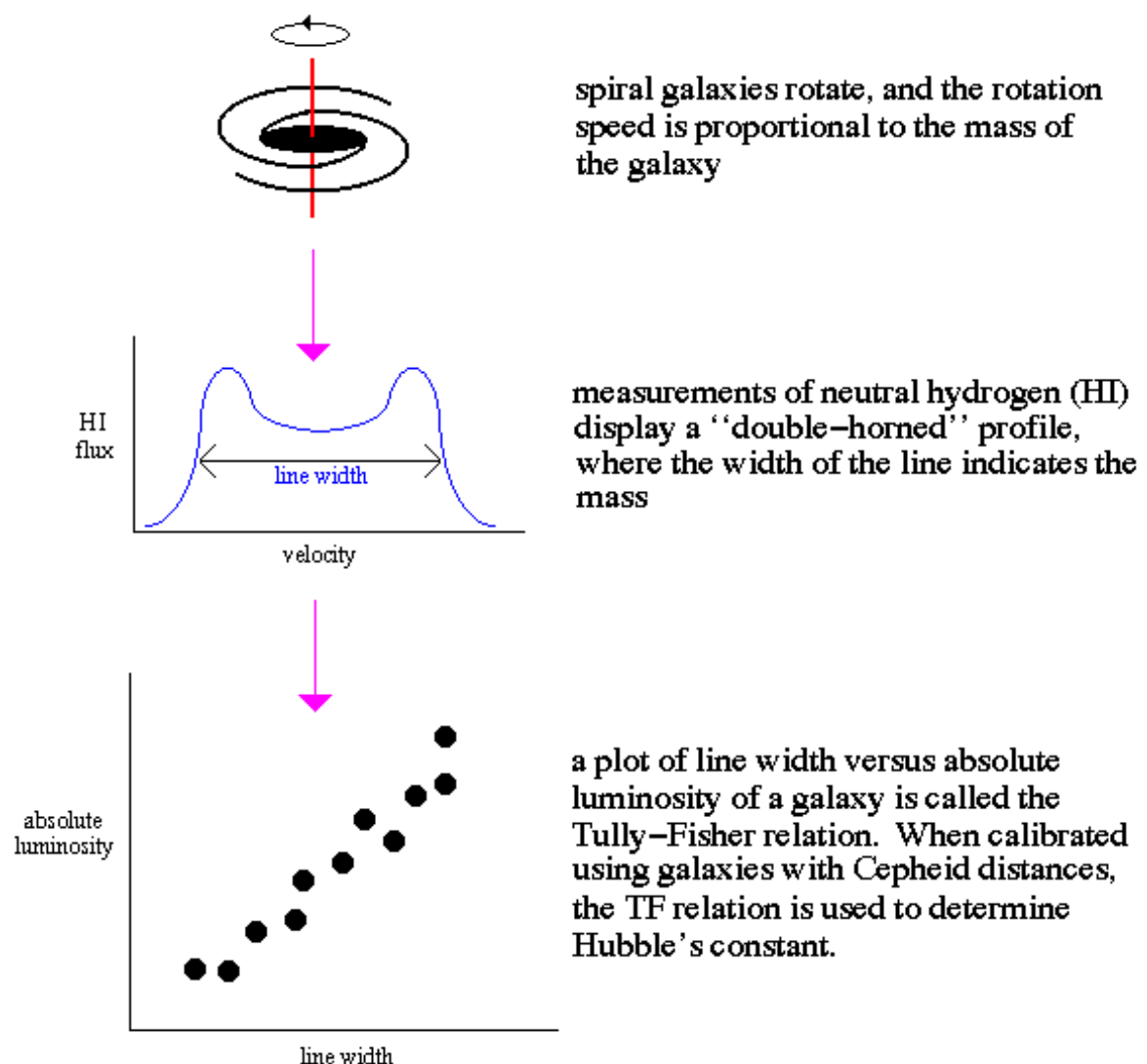
The result is that it is possible to measure the light curve of Cepheids in other galaxies and determine their distances.

Tertiary Calibrators:

The nearby region of the Universe, known as the [Local Group](#) and is located at the edge of what is known as the the Virgo supercluster of galaxies. The use of Cepheid variables is limited to within the volume of space outlined by Virgo system. Thus, the distances to nearby galaxies does not measure the true Hubble flow of the expanding Universe, but rather the gravitational infall into Virgo.

In order to determine Hubble's constant, we must measure the velocity of galaxies much farther away then the Local Group or the Virgo supercluster. But, at these distances we cannot see Cepheid stars, so we determine the total luminosity of the galaxy by the Tully-Fisher method, the last leg of the distance scale ladder.

Tully–Fisher relation



The Tully-Fisher relation is basically a plot of mass versus luminosity of a galaxy. Its not surprising that luminosity and mass are correlated since stars make up basically most of a galaxy's mass and all of the light. Missing mass would be in the form of gas, dust and dark matter.

The key parameter for this last leg of the distance scale are the calibrating galaxies to the Tully-Fisher relation, i.e. the galaxies where we know both the total luminosity from Cepheid

distances and the total luminosity from the Tully-Fisher relation.

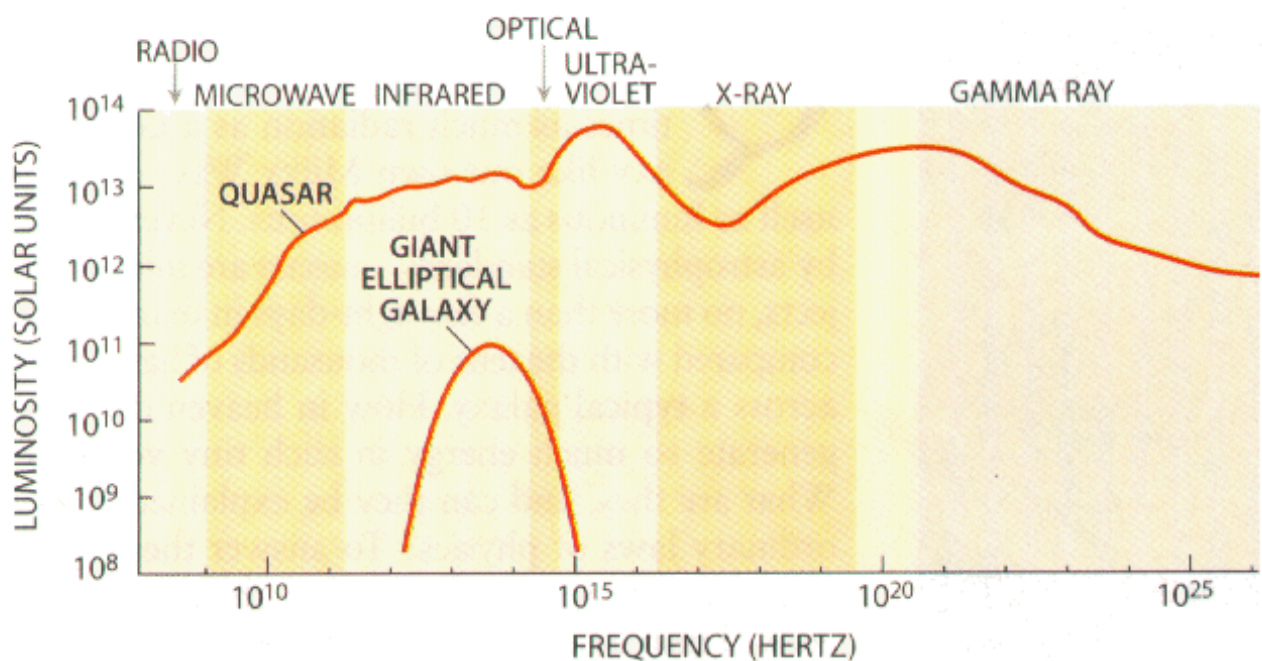
There is currently a strong debate on the value of the Hubble's constant fueled by new data from HST Cepheid studies of nearby galaxies. The community is divided into two schools of thought; 1) the old school which proposes a value for Hubble's constant around 50 to agree with the ages of the oldest stars in our Galaxy, and 2) a newer, and larger school which finds a higher Hubble's constant of 75. This higher value poses a problem for modern cosmology in that the age of the Universe from Hubble's constant is less than the age of the oldest stars as determined by nuclear physics.

So the dilemma is this, either something is wrong with nuclear physics or something is wrong with our understanding of the geometry of the Universe. One possible solution is the introduction of the [cosmological constant](#), once rejected as unnecessary to cosmology, it has now grown in importance due to the conflict of stellar ages and the age of the Universe.

Quasars:

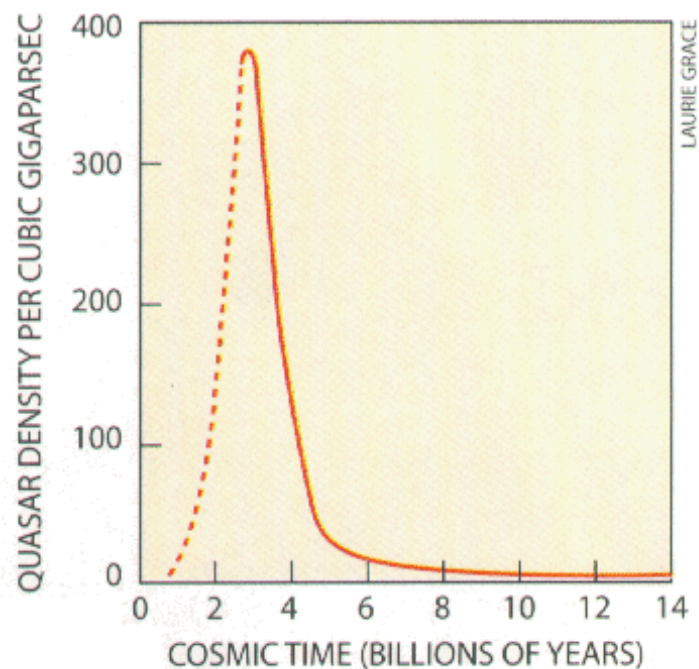
[Quasars](#) are the most luminous objects in the Universe. The typical quasar emits 100 to 1000 times the amount of radiation as our own Milky Way galaxy. However, quasars are also variable on the order of a few days, which means that the source of radiation must be contained in a volume of space on a few light-days across. How such amounts of energy can be generated in such small volumes is a challenge to our current physics.

Quasars were originally discovered in the radio region of the spectrum, even though they emit most of their radiation in the high energy x-ray and gamma-ray regions. Optical spectra of the first quasars in the 1960's showed them to be over two billion light-years away, meaning two billion years into the past as well.



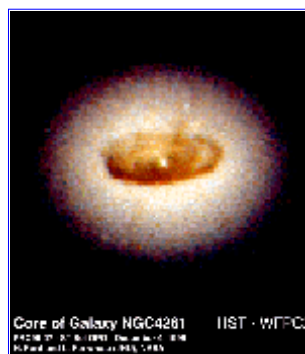
QUASAR SPECTRUM of 3C273—one of the brightest quasars and the first to be discovered—is far broader than the spectrum of a typical giant elliptical galaxy (*left*). In the optical range, the quasar is hundreds of times more luminous. Quasars were most numerous when the universe was two to four billion years old (*right*). Today quasars are 1 000 times less common. Quasars

(right). Today quasars are 1,000 times less common. Quasars were also rare in the very early history of the universe, but the exact numbers are uncertain.



Over a thousand quasars have been discovered, most having redshifts greater than 10 billion light-years away. The number density of quasars drops off very fast, such that they are objects associated with a time when galaxies were young.

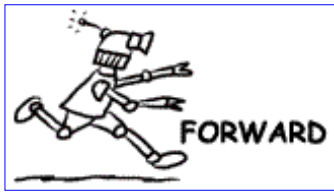
The large amount of radio and x-ray emission from quasars gives them similar properties to the class of what are called active galaxies, such as [Seyfert galaxies](#), originally recognized by the American astronomer Carl K. Seyfert from optical spectra. Seyfert galaxies have very bright nuclei with strong emission lines of hydrogen and other common elements, showing velocities of hundreds or thousands of kilometers per second, where the high energy emission is probably due to a Galactic mass black hole at the galaxies core (for example, NGC 4261 shown below). The idea is that quasars are younger, and brighter, versions of Seyfert galaxies.



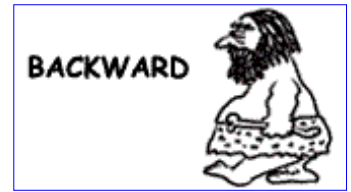
HST imaging showed that quasars are centered in the middle of [host galaxies](#), giving more support to the idea that the quasar phenomenon is associated with Galactic mass black holes in the middle of the host galaxies. Since a majority of the host galaxies are disturbed in appearance, the suspicion is that colliding galaxies cause stars and gas to be tidally pushed into the black hole to fuel the quasar.

This process would explain the occurrence of quasars with redshift. In the far distant past there were no galaxies, so no sites for quasars. In the early phases of galaxy formation, the galaxy

density was high, and there were many collisions producing many quasars. As time passed, the number of collisions decreased as space expanded and the number of quasar also dropped.



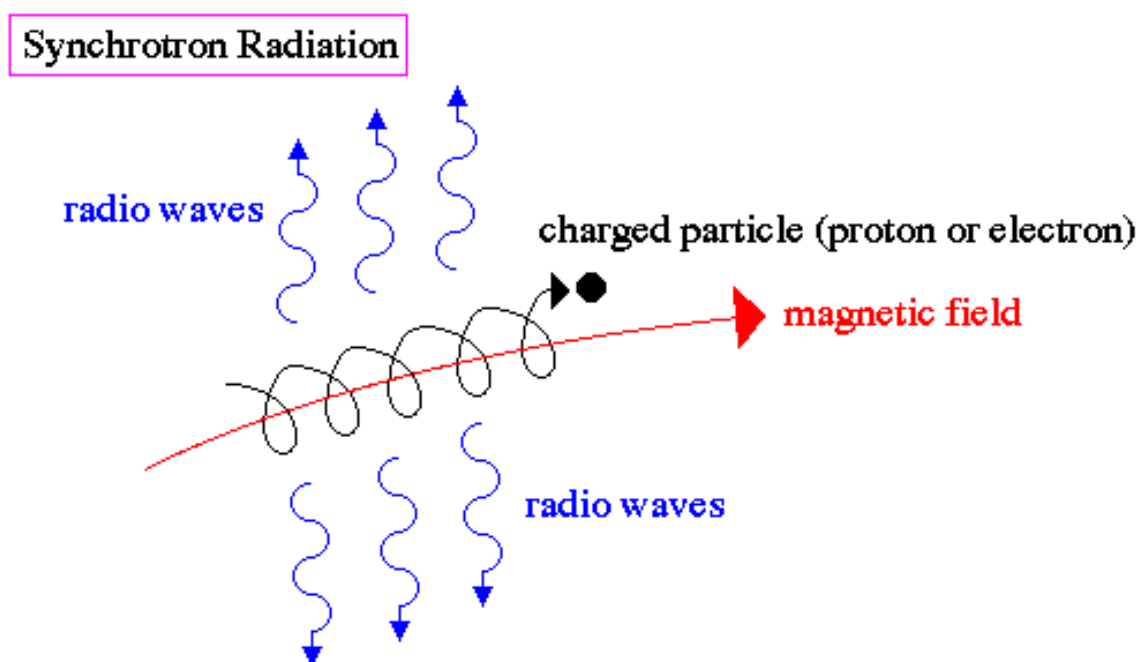
[Chapter 25 of Hartman and Impey](#)



Active Galaxies:

Most galaxies are 'normal' in that most of their light is generated by stars or heated gas. This energy is primarily radiated away as optical and infrared energy. However, there exists a subclass of galaxies, known as active galaxies, which radiate tremendous amounts of energy in the radio and x-ray regions of the spectrum. These objects often emit hundreds to thousands of times the energy emitted by our Galaxy and, because of this high luminosity, are visible to the edges of the Universe.

Active galaxies usually fall in three types; [Seyfert galaxies](#), [radio galaxies](#) and [quasars](#). Radio galaxies often have a [double-lobe](#) appearance, and the type of radio emission suggests that the origin is synchrotron radiation.



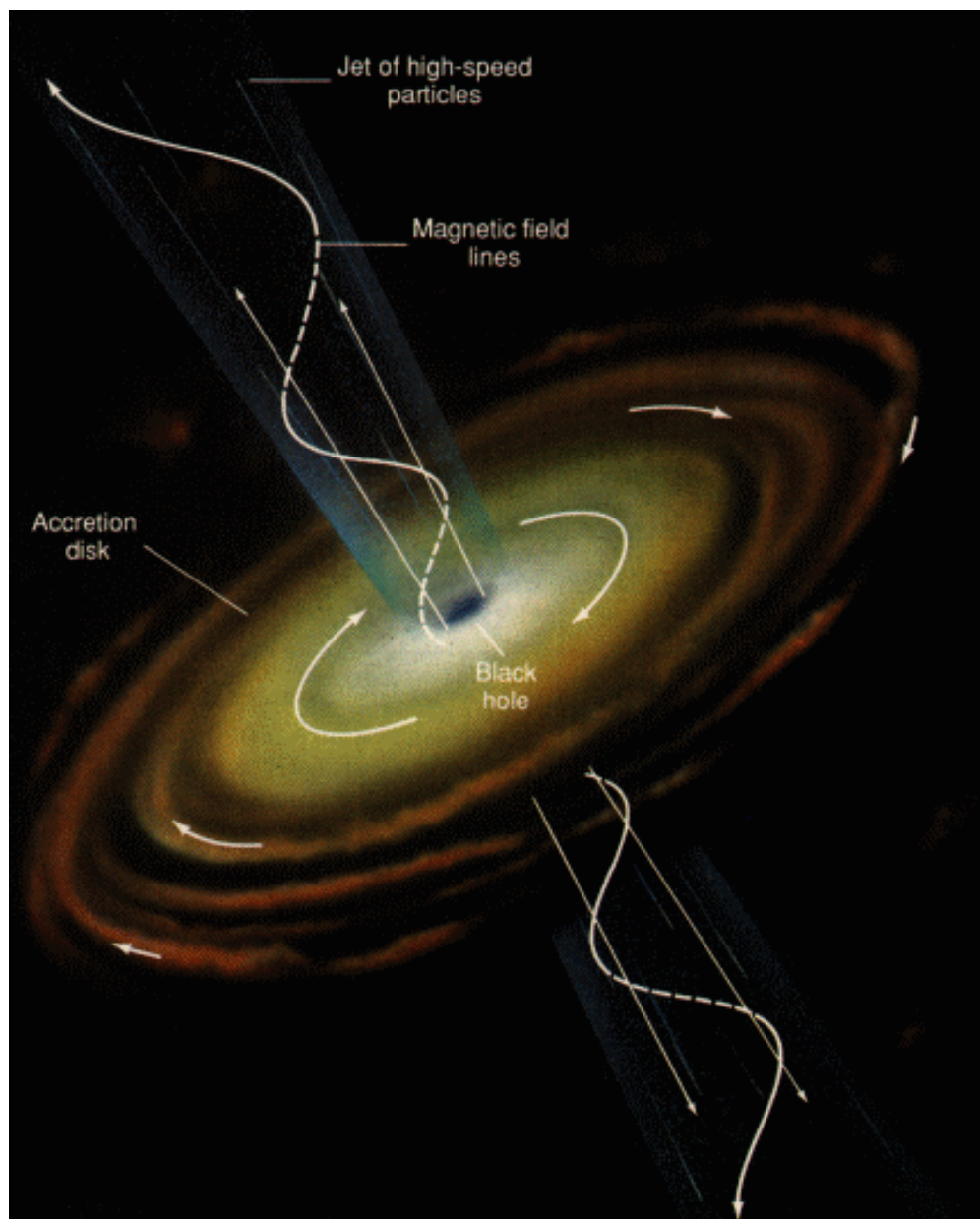
synchrotron radiation occurs when a charged particle encounters a strong magnetic field – the particle is accelerated along a spiral path following the magnetic field and emitting radio waves in the process – the result is a distinct radio signature that reveals the strength of the magnetic field

Active galaxies emit large amounts of x-rays and gamma-rays, extremely high energy forms of electromagnetic radiation. Strong magnetic fields (synchrotron radiation in the radiio) plus gamma-rays implies very violent events in the cores of active galaxies.

Although active galaxies different in their appearance, they are related in the mechanism that produces their huge amounts of energy, a Galactic mass black hole at the galaxy's center. The gas flowing towards the center of the galaxy forms a

thick circle of orbiting material called an [accretion disk](#) many hundreds of light-years across.

Since the infalling gas retains the direction of orbital motion of the companion, the stream of material forms a rotating disk.



[Friction](#) between the gas in neighboring orbits causes the material to spiral inward until it hits the event horizon of the central black hole. As the spiraling gas moves inward, gravitational energy is released as heat into the accretion disk. The release of energy is greatest at the inner edge of the accretion disk where temperatures can reach millions of degrees. It is from this region that the magnetic fields are produced for the synchrotron radiation and the collision between atoms to emit x-rays and gamma-rays.

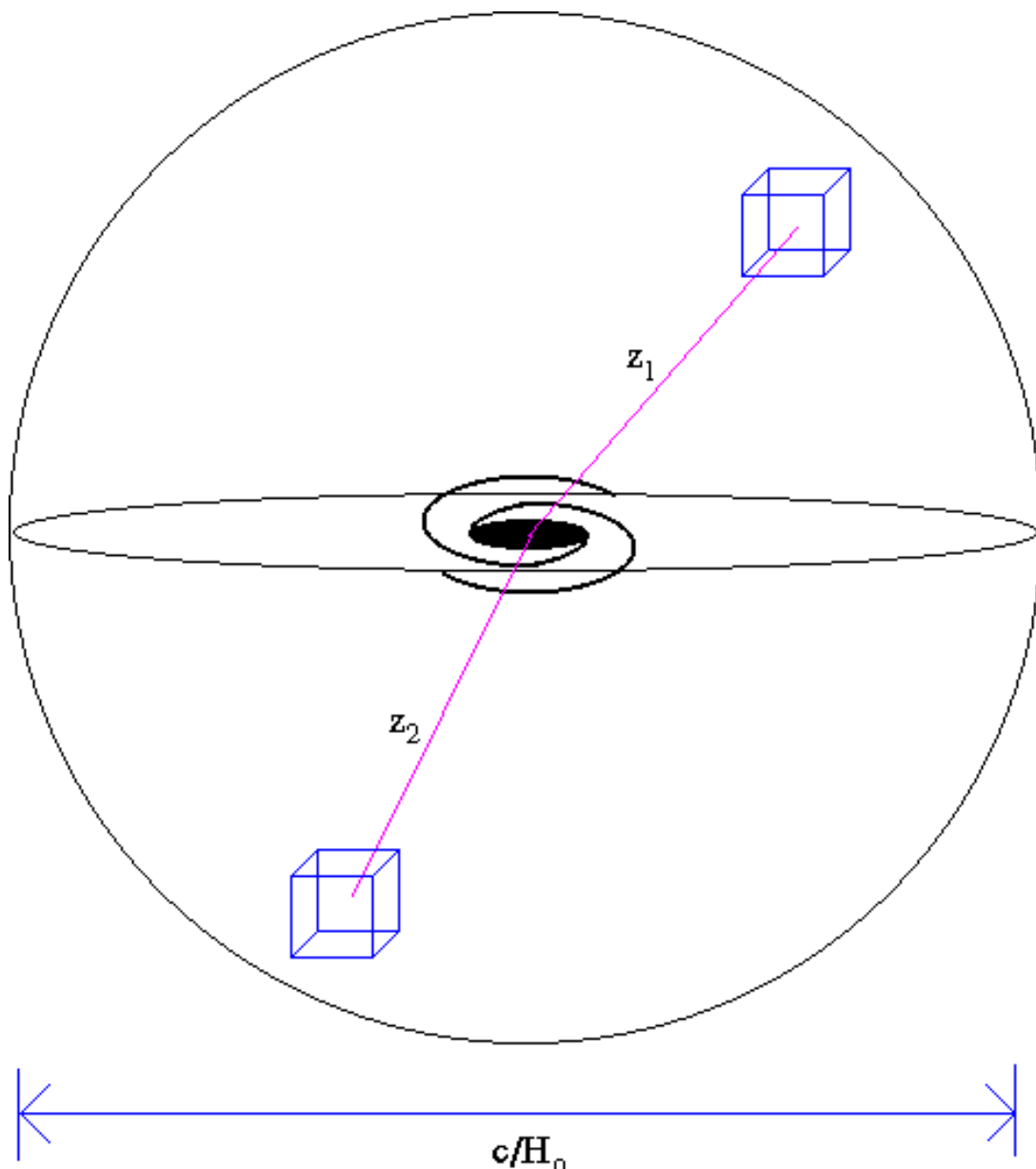
Our own Galaxy core may harbor a small active nuclei similar to those found in quasars. In fact, all galaxies may have dormant black holes, invisible because there is no accretion. Seyfert, radio galaxies and quasars may simply be normal galaxies

in an [active phase](#).

This hypothesis has been confirmed by HST imaging of distant [QSO hosts](#) which show the bright quasar core is in the center of fairly normal looking galaxies.

Lookback Time:

The large size of the Universe, combined with the finite speed for light, produces the phenomenon known as [lookback time](#). Lookback time means that the farther away an object is from the Earth, the longer it takes for its light to reach us. Thus, we are looking back in time as we look farther away.



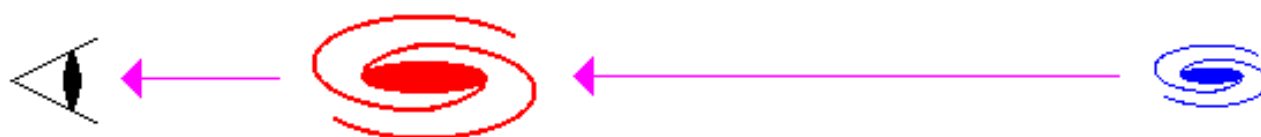
The galaxies we see at large distances are younger than the galaxies we see nearby. This allows us to study galaxies as they evolve. Note that we don't see the individuals evolve, but we can compare spirals nearby with spirals far away to see how the typical spiral has changed with time.

Galaxy Evolution:

The phenomenon of lookback time allows us to actually observe the evolution of galaxies. We are not seeing the same galaxies as today, but it is possible to trace the behavior of galaxies types with distance/time.

It is known that galaxies form from large clouds of gas in the early Universe. The gas collects under self-gravity and, at some point, the gas fragments into star cluster sized elements where star formation begins. Thus, we have the expectation that distant galaxies (i.e. younger galaxies) will be undergoing large amounts of star formation and producing hot stars = blue stars. The study of this phenomenon is called color evolution.

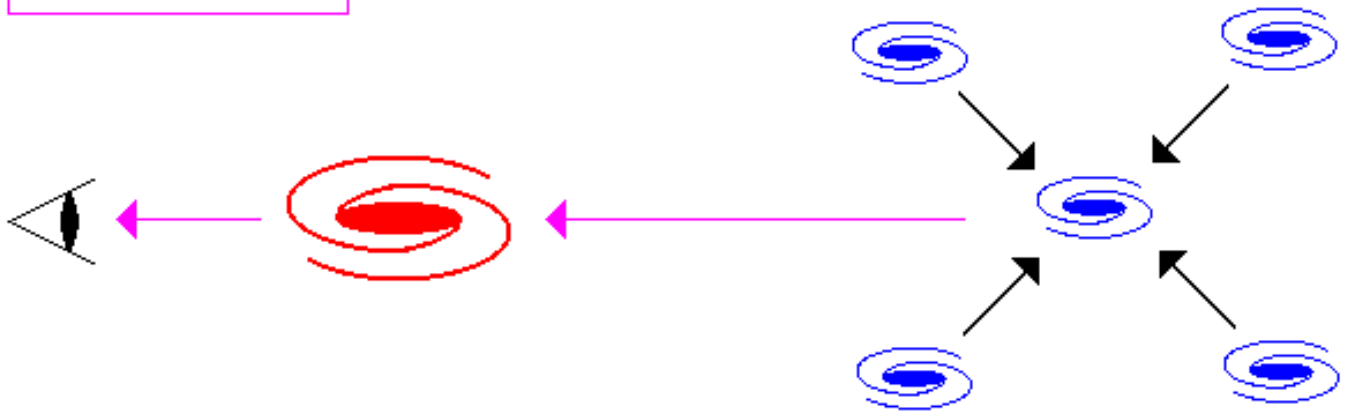
Color Evolution



distant galaxies are bluer since we are looking back in time, and are seeing them at a younger age, younger stars = hotter stars = bluer stars

Computer simulations also indicate that the epoch right after galaxy formation is a time filled with many encounters/collisions between young galaxies. Galaxies that pass near each other can be captured in their mutual self-gravity and merge into a new galaxy. Note that this is unlike cars, which after collisions are not new types of cars, because galaxies are composed of many individual stars, not solid pieces of matter. The evolution of galaxies by mergers and collisions is called number evolution.

Number Evolution

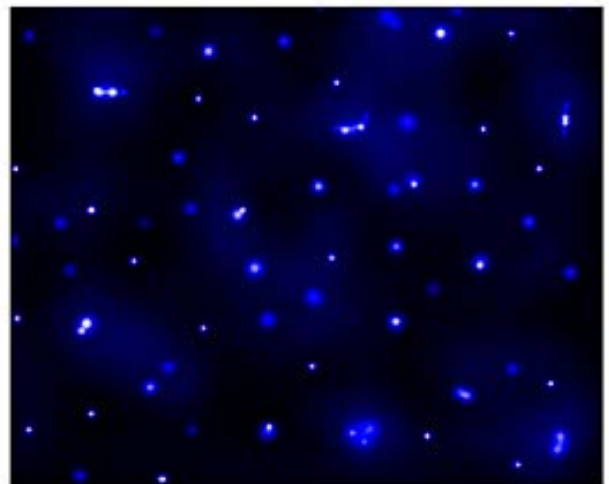


small galaxies merger at early epochs to form present-day galaxies. More galaxies are seen as we look back into the past.

Thus, our picture of galaxy evolution, incorporating both these principles, looks like the following:



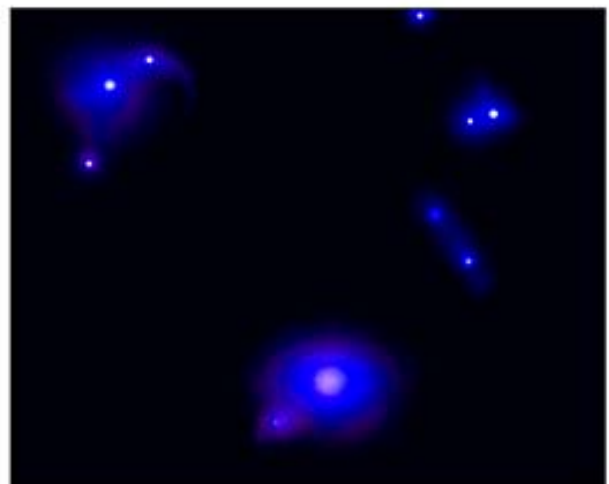
After 0 - 0.5 billion years -- starting out with a very smooth distribution of matter directly after the Big-Bang, gravity of the more massive clumps of stars starts to attract more matter.



After 0.5 - 1 billion years -- larger clumps grow from merging of smaller ones.



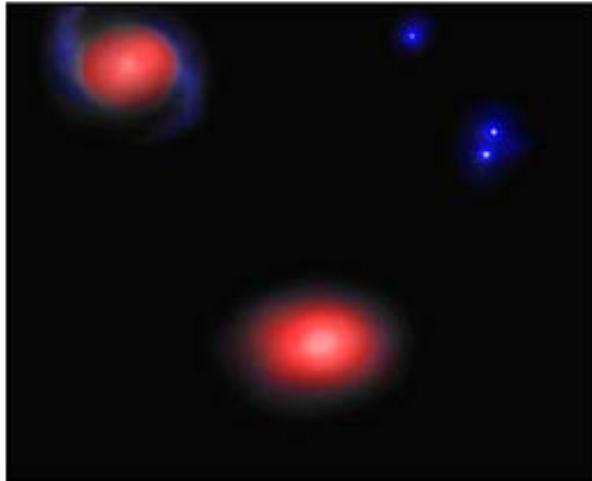
After 1 - 2 billions years -- after growing to a fraction



After 2-4 billion years -- larger irregular looking

of the size of our own Galaxy, the clumps are large enough for Hubble Space Telescope to see them.

objects form through collisions and mergers between these sub-galactic sized clumps.

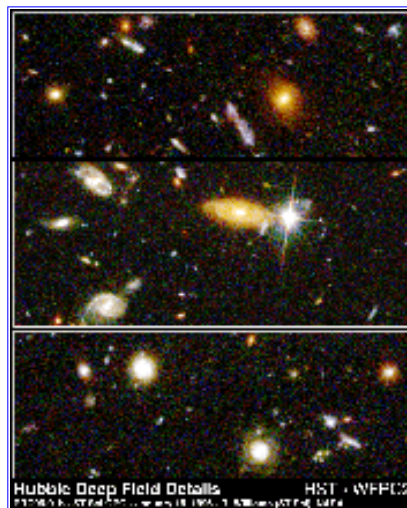


After 4-13 billion years -- galaxies as we see them today form, and take their final shapes. The elliptical and the spiral galaxies with old red stellar populations in their centers form first, and the spiral galaxy disks form later from infalling surrounding gas.

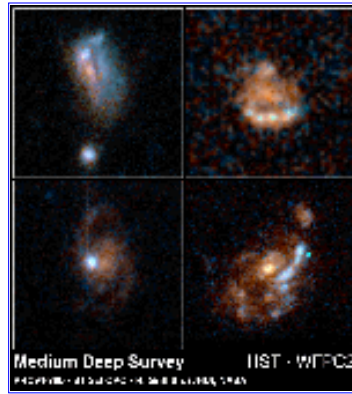
Credit: Sam Pascarella (Arizona State University)

Some types of galaxies are still forming stars at the present epoch (e.g. spiral and irregular galaxies). However, the past was marked by a much higher rate of star formation than the present-day average rate because there was more gas clouds in the past. Galaxies, themselves, were built in the past from high, initial rates of star formation.

The time of quasars is also during the time of first star formation in galaxies, so the two phenomenon are related, the past was a time of rapid change and violent activity in galaxies.



Space observations called the [Hubble Deep Field](#) produced images of faint galaxies and distant galaxies at high redshift which confirmed, quantitatively, our estimates of the style and amount of star formation. Nature lends a hand by providing images of distant galaxies by gravitational lensing, as seen in this HST image of [CL0024](#).



Interestingly enough, it is often easier to simulate the evolution of galaxies in a computer, then use the simulations to solve for various cosmological constants, such as Hubble's constant or the geometry of the Universe. The field of extragalactic studies is just such a process of iteration on the fundamental constants of the Universe and the behavior of galaxies with time (i.e. galaxy evolution).

Creation Event:

The debate about the origin of the Universe presupposes that there was an origin. Instead of a beginning, the Universe may be experiencing an endless number of cycles. Ancient Chinese believed that all events formed a periodic pattern driven by two basic forces, Yin and Yang.



The Hindu cosmological system consisted of cycles within cycles of immense duration (one lifecycle of Brahma is 311 trillion years). Cyclicity cosmologies, and their associated fatalism, is also found in Babylonian, Egyptian and Mayan cultures.

Judeo-Christian tradition was unique in its belief that God created the Universe at some specific moment in the past, and that events form an unfolding unidirectional sequence. Key to this philosophy is that the Creator is entirely separate from and independent of His creation. God brings order to a primordial chaos.



Belief that a divine being starts the Universe then 'sits back' and watches events unfold, taking no direct part in affairs, is known as deism. Here God is considered a cosmic engineer. In contrast, theism is the belief in a God who is creator of the Universe and who also remains directly involved in the day-to-day running of the world, especially the affairs of human beings. God maintains a personal and guiding role. In both deism and theism, God is regarded as wholly other than, and beyond, the physical Universe. In pantheism, no such separation is made between God and the physical Universe. God is identified with Nature itself: everything is a part of God and God is in everything.

A Creation event implies that everything came from nothing (creation ex nihilo) since if there were something before Creation, then an earlier Creation is needed to explain that something. God existed before Creation, and the definition is not limited to work with pre-existing matter or pre-existing physical laws either. In fact, the most obvious distinction between the Creator and the created Universe is that the Creator is eternal and the created Universe had a beginning.

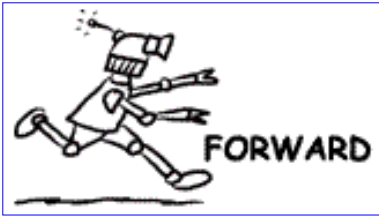
Hot Big Bang:

The discovery of an expanding Universe implies the obvious, that the Universe must have had an initial starting point, an alpha point or Creation. In other words, there existed a point in the past when the radius of the Universe was zero. Since all the matter in the Universe must have been condensed in a small region, along with all its energy, this moment of Creation is referred to as the [Big Bang](#).

A common question that is asked when considering a Creation point in time is ``What is before the Big Bang?`. This type of question is meaningless or without context since time was created with the Big Bang. It is similar to asking ``What is north of the North Pole?`. The question itself can not be phrased in a meaningful way.

The Big Bang theory has been supported by numerous observations and, regardless of the details in our final theories of the Universe, remains the core element to our understanding of the past. Note that an alpha point automatically implies two things: 1) the Universe has a finite age (about 15 billion years) and 2) the Universe

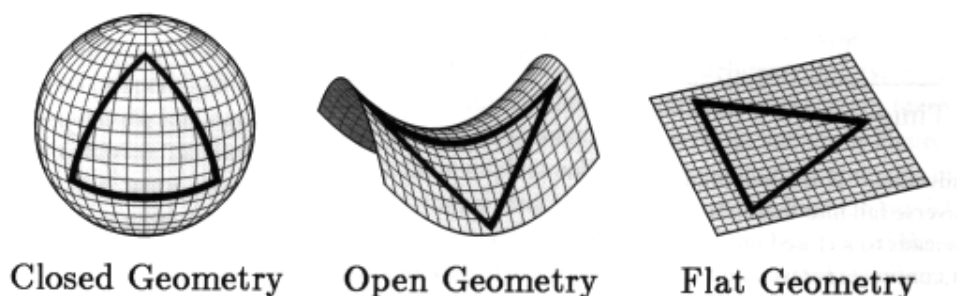
has a finite size (its expanding at a finite speed in a finite time).



Geometry of the Universe:

Can the Universe be finite in size? If so, what is ``outside" the Universe? The answer to both these questions involves a discussion of the intrinsic geometry of the Universe.

There are basically three possible shapes to the Universe; a flat Universe (Euclidean or zero curvature), a spherical or closed Universe (positive curvature) or a hyperbolic or open Universe (negative curvature). Note that this curvature is similar to spacetime curvature due to stellar masses except that the entire mass of the Universe determines the curvature. So a high mass Universe has positive curvature, a low mass Universe has negative curvature.

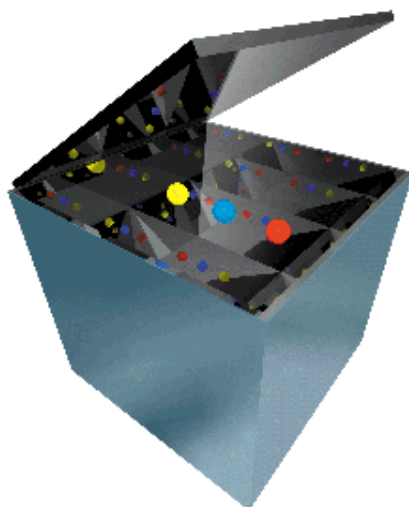


All three geometries are classes of what is called Riemannian geometry, based on three possible states for parallel lines

- never meeting (flat or Euclidean)
- must cross (spherical)
- always divergent (hyperbolic)

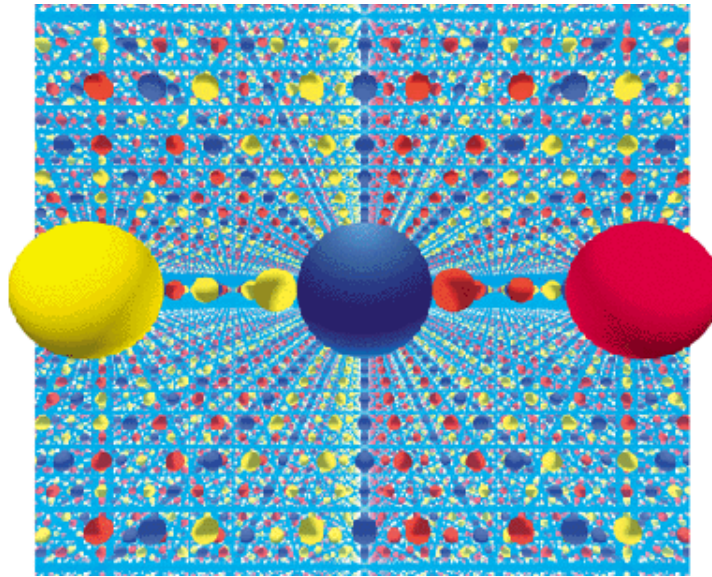
or one can think of triangles where for a flat Universe the angles of a triangle sum to 180 degrees, in a closed Universe the sum must be greater than 180, in an open Universe the sum must be less than 180.

Standard cosmological observations do not say anything about how those volumes fit together to give the universe its overall shape--its topology. The three plausible cosmic geometries are consistent with many different topologies. For example, relativity would describe both a torus (a doughnutlike shape) and a plane with the same equations, even though the torus is finite and the plane is infinite. Determining the topology requires some physical understanding beyond relativity.

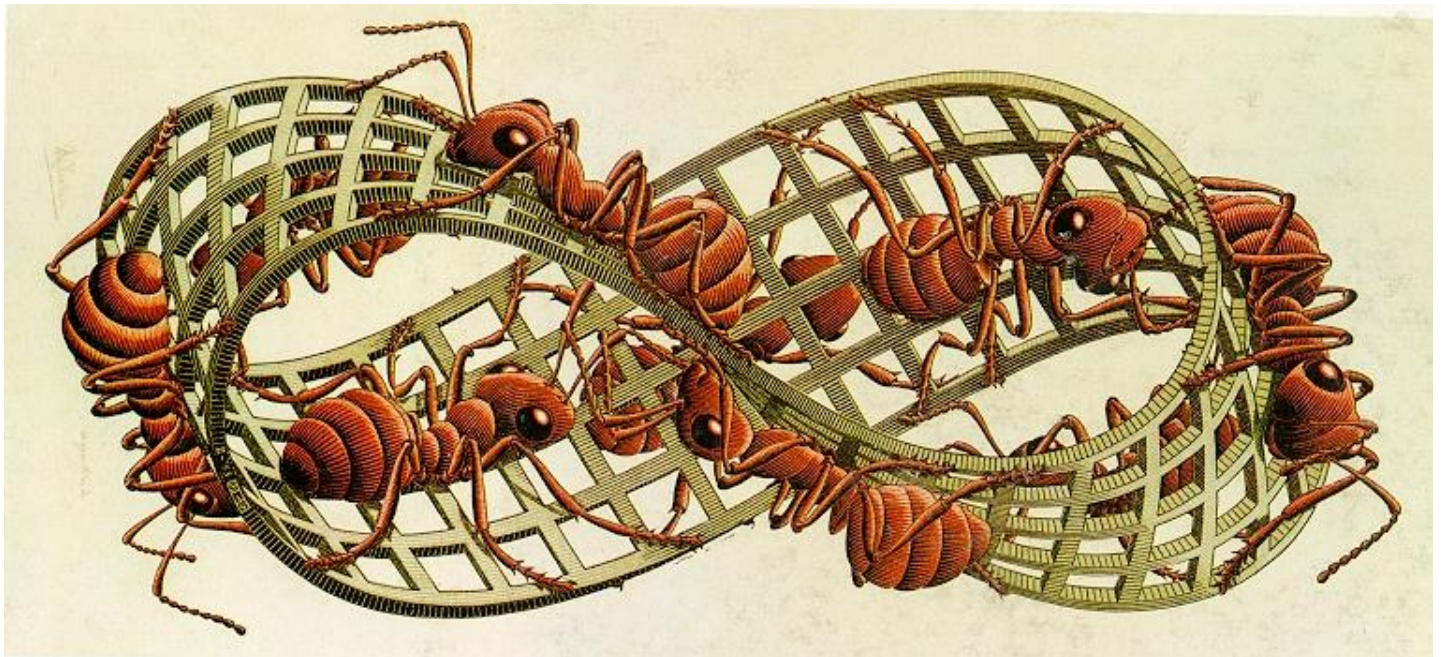


Like a hall of mirrors, the apparently endless universe might be deluding us. The cosmos could, in fact, be finite. The illusion of infinity would come about as light wrapped all the way around space, perhaps more than once--creating multiple images of each galaxy. A mirror box evokes a finite cosmos that looks

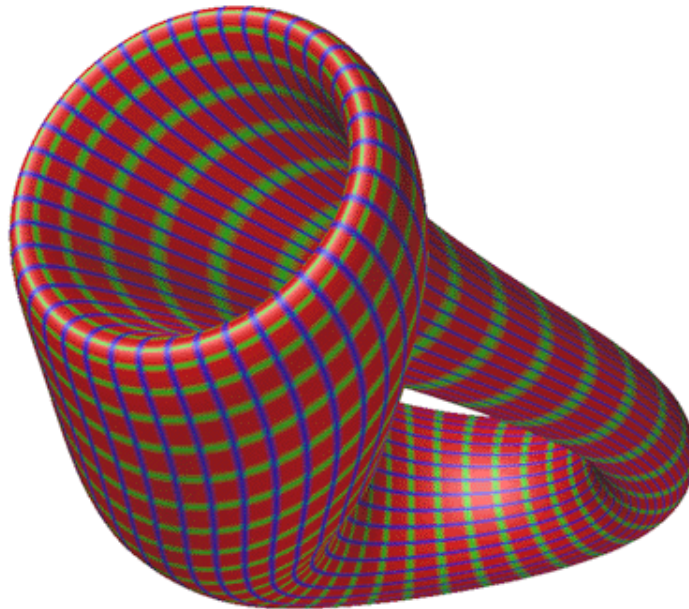
endless. The box contains only three balls, yet the mirrors that line its walls produce an infinite number of images. Of course, in the real universe there is no boundary from which light can reflect. Instead a multiplicity of images could arise as light rays wrap around the universe over and over again. From the pattern of repeated images, one could deduce the universe's true size and shape.



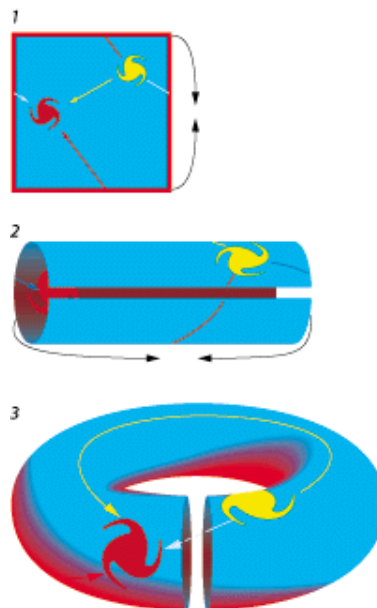
Topology shows that a flat piece of spacetime can be folded into a [torus](#) when the edges touch. In a similar manner, a flat strip of paper can be twisted to form a [Moebius Strip](#).



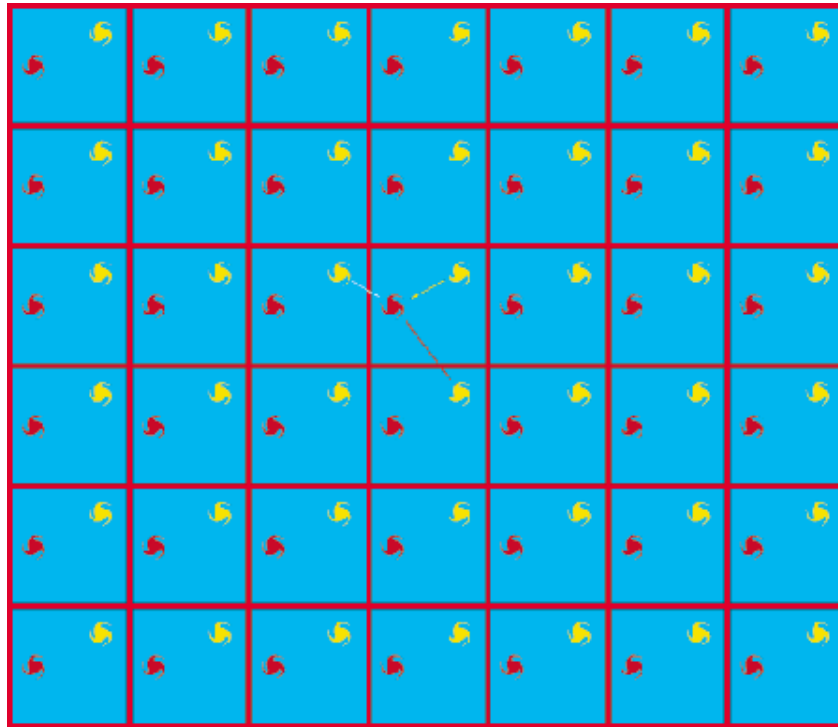
The 3D version of a moebius strip is a [Klein Bottle](#), where spacetime is distorted so there is no inside or outside, only one surface.



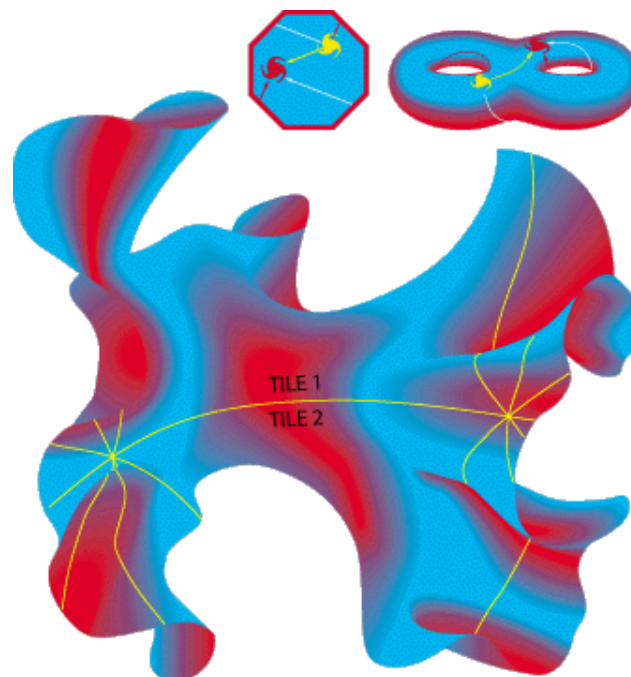
The usual assumption is that the universe is, like a plane, "simply connected," which means there is only one direct path for light to travel from a source to an observer. A simply connected Euclidean or hyperbolic universe would indeed be infinite. But the universe might instead be "multiply connected," like a torus, in which case there are many different such paths. An observer would see multiple images of each galaxy and could easily misinterpret them as distinct galaxies in an endless space, much as a visitor to a mirrored room has the illusion of seeing a huge crowd.



One possible finite geometry is donutspace or more properly known as the Euclidean 2-torus, is a flat square whose opposite sides are connected. Anything crossing one edge reenters from the opposite edge (like a video game see 1 above). Although this surface cannot exist within our three-dimensional space, a distorted version can be built by taping together top and bottom (see 2 above) and scrunching the resulting cylinder into a ring (see 3 above). For observers in the pictured red galaxy, space seems infinite because their line of sight never ends (below). Light from the yellow galaxy can reach them along several different paths, so they see more than one image of it. A Euclidean 3-torus is built from a cube rather than a square.



A finite hyperbolic space is formed by an octagon whose opposite sides are connected, so that anything crossing one edge reenters from the opposite edge (top left). Topologically, the octagonal space is equivalent to a two-holed pretzel (top right). Observers who lived on the surface would see an infinite octagonal grid of galaxies. Such a grid can be drawn only on a hyperbolic manifold--a strange floppy surface where every point has the geometry of a saddle (bottom).

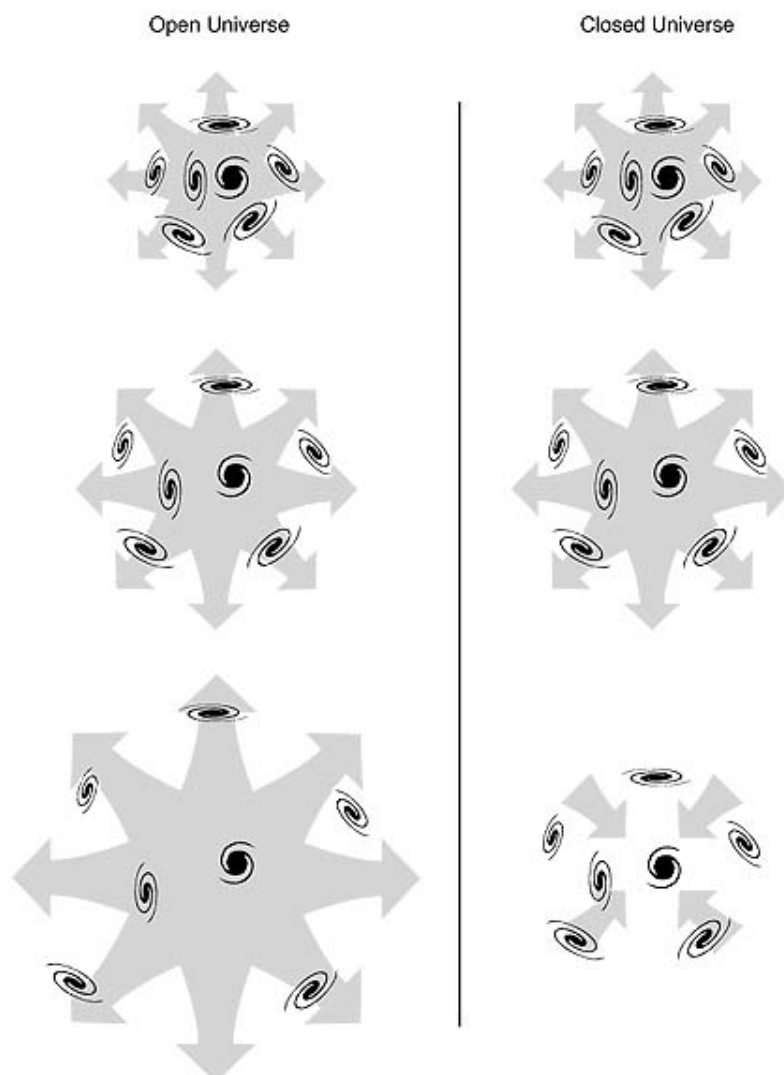


It's important to remember that the above images are 2D shadows of 4D space, it is impossible to draw the geometry of the Universe on a piece of paper (although we can come close with a [hypercube](#)), it can only be described by mathematics. All possible Universes are finite since there is only a finite age and, therefore, a limiting horizon. The geometry may be flat or open, and therefore infinite in possible size (it continues to grow forever), but the amount of mass and time in our Universe is finite.

Density of the Universe:

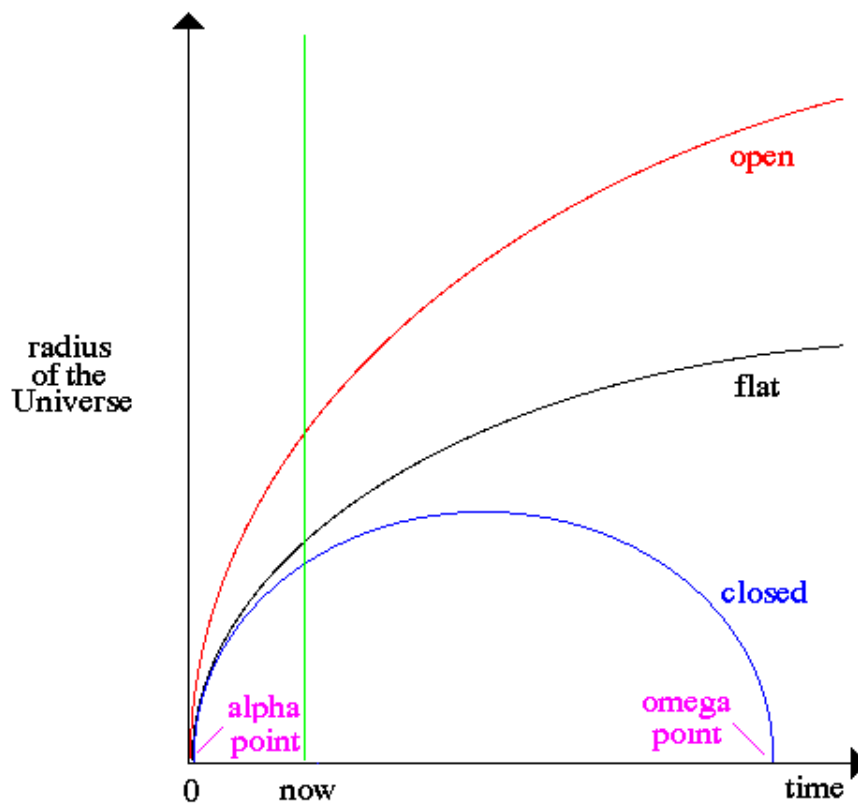
The description of the various geometries of the Universe (open, closed, flat) also relate to their futures.

There are two possible futures for our Universe, continual expansion (open and flat), turn-around and collapse (closed). Note that flat is the specific case of expansion to zero velocity.



An open universe expands forever because it does not contain enough mass, and so does not have enough gravity to slow the expansion of space. A closed universe contains enough mass to halt the expansion, and eventually collapses. A universe with a "critical density" of matter in space is exactly balanced between these two alternatives, and expands at an ever-slowing rate.

The key factor that determines which history is correct is the amount of mass/gravity for the Universe as a whole. If there is sufficient mass, then the expansion of the Universe will be slowed to the point of stopping, then retraction to collapse. If there is not a sufficient amount of mass, then the Universe will expand forever without stopping. The flat Universe is one where there is exactly the balance of mass to slow the expansion to zero, but not for collapse.



The parameter that is used to measure the mass of the Universe is the critical density, Omega. Omega is usually expressed as the ratio of the mean density observed to that of the density in a flat Universe.

Ω , the cosmic density parameter

the fate of the expansion of the Universe is determined by the amount of mass in the Universe, or the mean density of matter, ρ

a flat Universe is given a critical density, ρ_c

Ω is defined as the ratio of the observed density, ρ to the critical density, ρ_c such that

$$\Omega = \frac{\rho}{\rho_c}$$

a closed Universe has $\Omega > 1$

an open Universe has $\Omega < 1$

our current observations of the Universe indicate that Ω is between 0.1 and 0.3, but there is strong theoretical reasons to believe Ω is 1

Given all the range of values for the mean density of the Universe, it is strangely close to the density of a flat Universe. And our theories of the early Universe (see inflation) strongly suggest the value of Omega should be exactly equal to one. If so our measurements of the density by galaxy counts or dynamics are grossly in error and remains one of the key problems for modern astrophysics.

Cosmological Constants:

The size, age and fate of the Universe are determined by two constants:

Fundamental Cosmological Constants

The two most fundamental cosmological constants are Hubble's constant (H_0) which measures the expansion rate of the Universe and the density measure Ω_0 , where the "o" indicates these values today.

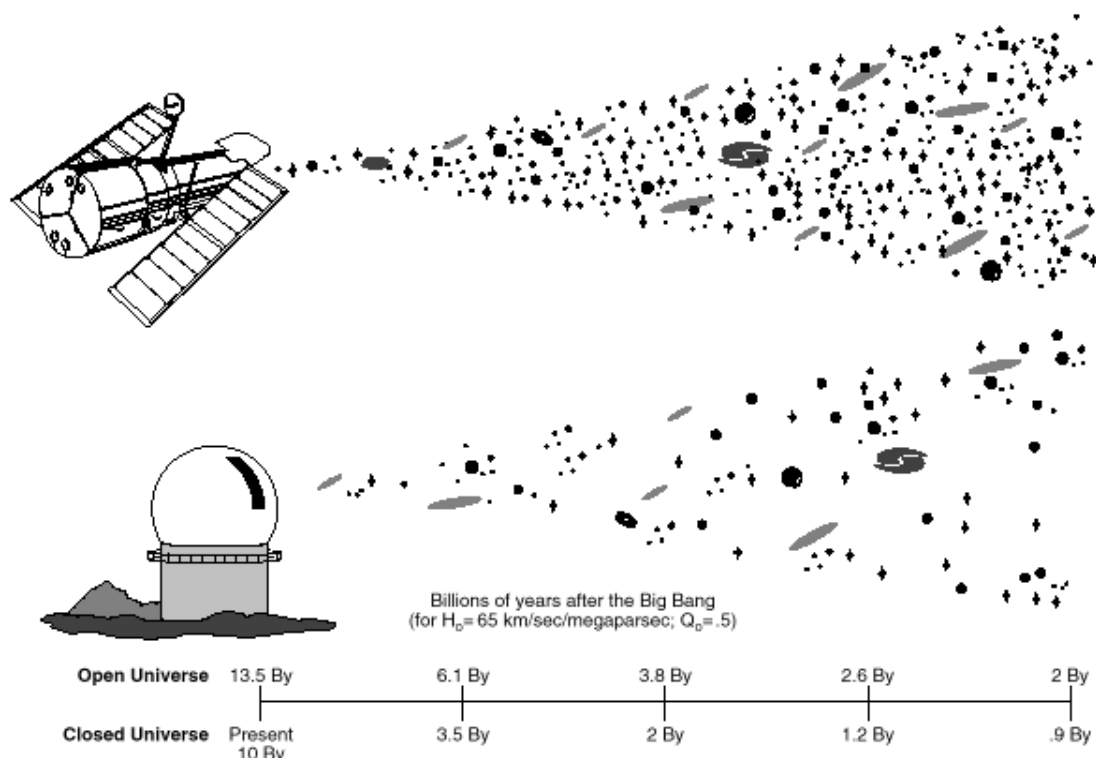
H_0 sets the scale and the age of the Universe, in the sense that the distance to galaxies is determined from their recessional velocity divided by H_0 and the age of the Universe is simply the size of the Universe divided by the expansion rate or $1/H_0$.

The density measure, Ω_0 determines the fate of the Universe (open, closed or flat)

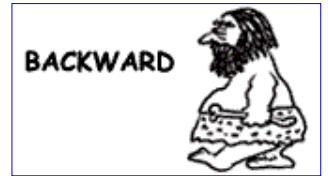
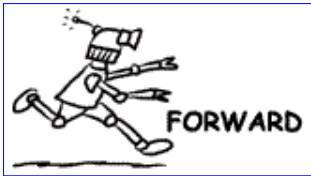
H_0 is measured by determining the distance scale of galaxies

Ω_0 is measured by estimating the local mass density from gravity forces between galaxies and clusters of galaxies

The measurement of these constants consumes major amounts of telescope time over all wavelengths. Both constants remain uncertain to about 30%; however, within this decade we can expect to measure highly accurate values for both due to the [Hubble Space Telescope](#) and the [Keck twins](#).



The Hubble Deep Field uncovered much fainter objects in the universe (down to nearly 30th magnitude) than seen previously from ground-based telescopes. Some of the dim objects along Hubble's line-of sight may be intrinsically faint, foreground galaxies. Others, however, are dim because they are extremely distant. Some of the faintest galaxies in the survey existed when the universe was a fraction of its present age.



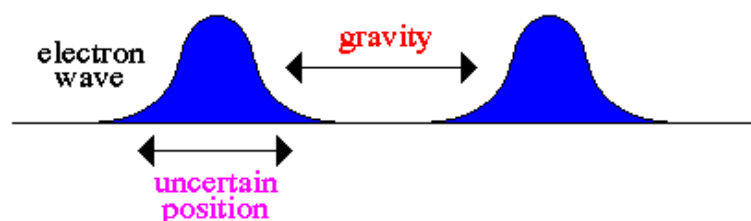
Birth of the Universe :

Physics of the early Universe is at the boundary of astronomy and philosophy since we do not currently have a complete theory that unifies all the fundamental forces of Nature at the moment of Creation. In addition, there is no possibility of linking observation or experimentation of early Universe physics to our theories (i.e. its not possible to 'build' another Universe). Our theories are rejected or accepted based on simplicity and aesthetic grounds, plus their power of prediction to later times, rather than an appeal to empirical results. This is a very different way of doing science from previous centuries of research.

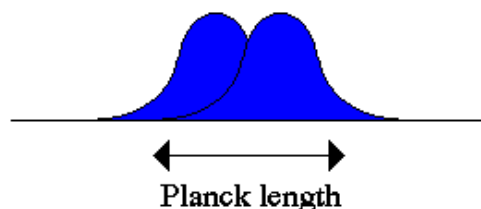
Our physics can explain most of the evolution of the Universe after the [Planck time](#) (approximately 10^{-43} seconds after the Big Bang).

Planck Time

The Planck time is the earliest moment in the history of the Universe where our physics still works. Similarly, it is the shortest amount of time that we can probe in the laboratory with our current knowledge of physics. This time is set by a combination of gravity, quantum mechanics and relativity, the typical scale lengths for these events divided by the speed of light.

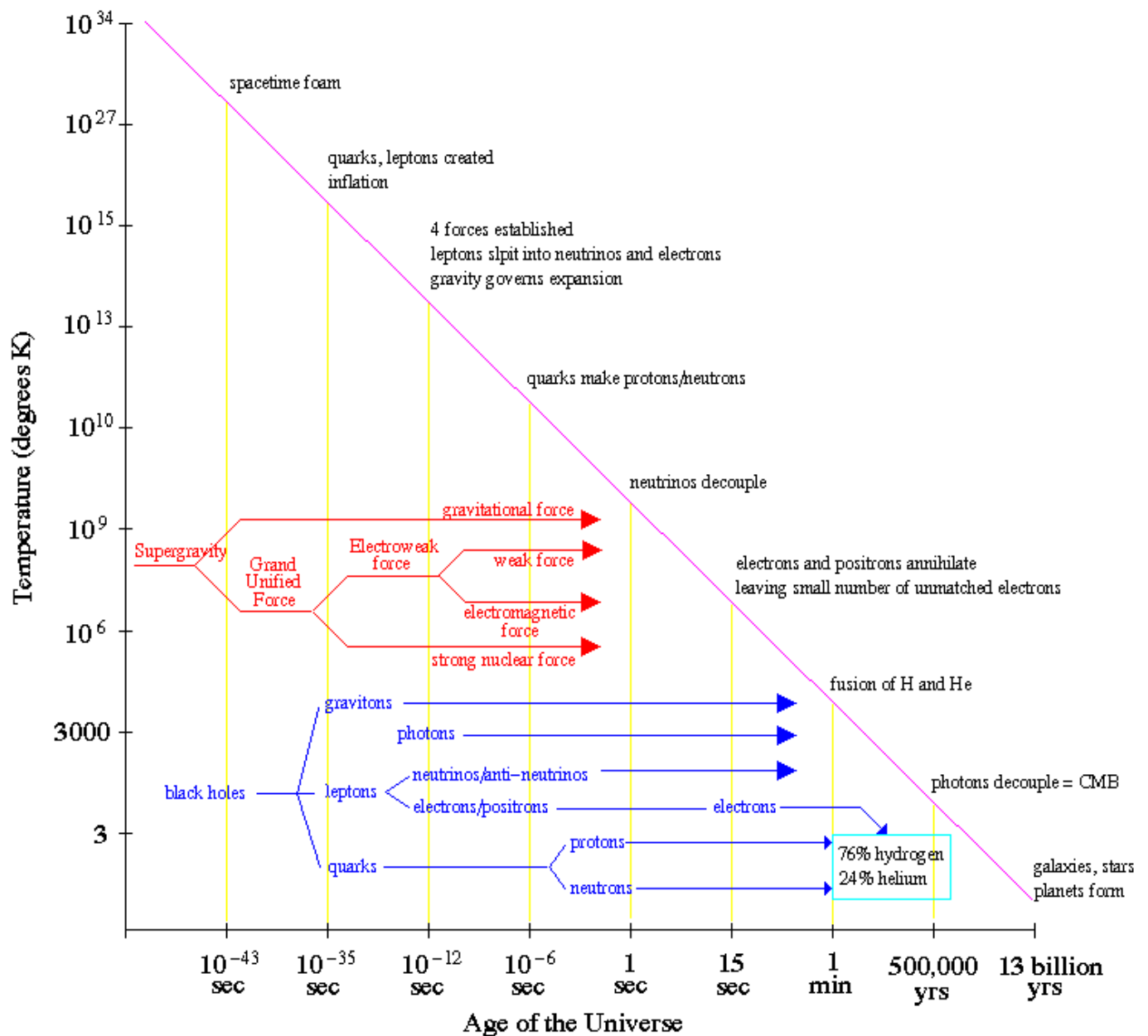


Every particle will have a wave function from quantum mechanics, and every particle exerts a gravitational force. But when the particles are close enough so the wave functions overlap, the meaning of gravity breaks down.



From special relativity we know that the speed of light is a maximum transfer of information. So we have no information for timescales less than the Planck length divided by the speed of light.

However, events before this time are undefined in our current science and, in particular, we have no solid understanding of the origin of the Universe (i.e. what started or 'caused' the Big Bang). At best, we can describe our efforts to date as probing around the 'edges' of our understanding in order to define what we don't understand, much like a blind person would explore the edge of a deep hole, learning its diameter without knowing its depth.



Cosmic Singularity :

One thing is clear in our framing of questions such as 'How did the Universe get started?' is that the Universe was self-creating. This is not a statement on a 'cause' behind the origin of the Universe, nor is it a statement on a lack of purpose or destiny. It is simply a statement that the Universe was emergent, that the actual of the Universe probably derived from a indeterminate sea of potentiality that we call the quantum vacuum, whose properties may always remain beyond our current understanding.

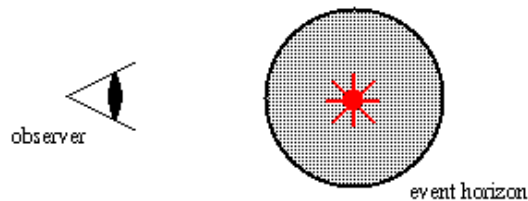
Extrapolation from the present to the moment of Creation implies an origin of infinite density and infinite temperature (all the Universe's mass and energy pushed to a point of zero volume). Such a point is called the cosmic singularity.

Singularity

a singularity is a point of infinite mass or density or energy



a singularity produces a paradox of infinite forces if observed or experienced



thus, a singularity only has a mathematical existence and is prevented from having a physical existence by the principle of cosmic censorship. In most cases, such as a black hole, the principle evokes itself by the construction of an event horizon, protecting the observer from experiencing the singularity

Infinities are unacceptable as physical descriptions, but our hypothetical observers back at the beginning of time are protected by the principle of cosmic censorship. What this means is that singularities exist only mathematically and not as a physical reality that we can observe or measure. Nature's solution to this problem are things like the event horizon around black holes. Barriers built by relativity to prevent observation of a singularity.

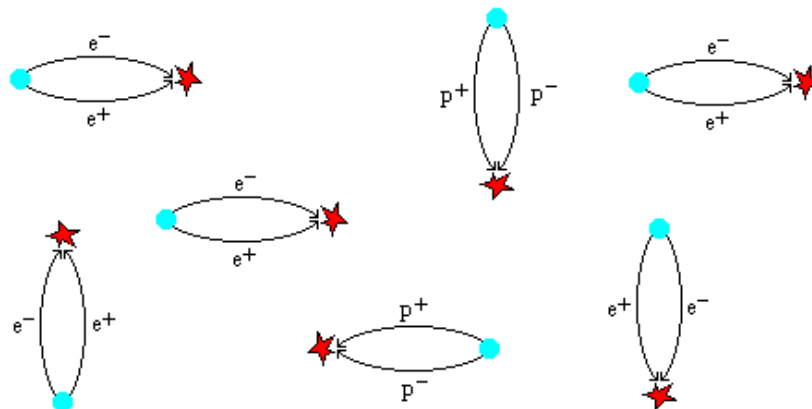
Quantum Vacuum:

The cosmic singularity, that was the Universe at the beginning of time, is shielded by the lack of any physical observers. But the next level of inquiry is what is the origin of the emergent properties of the Universe, the properties that become the mass of the Universe, its age, its physical constants, etc. The answer appears to be that these properties have their origin as the fluctuations of the [quantum vacuum](#).

The properties of the Universe come from 'nothing', where nothing is the quantum vacuum, which is a very different kind of nothing. If we examine a piece of 'empty' space we see it is not truly empty, it is filled with spacetime, for example. Spacetime has curvature and structure, and obeys the laws of quantum physics. Thus, it is filled with potential particles, pairs of virtual matter and anti-matter units, and potential properties at the quantum level.

Quantum Vacuum

the quantum vacuum cannot be perceived or measured directly since it appears to be empty, in fact it is filled with potentiality



within the quantum vacuum, pairs of virtual matter and anti-matter particles are continually created and destroyed, borrowing their mass/energy by the uncertainty principle. They do not exist as observable entities, but their existence is exerted on other particles as a subtle pressure (called the Casimir effect)

The creation of virtual pairs of particles does not violate the law of conservation of mass/energy because they only exist for times much less than the Planck time. There is a temporary violation of the law of conservation of mass/energy, but this violation occurs within the timescale of the uncertainty principle and, thus, has no impact on macroscopic laws.

The quantum vacuum is the ground state of energy for the Universe, the lowest possible level. Attempts to perceive the vacuum directly only lead to a confrontation with a void, a background that appears to be empty. But, in fact, the quantum vacuum is the source of all potentiality. For example, quantum entities have both wave and particle characteristics. It is the quantum vacuum that such characteristics emerge from, particles 'stand-out' from the vacuum, waves 'undulate' on the underlying vacuum, and leave their signature on objects in the real Universe.

In this sense, the Universe is not filled by the quantum vacuum, rather it is 'written on' it, the substratum of all existence.

With respect to the origin of the Universe, the quantum vacuum must have been the source of the laws of Nature and the properties that we observe today. How those laws and properties emerge is unknown at this time.

Quantum Fluctuations :

The fact that the Universe exists should not be a surprise in the context of what we know about quantum physics. The uncertainty and unpredictability of the quantum world is manifested in the fact that whatever *can* happen, *does* happen (this is often called the principle of totalitarianism, that if a quantum mechanical process is not strictly forbidden, then it must occur).

For example, radioactive decay occurs when two protons and two neutrons (an alpha particle) leap out of an atomic nuclei. Since the positions of the protons and neutrons is governed by the wave function, there is a small, but finite, probability that all four will quantum tunnel outside the nucleus, and therefore escape. The probability of this happening is small, but given enough time (tens of years) it *will* happen.

The same principles were probably in effect at the time of the Big Bang (although we can not test

this hypothesis within our current framework of physics). But as such, the fluctuations in the quantum vacuum effectively guarantee that the Universe would come into existence.

Planck Era :

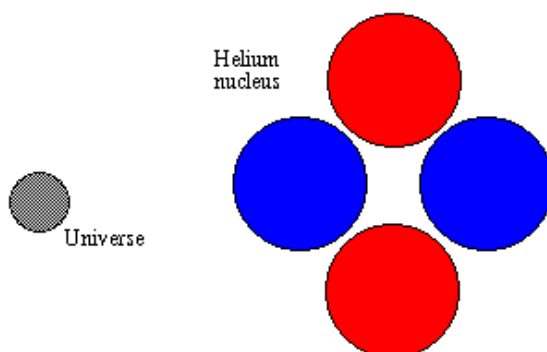
The earliest moments of Creation are where our modern physics breakdown, where 'breakdown' means that our theories and laws have no ability to describe or predict the behavior of the early Universe. Our everyday notions of space and time cease to be valid.

Although we have little knowledge of the Universe before the Planck time, only speculation, we can calculate when this era ends and when our physics begins. The hot Big Bang model, together with the ideas of modern particle physics, provides a sound framework for sensible speculation back to the Planck era. This occurs when the Universe is at the Planck scale in its expansion.

Planck Era

our cosmology commences when the Universe achieves a scale where our quantum physics has meaning. This scale when the smallest scale from quantum physics is set equal to the Schwarzschild radius from general relativity

This occurs at about 10^{-33} cm, much smaller than the nucleus of an atom



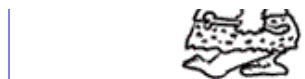
after this point, the Universe breaks free into the realm of normal 4D spacetime and unified physics

Remember, there is no 'outside' to the Universe. So one can only measure the size of the Universe much like you measure the radius of the Earth. You don't dig a hole in the Earth and lower a tape measure, you measure the circumference (take an airplane ride) of the Earth and divide by 2π (i.e. $C = 2 \times \pi \times \text{radius}$).

The Universe expands from the moment of the Big Bang, but until the Universe reaches the size of the Planck scale, there is no time or space. Time remains undefined, space is compactified. [String](#) theory maintains that the Universe had 10 dimensions during the Planck era, which collapses into 4 at the end of the Planck era (think of those extra 6 dimensions as being very, very small hyperspheres inbetween the space between elementary particles, 4 big dimensions and 6 little tiny ones).

During the Planck era, the Universe can be best described as a quantum foam of 10 dimensions containing Planck length sized black holes continuously being created and annihilated with no cause or effect. In other words, try not to think about this era in normal terms.



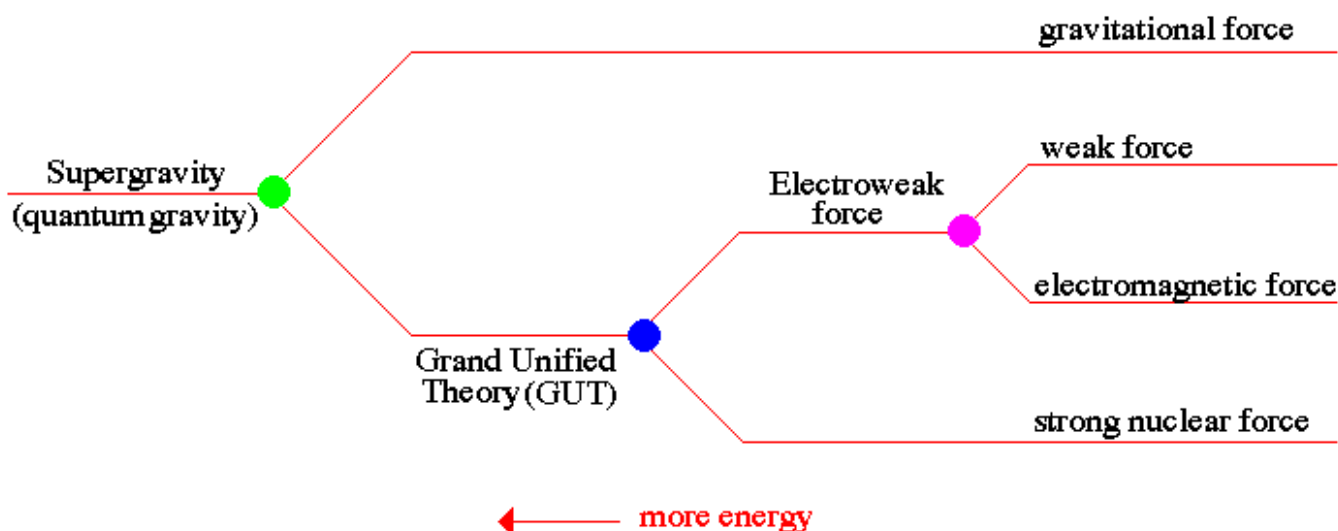


Unification:

One of the reasons our physics is incomplete during the Planck era is a lack of understanding of the unification of the forces of Nature during this time. At high energies and temperatures, the forces of Nature become symmetric. This means the forces resemble each other and become similar in strength, i.e. they unify.

Unification

all the forces of Nature should be capable of being described by a single theory. But only at high energies should the behavior of the forces combine, this is called unification



before the unification point, the forces are indistinguishable and have symmetry. After the unification point, the forces act differently and the symmetry is broken.

An example of unification is to consider the interaction of the weak and electromagnetic forces. At low energy, photons and W,Z particles are the force carriers for the electromagnetic and weak forces. The W and Z particles are very massive and, thus, require a lot of energy ($E=mc^2$). At high energies, photons take on similar energies to W and Z particles, and the forces become unified into the electroweak force.

There is the expectation that all the nuclear forces of matter (strong, weak and electromagnetic) unify at extremely high temperatures under a principle known as Grand Unified Theory, an extension of quantum physics using as yet undiscovered relationships between the strong and electroweak forces.

The final unification resolves the relationship between quantum forces and gravity (supergravity).

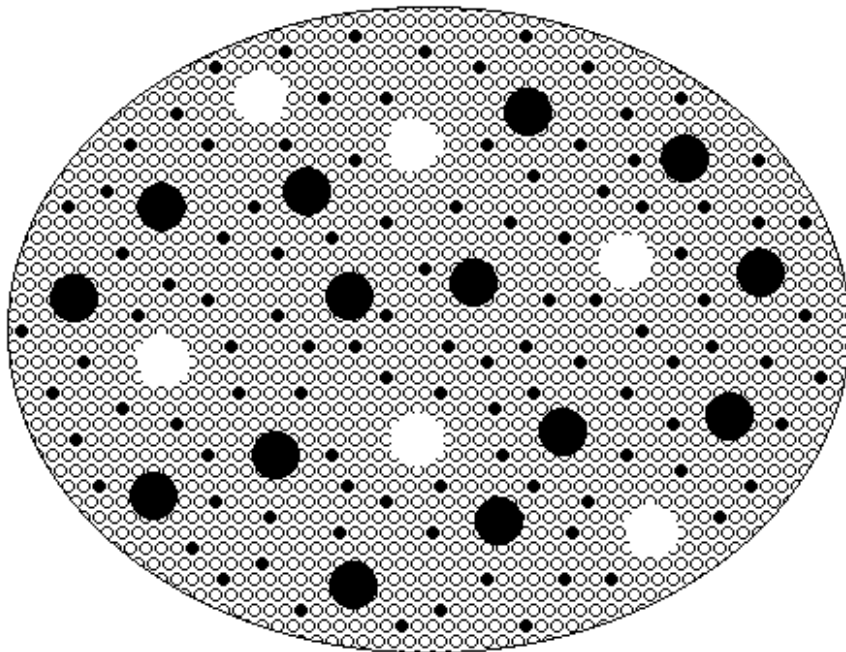
In the early Universe, the physics to predict the behavior of matter is determined by which forces are unified and the form that they take. The interactions just at the edge of the Planck era are ruled by supergravity, the quantum effects of mini-black holes. After the separation of gravity and nuclear forces, the spacetime of the Universe is distinct from matter and radiation.

Spacetime Foam :

The first moments after the Planck era are dominated by conditions where spacetime itself is twisted and distorted by the pressures of the extremely small and dense Universe.

Spacetime Foam

the immediate era after the Planck time is one of pressures and densities so high that spacetime itself is folded and meshed into foam of mini-black holes and wormholes



although the Planck time marks the beginning of time in the Universe, the convoluted nature of spacetime in this early phase, with numerous singularities and overlapping event horizons, makes it impossible for matter, photons or even causality to exist

Most of these black holes and wormholes are leftover from the Planck era, remnants of the event horizon that protected the cosmic singularity. These conditions are hostile to any organization or structure not protected by an event horizon. Thus, at this early time, black holes are the only units that can survive intact under these conditions, and serve as the first building blocks of structure in the Universe, the first 'things' that have individuality.

Based on computer simulations of these early moments of the Universe, there is the prediction that many small, primordial black holes were created at this time with no large black holes (the Universe was too small for them to exist). However, due to [Hawking radiation](#), the primordial black holes from this epoch have all decayed and disappeared by the present-day.

Matter arises at the end of the spacetime foam epoch as the result of strings, or loops in spacetime. The transformation is from ripping spacetime foam into black holes, which then transmute into elementary particles. Thus, there is a difference between something of matter and nothing of spacetime, but it is purely geometrical and there is nothing behind the geometry. Matter during this era is often called GUT matter to symbolize its difference from quarks and leptons and its existence under GUT forces.

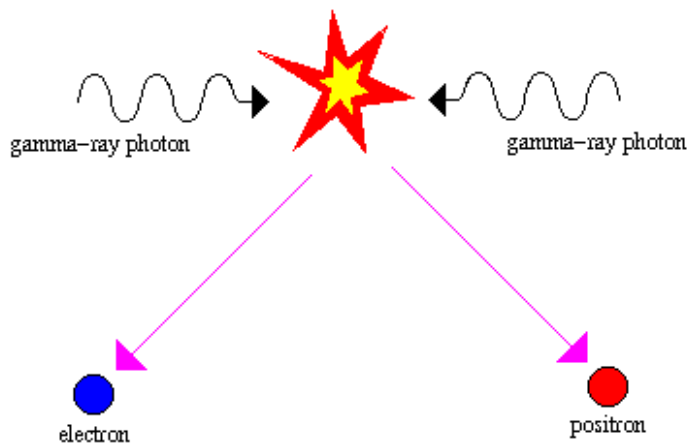
Hawking Radiation:

[Hawking](#), an English theoretical physicist, was one of the first to consider the details of the behavior of a black hole whose Schwarzschild radius was on the level of an atom. These black holes are not necessarily low mass, for example, it requires 1 billion tons of matter to make a black hole the size of

a proton. But their small size means that their behavior is a mix of quantum mechanics rather than relativity.

Before black holes were discovered it was known that the collision of two photons can cause [pair production](#). This is a direct example of converting energy into mass (unlike fission or fusion which turn mass into energy). Pair production is one of the primary methods of forming matter in the early Universe.

Pair Production



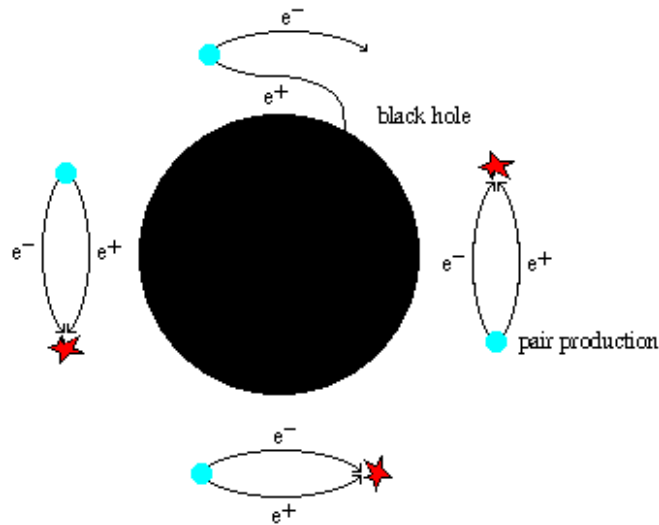
when two high energy gamma-ray photons collide, an electron/positron pair are produced (the energy is converted into mass, $E=mc^2$)

Note that pair production is symmetric in that a matter and antimatter particle are produced (an electron and an anti-electron (positron) in the above example).

Hawking showed that the strong gravitational gradients (tides) near black holes can also lead to pair production. In this case, the gravitational energy of the black hole is converted into particles.

Hawking Radiation

the strong gravitational field around a black hole causes pair production

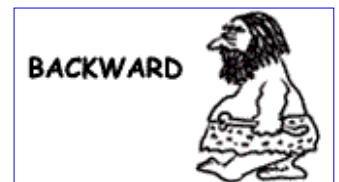
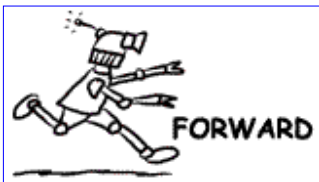


if a pair is produced outside the event horizon, then one member will fall back into the black hole, but the other member will escape and the black hole loses mass

the amount of mass lost is greater for small black holes, therefore quantum sized black holes disintegrate in very short timescales

If the matter/anti-matter particle pair is produced below the event horizon, then particles remain trapped within the black hole. But, if the pair is produced above the event horizon, it is possible for one member to fall back into the black hole, the other to escape into space. Thus, the black hole can lose mass by a quantum mechanical process of pair production outside of the event horizon.

The rate of pair production is stronger when the curvature of spacetime is high. Small black holes have high curvature, so the rate of pair production is inversely proportional to the mass of the black hole (this means its faster for smaller black holes). Thus, Hawking was able to show that the mini or primordial black holes expected to form in the early Universe have since disintegrated, resolving the dilemma of where all such mini-black holes are today.



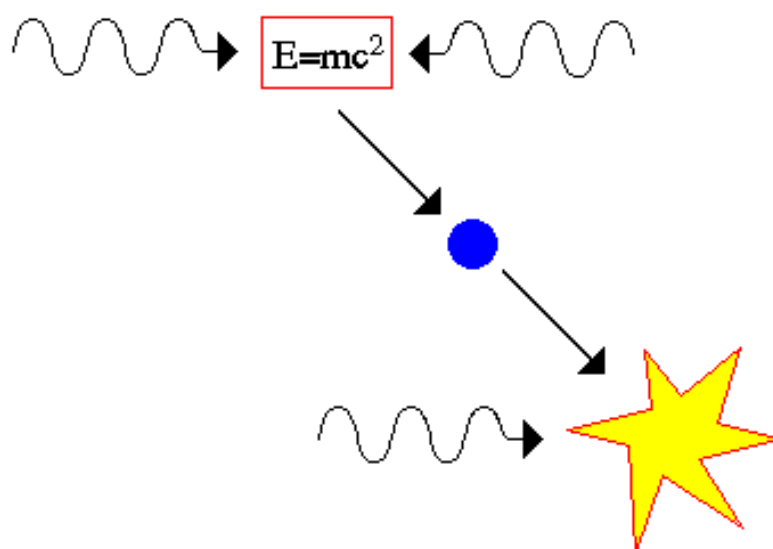
Symmetry Breaking:

In the early Universe, pressures and temperature prevented the permanent establishment of elementary particles. Even quarks and leptons were unable to form stable objects until the Universe had cooled beyond the supergravity phase. If the fundamental building blocks of Nature (elementary particles) or spacetime itself were not permanent then what remained the same? The answer is symmetry.

Matter Destruction

The formation of matter is inhibited in the early Universe due to collisions with high energy photons.

Matter forms when two photons collide, pair production.



But, when the energy density is high, the newly formed matter will quickly encounter another photon and be destroyed.

Often symmetry is thought of as a relationship, but in fact it has its own identical that is preserved during the chaos and flux of the early Universe. Even though virtual particles are created and destroyed, there is always a symmetry to the process. For example, for every virtual electron that is formed a virtual positron (anti-electron) is also formed. There is a time symmetric, mirror-like quality to every interaction in the early Universe.

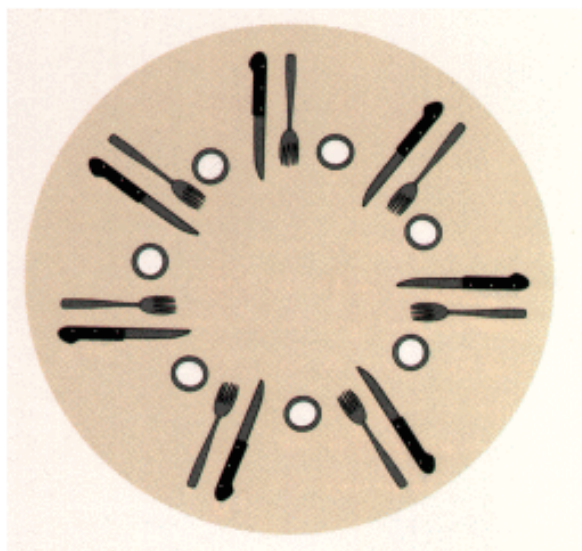
Symmetry also leads to conservation laws, and conservation laws limit the possible interactions between particles. Those imaginary processes that violate conservation laws are forbidden. So the existence of symmetry provides a source of order to the early Universe.

Pure symmetry is like a spinning coin. The coin has two states, but while spinning neither state is determined, and yet both states exist. The coin is in a state of both/or.

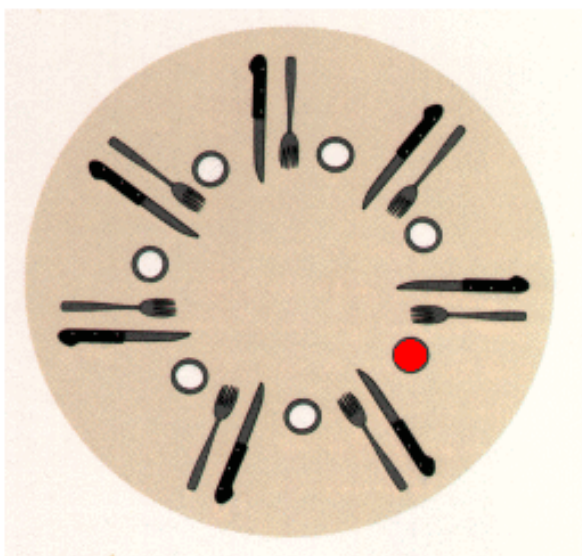
When the coin hits the floor the symmetry is broken (its either heads or tails) and energy is released in the process (the noise the coin makes as it hits the ground).

Symmetry Breaking

an example of symmetry is the place settings below



it is unclear which glass goes with any particular setting, until one is chosen

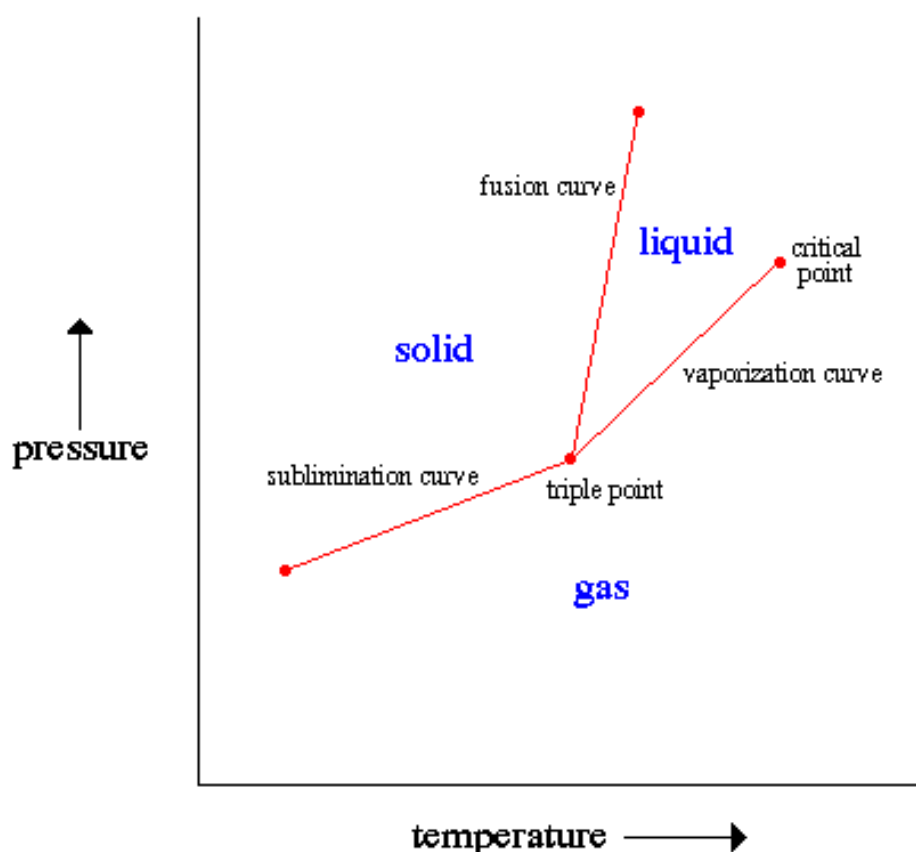


once a glass is chosen the symmetry is broken and the matching of glasses becomes unique

The effect of symmetry breaking in the early Universe was a series of [phase changes](#), much like when ice melts to water or water boils to steam. A phase change is the dramatic change in the internal order of a substance. When ice melts, the increased heat breaks the bonds in the lattice of water molecules, and the ice no longer holds its shape.

Phase change in the early Universe occurs at the unification points of fundamental

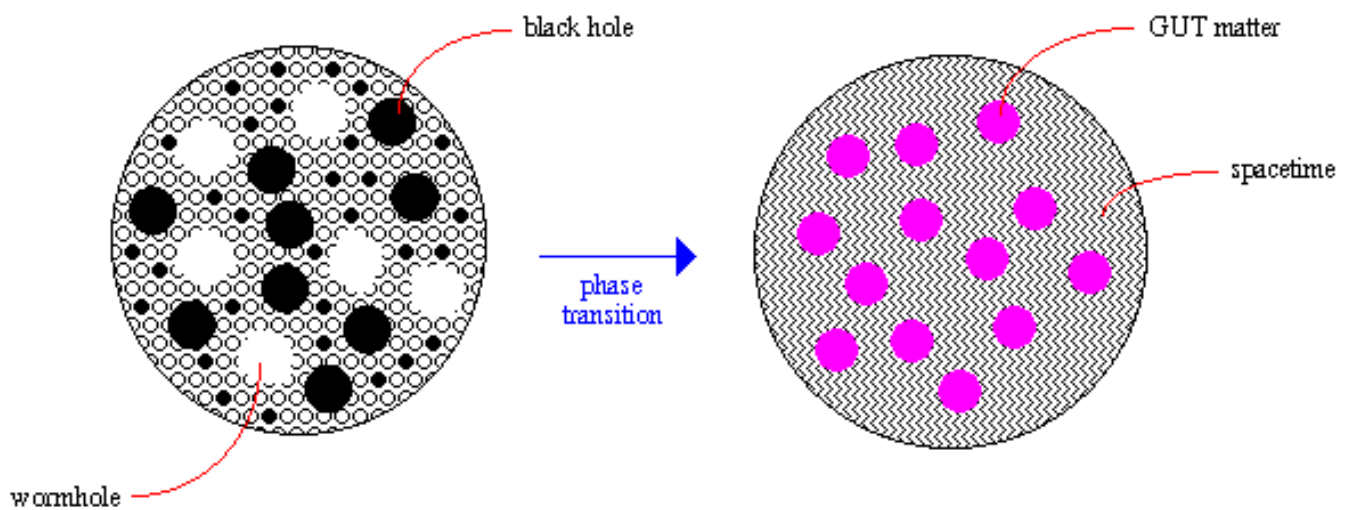
forces. The decoupling of those forces provides the energy input for phase changes in the Universe as a whole.



With respect to the Universe, a phase change during symmetry breaking is a point where the characteristics and the properties of the Universe make a radical shift. At the supergravity symmetry breaking, the Universe passed from the Planck era of total chaos to the era of spacetime foam. The energy release was used to create spacetime. During the GUT symmetry breaking, mass and spacetime separated and the energy released was used to create particles.

Asymmetric Universe

the process of symmetry breaking changes the Universe from a symmetric case to an asymmetric case



the spacetime foam was a symmetric case of the distribution of black holes and wormholes. After the GUT symmetry breaking, the Universe became an asymmetric case of lumps of GUT matter in a uniform spacetime.

Notice that as symmetry breaks, there is less order, more chaos. The march of entropy in the Universe applies to the laws of Nature as well as matter. The Universe at the time of the cosmic singularity was a time of pure symmetry, all the forces had equal strength, all the matter particles had the same mass (zero), spacetime was the same everywhere (although all twisted and convolved). As forces decouple, they lose their symmetry and the Universe becomes more disordered.

Inflation:

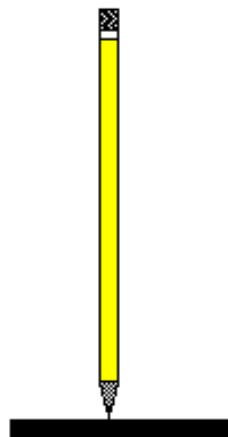
There are two major problems for the Big Bang model of the creation of the Universe. They are

- the flatness problem
- the horizon problem

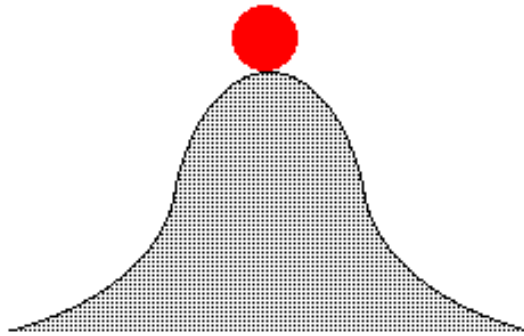
The flatness problem relates to the density parameter of the Universe, Ω . Values for Ω can take on any number between 0.01 and 5 (lower than 0.01 and galaxies can't form, more than 5 and the Universe is younger than the oldest rocks). The measured value is near 0.2. This is close to an Ω of 1, which is strange because Ω of 1 is an unstable point for the geometry of the Universe.

Instability

there are many examples in Nature of balance points that are **unstable**



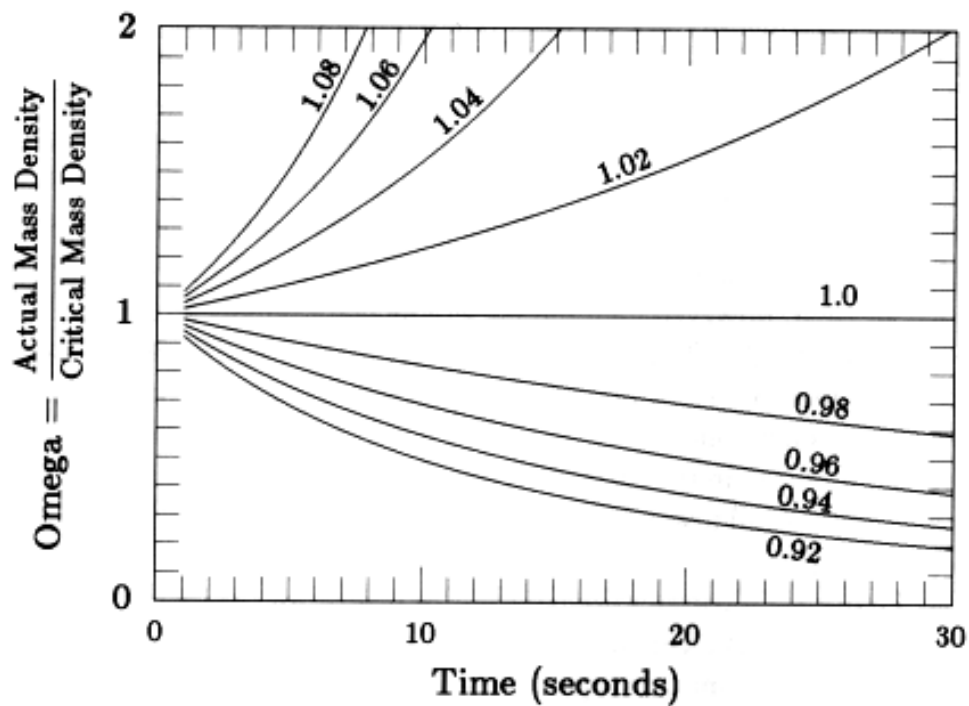
pencil on its point



ball at the top of a hill

any **perturbation** on these objects causes a runaway to a more stable point

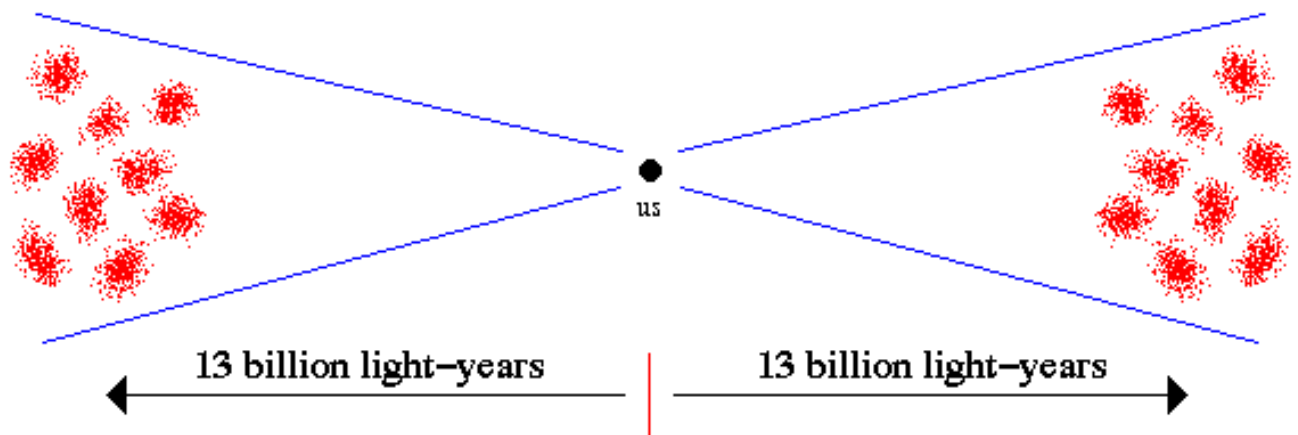
Values of Omega slightly below or above 1 in the early Universe rapidly grow to much less than 1 or much larger than 1 as time passes (like a ball at the top of a hill). After several billion years, Omega would have grown, or shrunk, to present-day values of much, much more, or much, much less than 1. So the fact that the measured value of 0.2 is so close to 1 that we expect to find that our measured value is too low and that the Universe must have a value of Ω exactly equal to 1 for stability. Therefore, the flatness problem is that some mechanism is needed to produce a value for Ω to be exactly one (to balance the pencil). A Universe of Ω of 1 is a flat Universe.



The horizon problem concerns the fact that the Universe is isotropic. No matter what distant corners of the Universe you look at, the sizes and distribution of objects is exactly the same (see the Cosmological Principle). But there is no reason to expect this since opposite sides of the Universe are not causally connected, any information that is be transmitted from one side would not reach the other side in the lifetime of the Universe (limited to travel at the speed of light).

Horizon Problem

the number and size of density fluctuations on both sides of the sky are similar, yet they are separated by a distance that is greater than the speed of light times the age of the Universe, i.e. they should have no knowledge of each other by special relativity



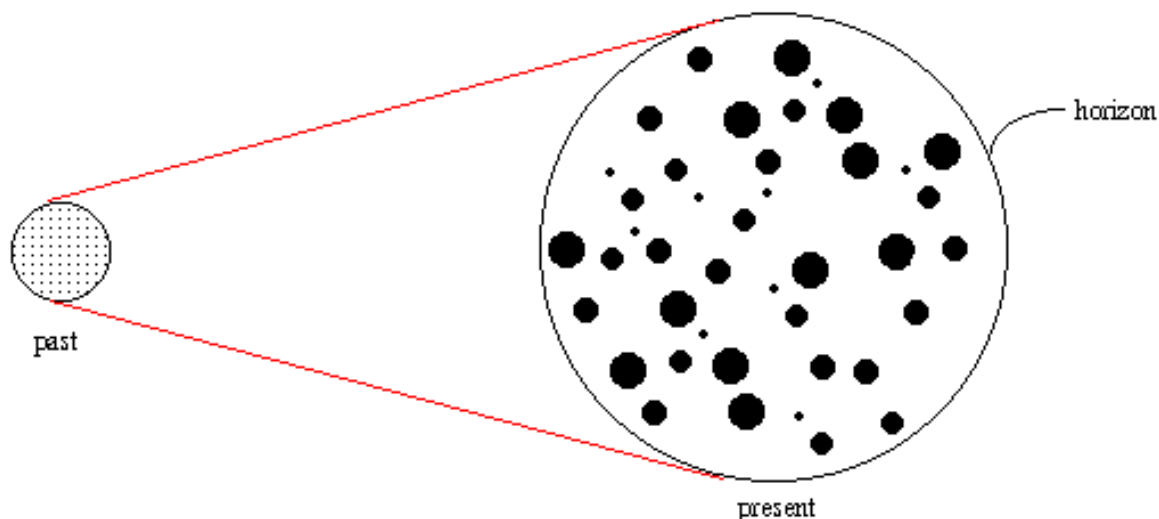
at some time in the early Universe, all parts of spacetime were causally connected, this must have happened after the spacetime foam era, and before the time where thermalization of matter occurred.

All of the Universe has an origin at the Big Bang, but time didn't exist until after the Planck era. By the end of that epoch, the Universe was already expanding so that opposite sides could not be causally connected.

The solution to both the flatness and horizon problems is found during a phase of the Universe called the inflation era. During the inflation era the Universe expanded a factor of 10^{54} , so that our horizon now only sees a small piece of what was once the total Universe from the Big Bang.

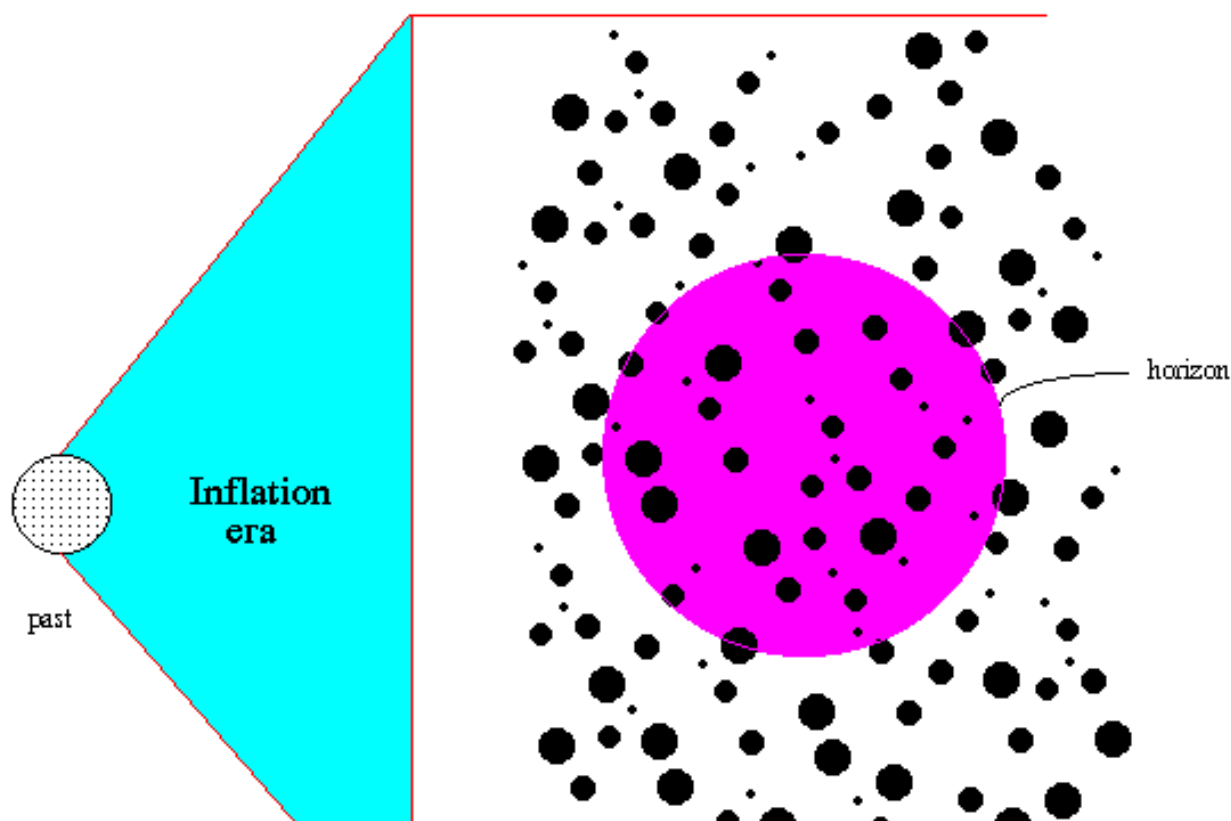
Normal Expansion

normal expansion is when the Universe expands at less than the speed of light such that all the Universe is within our horizon either now or sometime in the future



Inflation

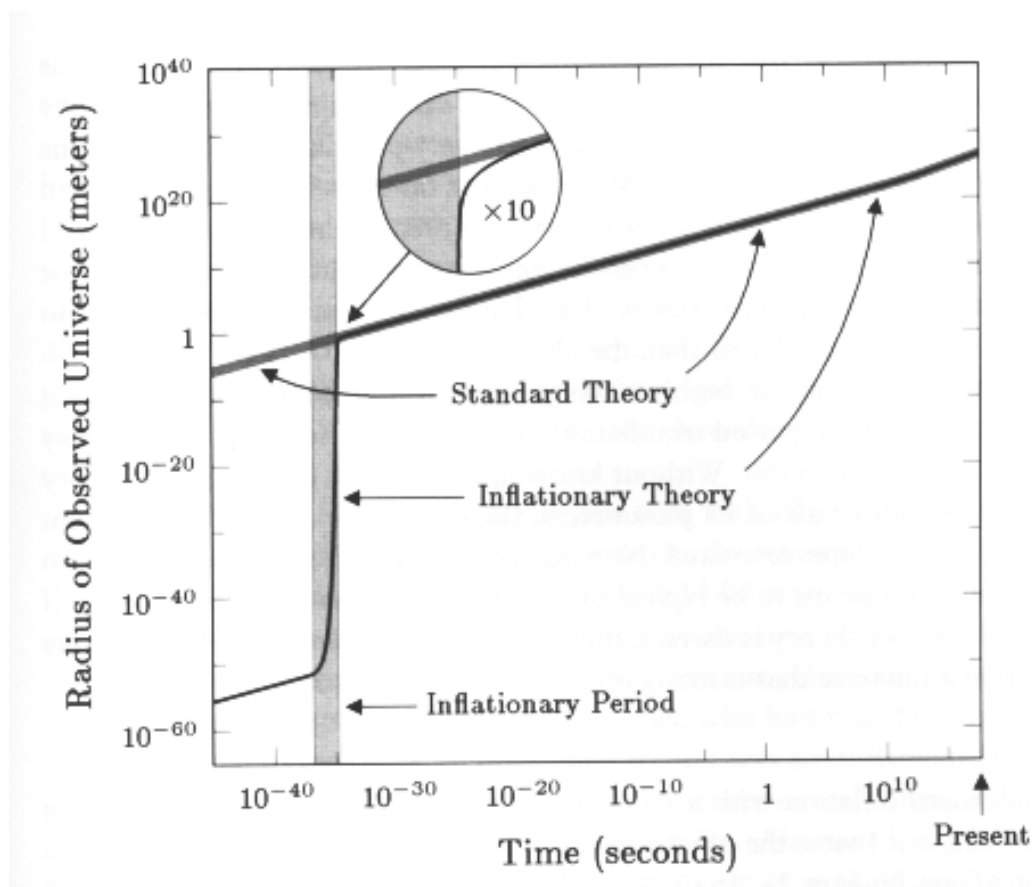
under inflationary cosmology, the Universe underwent a phase change at the GUT era and expanded faster than the speed of light (the spacetime itself expanded, so there is no violation of special relativity)





the result is that only a small part of the original Big Bang is within our horizon, what we call our Universe.

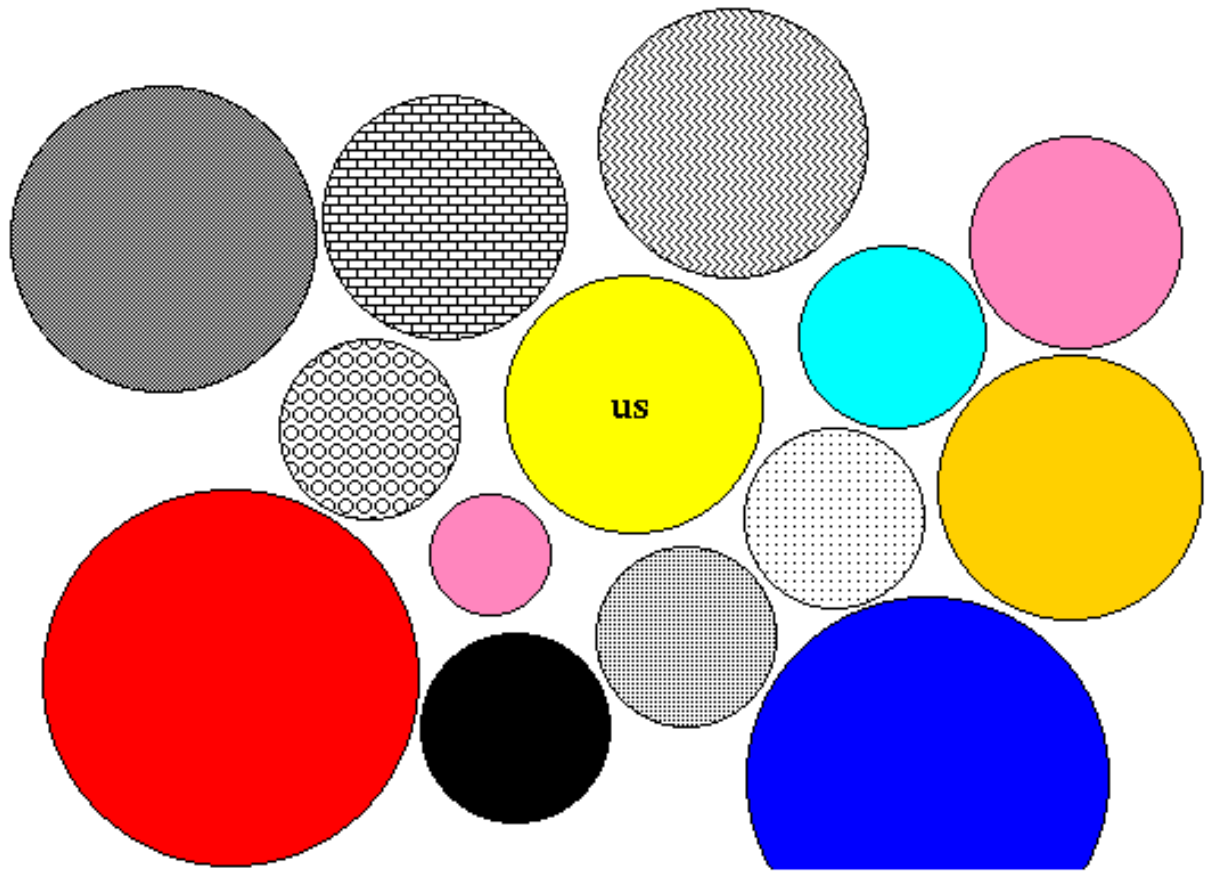
The cause of the inflation era was the symmetry breaking at the GUT unification point. At this moment, spacetime and matter separated and a tremendous amount of energy was released. This energy produced an overpressure that was applied not to the particles of matter, but to spacetime itself. Basically, the particles stood still as the space between them expanded at an exponential rate.



Note that this inflation was effectively at more than the speed of light, but since the expansion was on the geometry of the Universe itself, and not the matter, then there is no violation of special relativity. Our visible Universe, the part of the Big Bang within our horizon, is effectively a 'bubble' on the larger Universe. However, those other bubbles are not physically real since they are outside our horizon. We can only relate to them in an imaginary, theoretical sense. They are outside our horizon and we will never be able to communicate with those other bubble universes.

Bubble Universes

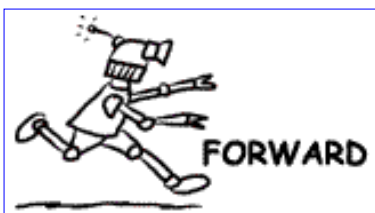
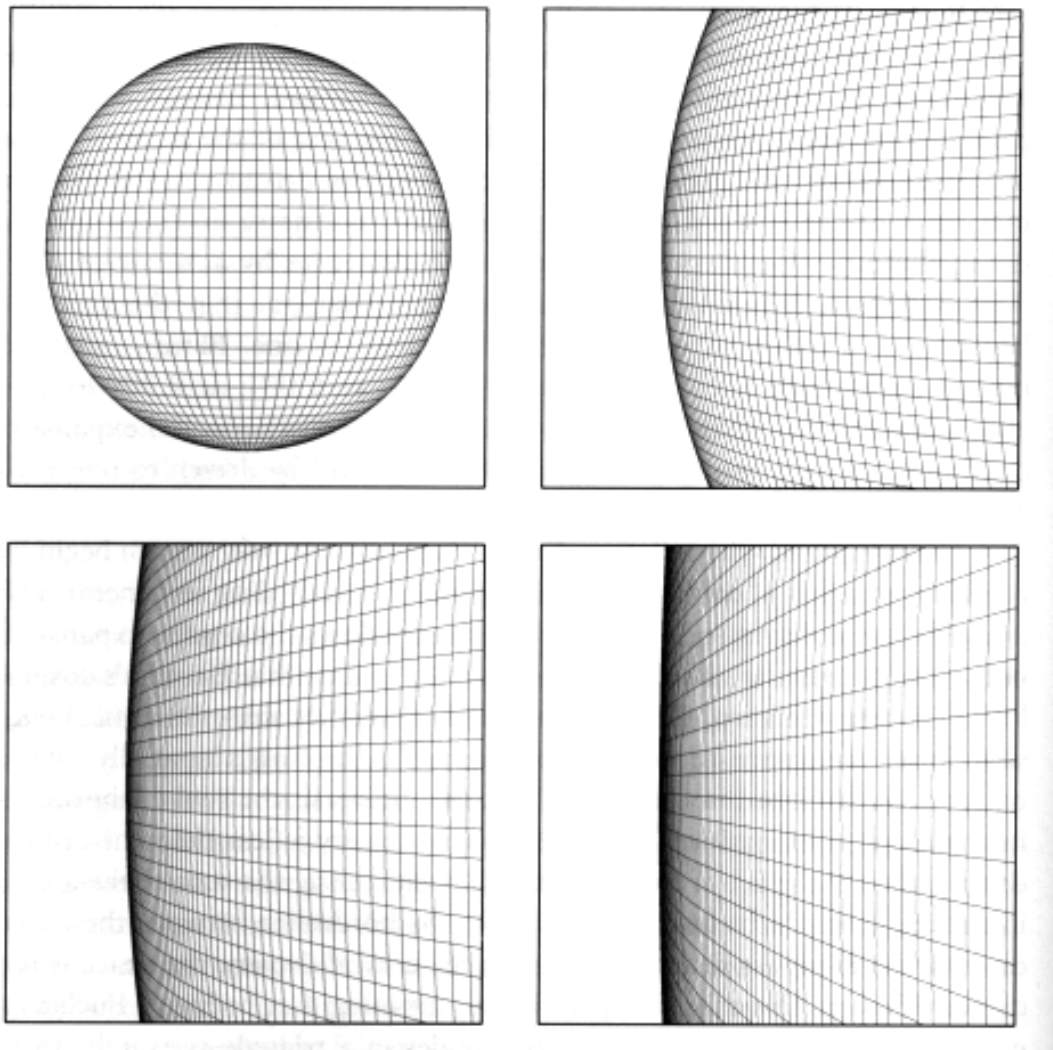
after the inflation era, our Universe became just one of many bubbles in the Big Bang substratum



although these other bubble universes may exist theoretically, we will never be able to observe or communicate with them since they are outside our horizon

Notice how this solves the horizon problem in that our present Universe was simply a small piece of a larger Big Bang universe that was all in causal connection before the inflation era. Other bubble universes might have very different constants and evolutionary paths, but our Universe is composed of a small, isotropic slice of the bigger Big Bang universe.

Inflation also solves the flatness problem because of the exponential growth. Imagine a highly crumpled piece of paper. This paper represents the Big Bang universe before inflation. Inflation is like zooming in of some very, very small section of the paper. If we zoom in to a small enough scale, the paper will appear flat. Our Universe must be exactly flat for the same reason, it is a very small piece of the larger Big Bang universe.



Anthropic Principle :

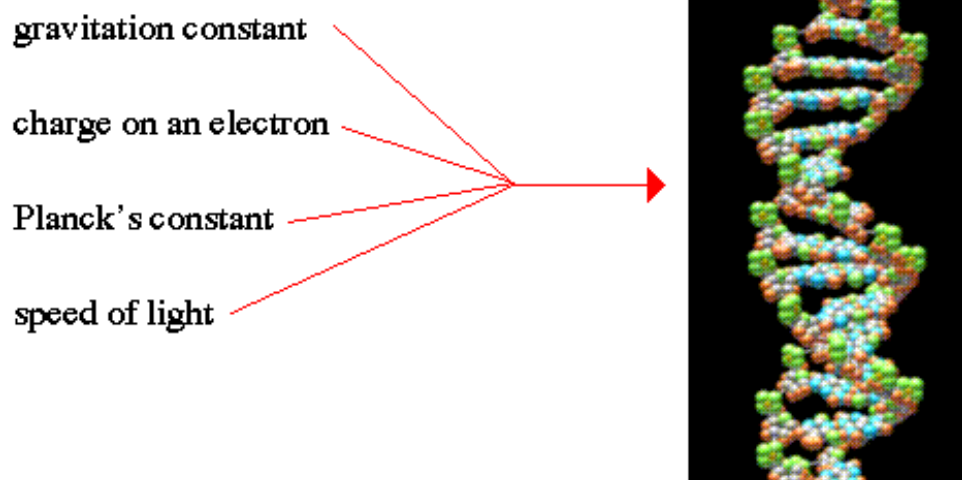
In the past 20 years our understanding of physics and biology has noted a peculiar specialness to our Universe, a specialness with regard to the existence of intelligent life. This sends up warning signs from the Copernican Principle, the idea that no scientific theory should invoke a special place or aspect to humans.

All the laws of Nature have particular constants associated with them, the gravitational constant, the speed of light, the electric charge, the mass of the electron, Planck's constant from quantum mechanics. Some are derived from physical laws (the speed of light, for example, comes from Maxwell's equations). However, for most, their values are arbitrary. The laws would still operate if the constants had different values, although the resulting interactions would be radically different.

Examples:

- gravitational constant: Determines strength of gravity. If lower than stars would have insufficient pressure to overcome Coulomb barrier to start thermonuclear fusion (i.e. stars would not shine). If higher, stars burn too fast, use up fuel before life has a chance to evolve.
- strong force coupling constant: Holds particles together in nucleus of atom. If weaker than multi-proton particles would not hold together, hydrogen would be the only element in the Universe. If stronger, all elements lighter than iron would be rare. Also radioactive decay would be less, which heats core of Earth.
- electromagnetic coupling constant: Determines strength of electromagnetic force that couples electrons to nucleus. If less, than no electrons held in orbit. If stronger, electrons will not bond with other atoms. Either way, no molecules.

All the above constants are critical to the formation of the basic building blocks of life. And, the range of possible values for these constants is very narrow, only about 1 to 5% for the combination of constants.



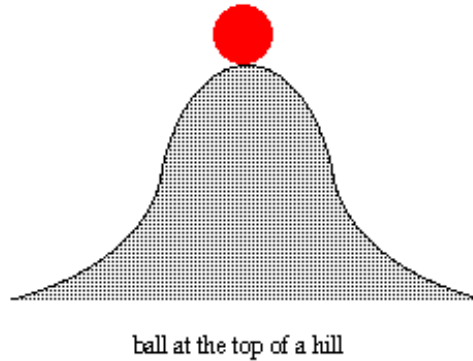
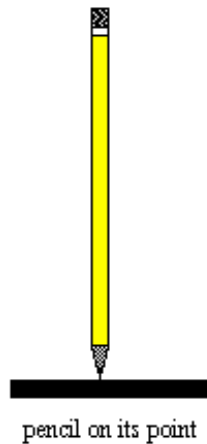
It is therefore possible to imagine whole different kinds of universes with different constants. For example, a universe with a lower gravitational constant would have a weaker force of gravity, where stars and planets might not form. Or a universe with a high strong force which would inhibit thermonuclear fusion, which would make the luminosity of stars be much lower, a darker universe, and life would have to evolve without sunlight.

The situation became worst with the cosmological discoveries of the 1980's. The two key cosmological parameters are the cosmic expansion rate (Hubble's constant, which determines the age of the Universe) and the cosmic density parameter (Ω), which determines the acceleration of the Universe and its geometry).

The flatness problem relates to the density parameter of the Universe, Ω . Values for Ω can take on any number, but it has to be between 0.01 and 5. If Ω is less than 0.01 the Universe is expanding so fast that the Solar System flies apart. And Ω has to be less than 5 or the Universe is younger than the oldest rocks. The measured value is near 0.2. This is close to an Ω of 1, which is strange because Ω of 1 is an unstable critical point for the geometry of the Universe.

Instability

there are many examples in Nature of balance points that are **unstable**



any **perturbation** on these objects causes a runaway to a more stable point

Values slightly below or above 1 in the early Universe rapidly grow to much less than 1 or much larger than 1 (like a ball at the top of a hill). So the fact that the measured value of 0.2 is so close to 1 that we expect to find in the future that our measured value is too low and that the Universe has a value of Ω exactly equal to 1 for stability.

This dilemma of the extremely narrow range of values for physical constants is allowed for the evolution of conscious creatures, such as ourselves, is called the anthropic principle, and has the form:

Anthropic Principle: The Universe must have those properties which allow life to develop within it at some stage in its history.

There are three possible alternatives from the anthropic principle;

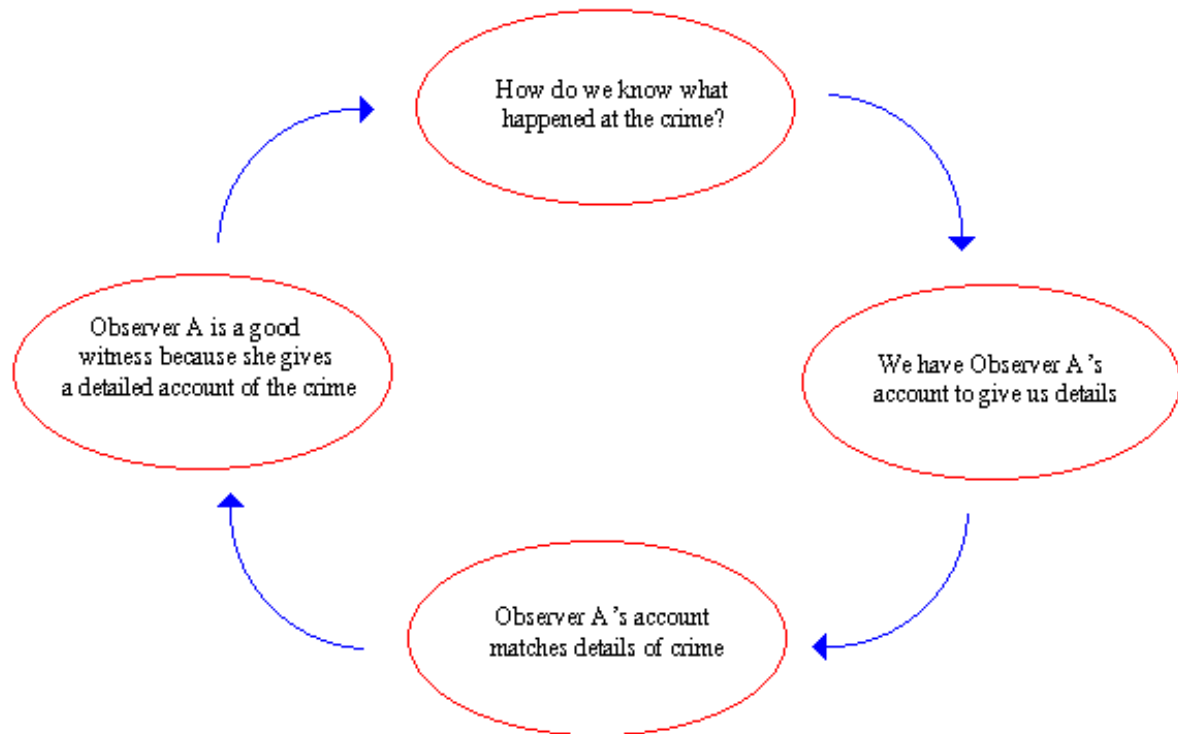
1. There exists one possible Universe 'designed' with the goal of generating and sustaining 'observers' (theological universe). Or...
2. Observers are necessary to bring the Universe into being (participatory universe). Or...
3. An ensemble of other different universes is necessary for the existence of our Universe (multiple universes)

Anthropic Principle and Circular Reasoning :

The usual criticism of any form of the anthropic principle is that it is guilty of a tautology or circular reasoning.

Circular Reasoning

A frequent error in arguments is to use the conclusion as one of the premises. This often 'begs the question' or involves circular reasoning, an argument which gives no support to the conclusion



The larger and more complex a circle of an argument, the harder to detect the fallacy

With the respect to our existence and the Universe, the error in reasoning is that because we are here, it must be possible that we can be here. In other words, we exist to ask the question of the anthropic principle. If we didn't exist then the question could not be asked. So there is nothing special to the anthropic principle, it simply states we exist to ask questions about the Universe.

An example of this style of question is whether life is unique to the Earth. There are many special qualities to the Earth (proper mass, distance from Sun for liquid water, position in Galaxy for heavy elements from nearby supernova explosion). But, none of these characteristics are unique to the Earth. There may exist hundreds to thousands of solar systems with similar characteristics where life would be possible, if not inevitable. We simply live on one of them, and we would not be capable of living on any other world.

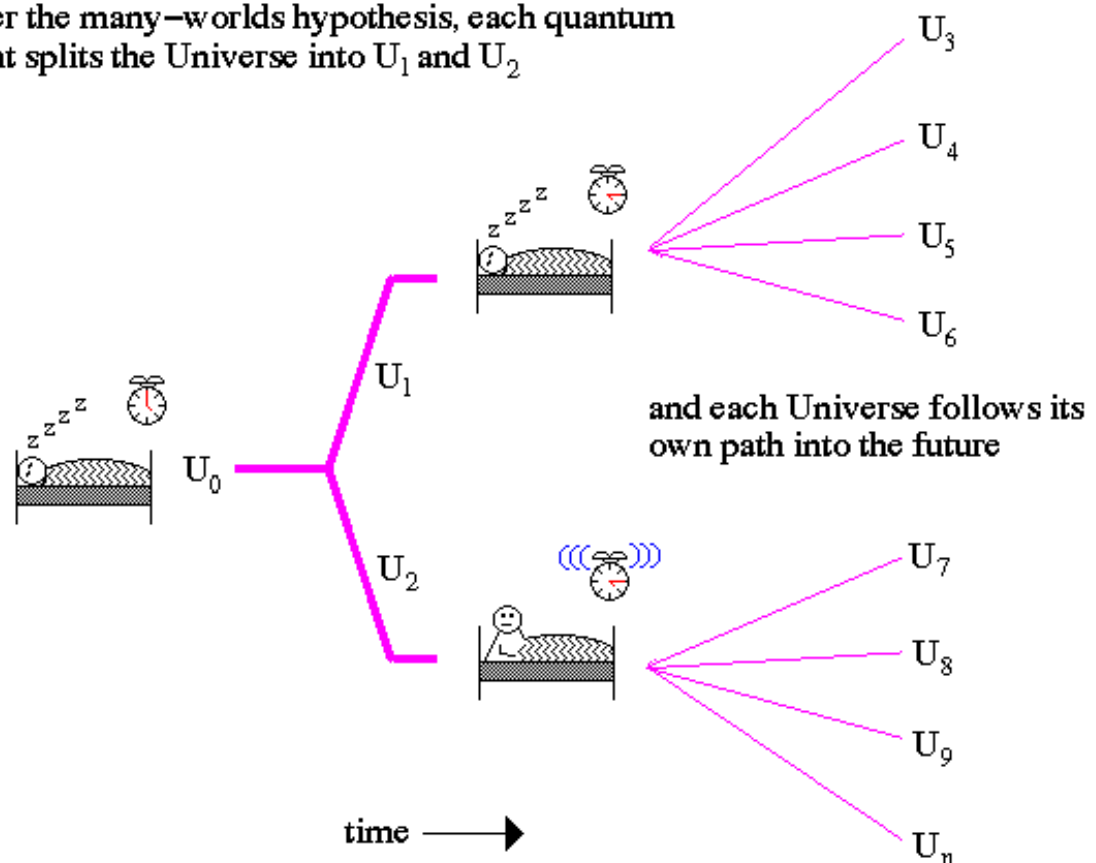
This solution is mildly unsatisfying with respect to physical constants since it implies some sort-of lottery system for the existence of life, and we have no evidence of previous Universes for the randomness to take place.

Anthropic Principle and Many-Worlds Hypothesis:

Another solution to the anthropic principle is that all possible universes, that can be imagined under the current laws of Nature, are possible and do have an existence as quantum

superpositions.

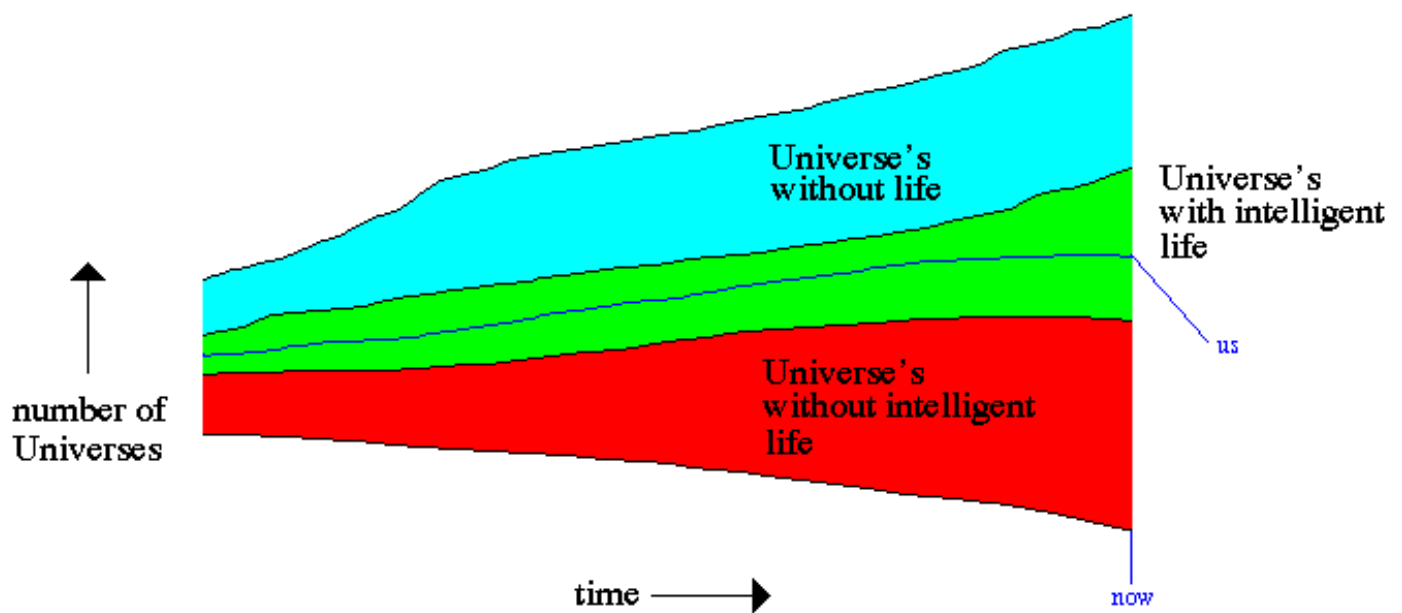
under the many-worlds hypothesis, each quantum event splits the Universe into U_1 and U_2



all the possible Universe's exist, but none can communicate with another.

This is the infamous many-worlds hypothesis used to explain how the position of an electron can be fuzzy or uncertainty. Its not uncertain, it actual exists in all possible positions, each one having its own separate and unique universe. Quantum reality is explained by the using of [infinite numbers of universes](http://zebu.uoregon.edu/~js/ast123/lectures/lec21.html) where every possible realization of position and energy of every particle actually exists.

The many-worlds hypothesis resolves the anthropic problem since we simply live in one of the many Universes which support intelligent life.



With respect to the anthropic principle, we simply exist in one of the many universes where intelligent life is possible and did evolve. There are many other universes where this is not the case, existing side by side with us in some super-reality of the many-worlds. Since the many-worlds hypothesis lacks the ability to test the existence of these other universes, it is not falsifiable and, therefore, borders on pseudo-science.

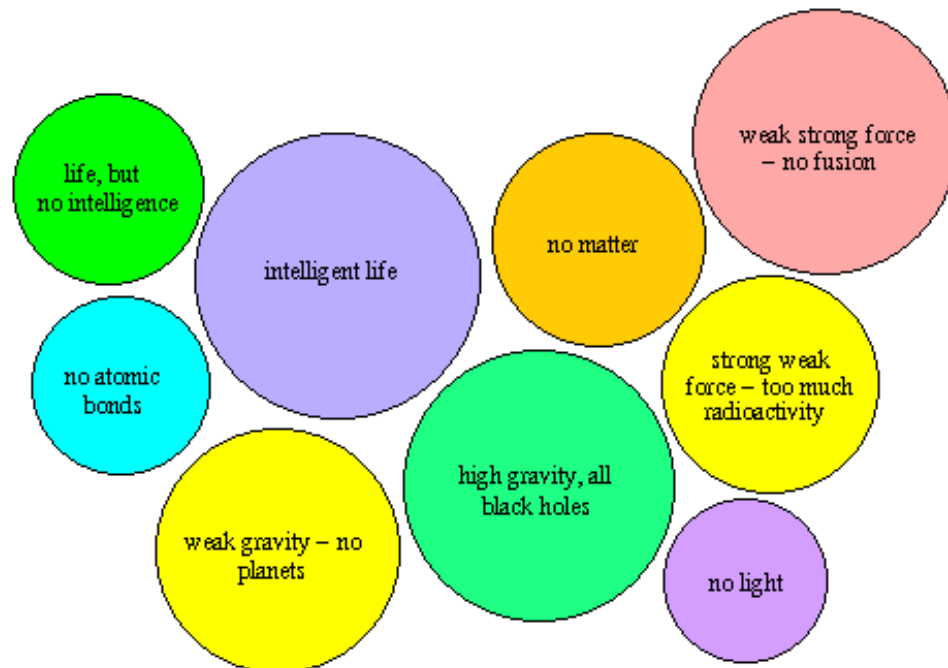
Anthropic Principle and Inflation :

Another avenue to understanding the anthropic principle is through inflation. Inflation theory shows that the fraction of the volume of the Universe with given properties does not depend on time. Each part evolves with time, but the Universe as a whole may be stationary and the properties of the parts do not depend on the initial conditions.

During the inflation era, the Universe becomes divided into exponentially large domains containing matter in all possible 'phases'. The distribution of volumes of different domains may provide some possibility to find the "most probable" values for universal constants. When the Universe inflated, these different domains separated, each with its own values for physical constants.

Anthropic Bubbles

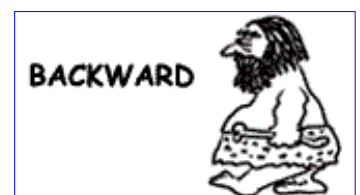
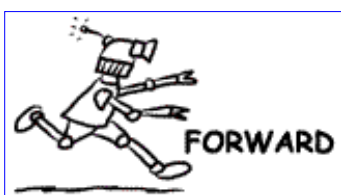
one possible solution to the anthropic dilemma is the numerous bubble universes produced by inflation. Each bubble universe may have its own physical constants, which determine the evolution within the bubble



the evolution of intelligent life is extremely sensitive to the initial conditions, but since number of bubble universes is also large, the possibility is finite and our existence is not a big mystery

Inflation's answer to the anthropic principle is that multiple universes were created from the Big Bang. Our Universe had the appropriate physical constants that lead to the evolution of intelligent life. However, that evolution was not determined or required. There may exist many other universes with similar conditions, but where the emergent property of life or intelligence did not develop.

Hopefully a complete Theory of Everything will resolve the 'how' questions on the origin of physical constants. But a complete physical theory may be lacking the answers to 'why' questions, which is one of the reasons that modern science is in a crisis phase of development, our ability to understand 'how' has outpaced our ability to answer if we 'should'.

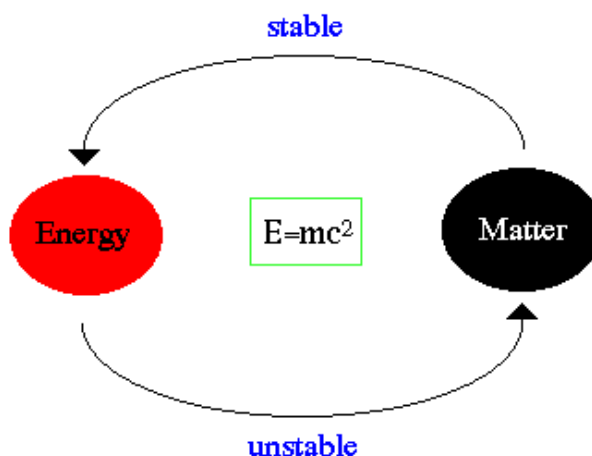


GUT matter :

Spacetime arrives when supergravity separates into the combined nuclear forces (strong, weak, electromagnetic) and gravitation. Matter makes its first appearance during this era as a composite form called Grand Unified Theory or GUT matter. GUT matter is a combination of what will become leptons, quarks and photons. In other words, it contains all the superpositions of future normal matter. But, during the GUT era, it is too hot and violent for matter to survive in the form of leptons and quarks.

Why can't matter remain stable at this point in the Universe's evolution? This involves the concept of [equilibrium](#), the balance between particle creation and annihilation.

at high temperatures, pressures and densities, it is more stable to produce energy than matter



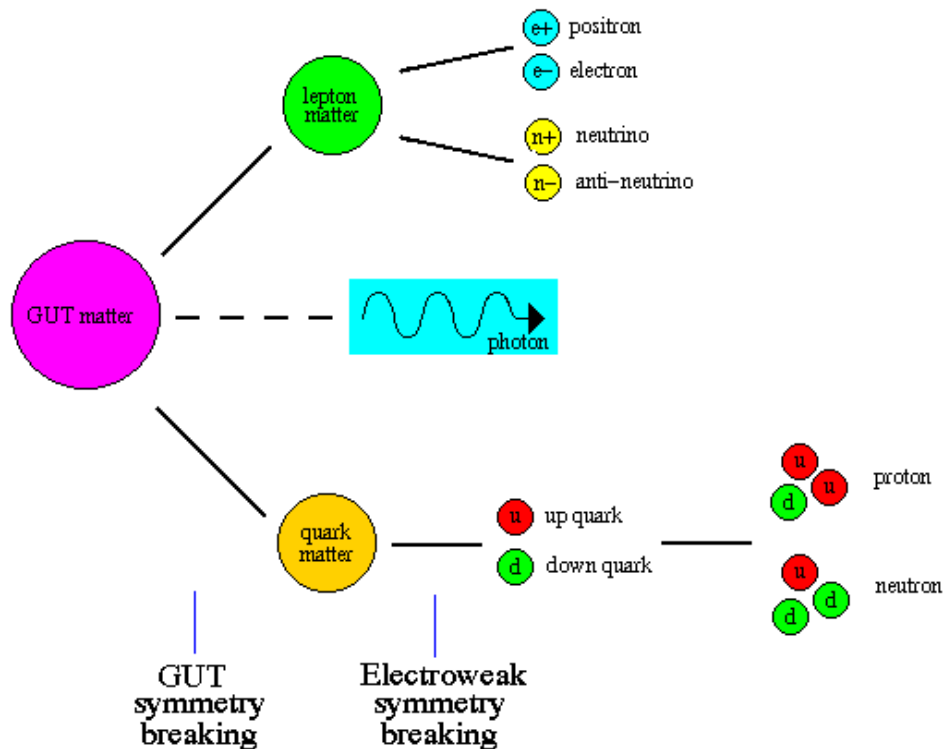
During pair production, energy is converted directly into mass in the form of a matter and anti-matter particle pair. The simplest particles are, of course, leptons such as an electron/positron pair. However, in high energy regimes, such as the early Universe, the conversion from energy to mass is unstable compared to the more probable mass to energy conversion (because the created mass must be so high in mass to match the energy used). In other words, when temperatures are high, matter is unstable and energy is stable.

Any matter that forms in the early Universe quickly collides with other matter or energy and is converted back into energy. The matter is in equilibrium with the surrounding energy and at this time the Universe is energy or radiation-dominated.

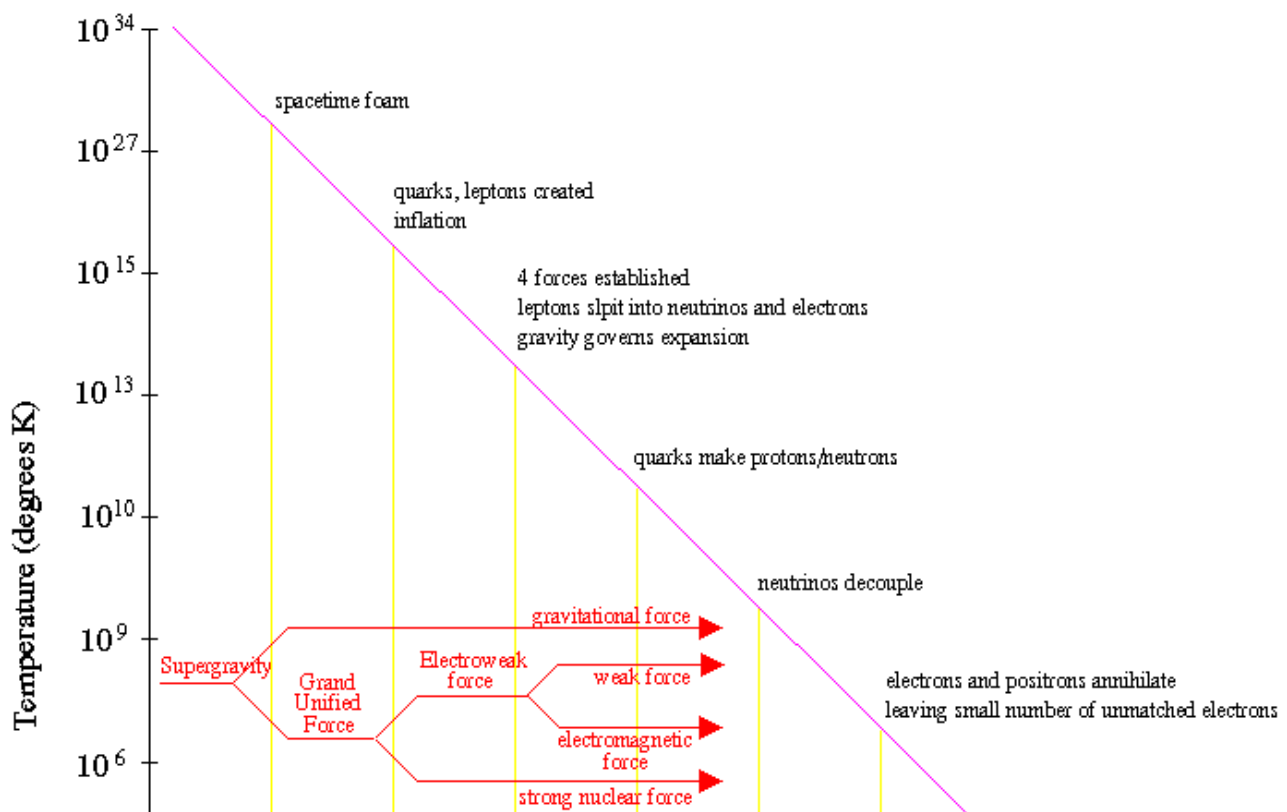
The type of matter that is created is dependent on the energy of its surroundings. Since the temperatures are so high in the early Universe, only very massive matter (= high energy) can form. However, massive particles are also unstable particles. As the Universe expands and cools, more stable, less massive forms of matter form.

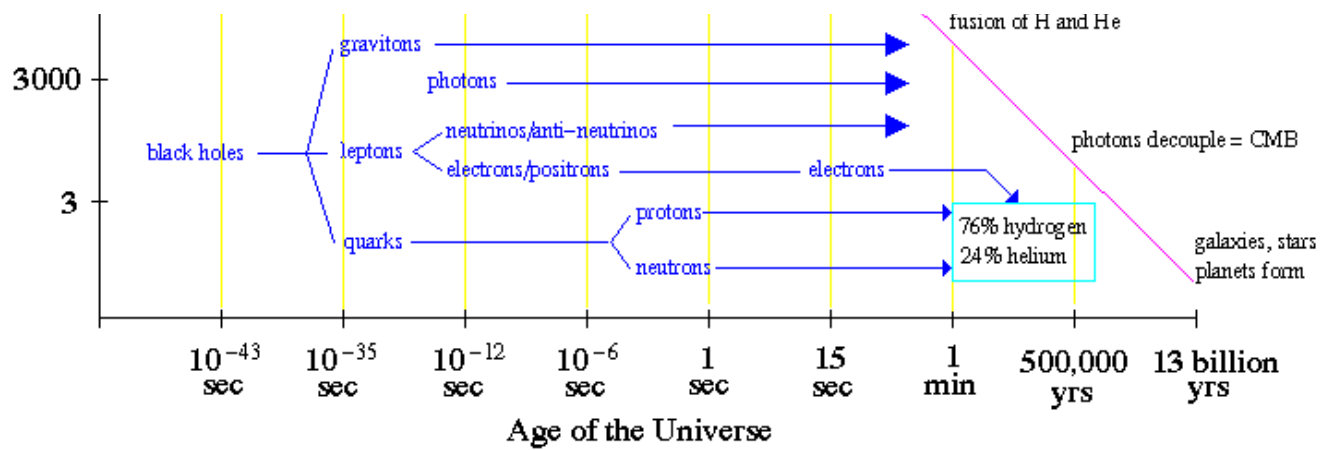
Matter Evolution

as the Universe expanded and cooled, matter began to condense starting with massive GUT matter. Each symmetry breaking produces a phase change and different forms of matter appear



As the Universe expands, matter is able to exist for longer periods of time without being broken down by energy. Eventually quarks and leptons are free to combine and form protons, neutrons and atoms, the ordinary matter of today.





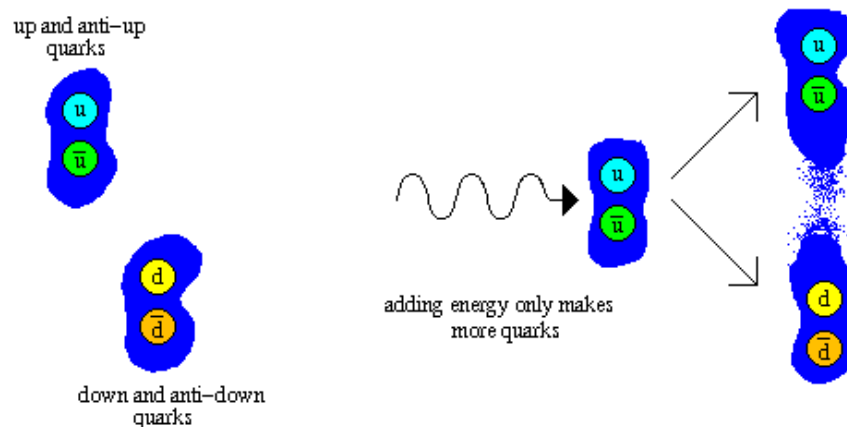
Quarks and Leptons :

After GUT matter forms, the next phase is for GUT matter to decay into lepton and quark matter. Lepton matter will become our old [friends](#) the electron and neutrino (and their anti-particles). But quark matter is unusual because of the property of quark confinement.

Quarks can never be found in isolation because the strong force becomes stronger with distance. Any attempt to separate pairs or triplets of quarks requires large amounts of energy, which are used to produce new groups of quarks.

Runaway Quark Production

soon after the GUT symmetry breaking, quarks pairs were produced in large numbers. Since the strong force has the strange property of becoming stronger with distance to the point where new quark pairs are produced instead of breaking quarks free (quark confinement), the number of quark pairs increases at a high rate



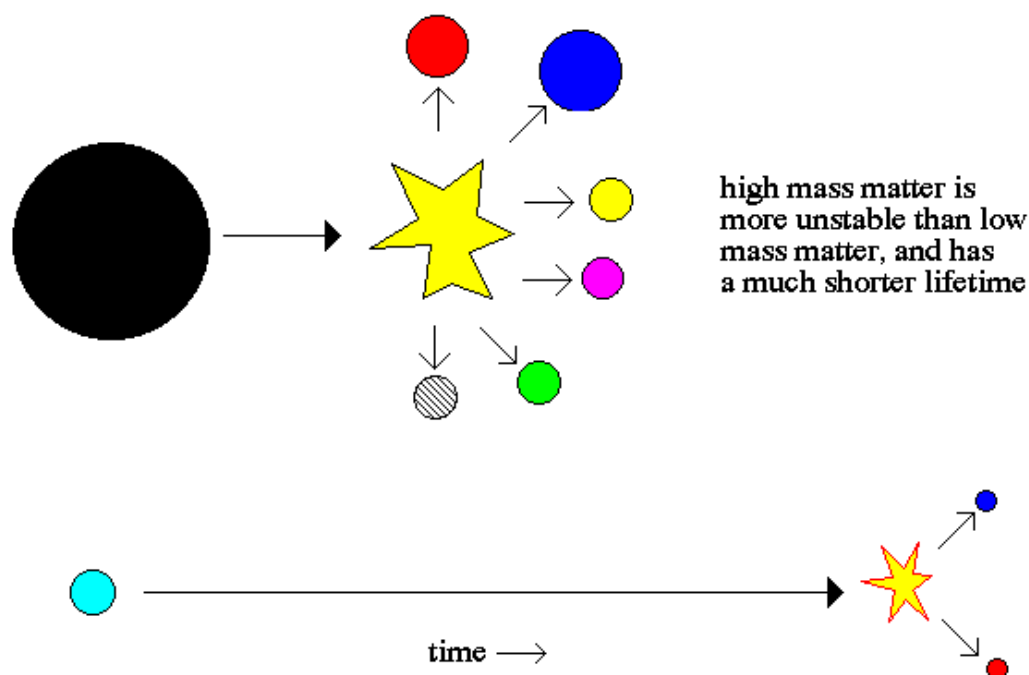
the resulting effect is to produce a 'quark soup', a rich collection of quarks and anti-quarks

With so much energy available in the early Universe, the endresult is a runaway production of quark and anti-quark pairs. Trillions of times the amounts we currently see in the Universe. The resulting soup of quark pairs will eventually suffer massive annihilation of its matter and anti-matter sides as soon as the Universe expands and cools sufficiently for quark production to stop.

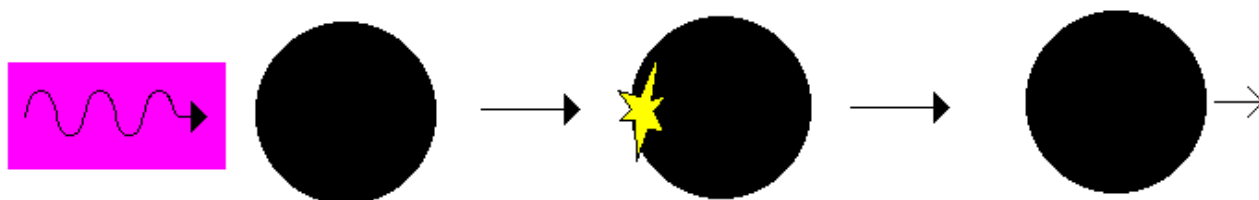
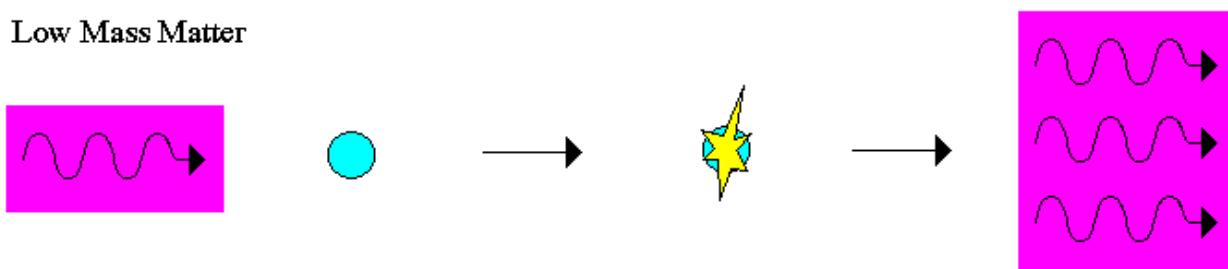
Notice that quark pairs are more stable than triplets, so that most of the quark production is done in pairs. Later, pairs will interact to form triplets, which are called baryons.

Baryogenesis :

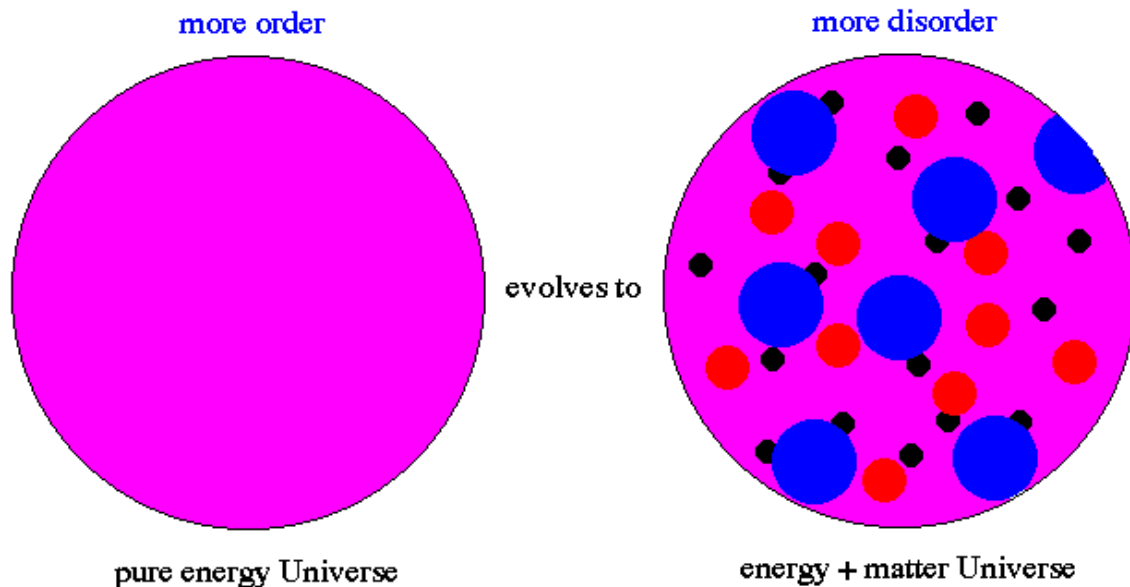
As the Universe cools a weak asymmetry in the direction towards matter becomes evident. Matter that is massive is unstable, particularly at the high temperature in the early Universe. Low mass matter is stable, but susceptible to destruction by high energy radiation (photons).



As the volume of the Universe increases, the lifetime of stable matter (its time between collisions with photons) increases. This also means that the time available for matter to interact with matter also increases.

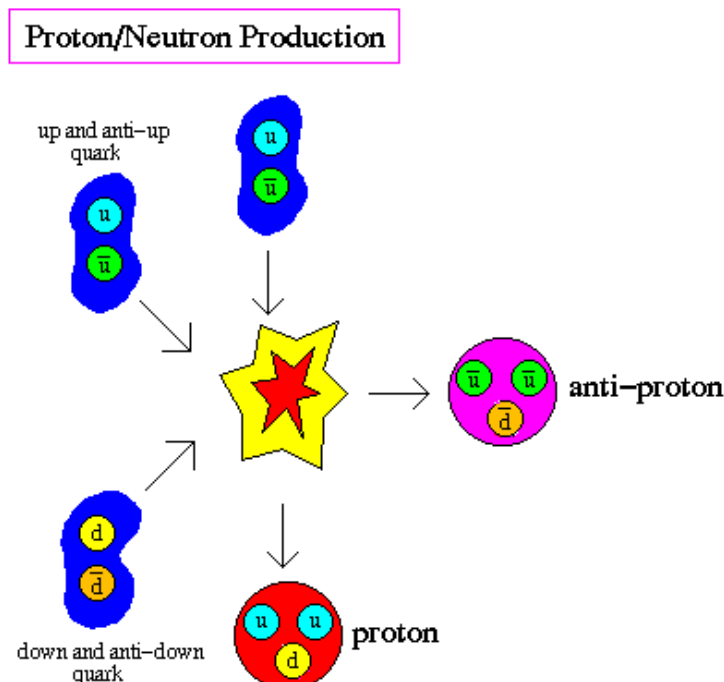
High Mass Matter**Low Mass Matter**

The Universe evolves from a pure, energy dominated domain to a more disordered, matter dominated domain, i.e. entropy marches on.



The last two stages of matter construction is the combining of three quark groups into baryons (protons and neutrons), then the collection of electrons by proton/neutron atomic nuclei to form atoms. The construction of baryons is called baryogenesis.

Baryogenesis begins around 1 second after the Big Bang. The equilibrium process at work is the balance between the strong force binding quarks into protons and neutrons versus the splitting of quark pairs into new quark pairs. When the temperature of the Universe drops to the point that there is not enough energy to form new quarks, the current quarks are able to link into stable triplets.



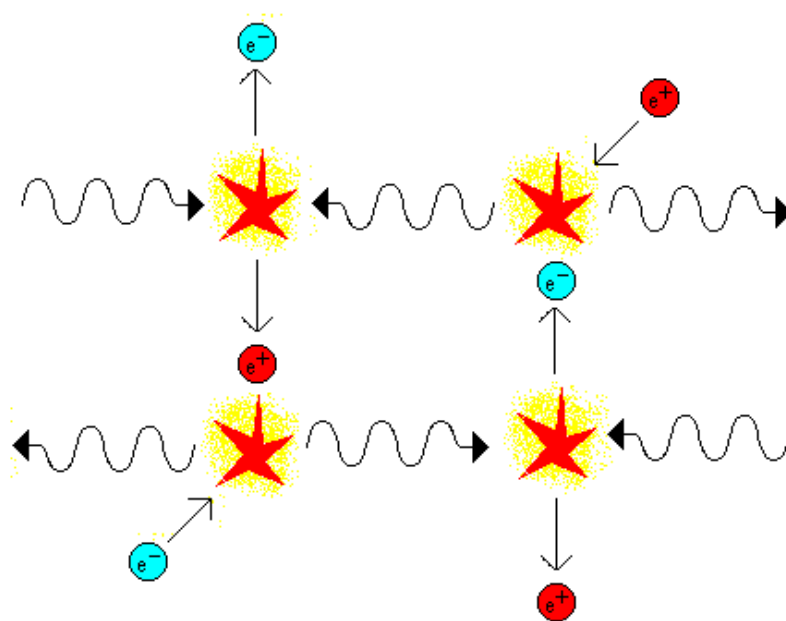
As all the anti-particles annihilate by colliding with their matter counterparts (leaving the small percentage of matter particles, see next lecture) leaving the remaining particles in the Universe to be photons, electrons, protons and neutrons. All quark pairs have reformed into baryons (protons and neutrons). Only around exotic objects, like black holes, do we find any anti-matter or mesons (quark pairs) or any of the other strange matter that was once found throughout the early Universe.

Matter versus Anti-Matter :

Soon after the second symmetry breaking (the GUT era), there is still lots of energy available to produce matter by [pair production](#), rather than quark confinement. However, the densities are so high that every matter and anti-matter particle produced is soon destroyed by collisions with other particles, in a cycle of equilibrium.

Particle Equilibrium

a state of particle equilibrium exists when the number of particle creations exactly matches the number of annihilations. Usually this is because there is no time for matter to decay or combine into new forms before a collision with an anti-particle



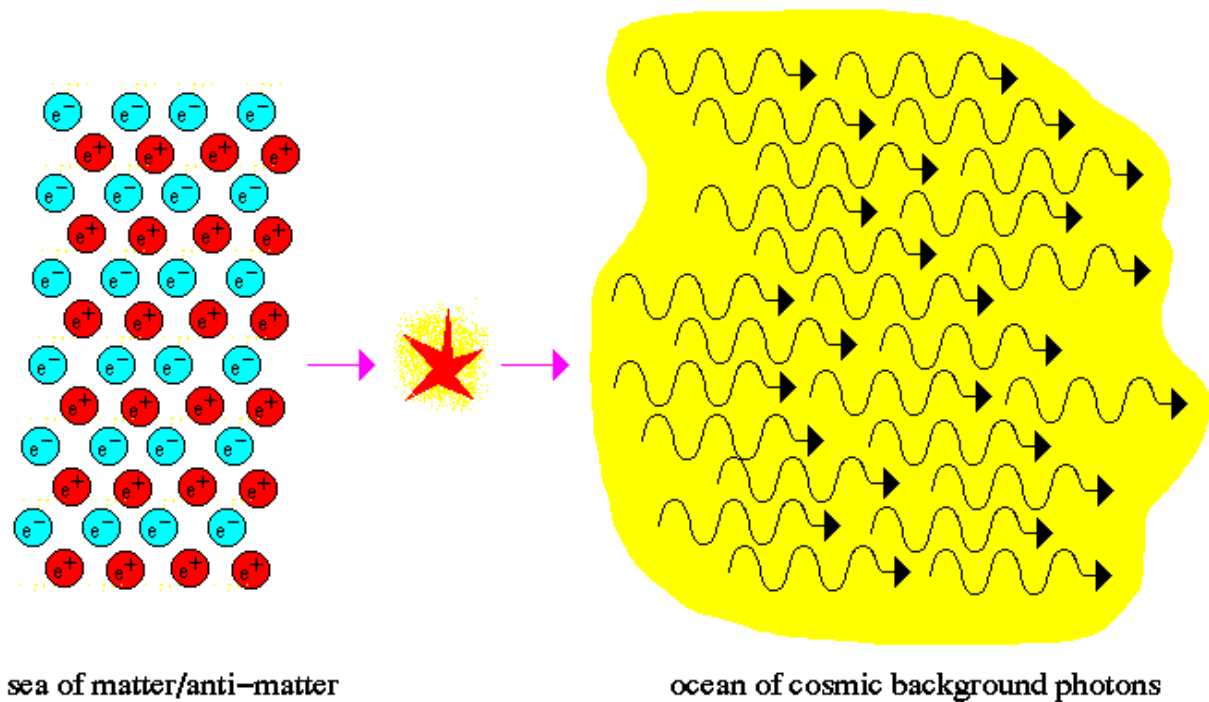
Notice that an equilibrium process keeps the number of matter and anti-matter particles equal.

Note that this process (and quark confinement) produces an equal number of matter and anti-matter particles, and that any particular time, if the process of pair production or quark confinement were to stop, then all matter and anti-matter would eventually collide and the Universe will be composed only of photons. In other words, since there are equal numbers of matter and anti-matter particles created by pair production, then why is the Universe made mostly of matter? Anti-matter is extremely rare at the present time, yet matter is very abundant.

This asymmetry is called the matter/anti-matter puzzle. Why if particles are created symmetrically as matter and anti-matter does matter dominate the Universe today. In theory, all the matter and anti-matter should have canceled out and the Universe should be a ocean of photons.

Annihilation

since matter and anti-matter are produced in equal numbers, the endresult will be a massive annihilation of all particles

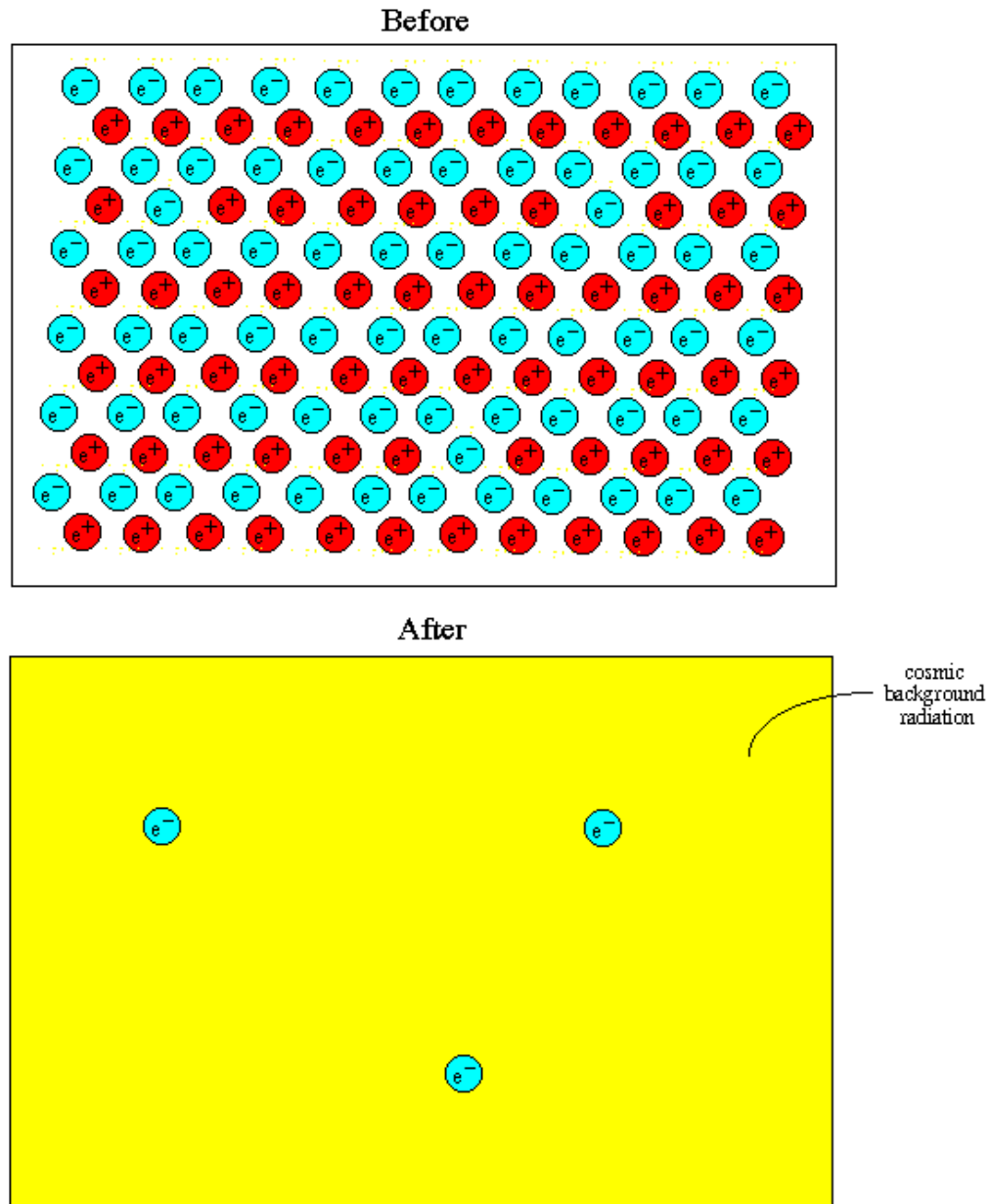


It is not the case that the Universe is only filled with photons (look around the room). And it is not the case that 1/2 the Universe is matter and the other half is anti-matter (there would be a lot of explosions).

Therefore, some mechanism produced more matter particles than anti-matter particles. How strong was this asymmetry? We can't go back in time and count the number of matter/anti-matter pairs, but we can count the number of cosmic background photons that remain after the annihilations. That counting yields a value of 1 matter particle for every 10^{10} photons, which means the asymmetry between matter and anti-matter was only 1 part in 10,000,000,000.

This means that for every 10,000,000,000 anti-matter particles there are 10,000,000,001 matter particles, an asymmetry of 1 particle out of 10 billion. And the endresult is that every 10 billion matter/anti-matter pairs annihilated each other leaving behind 1 matter particle and 10 billion photons that make up the cosmic background radiation, the echo of the Big Bang we measure today. This ratio of matter to photons is called the baryon number.

Baryon Number



Even though the baryon number is extremely small (10^{-10}) why isn't it zero? In Nature, there are only three natural numbers, 0, 1 and infinity. All other numbers require explanation. What caused the asymmetry of even one extra matter particle for every 10 billion matter/anti-matter pairs?

One answer is that the asymmetry occurs because the Universe is out of equilibrium. This is clearly true because the Universe is expanding, and a dynamic thing is out of equilibrium (only static things are stable). And there are particular points in the history of the Universe when the system is out of equilibrium, the symmetry breaking moments. Notice also that during the inflation era, any asymmetries in the microscopic world would be magnified into the macroscopic world. One such quantum asymmetry is CP violation.

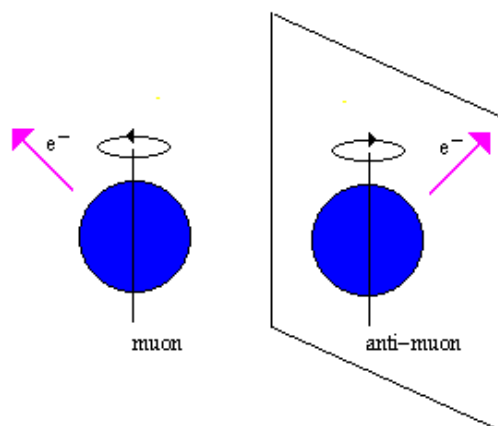
CP Violation:

As the Universe expands and cools and the process of creation and annihilation of matter/anti-matter pairs slows down. Soon matter and anti-matter has time to undergo other nuclear processes, such as nuclear decay. Many exotic particles, massive bosons or mesons, can undergo decay into smaller particles. If the Universe is out of equilibrium, then the decay process, fixed by the emergent laws of Nature, can become out of balance if there exists some asymmetry in the rules of particle interactions. This would result in the

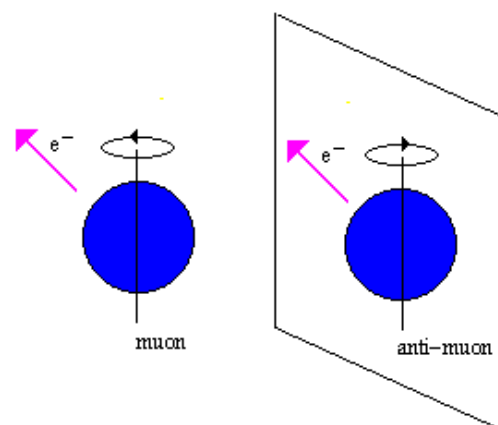
production of extra matter particles, rather than equal numbers of matter and anti-matter.

In the quantum world, there are large numbers of symmetric relationships. For example, there is the symmetry between matter and anti-matter. For every matter particle, there is a corresponding anti-matter particle of opposite charge. In the 1960's, it was found that some types of particles did not conserve left or right-handedness during their decay into other particles. This property, called [parity](#), was found to be broken in a small number of interactions at the same time the charge symmetry was also broken and became known as [CP violation](#).

Parity, for elementary particles, can be defined with respect to the direction in which they emit decay particles. A muon will typically emit an electron to the right. If the parity is reversed, then the muon should emit the electron to the left, the mirror image.



One time out of a thousand, an anti-muon will decay to the right, a violation of the charge-parity or CP rule.



The symmetry is restored when particle interactions are considered under the global CPT rule (charge - parity - time reversal), which states that that a particle and its anti-particle may be different, but will behave the same in a mirror-reflected, time-reversed study. During the inflation era, the rapid expansion of spacetime would have thrown the T in CPT symmetry out of balance, and the CP violation would have produced a small asymmetry in the baryon number.

This is another example of how quantum effects can be magnified to produce large consequences in the macroscopic world.



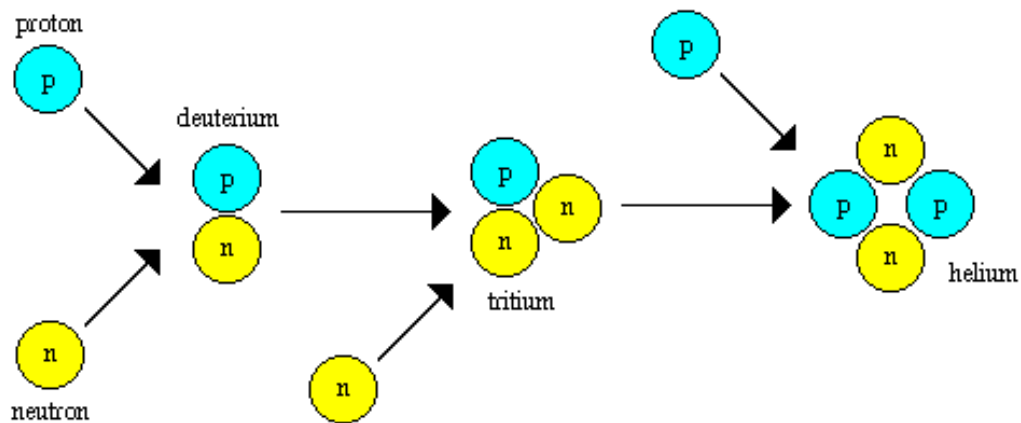


Nucleosynthesis:

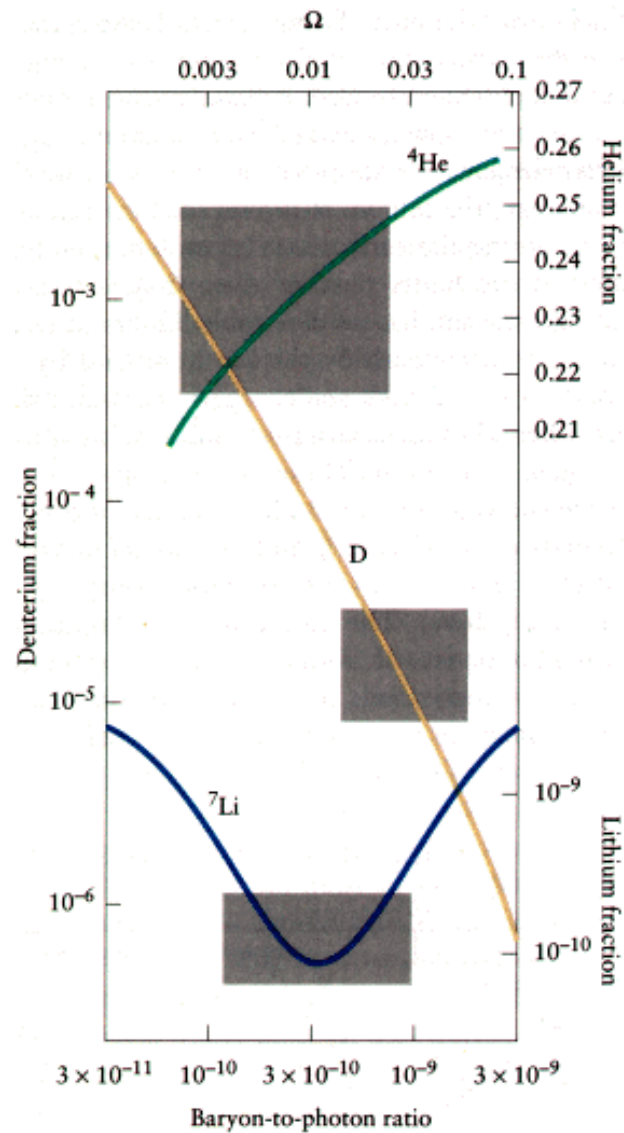
The Universe is now 1 minute old, and all the anti-matter has been destroyed by annihilation with matter. The leftover matter is in the form of electrons, protons and neutrons. As the temperature continues to drop, protons and neutrons can undergo fusion to form heavier atomic nuclei. This process is called nucleosynthesis.

Nucleosynthesis

as the Universe cools, protons and neutrons can fuse to form heavier atomic nuclei



It's harder and harder to make nuclei with higher masses. So the most common substance in the Universe is hydrogen (one proton), followed by helium, lithium, beryllium and boron (the first elements on the [periodic table](#)). [Isotopes](#) are formed, such as deuterium and tritium, but these elements are unstable and decay into free protons and neutrons.



Note that this above diagram refers to the density parameter, Ω , of baryons, which is close to 0.1. However, much of the Universe is in the form of dark matter (see later lecture).

A key point is that the ratio of hydrogen to helium is extremely sensitive to the density of matter in the Universe (the parameter that determines if the Universe is open, flat or closed). The higher the density, the more helium produced during the nucleosynthesis era. The current measurements indicate that 75% of the mass of the Universe is in the form of hydrogen, 24% in the form of helium and the remaining 1% in the rest of the periodic table (note that your body is made mostly of these 'trace' elements). Note that since helium is 4 times the mass of hydrogen, the number of hydrogen atoms is 90% and the number of helium atoms is 9% of the total number of atoms in the Universe.

Periodic Table of the Elements

1 H																	2 He																																		
3 Li	4 Be																	5 B	6 C	7 N	8 O	9 F	10 Ne																												
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar											19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr																
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe											55 Cs	56 Ba	57 *La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu							
87 Fr	88 Ra	89 +Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr											104 Rf	105 Ha	106 106	107 107	108 108	109 109	110 110	111 111	112 112											113 113	114 114	115 115	116 116	117 117	118 118

Naming conventions of new elements

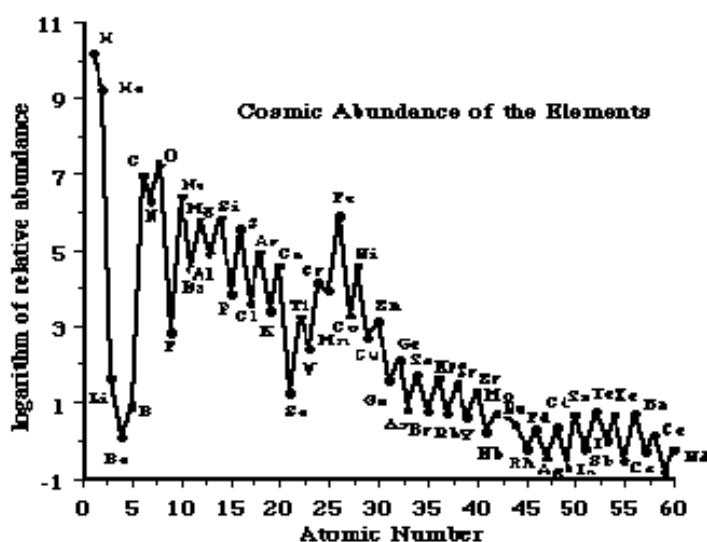
* Lanthanide Series
+ Actinide Series

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

There are over 100 naturally occurring elements in the Universe and classification makes up the periodic table. The very lightest elements are made in the early Universe. The elements between boron and iron (atomic number 26) are made in the cores of stars by thermonuclear fusion, the power source for all stars.

The fusion process produces energy, which keeps the temperature of a stellar core high to keep the reaction rates high. The fusing of new elements is balanced by the destruction of nuclei by high energy gamma-rays. Gamma-rays in a stellar core are capable of disrupting nuclei, emitting free protons and neutrons. If the reaction rates are high, then a net flux of energy is produced.

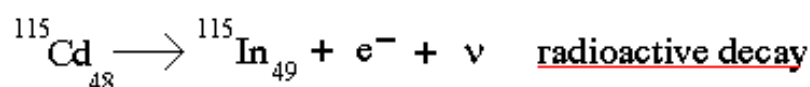
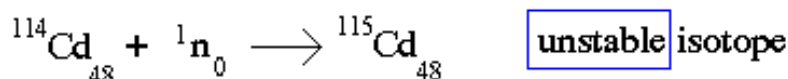
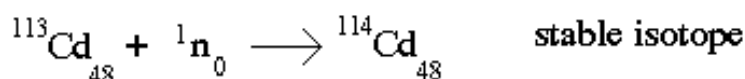
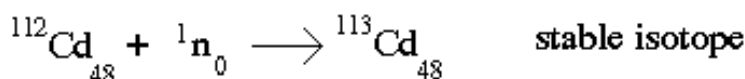
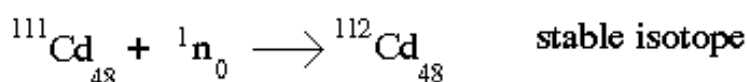
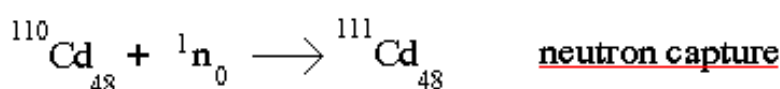
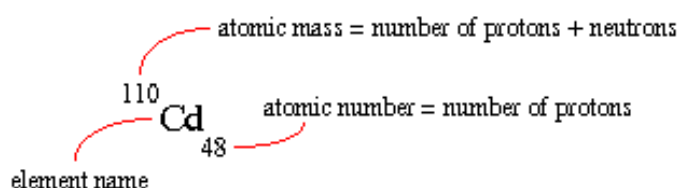
Fusion of elements with atomic numbers (the number of protons) greater than 26 uses up more energy than is produced by the reaction. Thus, elements heavier than iron cannot be fuel sources in stars. And, likewise, elements heavier than iron are not produced in stars, so what is their origin?.



The construction of elements heavier than involves nucleosynthesis by neutron capture. A nuclei can capture or fuse with a neutron because the neutron is electrically neutral and, therefore, not repulsed like the proton. In everyday life, free neutrons are rare because they have short [half-life's](#) before they [radioactively decay](#). Each neutron capture produces an [isotope](#), some are stable, some are unstable. Unstable isotopes will decay by emitting a positron and a neutrino to make a new element.

Nucleosynthesis by Neutron Capture

construction of elements beyond iron involves the capture of a neutron to produce isophotes. Unstable isotopes decay into new elements



Neutron capture can happen by two methods, the s and r-processes, where s and r stand for slow and rapid. The s-process happens in the inert carbon core of a star, the slow capture of neutrons. The s-process works as long as the decay time for unstable isotopes is longer than the capture time. Up to the element bismuth (atomic number 83), the s-process works, but above this point the more massive nuclei that can be built from bismuth are unstable.

The second process, the r-process, is what is used to produce very heavy, neutron rich nuclei. Here the capture of neutrons happens in such a dense environment that the unstable isotopes do not have time to decay. The high density of neutrons needed is only

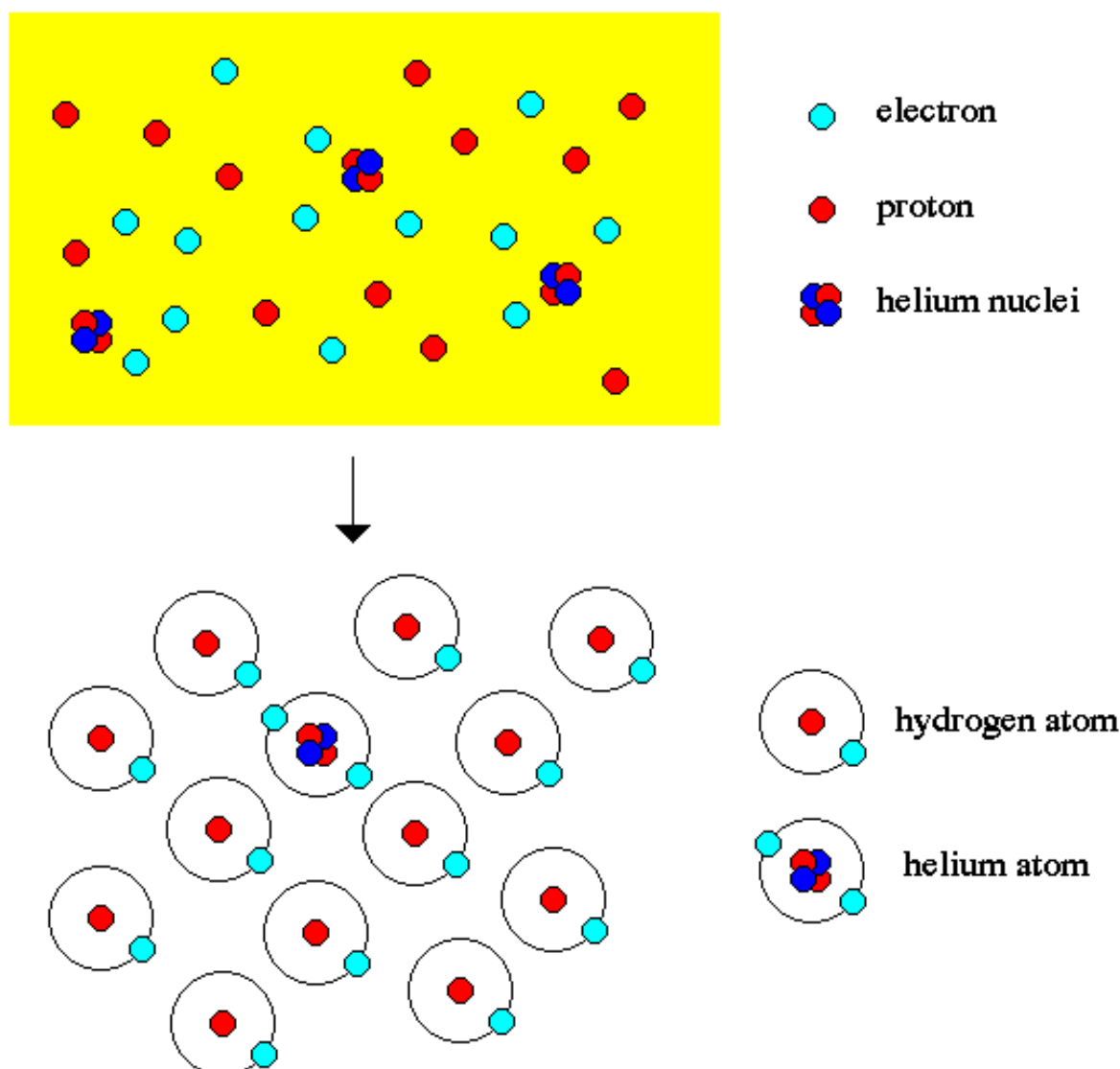
found during a supernova explosion and, thus, all the heavy elements in the Universe (radium, uranium and plutonium) are produced this way. The supernova explosion also has the side benefit of propelling the new created elements into space to seed molecular clouds which will form new stars and solar systems.

Ionization:

The last stage in matter production is when the Universe cools sufficiently for electrons to combine with the proton/neutron nuclei and form atoms. Constant impacts by photons knock electrons off of atoms which is called ionization. Lower temperatures mean photons with less energy and fewer collisions. Thus, atoms become stable at about 15 minutes after the Big Bang.

Recombination

As the Universe expands and cools, protons and electrons combine to form hydrogen (the most abundant element). And helium nuclei combine with electrons to form helium atoms. This process is called recombination.



These atoms are now free to bond together to form simple compounds, molecules, etc. And these are the building blocks for galaxies and stars.

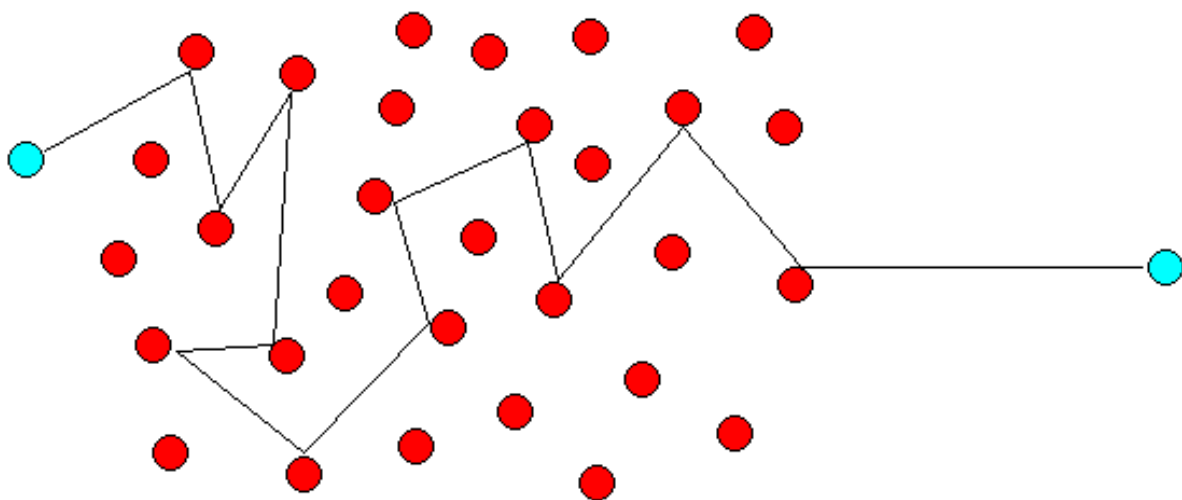
Radiation/Matter Dominance :

Even after the annihilation of anti-matter and the formation of protons, neutrons and electrons, the Universe is still a violent and extremely active environment. The photons created by the matter/anti-matter annihilation epoch exist in vast numbers and have energies at the x-ray level.

Radiation, in the form of photons, and matter, in the form of protons, neutrons and electron, can interact by the process of scattering. Photons bounce off of elementary particles, much like billiard balls. The energy of the photons is transferred to the matter particles. The distance a photon can travel before hitting a matter particle is called the mean free path.

Mean Free Path

all particles, including photons, suffer from collisions with other particles such that their path through space is very short the higher the densities. This typical path length is called the mean free path.



the Universe is opaque at high densities (the mean free path of a photon is very short), as the density drops with time, the Universe becomes transparent (the mean free path of a photon becomes very large).

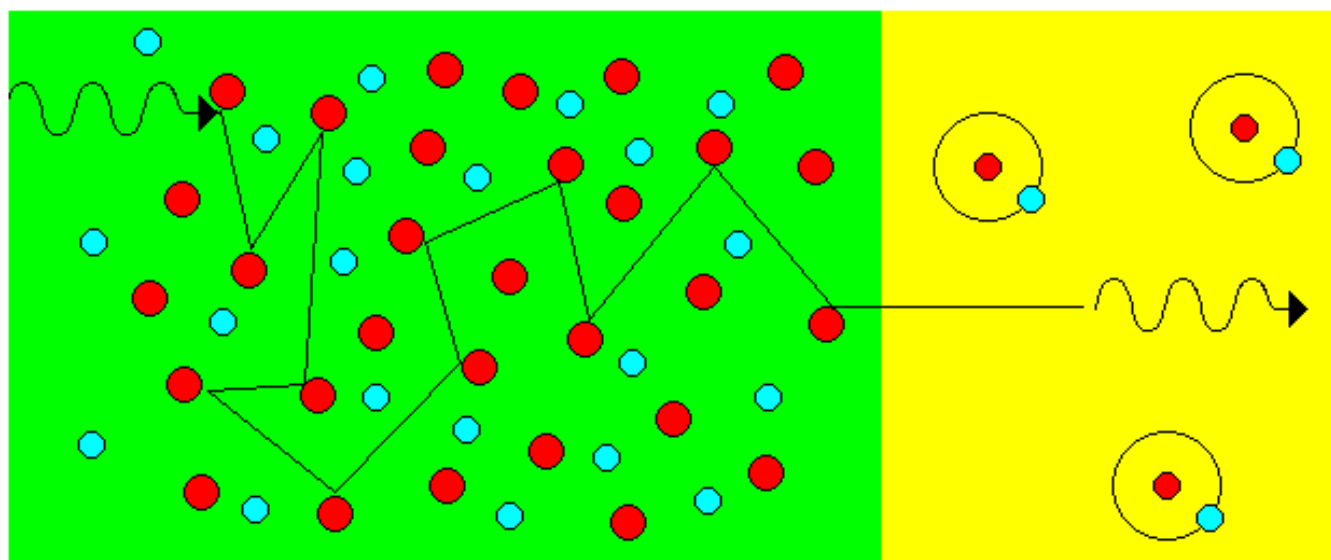
Since matter and photons were in constant contact, their temperatures were the same, a process called thermalization. Note also that the matter can not clump together by gravity. The impacts by photons keep the matter particles apart and smoothly distributed.

The density and the temperature for the Universe continues to drop as it expands. At some point about 15 minutes after the Big Bang, the temperature has dropped to the point where ionization no longer takes places. Neutral atoms can form, atomic nuclei surround by

electron clouds. The number of free particles drops by a large fraction (all the protons, neutrons and electron form atoms). And suddenly the photons are free to travel without collisions, this is called decoupling.

Last Scattering Epoch

As the Universe cooled, the free electrons and protons could finally bond together to form hydrogen atoms. At the same time, the Universe went from a rich plasma to a gas of neutral hydrogen.



hydrogen plasma

atomic hydrogen

In a plasma, the mean free path of a photon is very short. In a gas of atomic hydrogen, the mean free path is very long, as long as the size of the Universe. Thus, the transition from the early plasma to atomic hydrogen is the epoch of last scattering, the point in time when the photons became free to travel without hindrance.

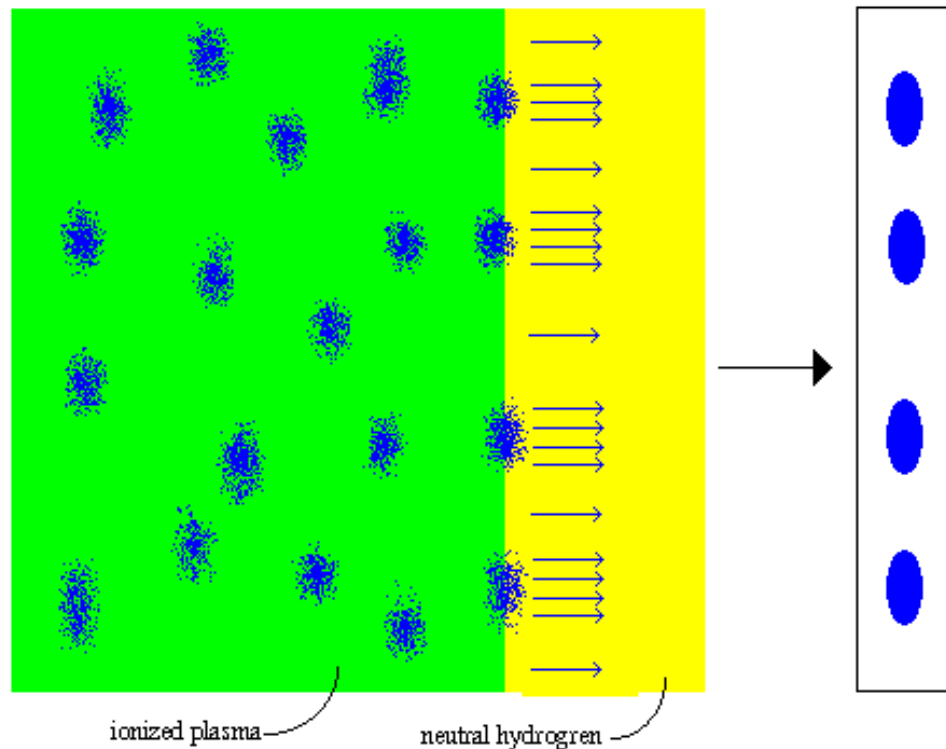
The Universe becomes transparent at this point. Before this epoch, a photon couldn't travel more than a few inches before a collision. So an observer's line-of-sight was only a few inches and the Universe was opaque, matter and radiation were coupled. This is the transition from the radiation era to the matter era.

Density Fluctuations:

The time of neutral atom construction is called recombination, this is also the first epoch we can observe in the Universe. Before recombination, the Universe was too dense and opaque. After recombination, photons are free to travel through all of space. Thus, the limit to our observable Universe is back in time (outward in space) to the moment of recombination.

Density Fluctuations

at recombination, density fluctuations (future galaxy seeds) emit an extra amount of cosmic background radiation. Irregularities in the CMB reflect density fluctuations from the early Universe



these fluctuations are the earliest observations possible. Earlier times are hidden behind the wall of ionized plasma at recombination

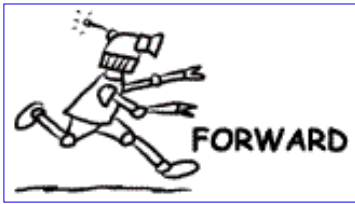
The time of recombination is also where the linked behavior between photons and matter decouples or breaks, and is also the last epoch where radiation traces the mass density. Photon/matter collisions become rare and the evolution of the Universe is dominated by the behavior of matter (i.e. gravity), so this time, and until today, is called the matter era.

Today, radiation in the form of photons have a very passive role in the evolution of the Universe. They only serve to illuminate matter in the far reaches of the Galaxy and other galaxies. Matter, on the other hand, is free to interact without being jostled by photons. Matter becomes the organizational element of the Universe, and its controlling force is gravity.

Notice that as the Universe ages it moves to more stable elements. High energy radiation (photons) are unstable in their interactions with matter. But, as matter condenses out of the cooling Universe, a more stable epoch is entered, one where the slow, gentle force of gravity dominates over the nuclear forces of earlier times.

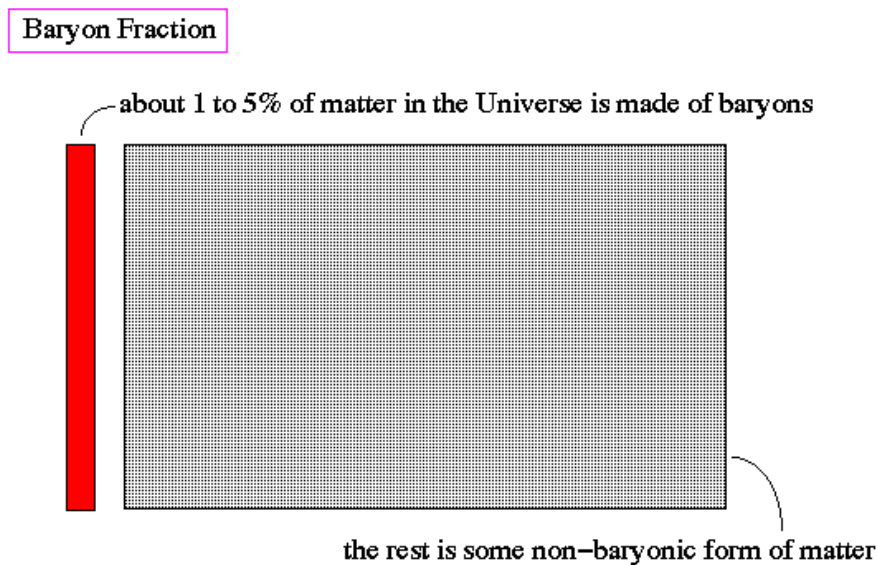
Much of the hydrogen that was created at recombination was used up in the formation of galaxies, and converted into stars. There is very little remnant hydrogen between galaxies, the so-called intergalactic medium, except in clusters of galaxies. Clusters of galaxies frequently have a hot hydrogen gas surrounding the core, this is leftover gas from

the formation of the cluster galaxies that has been heated by the motions of the cluster members.



Baryon Fraction:

The amount of hydrogen in the Universe today, either in stars and galaxies, or hot gas between galaxies, is called the baryon fraction. The current measurements indicate that the baryon fraction is about 3% (0.03) the value of closure for the Universe (the critical density). Remember the value from the abundance of light elements is 10% (0.10) the closure value.



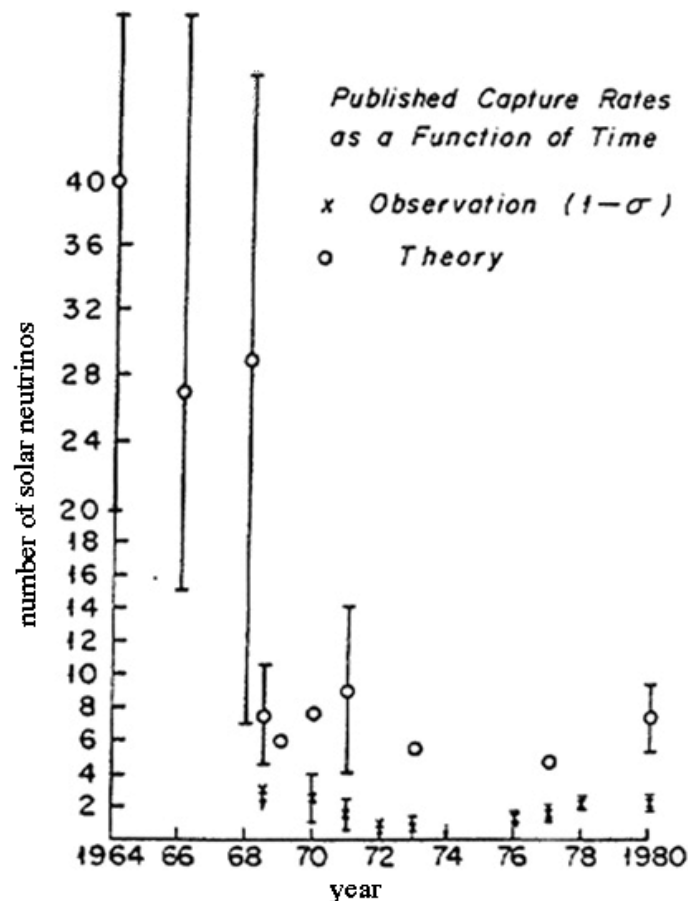
The most immediate result here is that the mass density of the Universe appears to be much less than the closure value, i.e. we live in an open Universe. However, the inflation model demands that we live in a Universe with exactly the critical density, Omega of 1. This can only be true if about 90% of the mass of the Universe is not in baryons.

Neutrinos :

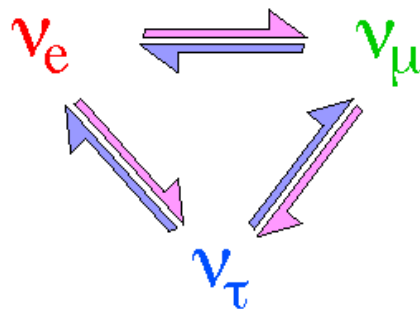
There are two types of leptons, the electron and the [neutrino](#). The neutrino is a strange particle, not discovered directly, but by inference from the decay of other particles by Wolfgang Pauli in 1930. It has no charge and a very small mass. It interacts with other particles only through the weak force (i.e. it is immune to the strong and electromagnetic forces). The weak force is so weak, that a neutrino can pass through several Earth's of lead with only a 50/50 chance of interacting with an atom, i.e. they are effectively transparent to matter.

The weakly interacting nature of neutrinos makes them very difficult to detect, and therefore measure, in experiments. Plus, the only sources of large amounts of neutrinos are high energy events such as supernova, relics from the early Universe and nuclear power plants. However, they are extremely important to our understanding of nuclear reactions since practically every fusion reaction produces a neutrino. In fact, a majority of the energy produced by stars and supernova are carried away in the form of neutrinos (the Sun produces 100 trillion trillion trillion neutrinos every second).

Detecting neutrinos from the Sun was an obvious first experiment to measure neutrinos. The pioneering experiment was Ray Davis's 600 tonne chlorine tank (actually dry cleaning fluid) in the Homestake mine, South Dakota. His experiment, begun in 1967, found evidence for only one third of the expected number of neutrino events. A light water Cherenkov experiment at Kamioka, Japan, upgraded to detect solar neutrinos in 1986, finds one half of the expected events for the part of the solar neutrino spectrum for which they are sensitive. Two recent gallium detectors (SAGE and GALLEX), which have lower energy thresholds, find about 60-70% of the expected rate.

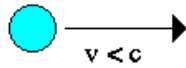


The clear trend is that the measured flux is found to be dramatically less than is possible for our present understanding of the reaction processes in stars. There are two possible answers to this problem: 1) The structure and constitution of stars, and hence the reaction mechanisms are not correctly understood (this would be a real blow for models that have otherwise been very successful), or 2) something happens to the neutrinos in transit to earth; in particular, they might change into another type of neutrino, call oscillation (this idea is not as crazy as it sounds, as a similar phenomenon is well known to occur with the meson particles). An important consequence to oscillation is that the neutrino must have mass (unlike the photon which has zero mass).

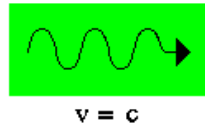


By the late 1990s, the oscillation hypothesis is shown to be correct. In addition, analysis of the neutrino events from the supernova 1987A indicates that the neutrinos traveled at slightly less than the speed of light. This is an important result since the neutrino is so light that it was unclear if its mass was very small or exact zero. Zero mass particles (like the photon) must travel exactly the speed of light (no faster, no slower). But objects with mass must travel at less than the speed of light as stated by special relativity.

if the neutrino has mass, then it is a particle that moves with a velocity near, but less than, the speed of light



if the neutrino is massless, then it acts like a photon and moves exactly at the speed of light since all massless objects must travel at only the speed of light



Since neutrinos interact very weakly, they are the first particles to decouple from other particles, at about 1 sec after the Big Bang. The early Universe is so dense that even neutrinos are trapped in their interactions. But as the Universe expands, its density drops to the point where the neutrinos are free to travel. This happens when the rate at which neutrinos are absorbed and emitted (the weak interaction rate) becomes slower than the expansion rate of the Universe. At this point the Universe expands faster than the neutrinos are absorbed and they take off into space (the expanding space).

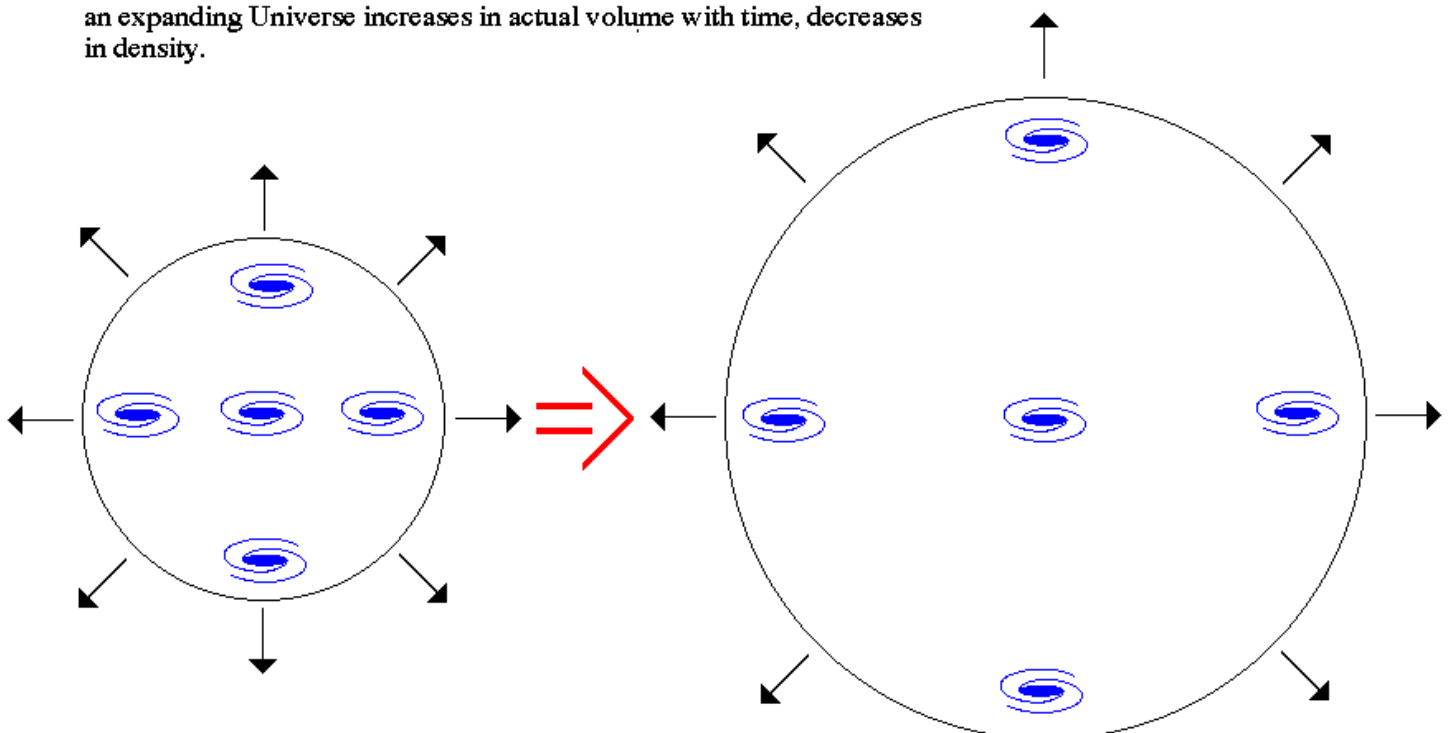
Now that neutrinos have been found to have mass, they also are important to our cosmology as a component of the cosmic density parameter. Even though each individual neutrino is much less massive than an electron, trillions of them are produced for every electron in the early Universe. Thus, neutrinos must make up some fraction of the non-baryonic matter in the Universe (although not a lot of it, see lecture on the large scale structure of the Universe).

Cosmic Background Radiation :

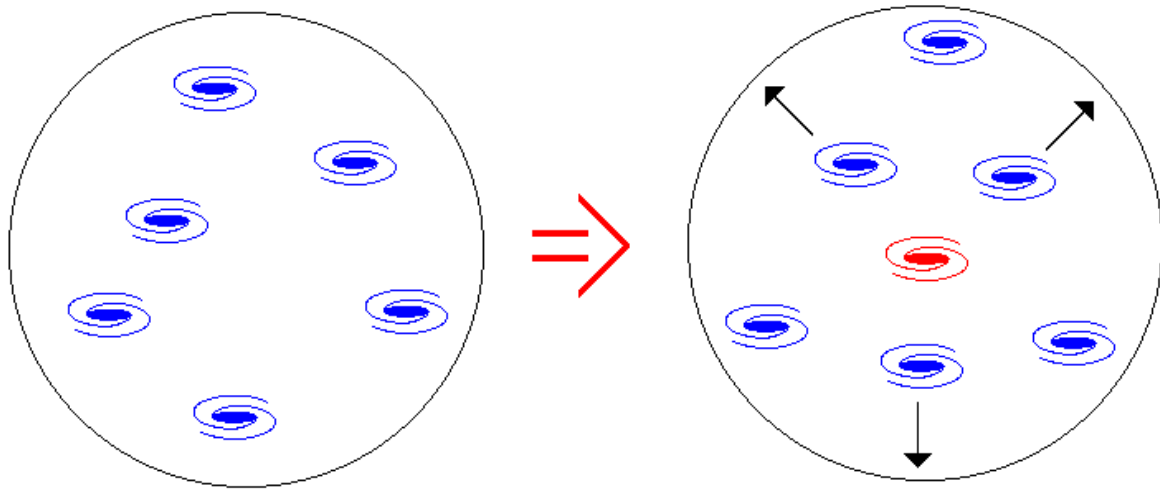
One of the foremost cosmological discoveries was the detection of the cosmic background radiation. The discovery of an expanding Universe by Hubble was critical to our understanding of the origin of the Universe, known as the [Big Bang](#). However, a dynamic Universe can also be explained by the [steady state theory](#).

The steady state theory avoids the idea of Creation by assuming that the Universe has been expanding forever. Since this would mean that the density of the Universe would get smaller and smaller with each passing year (and surveys of galaxies out to distant volumes shows this is not the case), the steady-state theory requires that new matter be produced to keep the density constant.

an expanding Universe increases in actual volume with time, decreases in density.



the steady state theory proposes that new matter is formed which pushes galaxies apart while keeping the density of the Universe constant



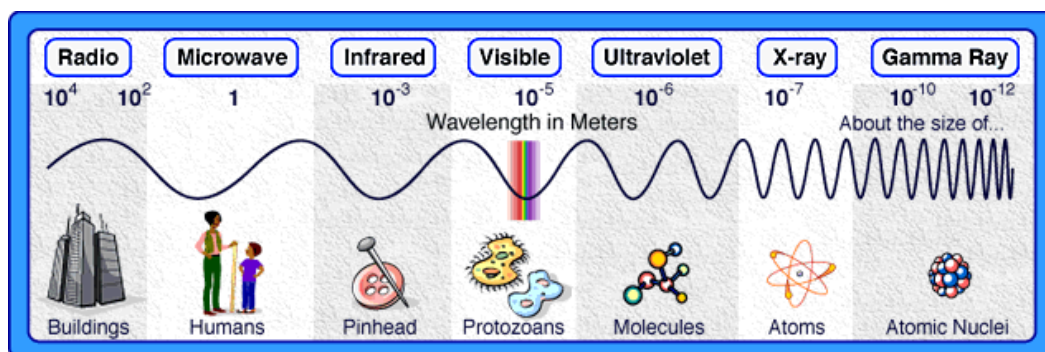
The creation of new matter would violate the conservation of matter principle, but the amount needed would only be one atom per cubic meter per 100 years to match the expansion rate given by Hubble's constant.

The discovery of the [cosmic microwave background \(CMB\)](#) confirmed the explosive nature to the origin of our Universe. For every matter particle in the Universe there are 10 billion more photons. This is the baryon number that reflects the asymmetry between matter and anti-matter in the early Universe. Looking around the Universe it's obvious that there is a great deal of matter. By the same token, there are even many, many more photons from the initial annihilation of matter and anti-matter.

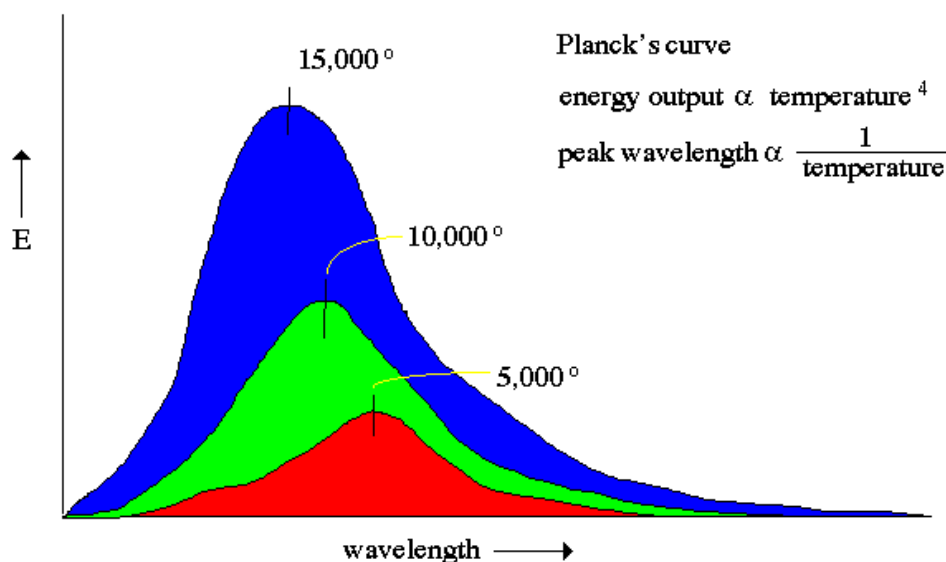
Most of the photons that you see with your naked eye at night come from the centers of stars. Photons created by nuclear fusion at the cores of stars then scatter their way out from a star's center to its surface, to shine in the night sky. But these photons only make up a very small fraction of the total number of photons in the Universe. Most photons in the Universe are cosmic background radiation, invisible to the eye.

Cosmic background photons have their origin at the matter/anti-matter annihilation era and, thus, were formed as gamma-rays. But, since then, they have found themselves scattering off particles during the radiation era. At recombination, these cosmic background photons escaped from the interaction with matter to travel freely through the Universe.

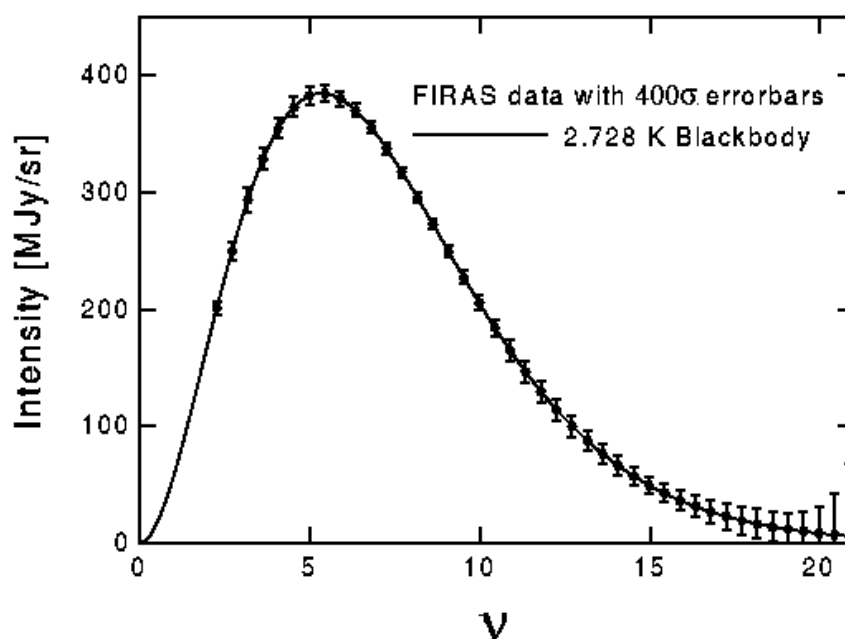
As the Universe continued to expand over the last 15 billion years, these cosmic background photons also 'expanded', meaning their wavelengths increased. The original gamma-ray energies of cosmic background photons have since cooled to microwave wavelengths. Thus, this microwave radiation that we see today is an 'echo' of the Big Bang.



The discovery of the cosmic microwave background (CMB) in the early 1960's was powerful confirmation of the Big Bang theory. Since the time of recombination, cosmic background photons have been free to travel uninhibited by interactions with matter. Thus, we expect their distribution of energy to be a perfect [blackbody](#) curve. A blackbody is the curve expected from a thermal distribution of photons, in this case from the thermalization era before recombination.



Today, based on space-based observations because the microwave region of the spectrum is blocked by the Earth's atmosphere, we have an accurate map of the CMB's energy curve. The peak of the curve represents the mean temperature of the CMB, 2.7 degrees about absolute zero, the temperature the Universe has dropped to 15 billion years after the Big Bang.



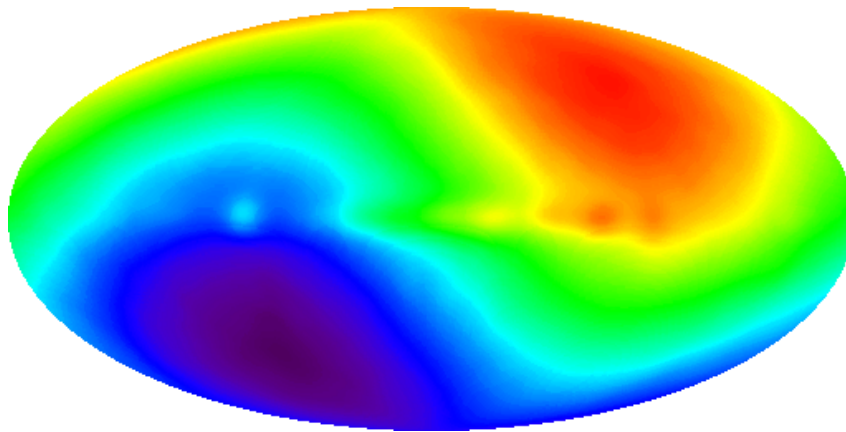
Where are the CMB photons at the moment? The answer is 'all around you'. CMB photons fill the Universe, and this lecture hall, but their energies are so weak after 15 billion years that they are difficult to detect without very sensitive microwave antennas.

CMB Fluctuations :

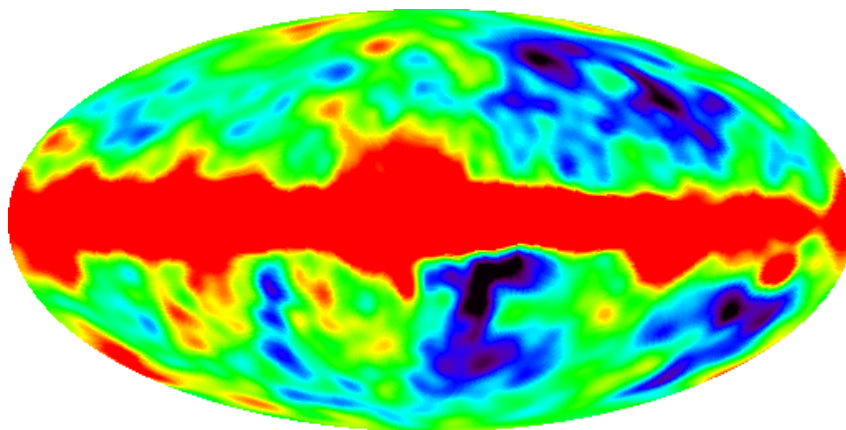
The CMB is highly isotropic, uniform to better than 1 part in 100,000. Any deviations from uniformity are measuring the fluctuations that grew by [gravitational instability](#) into galaxies and clusters of galaxies.

Images of the CMB are a full sky image, meaning that it looks like a map of the Earth unfolded from a globe. In this case, the globe is the celestial sphere and we are looking at a flat map of the sphere.

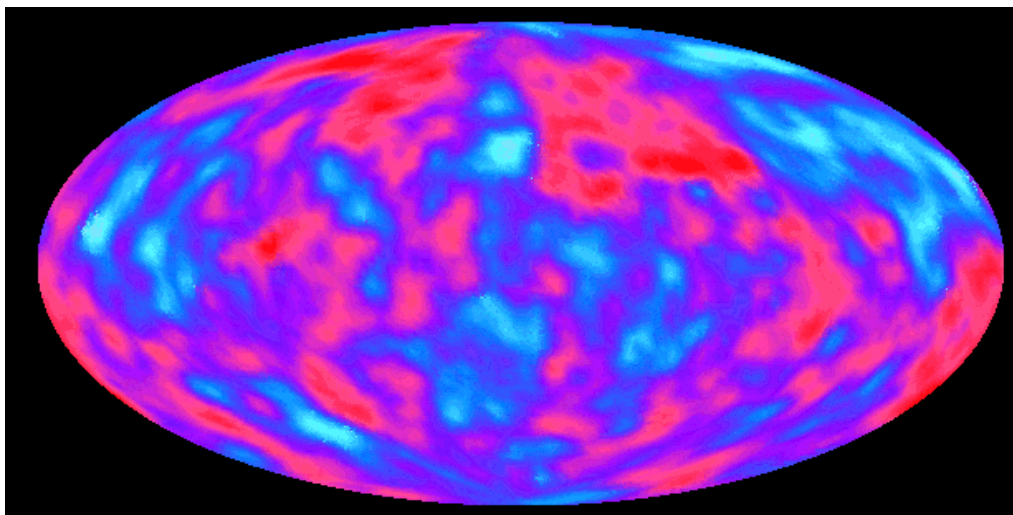
Maps of the CMB have to go through three stages of analysis to reveal the fluctuations associated with the early Universe. The raw image of the sky looks like the following, where red is hotter and blue is cooler:



The above image has a typical dipole appearance because our Galaxy is moving in a particular direction. The result is one side of the sky will appear [redshifted](#) and the other side of the sky will appear blueshifted. In this case, redshifting means the photons are longer in wavelength = cooler (so backwards from their name, they look blue in the above diagram). Removing the Galaxy's motion produces the following map:



This map is dominated by the far-infrared emission from gas in our own Galaxy. This gas is predominately in the plane of our Galaxy's disk, thus the dark red strip around the equator. The gas emission can be removed, with some assumptions about the distribution of matter in our Galaxy, to reveal the following map:



This CMB image is a picture of the last scattering epoch, i.e. it is an image of the moment when matter and photons decoupled, literally an image of the recombination wall. This is the last barrier to our observations about the early Universe, where the early epochs behind this barrier are not visible to us.

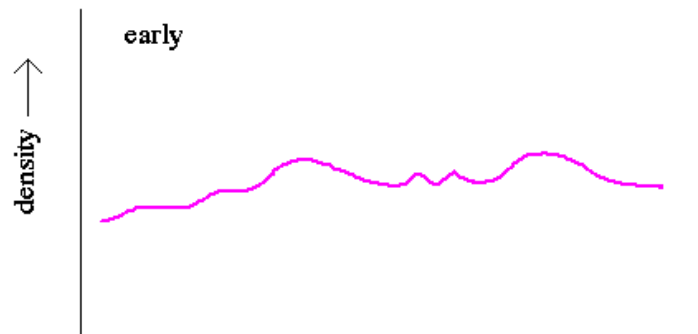
The clumpiness of the CMB image is due to fluctuations in temperature of the CMB photons. Changes in temperature are due to changes in density of the gas at the moment of recombination (higher densities equal higher temperatures). Since these photons are coming to us from the last scattering epoch, they represent fluctuations in density at that time.

The origin of these fluctuations are primordial quantum fluctuations from the very earliest moments of are echo'd in the CMB at recombination. Currently, we believe that these quantum fluctuations grew to greater than galaxy-size during the inflation epoch, and are the source of structure in the Universe.

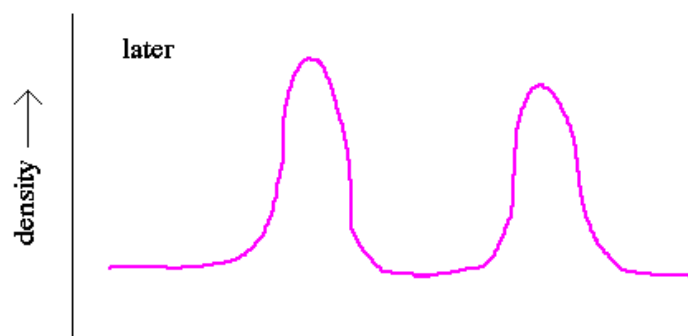
Fluctuations and the Origin of Galaxies :

The density fluctuations at recombination, as measured in the CMB, are too large and too low in amplitude to form galaxy sized clumps. Instead, they are the seeds for galaxy cluster-sized clouds that will then later break up into galaxies. However, in order to form cluster-sized lumps, they must grow in amplitude (and therefore mass) by gravitational instability, where the self-gravity of the fluctuation overcomes the gas pressure.

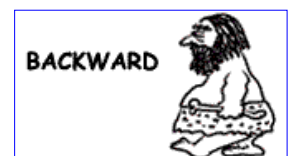
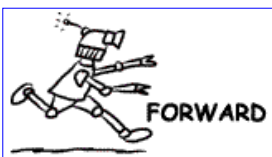
large scale fluctuations become gravitationally unstable and grow in amplitude



small scale fluctuations damp out with time



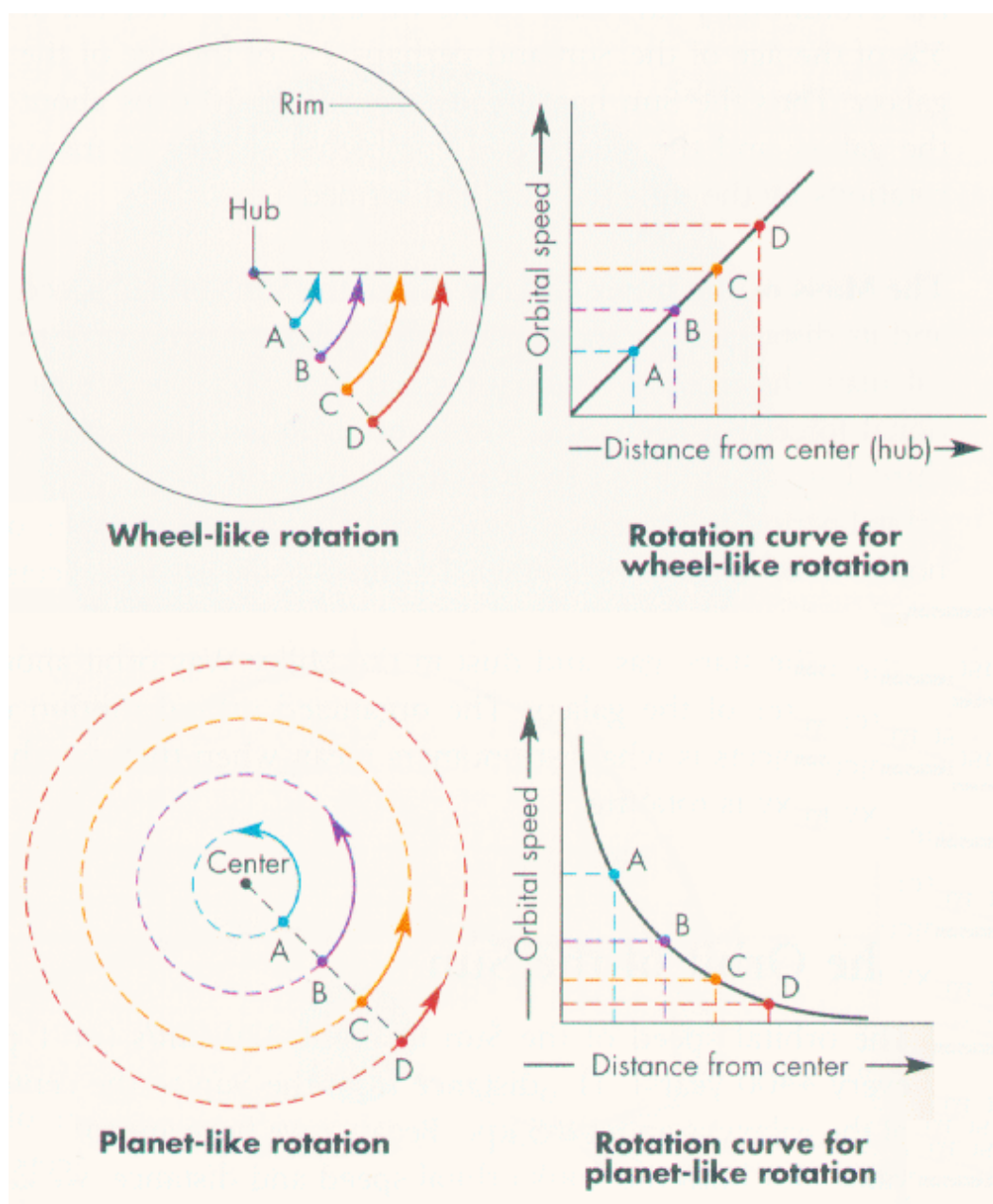
The CMB fluctuations are a link between Big Bang and the [large scale structure](#) of galaxies in the Universe, their distribution in terms of clusters of galaxies and filaments of galaxies that we observe around the Milky Way today.



Rotation Curve of Galaxy:

Dynamical studies of the Universe began in the late 1950's. This meant that instead of just looking and classifying galaxies, astronomers began to study their internal motions (rotation for disk galaxies) and their interactions with each other, as in clusters. The question was soon developed of whether we were observing the mass or the light in the Universe. Most of what we see in galaxies is starlight. So clearly, the brighter the galaxy, the more stars, therefore the more massive the galaxy. By the early 1960's, there were indications that this was not always true, called the missing mass problem.

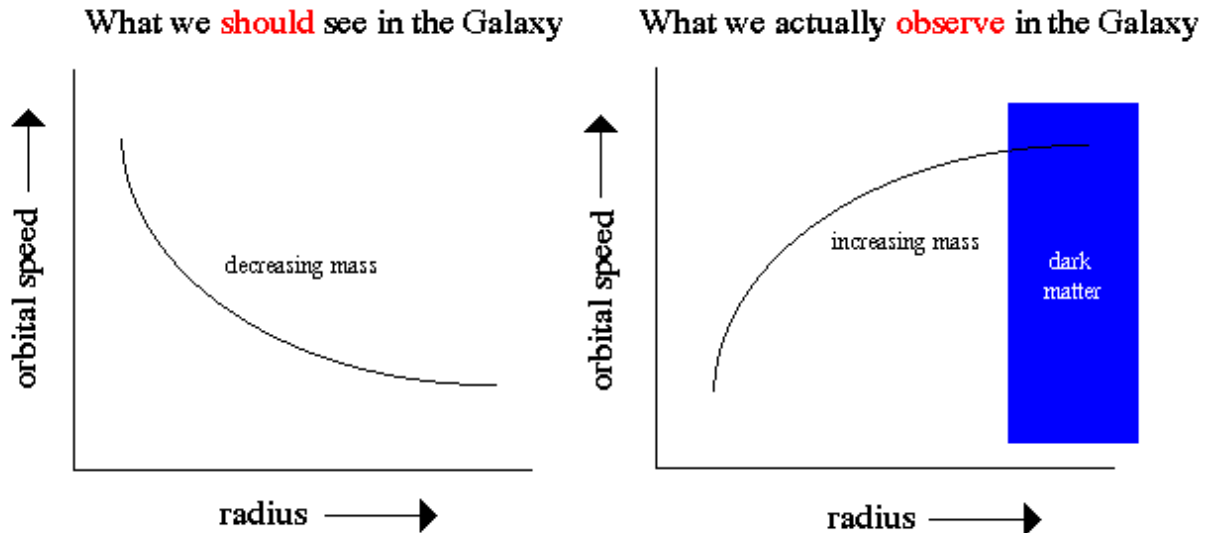
The first indications that there is a significant fraction of missing matter in the Universe was from studies of the rotation of our own Galaxy, the Milky Way. The orbital period of the Sun around the Galaxy gives us a mean mass for the amount of material inside the Sun's orbit. But, a detailed plot of the orbital speed of the Galaxy as a function of radius reveals the distribution of mass within the Galaxy. The simplest type of rotation is wheel rotation shown below.



Rotation following Kepler's 3rd law is shown above as planet-like or differential rotation. Notice that the orbital speeds falls off as you go to greater radii within the Galaxy. This is called a Keplerian rotation curve.

To determine the rotation curve of the Galaxy, stars are not used due to interstellar extinction. Instead, 21-cm maps of neutral hydrogen are used. When this is done, one finds that the [rotation curve of the Galaxy](#) stays flat out to large distances, instead of falling off as in the figure above. This means that the mass of the Galaxy increases with increasing distance from the center.

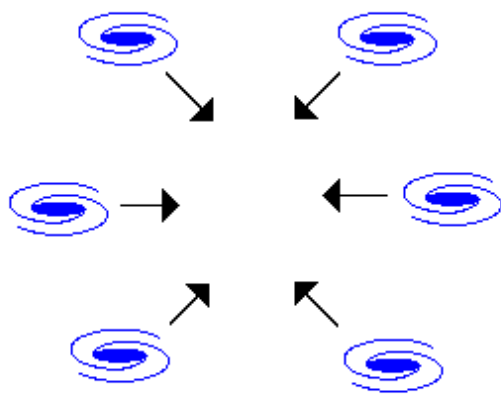
Rotation Curve of the Galaxy



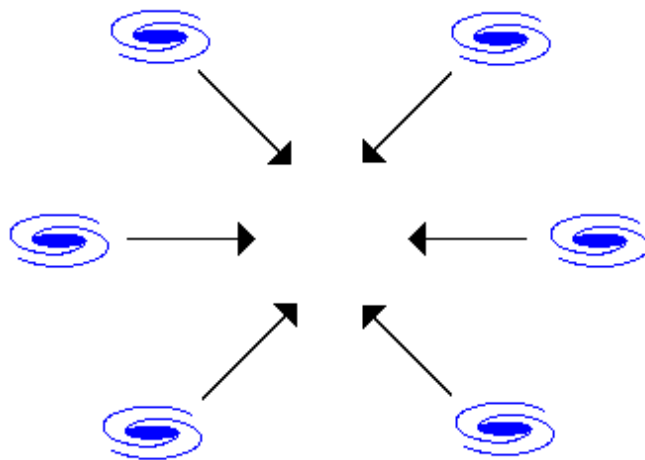
The surprising thing is there is very little visible matter beyond the Sun's orbital distance from the center of the Galaxy. So, the rotation curve of the Galaxy indicates a great deal of mass, but there is no light out there. In other words, the halo of our Galaxy is filled with a mysterious dark matter of unknown composition and type.

Cluster Masses:

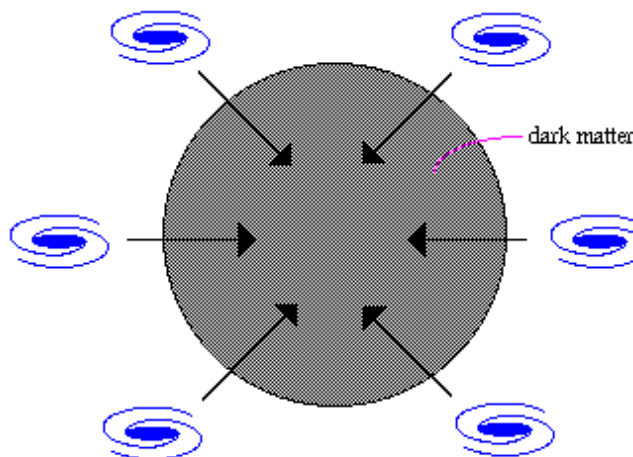
Most galaxies occupy groups or clusters with membership ranging from 10 to hundreds of galaxies. Each cluster is held together by the gravity from each galaxy. The more mass, the higher the velocities of the members, and this fact can be used to test for the presence of unseen matter.



The members of a cluster of galaxies move because of their mutual gravitational attraction.

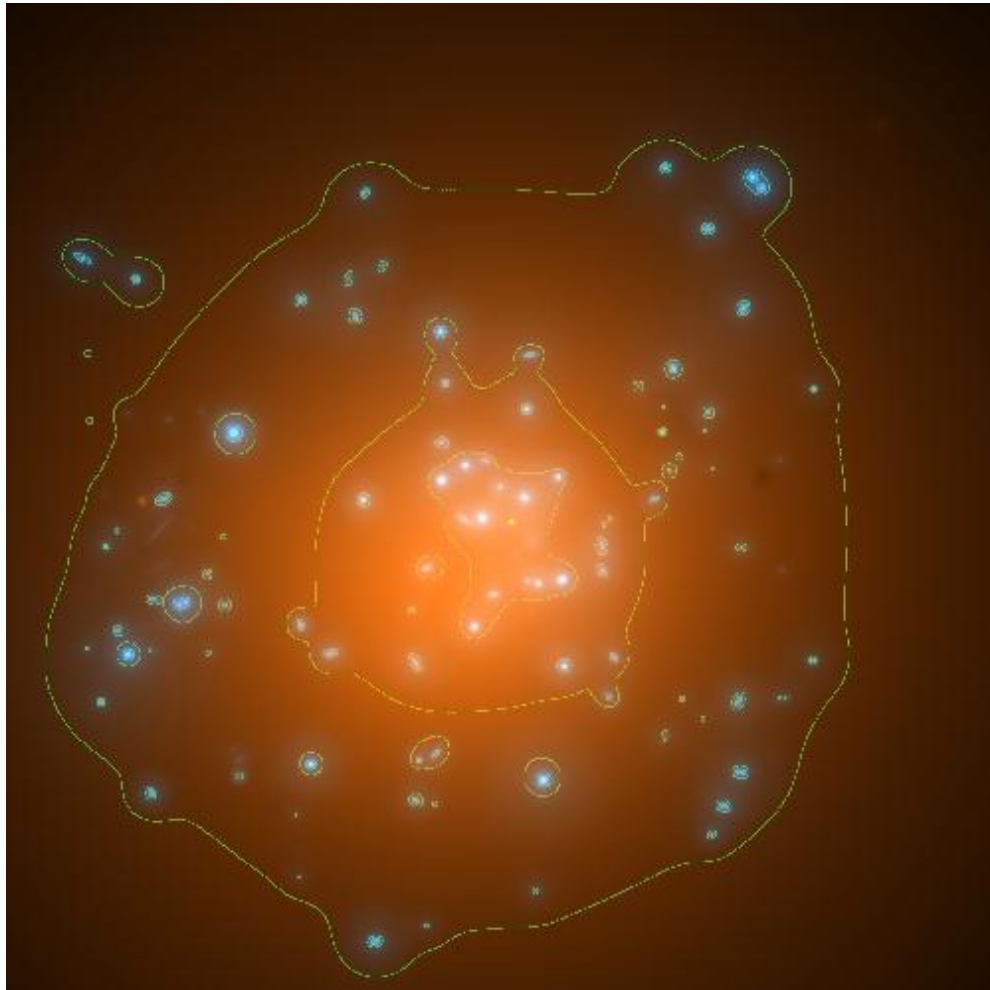


In most clusters, the velocity of the cluster galaxies is much higher than can be accounted for from the individual galaxy masses.



The result is there must be an unseen core of dark matter attracting the galaxies with more gravity and, therefore, more velocity.

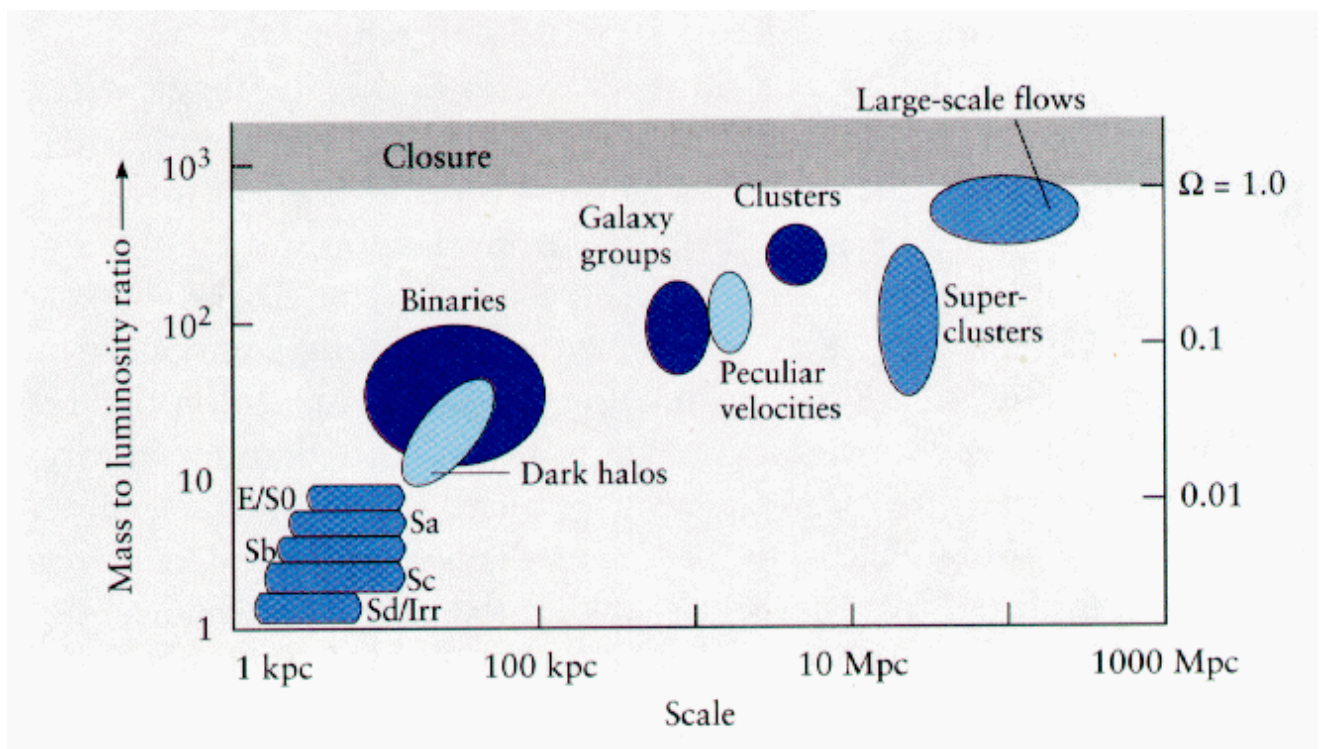
When these measurements were performed, it was found that up to 95% of the mass in clusters is not seen, i.e. dark. Since the physics of the motions of galaxies is so basic (pure Newtonian physics), there is no escaping the conclusion that a majority of the matter in the Universe has not been identified, and that the matter around us that we call 'normal' is special. The question that remains is whether dark matter is baryonic (normal) or a new substance, non-baryonic.



Mass-to-Luminosity Ratios:

Exactly how much of the Universe is in the form of dark matter is a mystery and difficult to determine, obviously because it's not visible. It has to be inferred by its gravitational effects on the luminous matter in the Universe (stars and gas) and is usually expressed as the mass-to-luminosity ratio (M/L). A high M/L indicates lots of dark matter, a low M/L indicates that most of the matter is in the form of baryonic matter, stars and stellar remnants plus gas.

An important point to the study of dark matter is how it is distributed. If it is distributed like the luminous matter in the Universe, that most of it is in galaxies. However, studies of M/L for a range of scales shows that dark matter becomes more dominant on larger scales.



Most importantly, on very large scales of 100 Mpc's (Mpc = megaparsec, one million [parsecs](#) and kpc = 1000 parsecs) the amount of dark matter inferred is near the value needed to close the Universe. Thus, it is for two reasons that the dark matter problem is important, one to determine what is the nature of dark matter, is it a new form of undiscovered matter? The second is to determine if the amount of dark matter is sufficient to close the Universe.

Baryonic Dark Matter:

We know of the presence of dark matter from dynamical studies. But we also know from the abundance of light elements that there is also a problem in our understanding of the fraction of the mass of the Universe that is in normal matter or baryons. The fraction of light elements (hydrogen, helium, lithium, boron) indicates that the density of the Universe in baryons is only 2 to 4% what we measure as the observed density.

It is not too surprising to find that at least some of the matter in the Universe is dark since it requires energy to observe an object, and most of space is cold and low in energy. Can dark matter be some form of normal matter that is cold and does not radiate any energy? For example, dead stars?

Once a normal star has used up its hydrogen fuel, it usually ends its life as a white dwarf star, slowly cooling to become a black dwarf. However, the timescale to cool to a black dwarf is thousands of times longer than the age of the Universe. High mass stars will explode and their cores will form neutron stars or black holes. However, this is rare and we would need 90% of all stars to go supernova to explain all of the dark matter.

Baryonic Dark Matter Candidates

	neutron stars	<u>problems</u> Universe too young
	black holes	how formed?
	black dwarf stars	Universe too young
	brown dwarf stars	not seen nearby
	planets	not seen nearby
	rocks	not seen nearby

Another avenue of thought is to consider low mass objects. Stars that are very low in mass fail to produce their own light by thermonuclear fusion. Thus, many, many brown dwarf stars could make up the dark matter population. Or, even smaller, numerous Jupiter-sized planets, or even plain rocks, would be completely dark outside the illumination of a star. The problem here is that to make-up the mass of all the dark matter requires huge numbers of brown dwarfs, and even more Jupiter's or rocks. We do not find many of these objects nearby, so to presume they exist in the dark matter halos is unsupported.

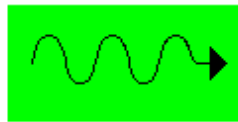
Non-Baryonic Dark Matter:

An alternative idea is to consider forms of dark matter not composed of quarks or leptons, rather made from some exotic material. If the neutrino has mass, then it would make a good dark matter candidate since it interacts weakly with matter and, therefore, is very hard to detect. However, neutrinos formed in the early Universe would also have mass, and that mass would have a predictable effect on the cluster of galaxies, which is not seen.

Another suggestion is that some new particle exists similar to the neutrino, but more massive and, therefore, more rare. This Weakly Interacting Massive Particle (WIMP) would escape detection in our modern particle accelerators, but no other evidence of its existence has been found.

Non-Baryonic Dark Matter Candidates

problems



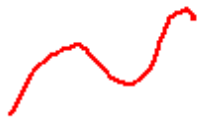
massive neutrino

they exist, but very low mass



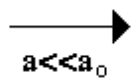
Weakly Interacting Massive Particles (WIMPS)

little to no evidence of their existence



cosmic strings

little to no evidence of their existence



modified gravity

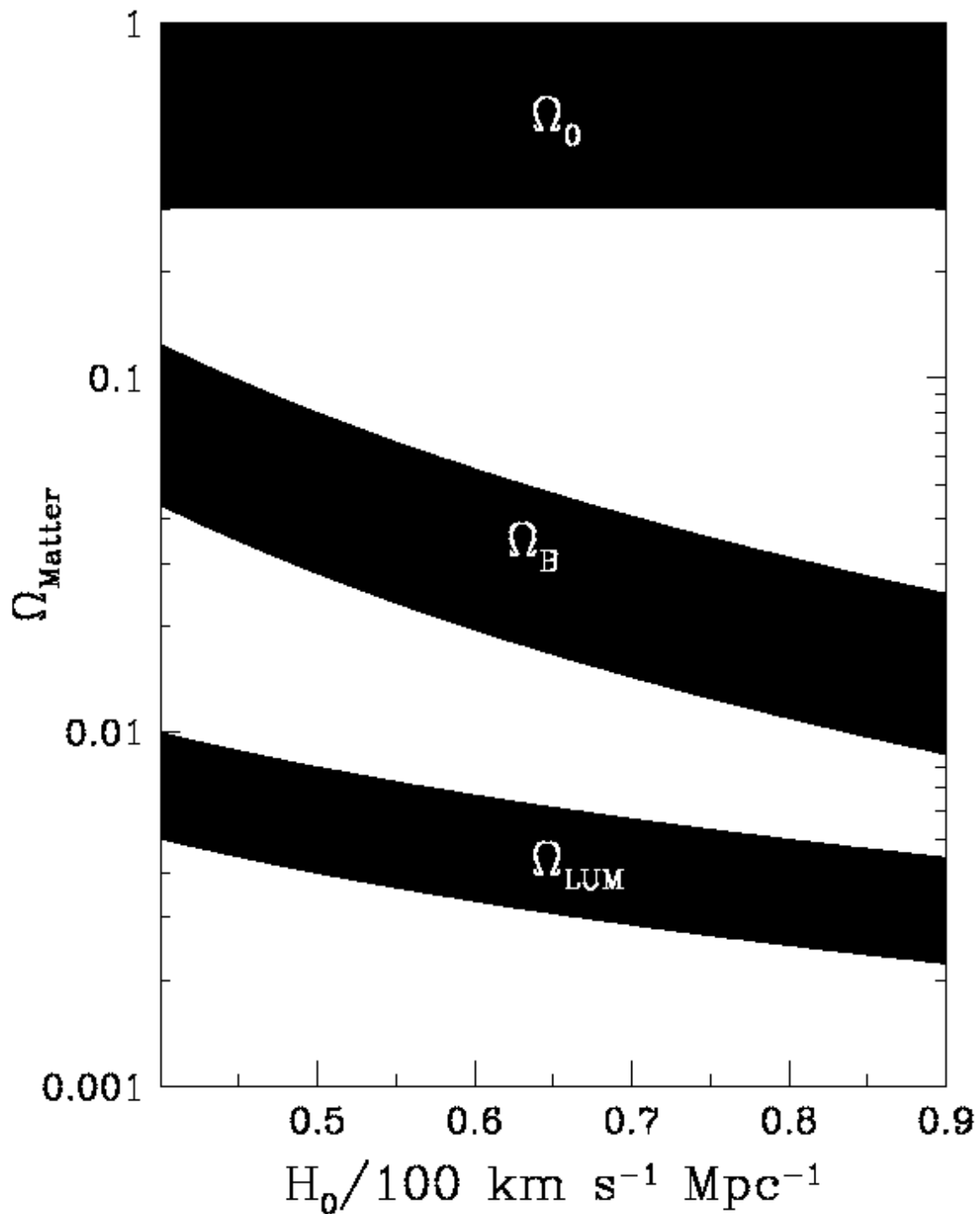
little to no evidence of their existence

The more bizarre proposed solutions to the dark matter problem require the use of little understood relics or defects from the early Universe. One school of thought believes that topological defects may have appeared during the phase transition at the end of the GUT era. These defects would have had a string-like form and, thus, are called cosmic strings. Cosmic strings would contain the trapped remnants of the earlier dense phase of the Universe. Being high density, they would also be high in mass but are only detectable by their gravitational radiation.

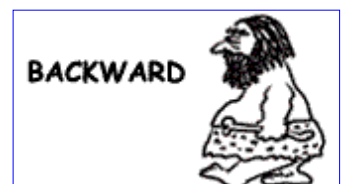
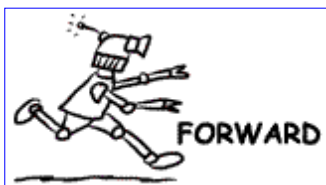
Lastly, the dark matter problem may be an illusion. Rather than missing matter, gravity may operate differently on scales the size of galaxies. This would cause us to overestimate the amount of mass, when it is the weaker gravity to blame. This is no evidence of modified gravity in our laboratory experiments to date.

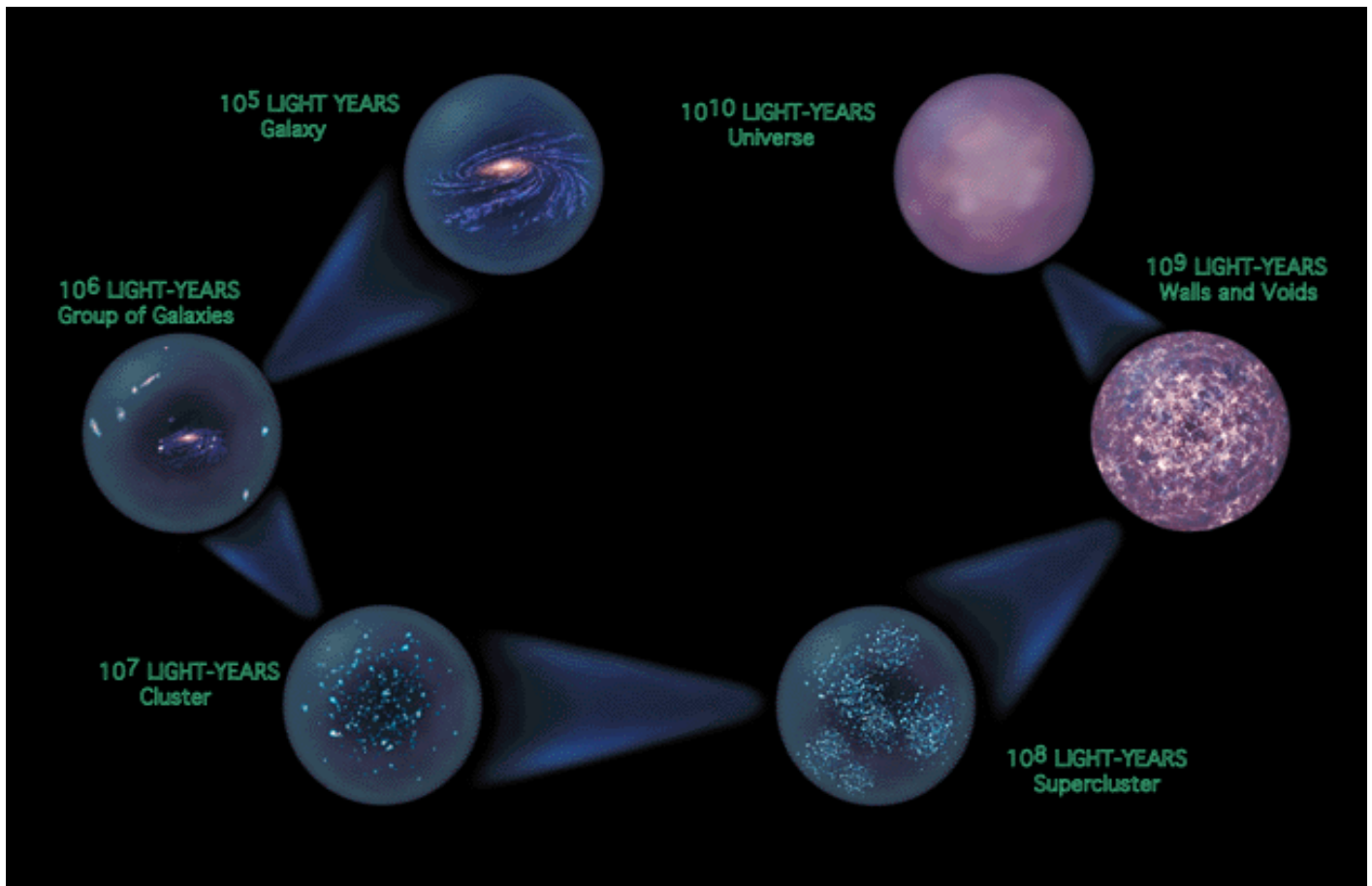
Current View of Dark Matter:

The current observations and estimates of dark matter is that 1/2 of dark matter is probably in the form of massive neutrinos, even though that mass is uncertain. The other 1/2 is in the form of stellar remnants and low mass, brown dwarfs. However, the combination of both these mixtures only makes 10 to 20% the amount mass necessary to close the Universe. Thus, the Universe appears to be open.



Summary of knowledge of Ω . The lowest band is luminous matter, in the form of bright stars and related material; the middle band is the big-bang nucleosynthesis determination of the density of baryons; the upper region is the estimate of Ω_{Matter} based upon the peculiar velocities of galaxies. The gaps between the bands illustrate the two dark matter problems: most of the ordinary matter is dark and most of the matter is nonbaryonic.





Origin of Structure :

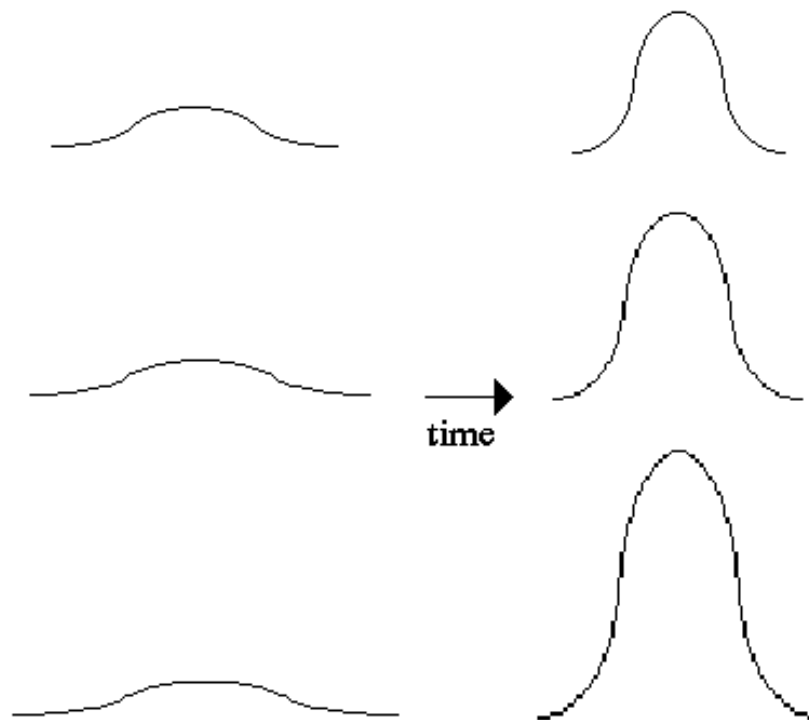
As we move forward in time from the beginning of the Universe we pass through the inflation era, baryogenesis, nucleosynthesis and radiation decoupling. The culmination is the formation of the structure of matter, the distribution of galaxies in the Universe.

During radiation era growth of structure is suppressed by the tight interaction of photons and matter. Matter was not free to respond to its own gravitational force, so density enhancements from the earliest times could not grow.

Density enhancements at the time of recombination (having their origin in quantum fluctuations that expanded to galaxy-sized objects during the inflation era) have two routes to go. They can grow or disperse.

Fluctuation Growth

small scale, low mass density enhancements disperse over time due to pressure effects



large mass density enhancements grow due to the attractive force of gravity which overcomes outward pressure effects

The 'pressure effects' that density enhancements experience are due to the expanding Universe. The space itself between particles is expanding. So each particle is moving away from each other. Only if there is enough matter for the force of gravity to overcome the expansion do density enhancements collapse and grow.

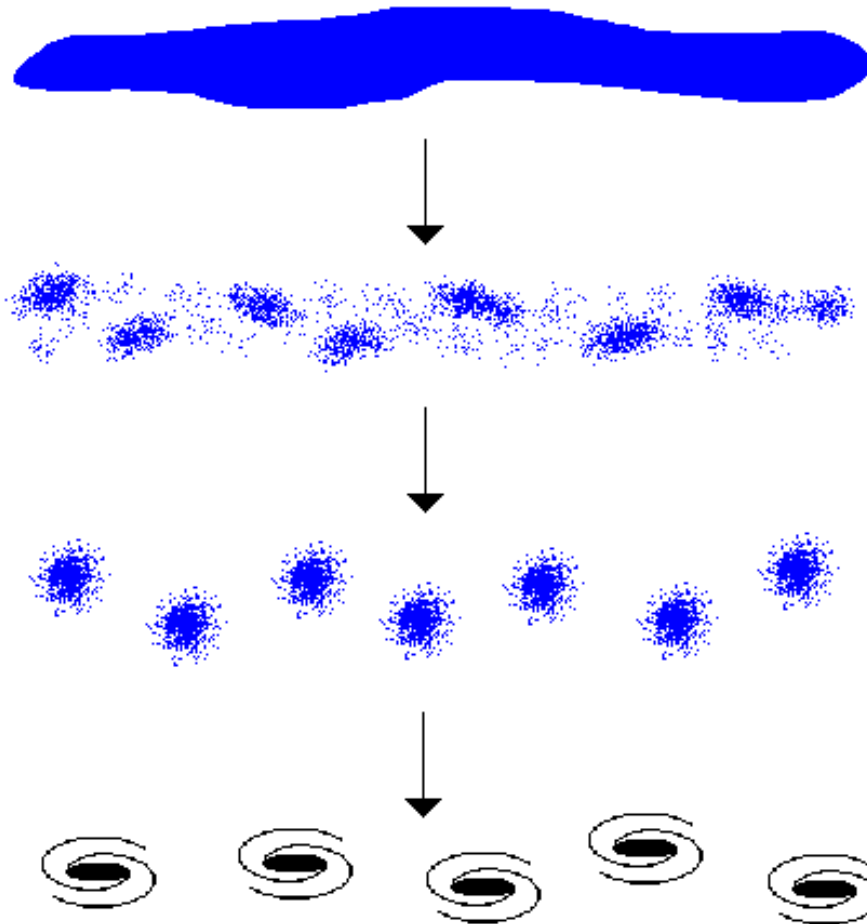
Top-Down Scenario:

Structure could have formed in one of two sequences: either large structures the size of galaxy clusters formed first, then latter fragmented into galaxies, or dwarf galaxies formed first, then merged to produce larger galaxies and galaxy clusters.

The former sequence is called the top-down scenario, and is based on the principle that radiation smoothed out the matter density fluctuations to produce large [pancakes](#). These pancakes accrete matter after recombination and grow until they collapse and fragment into galaxies.

Top-Down Structure Formation

in a top-down scenario, large pancakes of matter form first, then fragment into galaxy-sized lumps



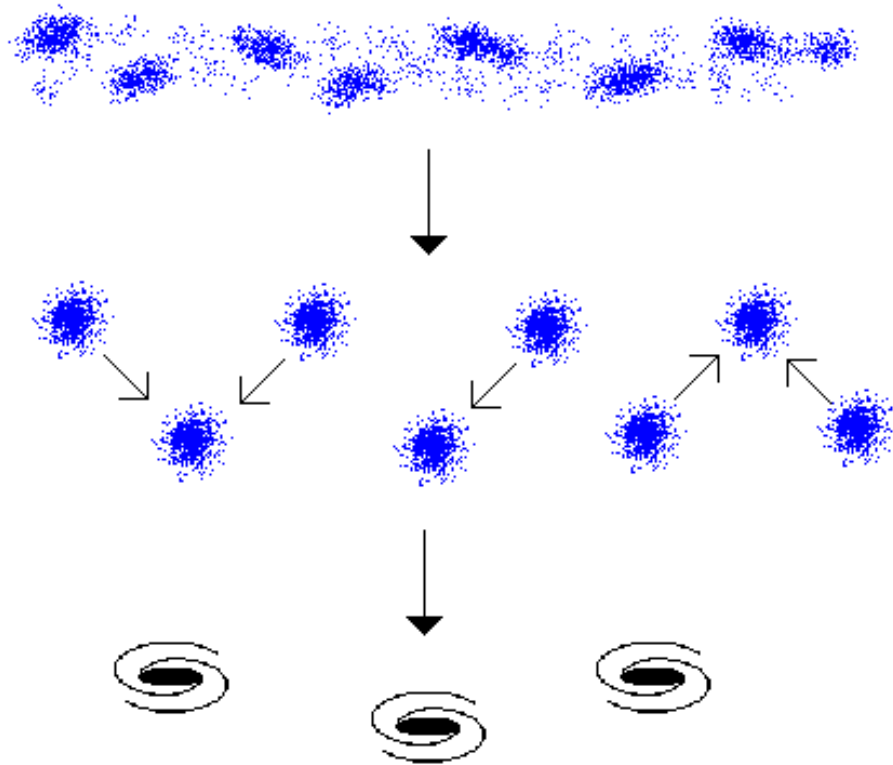
This scenario has the advantage of predicting that there should be large sheets of galaxies with low density voids between the sheets. Clusters of galaxies form where the sheets intersect.

Bottom-Up Scenario:

The competing scenario is one where galaxies form first and merge into clusters, called the bottom-up scenario. In this scenario, the density enhancements at the time of recombination were close to the size of small galaxies today. These enhancements collapsed from self-gravity into dwarf galaxies.

Bottom-Up Structure Formation

in a bottom-up scenario, small, dwarf galaxy-sized lumps form first, then merge to make galaxies and clusters of galaxies



Once the small galaxies are formed, they attract each other by gravity and [merge](#) to form larger galaxies. The galaxies can then, by gravity, cluster together to form filaments and clusters. Thus, gravity is the mechanism to form larger and larger structures.

Hot Dark Matter vs. Cold Dark Matter :

Each scenario of structure formation has its own predictions for the appearance of the Universe today. Both require a particular form for dark matter, a particular type of particle that makes up the 90% of the Universe not visible to our instruments. These two forms of dark matter are called Hot and Cold.

Hot Dark Matter (HDM)

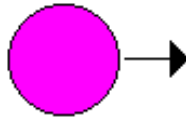
top-down scenarios require that dark matter be composed of a weakly interacting, high velocity particle



a massive neutrino is a good candidate for an HDM particle

Cold Dark Matter (CDM)

bottom-up scenarios require that dark matter be composed of a highly massive, slow moving particles



note that neither of these particles are baryons, the ordinary matter makes up stars or planets

HDM produces large, smooth features since it travels at high velocity. Massive neutrinos move at near the speed of light, yet interact very weakly with matter so can serve to smooth out large density enhancements.

CDM, on the other hand, is slow moving and, therefore, clumps into small regions. Large scale features are suppressed since the small clumps grow to form small galaxies.

There is strong evidence that galaxies formed before clusters, in the sense that the stars in galaxies are 10 to 14 billion years old, but many clusters of galaxies are still forming today. This would rule against the top-down scenario and support the bottom-up process.



Large Scale Structure :

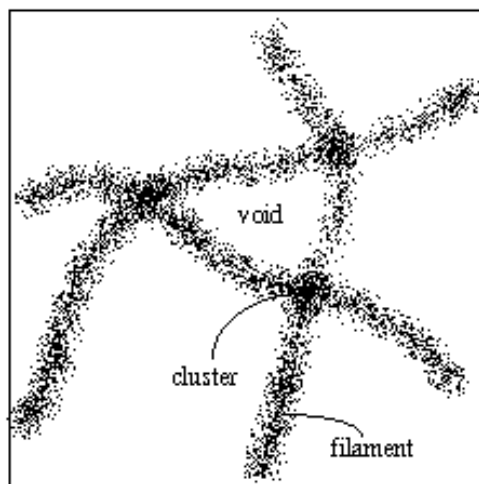
Galaxies in the Universe are not distributed evenly, i.e. like dots in a grid. Surveys of

galaxy positions, e.g. [maps of galaxies](#), have shown that galaxies have large scale structure in terms of clusters, filaments and voids.

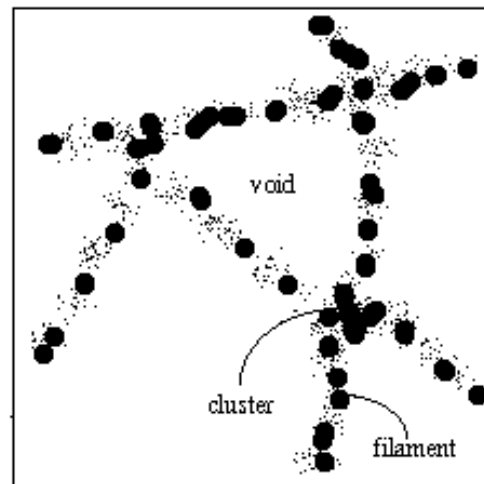
The clusters, filaments and voids reflect the initial fluctuations at recombination, plus any further evolution as predicted by HDM or CDM models. CDM and HDM models have particular predictions that can be tested by maps or redshift surveys that cover 100's of millions of light-years.

Large Scale Structure

HDM and the top-down scenario predict smooth, weak features in the large scale distribution of galaxies



HDM

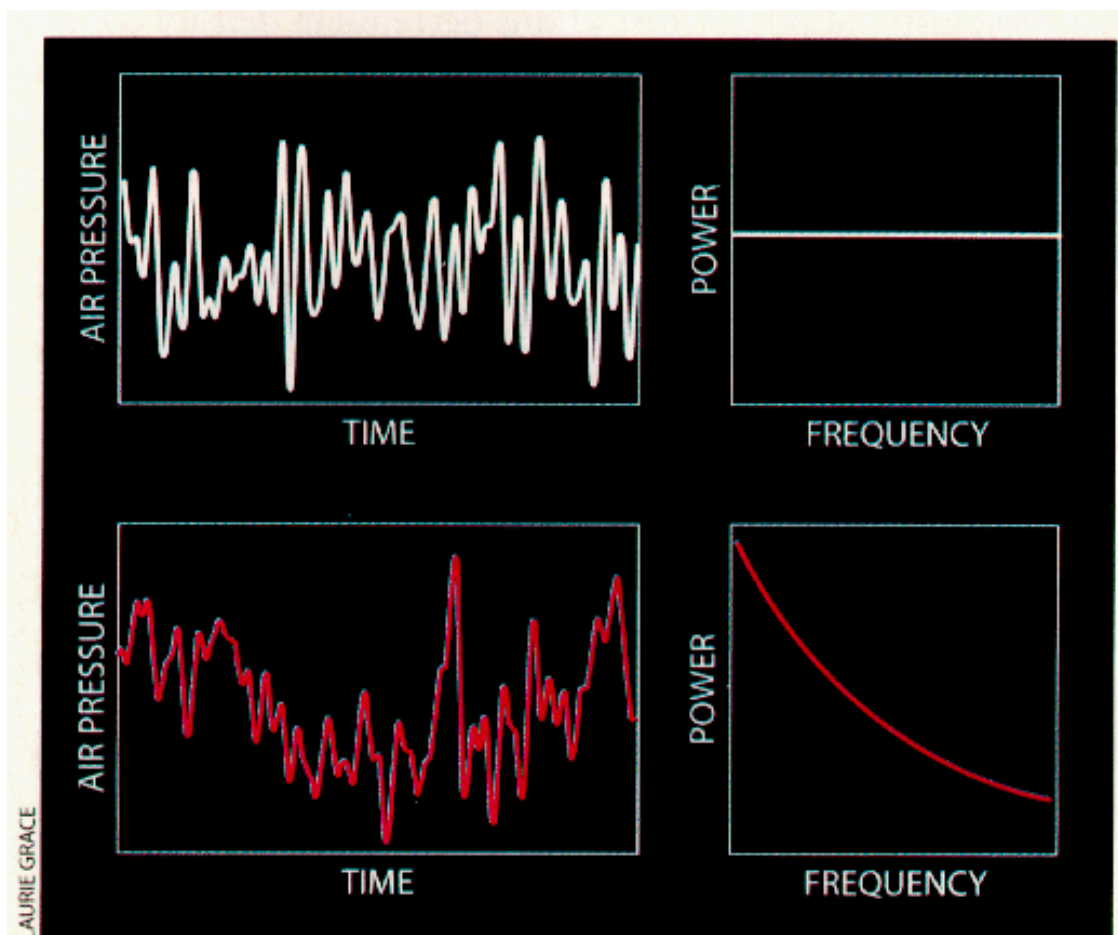


CDM

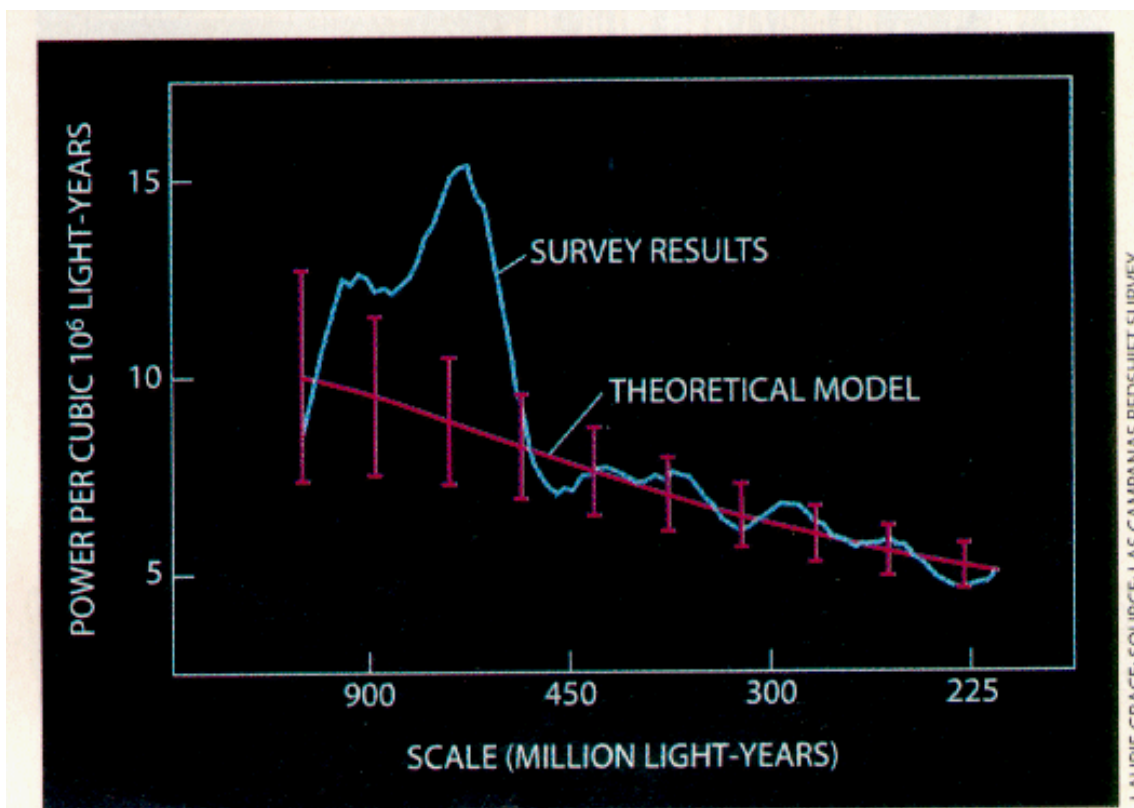
CDM and the bottom-up scenario predict sharp features with weak connecting filaments

Interestingly enough, the real distribution of galaxies from redshift surveys is exactly in-between the HDM and CDM predictions, such that a hybrid model of both HDM and CDM is needed to explain what we see.

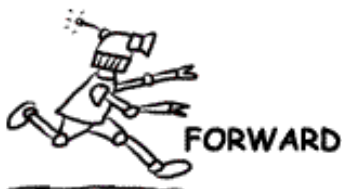
The mapping of large scale structure also has an impact on determining if the Universe is open or closed. Galaxies on the edges of the filaments will move in bulk motion towards concentrations of other galaxies and dark matter. These large scale flows can be used to determine the density of large regions of space, then extrapolated to determine the mean density of the Universe.



WHITE AND PINK NOISE surround us. White noise, the grating sound of static on a badly tuned radio or television, is completely random. The sound fluctuates from instant to instant without any pattern (*top*). Pink noise, the sound of a waterfall or waves crashing on the beach, is fractal (*bottom*). This distinction is reflected in the power spectra (*graphs at right*): white noise has equal power at all frequencies, but pink noise has more power in the bass than in the treble, in inverse proportion to the frequency.



POWER SPECTRUM of the cosmos, as measured by the Las Campanas survey (*blue line*), generally follows the prediction of the cold dark matter model (*pink*). But the power increases dramatically on scales of 600 million to 900 million light-years. This discrepancy means that the universe is much clumpier on those scales than current theories can explain.



Galaxy Formation :

Galaxies are the basic unit of cosmology. They contain stars, gas, dust and alot of dark matter. They are the only `signposts' from here to the edge of the Universe and contain the fossil clues to this earlier time.

The physics of galaxy formation is complicated because it deals with the dynamics of stars (gravitational interaction), thermodynamics of gas and energy production of stars. For example, stars form from gas clouds, but new stars heat these clouds, which dissipates them and stops the formation of other stars.

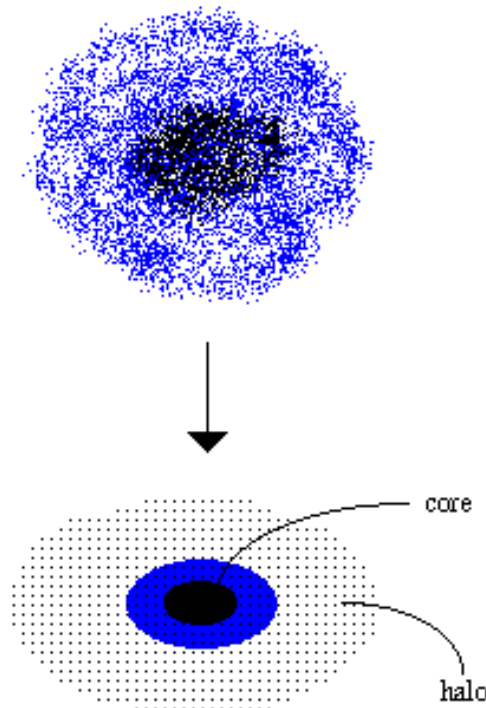
Protogalaxies:

After recombination, density enhancements either grew or dispersed. According to our hybrid top-down/bottom-up scenario, an assortment of enhancements formed of various sizes. Small, dense ones collapsed first, large ones formed slower and fragmented as they collapsed.

The first lumps that broke free of the Universe's expansion were mostly dark matter and some neutral hydrogen with a dash of helium. Once this object begins to collapse under its own gravity, it is called a [protogalaxy](#). The first protogalaxies appeared about 14 billion years ago.

Protogalaxy

a lump of dark matter and gas from the time of recombination collapses under its own gravity to form a protogalaxy



gravity separates out the protogalaxy into a core and halo. The baryons that make up the gas can interact to lose energy and fall to the core of the protogalaxy. The dark matter, which only weakly interacts, remains in the halo.

Note that [dark matter](#) and ordinary matter (in the form of hydrogen and helium gas at this time) separate at this time. Gas can dissipate its energy through collisions. The atoms in the gas collide and heat up, the heat is radiated in the infrared (light) and the result is the gas loses energy, moves slowly = collapses to the center. Dark matter does not interact this way and continues to orbit in the halo.

Even though there are no stars yet, protogalaxies should be detectable by their infrared emission (i.e. their heat). However, they are very faint and very far away (long time ago), so our technology has not been successful in discovering any at this time.

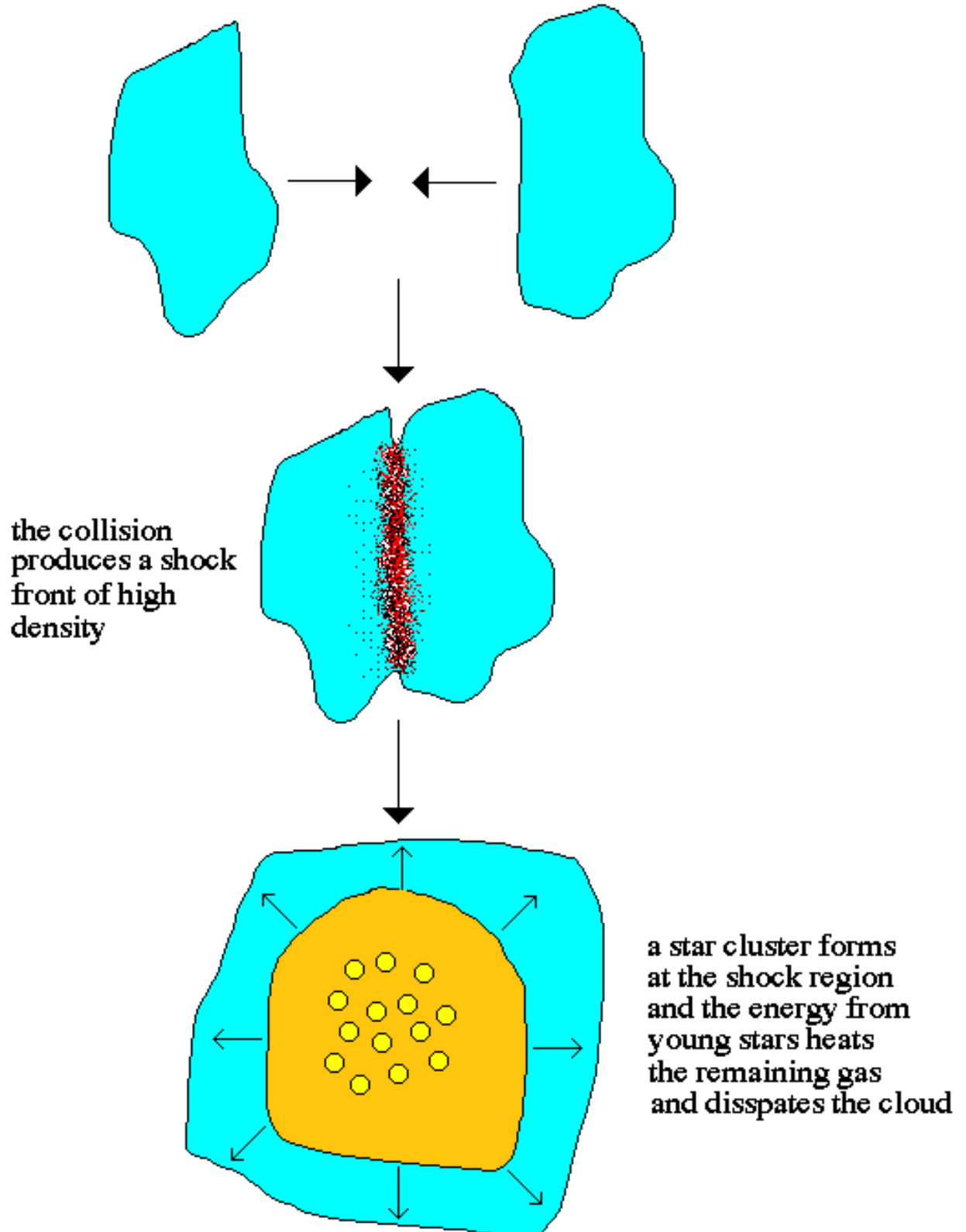
Formation of the First Stars :

As the gas in the protogalaxy loses energy, its density goes up. Gas clouds form

and move around in the protogalaxy on orbits. When two clouds collide, the gas is compressed into a shock front.

Star Formation

star formation begins with the collision of two gas clouds



The first stars in a galaxy form in this manner. With the production of its first

photons by thermonuclear fusion, the galaxy becomes a primeval galaxy.

Star formation sites in primeval galaxies are similar to [star forming regions](#) in present-day galaxies. A grouping of young stars embedded in a cloud of heated gas. The gas will eventually be pushed away from the stars to leave a [star cluster](#).

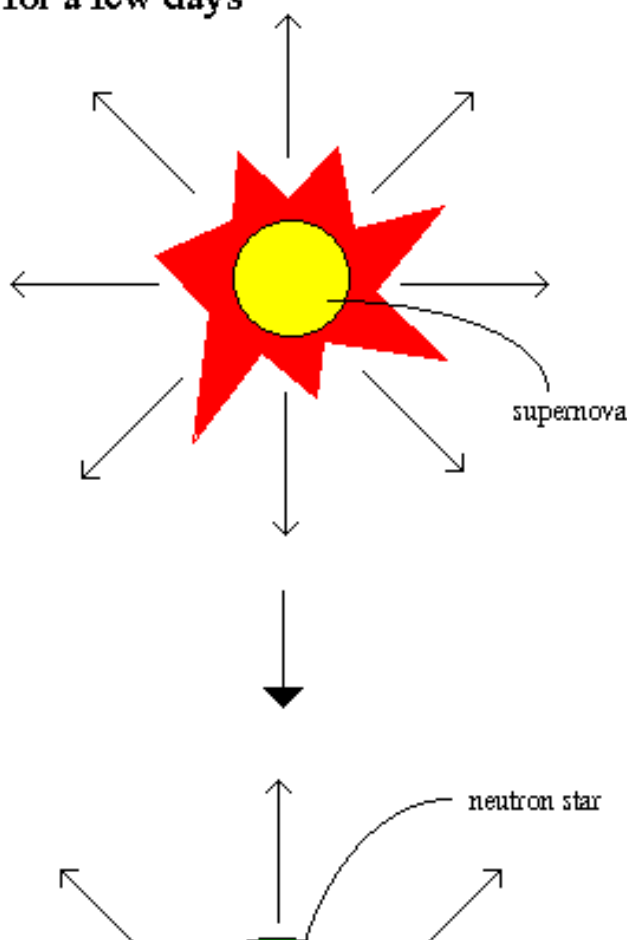
The first stars in our Galaxy are the [globular star clusters](#) orbiting outside the stellar disk which contains the spiral arms. Most galaxies with current star formation have an underlying distribution of old stars from the first epoch of star formation 14 billion years ago.

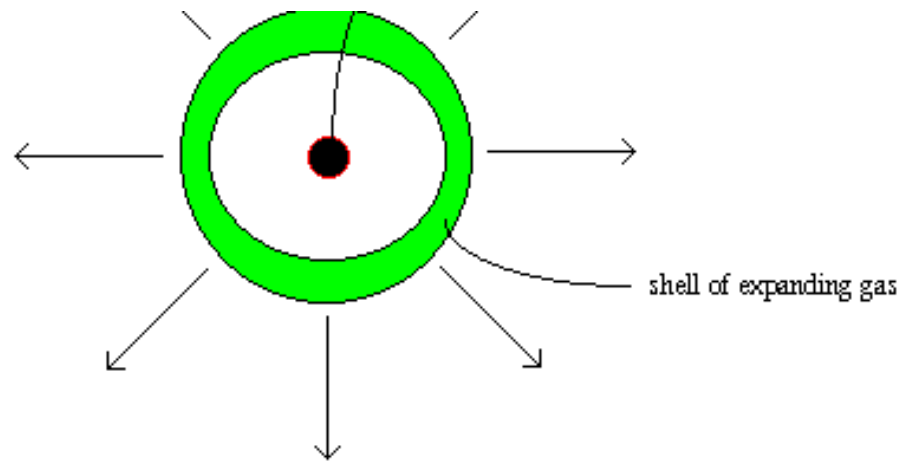
Stellar Death :

The most massive stars end their lives as [supernova](#), the explosive destruction of a star. Supernova's occur when a star uses up its interior fuel of hydrogen and collapses under its own weight. The infalling hydrogen from the star's outer envelope hits the core and ignites explosively.

Supernova

very massive stars collapse under their own weight and explode into supernovae. The supernova will accede the brightness of a million stars for a few days





a gas shell, enriched in heavy elements, is ejected into the galaxy

During the explosion, runaway fusion occurs and all the elements in the periodic table past lithium are produced. This is the only method of producing the heavy elements and is the origin to all the elements in your body.

This shell of enriched gas is ejected into the galaxy's gas supply. Thus, the older a galaxy, the more rich its gas is in heavy elements, a process called chemical evolution.

Ellipticals vs. Spirals :

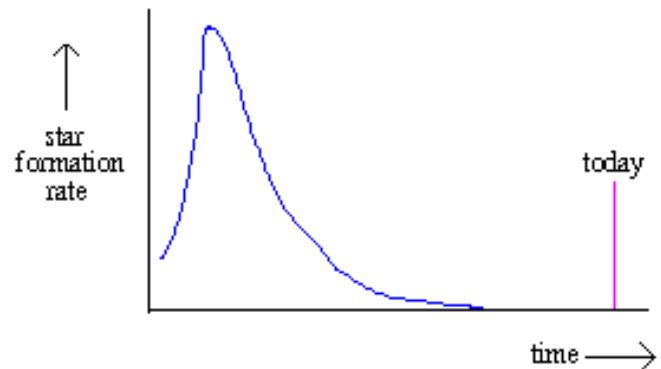
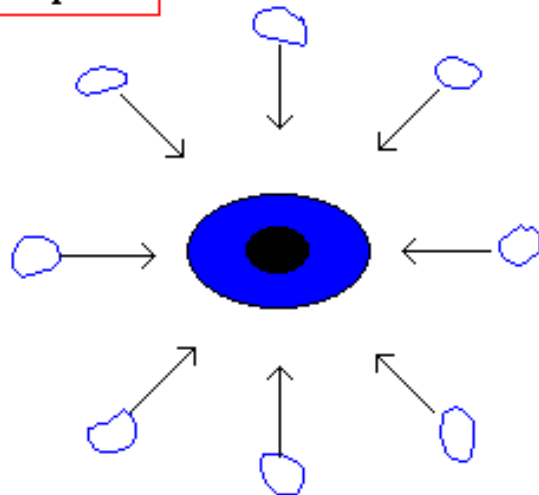
The two most distinct galaxy types are [ellipticals](#) and [spirals](#). Ellipticals have no ongoing star formation today, spirals have a lot. Assuming that ellipticals and spirals are made from the same density enhancements at the time of recombination, why did they evolve into very different appearances and star formation rates?

The answer is how rapid their initial star formation was when they formed. If star formation proceeds slowly, the gas undergoes collisions and conservation of angular momentum forms a disk (a spiral). If star formation is rapid and all the gas is used up in an initial burst, the galaxy forms as a smooth round shape, an elliptical.

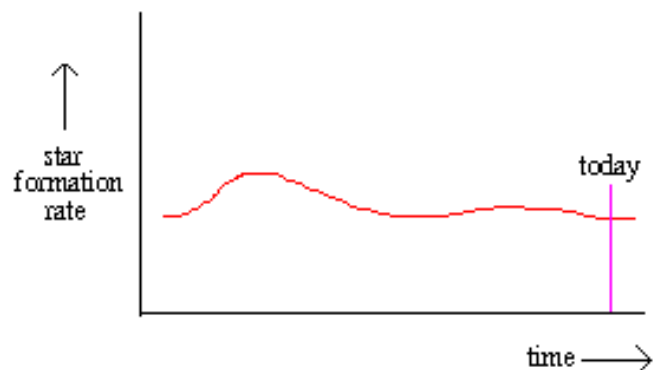
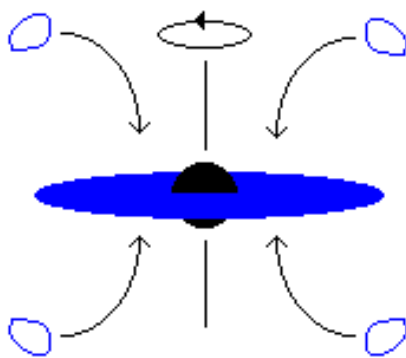
Initial Star Formation Rates

the appearance of a galaxy is determined by its initial star formation rate, which determined if all its gas was used to make stars all in one burst, or slowly over billions of years

Elliptical



Spiral



Gas falling into a spiral disk is slowed by collisions and star formation continues till today. The spiral arms and patterns are due to ongoing star formation, whereas ellipticals used all their gas supplies in an initial burst 14 billion years ago and now have no ongoing star formation.

Galaxy Mergers/Interactions :

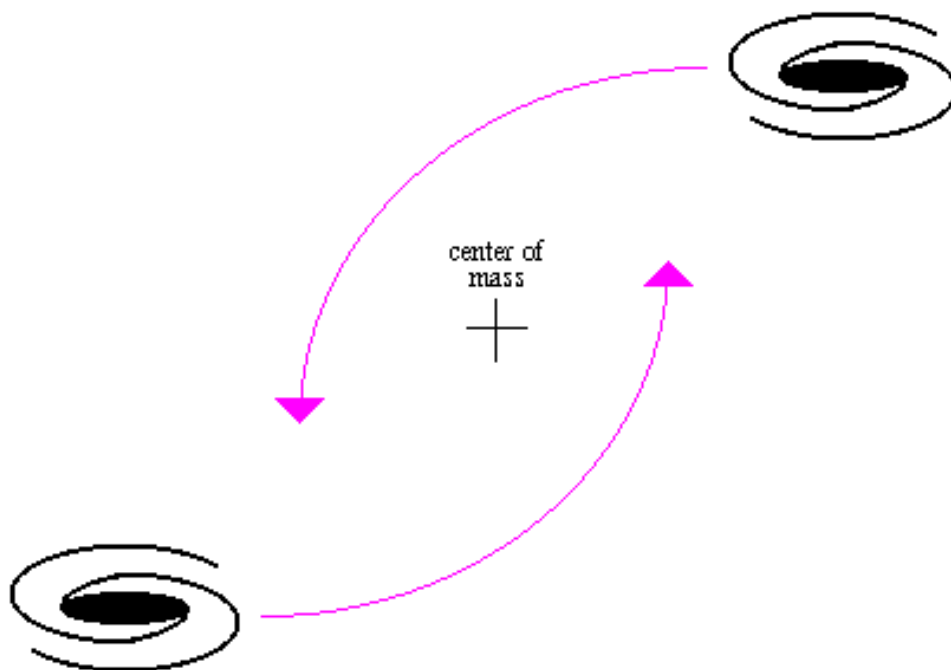
After their formation, galaxies can still change their appearance and star formation rates by interactions with other galaxies. Galaxies orbit each other in

clusters. Those orbits can sometimes cause two galaxies to pass quite close to each other to produce interesting results.

Solid objects, like planets, can pass near each other with no visible effects. However, galaxies are not solid, and can undergo inelastic collisions, which means some of the energy of the collision is transferred internally to the stars and gas in each galaxy.

Galaxy Collisions

galaxies in clusters will often pass near each other and go into near orbit around a common center of mass



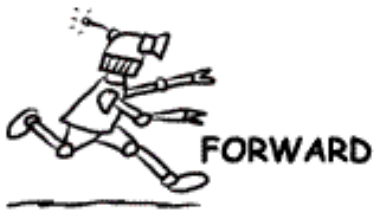
the tides produced in a close passage will cause cloud collisions and an increase in star formation



The tidal forces will often induce star formation and distort the spiral pattern in both galaxies.

If enough energy is transferred internally to the stars, then galaxies may [merge](#).

[Galaxy mergers](#) are most frequent in dense environments, such as galaxy clusters.



BACKWARD



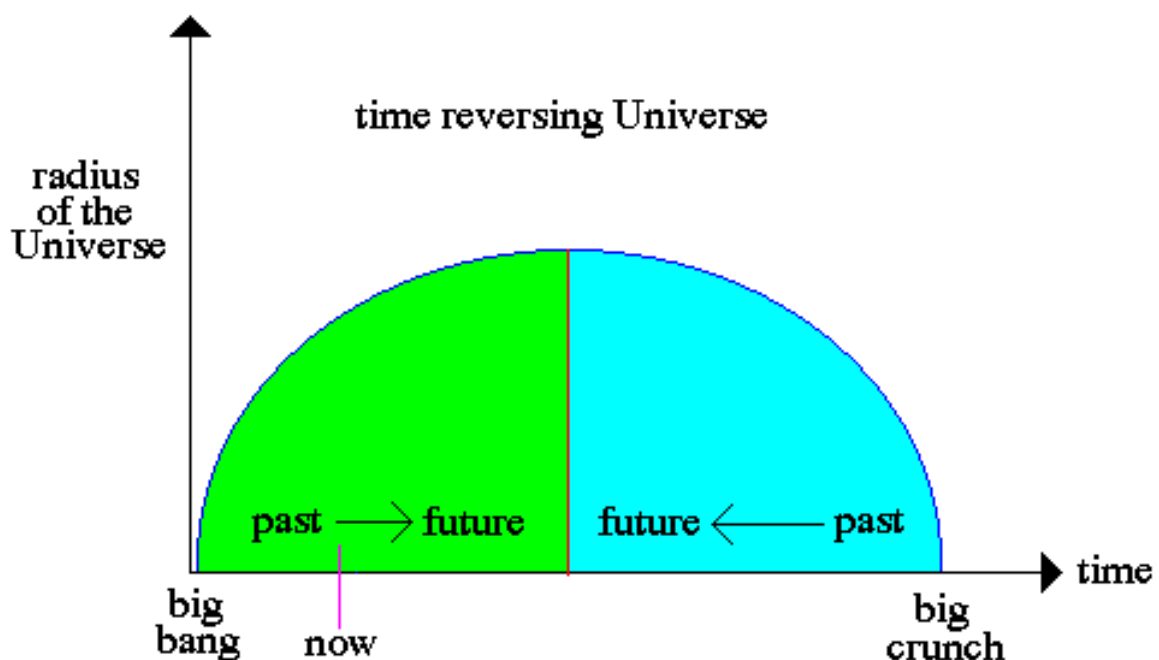
Universe Today :

The present-day Universe is a rich collection of galaxies of many types, clusters of galaxies, large scale structure and exotic phenomenon (e.g. Galactic black holes). The galaxies themselves contain stars of all sizes, luminosities and colors, as well as regions of gas and dust where new stars form. We suspect that many stars have planets, solar systems in their own right, possible harbors of life.

So what's going to happen in the future??

Time Reversal:

If the Universe is closed, then we might expect the arrow of time, as defined by entropy to reverse. There appears to be a natural connection between the expanding Universe and the fact that heat moves from hot areas (like stars) to cold areas (like outer space). So if the expansion of space were to reverse, then would entropy run the other way?



This kind of Universe has no real beginning or end, and is referred to as an oscillating Universe. Notice that it's impossible to determine which side you currently are on since time reverses and all appears normal to the observer.

The Fate of the Universe :

The past history of the Universe is one of an early, energetic time. As the Universe expanded and cooled, phenomenon became less violent and more stable.

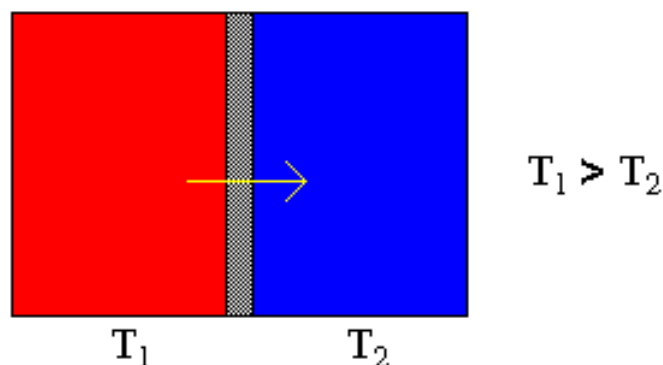
This ruling law of Nature during the evolution of the Universe has been entropy, the fact that objects go from order to disorder. There are local regions of high order, such as our planet, but only at the cost of greater disorder somewhere nearby.

If the Universe is open or flat (as our current measurements and theories suggest) then the march of entropy will continue and the fate of our Universe is confined to the principle of heat death, the flow of energy from high regions to low regions.

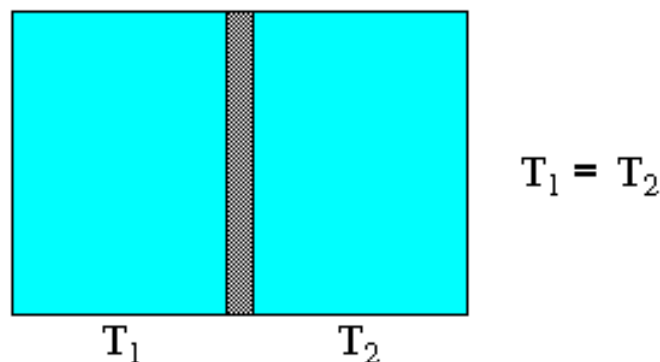
Heat Flow

energy flows from high temperature regions to low temperature regions

Order



Disorder



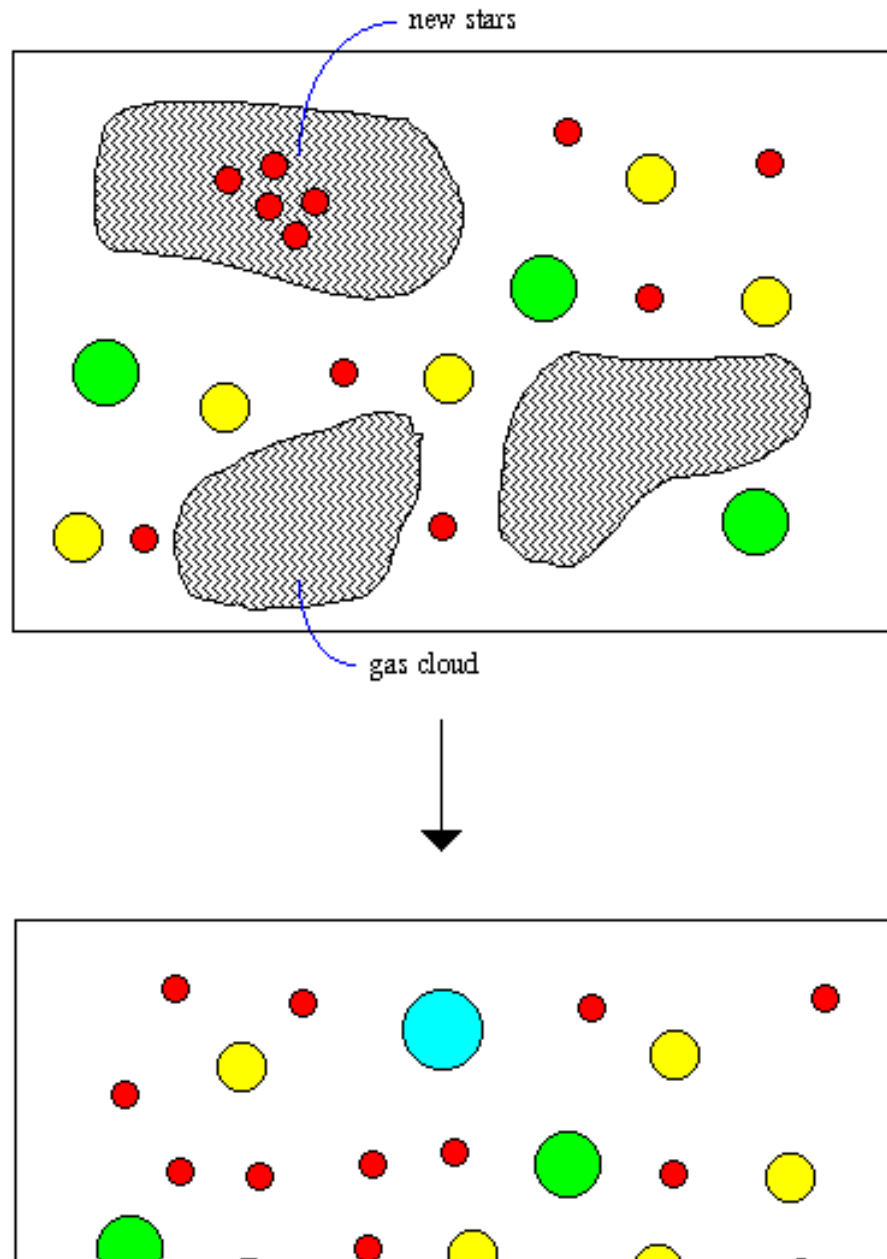
heat flow is the process that converts a low entropy region into a maximal entropy region

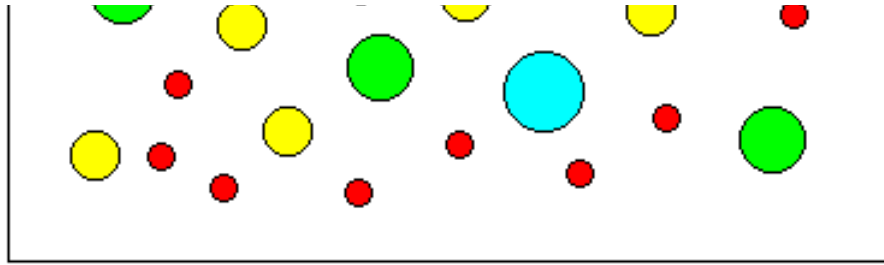
With this principle in mind, we predict the future of the Universe will pass through four stages as it continues to expand.

Stellar Era :

The Stellar Era is the time we currently live in, where most of the energy of the Universe comes from thermonuclear fusion in the cores of stars. The lifetime of the era is set by the time it takes for the smallest, lowest mass stars to use up their hydrogen fuel.

The lower mass a star is, the cooler its core and the slower it burns its hydrogen fuel (also the dimmer the star is). The slower it burns its fuel, the longer it lives (where 'live' is defined as still shining). The longest lifetime of stars less than 1/10 a solar mass (the mass of our Sun) is 10^{14} years.



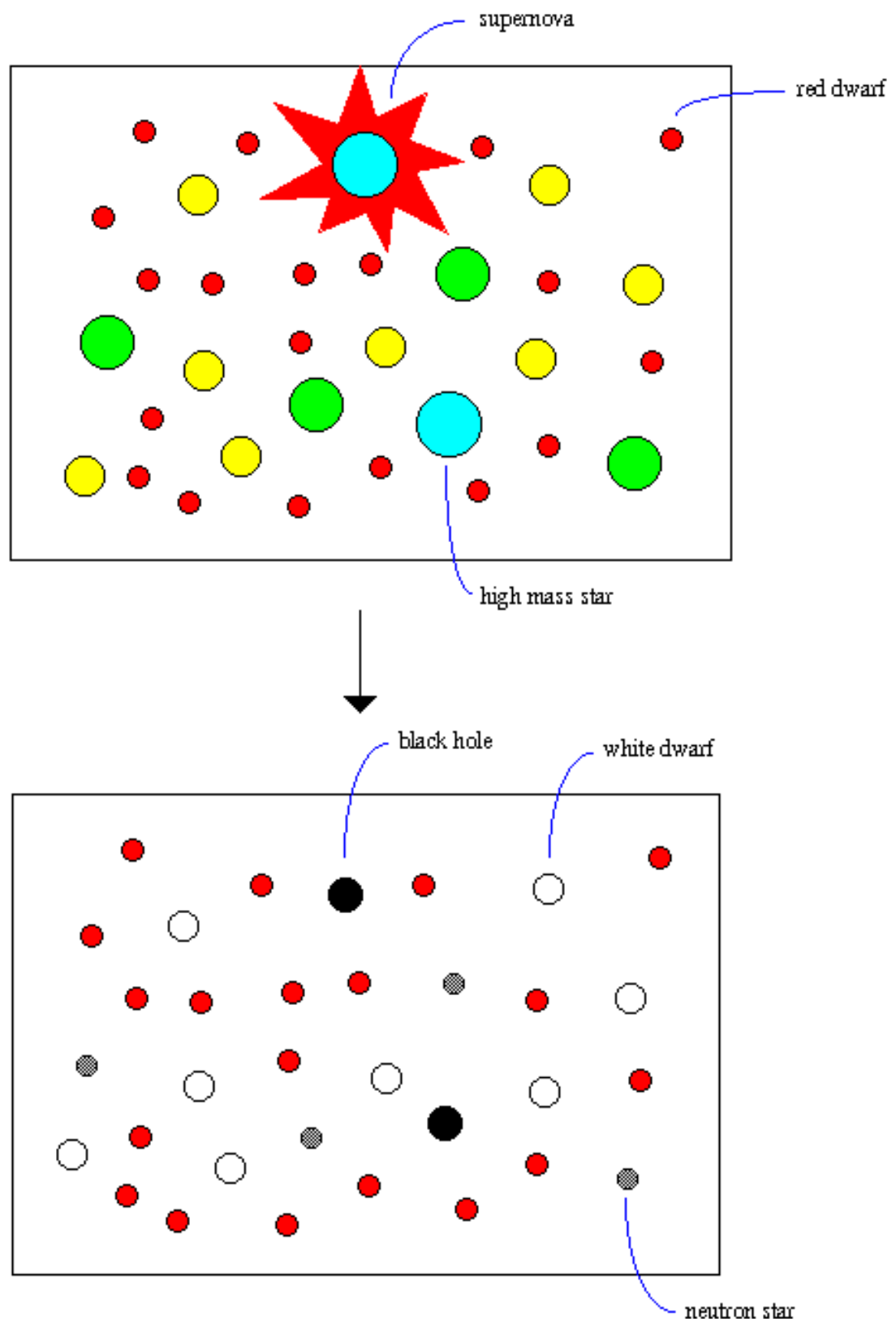


New stars are produced from gas clouds in galaxies. However, 10^{14} years is more than a sufficiently long enough time for all the gas to be used up in the Universe. Once the gas clouds are gone, all the matter in the Universe is within stars.

Degenerate Era :

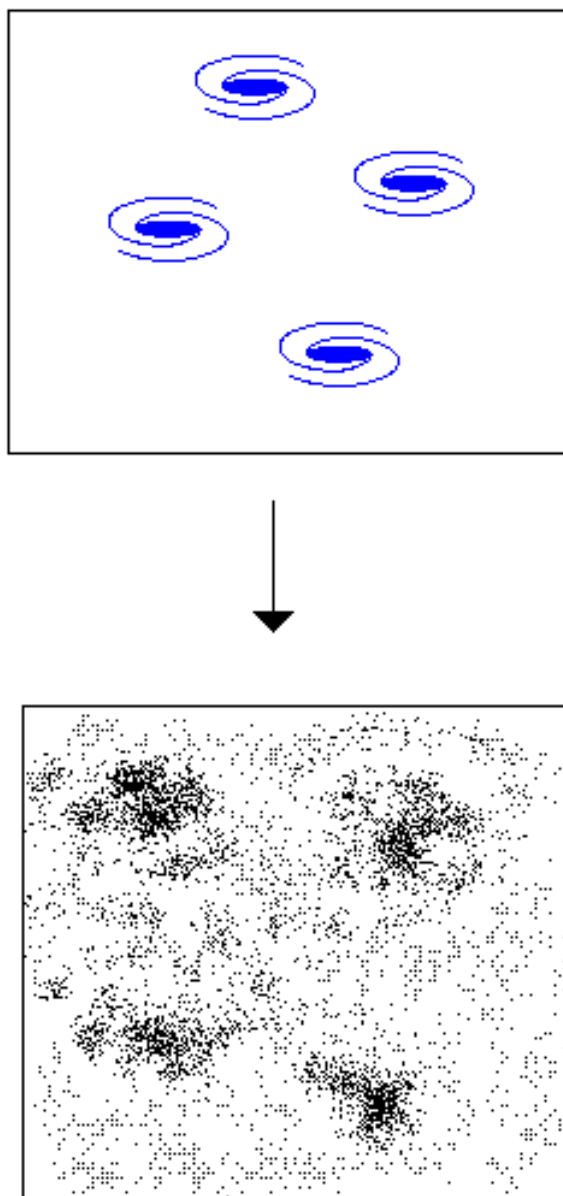
Once all the matter has been converted into stars, and the hydrogen fuel in the center of those stars has been exhausted, the Universe enters its second era, the Degenerate Era. The use of the word degenerate here is not a comment on the moral values of the Universe, rather [degenerate](#) is a physical word to describe the state of matter that has cooled to densities where all the electron shell orbits are filled and in their lowest states.

During this phase all stars are in the form of white or brown dwarfs, or neutron stars and black holes from previous explosions. White and brown dwarfs are degenerate in their matter, slowly cooling and turning into black dwarfs.



During this era, galaxies dissolve as stars go through two-body relaxation. Two-body relaxation is when two stars pass close to one another, one is kicked to high velocity and leaves the galaxy, the other is slowed down and merges with the Galactic black hole in the center of the galaxy's core. In the end, the

Universe becomes filled with free stars and giant black holes, leftover from the galaxy cores.

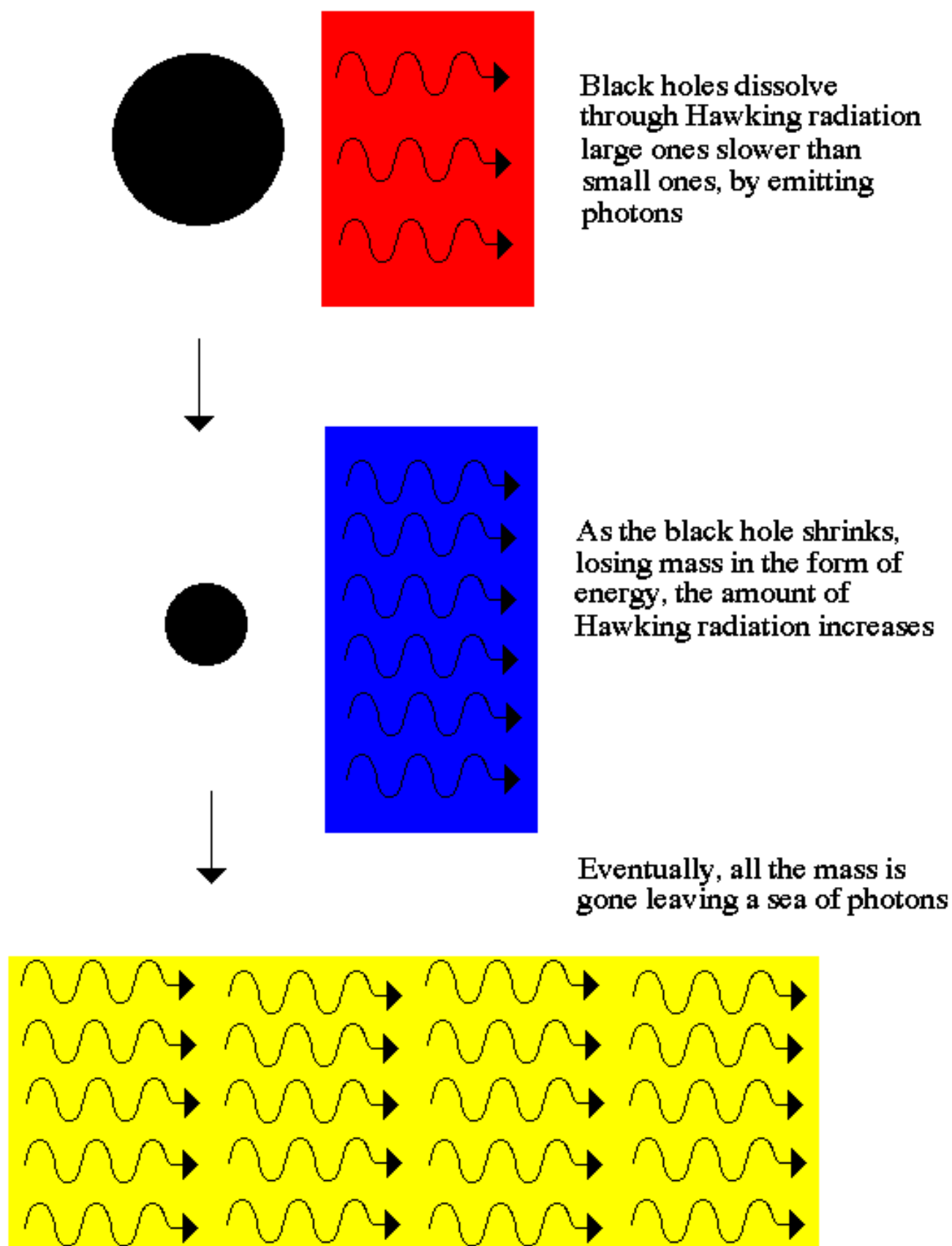


The Universe would evolve towards a vast soup of black dwarf stars except for process known as proton decay. The proton is one of the most stable elementary particles, yet even the proton decays into a positron and a meson on the order of once per 10^{32} years. Thus, the very protons that make up black dwarf stars and planets will decay and the stars and planets will dissolve into free leptons. This all takes about 10^{37} years.

Black Hole Era :

Once all the protons in the Universe have decayed into leptons, the only

organized units are black holes. From Hawking radiation, we know that even black holes are unstable and evaporate into electrons and positrons.



This process is extremely slow, varying inversely as the mass of the black hole. For Galactic mass black holes the time to dissolve can last up to 10^{100} years. The result is a bunch of photons, slowly cooling in the expanding Universe.

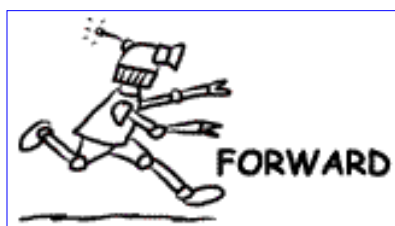
Dark Era :

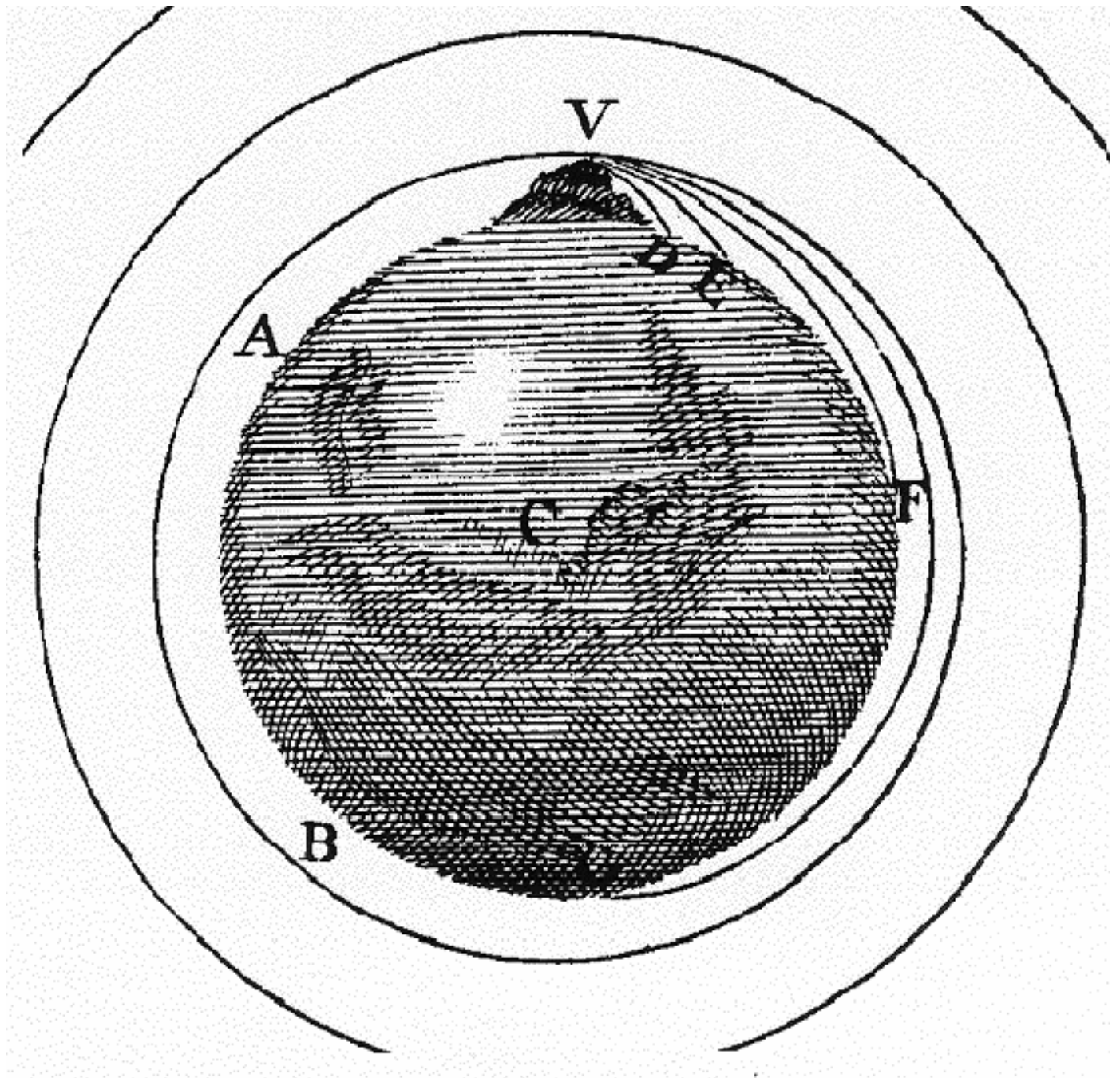
After all the black holes have evaporated, the Universe consists of an expanding sea of very long wavelength photons and neutrinos. This is a system of maximum disorder, no coherent structures or objects. No sources of energy, and no sinks as well. The rest of time is simply a continual lower of energy until the state of quantum vacuum is reached.

End of Time :

This course has been an exploration into modern cosmology and the search for the final laws of Nature (a theory of Everything) and the origin of Universe. Although there are many, many unsolved riddles to the Universe, the basic picture known as the Big Bang model is, at the very least, the foundation whose basic properties will always remain unchanged.

Although many of the concepts discussed in this course are strange, they are all based on rational scientific thought (the real world is stranger than anything you can imagine). A proper scientific model leaves less room for irrational beliefs. Understanding within the scientific method removes the blank areas on our maps, the place where we once drew monsters and golden cities. This knowledge dampens our fears like a candle in the dark.





Laws of Nature:

Laws of Nature are a stated regularity in the relations or order of phenomena in the world that holds, under a stipulated set of conditions, either universally or in a stated proportion of instances.

Laws of nature are of two basic forms: (1) a law is universal if it states that some conditions, so far as are known, invariably are found together with certain other conditions; and (2) a law is probabilistic if it affirms that, on the average, a stated fraction of cases displaying a given condition will display a certain other condition as well. In either case, a law may be valid even though it obtains only under special circumstances or as a convenient approximation. Moreover, a law of nature has no logical necessity; rather, it rests directly or indirectly upon the evidence of experience.

Laws of universal form must be distinguished from generalizations, such as "All chairs in this office are gray," which appear to be accidental. Generalizations, for example, cannot support counterfactual conditional statements such as "If this chair had been in my office, it would be gray" nor subjunctive conditionals such as "If this chair were put in my office, it would be gray." On the other hand, the statement "All planetary objects move in nearly elliptical paths about their star" does provide this support. All scientific laws appear to give similar results. The class of universal statements that can be candidates for the status of laws, however, is determined at any time in history by the theories of science then current.

Several positive attributes are commonly required of a natural law. Statements about things or events limited to one location or one date cannot be lawlike. Also, most scientists hold that the predicate must apply to evidence not used in deriving the law: though the law is founded upon experience, it must predict or help one to understand matters not included among these experiences. Finally, it is normally expected that a law will be explainable by more embracing laws or by some theory. Thus, a regularity for which there are general theoretical grounds for expecting it will be more readily called a natural law than an empirical regularity that cannot be subsumed under more general laws or theories.

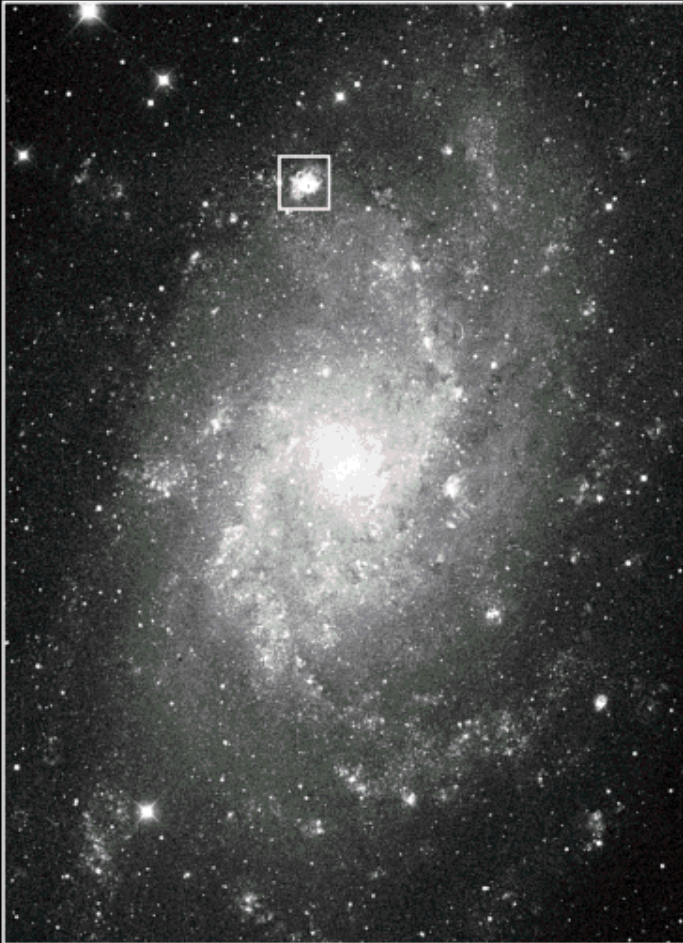
Universal laws are of several types. Many assert a dependence between varying quantities measuring certain properties, as in the law that the pressure of a gas under steady temperature is inversely proportional to its volume. Others state that events occur in an invariant order, as in "Vertebrates always occur in the fossil record after the rise of invertebrates." Lastly, there are laws affirming that if an object is of a stated sort it will have certain observable properties. Part of the reason for the ambiguity of the term law of nature lies in the temptation to apply the term only to statements of one of these sorts of laws, as in the claim that science deals solely with cause and effect relationships, when in fact all three kinds are equally valid.

Everyone is subject to the laws of Nature whether or not they believe in them, agree with them, or accept them. There is no trial, no jury, no argument, and no appeal.

Objectivity :

Fundamental issues concerning the status of historical enquiry of the kind just mentioned have arisen in another crucial area of discussion, centering upon the question of whether--and, if so, in what sense--science can be said to be an objective discipline. Some modern philosophers have inclined to the view that the entirely general problem of whether science is objective cannot sensibly be raised; legitimate questions regarding objectivity are only in place where some particular piece of historical work is under consideration, and in that case there are accepted standards available, involving such matters as documentation and accuracy, by which they can be settled. To others, however, things have not seemed so clear, and they have drawn attention to the doubts that may be felt when history is compared with different branches of investigation, such as chemistry or biology: by contrast with such enquiries, the historian's procedure, including the manner in which he conceptualizes his data and the principles of argument he employs, may appear to be governed by subjective or culturally determined predilections that are essentially contestable and, therefore, out of place in a supposedly reputable form of knowledge.

Palomar



HST

HST • WFPC2

NGC 604 in Galaxy M33

PRC96-27 • ST ScI OPO • August 7, 1996 • Hui Yang (U.IL) and NASA

Euclid:

Euclid (fl. c. 300 BC, Alexandria), the most prominent mathematician of Greco-Roman antiquity, best known for his treatise on geometry, the Elements.

Life and work. Of Euclid's life it is known only that he taught at and founded a school at Alexandria in the time of Ptolemy I Soter, who reigned from 323 to 285/283 BC. Medieval translators and editors often confused him with the philosopher Eucleides of Megara, a contemporary of Plato about a century before, and therefore called him Megarensis. Writing in the 5th century AD, the Greek philosopher Proclus told the story of Euclid's reply to Ptolemy, who asked whether there was any shorter way in geometry than that of the Elements--"There is no royal road to geometry." Another anecdote relates that a student, probably in Alexandria, after learning the very first proposition in geometry, wanted to know what he would get by learning these things, whereupon Euclid called his slave and said, "Give him threepence since he must needs make gain by what he learns."

Euclid compiled his Elements from a number of works of earlier men. Among these are Hippocrates of Chios (5th century BC), not to be confused with the physician Hippocrates of Cos (flourished 400 BC). The latest compiler before Euclid was Theudius, whose textbook was used in the Academy and was probably the one used by Aristotle. The older elements were at once superseded by Euclid's and then forgotten. For his subject matter Euclid doubtless drew upon all his predecessors, but it is clear that the whole design of his work was his own. He evidently altered the arrangement of the books, redistributed propositions among them and invented new proofs if the new order made the earlier proofs inapplicable. Thus, while Book X was mainly the work of the Pythagorean Theaetetus (flourished 369 BC), the proofs of several theorems in this book had to be changed in order to adapt them to the new definition of proportion developed by Eudoxus (q.v.). According to Proclus, Euclid incorporated into his work many discoveries of Eudoxus and Theaetetus. Most probably Books V and XII are the work of Eudoxus, X and XIII of Theaetetus. Book V expounds the very influential theory of proportion that is applicable to commensurable and incommensurable magnitudes alike (those whose ratios can be expressed as the quotient of two integers and those that cannot). The main theorems of Book XII state that circles are to one another as the squares of their diameters and that spheres are to each other as the cubes of their diameters. These theorems are certainly the work of Eudoxus, who proved them with his "method of exhaustion," by which he continuously subdivided a known magnitude until it approached the properties of an unknown. Book X deals with irrationals of different classes. Apart from some new proofs and additions, the contents of Book X are the work of Theaetetus; so is most of Book XIII, in which are described the five regular solids, earlier identified by the Pythagoreans. Euclid seems to have incorporated a finished treatise of Theaetetus on the regular solids into his Elements. Book VII, dealing with the foundations of arithmetic, is a self-consistent treatise, written most probably before 400 BC.



Plato, Roman herm probably copied from a Greek original, 4th century BC. In the Staatliche Museen, Berlin.

Plato:

Plato was born, the son of Ariston and Perictione, in Athens, or perhaps in Aegina, in about 428 BC, the year after the death of the great statesman Pericles. His family, on both sides, was among the most distinguished in Athens. Ariston is said to have claimed descent from the god Poseidon through Codrus, the last king of Athens; on the mother's side, the family was related to the early Greek lawmaker Solon. Nothing is known about Plato's father's death. It is assumed that he died when Plato was a boy. Perictione apparently married as her second husband her uncle Pyrilampes, a prominent supporter of Pericles; and Plato was probably brought up chiefly in his house. Critias and Charmides, leaders among the extremists of the oligarchic terror of 404, were, respectively, cousin and brother of Perictione; both were friends of Socrates, and through them Plato must have known the philosopher from boyhood.

The most important formative influence to which the young Plato was exposed was Socrates. It does not appear, however, that Plato belonged as a "disciple" to the circle of Socrates' intimates. The Seventh Letter speaks of Socrates not as a "master" but as an older "friend," for whose character Plato had a profound respect; and he has recorded his own absence (through indisposition) from the death scene of the Phaedo. It may well be that his own vocation to philosophy dawned on him only afterward, as he reflected on the treatment of Socrates by the democratic leaders. Plato owed to Socrates his commitment to philosophy, his rational method, and his concern for ethical questions. Among other philosophical influences the most significant were those of Heracleitus and his followers, who disparaged the phenomenal world as an arena of constant change and flux, and of the Pythagoreans, with whose metaphysical and mystical notions Plato had great sympathy.

Plato's Theory of Forms:

Plato believed that there exists an immaterial Universe of `forms', perfect aspects of everyday things such as a table, bird, and ideas/emotions, joy, action, etc. The objects and ideas in our material world are `shadows' of the forms (see [Plato's Allegory of the Cave](#)).

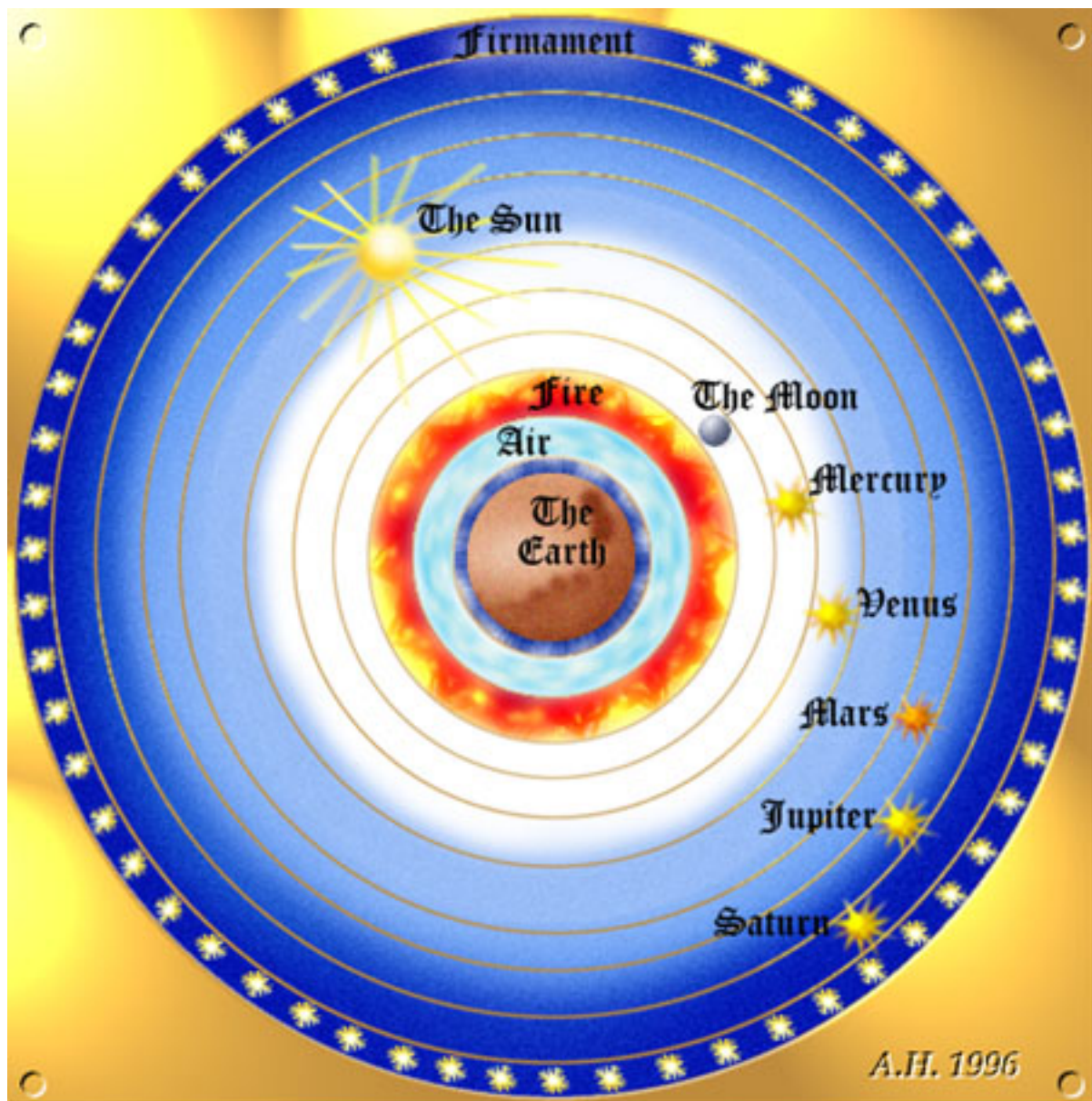
This solves the problem of how objects in the material world are all distinct (no two tables are exactly the same) yet they all have `tableness' in common. There are different objects reflecting the `tableness' from the Universe of Forms. Plato refused to write his own metaphysics, knowledge of its final shape has to be derived from hints in the dialogues and statements by Aristotle and, to a far lesser extent, other ancient authorities. According to these, Plato's doctrine of Forms was, in its general character, highly mathematical, the Forms being somehow identified with, or explained in terms of, numbers. Here may be seen the influence of the Pythagoreans, though, as Aristotle says, the details of Plato's views on the mathematical constituents of being were not the same as theirs. In addition Aristotle states that Plato introduced a class of "mathematicals," or "intermediates," positioned between sensible objects and Forms. These differ from sensible objects in being immaterial (e.g., the geometer's triangles ABC and XYZ) and from the Forms in being plural, unlike the Triangle itself.

Aristotle himself had little use for this sort of mathematical metaphysics and rejected Plato's doctrine of transcendent eternal Forms altogether. Something of Platonism, nonetheless, survived in Aristotle's system in his beliefs that the reality of anything lay in a changeless (though wholly immanent) form or essence comprehensible and definable by reason and that the highest realities were eternal, immaterial, changeless self-sufficient intellects which caused the ordered movement of the universe. It was the desire to give expression to their transcendent perfection that kept the heavenly spheres rotating. Man's intellect at its highest was akin to them. This Aristotelian doctrine of Intellect (nous) was easily recombined with Platonism in later antiquity.

Plato's cosmology derives from a mathematical discovery by Empedocles. He found that there are only five solid shapes whose sides are made from regular polygons (triangles, squares, pentagons, hexagons, etc) - for example, the cube.

Plato was so impressed with this discovery that he was convinced that atoms of matter must derive from these five fundamental solids. But at the time the Greek periodic table consisted only of earth, water, air and fire (i.e. four atomic types). Therefore, Plato postulated that a fifth atomic type must exist which Aristotle later called `ether'. The heavens, and objects in the heavens (stars, planets, Sun) are composed of atoms of ether.

This is perhaps the first example of the use of theoretical thought experiments to predict or postulate new concepts. In this case, the existence of a new form of matter, ether. And led to the formulation of a Universe that looked like the following:



Physical Model :

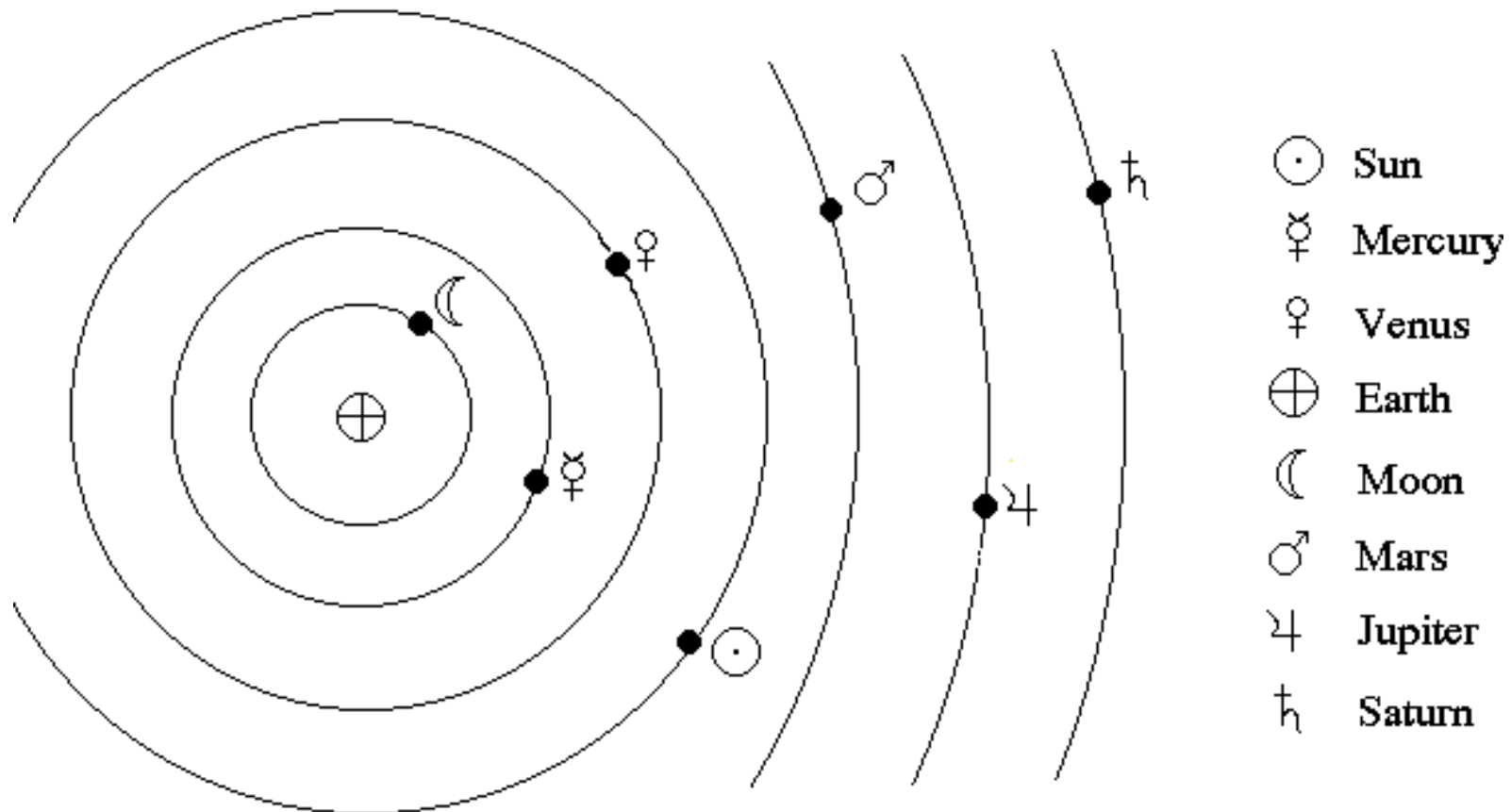
A physical model is a framework of ideas and concepts from which we interpret our observations and experimental results.

In its highest form, a physical model is expressed as a set of natural laws, e.g. Newton's laws of motion or Darwin's law of evolution. Often new discoveries produce a paradox for a current model, such as the [photoelectric effect](#) in the early 1900's. New aspects of nature can lead to a new model, or corrections to the old model. For example, special relativity was a modification to Newton's laws of motion to incorporate the effects seen when moving at high velocities and the result of a barrier to velocity of the speed of light.

Geocentric Theory:

Heraclides (330 B.C.) developed the first Solar System model, beginning of the geocentric versus heliocentric debate

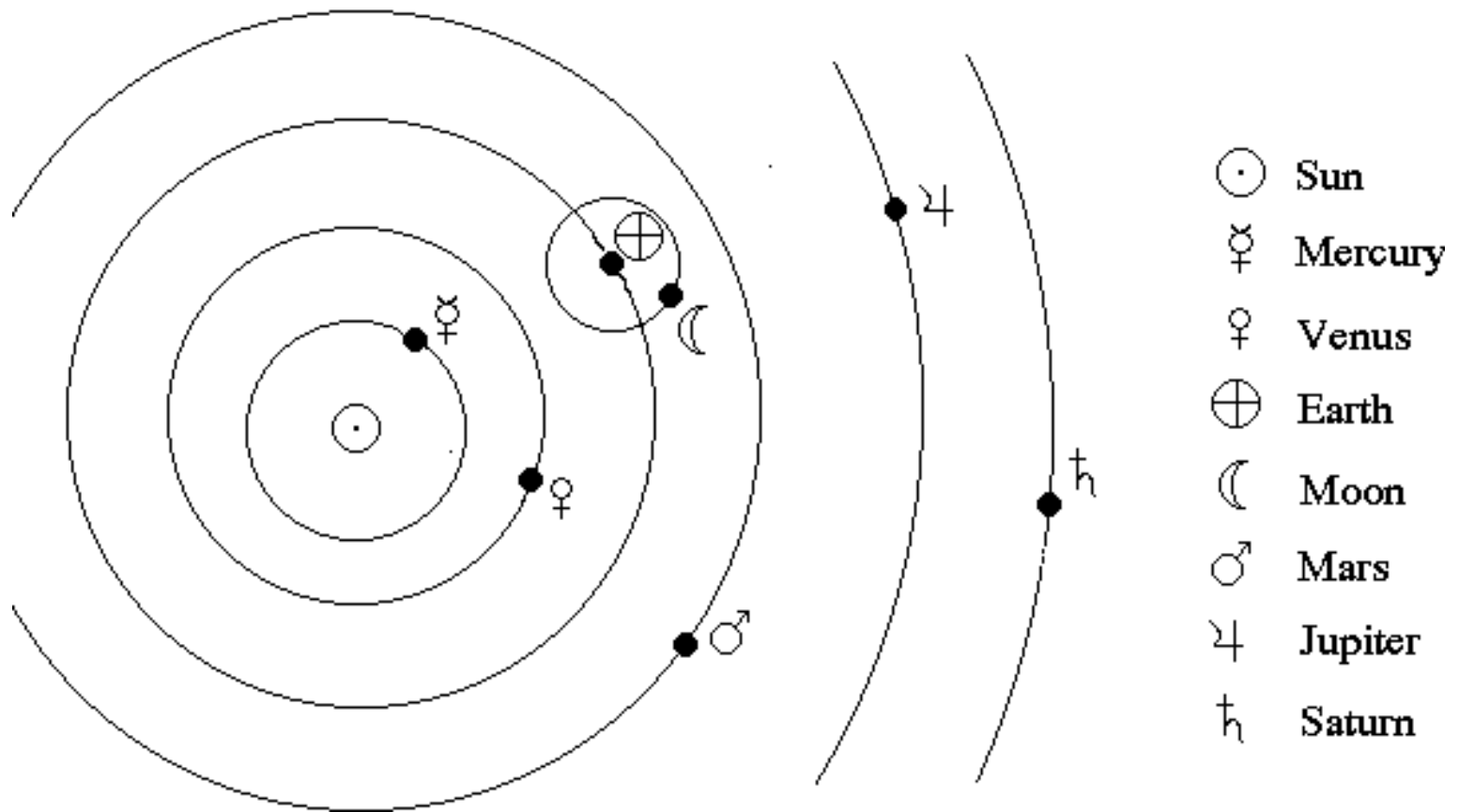
Geocentric Solar System Model



Note that orbits are perfect circles (for philosophical reasons = all things in the Heavens are "perfect")

[Aristarchus](http://zebu.uoregon.edu/~js/glossary/geocentric_theory.html) (270 B.C.) developed the heliocentric theory

Heliocentric Solar System Model



Problems for heliocentric theory:

- Earth in motion??? can't feel it
- no [parallax](#) seen in stars
- geocentric = ego-centric = more "natural"

Dante Alighieri:

Dante Alighieri (1265-1321) is Italy's greatest poet and also one of the towering figures in western European literature. He is best known for his monumental epic poem, *La commedia*, later named *La divina commedia* (The Divine Comedy). This great work of medieval literature is a profound Christian vision of man's temporal and eternal destiny. On its most personal level, it draws on the poet's own experience of exile from his native city of Florence; on its most comprehensive level, it may be read as an allegory, taking the form of a journey through hell, purgatory, and paradise. The poem amazes by its array of learning, its penetrating and comprehensive analysis of contemporary problems, and its inventiveness of language and imagery. By choosing to write his poem in Italian rather than in Latin, Dante decisively influenced the course of literary development. Not only did he lend a voice to the emerging lay culture of his own country, but Italian became the literary language in western Europe for several centuries. In addition to poetry Dante wrote important theoretical works ranging from discussions of rhetoric to moral philosophy and political thought. He was fully conversant with the classical tradition, drawing for his own purposes on such writers as Virgil, Cicero, and Boethius. But, most unusual for a layman, he also had an impressive command of the most recent scholastic philosophy and of theology. His learning and his personal involvement in the heated political controversies of his age led him to the composition of *De monarchia*, one of the major tracts of medieval political philosophy.

Dante's *La Divina Commedia* was written c. 1310-14 and has three major sections--*Inferno*, *Purgatorio*, and *Paradiso*--the narrative traces the journey of Dante from darkness and error to the revelation of the divine light, culminating in the Beatific Vision of God. Dante is guided by the Roman poet Virgil, who represents the epitome of human knowledge, from the dark wood through the descending circles of the pit of Hell (*Inferno*). Passing Lucifer at the pit's bottom, at the dead-centre of the world, Dante and Virgil emerge on the beach of the island mountain of Purgatory. At the summit of Purgatory, where repentant sinners are purged of their sins, Virgil departs, having led Dante as far as human knowledge is able, to the threshold of Paradise. There Dante is met by Beatrice, embodying the knowledge of divine mysteries bestowed by Grace, who leads him through the successive ascending levels of heaven to the Empyrean, where he is allowed to glimpse, for a moment, the glory of God.

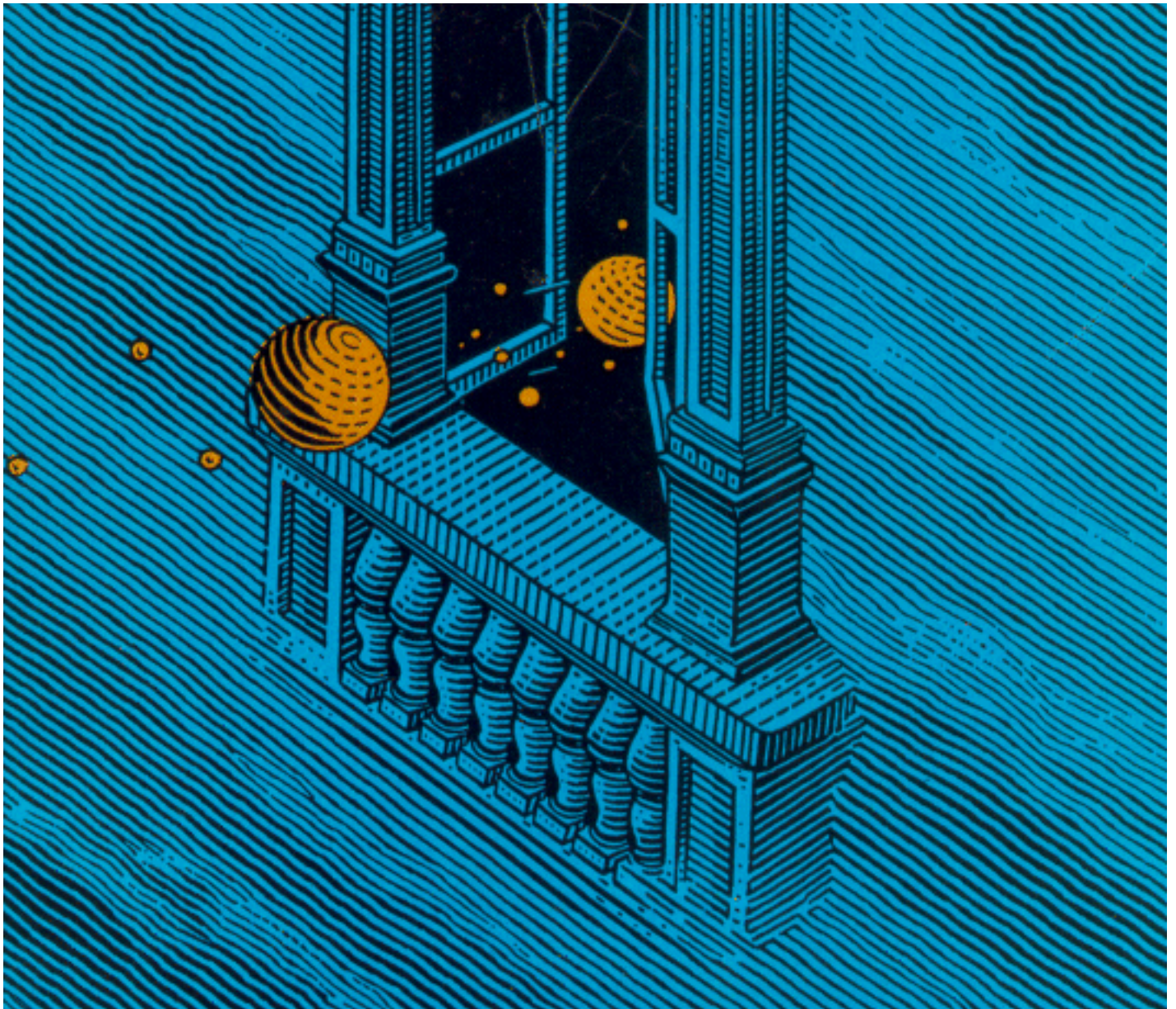
``A paradox is not a conflict within reality. It is a conflict between reality and your feeling of what reality should be like.''

- Richard Feynman

Paradox:

A paradox is an apparently self-contradictory statement, the underlying meaning of which is revealed only by careful scrutiny. The purpose of a paradox is to arrest attention and provoke fresh thought. The statement "Less is more" is an example. Francis Bacon's saying, "The most corrected copies are commonly the least correct," is an earlier literary example. In George Orwell's anti-utopian satire *Animal Farm* (1945), the first commandment of the animals' commune is revised into a witty paradox: "All animals are equal, but some animals are more equal than others." Paradox has a function in poetry, however, that goes beyond mere wit or attention-getting. Modern critics view it as a device, integral to poetic language, encompassing the tensions of error and truth simultaneously, not necessarily by startling juxtapositions but by subtle and continuous qualifications of the ordinary meaning of words.





Rationalism:

Rationalism is a method of inquiry that regards reason as the chief source and test of knowledge and, in contrast to empiricism, tends to discountenance sensory experience. It holds that, because reality itself has an inherently rational structure, there are truths--especially in logic and mathematics but also in ethics and metaphysics--that the intellect can grasp directly. In ethics, rationalism relies on a "natural light," and in theology it replaces supernatural revelation with reason.

The inspiration of rationalism has always been mathematics, and rationalists have stressed the superiority of the deductive over all other methods in point of certainty. According to the extreme rationalist doctrine, all the truths of physical science and even history could in principle be discovered by pure thinking and set forth as the consequences of self-evident premises. This view is opposed to the various systems which regard the mind as a *tabula rasa* (blank tablet) in which the outside world, as it were, imprints itself through the senses.

The opposition between rationalism and empiricism is, however, rarely so simple and direct, inasmuch as many thinkers have admitted both sensation and reflection. Locke, for example, is a rationalist in the weakest sense, holding that the materials of human knowledge (ideas) are supplied by sense experience or introspection, but that knowledge consists in seeing necessary connections between them, which is the function of reason.

Most philosophers who are called rationalists have maintained that the materials of knowledge are derived not from experience but deductively from fundamental elementary concepts. This attitude may be studied in Ren Descartes, Gottfried Wilhelm Leibniz, and Christian von Wolff. It is based on Descartes's fundamental principle that knowledge must be clear, and seeks to give to philosophy the certainty and demonstrative character of mathematics, from the a priori principle of which all its claims are derived. The attack made by David Hume on the causal relation led directly to the new rationalism of Kant, who argued that it was wrong to regard thought as mere analysis. In Kant's views, a priori concepts do exist, but if they are to lead to the amplification of knowledge, they must be brought into relation with empirical data.

Logical Systems :

Logical systems are idealized, abstract languages originally developed by modern logicians as a means of analyzing the concept of deduction. Logical models are structures which may be used to provide an interpretation of the symbolism embodied in a formal system. Together the concepts of formal system and model constitute one of the most fundamental tools employed in modern physical theories.



A formal logical system is a collection of abstract symbols, together with a set of rules for assembling the symbols into strings. Such a system has four components: 1) an alphabet, a set of abstract symbols, 2) grammar, rules which specify the valid ways one can combine the symbols, 3) axioms, a set of well-formed statements accepted as true without proof, and 4) rules of inference, procedures by which one can combine and change axioms into new strings.

How does a formal system relate to the mathematical world that we use to describe Nature? One can use a process of dictionary construction to attach meaning to the abstract, purely syntactic structure of the symbols and strings of a formal system to the semantics of a mathematical one.

Scientist:

A physical scientist is a seeker of harmonies and constancies in the jungle of experience. He aims at knowledge and prediction, particularly through discovery of mathematical laws.

Science has two aspects: one is the speculative, creative element, the continual flow of contributions by many individuals, each working on his own task by his own usually unexamined methods, motivated in his own way, and generally uninterested in attending to long-range philosophical problems. The second aspect is science as the evolving compromise, science as a growing network synthesized from these individual contributions by accepting or adopting those ideas which do indeed prove meaningful and useful to generation after generation of scientists.

Locality :

Although people gain much information from their impressions, most matters of fact depend upon reasoning about causes and effects, even though people do not directly experience causal relations. What, then, are causal relations? According to Hume they have three components: contiguity of time and place, temporal priority of the cause, and constant conjunction.

In order for x to be the cause of y, x and y must exist adjacent to each other in space and time, x must precede y, and x and y must invariably exist together. There is nothing more to the idea of causality than this; in particular, people do not experience and do not know of any power, energy, or secret force that causes possess and that they transfer to the effect. Still, all judgments about causes and their effects are based upon experience.

To cite examples from *An Enquiry Concerning Human Understanding* (1748), since there is nothing in the experience of seeing a fire close by which logically requires that one will feel heat, and since there is nothing in the experience of seeing one rolling billiard ball contact another that logically requires the second one to begin moving, why does one expect heat to be felt and the second ball to roll? The explanation is custom. In previous experiences, the feeling of heat has regularly accompanied the sight of fire, and the motion of one billiard ball has accompanied the motion of another. Thus the mind becomes accustomed to certain expectations. "All inferences from experience, therefore, are effects of custom, not of reasoning." Thus it is that custom, not reason, is the great guide of life. In short, the idea of cause and effect is neither a relation of ideas nor a matter of fact. Although it is not a perception and not rationally justified, it is crucial to human survival and a central aspect of human survival and a central aspect of human cognition.

Regularities, even when expressed mathematically as laws of nature, are not fully satisfactory to everyone. Some insist that genuine understanding demands explanations of the causes of the laws, but it is in the realm of causation that there is the greatest disagreement. Modern quantum mechanics, for example, has given up the quest for causation and today rests only on mathematical description. Modern biology, on the other hand, thrives on causal chains that permit the understanding of physiological and evolutionary processes in terms of the physical activities of entities such as molecules, cells, and organisms. But even if causation and explanation are admitted as necessary, there is little agreement on the kinds of causes that are permissible, or possible, in science. If the history of science is to make any sense whatsoever, it is necessary to deal with the past on its own terms, and the fact is that for most of the history of science natural philosophers appealed to causes that would be summarily rejected by modern scientists. Spiritual and divine forces were accepted as both real and necessary until the end of the 18th century and, in areas such as biology, deep into the 19th century as well.

Certain conventions governed the appeal to God or the gods or to spirits. Gods and spirits, it was held, could not be completely arbitrary in their actions; otherwise the proper response would be propitiation, not rational investigation. But since the deity or deities were themselves rational, or bound by rational principles, it was possible for humans to uncover the rational order of the world. Faith in the ultimate

rationality of the creator or governor of the world could actually stimulate original scientific work. Kepler's laws, Newton's absolute space, and Einstein's rejection of the probabilistic nature of quantum mechanics were all based on theological, not scientific, assumptions. For sensitive interpreters of phenomena, the ultimate intelligibility of nature has seemed to demand some rational guiding spirit. A notable expression of this idea is Einstein's statement that the wonder is not that mankind comprehends the world, but that the world is comprehensible.

Science, then, is to be considered in this context as knowledge of natural regularities that is subjected to some degree of skeptical rigor and explained by rational causes. One final caution is necessary. Nature is known only through the senses, of which sight, touch, and hearing are the dominant ones, and the human notion of reality is skewed toward the objects of these senses. The invention of such instruments as the telescope, the microscope, and the Geiger counter has brought an ever-increasing range of phenomena within the scope of the senses. Thus, scientific knowledge of the world is only partial, and the progress of science follows the ability of humans to make phenomena perceivable.

The first entanglement of three photons has been experimentally demonstrated by researchers at the University of Innsbruck. Individually, an entangled particle has properties (such as momentum) that are indeterminate and undefined until the particle is measured or otherwise disturbed. Measuring one entangled particle, however, defines its properties and seems to influence the properties of its partner or partners instantaneously, even if they are light years apart.

In the present experiment, sending individual photons through a special crystal sometimes converted a photon into two pairs of entangled photons. After detecting a "trigger" photon, and interfering two of the three others in a beamsplitter, it became impossible to determine which photon came from which entangled pair. As a result, the respective properties of the three remaining photons were indeterminate, which is one way of saying that they were entangled (the first such observation for three physically separated particles).

The researchers deduced that this entangled state is the long-coveted GHZ state proposed by physicists Daniel Greenberger, Michael Horne, and Anton Zeilinger in the late 1980s. In addition to facilitating more advanced forms of quantum cryptography, the GHZ state will help provide a nonstatistical test of the foundations of quantum mechanics. Albert Einstein, troubled by some implications of quantum science, believed that any rational description of nature is incomplete unless it is both a local and realistic theory: "realism" refers to the idea that a particle has properties that exist even before they are measured, and "locality" means that measuring one particle cannot affect the properties of another, physically separated particle faster than the speed of light.

But quantum mechanics states that realism, locality--or both--must be violated. Previous experiments have provided highly convincing evidence against local realism, but these "Bell's inequalities" tests require the measurement of many pairs of entangled photons to build up a body of statistical evidence against the idea. In contrast, studying a single set of properties in the GHZ particles (not yet reported) could verify the predictions of quantum mechanics while contradicting those of local realism.

Energy :

Energy is the capacity for doing [work](#). It may exist in potential, kinetic, thermal, electrical, chemical, nuclear, or other various forms. There are, moreover, heat and work; i.e., energy in the process of transfer from one body to another. After it has been transferred, energy is always designated according to its nature. Hence, heat transferred may become thermal energy, while work done may manifest itself in the form of mechanical energy.

All forms of energy are associated with motion. For example, any given body has kinetic energy if it is in motion. A tensioned device such as a bow or spring, though at rest, has the potential for creating motion; it contains potential energy because of its configuration. Similarly, nuclear energy is potential energy because it results from the configuration of subatomic particles in the nucleus of an atom.

Potential Energy :

Potential energy is stored energy that depends upon the relative position of various parts of a system. A spring has more potential energy when it is compressed or stretched. A steel ball has more potential energy raised above the ground than it has after falling to the Earth. In the raised position it is capable of doing more work. Potential energy is a property of a system and not of an individual body or particle; the system composed of the Earth and the raised ball, for example, has more potential energy as the two are farther separated.

Potential energy arises in systems with parts that exert forces on each other of a magnitude dependent on the configuration, or relative position, of the parts. In the case of the Earth-ball system, the force of gravity between the two depends only on the distance separating them. The work done in separating them farther, or in raising the ball, transfers additional energy to the system, where it is stored as gravitational potential energy.

Potential energy also includes other forms. The energy stored between the plates of a charged capacitor is electrical potential energy. What is commonly known as chemical energy, the capacity of a substance to do work or to evolve heat by undergoing a change of composition, may be regarded as potential energy resulting from the mutual forces among its molecules and atoms. Nuclear energy is also a form of potential energy.

The potential energy of a system of particles depends only on their initial and final configurations; it is independent of the path the particles travel. In the case of the steel ball and the earth, if the initial position of the ball is ground level and the final position is ten feet above the ground, the potential energy is the same, no matter how or by what route the ball was raised. The value of potential energy is arbitrary and relative to the choice of reference point. In the case given above, the system would have twice as much potential energy if the initial position were the bottom of a ten-foot-deep hole.

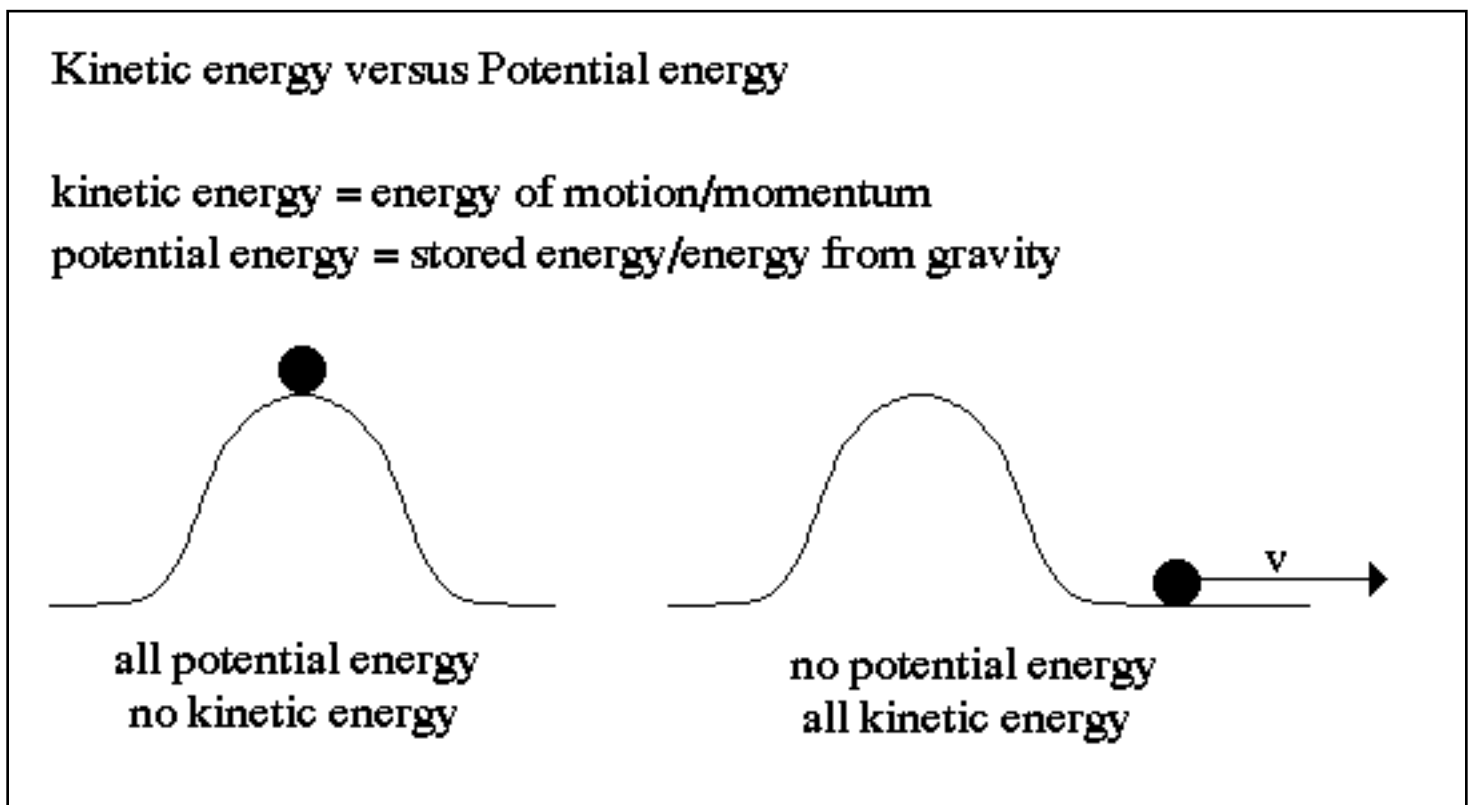
Gravitational potential energy near the Earth's surface may be computed by multiplying the weight of an

object by its distance above the reference point. In bound systems, such as atoms, in which electrons are held by the electric force of attraction to nuclei, the zero reference for potential energy is a distance from the nucleus so great that the electric force is not detectable. In this case, bound electrons have negative potential energy, and those just free of the nucleus and at rest have zero potential energy.

Kinetic Energy :

Potential energy may be converted into energy of motion, called kinetic energy, and in turn to other forms such as electrical energy. Thus, water behind a dam flows to lower levels through turbines that turn electric generators, producing electric energy plus some unusable heat energy resulting from turbulence and friction.

Historically, potential energy was included with kinetic energy as a form of mechanical energy so that the total energy in gravitational systems could be calculated as a constant.



Appearance and Reality :

Metaphysics is the science that seeks to define what is ultimately real as opposed to what is merely apparent.

The contrast between appearance and reality, however, is by no means peculiar to metaphysics. In everyday life people distinguish between the real size of the Sun and its apparent size, or again between the real color of an object (when seen in standard conditions) and its apparent color (nonstandard conditions). A cloud appears to consist of some white, fleecy substance, although in reality it is a concentration of drops of water. In general, men are often (though not invariably) inclined to allow that the scientist knows the real constitution of things as opposed to the surface aspects with which ordinary men are familiar. It will not suffice to define metaphysics as knowledge of reality as opposed to appearance; scientists, too, claim to know reality as opposed to appearance, and there is a general tendency to concede their claim.

It seems that there are at least two components in the metaphysical conception of reality. One characteristic, which has already been illustrated by Plato, is that reality is genuine as opposed to deceptive. The ultimate realities that the metaphysician seeks to know are precisely things as they are--simple and not variegated, exempt from change and therefore stable objects of knowledge. Plato's own assumption of this position perhaps reflects certain confusions about the knowability of things that change; one should not, however, on that ground exclude this aspect of the concept of reality from metaphysical thought in general. Ultimate reality, whatever else it is, is genuine as opposed to sham.

Second, and perhaps most important, reality for the metaphysician is intelligible as opposed to opaque. Appearances are not only deceptive and derivative, they also make no sense when taken at their own level. To arrive at what is ultimately real is to produce an account of the facts that does them full justice. The assumption is, of course, that one cannot explain things satisfactorily if one remains within the world of common sense, or even if one advances from that world to embrace the concepts of science. One or the other of these levels of explanation may suffice to produce a sort of local sense that is enough for practical purposes or that forms an adequate basis on which to make predictions. Practical reliability of this kind, however, is very different from theoretical satisfaction; the task of the metaphysician is to challenge all assumptions and finally arrive at an account of the nature of things that is fully coherent and fully thought-out.

Science:

Science is the organized systematic enterprise that gathers knowledge about the world and condenses the knowledge into testable laws and principles. Its defining traits are

- ✦ first, the confirmation of discoveries and support of hypotheses through repetition by independent investigators, preferably with different tests and analyses;
- ✦ second, mensuration, the quantitative description of the phenomena on universally accepted scales;
- ✦ third, economy, by which the largest amount of information is abstracted into a simple and precise form, which can be unpacked to re-create detail;
- ✦ fourth, heuristics, the opening of avenues to new discovery and interpretation.

Physical science, like all the natural sciences, is concerned with describing and relating to one another those experiences of the surrounding world that are shared by different observers and whose description can be agreed upon. One of its principal fields, physics, deals with the most general properties of matter, such as the behaviour of bodies under the influence of forces, and with the origins of those forces. In the discussion of this question, the mass and shape of a body are the only properties that play a significant role, its composition often being irrelevant. Physics, however, does not focus solely on the gross mechanical behaviour of bodies, but shares with chemistry the goal of understanding how the arrangement of individual atoms into molecules and larger assemblies confers particular properties. Moreover, the atom itself may be analyzed into its more basic constituents and their interactions.

The present opinion, rather generally held by physicists, is that these fundamental particles and forces, treated quantitatively by the methods of quantum mechanics, can reveal in detail the behaviour of all material objects. This is not to say that everything can be deduced mathematically from a small number of fundamental principles, since the complexity of real things defeats the power of mathematics or of the largest computers. Nevertheless, whenever it has been found possible to calculate the relationship between an observed property of a body and its deeper structure, no evidence has ever emerged to suggest that the more complex objects, even living organisms, require that special new principles be invoked, at least so long as only matter, and not mind, is in question. The physical scientist thus has two very different roles to play: on the one hand, he has to reveal the most basic constituents and the laws that govern them; and, on the other, he must discover techniques for elucidating the peculiar features that arise from complexity of structure without having recourse each time to the fundamentals.

This modern view of a unified science, embracing fundamental particles, everyday phenomena, and the vastness of the Cosmos, is a synthesis of originally independent disciplines, many of which grew out of useful arts. The extraction and refining of metals, the occult manipulations of alchemists, and the astrological interests of priests and politicians all played a part in initiating systematic studies that expanded in scope until their mutual relationships became clear, giving rise to what is customarily

recognized as modern physical science.

Superstition:

Superstition is a belief, half-belief, or practice for which there appears to be no rational substance. Those who use the term imply that they have certain knowledge or superior evidence for their own scientific, philosophical, or religious convictions. An ambiguous word, it probably cannot be used except subjectively. With this qualification in mind, superstitions may be classified roughly as religious, cultural, and personal.

Every religious system tends to accumulate superstitions as peripheral beliefs--a Christian, for example, may believe that in time of trouble he will be guided by the Bible if he opens it at random and reads the text that first strikes his eye. Often one person's religion is another one's superstition: Constantine called paganism superstition; Tacitus called Christianity a pernicious superstition; Roman Catholic veneration of relics, images, and the saints is dismissed as superstitious to many Protestants; Christians regard many Hindu practices as superstitious; and adherents of all "higher" religions may consider the Australian Aborigine's relation to his totem superstitious. Finally, all religious beliefs and practices may seem superstitious to the person without religion.

Superstitions that belong to the cultural tradition (in some cases inseparable from religious superstition) are enormous in their variety. Many persons, in nearly all times, have held, seriously or half-seriously, irrational beliefs concerning methods of warding off ill or bringing good, foretelling the future, and healing or preventing sickness or accident. A few specific folk traditions, such as belief in the evil eye or in the efficacy of amulets, have been found in most periods of history and in most parts of the world. Others may be limited to one country, region, or village, to one family, or to one social or vocational group.

Finally, people develop personal superstitions: a schoolboy writes a good examination paper with a certain pen, and from then on that pen is lucky; a horseplayer may be convinced that gray horses run well for him.

Superstition has been deeply influential in history. Even in so-called modern times, in a day when objective evidence is highly valued, there are few people who would not, if pressed, admit to cherishing secretly one or two irrational beliefs or superstitions.

Science and other kinds of knowledge			
	<u>Religious Knowledge</u>	<u>Artistic/Mystic Knowledge</u>	<u>Scientific Knowledge</u>
Outrageous stereotype of user	Bible-thumping fundamentalist or robe-draped monk; fond of Sunday-morning radio.	Crystal-hugging wearer of tie-dyed T-shirts; listens to new-age music.	Geek with pocket protector and calculator; watches <i>Discovery Channel</i> a lot.

How one discovers knowledge	From ancient texts or revelations of inspired individuals.	From personal insight, or insight of others	From evidence generated by observation of nature or by experimentation.
Extent to which knowledge changes through time	Little.	May be considerable.	Considerable.
Extent to which future changes in knowledge are expected by user	None.	Can be expected, to the degree that the user expects personal development	Considerable.
How knowledge changes through time	Unchangeable except by reinterpretation by authorities, or by new inspired revelations, or by divergence of mavericks.	As user changes or as user encounters ideas of others	By new observations or experiments, and/or by reinterpretation of existing data.
Certainty of the user	High, given sufficient faith; can be complete.	High	Dependent on quality and extent of evidence; should never be complete.
Assumptions	That ancient texts or inspired revelation have meaning to modern or future conditions.	That personal feelings and insights reflect nature.	That nature has discernible, predictable, and explainable patterns of behavior.
Where users put their faith	In the supernatural beings that they worship or in the authorities who interpret texts and events.	In their own perceptions.	In the honesty of the people reporting scientific data (the incomes of whom depend on generation of that data), and in the human ability to understand nature.

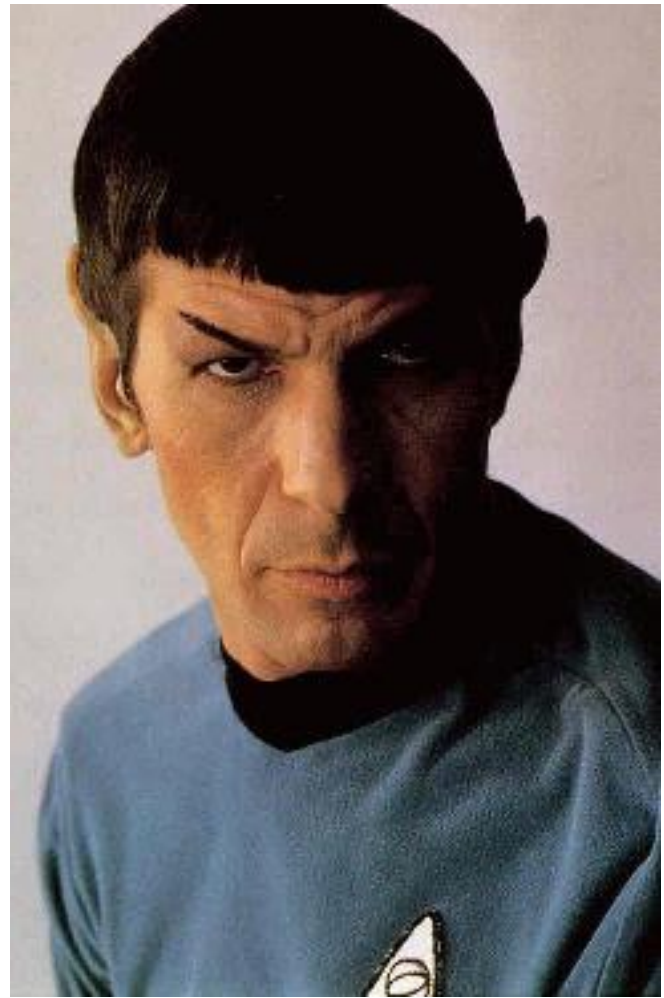
Sources of contradiction	Between different religions; between different texts and/or authorities within one religion; within individual texts (as in the two accounts of human origin in the Judeo-Christian Genesis).	Between users, who each draw on their own personal insights	Across time, as understanding changes; between fields, which use different approaches and materials; and between individuals, who use different approaches and materials.
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Logic:

Logic is the the study of propositions and of their use in argumentation. This study may be carried on at a very abstract level, as in formal logic, or it may focus on the practical art of right reasoning, as in applied logic.

Valid arguments have two basic forms. Those that draw some new proposition (the conclusion) from a given proposition or set of propositions (the premises) in which it may be thought to lie latent are called deductive. These arguments make the strong claim that the conclusion follows by strict necessity from the premises, or in other words that to assert the premises but deny the conclusion would be inconsistent and self-contradictory. Arguments that venture general conclusions from particular facts that appear to serve as evidence for them are called inductive. These arguments make the weaker claim that the premises lend a certain degree of probability or reasonableness to the conclusion. The logic of inductive argumentation has become virtually synonymous with the methodology of the physical, social, and historical sciences and is no longer treated under logic.

Logic as currently understood concerns itself with deductive processes. As such it encompasses the principles by which propositions are related to one another and the techniques of thought by which these relationships can be explored and valid statements made about them.



In its narrowest sense deductive logic divides into the logic of propositions (also called sentential logic) and the logic of predicates (or noun expressions). In its widest sense it embraces various theories of language (such as logical syntax and semantics), metalogic (the methodology of formal systems), theories of modalities (the analyses of the notions of necessity, possibility, impossibility, and contingency), and the study of paradoxes and logical fallacies. Both of these senses may be called formal or pure logic, in that they construct and analyze an abstract body of symbols, rules for stringing these symbols together into formulas, and rules for manipulating these formulas. When certain meanings are attached to these symbols and formulas, and this machinery is adapted and deployed over the concrete issues of a certain range of special subjects, logic is said to be applied. The analysis of questions that transcend the formal concerns of either pure or applied logic, such as the examination of the meaning and implications of the concepts and assumptions of either discipline, is the domain of the philosophy of logic.

Logic was developed independently and brought to some degree of systematization in China (5th to 3rd century BC) and India (from the 5th century BC through the 16th and 17th centuries AD). Logic as it is known in the West comes from Greece. Building on an important tradition of mathematics and rhetorical and philosophical argumentation, Aristotle in the 4th century BC worked out the first system of the logic of noun expressions. The logic of propositions originated in the work of Aristotle's pupil Theophrastus

and in that of the 4th-century Megarian school of dialecticians and logicians and the school of the Stoics. After the decline of Greek culture, logic reemerged first among Arab scholars in the 10th century. Medieval interest in logic dated from the work of St. Anselm of Canterbury and Peter Abelard. Its high point was the 14th century, when the Scholastics developed logic, especially the analysis of propositions, well beyond what was known to the ancients. Rhetoric and natural science largely eclipsed logic during the Renaissance. Modern logic began to develop with the work of the mathematician G.W. Leibniz, who attempted to create a universal calculus of reason. Great strides were made in the 19th century in the development of symbolic logic, leading to the highly fruitful merging of logic and mathematics in formal analysis.

Modern formal logic is the study of inference and proposition forms. Its simplest and most basic branch is that of the propositional calculus (or PC). In this logic, propositions or sentences form the only semantic category. These are dealt with as simple and remain unanalyzed; attention is focused on how they are related to other propositions by propositional connectives (such as "if . . . then," "and," "or," "it is not the case that," etc.) and thus formed into arguments. By representing propositions with symbols called variables and connectives with symbolic operators, and by deciding on a set of transformation rules (axioms that define validity and provide starting points for the derivation of further rules called theorems), it is possible to model and study the abstract characteristics and consequences of this formal system in a way similar to the investigations of pure mathematics. When the variables refer not to whole propositions but to noun expressions (or predicates) within propositions, the resulting formal system is known as a lower predicate calculus (or LPC).

Changing the operators, variables, or rules of such formal systems yields different logics. Certain systems of PC, for example, add a third "neuter" value to the two traditional possible values--true or false--of propositions. A major step in modern logic is the discovery that it is possible to examine and characterize other formal systems in terms of the logic resulting from their elements, operations, and rules of formation; such is the study of the logical foundations of mathematics, set theory, and logic itself.

Logic is said to be applied when it systematizes the forms of sound reasoning or a body of universal truths in some restricted field of thought or discourse. Usually this is done by adding extra axioms and special constants to some preestablished pure logic such as PC or LPC. Examples of applied logics are practical logic, which is concerned with the logic of choices, commands, and values; epistemic logic, which analyzes the logic of belief, knowing, and questions; the logics of physical application, such as temporal logic and mereology; and the logics of correct argumentation, fallacies, hypothetical reasoning, and so on.

Varieties of logical semantics have become the central area of study in the philosophy of logic. Some of the more important contemporary philosophical issues concerning logic are the following: What is the relation between logical systems and the real world? What are the limitations of logic, especially with regard to some of the assumptions of its wider senses and the incompleteness of first-order logic? What consequences stem from the nonrecursive nature of many mathematical functions?

Skepticism:

Skepticism is the philosophical attitude of doubting the knowledge claims set forth in various areas and asking what they are based upon, what they actually establish, and whether they are indubitable or necessarily true. Skeptics have thus challenged the alleged grounds of accepted assumptions in metaphysics, in science, in morals and manners, and especially in religion.

Skeptical philosophical attitudes are prominent throughout the course of Western philosophy; as early as the 5th century BC the Eleatic school of thinkers denied that reality could be described in terms of ordinary experience. Evidence of Skeptical thought appears even earlier in non-Western philosophy, in particular in the Upanisads, philosophic texts of the later Vedic period (c. 1000-c. 600 BC) in India.

Pyrrhon of Elis (c. 360-c. 272 BC), credited with founding Greek Skepticism, sought mental peace by avoiding commitment to any particular view; his approach gave rise in the 1st century BC to Pyrrhonism, proponents of which sought to achieve *epoche* (suspension of judgment) by systematically opposing various kinds of knowledge claims. One of its later leaders, Sextus Empiricus (2nd or 3rd century BC), challenged the claims of dogmatic philosophers to know more than what is evident. His goal was the state of *ataraxia*, wherein a person willing to suspend judgment would be relieved of the frustration of not knowing reality and would live, without dogma, according to appearances, customs, and natural inclination. The Pyrrhonians criticized Academic Skepticism, first developed in Plato's Academy in Greece in the 3rd century BC; the Academics argued that nothing could be known, and that only reasonable or probable standards could be established for knowledge.

Academic Skepticism survived into the Middle Ages in Europe and was considered and refuted by St. Augustine, whose conversion to Christianity convinced him that faith could lead to understanding. Among Islamic philosophers also there arose an antirational Skepticism that encouraged the acceptance of religious truths by faith.

Modern Skepticism dates from the 16th century, when the accepted Western picture of the world was radically altered by the rediscovery of ancient learning, by newly emerging science, and by voyages of exploration, as well as by the Reformation, which manifested fundamental disagreement among Roman Catholics and Protestants about the bases and criteria of religious knowledge. Prominent among modern Skeptical philosophers is Michel de Montaigne, who in the 17th century opposed science and all other disciplines and encouraged acceptance, instead, of whatever God reveals. His view was refuted in part by Pierre Gassendi, who remained doubtful about knowledge of reality but championed science as useful and informative. Reni Descartes also refuted Montaigne's Skepticism, maintaining that by doubting all beliefs that could possibly be false, a person can discover one genuinely indubitable truth: "I think, therefore I am" (*cogito ergo sum*), and that from that truth one can establish the existence of God and the existence of the external world, which Descartes claimed can be known through mathematical principles. At the end of the 17th century Pierre Bayle employed Skeptical arguments to urge that rational activity be abandoned in favour of pursuit of the conscience.

In the 18th century David Hume assembled some of the most important and enduring Skeptical arguments. He claimed that the very basis of modern science, the method of induction--by which regularities observed in the past justify the prediction that they will continue--is based on the uniformity of nature, itself an unjustifiable metaphysical assumption. Hume also sought to demonstrate that the notion of causality, the identity of the self, and the existence of an external world lacked any basis. In rebuttal, Immanuel Kant maintained that, in order to have and describe even the simplest experience, certain universal and necessary conditions must prevail.

In the 19th century Soren Kierkegaard developed religious Existentialist thought from an irrational Skepticism, asserting that certainty can be found only by making an unjustifiable "leap into faith." Nonreligious Existentialist writers, such as Albert Camus in the 20th century, have claimed that rational and scientific examination of the world shows it to be unintelligible and absurd, but that it is necessary for the individual to struggle with that absurdity. In the 20th century other forms of Skepticism have been expressed among Logical Positivist and Linguistic philosophers.

Fallacies:

Here is a list of everyday fallacies take from Peter A. Angeles Dictionary of Philosophy-- published by Barnes and Noble, copyright 1981.

Fallacy, classification of informal. Informal fallacies may be classified in a variety of ways. Three general categories: (a) Material fallacies have to do with the facts (the matter, the content) of the argument in question. Two subcategories of material fallacies are: (1) fallacies of evidence, which refer to arguments that do not provide the required factual support (ground, evidence) for their conclusions, and (2) fallacies of irrelevance (or relevance) which refer to arguments that have supporting statements that are irrelevant to the conclusion being asserted and therefore cannot establish the truth of that conclusion. (b) Linguistic fallacies have to do with defects in arguments such as ambiguity (in which careless shifts of meanings or linguistic imprecisions lead to erroneous conclusions), vagueness, incorrect use of words, lack of clarity, linguistic inconsistencies, circularities. (c) Fallacies of irrelevant emotional appeal have to do with affecting behavior (responses, attitudes). That is, arguments are presented in such a way as to appeal to one's prejudices, biases, loyalty, dedication, fear, guilt, and so on. They persuade, cajole, threaten, or confuse in order to win assent to an argument.

Fallacy, types of informal. Sometimes semi-formal or quasi-formal fallacies. The following is a list of 40 informal fallacies which is by no means exhaustive. No attempt has been made to subsume them under general categories such as Fallacies, Classification of Informal [which I will also include].

1. Black-and-white fallacy. Arguing (a) with the use of sharp ("black-and-white") distinctions despite any factual or theoretical support for them, or (b) by classifying any middle point between the extremes ("black-and-white") as one of the extremes. Examples: "If he is an atheist then he is a decent person." "He is either a conservative or a liberal." "He must not be peace-loving, since he participated in picketing the American embassy."

2. Fallacy of argumentum ad baculum (argument from power or force.) The Latin means "an argument according to the stick." "argument by means of the rod," "argument using force." Arguing to support the acceptance of an argument by a threat, or use of force. Reasoning is replaced by force, which results in the termination of logical argumentation, and elicits other kinds of behavior (such as fear, anger, reciprocal use of force, etc.).

3. Fallacy of argumentum ad hominem (argument against the man) [a personal favorite of mine]. The Latin means "argument to the man." (a) Arguing against, or rejecting a person's views by attacking or abusing his personality, character, motives, intentions, qualifications, etc. as opposed to providing evidence why the views are incorrect. Example: "What John said should not be believed because he was a Nazi sympathizer." [Well, there goes Heidegger.]

4. Fallacy of argumentum ad ignorantiam (argument from ignorance). The Latin means "argument to ignorance." (a) Arguing that something is true because no one has proved it to be false, or (b) arguing

that something is false because no one has proved it to be true. Examples: (a) Spirits exist since no one has as yet proved that there are not any. (b) Spirits do not exist since no one has as yet proved their existence. Also called the appeal to ignorance: the lack of evidence (proof) for something is used to support its truth.

5. Fallacy of argumentum ad misericordiam (argument to pity). Arguing by appeal to pity in order to have some point accepted. Example: "I've got to have at least a B in this course, Professor Angeles. If I don't I won't stand a chance for medical school, and this is my last semester at the university." Also called the appeal to pity.

6. Fallacy of argumentum ad personam (appeal to personal interest). Arguing by appealing to the personal likes (preferences, prejudices, predispositions, etc.) of others in order to have an argument accepted.

7. Fallacy of argumentum ad populum (argument to the people). Also the appeal to the gallery, appeal to the majority, appeal to what is popular, appeal to popular prejudice, appeal to the multitude, appeal to the mob instinct [appeal to the stupid, stinking masses]. Arguing in order to arouse an emotional, popular acceptance of an idea without resorting to logical justification of the idea. An appeal is made to such things as biases, prejudices, feelings, enthusiasms, attitudes of the multitude in order to evoke assent rather than to rationally support the idea.

8. Fallacy of argumentum ad verecundiam (argument to authority or to veneration) [another of my personal favorites]. (a) appealing to authority (including customs, traditions, institutions, etc.) in order to gain acceptance of a point at issue and/or (b) appealing to the feelings of reverence or respect we have of those in authority, or who are famous. Example: "I believe that the statement 'YOU cannot legislate morality' is true, because President Eisenhower said it."

9. Fallacy of accent. Sometimes classified as ambiguity of accent. Arguing to conclusions from undue emphasis (accent, tone) upon certain words or statements. Classified as a fallacy of ambiguity whenever this emphasis creates an ambiguity or AMPHIBOLY in the words or statements used in an argument. Example: "The queen cannot but be praised." [also "We are free iff we could have done otherwise."-- as this statement is used by incompatibilists about free-will and determinism.]

10. Fallacy of accident. Also called by its Latin name a dicto simpliciter ad dictum secundum quid. (a) Applying a general rule or principle to a particular instance whose circumstances by "accident" do not allow the proper application of that generalization. Example: "It is a general truth that no one should lie. Therefore, no one should lie if a murderer at the point of a knife asks you for information you know would lead to a further murder." (b) The error in argumentation of applying a general statement to a situation to which it cannot, and was not necessarily intended to, be applied.

11. Fallacy of ambiguity. An argument that has at least one ambiguous word or statement from which a misleading or wrong conclusion is drawn.

12. Fallacy of amphiboly. Arguing to conclusions from statements that themselves are amphibolous--ambiguous because of their syntax (grammatical construction). Sometimes classified as a fallacy of ambiguity.

13. Fallacy of begging the question. (a) Arriving at a conclusion from statements that themselves are questionable and have to be proved but are assumed true. Example: The universe has a beginning. Every thing that has a beginning has a beginner. Therefore, the universe has a beginner called God. This assumes (begs the question) that the universe does indeed have a beginning and also that all things that have a beginning have a beginner. (b) Assuming the conclusion as part of the conclusion in the premises of an argument. Sometimes called circular reasoning, vicious circularity, vicious circle fallacy [Continental Philosophy-- sorry, I just couldn't resist]. Example: "Everything has a cause. The universe is a thing. Therefore, the universe is a thing that has a cause." (c) Arguing in a circle. One statement is supported by reference to another statement which is itself supported by reference to the first statement [such as a coherentist account of knowledge/truth]. Example: "Aristocracy is the best form of government because the best form of government is that which has strong aristocratic leadership."

14. Fallacy of complex question (or loaded question). (a) Asking questions for which either a yes or no answer will incriminate the respondent. The desired answer is already tacitly assumed in the question and no qualification of the simple answer is allowed. Example: "Have you discontinued the use of opiates?" (b) Asking questions that are based on unstated attitudes or questionable (or unjustified) assumptions. These questions are often asked rhetorically of the respondent in such a way as to elicit an agreement with those attitudes or assumptions from others. Example: "How long are you going to put up with this brutality?"

15. Fallacy of composition. Arguing (a) that what is true of each part of a whole is also (necessarily) true of the whole itself, or (b) what is true of some parts is also (necessarily) true of the whole itself. Example: "Each member (or some members) of the team is married, therefore the team also has (must have) a wife." [A less silly example-- you promise me that you will come to Portland tomorrow, you also promise someone else that you will go to Detroit tomorrow. Now, you ought to be in Portland tomorrow, and you ought to be in Detroit tomorrow (because you ought to keep your promises). However, it does not follow that you ought to be in both Portland and Detroit tomorrow (because ought implies can).] Inferring that a collection has a certain characteristic merely on the basis that its parts have them erroneously proceeds from regarding the collection DISTRIBUTIVELY to regarding it COLLECTIVELY.

16. Fallacy of consensus gentium. Arguing that an idea is true on the basis (a) that the majority of people believe it and/or (b) that it has been universally held by all men at all times. Example: "God exists because all cultures have had some concept of a God."

17. Fallacy of converse accident. Sometimes converse fallacy of accident. Also called by its Latin name a dicto secundum quid ad dictum simpliciter. The error of generalizing from atypical or exceptional instances. Example: "A shot of warm brandy each night helps older people relax and sleep better. People

in general ought to drink warm brandy to relieve their tension and sleep better."

18. Fallacy of division. Arguing that what is true of a whole is (a) also (necessarily) true of its parts and/or (b) also true of some of its parts. Example: "The community of Pacific Palisades is extremely wealthy. Therefore, every person living there is (must be) extremely wealthy (or therefor Adma, who lives there, must be extremely wealthy." Inferring that the parts of a collection have certain characteristic merely on the basis that their collection has them erroneously proceeds from regarding the collection collectively to regarding it distributively.

19. Fallacy of equivocation. An argument in which a word is used with one meaning in one part of the argument and with another meaning in another part. A common example: "The end of a thing is its perfection; death is the end of life; hence, death is the perfection of life." 20. Fallacy of non causa pro causa. the Latin may be translated as "there is no cause of the sort that has been given as the cause." (a) Believing that something is the cause of an effect when in reality it is not. Example: "My incantations caused it to rain." (b) Arguing so that a statement appears unacceptable because it implies another statement that is false (but in reality does not).

21. Fallacy of post hoc ergo propter hoc. The Latin means "after this therefore the consequence (effect) of this," or "after this therefore because of this." Sometimes simply fallacy of false cause. Concluding that one thing is the cause of another thing because it precedes it in time. A confusion between the concept of succession and that of causation. Example: "A black cat ran across my path. Ten minutes later I was hit by a truck. Therefore, the cat's running across my path was the cause of my being hit by a truck."

22. Fallacy of hasty generalization. Sometimes fallacy of hasty induction. An error of reasoning whereby a general statement is asserted (inferred) based on (a) limited information or (b) inadequate evidence, or (c) an unrepresentative sampling.

23. Fallacy of ignoratio elenchi (irrelevant conclusion). An argument that is irrelevant; that argues for something other than that which is to be proved and thereby in no way refutes (or supports) the points at issue. Example: A lawyer in defending his alcoholic client who has murdered three people in a drunken spree argues that alcoholism is a terrible disease and attempts should be made to eliminate it. IGNORATIO ELENCHI is sometimes used as a general name for all fallacies that are based on irrelevancy (such as ad baculum, ad hominem, as misericordiam, as populum, ad verecundiam, consensus gentium, etc.)

24. Fallacy of inconsistency. Arguing from inconsistent statements, or to conclusions that are inconsistent with the premises. See fallacy of tu quoque below.

25. Fallacy of irrelevant purpose. Arguing against something on the basis that it has not fulfilled its purpose (although in fact that was not its intended purpose).

- 26 Fallacy of 'is' to 'ought.' Arguing from premises that have only descriptive statements (is) to a conclusion that contains an ought, or a should.
27. Fallacy of limited (or false) alternatives. The error of insisting without full inquiry or evidence that the alternatives to a course of action have been exhausted and/or are mutually exclusive.
28. Fallacy of many questions. Sometimes fallacy of the false question. Asking a question for which a single and simple answer is demanded yet the question (a) requires a series of answers, and/or (b) requires answers to a host of other questions, each of which have to be answered separately. Example: "Have you left school?"
29. Fallacy of misleading context. Arguing by misrepresenting, distorting, omitting or quoting something out of context.
30. Fallacy of prejudice. Arguing from a bias or emotional identification or involvement with an idea (argument, doctrine, institution, etc.).
31. Fallacy of red herring. Ignoring criticism of an argument by changing attention to another subject. Examples: "You believe in abortion, yet you don't believe in the right-to-die-with-dignity bill before the legislature."
32. Fallacy of slanting. Deliberately omitting, deemphasizing, or overemphasizing certain points to the exclusion of others in order to hide evidence that is important and relevant to the conclusion of the argument and that should be taken into account in an argument.
33. Fallacy of special pleading. (a) Accepting an idea or criticism when applied to an opponent's argument but rejecting it when applied to one's own argument. (b) rejecting an idea or criticism when applied to an opponent's argument but accepting it when applied to one's own.
34. Fallacy of the straw man. Presenting an opponent's position in as weak or misrepresented a version as possible so that it can be easily refuted. Example: "Darwinism is in error. It claims that we are all descendents from an apelike creature, from which we evolved according to natural selection. No evidence of such a creature has been found. No adequate and consistent explanation of natural selection has been given. Therefore, evolution according to Darwinism has not taken place."
35. Fallacy of the beard. Arguing (a) that small or minor differences do not (or cannot) make a difference, or are not (or cannot be) significant, or (b) arguing so as to find a definite point at which something can be named. For example, insisting that a few hairs lost here and there do not indicate anything about my impending baldness; or trying to determine how many hairs a person must have before he can be called bald (or not bald).
36. Fallacy of tu quoque (you also). (a) Presenting evidence that a person's actions are not consistent with

that for which he is arguing. Example: "John preaches that we should be kind and loving. He doesn't practice it. I've seen him beat up his kids." (b) Showing that a person's views are inconsistent with what he previously believed and therefore (1) he is not to be trusted, and/or (2) his new view is to be rejected. Example: "Judge Egner was against marijuana legislation four years ago when he was running for office. Now he is for it. How can you trust a man who can change his mind on such an important issue? His present position is inconsistent with his earlier view and therefore it should not be accepted." (c) Sometimes related to the Fallacy of two wrongs make a right. Example: The Democrats for years used illegal wiretapping; therefore the Republicans should not be condemned for their use of illegal wiretapping.

37. Fallacy of unqualified source. Using as support in an argument a source of authority that is not qualified to provide evidence.

38. Gambler's fallacy. (a) Arguing that since, for example, a penny has fallen tails ten times in a row then it will fall heads the eleventh time or (b) arguing that since, for example, an airline has not had an accident for the past ten years, it is then soon due for an accident. The gambler's fallacy rejects the assumption in probability theory that each event is independent of its previous happening. the chances of an event happening are always the same no matter how many times that event has taken place in the past. Given those events happening over a long enough period of time then their frequency would average out to 1/2. Sometimes referred to as the Monte Carlo fallacy (a generalized form of the gambler's fallacy): The error of assuming that because something has happened less frequently than expected in the past, there is an increased chance that it will happen soon.

39. Genetic fallacy. Arguing that the origin of something is identical with that thing with that from which it originates. Example: 'Consciousness originates in neural processes. Therefore, consciousness is (nothing but) neural processes. Sometimes referred to as the nothing-but fallacy, or the REDUCTIVE FALLACY. (b) Appraising or explaining something in terms of its origin, or source, or beginnings. (c) Arguing that something is to be rejected because its origins are [unknown] and/or suspicious.

40. Pragmatic fallacy. Arguing that something is true because it has practical effects upon people: it makes them happier, easier to deal with, more moral, loyal, stable. Example: "An immortal life exists because without such a concept men would have nothing to live for. There would be no meaning or purpose in life and everyone would be immoral."

41. Pathetic fallacy. Incorrectly projecting (attributing) human emotions, feeling, intentions, thoughts, traits upon events or objects which do not possess the capacity for such qualities.

42. Naturalist fallacy (ethics). 1. The fallacy of reducing ethical statements to factual statements, to statements about natural events. 2. The fallacy of deriving (deducing) ethical statements from nonethical statements. [is/ought fallacy]. 3. The fallacy of defining ethical terms in nonethical (descriptive, naturalistic, or factual) terms [ought/is fallacy].

Problem Solving:

Still more complex forms of realistic thinking seem to occur when tasks are presented in which the goal is impossible (or very difficult) to achieve directly. In such situations, people commonly appear to pass through intermediate stages of exploring and organizing their resources; indeed, one may first need to exert himself in understanding the problem itself before he can begin to seek possible directions toward a solution. Familiar examples of problem-solving tasks include anagrams (e.g., rearrange "lpepa" to spell "apple"); mathematical problems; mechanical puzzles; verbal "brain teasers" (e.g., Is it legal for a man to marry his widow's sister? answer below); and, in a more practical sense, design and construction problems. Also of interest are issues of human relations, games, and questions pertinent to economics and politics.

Trial and error.

Problem-solving activity falls broadly into two categories: one emphasizes simple trial and error; the other requires some degree of insight. In trial and error, the individual proceeds mainly by exploring and manipulating elements of the problem situation in an effort to sort out possibilities and to run across steps that might carry him closer to the goal. This behaviour is most likely to be observed when the problem solver lacks advance knowledge about the character of the solution, or when no single rule seems to underlie the solution. Trial-and-error activity is not necessarily overt (as in one's observable attempts to fit together the pieces of a mechanical puzzle); it may be implicit or vicarious as well, the individual reflecting on the task and symbolically testing possibilities by thinking about them.

Solutions through insight.

In striving toward insight, a person tends to exhibit a strong orientation toward understanding principles that might bear on the solution sought. The person actively considers what is required by the problem, noting how its elements seem to be interrelated, and seeks some rule that might lead directly to the goal. The insightful thinker is likely to centre on the problem to understand what is needed, to take the time to organize his resources, and to recentre on the problem (reinterpret the situation) in applying any principle that seems to hold promise.

Direction and flexibility characterize insightful problem solving. The thinker directs or guides his steps toward solution according to some plan; he exhibits flexibility in his ability to modify or to adapt procedures as required by his plan and in altering the plan itself. Both characteristics are influenced by the thinker's attitudes and by environmental conditions. If, for example, the task is to empty a length of glass tubing of water (without breaking it) by removing wax plugs about a half-inch up the tube from each end, and the only potential tools are a few objects ordinarily found on a desk top, the usual appearance and functions of such common objects may make it difficult for the problem solver to see how they can be adapted to fit task requirements. If a paper clip is perceived as holding a sheaf of papers in the usual way, such perception would tend to interfere with the individual's ability to employ the principle that the clip's shape could be changed: straightened out for use in poking a hole in the wax.

Formal, logical processes.

A special form of problem solving employs formal, systematic, logical thinking. The thinker develops a series of propositions, often as postulates; e.g., the shortest distance between two points is a straight line. He builds a structure of arguments in which statements are consistent with each other in reaching some goal, such as defining the area of a triangle. This kind of logical, mathematical reasoning applies formal rules in supporting the validity of successive propositions.

Both inductive and deductive processes may be used by a problem solver. In inductive thinking one considers a number of particular or specific items of information to develop more inclusive (or general) conceptions. After aspirin was synthesized, for example, some people who swallowed the substance reported that it relieved their particular headaches. Through induction, the reports of these specific individuals were the basis for developing a more inclusive notion: aspirin may be helpful in relieving headaches in general.

Deduction is reasoning from general propositions--or hypotheses--to more specific instances or statements. Thus, after the general hypothesis about the effectiveness of aspirin had been put forward, physicians began to apply it to specific, newly encountered headache cases. The deduction was that, if aspirin is generally useful in managing pains in the head, it might also be helpful in easing pains elsewhere in the body. Although a person may deliberately choose to use induction or deduction, people typically shift from one to the other, depending on the exigencies of the reasoning process.

Students of problem solving almost invariably have endorsed some variety of mediation theory in their efforts to understand realistic thinking. The assumptions in that kind of theory are that implicit (internal) representations of experience are stored in and elicited from memory and are linked together during the period between the presentation of a stimulus and the implementation of a response. Those theorists who prefer to avoid the use of unobservable "entities" (e.g., "mind") increasingly have been invoking the nervous system (particularly the brain) as the structure that mediates such functions.

Answer to brain teaser: dead men can't marry

Principle of Falsification:

Being unrestricted, scientific theories cannot be verified by any possible accumulation of observational evidence. The formation of hypothesis is a creative process of the imagination and is not a passive reaction to observed regularities. A scientific test consists in a persevering search for negative, falsifying instances. If a hypothesis survives continuing and serious attempts to falsify it, then it has "proved its mettle" and can be provisionally accepted, but it can never be established conclusively. Later corroboration generates a series of hypothesis into a scientific theory.

Thus, the core element of a scientific hypothesis is that it must be capability of being proven false. For example, the hypothesis that "atoms move because they are pushed by small, invisible, immaterial demons" is pseudo-science since the existence of the demons cannot be proven false (i.e. cannot be tested at all).

Reductionism :

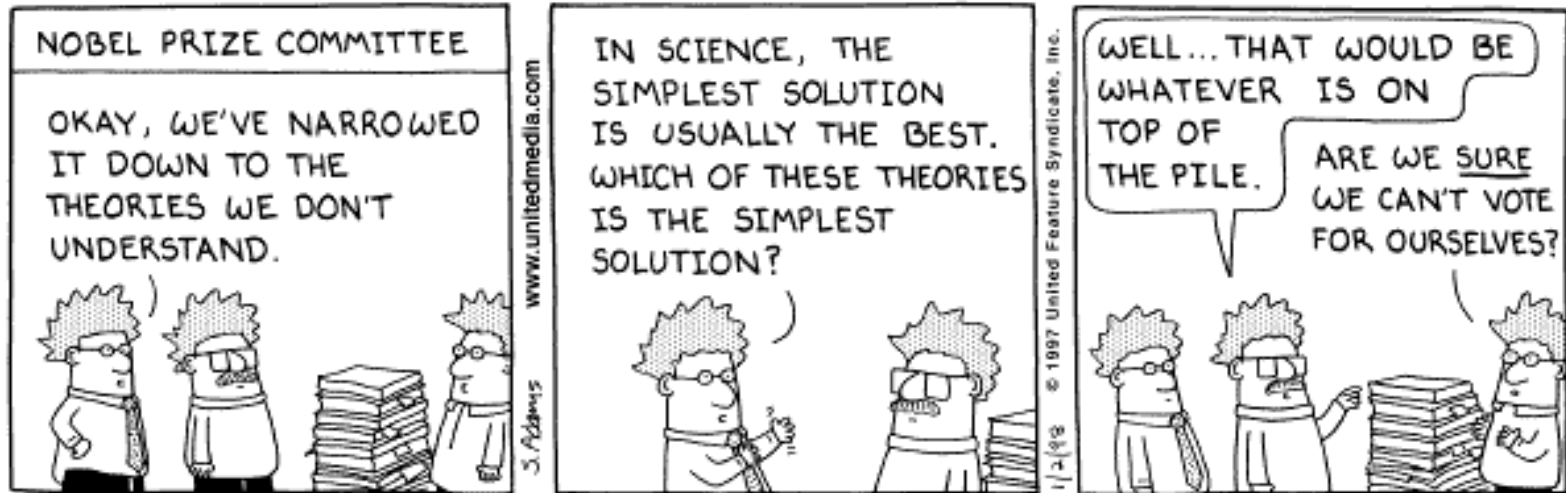
Reductionism is a view that asserts that entities of a given kind are collections or combinations of entities of a simpler or more basic kind or that expressions denoting such entities are definable in terms of expressions denoting the more basic entities. Thus, the ideas that physical bodies are collections of atoms or that thoughts are combinations of sense impressions are forms of reductionism.

Two very general forms of reductionism have been held by philosophers in the 20th century: (1) Logical positivists have maintained that expressions referring to existing things or to states of affairs are definable in terms of directly observable objects, or sense-data, and, hence, that any statement of fact is equivalent to some set of empirically verifiable statements. In particular, it has been held that the theoretical entities of science are definable in terms of observable physical things, so that scientific laws are equivalent to combinations of observation reports. (2) Proponents of the unity of science have held the position that the theoretical entities of particular sciences, such as biology or psychology, are definable in terms of those of some more basic science, such as physics; or that the laws of these sciences can be explained by those of the more basic science.

The logical positivist version of reductionism also implies the unity of science insofar as the definability of the theoretical entities of the various sciences in terms of the observable would constitute the common basis of all scientific laws. Although this version of reductionism is no longer widely accepted, primarily because of the difficulty of giving a satisfactory characterization of the distinction between theoretical and observational statements in science, the question of the reducibility of one science to another remains controversial.

Occam's Razor :

William of Occam (1284-1347) was an English philosopher and theologian. His work on knowledge, logic and scientific inquiry played a major role in the transition from medieval to modern thought. He based scientific knowledge on experience and self-evident truths, and on logical propositions resulting from those two sources. In his writings, Occam stressed the Aristotelian principle that *entities must not be multiplied beyond what is necessary*. This principle became known as Occam's Razor, a problem should be stated in its basic and simplest terms. In science, the simplest theory that fits the facts of a problem is the one that should be selected.



Spencer's Scientism :

The English sociologist Herbert Spencer was perhaps the most important popularizer of science and philosophy in the 19th century. Presenting a theory of evolution prior to Charles Darwin's ``On the Origin of Species by Means of Natural Selection'', Spencer argued that all of life, including education, should take its essential lessons from the findings of the sciences. In ``Education: Intellectual, Moral, and Physical" (1860) he insisted that the answer to the question "What knowledge is of most worth?" is the knowledge that the study of science provides.

While the educational methodology Spencer advocated was a version of the sense realism espoused by reformers from Ratke and Comenius down to Pestalozzi, Spencer himself was a social conservative. For him, the value of science lies not in its possibilities for making a better world but in the ways science teaches man to adjust to an environment that is not susceptible to human engineering. Spencer's advocacy of the study of science was an inspiration to the American Edward Livingston Youmans and others who argued that a scientific education could provide a culture for modern times superior to that of classical education.

Paradigm Shift:

In Thomas Kuhn's landmark book, *The Structure of Scientific Revolutions*, he argued that scientific research and thought are defined by "paradigms," or conceptual world-views, that consist of formal theories, classic experiments, and trusted methods. Scientists typically accept a prevailing paradigm and try to extend its scope by refining theories, explaining puzzling data, and establishing more precise measures of standards and phenomena. Eventually, however, their efforts may generate insoluble theoretical problems or experimental anomalies that expose a paradigm's inadequacies or contradict it altogether. This accumulation of difficulties triggers a crisis that can only be resolved by an intellectual revolution that replaces an old paradigm with a new one. The overthrow of Ptolemaic cosmology by Copernican heliocentrism, and the displacement of Newtonian mechanics by quantum physics and general relativity, are both examples of major paradigm shifts.

Kuhn questioned the traditional conception of scientific progress as a gradual, cumulative acquisition of knowledge based on rationally chosen experimental frameworks. Instead, he argued that the paradigm determines the kinds of experiments scientists perform, the types of questions they ask, and the problems they consider important. A shift in the paradigm alters the fundamental concepts underlying research and inspires new standards of evidence, new research techniques, and new pathways of theory and experiment that are radically incommensurate with the old ones.

Kuhn's book revolutionized the history and philosophy of science, and his concept of paradigm shifts was extended to such disciplines as political science, economics, sociology, and even to business management.

Materialism :

Materialism in philosophy, the view that all facts (including facts about the human mind and will and the course of human history) are causally dependent upon physical processes, or even reducible to them.

The many materialistic philosophies that have arisen from time to time may be said to maintain one or more of the following theses: (1) that what are called mental events are really certain complicated physical events, (2) that mental processes are entirely determined by physical processes (e.g., that "making up one's mind," while it is a real process that can be introspected, is caused by bodily processes, its apparent consequences following from the bodily causes), (3) that mental and physical processes are two aspects of what goes on in a substance at once mental and bodily (this thesis, whether called "materialistic" or not, is commonly opposed by those who oppose materialism), and (4) that thoughts and wishes influence an individual's life, but that the course of history is determined by the interaction of masses of people and masses of material things, in such a way as to be predictable without reference to the "higher" processes of thought and will.

Materialism is thus opposed to philosophical dualism or idealism and, in general, to belief in God, in disembodied spirits, in free will, or in certain kinds of introspective psychology. Materialistic views insist upon settling questions by reference to public observation and not to private intuitions. Since this is a maxim which scientists must profess within the limits of their special inquiries, it is natural that philosophies which attach the highest importance to science should lean toward materialism. But none of the great empiricists have been satisfied (at least for long) with systematic materialism.

The Greek atomists of the 5th century BC (Leucippus and Democritus) offered simple mechanical explanations of perception and thought--a view that was condemned by Socrates in the *Phaedo*. In the 17th century Thomas Hobbes and Pierre Gassendi, inspired by the Greek atomists, used materialistic arguments in defense of science against Aristotle and against the orthodox tradition, and in the next century the materialists of the Enlightenment (Julien de Lamettrie, Paul d'Holbach, and others) attempted to provide a detailed account of psychology.

During the modern period, the question of materialism came to be applied on the one hand to problems of method and interpretation in science (Henri Bergson, Samuel Alexander, A.N. Whitehead) and on the other hand to the interpretation of human history (G.W.F. Hegel, Auguste Comte, Karl Marx). Marx offered a new kind of materialism, dialectic and not mechanistic, and embracing all sciences.

In the 20th century, materialistic thought faced novel developments in the sciences and in philosophy. In physics, relativity and quantum theory modified, though they did not abandon, the notions of cause and of universal determinism. In psychology, J.B. Watson's behaviourism, an extreme form of materialism, did not find general acceptance; and researches both in psychology and in psychoanalysis made it impossible to hold any simple direct view of the mind's dependence on the processes and mechanisms of the nervous system. In philosophy, further reflection suggested to many that it is futile to try to erect a system of belief, whether materialistic or otherwise, on the basis of the concepts of science and of

common sense (especially those of cause and of explanation).

Relativism :

Relativism is the view that what is right or wrong and good or bad is not absolute but variable and relative, depending on the person, circumstances, or social situation. The view is as ancient as Protagoras, a leading Greek Sophist of the 5th century BC, and as modern as the scientific approaches of sociology and anthropology.

Many people's understanding of this view is often vague and confused. It is not simply the belief, for example, that what is right depends on the circumstances, because everyone, including the absolutists, agrees that circumstances can make a difference; it is acknowledged that whether it is right for a man to enter a certain house depends upon whether he is the owner, a guest, a police officer with a warrant, or a burglar. Nor is it the belief that what someone thinks is right is relative to his social conditioning, for again anyone can agree that there are causal influences behind what people think is right. Relativism is, rather, the view that what is really right depends solely upon what the individual or the society thinks is right. Because what one thinks will vary with time and place, what is right will also vary accordingly. Relativism is, therefore, a view about the truth status of moral principles, according to which changing and even conflicting moral principles are equally true, so that there is no objective way of justifying any principle as valid for all people and all societies.

The sociological argument for relativism proceeds from the diversity of different cultures. Ruth Benedict, an American anthropologist, suggested, for example, in *Patterns of Culture* (1934) that the differing and even conflicting moral beliefs and behavior of the North American Indian Kwakiutl, Pueblo, and Dobu cultures provided standards that were sufficient within each culture for its members to evaluate correctly their own individual actions. Thus, relativism does not deprive one of all moral guidance. However, some anthropologists, such as Clyde Kluckhohn and Ralph Linton, have pointed up certain "ethical universals," or cross-cultural similarities, in moral beliefs and practices--such as prohibitions against murder, incest, untruth, and unfair dealing--that are more impressive than the particularities of moral disagreement, which can be interpreted as arising within the more basic framework that the universals provide. Some critics point out, further, that a relativist has no grounds by which to evaluate the social criticism arising within a free or open society, that his view appears in fact to undercut the very idea of social reform.

A second argument for relativism is that of the skeptic who holds that moral utterances are not cognitive statements, verifiable as true or false, but are, instead, emotional expressions of approval or disapproval or are merely prescriptions for action. In this view, variations and conflicts between moral utterances are relative to the varying conditions that occasion such feelings, attitudes, or prescriptions, and there is nothing more to be said. Critics of the skeptical view may observe that classifying moral utterances as emotive expressions does not in itself disqualify them from functioning simultaneously as beliefs with cognitive content. Or again, they may observe that, even if moral utterances are not cognitive, it does not follow that they are related, as the relativist suggests, only to the changeable elements in their background; they may also be related in a special way to needs and wants that are common and essential to human nature and society everywhere and in every age. If so, the criticism continues, these needs can provide good reasons for the justification of some moral utterances over others. The relativist will then

have to reply either that human nature has no such common, enduring needs or that, if it does, they cannot be discovered and employed to ground man's moral discourse.

Determinism :

Determinism is the theory that all events, including moral choices, are completely determined by previously existing causes that preclude free will and the possibility that humans could have acted otherwise. The theory holds that the Universe is utterly rational because complete knowledge of any given situation assures that unerring knowledge of its future is also possible. Pierre-Simon, Marquis de Laplace, in the 18th century framed the classical formulation of this thesis. For him, the present state of the Universe is the effect of its previous state and the cause of the state that follows it. If a mind, at any given moment, could know all of the forces operating in nature and the respective positions of all its components, it would thereby know with certainty the future and the past of every entity, large or small. The Persian poet Omar Khayyam expressed a similar deterministic view of the world in the concluding half of one of his quatrains: "And the first Morning of Creation wrote / What the Last Dawn of Reckoning shall read."

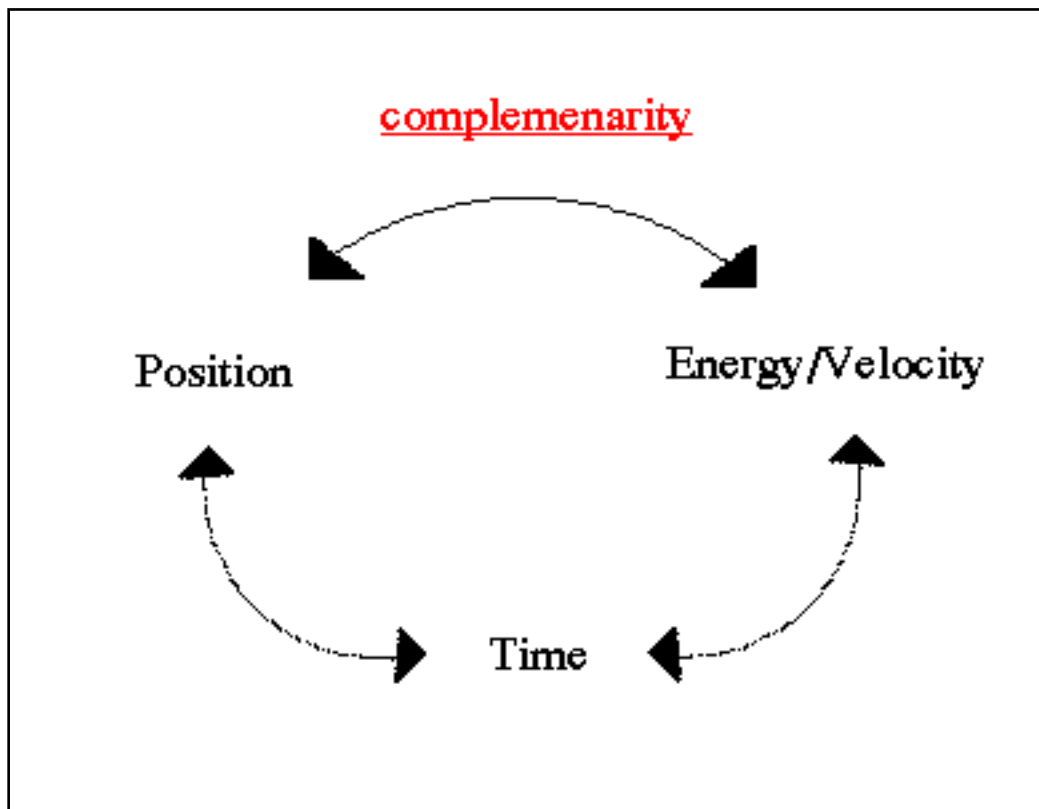
Indeterminism, on the other hand, though not denying the influence of behavioral patterns and certain extrinsic forces on human actions, insists on the reality of free choice. Exponents of determinism strive to defend their theory as compatible with moral responsibility by saying, for example, that evil results of certain actions can be foreseen, and this in itself imposes moral responsibility and creates a deterrent external cause that can influence actions.

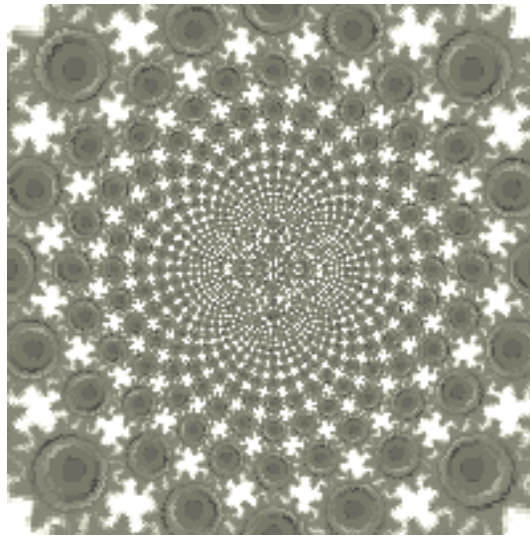
Newtonian Physics :

The publication, in 1687, of *Mathematical Principles of Natural Philosophy* by English scientist Sir Issac Newton was the culmination of a reduction philosophy of science that used force and action to explain the fundamentals of motion/energy, time and space/position. Newton showed how both the motions of heavenly bodies and the motions of objects on or near the surface of the Earth could be explained by four simple laws; the three [laws of motion](#) and the [law of universal gravitation](#).

This brilliant synthesis of several apparently different topics was an extension of the work of Galileo's law of falling bodies and Kepler's law of planetary motion. Newton developed a new form of mathematics, calculus, as a framework to his new physics.

Newtonian physics is often referred to as classical physics after the development of modern physics (quantum physics) in the 1920's.





Clockwork Universe :

The 17th century was a time of intense religious feeling, and nowhere was that feeling more intense than in Great Britain. There a devout young man, Isaac Newton, was finally to discover the way to a new synthesis in which truth was revealed and [God](#) was preserved.

Newton was both an experimental and a mathematical genius, a combination that enabled him to establish both the Copernican system and a new mechanics. His method was simplicity itself: "from the phenomena of motions to investigate the forces of nature, and then from these forces to demonstrate the other phenomena." Newton's genius guided him in the selection of phenomena to be investigated, and his creation of a fundamental mathematical tool--the calculus (simultaneously invented by Gottfried Leibniz)--permitted him to submit the forces he inferred to calculation. The result was *Philosophiae Naturalis Principia Mathematica* (Mathematical Principles of Natural Philosophy, usually called simply the Principia), which appeared in 1687. Here was a new physics that applied equally well to terrestrial and celestial bodies. Copernicus, Kepler, and Galileo were all justified by Newton's analysis of forces. Descartes was utterly routed.

Newton's three laws of motion and his principle of universal gravitation sufficed to regulate the new cosmos, but only, Newton believed, with the help of [God](#). Gravity, he more than once hinted, was direct divine action, as were all forces for order and vitality. Absolute space, for Newton, was essential, because space was the "sensorium of [God](#)," and the divine abode must necessarily be the ultimate coordinate system.

Mechanics came to be regarded as the ultimate explanatory science: phenomena of any kind, it was believed, could and should be explained in terms of mechanical conceptions. Newtonian physics was used to support the deistic view that [God](#) had created the world as a perfect machine that then required no further interference from Him, the Newtonian world machine or Clockwork Universe. These ideals were typified in Laplace's view that a Supreme Intelligence, armed with a knowledge of Newtonian laws of nature and a knowledge of the positions and velocities of all particles in the Universe at any moment,

could deduce the state of the Universe at any time.

To the eighteenth and much of the nineteenth centuries, Newton himself became idealized as the perfect scientist: cool, objective and never going beyond what the facts warrent to speculative hypothesis. The *Principia* became the model of scientific knowledge, a synthesis expressing the Enlightenment conception of the Universe as a rationally ordered machine governed by simple mathematical laws. To some, even the fundamental principles from which this system was deduced seemed to be a priori truths, attainable by reason alone.

Metaphysics:

Metaphysics means the study of topics about physics (or science in general), as opposed to the scientific subject itself. Metaphysics has come to mean the study of 'theories about theories' and to include the discussion of how our science describes reality.

Hume argued that meaning can be attached only to those ideas that stem directly from our observations of the world, or from deductive schemes such as mathematics. This was called empiricism and treats the facts of experience as the foundation for what we can know.

Later, Kant proposed that there exists two forms of knowledge, sense data obtained by direct perception and 'a priori' knowledge obtained by reasoning and higher intellectual functions. Our reasoning can only be applied to the realm of experience, or things-as-we-see-them, and this can tell us nothing about the things-in-themselves, called idealism.

Metaphysics, by its inquiry into the knowledge building process, also addresses what we mean by reality. In particular, science has been the pursuit of those aspects of reality which endure and are absolutely constant. Thus, the oldest question raised by our philosophies of science is how can the changing world of experience be connected to the unchanging world of abstract concepts?

The earliest attempt at a solution to this question comes from Plato and his [Theory of Forms](#). In Plato's philosophy, the true reality lay in the transcendent world of unchanging, perfect, abstract Ideas or Forms, a domain of mathematical relationships and fixed geometrical structures. The world of Forms was eternal and immutable, beyond space and time, home to the Good, a timeless Deity. The changing world of material objects and forces was the domain of the Demiurge, whose task was to fashion existing matter into an ordered state using the Forms as templates. Being less than perfect, the material world is continually disintegrating and being reassembled, a state of flux to our sense impressions.

Platonic realm is beyond space and time, inhabited by God who has a set of well-defined qualities (perfection, simplicity, timelessness, omnipotence, omniscience) It is a dilemma on whether when we talk about aspects of physics, such as sub-atomic particles, as having independent existence apart from the theory or model. Quantum physics enables us to relate to different observations made on sub-atomic particles. But quantum physics is a procedure for connecting these observations into a consistent logical scheme. It is helpful to encapsulate the abstract concept into physical language, but that may not mean that the sub-atomic particles are actually there as well-defined entities. It is this ill-defined view of reality that causes many individuals to reject the scientific view as too vague or malleable. However, they fail to see that the strength to science is its uncompromising standards of skepticism and objectivity. Better to have a partial description of reality than to retreat into an uncritical acceptance of dogma. A pragmatic approach of inquiring what is observed with a phenomenon and not trying to formulate a model of what is, is called positivism.



Galileo Galilei :

Italian mathematician, astronomer, and physicist, made several significant contributions to modern scientific thought. As the first man to use the telescope to study the skies, he amassed evidence that proved the Earth revolves around the Sun and is not the centre of the universe, as had been believed. His position represented such a radical departure from accepted thought that he was tried by the Inquisition in Rome, ordered to recant, and forced to spend the last eight years of his life under house arrest. He informally stated the principles later embodied in Newton's first two laws of motion. Because of his pioneer work in gravitation and motion and in combining mathematical analysis with experimentation, Galileo often is referred to as the founder of modern mechanics and experimental physics. Perhaps the most far-reaching of his achievements was his reestablishment of mathematical rationalism against Aristotle's logico-verbal approach and his insistence that the "Book of Nature is . . . written in mathematical characters." From this base, he was able to found the modern experimental method.

Galileo was born at Pisa on February 15, 1564, the son of Vincenzo Galilei, a musician. He received his early education at the monastery of Vallombrosa near Florence, where his family had moved in 1574. In 1581 he entered the University of Pisa to study medicine. While in the Pisa cathedral during his first year at the university, Galileo supposedly observed a lamp swinging and found that the lamp always required the same amount of time to complete an oscillation, no matter how large the range of the swing. Later in life Galileo verified this observation experimentally and suggested that the principle of the pendulum might be applied to the regulation of clocks.

Until he supposedly observed the swinging lamp in the cathedral, Galileo had received no instruction in mathematics. Then a geometry lesson he overheard by chance awakened his interest, and he began to study mathematics and science with Ostilio Ricci, a teacher in the Tuscan court. But in 1585, before he had received a degree, he was withdrawn from the university because of lack of funds. Returning to Florence, he lectured at the Florentine academy and in 1586 published an essay describing the

hydrostatic balance, the invention of which made his name known throughout Italy. In 1589 a treatise on the centre of gravity in solids won for Galileo the honourable, but not lucrative, post of mathematics lecturer at the University of Pisa.

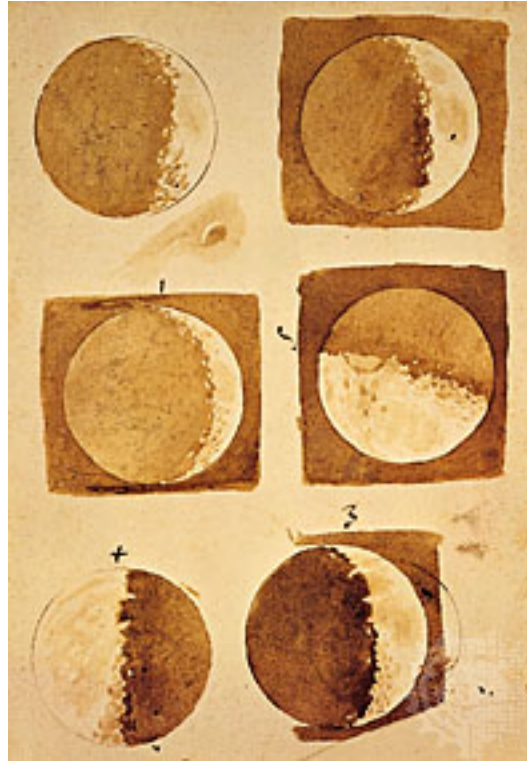
Galileo then began his research into the theory of motion, first disproving the Aristotelian contention that bodies of different weights fall at different speeds. Because of financial difficulties, Galileo, in 1592, applied for and was awarded the chair of mathematics at Padua, where he was to remain for 18 years and perform the bulk of his most outstanding work. At Padua he continued his research on motion and proved theoretically (about 1604) that falling bodies obey what came to be known as the law of uniformly accelerated motion (in such motion a body speeds up or slows down uniformly with time). Galileo also gave the law of parabolic fall (e.g., a ball thrown into the air follows a parabolic path). The legend that he dropped weights from the leaning tower of Pisa apparently has no basis in fact.

Galileo became convinced early in life of the truth of the Copernican theory (i.e., that the planets revolve about the Sun) but was deterred from avowing his opinions--as shown in his letter of April 4, 1597, to Kepler--because of fear of ridicule. While in Venice in the spring of 1609, Galileo learned of the recent invention of the telescope. After returning to Padua he built a telescope of threefold magnifying power and quickly improved it to a power of 32. Because of the method Galileo devised for checking the curvature of the lenses, his telescopes were the first that could be used for astronomical observation and soon were in demand in all parts of Europe.



As the first person to apply the telescope to a study of the skies, Galileo in late 1609 and early 1610 announced a series of astronomical discoveries. He found that the surface of the Moon was irregular and not smooth, as had been supposed; he observed that the Milky Way system was composed of a collection of stars; he discovered the satellites of Jupiter and named them Sidera Medicea (Medicean Stars) in honour of his former pupil and future employer, Cosimo II, grand duke of Tuscany. He also observed

Saturn, spots on the Sun, and the phases of Venus . His first decisive astronomical observations were published in 1610 in *Sidereus Nuncius* ("The Starry Messenger").



Although the Venetian senate had granted Galileo a lifetime appointment as professor at Padua because of his findings with the telescope, he left in the summer of 1610 to become "first philosopher and mathematician" to the grand duke of Tuscany, an appointment that enabled him to devote more time to research.

In 1611 Galileo visited Rome and demonstrated his telescope to the most eminent personages at the pontifical court. Encouraged by the flattering reception accorded to him, he ventured, in three letters on the sunspots printed at Rome in 1613 under the title *Istoria e dimostrazioni intorno alle macchie solari e loro accidenti* . . . , to take up a more definite position on the Copernican theory. Movement of the spots across the face of the Sun, Galileo maintained, proved Copernicus was right and Ptolemy wrong.

His great expository gifts and his choice of Italian, in which he was an acknowledged master of style, made his thoughts popular beyond the confines of the universities and created a powerful movement of opinion. The Aristotelian professors, seeing their vested interests threatened, united against him. They strove to cast suspicion upon him in the eyes of ecclesiastical authorities because of contradictions between the Copernican theory and the Scriptures. They obtained the cooperation of the Dominican preachers, who fulminated from the pulpit against the new impiety of "mathematicians" and secretly denounced Galileo to the Inquisition for blasphemous utterances, which, they said, he had freely invented. Gravely alarmed, Galileo agreed with one of his pupils, B. Castelli, a Benedictine monk, that something should be done to forestall a crisis. He accordingly wrote letters meant for the Grand Duke and for the Roman authorities (letters to Castelli, to the Grand Duchess Dowager, to Monsignor Dini) in

which he pointed out the danger, reminding the church of its standing practice of interpreting Scripture allegorically whenever it came into conflict with scientific truth, quoting patristic authorities and warning that it would be "a terrible detriment for the souls if people found themselves convinced by proof of something that it was made then a sin to believe." He even went to Rome in person to beg the authorities to leave the way open for a change. A number of ecclesiastical experts were on his side. Unfortunately, Cardinal Robert Bellarmine, the chief theologian of the church, was unable to appreciate the importance of the new theories and clung to the time-honoured belief that mathematical hypotheses have nothing to do with physical reality. He only saw the danger of a scandal, which might undermine Catholicity in its fight with Protestantism. He accordingly decided that the best thing would be to check the whole issue by having Copernicanism declared "false and erroneous" and the book of Copernicus suspended by the congregation of the Index. The decree came out on March 5, 1616. On the previous February 26, however, as an act of personal consideration, Cardinal Bellarmine had granted an audience to Galileo and informed him of the forthcoming decree, warning him that he must henceforth neither "hold nor defend" the doctrine, although it could still be discussed as a mere "mathematical supposition."

For the next seven years Galileo led a life of studious retirement in his house in Bellosguardo near Florence. At the end of that time (1623), he replied to a pamphlet by Orazio Grassi about the nature of comets; the pamphlet clearly had been aimed at Galileo. His reply, titled *Saggiatore . . .* ("Assayer . . ."), was a brilliant polemic on physical reality and an exposition of the new scientific method. In it he distinguished between the primary (i.e., measurable) properties of matter and the others (e.g., odour) and wrote his famous pronouncement that the "Book of Nature is . . . written in mathematical characters." The book was dedicated to the new pope, Urban VIII, who as Maffeo Barberini had been a longtime friend and protector of Galileo. Pope Urban received the dedication enthusiastically.

In 1624 Galileo again went to Rome, hoping to obtain a revocation of the decree of 1616. This he did not get, but he obtained permission from the Pope to write about "the systems of the world," both Ptolemaic and Copernican, as long as he discussed them noncommittally and came to the conclusion dictated to him in advance by the pontiff--that is, that man cannot presume to know how the world is really made because [God](#) could have brought about the same effects in ways unimagined by him, and he must not restrict God's omnipotence. These instructions were confirmed in writing by the head censor, Monsignor Niccol Riccardi.

Galileo returned to Florence and spent the next several years working on his great book *Dialogo sopra i due massimi sistemi del mondo, tolemaico e copernicano* (Dialogue Concerning the Two Chief World Systems--Ptolemaic and Copernican). As soon as it came out, in the year 1632, with the full and complete imprimatur of the censors, it was greeted with a tumult of applause and cries of praise from every part of the European continent as a literary and philosophical masterpiece.

On the crisis that followed there remain now only inferences. It was pointed out to the Pope that despite its noncommittal title, the work was a compelling and unabashed plea for the Copernican system. The strength of the argument made the prescribed conclusion at the end look anticlimactic and pointless. The Jesuits insisted that it could have worse consequences on the established system of teaching "than Luther

and Calvin put together." The Pope, in anger, ordered a prosecution. The author being covered by license, the only legal measures would be to disavow the licensers and prohibit the book. But at that point a document was "discovered" in the file, to the effect that during his audience with Bellarmine on February 26, 1616, Galileo had been specifically enjoined from "teaching or discussing Copernicanism in any way," under the penalties of the Holy Office. His license, it was concluded, had therefore been "extorted" under false pretenses. (The consensus of historians, based on evidence made available when the file was published in 1877, has been that the document had been planted and that Galileo was never so enjoined.) The church authorities, on the strength of the "new" document, were able to prosecute him for "vehement suspicion of heresy." Notwithstanding his pleas of illness and old age, Galileo was compelled to journey to Rome in February 1633 and stand trial. He was treated with special indulgence and not jailed. In a rigorous interrogation on April 12, he steadfastly denied any memory of the 1616 injunction. The commissary general of the Inquisition, obviously sympathizing with him, discreetly outlined for the authorities a way in which he might be let off with a reprimand, but on June 16 the congregation decreed that he must be sentenced. The sentence was read to him on June 21: he was guilty of having "held and taught" the Copernican doctrine and was ordered to recant. Galileo recited a formula in which he "abjured, cursed and detested" his past errors. The sentence carried imprisonment, but this portion of the penalty was immediately commuted by the Pope into house arrest and seclusion on his little estate at Arcetri near Florence, where he returned in December 1633. The sentence of house arrest remained in effect throughout the last eight years of his life.

Although confined to his estate, Galileo's prodigious mental activity continued undiminished to the last. In 1634 he completed *Discorsi e dimostrazioni matematiche intorno a due nuove scienze attenenti alla meccanica* (Dialogue Concerning Two New Sciences), in which he recapitulated the results of his early experiments and his mature meditations on the principles of mechanics. This, in many respects his most valuable work, was printed by Louis Elzevirs at Leiden in 1638. His last telescopic discovery--that of the Moon's diurnal and monthly librations (wobbling from side to side)--was made in 1637, only a few months before he became blind. But the fire of his genius was not even yet extinct. He continued his scientific correspondence with unbroken interest and undiminished acumen; he thought out the application of the pendulum to the regulation of clockwork, which the Dutch scientist Christiaan Huygens put into practice in 1656; he was engaged in dictating to his disciples, Vincenzo Viviani and Evangelista Torricelli, his latest ideas on the theory of impact when he was seized with the slow fever that resulted in his death at Arcetri on January 8, 1642.

The direct services of permanent value that Galileo rendered to astronomy are virtually summed up in his telescopic discoveries. His name is justly associated with a vast extension of the bounds of the visible universe, and his telescopic observations are a standing monument of his ability. Within two years after their discovery, he had constructed approximately accurate tables of the revolutions of Jupiter's satellites and proposed their frequent eclipses as a means of determining longitudes on land and at sea. The idea, though ingenious, has been found of little use at sea. His observations on sunspots are noteworthy for their accuracy and for the deductions he drew from them with regard to the rotation of the Sun and the revolution of the Earth.

A puzzling circumstance is Galileo's neglect of Kepler's laws, which were discovered during his lifetime.

But then he believed strongly that orbits should be circular (not elliptical, as Kepler discovered) in order to keep the fabric of the cosmos in its perfect order. This preconception prevented him from giving a full formulation of the inertial law, which he himself discovered, although it usually is attributed to the French mathematician Ren Descartes. Galileo believed that the inertial path of a body around the Earth must be circular. Lacking the idea of Newtonian gravitation, he hoped this would allow him to explain the path of the planets as circular inertial orbits around the Sun.

The idea of a universal force of gravitation seems to have hovered on the borders of this great man's mind, but he refused to entertain it because, like Descartes, he considered it an "occult" quality. More valid instances of the anticipation of modern discoveries may be found in his prevision that a small annual parallax would eventually be found for some of the fixed stars and that extra-Saturnian planets would at some future time be ascertained to exist and in his conviction that light travels with a measurable although extremely great velocity. Although Galileo discovered, in 1610, a means of adapting his telescope to the examination of minute objects, he did not become acquainted with the compound microscope until 1624, when he saw one in Rome and, with characteristic ingenuity, immediately introduced several improvements into its construction.

A most substantial part of his work consisted undoubtedly of his contributions toward the establishment of mechanics as a science. Some valuable but isolated facts and theorems had previously been discovered and proved, but it was Galileo who first clearly grasped the idea of force as a mechanical agent. Although he did not formulate the interdependence of motion and force into laws, his writings on dynamics are everywhere suggestive of those laws, and his solutions of dynamical problems involve their recognition. In this branch of science he paved the way for the English physicist and mathematician Isaac Newton later in the century. The extraordinary advances made by him were due to his application of mathematical analysis to physical problems.

Galileo was the first man who perceived that mathematics and physics, previously kept in separate compartments, were going to join forces. He was thus able to unify celestial and terrestrial phenomena into one theory, destroying the traditional division between the world above and the world below the Moon. The method that was peculiarly his consisted in the combination of experiment with calculation--in the transformation of the concrete into the abstract and the assiduous comparison of results. He created the modern idea of experiment, which he called *cimento* ("ordeal"). This method was applied to check theoretical deductions in the investigation of the laws of falling bodies, of equilibrium and motion on an inclined plane, and of the motion of a projectile. The latter, together with his definition of momentum and other parts of his work, implied a knowledge of the laws of motion as later stated by Newton. In his *Discorso intorno alle cose che stanno in su l'acqua* ("Discourse on Things That Float"), published in 1612, he used the principle of virtual velocities to demonstrate the more elementary theorems of hydrostatics, deducing the equilibrium of fluid in a siphon, and worked out the conditions for the flotation of solid bodies in a liquid. He also constructed, in 1607, an elementary form of air thermometer.

Velocity :

Velocity is a quantity that designates how fast and in what direction a point is moving. Because it has direction as well as magnitude, velocity is known as a vector quantity and cannot be specified completely by a number, as can be done with time or length, which are scalar quantities. Like all vectors, velocity is represented graphically by a directed line segment (arrow) the length of which is proportional to its magnitude.

A point always moves in a direction that is tangent to its path; for a circular path, for example, its direction at any instant is perpendicular to a line from the point to the centre of the circle (a radius). The magnitude of the velocity (i.e., the speed) is the time rate at which the point is moving along its path.

Interrupt If a point moves a certain distance along its path in a given time interval, its average speed during the interval is equal to the distance moved divided by the time taken. A train that travels 100 km in 2 hours, for example, has an average speed of 50 km per hour.

During the two-hour interval, the speed of the train in the previous example may have varied considerably around the average. The speed of a point at any instant may be approximated by finding the average speed for a short time interval including the instant in question. The differential calculus, which was invented by Isaac Newton for this specific purpose, provides means for determining exact values of the instantaneous velocity.

Force :

Force is any action that tends to maintain or alter the position of a body or to distort it. The concept of force is commonly explained in terms of Newton's three laws of motion set forth in his *Principia Mathematica* (1687). According to Newton's first principle, a body that is at rest or moving at a uniform rate in a straight line will remain in that state until some force is applied to it. The second law says that when an external force acts on a body, it produces an acceleration (change in velocity) of the body in the direction of the force. The magnitude of the acceleration is directly proportional to the magnitude of the external force and inversely proportional to the quantity of matter in the body. Newton's third law states that when one body exerts a force on another body, the second body exerts an equal force on the first body. This principle of action and reaction explains why a force tends to deform a body (i.e., change its shape) whether or not it causes the body to move. The deformation of a body can usually be neglected when investigating its motion.

Because force has both magnitude and direction, it is a vector quantity and can be represented graphically as a directed line segment; that is, a line with a length equal to the magnitude of the force, to some scale, inclined at the proper angle, with an arrowhead at one end to indicate direction. The representation of forces by vectors implies that they are concentrated either at a single point or along a single line. This is, however, physically impossible. On a loaded component of a structure, for example, the applied force produces an internal force, or stress, that is distributed over the cross section of the component. The force of gravity is invariably distributed throughout the volume of a body. Nonetheless, when the equilibrium of a body is the primary consideration, it is generally valid as well as convenient to assume that the forces are concentrated at a single point. In the case of gravitational force, the total weight of a body may be assumed to be concentrated at its centre of gravity.

Physicists use the newton, a unit of the International System (SI), for measuring force. A newton is the force needed to accelerate a body weighing one kilogram by one metre per second per second. The formula $F = ma$ is employed to calculate the number of newtons required to increase or decrease the velocity of a given body. In countries still using the English system of measurement, engineers commonly measure force in pounds. One pound of force imparts to a one-pound object an acceleration of 32.17 feet per second squared.

Inertia :

Inertia is the property of a body by virtue of which it opposes any agency that attempts to put it in motion or, if it is moving, to change the magnitude or direction of its velocity. Inertia is a passive property and does not enable a body to do anything except oppose such active agents as forces and torques. A moving body keeps moving not because of its inertia but only because of the absence of a force to slow it down, change its course, or speed it up.

There are two numerical measures of the inertia of a body: its mass, which governs its resistance to the action of a force, and its moment of inertia about a specified axis, which measures its resistance to the action of a torque about the same axis.

Evolution of Scientific Thought:

The Babylonians (c1000 BC) recorded the comings and goings of the Moon arithmetically without understanding the geometry. The Greeks (c200 BC) went further; they viewed the solar system as sitting in an immense vacuum surrounded by the fixed stars. But even the clever Greeks knew nothing about the underlying physics of the solar system. This fell to Newton (1687) in the "Principia," and the 18th century mathematician/physicists such as Laplace. These thinkers proposed the principle of universal gravitation and tried to check it out on the complicated Moon-Earth-Sun system. In many physics problems, the dynamics of two interacting bodies (a planet and a star or two electrical charges, say) is easy. Add a third body and things get complicated, indeed chaotic, which is why Newton and his 18-century followers were largely stumped in their efforts to nail down the Earth-Sun-Moon dynamics.

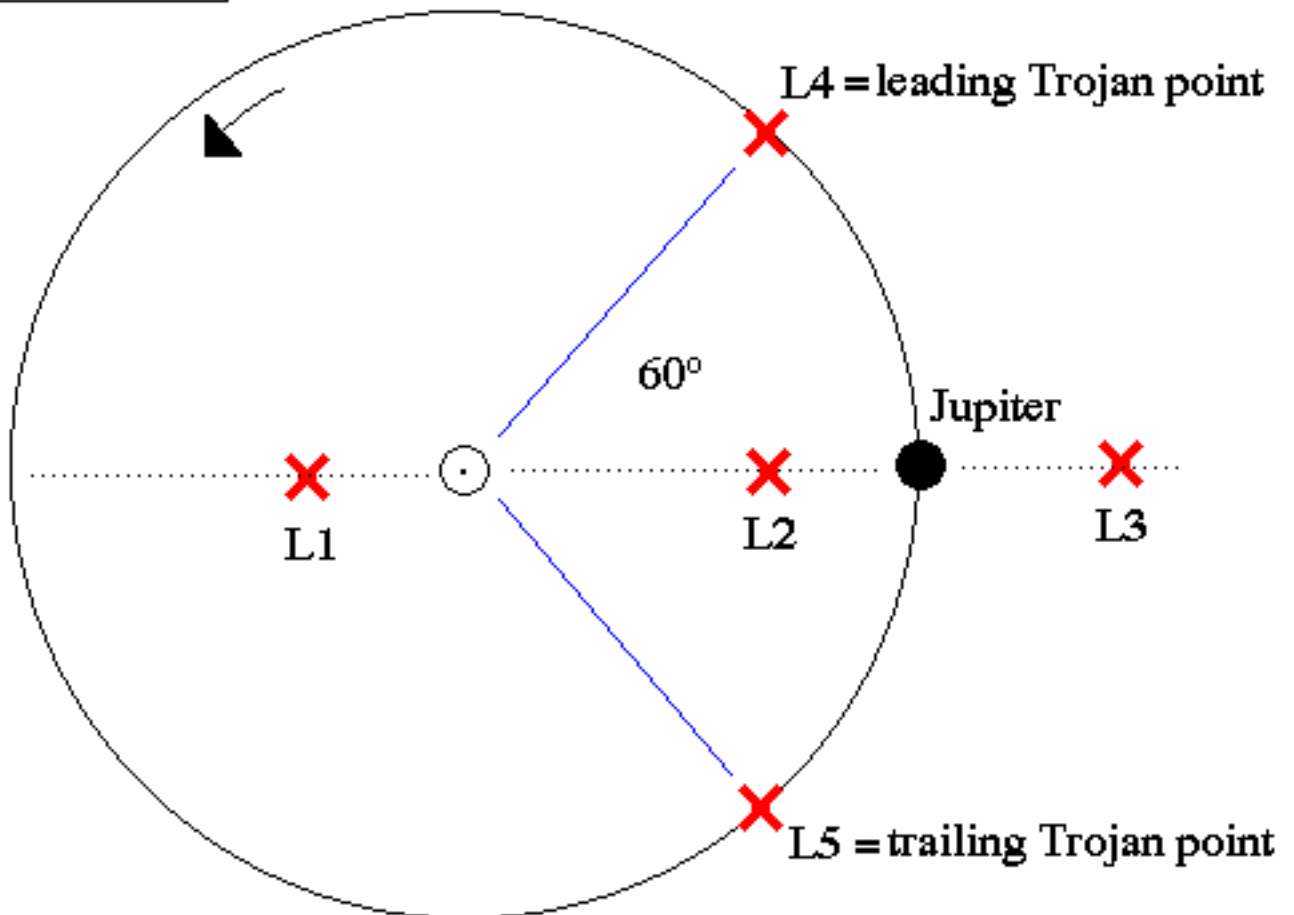
The amassing of positions, orbits, times (the kinds of things published in tables) corresponds to the "Babylonian phase," while the advent of a model of the solar system represents the "Greek phase." The third, or Newtonian, age is when the underlying forces are deduced.

Three Body Problem:

The problem of determining the motion of three celestial bodies moving under no influence other than that of their mutual gravitation. No general solution of this problem (or the more general problem involving more than three bodies) is possible.

As practically attacked, it consists of the problem of determining the perturbations (disturbances) in the motion of one of the bodies around the principal, or central, body that are produced by the attraction of the third. Examples are the motion of the Moon around the Earth, as disturbed by the action of the Sun, and of one planet around the Sun, as disturbed by the action of another planet. The problem can be solved for some special cases; for example, those in which the mass of one body, as a spacecraft, can be considered infinitely small, and in the Lagrangian and Eulerian cases.

Lagrangian points



L1,2,3 are unstable, L4,5 are stable

Deterministic Chaos :

Chaos is where apparently random or unpredictable behaviour in systems governed by deterministic laws. A more accurate term, "deterministic chaos," suggests a paradox because it connects two notions that are familiar and commonly regarded as incompatible. The first is that of randomness or unpredictability, as in the trajectory of a molecule in a gas or in the voting choice of a particular individual from out of a population. In conventional analyses, randomness was considered more apparent than real, arising from ignorance of the many causes at work. In other words, it was commonly believed that the world is unpredictable because it is complicated. The second notion is that of deterministic motion, as that of a pendulum or a planet, which has been accepted since the time of Isaac Newton as exemplifying the success of science in rendering predictable that which is initially complex.

In recent decades, however, a diversity of systems have been studied that behave unpredictably despite their seeming simplicity and the fact that the forces involved are governed by well-understood physical laws. The common element in these systems is a very high degree of sensitivity to initial conditions and to the way in which they are set in motion. For example, the meteorologist Edward Lorenz discovered that a simple model of heat convection possesses intrinsic unpredictability, a circumstance he called the "butterfly effect," suggesting that the mere flapping of a butterfly's wing can change the weather. A more homely example is the pinball machine: the ball's movements are precisely governed by laws of gravitational rolling and elastic collisions--both fully understood--yet the final outcome is unpredictable.

In classical mechanics the behaviour of a dynamical system can be described geometrically as motion on an "attractor." The mathematics of classical mechanics effectively recognized three types of attractor: single points (characterizing steady states), closed loops (periodic cycles), and tori (combinations of several cycles). In the 1960s a new class of "strange attractors" was discovered by the American mathematician Stephen Smale. On strange attractors the dynamics is chaotic. Later it was recognized that strange attractors have detailed structure on all scales of magnification; a direct result of this recognition was the development of the concept of the fractal (q.v.; a class of complex geometric shapes that commonly exhibit the property of self-similarity), which led in turn to remarkable developments in computer graphics.

Applications of the mathematics of chaos are highly diverse, including the study of turbulent flow of fluids, irregularities in heartbeat, population dynamics, chemical reactions, plasma physics, and the motion of groups and clusters of stars.

Electricity :

The first proper understanding of the electricity dates to the 18th century, when a French physicist, Charles Coulomb, showed that the electrostatic force between electrically charged objects follows a law similar to Newton's law of gravitation. Coulomb found that the force F between one charge q_1 and a second charge q_2 is equal to the product of the charges divided by the square of the distance r between them, or $F = q_1 q_2 / r^2$. The force can be either attractive or repulsive, because the source of the force, electric charge, exists in two varieties, positive and negative. The force between opposite charges is attractive, whereas bodies with the same kind of charge experience a repulsive force. The science of electricity is concerned with the behavior of aggregates of charge, including the distribution of charge within matter and the motion of charge from place to place. Different types of materials are classified as either conductors or insulators on the basis of whether charges can move freely through their constituent matter. Electric current is the measure of the flow of charges; the laws governing currents in matter are important in technology, particularly in the production, distribution, and control of energy.

The concept of voltage, like those of charge and current, is fundamental to the science of electricity. Voltage is a measure of the propensity of charge to flow from one place to another; positive charges generally tend to move from a region of high voltage to a region of lower voltage. A common problem in electricity is determining the relationship between voltage and current or charge in a given physical situation.

Magnetism :

Coulomb also showed that the force between magnetized bodies varies inversely as the square of the distance between them. Again, the force can be attractive (opposite poles) or repulsive (like poles).

Particles with electric charge interact by an electric force, while charged particles in motion produce and respond to magnetic forces as well. Many subatomic particles, including the electrically charged electron and proton and the electrically neutral neutron, behave like elementary magnets. On the other hand, in spite of systematic searches undertaken, no magnetic monopoles, which would be the magnetic analogues of electric charges, have ever been found.

Electromagnetism :

Magnetism and electricity are not separate phenomena; they are the related manifestations of an underlying electromagnetic force. Experiments in the early 19th century by, among others, Hans Orsted (in Denmark), Andr-Marie Ampre (in France), and Michael Faraday (in England) revealed the intimate connection between electricity and magnetism and how the one can give rise to the other. The results of these experiments were synthesized in the 1850s by the Scottish physicist James Clerk Maxwell in his electromagnetic theory. Maxwell's theory predicted the existence of electromagnetic waves--undulations in intertwined electric and magnetic fields, traveling with the velocity of light.

The final steps in synthesizing electricity and magnetism into one coherent theory were made by Maxwell. He was deeply influenced by Faraday's work, having begun his study of the phenomena by translating Faraday's experimental findings into mathematics. (Faraday was self-taught and had never mastered mathematics.) In 1856 Maxwell developed the theory that the energy of the electromagnetic field is in the space around the conductors as well as in the conductors themselves. By 1864 he had formulated his own electromagnetic theory of light, predicting that both light and radio waves are electric and magnetic phenomena. While Faraday had discovered that changes in magnetic fields produce electric fields, Maxwell added the converse: changes in electric fields produce magnetic fields even in the absence of electric currents. Maxwell predicted that electromagnetic disturbances traveling through empty space have electric and magnetic fields at right angles to each other and that both fields are perpendicular to the direction of the wave. He concluded that the waves move at a uniform speed equal to the speed of light and that light is one form of electromagnetic wave. Their elegance notwithstanding, Maxwell's radical ideas were accepted by few outside England until 1886, when the German physicist Heinrich Hertz verified the existence of electromagnetic waves traveling at the speed of light; the waves he discovered are known now as radio waves.



Faraday:

Michael Faraday, who became one of the greatest scientists of the 19th century, began his career as a chemist. He wrote a manual of practical chemistry that reveals his mastery of the technical aspects of his art, discovered a number of new organic compounds, among them benzene, and was the first to liquefy a "permanent" gas (i.e., one that was believed to be incapable of liquefaction). His major contribution, however, was in the field of electricity and magnetism. He was the first to produce an electric current from a magnetic field, invented the first electric motor and dynamo, demonstrated the relation between electricity and chemical bonding, discovered the effect of magnetism on light, and discovered and named diamagnetism, the peculiar behaviour of certain substances in strong magnetic fields. He provided the experimental, and a good deal of the theoretical, foundation upon which James Clerk Maxwell erected classical electromagnetic field theory.

Michael Faraday was born on September 22, 1791, in the country village of Newington, Surrey, now a part of South London. His father was a blacksmith who had migrated from the north of England earlier in 1791 to look for work. His mother was a country woman of great calm and wisdom who supported her son emotionally through a difficult childhood. Faraday was one of four children, all of whom were hard put to get enough to eat, since their father was often ill and incapable of working steadily. Faraday later recalled being given one loaf of bread that had to last him for a week. The family belonged to a small Christian sect, called Sandemanians, that provided spiritual sustenance to Faraday throughout his life. It was the single most important influence upon him and strongly affected the way in which he approached and interpreted nature.

Faraday received only the rudiments of an education, learning to read, write, and cipher in a church Sunday school. At an early age he began to earn money by delivering newspapers for a book dealer and bookbinder, and at the age of 14 he was apprenticed to the man. Unlike the other apprentices, Faraday took the opportunity to read some of the books brought in for rebinding. The article on electricity in the

third edition of the Encyclopædia Britannica particularly fascinated him. Using old bottles and lumber, he made a crude electrostatic generator and did simple experiments. He also built a weak voltaic pile with which he performed experiments in electrochemistry.

Faraday's great opportunity came when he was offered a ticket to attend chemical lectures by Sir Humphry Davy at the Royal Institution of Great Britain in London. Faraday went, sat absorbed with it all, recorded the lectures in his notes, and returned to bookbinding with the seemingly unrealizable hope of entering the temple of science. He sent a bound copy of his notes to Davy along with a letter asking for employment, but there was no opening. Davy did not forget, however, and, when one of his laboratory assistants was dismissed for brawling, he offered Faraday a job. Faraday began as Davy's laboratory assistant and learned chemistry at the elbow of one of the greatest practitioners of the day. It has been said, with some truth, that Faraday was Davy's greatest discovery.

When Faraday joined Davy in 1812, Davy was in the process of revolutionizing the chemistry of the day. Antoine-Laurent Lavoisier, the Frenchman generally credited with founding modern chemistry, had effected his rearrangement of chemical knowledge in the 1770s and 1780s by insisting upon a few simple principles. Among these was that oxygen was a unique element, in that it was the only supporter of combustion and was also the element that lay at the basis of all acids. Davy, after having discovered sodium and potassium by using a powerful current from a galvanic battery to decompose oxides of these elements, turned to the decomposition of muriatic (hydrochloric) acid, one of the strongest acids known. The products of the decomposition were hydrogen and a green gas that supported combustion and that, when combined with water, produced an acid. Davy concluded that this gas was an element, to which he gave the name chlorine, and that there was no oxygen whatsoever in muriatic acid. Acidity, therefore, was not the result of the presence of an acid-forming element but of some other condition. What else could that condition be but the physical form of the acid molecule itself? Davy suggested, then, that chemical properties were determined not by specific elements alone but also by the ways in which these elements were arranged in molecules. In arriving at this view he was influenced by an atomic theory that was also to have important consequences for Faraday's thought. This theory, proposed in the 18th century by Ruggero Giuseppe Boscovich, argued that atoms were mathematical points surrounded by alternating fields of attractive and repulsive forces. A true element comprised a single such point, and chemical elements were composed of a number of such points, about which the resultant force fields could be quite complicated. Molecules, in turn, were built up of these elements, and the chemical qualities of both elements and compounds were the results of the final patterns of force surrounding clumps of point atoms. One property of such atoms and molecules should be specifically noted: they can be placed under considerable strain, or tension, before the "bonds" holding them together are broken. These strains were to be central to Faraday's ideas about electricity.

Faraday's second apprenticeship, under Davy, came to an end in 1820. By then he had learned chemistry as thoroughly as anyone alive. He had also had ample opportunity to practice chemical analyses and laboratory techniques to the point of complete mastery, and he had developed his theoretical views to the point that they could guide him in his researches. There followed a series of discoveries that astonished the scientific world.

Faraday achieved his early renown as a chemist. His reputation as an analytical chemist led to his being called as an expert witness in legal trials and to the building up of a clientele whose fees helped to support the Royal Institution. In 1820 he produced the first known compounds of carbon and chlorine, C_2Cl_6 and C_2Cl_4 . These compounds were produced by substituting chlorine for hydrogen in "olefiant gas" (ethylene), the first substitution reactions induced. (Such reactions later would serve to challenge the dominant theory of chemical combination proposed by Jns Jacob Berzelius.) In 1825, as a result of research on illuminating gases, Faraday isolated and described benzene. In the 1820s he also conducted investigations of steel alloys, helping to lay the foundations for scientific metallurgy and metallography. While completing an assignment from the Royal Society of London to improve the quality of optical glass for telescopes, he produced a glass of very high refractive index that was to lead him, in 1845, to the discovery of diamagnetism. In 1821 he married Sarah Barnard, settled permanently at the Royal Institution, and began the series of researches on electricity and magnetism that was to revolutionize physics.

In 1820 Hans Christian rsted had announced the discovery that the flow of an electric current through a wire produced a magnetic field around the wire. Andr-Marie Ampre showed that the magnetic force apparently was a circular one, producing in effect a cylinder of magnetism around the wire. No such circular force had ever before been observed, and Faraday was the first to understand what it implied. If a magnetic pole could be isolated, it ought to move constantly in a circle around a current-carrying wire. Faraday's ingenuity and laboratory skill enabled him to construct an apparatus that confirmed this conclusion. This device, which transformed electrical energy into mechanical energy, was the first electric motor.

This discovery led Faraday to contemplate the nature of electricity. Unlike his contemporaries, he was not convinced that electricity was a material fluid that flowed through wires like water through a pipe. Instead, he thought of it as a vibration or force that was somehow transmitted as the result of tensions created in the conductor. One of his first experiments after his discovery of electromagnetic rotation was to pass a ray of polarized light through a solution in which electrochemical decomposition was taking place in order to detect the intermolecular strains that he thought must be produced by the passage of an electric current. During the 1820s he kept coming back to this idea, but always without result.

In the spring of 1831 Faraday began to work with Charles (later Sir Charles) Wheatstone on the theory of sound, another vibrational phenomenon. He was particularly fascinated by the patterns (known as Chladni figures) formed in light powder spread on iron plates when these plates were thrown into vibration by a violin bow. Here was demonstrated the ability of a dynamic cause to create a static effect, something he was convinced happened in a current-carrying wire. He was even more impressed by the fact that such patterns could be induced in one plate by bowing another nearby. Such acoustic induction is apparently what lay behind his most famous experiment. On August 29, 1831, Faraday wound a thick iron ring on one side with insulated wire that was connected to a battery. He then wound the opposite side with wire connected to a galvanometer. What he expected was that a "wave" would be produced when the battery circuit was closed and that the wave would show up as a deflection of the galvanometer in the second circuit. He closed the primary circuit and, to his delight and satisfaction, saw the galvanometer needle jump. A current had been induced in the secondary coil by one in the primary.

When he opened the circuit, however, he was astonished to see the galvanometer jump in the opposite direction. Somehow, turning off the current also created an induced current in the secondary circuit, equal and opposite to the original current. This phenomenon led Faraday to propose what he called the "electrotonic" state of particles in the wire, which he considered a state of tension. A current thus appeared to be the setting up of such a state of tension or the collapse of such a state. Although he could not find experimental evidence for the electrotonic state, he never entirely abandoned the concept, and it shaped most of his later work.

In the fall of 1831 Faraday attempted to determine just how an induced current was produced. His original experiment had involved a powerful electromagnet, created by the winding of the primary coil. He now tried to create a current by using a permanent magnet. He discovered that when a permanent magnet was moved in and out of a coil of wire a current was induced in the coil. Magnets, he knew, were surrounded by forces that could be made visible by the simple expedient of sprinkling iron filings on a card held over them. Faraday saw the "lines of force" thus revealed as lines of tension in the medium, namely air, surrounding the magnet, and he soon discovered the law determining the production of electric currents by magnets: the magnitude of the current was dependent upon the number of lines of force cut by the conductor in unit time. He immediately realized that a continuous current could be produced by rotating a copper disk between the poles of a powerful magnet and taking leads off the disk's rim and centre. The outside of the disk would cut more lines than would the inside, and there would thus be a continuous current produced in the circuit linking the rim to the centre. This was the first dynamo. It was also the direct ancestor of electric motors, for it was only necessary to reverse the situation, to feed an electric current to the disk, to make it rotate.

While Faraday was performing these experiments and presenting them to the scientific world, doubts were raised about the identity of the different manifestations of electricity that had been studied. Were the electric "fluid" that apparently was released by electric eels and other electric fishes, that produced by a static electricity generator, that of the voltaic battery, and that of the new electromagnetic generator all the same? Or were they different fluids following different laws? Faraday was convinced that they were not fluids at all but forms of the same force, yet he recognized that this identity had never been satisfactorily shown by experiment. For this reason he began, in 1832, what promised to be a rather tedious attempt to prove that all electricities had precisely the same properties and caused precisely the same effects. The key effect was electrochemical decomposition. Voltaic and electromagnetic electricity posed no problems, but static electricity did. As Faraday delved deeper into the problem, he made two startling discoveries. First, electrical force did not, as had long been supposed, act at a distance upon chemical molecules to cause them to dissociate. It was the passage of electricity through a conducting liquid medium that caused the molecules to dissociate, even when the electricity merely discharged into the air and did not pass into a "pole" or "centre of action" in a voltaic cell. Second, the amount of the decomposition was found to be related in a simple manner to the amount of electricity that passed through the solution. These findings led Faraday to a new theory of electrochemistry. The electric force, he argued, threw the molecules of a solution into a state of tension (his electrotonic state). When the force was strong enough to distort the fields of forces that held the molecules together so as to permit the interaction of these fields with neighbouring particles, the tension was relieved by the migration of particles along the lines of tension, the different species of atoms migrating in opposite directions. The

amount of electricity that passed, then, was clearly related to the chemical affinities of the substances in solution. These experiments led directly to Faraday's two laws of electrochemistry: (1) The amount of a substance deposited on each electrode of an electrolytic cell is directly proportional to the quantity of electricity passed through the cell. (2) The quantities of different elements deposited by a given amount of electricity are in the ratio of their chemical equivalent weights.

Faraday's work on electrochemistry provided him with an essential clue for the investigation of static electrical induction. Since the amount of electricity passed through the conducting medium of an electrolytic cell determined the amount of material deposited at the electrodes, why should not the amount of electricity induced in a nonconductor be dependent upon the material out of which it was made? In short, why should not every material have a specific inductive capacity? Every material does, and Faraday was the discoverer of this fact.

By 1839 Faraday was able to bring forth a new and general theory of electrical action. Electricity, whatever it was, caused tensions to be created in matter. When these tensions were rapidly relieved (i.e., when bodies could not take much strain before "snapping" back), then what occurred was a rapid repetition of a cyclical buildup, breakdown, and buildup of tension that, like a wave, was passed along the substance. Such substances were called conductors. In electrochemical processes the rate of buildup and breakdown of the strain was proportional to the chemical affinities of the substances involved, but again the current was not a material flow but a wave pattern of tensions and their relief. Insulators were simply materials whose particles could take an extraordinary amount of strain before they snapped. Electrostatic charge in an isolated insulator was simply a measure of this accumulated strain. Thus, all electrical action was the result of forced strains in bodies.

The strain on Faraday of eight years of sustained experimental and theoretical work was too much, and in 1839 his health broke down. For the next six years he did little creative science. Not until 1845 was he able to pick up the thread of his researches and extend his theoretical views.

Since the very beginning of his scientific work, Faraday had believed in what he called the unity of the forces of nature. By this he meant that all the forces of nature were but manifestations of a single universal force and ought, therefore, to be convertible into one another. In 1846 he made public some of the speculations to which this view led him. A lecturer, scheduled to deliver one of the Friday evening discourses at the Royal Institution by which Faraday encouraged the popularization of science, panicked at the last minute and ran out, leaving Faraday with a packed lecture hall and no lecturer. On the spur of the moment, Faraday offered "Thoughts on Ray Vibrations." Specifically referring to point atoms and their infinite fields of force, he suggested that the lines of electric and magnetic force associated with these atoms might, in fact, serve as the medium by which light waves were propagated. Many years later, Maxwell was to build his electromagnetic field theory upon this speculation.

When Faraday returned to active research in 1845, it was to tackle again a problem that had obsessed him for years, that of his hypothetical electrotonic state. He was still convinced that it must exist and that he simply had not yet discovered the means for detecting it. Once again he tried to find signs of

intermolecular strain in substances through which electrical lines of force passed, but again with no success. It was at this time that a young Scot, William Thomson (later Lord Kelvin), wrote Faraday that he had studied Faraday's papers on electricity and magnetism and that he, too, was convinced that some kind of strain must exist. He suggested that Faraday experiment with magnetic lines of force, since these could be produced at much greater strengths than could electrostatic ones.

Faraday took the suggestion, passed a beam of plane-polarized light through the optical glass of high refractive index that he had developed in the 1820s, and then turned on an electromagnet so that its lines of force ran parallel to the light ray. This time he was rewarded with success. The plane of polarization was rotated, indicating a strain in the molecules of the glass. But Faraday again noted an unexpected result. When he changed the direction of the ray of light, the rotation remained in the same direction, a fact that Faraday correctly interpreted as meaning that the strain was not in the molecules of the glass but in the magnetic lines of force. The direction of rotation of the plane of polarization depended solely upon the polarity of the lines of force; the glass served merely to detect the effect.

This discovery confirmed Faraday's faith in the unity of forces, and he plunged onward, certain that all matter must exhibit some response to a magnetic field. To his surprise he found that this was in fact so, but in a peculiar way. Some substances, such as iron, nickel, cobalt, and oxygen, lined up in a magnetic field so that the long axes of their crystalline or molecular structures were parallel to the lines of force; others lined up perpendicular to the lines of force. Substances of the first class moved toward more intense magnetic fields; those of the second moved toward regions of less magnetic force. Faraday named the first group paramagnetics and the second diamagnetics. After further research he concluded that paramagnetics were bodies that conducted magnetic lines of force better than did the surrounding medium, whereas diamagnetics conducted them less well. By 1850 Faraday had evolved a radically new view of space and force. Space was not "nothing," the mere location of bodies and forces, but a medium capable of supporting the strains of electric and magnetic forces. The energies of the world were not localized in the particles from which these forces arose but rather were to be found in the space surrounding them. Thus was born field theory. As Maxwell later freely admitted, the basic ideas for his mathematical theory of electrical and magnetic fields came from Faraday; his contribution was to mathematize those ideas in the form of his classical field equations.

From about 1855, Faraday's mind began to fail. He still did occasional experiments, one of which involved attempting to find an electrical effect of raising a heavy weight, since he felt that gravity, like magnetism, must be convertible into some other force, most likely electrical. This time he was disappointed in his expectations, and the Royal Society refused to publish his negative results. More and more, Faraday began to sink into senility. Queen Victoria rewarded his lifetime of devotion to science by granting him the use of a house at Hampton Court and even offered him the honour of a knighthood. Faraday gratefully accepted the cottage but rejected the knighthood; he would, he said, remain plain Mr. Faraday to the end. He died on August 25, 1867, and was buried in Highgate Cemetery, London, leaving as his monument a new conception of physical reality.

Maxwell, James Clerk :

James Clerk Maxwell is regarded by most modern physicists as the scientist of the 19th century who had the greatest influence on 20th-century physics; he is ranked with Sir Isaac Newton and Albert Einstein for the fundamental nature of his contributions. In 1931, at the 100th anniversary of Maxwell's birth, Einstein described the change in the conception of reality in physics that resulted from Maxwell's work as "the most profound and the most fruitful that physics has experienced since the time of Newton." The concept of electromagnetic radiation originated with Maxwell, and his field equations, based on Michael Faraday's observations of the electric and magnetic lines of force, paved the way for Einstein's special theory of relativity, which established the equivalence of mass and energy. Maxwell's ideas also ushered in the other major innovation of 20th-century physics, the quantum theory. His description of electromagnetic radiation led to the development (according to classical theory) of the ultimately unsatisfactory law of heat radiation, which prompted Max Planck's formulation of the quantum hypothesis--i.e., the theory that radiant-heat energy is emitted only in finite amounts, or quanta. The interaction between electromagnetic radiation and matter, integral to Planck's hypothesis, in turn has played a central role in the development of the theory of the structure of atoms and molecules.

Maxwell came from a comfortable middle-class background. The original family name was Clerk, the additional surname being added by his father after he had inherited the Middlebie estate from Maxwell ancestors. James, an only child, was born on June 13, 1831, in Edinburgh, where his father was a lawyer; his parents had married late in life, and his mother was 40 years old at his birth. Shortly afterward the family moved to Glenlair, the country house on the Middlebie estate.

His mother died in 1839 from abdominal cancer, the very disease to which Maxwell was to succumb at exactly the same age. A dull and uninspired tutor was engaged who claimed that James was slow at learning, though in fact he displayed a lively curiosity at an early age and had a phenomenal memory. Fortunately he was rescued by his aunt Jane Cay and from 1841 was sent to school at the Edinburgh Academy. Among the other pupils were his biographer Lewis Campbell and his friend Peter Guthrie Tait.

Maxwell's interests ranged far beyond the school syllabus, and he did not pay particular attention to examination performance. His first scientific paper, published when he was only 14 years old, described a generalized series of oval curves that could be traced with pins and thread by analogy with an ellipse. This fascination with geometry and with mechanical models continued throughout his career and was of great help in his subsequent research.

At the age of 16 he entered the University of Edinburgh, where he read voraciously on all subjects and published two more scientific papers. In 1850 he went to the University of Cambridge, where his exceptional powers began to be recognized. His mathematics teacher, William Hopkins, was a well-known "wrangler maker" (a wrangler is one who takes first class honours in the mathematics examinations at Cambridge) whose students included Tait, George Gabriel (later Sir George) Stokes, William Thomson (later Lord Kelvin), Arthur Cayley, and Edward John Routh. Of Maxwell, Hopkins is

reported to have said that he was the most extraordinary man he had met with in the whole course of his experience, that it seemed impossible for him to think wrongly on any physical subject, but that in analysis he was far more deficient. (Other contemporaries also testified to Maxwell's preference for geometrical over analytical methods.) This shrewd assessment was later borne out by several important formulas advanced by Maxwell that obtained correct results from faulty mathematical arguments.

In 1854 Maxwell was second wrangler and first Smith's prizeman (the Smith's prize is a prestigious competitive award for an essay that incorporates original research). He was elected to a fellowship at Trinity, but, because his father's health was deteriorating, he wished to return to Scotland. In 1856 he was appointed to the professorship of natural philosophy at Marischal College, Aberdeen, but before the appointment was announced his father died. This was a great personal loss, for Maxwell had had a close relationship with his father. In June 1858 Maxwell married Katherine Mary Dewar, daughter of the principal of Marischal College. The union was childless and was described by his biographer as a "married life . . . of unexampled devotion."

In 1860 the University of Aberdeen was formed by a merger between King's College and Marischal College, and Maxwell was declared redundant. He applied for a vacancy at the University of Edinburgh, but he was turned down in favour of his school friend Tait. He then was appointed to the professorship of natural philosophy at King's College, London.

The next five years were undoubtedly the most fruitful of his career. During this period his two classic papers on the electromagnetic field were published, and his demonstration of colour photography took place. He was elected to the Royal Society in 1861. His theoretical and experimental work on the viscosity of gases also was undertaken during these years and culminated in a lecture to the Royal Society in 1866. He supervised the experimental determination of electrical units for the British Association for the Advancement of Science, and this work in measurement and standardization led to the establishment of the National Physical Laboratory. He also measured the ratio of electromagnetic and electrostatic units of electricity and confirmed that it was in satisfactory agreement with the velocity of light as predicted by his theory.

In 1865 he resigned his professorship at King's College and retired to the family estate in Glenlair. He continued to visit London every spring and served as external examiner for the Mathematical Tripos (exams) at Cambridge. In the spring and early summer of 1867 he toured Italy. But most of his energy during this period was devoted to writing his famous treatise on electricity and magnetism.

It was Maxwell's research on electromagnetism that established him among the great scientists of history. In the preface to his *Treatise on Electricity and Magnetism* (1873), the best exposition of his theory, Maxwell stated that his major task was to convert Faraday's physical ideas into mathematical form. In attempting to illustrate Faraday's law of induction (that a changing magnetic field gives rise to an induced electromagnetic field), Maxwell constructed a mechanical model. He found that the model gave rise to a corresponding "displacement current" in the dielectric medium, which could then be the seat of transverse waves. On calculating the velocity of these waves, he found that they were very close to the

velocity of light. Maxwell concluded that he could "scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena."

Maxwell's theory suggested that electromagnetic waves could be generated in a laboratory, a possibility first demonstrated by Heinrich Hertz in 1887, eight years after Maxwell's death. The resulting radio industry with its many applications thus has its origin in Maxwell's publications.

In addition to his electromagnetic theory, Maxwell made major contributions to other areas of physics. While still in his 20s, Maxwell demonstrated his mastery of classical physics by writing a prizewinning essay on Saturn's rings, in which he concluded that the rings must consist of masses of matter not mutually coherent--a conclusion that was corroborated more than 100 years later by the first Voyager space probe to reach Saturn.

The Maxwell relations of equality between different partial derivatives of thermodynamic functions are included in every standard textbook on thermodynamics. Though Maxwell did not originate the modern kinetic theory of gases, he was the first to apply the methods of probability and statistics in describing the properties of an assembly of molecules. Thus he was able to demonstrate that the velocities of molecules in a gas, previously assumed to be equal, must follow a statistical distribution (known subsequently as the Maxwell-Boltzmann distribution law). In later papers Maxwell investigated the transport properties of gases--i.e., the effect of changes in temperature and pressure on viscosity, thermal conductivity, and diffusion.

Maxwell was far from being an abstruse theoretician. He was skillful in the design of experimental apparatus, as was shown early in his career during his investigations of colour vision. He devised a colour top with adjustable sectors of tinted paper to test the three-colour hypothesis of Thomas Young and later invented a colour box that made it possible to conduct experiments with spectral colours rather than pigments. His investigations of the colour theory led him to conclude that a colour photography could be produced by photographing through filters of the three primary colours and then recombining the images. He demonstrated his supposition in a lecture to the Royal Institution of Great Britain in 1861 by projecting through filters a colour photograph of a tartan ribbon that had been taken by this method.

In addition to these well-known contributions, a number of ideas that Maxwell put forward quite casually have since led to developments of great significance. The hypothetical intelligent being known as Maxwell's demon was a factor in the development of information theory. Maxwell's analytic treatment of speed governors is generally regarded as the founding paper on cybernetics, and his "equal areas" construction provided an essential constituent of the theory of fluids developed by Johannes Diederik van der Waals. His work in geometrical optics led to the discovery of the fish-eye lens. From the start of his career to its finish his papers are filled with novelty and interest. He also was a contributor to the ninth edition of Encyclopædia Britannica.

In 1871 Maxwell was elected to the new Cavendish professorship at Cambridge. He set about designing the Cavendish Laboratory and supervised its construction. Maxwell had few students, but they were of

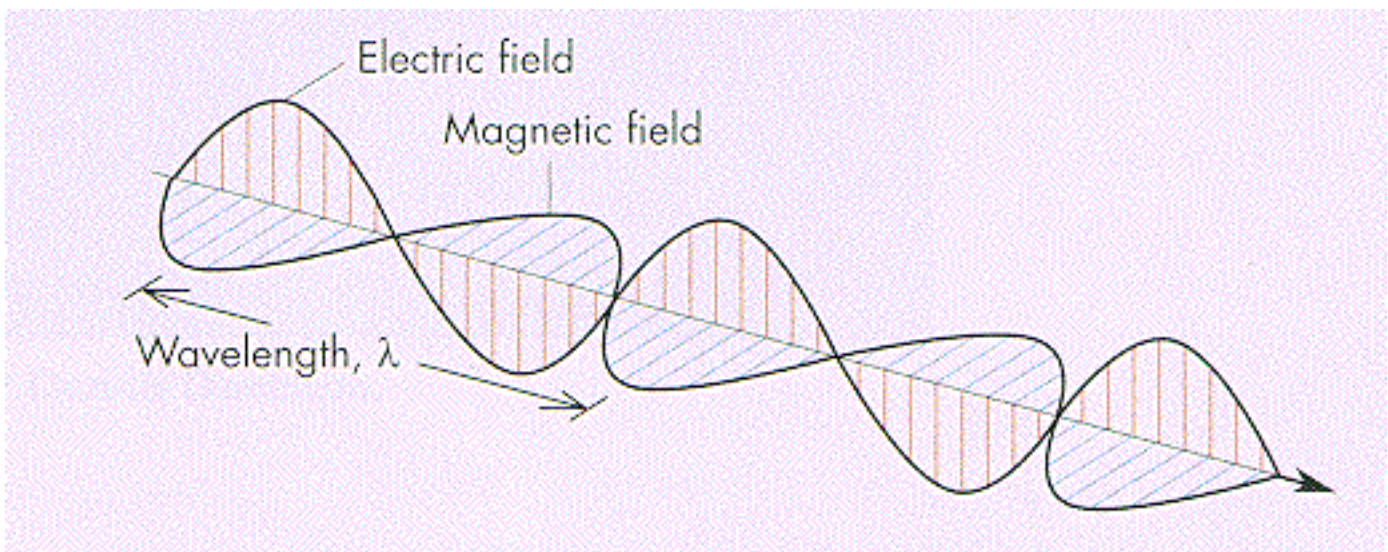
the highest calibre and included William D. Niven, Ambrose (later Sir Ambrose) Fleming, Richard Tetley Glazebrook, John Henry Poynting, and Arthur Schuster.

During the Easter term of 1879 Maxwell took ill on several occasions; he returned to Glenlair in June but his condition did not improve. He died after a short illness on Nov. 5, 1879. Maxwell received no public honours and was buried quietly in a small churchyard in the village of Parton, in Scotland.

Electromagnetic Radiation:

Electromagnetic radiation is energy that is propagated through free space or through a material medium in the form of electromagnetic waves, such as radio waves, visible light, and gamma rays. The term also refers to the emission and transmission of such radiant energy.

The Scottish physicist James Clerk Maxwell was the first to predict the existence of electromagnetic waves. In 1864 he set forth his electromagnetic theory, proposing that light--including various other forms of radiant energy--is an electromagnetic disturbance in the form of waves. In 1887 Heinrich Hertz, a German physicist, provided experimental confirmation by producing the first man-made electromagnetic waves and investigating their properties. Subsequent studies resulted in a broader understanding of the nature and origin of radiant energy.



It has been established that time-varying electric fields can induce magnetic fields and that time-varying magnetic fields can in like manner induce electric fields. Because such electric and magnetic fields generate each other, they occur jointly, and together they propagate as electromagnetic waves. An electromagnetic wave is a transverse wave in that the electric field and the magnetic field at any point and time in the wave are perpendicular to each other as well as to the direction of propagation. In free space (i.e., a space that is absolutely devoid of matter and that experiences no intrusion from other fields or forces), electromagnetic waves always propagate with the same speed--that of light (299,792,458 m per second, or 186,282 miles per second)--independent of the speed of the observer or of the source of the waves.

Electromagnetic radiation has properties in common with other forms of waves such as reflection, refraction, diffraction, and interference. Moreover, it may be characterized by the frequency with which it varies over time or by its wavelength. Electromagnetic radiation, however, has particle-like properties in addition to those associated with wave motion. It is quantized in that for a given frequency, its energy occurs as an integer times h , in which h is a fundamental constant of nature known as Planck's constant. A quantum of electromagnetic energy is called a photon. Visible light and other forms of electromagnetic

radiation may be thought of as a stream of photons, with photon energy directly proportional to frequency.

Electromagnetic radiation spans an enormous range of frequencies or wavelengths, as is shown by the electromagnetic spectrum. Customarily, it is designated by fields, waves, and particles in increasing magnitude of frequencies--radio waves, microwaves, infrared rays, visible light, ultraviolet light, X rays, and gamma rays. The corresponding wavelengths are inversely proportional, and both the frequency and wavelength scales are logarithmic.

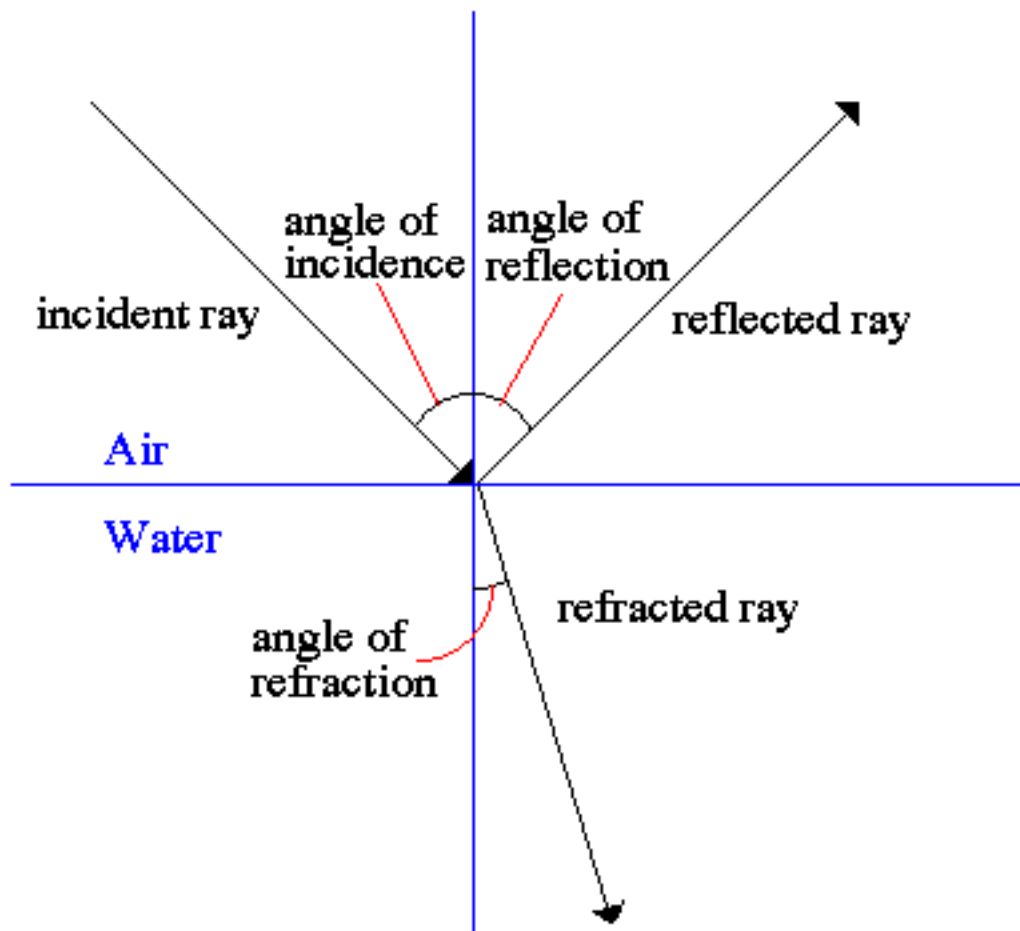
Electromagnetic radiation of different frequencies interacts with matter differently. A vacuum is the only perfectly transparent medium, and all material media absorb strongly some regions of the electromagnetic spectrum. For example, molecular oxygen (O₂), ozone (O₃), and molecular nitrogen (N₂) in the Earth's atmosphere are almost perfectly transparent to infrared rays of all frequencies, but they strongly absorb ultraviolet light, X rays, and gamma rays. The frequency (or energy equal to $h\nu$) of X rays is substantially higher than that of visible light, and so X rays are able to penetrate many materials that do not transmit light. Moreover, absorption of X rays by a molecular system can cause chemical reactions to occur. When X rays are absorbed in a gas, for instance, they eject photoelectrons from the gas, which in turn ionize its molecules. If these processes occur in living tissue, the photoelectrons emitted from the organic molecules destroy the cells of the tissue. Gamma rays, though generally of somewhat higher frequency than X rays, have basically the same nature. When the energy of gamma rays is absorbed in matter, its effect is virtually indistinguishable from the effect produced by X rays.

There are many sources of electromagnetic radiation, both natural and man-made. Radio waves, for example, are produced by cosmic objects such as pulsars and quasars and by electronic circuits. Sources of ultraviolet radiation include mercury vapour lamps and high-intensity lights, as well as the Sun. The latter also generates X rays, as do certain types of particle accelerators and electronic devices.

Reflection:

Reflection is the abrupt change in the direction of propagation of a wave that strikes the boundary between different mediums. At least part of the oncoming wave disturbance remains in the same medium. Regular reflection, which follows a simple law, occurs at plane boundaries. The angle between the direction of motion of the oncoming wave and a perpendicular to the reflecting surface (angle of incidence) is equal to the angle between the direction of motion of the reflected wave and a perpendicular (angle of reflection). Reflection at rough, or irregular, boundaries is diffuse. The reflectivity of a surface material is the fraction of energy of the oncoming wave that is reflected by it.

Reflection and Refraction

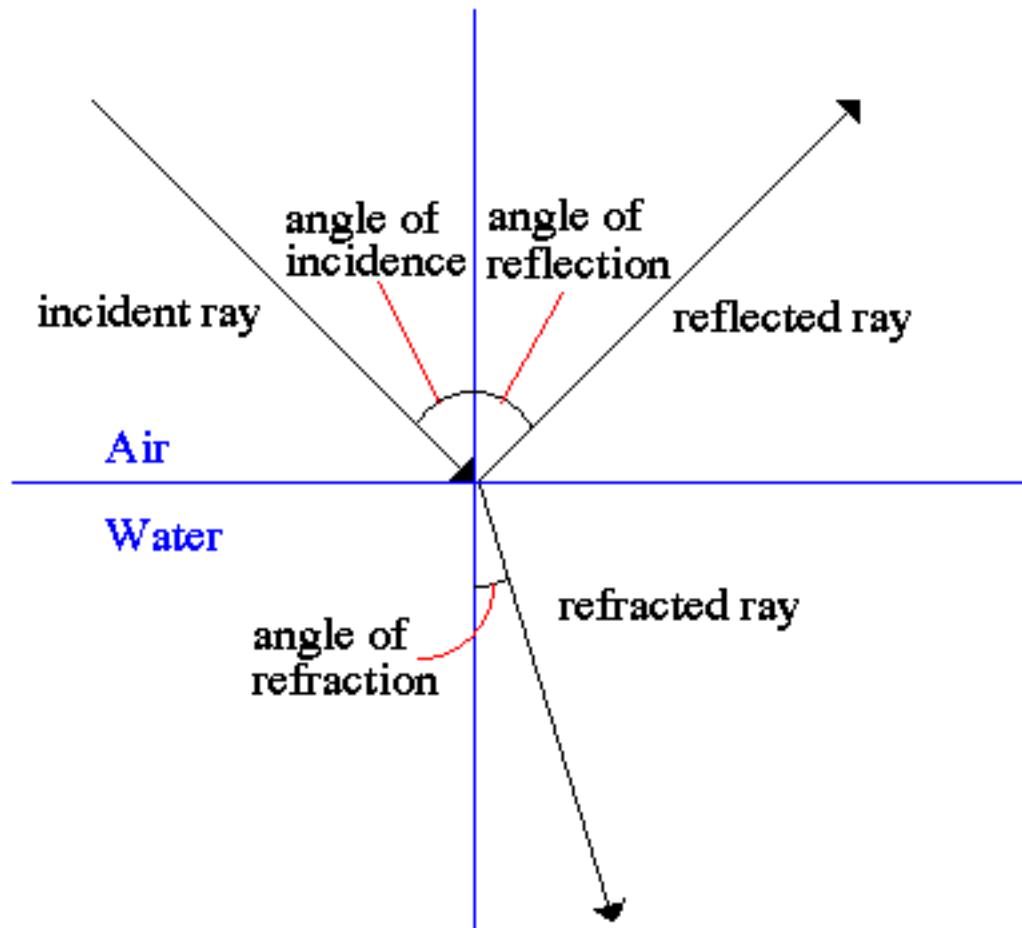


Refraction :

Refraction is the change in direction of a wave passing from one medium to another caused by its change in speed. For example, waves in deep water travel faster than in shallow; if an ocean wave approaches a beach obliquely, the part of the wave farther from the beach will move faster than that closer in, and so the wave will swing around until it moves in a direction perpendicular to the shoreline. The speed of sound waves is greater in warm air than in cold; at night, air is cooled at the surface of a lake, and any sound that travels upward is refracted down by the higher layers of air that still remain warm. Thus, sounds, such as voices and music, can be heard much farther across water at night than in the daytime.

The electromagnetic waves constituting light are refracted when crossing the boundary from one transparent medium to another because of their change in speed. A straight stick appears bent when partly immersed in water and viewed at an angle to the surface other than 90. A ray of light of one wavelength, or colour (different wavelengths appear as different colours to the human eye), in passing from air to glass is refracted, or bent, by an amount that depends on its speed in air and glass, the two speeds depending on the wavelength. A ray of sunlight is composed of many wavelengths that in combination appear to be colourless; upon entering a glass prism, the different refractions of the various wavelengths spread them apart as in a rainbow.

Reflection and Refraction



[Click here for refraction tool](#)

Diffraction :

Diffraction is the spreading of waves around obstacles. Diffraction takes place with sound; with electromagnetic radiation, such as light, X-rays, and gamma rays; and with very small moving particles such as atoms, neutrons, and electrons, which show wavelike properties. One consequence of diffraction is that sharp shadows are not produced. The phenomenon is the result of interference (i.e., when waves are superimposed, they may reinforce or cancel each other out) and is most pronounced when the wavelength of the radiation is comparable to the linear dimensions of the obstacle. When sound of various wavelengths or frequencies is emitted from a loudspeaker, the loudspeaker itself acts as an obstacle and casts a shadow to its rear so that only the longer bass notes are diffracted there. When a beam of light falls on the edge of an object, it will not continue in a straight line but will be slightly bent by the contact, causing a blur at the edge of the shadow of the object; the amount of bending will be proportional to the wavelength. When a stream of fast particles impinges on the atoms of a crystal, their paths are bent into a regular pattern, which can be recorded by directing the diffracted beam onto a photographic film.

Interference Movie

Please wait for images to be computed.
Click [here](#) for small movie.

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Interference Movie

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Ideal Gas Law:

An ideal gas is a gas that conforms, in physical behaviour, to a particular, idealized relation between pressure, volume, and temperature called the ideal gas law. This law is a generalization containing both Boyle's law and Charles's law as special cases and states that for a specified quantity of gas, the product of the volume, V , and pressure, P , is proportional to the absolute temperature T ; i.e., in equation form, $PV = kT$, in which k is a constant. Such a relation for a substance is called its equation of state and is sufficient to describe its gross behaviour.

The ideal gas law can be derived from the kinetic theory of gases and relies on the assumptions that (1) the gas consists of a large number of molecules, which are in random motion and obey Newton's laws of motion; (2) the volume of the molecules is negligibly small compared to the volume occupied by the gas; and (3) no forces act on the molecules except during elastic collisions of negligible duration.

Although no gas has these properties, the behaviour of real gases is described quite closely by the ideal gas law at sufficiently high temperatures and low pressures, when relatively large distances between molecules and their high speeds overcome any interaction. A gas does not obey the equation when conditions are such that the gas, or any of the component gases in a mixture, is near its condensation point.

The ideal gas law may be written in a form applicable to any gas, according to Avogadro's law (q.v.), if the constant specifying the quantity of gas is expressed in terms of the number of molecules of gas. This is done by using as the mass unit the gram-mole; i.e., the molecular weight expressed in grams. The equation of state of n gram-moles of a perfect gas can then be written as $pV/T = nR$, in which R is called the universal gas constant. This constant has been measured for various gases under nearly ideal conditions of high temperatures and low pressures, and it is found to have the same value for all gases: $R = 8.314$ joules per gram-mole-kelvin.

Thermodynamics :

A central consideration of thermodynamics is that any physical system, whether or not it can exchange energy and material with its environment, will spontaneously approach a stable condition (equilibrium) that can be described by specifying its properties, such as pressure, temperature, or chemical composition. If the external constraints are changed (for example, if the system is allowed to expand), then these properties will generally alter. The science of thermodynamics attempts to describe mathematically these changes and to predict the equilibrium conditions of the system.

Entropy :

There is one more influence of cosmological relationships upon macroscopic physics, which arises in connection with thermodynamics. The existence of irreversible processes in thermodynamics indicates a distinction between the positive and negative directions in time. As Clausius recognized in the 19th century, this irreversibility reflects a quantity, first defined by him, called entropy, which measures the degree of randomness evolving from all physical processes by which their energies tend to degrade into heat. Entropy can only increase in the positive direction of time. In fact, the increase in entropy during a process is a measure of the irreversibility of that process.



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The Lab

The Lecture Room

The Undamped and Undriven Pendulum Damped and Driven Pendula

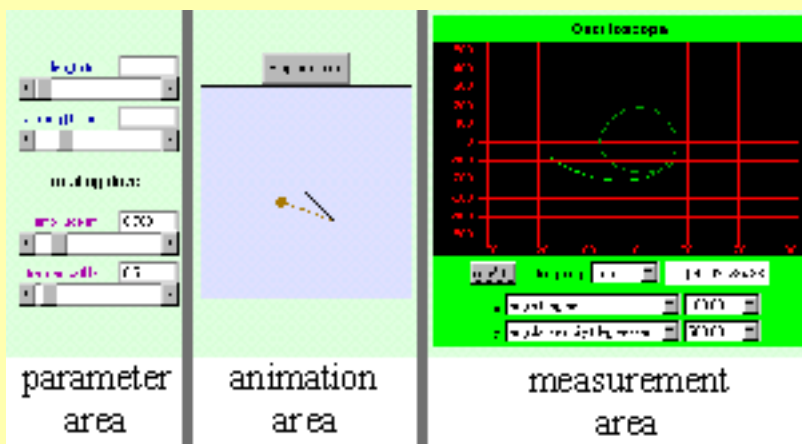
The Lab

In the lab you can do hands-on experiments. There are five experiments:

- [an undamped and undriven pendulum](#),
- [a pendulum driven by a sinusoidal force](#),
- [a horizontally driven pendulum](#),
- [a vertically driven pendulum](#),
- [a pendulum with a rotating suspension point](#).

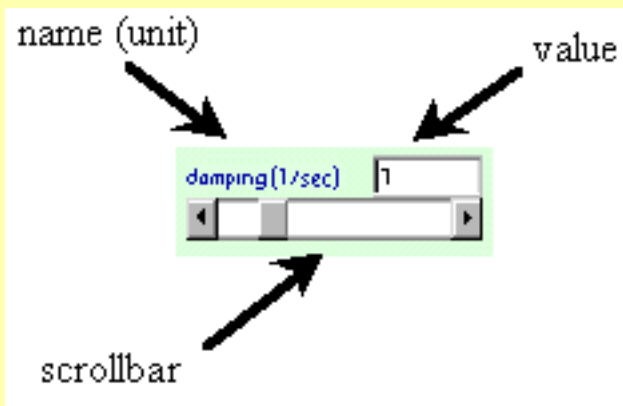
All experiments are realized with Java applets. They should run on any Web browser supporting Java. Sometimes the Java option is switched off. Be sure that it is switch on before running the experiments! Working with the applets should be rather intuitive. Nevertheless, reading the following instructions is recommended.

Instructions for the Java Applets



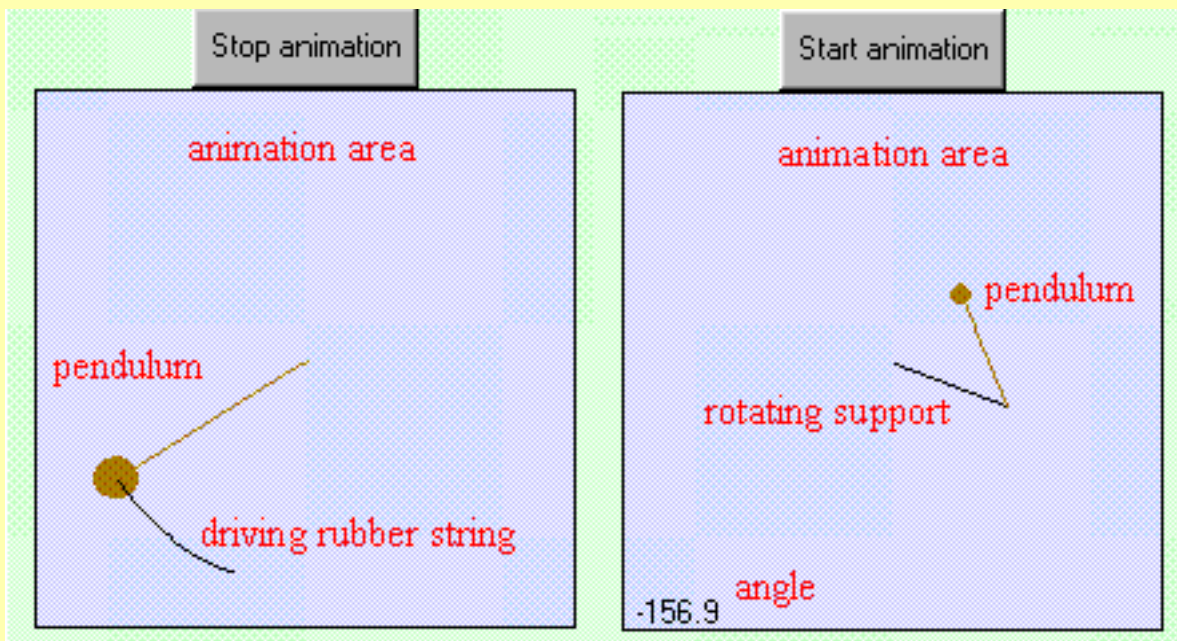
All Java applets in the lab are organized like in this screen snapshot. In the center there is the [animation area](#) where the motion of the pendulum is shown. On the left-hand side in the [parameter area](#), various parameters can be changed. In the [measurement area](#) either a stopwatch or an oscilloscope is available to measure certain properties of the system.

The parameter area



The parameter area shows control panels for several parameters. Each panel shows its verbal name (not the mathematical one!) together with its physical units. Its actual value is shown in a text field. It can be changed either by manipulating the scrollbar or by putting some numbers directly into the text field. Note, that parameters with fixed values are not shown.

The animation area



The animation is stopped by pressing the stop button. The motion is resumed by pressing it again. Note, that stopping the animation also stops the stopwatch and the oscilloscope. The motion is also stopped by clicking and dragging with the mouse pointer inside the animation area. In addition, the

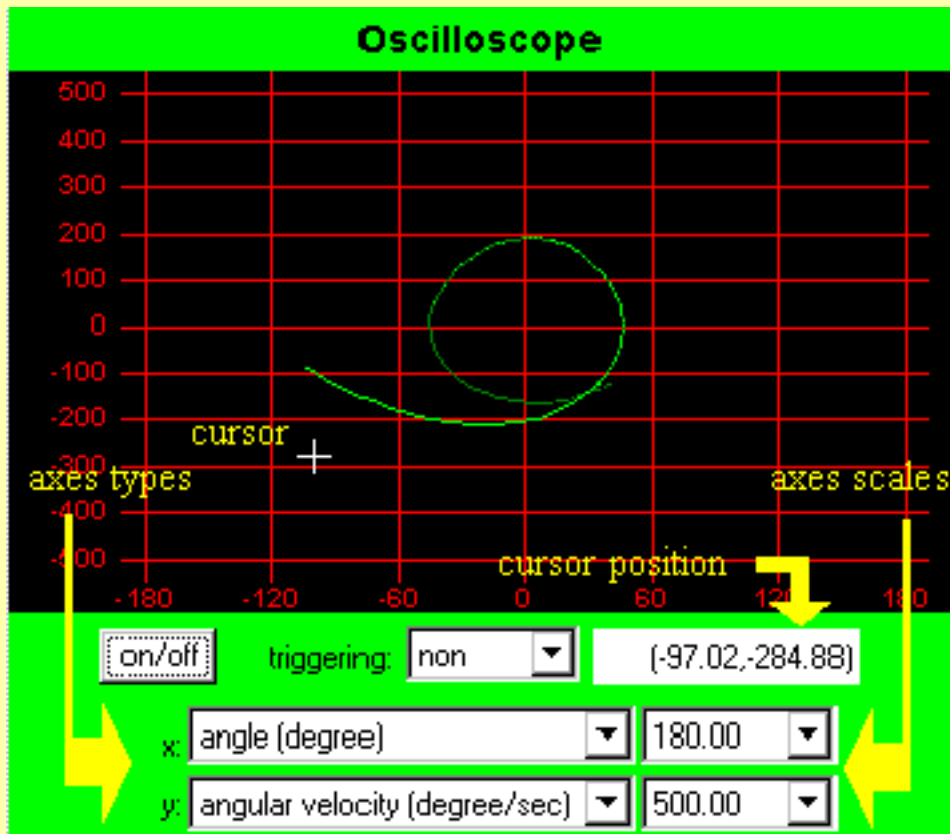
position of the pendulum is also changed according to the mouse pointer. Thereby, the angular velocity is put to zero. The actual value of the angle (in degree) is shown in the lower left corner. The driving mechanism is visualized in the following way: A periodically stretched rubber band of zero equilibrium length shows driving by a sinusoidal force (left part of the figure). A driven support is visualized by showing the lever mechanism (right part of the figure).

The measurement area



Stopwatch: It works similar to an ordinary stopwatch. It has three different phases which are indicated by the label of the leftmost button. In the stopped state it shows "Start". In the running state it shows "Lap" or "Run" depending on whether the display is updated constantly or not. In the stopped state the stopwatch is started by pressing the leftmost button. The stop and the reset

buttons stop the stopwatch. In addition the reset button resets the display to zero. The simulation time runs parallel to the physical time. Nevertheless, the stopwatch doesn't show the physical time because stopping the animation also stops the stopwatch. Its like freezing the time. After animation is restarted, the stopwatch runs again. The time increment is (roughly) given by the inverse frame rate which is 10Hz.



Oscilloscope: It is switch on by the on/off button. The type of triggering, the type of the x- and y-axis, and the scales of the x- and y-axis can be chosen from choice menus. Putting the mouse into the black screen turns the default cursor into a cross-hair cursor. In addition, its coordinates are shown.

When the x-axis is the time, the drawing of the curve is restarted at the left side of the axes box after it has leaved the box on the right side. If triggering is activated, drawing is restarted only if the trigger signal just crosses zero from negative values to positive values. In the case of internal triggering, the trigger signal is what is shown on the y-axis

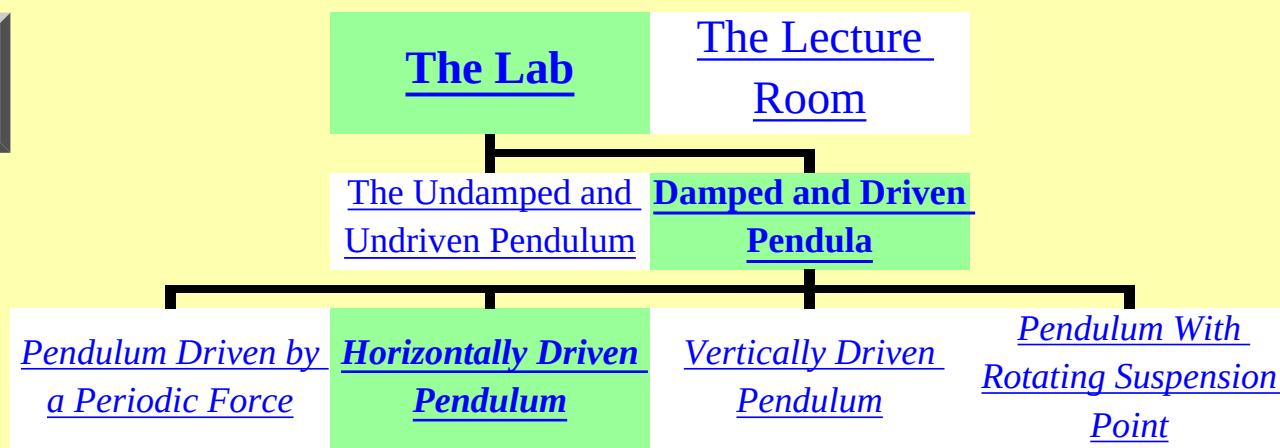
(i.e. the angle or the angular velocity). For external triggering it is the phase of driving. Note, that sometimes the pendulum may be in a dynamical state which never fulfills the trigger condition. In this case no curve is drawn. You may also miss the curve if the scale of the x-axis and/or y-axis are too small.

When the x-axis is the angle, external triggering turns the oscilloscope into a [Poincaré map](#) (more precisely: a stroboscopic map). The Poincaré condition is just the trigger condition. Thus, each point on the screen corresponds to the angle and angular velocity of pendulum at always the same phase of driving.



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The Horizontally Driven Pendulum

acceleration of gravity = 9.81 m/sec^2

Either your Web browser isn't able to run

Java applets

or you have forgotten to switch on this feature.



For *instructions* click on [here](#).

For the *equation of motion* click on [here](#)

Suggestions for EXPERIMENTS

1. Choose *length* = 1 m, *damping* = 1 sec^{-1} , and *amplitude* = 0.2 m (this are the default settings when the applet is started). Now, change the *frequency* of driving very slowly from 0.2 Hz to 0.8 Hz. Observe the change in the amplitude of oscillation and the change in the phase between driving and pendulum oscillations. You can measure it by using the [oscilloscope](#).

Related topics in the lecture room: [Resonance](#).

2. Choose the parameters as above but now change the *frequency* between 3 Hz to 4 Hz. Observe a shift of the center of oscillation either to the left or to the right (this is a [pitchfork bifurcation](#)). This bifurcation does not occur for the [pendulum driven by a periodic force](#). Why not?

Related topics in the lecture room: The [upside-down pendulum](#) and the [harmonic oscillator](#).

3. Choose *length* = 1 m, *damping* = 1 sec^{-1} , *amplitude* = 0.85 m, and *frequency* = 1 Hz. You will observe an irregular motion called [deterministic chaos](#) with a seemingly random number of left and right turns. On the oscilloscope select the *angle* as the x-axis and the *angular velocity* as the y-axis. For the scaling of the axes, choose 180 and 1000, respectively. Now, switch the oscilloscope on and you will be fascinated by the beautiful curves drawn on the screen. Turning the oscilloscope into a [Poincaré map](#) by selecting *external triggering* in order to see the irregularity better.



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Time :

Time is a measured or measurable period, a continuum that lacks spatial dimensions. "What then, is time? If no one asks me, I know what it is. If I wish to explain it to him who asks me, I do not know." In this remark St. Augustine in the 5th century AD drew attention to the fact that while time is the most familiar of concepts used in the organization of thought and action, it is also the most elusive. It cannot be given any simple, illuminating definition. In the face of this problem philosophers have sought an understanding of time by focussing on two broad questions: What is the relation between time and the physical world? And what is the relation between time and consciousness?

According to those, such as Sir Isaac Newton, who adopt an absolutist theory of time, the answer to the former question is that time is like a container within which the universe exists and change takes place. Time's existence and properties are independent of the physical universe. Time would have existed even if the universe had not. Time is held to be nonending, nonbeginning, linear, and continuous. That time has these properties is established philosophically, without reference to scientific investigation. According to the rival relationist theory, time can be reduced to change. Time is nothing over and above change in the physical universe; all hypotheses about time can be translated into hypotheses about the physical universe. Consequently the question "Has time a beginning?" becomes "Was there a first event in the history of the universe?" Also, investigating the properties of time is to be done scientifically. Relationists explore the possibility that physics could show time to have structure: it might consist of discrete particles (chronons), for instance, or it might be cyclical.

It has been realized in the 20th century that time cannot be treated in isolation from space. Consequently philosophers now tend to focus attention on space-time, conceived, after Einstein, as a continuum. While the temporal aspects of space-time remain importantly different from its spatial aspects, there is an interdependence that is shown in the case of measurement: the measure of an interval of time assigned by a clock depends on the path and speed with which it is moved. The fundamental controversy between the absolutist and the relationist remains; some philosophers argue that Einstein's theories of relativity vindicate relationist theories, others that they vindicate the absolutist theory.

Quantum :

Quantum, in physics, discrete natural unit, or packet, of energy, charge, angular momentum, or other physical property. Light, for example, appearing in some respects as a continuous electromagnetic wave, on the submicroscopic level is emitted and absorbed in discrete amounts, or quanta; and for light of a given wavelength, the magnitude of all the quanta emitted or absorbed is the same in both energy and momentum. These particle-like packets of light are called photons, a term also applicable to quanta of other forms of electromagnetic energy such as X rays and gamma rays.

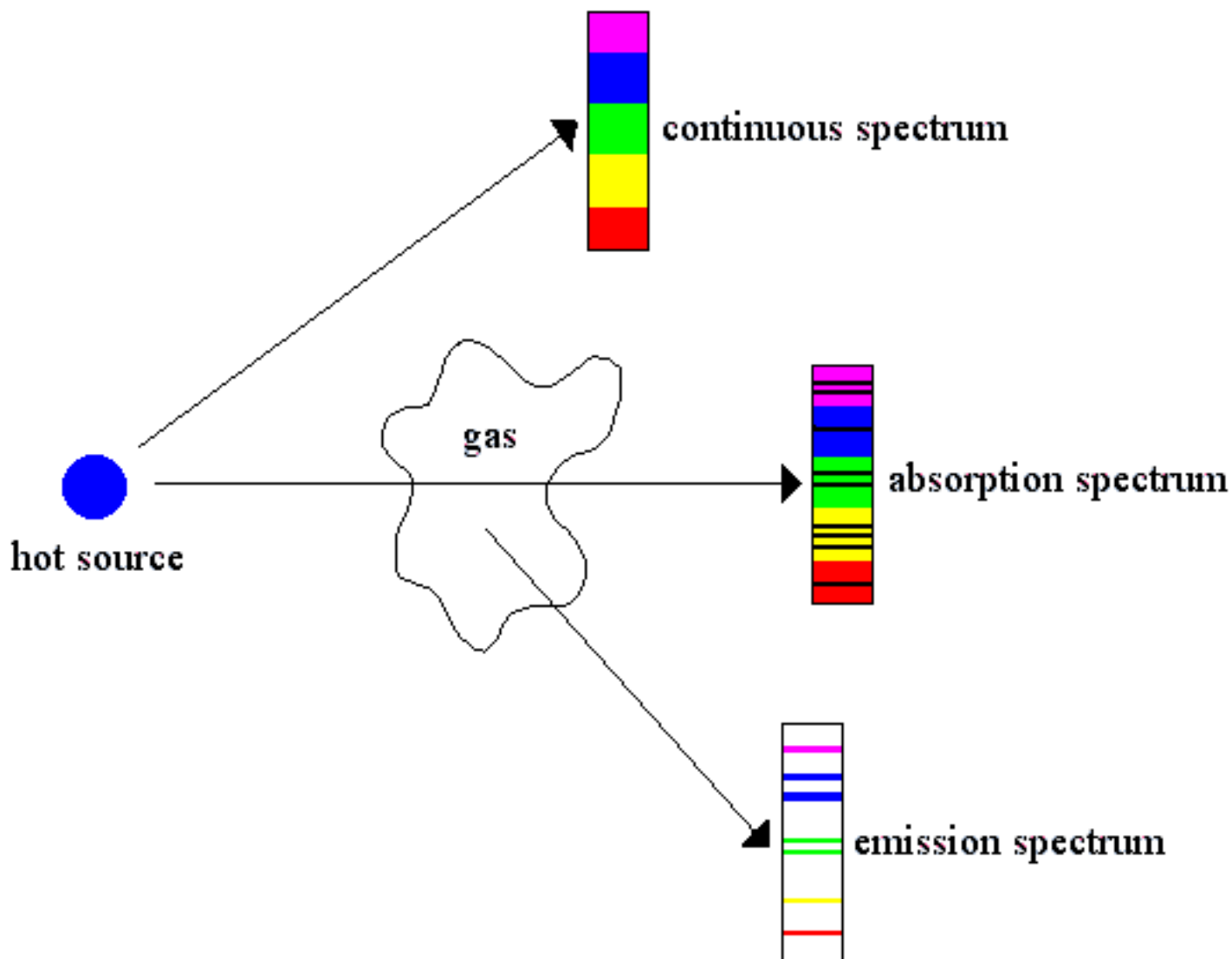
All phenomena in submicroscopic systems (the realm of quantum mechanics) exhibit quantization: observable quantities are restricted to a natural set of discrete values. When the values are multiples of a constant least amount, that amount is referred to as a quantum of the observable. Thus Planck's constant h is the quantum of action, and $h/2\pi$ (i.e., $h/2$) is the quantum of angular momentum, or spin.



Kirchhoff, Gustav :

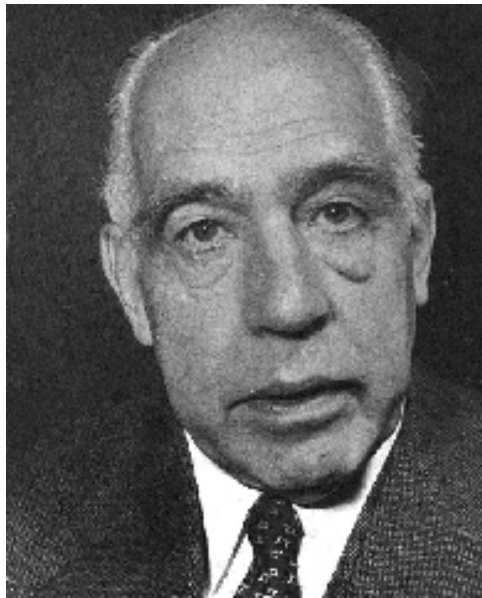
Gustav Kirchhoff (b. March 12, 1824, Knigsberg, Prussia [now Kaliningrad, Russia]--d. Oct. 17, 1887, Berlin, Ger.), German physicist who, with the chemist Robert Bunsen, firmly established the theory of spectrum analysis (a technique for chemical analysis by analyzing the light emitted by a heated material), which Kirchhoff applied to determine the composition of the Sun.

In 1847 Kirchhoff became Privatdozent (unsalaried lecturer) at the University of Berlin and three years later accepted the post of extraordinary professor of physics at the University of Breslau. In 1854 he was appointed professor of physics at the University of Heidelberg, where he joined forces with Bunsen and founded spectrum analysis. They demonstrated that every element gives off a characteristic coloured light when heated to incandescence. This light, when separated by a prism, has a pattern of individual wavelengths specific for each element. Applying this new research tool, they discovered two new elements, cesium (1860) and rubidium (1861).



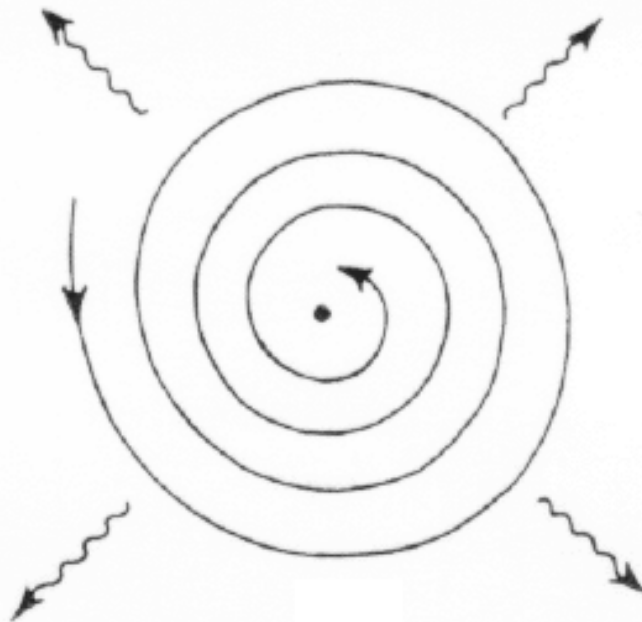
Kirchhoff went further to apply spectrum analysis to study the composition of the Sun. He found that when light passes through a gas, the gas absorbs those wavelengths that it would emit if heated. He used this principle to explain the numerous dark lines (Fraunhofer lines) in the Sun's spectrum. That discovery marked the beginning of a new era in astronomy.

In 1875 Kirchhoff was appointed to the chair of mathematical physics at the University of Berlin. Most notable of his published works are *Vorlesungen ber mathematische Physik* (4 vol., 1876-94; "Lectures on Mathematical Physics") and *Gesammelte Abhandlungen* (1882; supplement, 1891; "Collected Essays").



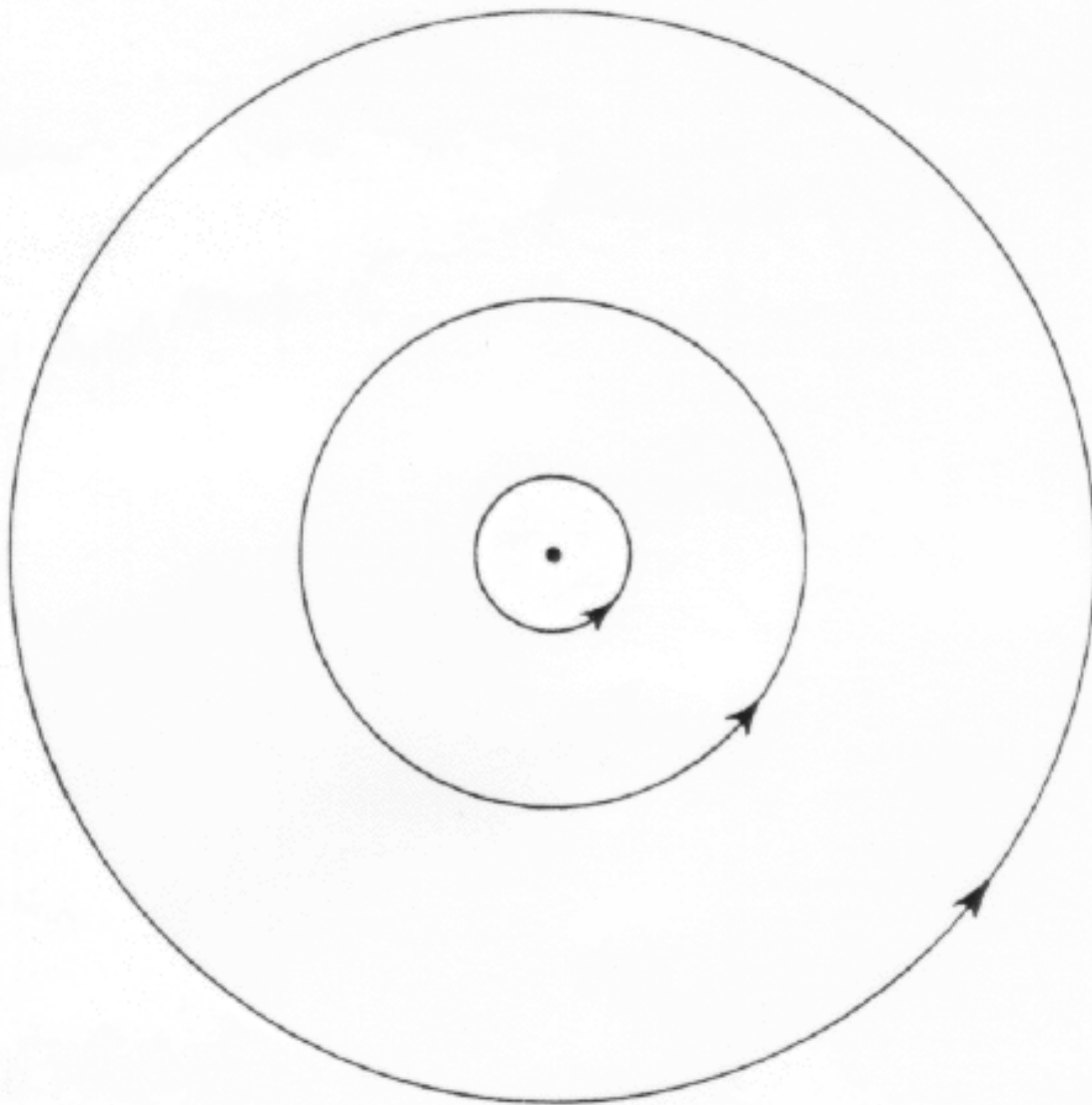
Bohr Atomic Model :

In 1913 Bohr proposed his quantized shell model of the atom to explain how electrons can have stable orbits around the nucleus. The motion of the electrons in the Rutherford model was unstable because, according to classical mechanics and electromagnetic theory, any charged particle moving on a curved path emits electromagnetic radiation; thus, the electrons would lose energy and spiral into the nucleus. To remedy the stability problem, Bohr modified the Rutherford model by requiring that the electrons move in orbits of fixed size and energy. The energy of an electron depends on the size of the orbit and is lower for smaller orbits. Radiation can occur only when the electron jumps from one orbit to another. The atom will be completely stable in the state with the smallest orbit, since there is no orbit of lower energy into which the electron can jump.



According to classical physics, an electron in orbit around an atomic nucleus should emit electromagnetic radiation (photons) continuously, because it is continually accelerating in a curved path. The resulting loss of energy implies that

continually accelerating in a curved path. The resulting loss of energy implies that the electron should spiral into the nucleus in a very short time (i.e. atoms can not exist).

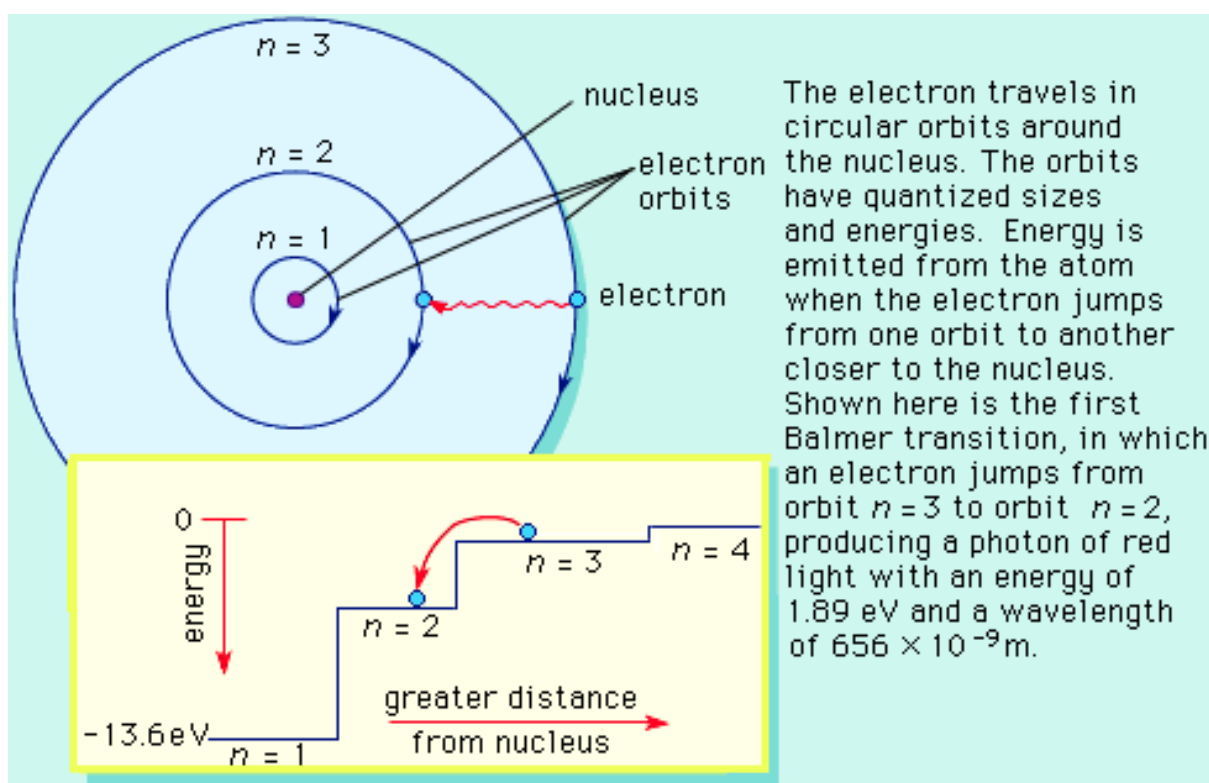


Bohr proposed that electrons are restricted to certain fixed (quantized) orbits. An electron can jump, suddenly, between these orbits by absorbing or emitting a photon with the appropriate precise wavelength.

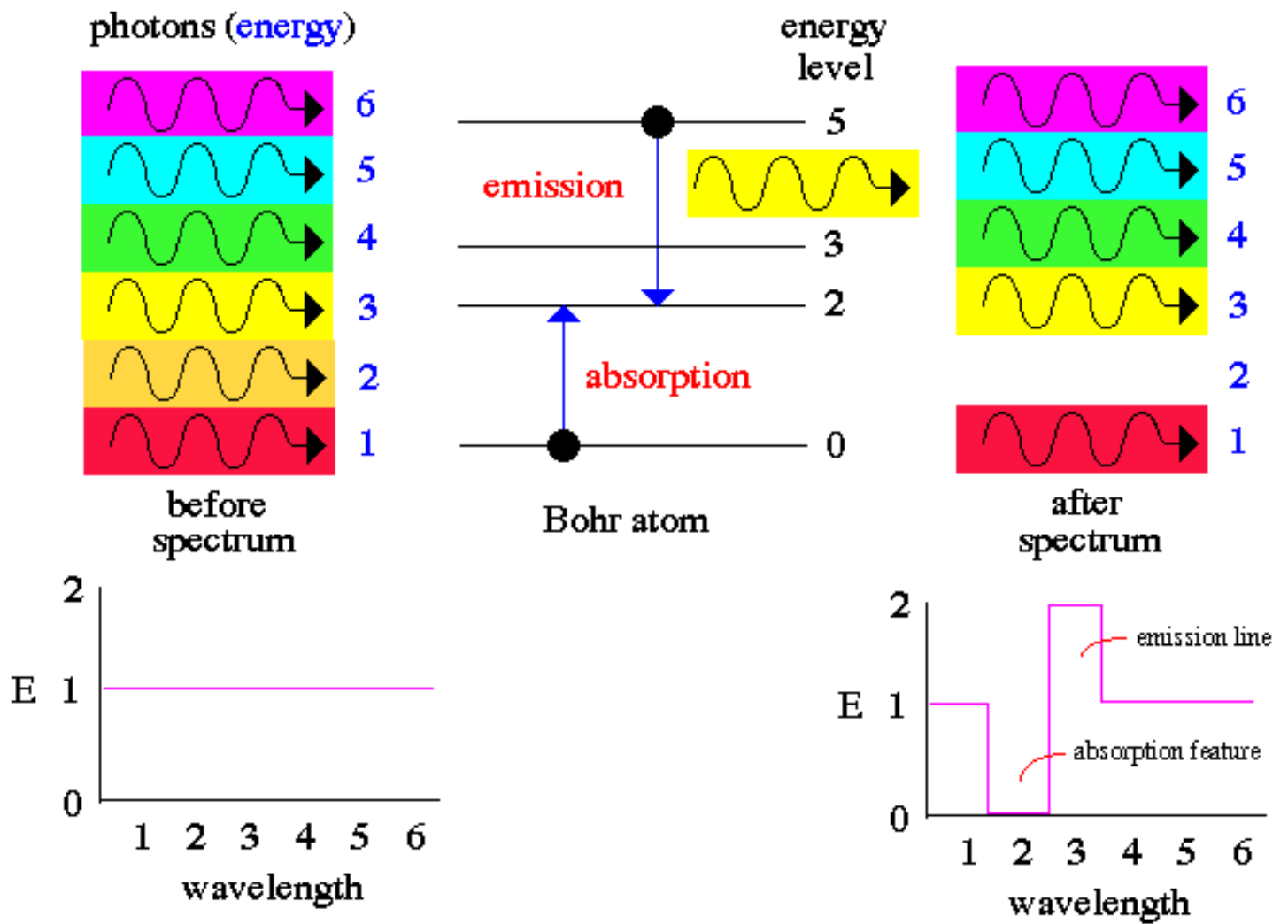
Bohr's starting point was to realize that classical mechanics by itself could never explain the atom's stability. A stable atom has a certain size so that any equation describing it must contain some fundamental constant or combination of constants with a dimension of length. The classical fundamental constants--namely, the charges and the masses of the electron and the nucleus--cannot be combined to make a length. Bohr noticed, however, that the quantum constant formulated by the German physicist Max Planck has dimensions which, when combined with the mass and charge of the electron, produce a measure of length. Numerically, the measure is close to the known size of atoms. This encouraged Bohr to use Planck's constant in searching for a theory of the atom.

Planck had introduced his constant in 1900 in a formula explaining the light radiation emitted from heated bodies. According to classical theory, comparable amounts of light energy should be produced at all frequencies. This is not only contrary to observation but also implies the absurd result that the total energy radiated by a heated body should be infinite. Planck postulated that energy can only be emitted or absorbed in discrete amounts, which he called quanta (the Latin word for "how much"). The energy quantum is related to the frequency of the light by a new fundamental constant, h . When a body is heated, its radiant energy in a particular frequency range is, according to classical theory, proportional to the temperature of the body. With Planck's hypothesis, however, the radiation can occur only in quantum amounts of energy. If the radiant energy is less than the quantum of energy, the amount of light in that frequency range will be reduced. Planck's formula correctly describes radiation from heated bodies. Planck's constant has the dimensions of action, which may be expressed as units of energy multiplied by time, units of momentum multiplied by length, or units of angular momentum. For example, Planck's constant can be written as $h = 6.6 \times 10^{-34}$ joule seconds.

Using Planck's constant, Bohr obtained an accurate formula for the energy levels of the hydrogen atom. He postulated that the angular momentum of the electron is quantized--i.e., it can have only discrete values. He assumed that otherwise electrons obey the laws of classical mechanics by traveling around the nucleus in circular orbits. Because of the quantization, the electron orbits have fixed sizes and energies. The orbits are labeled by an integer, the quantum number n .

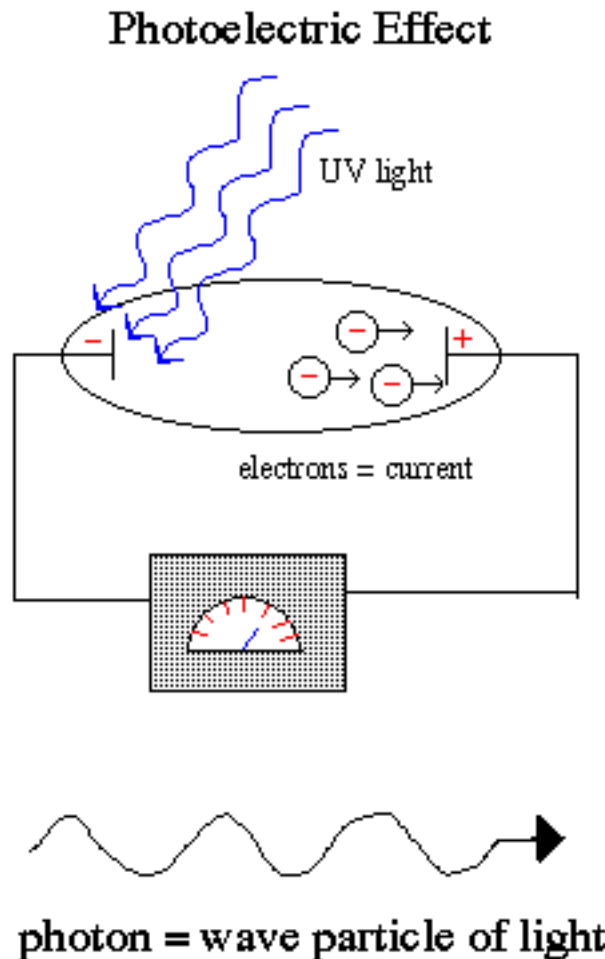


With his model, Bohr explained how electrons could jump from one orbit to another only by emitting or absorbing energy in fixed quanta. For example, if an electron jumps one orbit closer to the nucleus, it must emit energy equal to the difference of the energies of the two orbits. Conversely, when the electron jumps to a larger orbit, it must absorb a quantum of light equal in energy to the difference in orbits.



Photoelectric Effect :

An unusual phenomenon was discovered in the early 1900's. If a beam of light is pointed at the negative end of a pair of charged plates, a current flow is measured. A current is simply a flow of electrons in a metal, such as a wire. Thus, the beam of light must be liberating electrons from one metal plate, which are attracted to the other plate by electrostatic forces. This results in a current flow.



In classical physics, one would expect the current flow to be proportional to the strength of the beam of light (more light = more electrons liberated = more current). However, the observed phenomenon was that the current flow was basically constant with light strength, yet varied strong with the wavelength of light such that there was a sharp cutoff and no current flow for long wavelengths.

Einstein successfully explained the photoelectric effect within the context of the new physics of the time, quantum physics. In his scientific paper, he showed that light was made of packets of energy quantum called photons. Each photon carries a specific energy related to its wavelength, such that photons of short wavelength (blue light) carry more energy than long wavelength (red light) photons. To release an electron from a metal plate required a minimal energy which could only be transferred by a photon of energy equal or greater than that minimal threshold energy (i.e. the wavelength of the light had to be a

sufficiently short). Each photon of blue light released an electron. But all red photons were too weak. The result is no matter how much red light was shown on the metal plate, there was no current.

The photoelectric earned Einstein the Nobel Prize, and introduced the term "photon" of light into our terminology.

Einstein:

Recognized in his own time as one of the most creative intellects in human history, Albert Einstein, in the first 15 years of the 20th century, advanced a series of theories that for the first time asserted the equivalence of mass and energy and proposed entirely new ways of thinking about space, time, and gravitation. His theories of relativity and gravitation were a profound advance over the old Newtonian physics and revolutionized scientific and philosophic inquiry.

Herein lay the unique drama of Einstein's life. He was a self-confessed lone traveller; his mind and heart soared with the cosmos, yet he could not armour himself against the intrusion of the often horrendous events of the human community. Almost reluctantly he admitted that he had a "passionate sense of social justice and social responsibility." His celebrity gave him an influential voice that he used to champion such causes as pacifism, liberalism, and Zionism. The irony for this idealistic man was that his famous postulation of an energy-mass equation, which states that a particle of matter can be converted into an enormous quantity of energy, had its spectacular proof in the creation of the atomic and hydrogen bombs, the most destructive weapons ever known.

Albert Einstein was born in Ulm, Germany, on March 14, 1879. The following year his family moved to Munich, where Hermann Einstein, his father, and Jakob Einstein, his uncle, set up a small electrical plant and engineering works. In Munich Einstein attended rigidly disciplined schools. Under the harsh and pedantic regimentation of 19th-century German education, which he found intimidating and boring, he showed little scholastic ability. At the behest of his mother, Einstein also studied music; though throughout life he played exclusively for relaxation, he became an accomplished violinist. It was then only Uncle Jakob who stimulated in Einstein a fascination for mathematics and Uncle Cdsar Koch who stimulated a consuming curiosity about science.

By the age of 12 Einstein had decided to devote himself to solving the riddle of the "huge world." Three years later, with poor grades in history, geography, and languages, he left school with no diploma and went to Milan to rejoin his family, who had recently moved there from Germany because of his father's business setbacks. Albert Einstein resumed his education in Switzerland, culminating in four years of physics and mathematics at the renowned Federal Polytechnic Academy in Z|rich.

After his graduation in the spring of 1900, he became a Swiss citizen, worked for two months as a mathematics teacher, and then was employed as examiner at the Swiss patent office in Bern. With his newfound security, Einstein married his university sweetheart, Mileva Maric, in 1903.

Early in 1905 Einstein published in the prestigious German physics monthly *Annalen der Physik* a thesis, "A New Determination of Molecular Dimensions," that won him a Ph.D. from the University of Z|rich. Four more important papers appeared in *Annalen* that year and forever changed man's view of the universe.

The first of these, "\ber die von der molekularkinetischen Theorie der Wdrme geforderte Bewegung von

in ruhenden Flüssigkeiten suspendierten Teilchen" ("On the Motion--Required by the Molecular Kinetic Theory of Heat--of Small Particles Suspended in a Stationary Liquid"), provided a theoretical explanation of Brownian motion. In "Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt" ("On a Heuristic Viewpoint Concerning the Production and Transformation of Light"), Einstein postulated that light is composed of individual quanta (later called photons) that, in addition to wavelike behaviour, demonstrate certain properties unique to particles. In a single stroke he thus revolutionized the theory of light and provided an explanation for, among other phenomena, the emission of electrons from some solids when struck by light, called the photoelectric effect.

Einstein's special theory of relativity, first printed in "Zur Elektrodynamik bewegter Körper" ("On the Electrodynamics of Moving Bodies"), had its beginnings in an essay Einstein wrote at age 16. The precise influence of work by other physicists on Einstein's special theory is still controversial. The theory held that, if, for all frames of reference, the speed of light is constant and if all natural laws are the same, then both time and motion are found to be relative to the observer.

In the mathematical progression of the theory, Einstein published his fourth paper, "Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?" ("Does the Inertia of a Body Depend Upon Its Energy Content?"). This mathematical footnote to the special theory of relativity established the equivalence of mass and energy, according to which the energy E of a quantity of matter, with mass m , is equal to the product of the mass and the square of the velocity of light, c . This relationship is commonly expressed in the form $E = mc^2$.

Public understanding of this new theory and acclaim for its creator were still many years off, but Einstein had won a place among Europe's most eminent physicists, who increasingly sought his counsel, as he did theirs. While Einstein continued to develop his theory, attempting now to encompass with it the phenomenon of gravitation, he left the patent office and returned to teaching--first in Switzerland, briefly at the German University in Prague, where he was awarded a full professorship, and then, in the winter of 1912, back at the Polytechnic in Zürich. He was later remembered from this time as a very happy man, content in his marriage and delighted with his two young sons, Hans Albert and Edward.

In April 1914 the family moved to Berlin, where Einstein had accepted a position with the Prussian Academy of Sciences, an arrangement that permitted him to continue his researches with only the occasional diversion of lecturing at the University of Berlin. His wife and two sons vacationed in Switzerland that summer and, with the eruption of World War I, were unable to return to Berlin. A few years later this enforced separation was to lead to divorce. Einstein abhorred the war and was an outspoken critic of German militarism among the generally acquiescent academic community in Berlin, but he was primarily engrossed in perfecting his general theory of relativity, which he published in *Annalen der Physik* as "Die Grundlagen der allgemeinen Relativitätstheorie" ("The Foundation of the General Theory of Relativity") in 1916. The heart of this postulate was that gravitation is not a force, as Newton had said, but a curved field in the space-time continuum, created by the presence of mass. This notion could be proved or disproved, he suggested, by measuring the deflection of starlight as it travelled close by the Sun, the starlight being visible only during a total eclipse. Einstein predicted twice the light

deflection that would be accountable under Newton's laws.

His new equations also explained for the first time the puzzling irregularity--that is, the slight advance--in the planet Mercury's perihelion, and they demonstrated why stars in a strong gravitational field emitted light closer to the red end of the spectrum than those in a weaker field.

While Einstein awaited the end of the war and the opportunity for his theory to be tested under eclipse conditions, he became more and more committed to pacifism, even to the extent of distributing pacifist literature to sympathizers in Berlin. His attitudes were greatly influenced by the French pacifist and author Romain Rolland, whom he met on a wartime visit to Switzerland. Rolland's diary later provided the best glimpse of Einstein's physical appearance as he reached his middle 30s:

Einstein is still a young man, not very tall, with a wide and long face, and a great mane of crispy, frizzled and very black hair, sprinkled with gray and rising high from a lofty brow. His nose is fleshy and prominent, his mouth small, his lips full, his cheeks plump, his chin rounded. He wears a small cropped mustache. (By permission of Madame Marie Romain Rolland.)

Einstein's view of humanity during the war period appears in a letter to his friend, the Austrian-born Dutch physicist Paul Ehrenfest:

The ancient Jehovah is still abroad. Alas, he slays the innocent along with the guilty, whom he strikes so fearsomely blind that they can feel no sense of guilt. . . . We are dealing with an epidemic delusion which, having caused infinite suffering, will one day vanish and become a monstrous and incomprehensible source of wonderment to later generations. (From Otto Nathan and Heinz Norden [eds.], *Einstein on Peace*; Simon and Schuster, 1960.)

It would be said often of Einstein that he was naive about human affairs; for example, with the proclamation of the German Republic and the armistice in 1918, he was convinced that militarism had been thoroughly abolished in Germany.

International fame came to Einstein in November 1919, when the Royal Society of London announced that its scientific expedition to Principe Island, in the Gulf of Guinea, had photographed the solar eclipse on May 29 of that year and completed calculations that verified the predictions made in Einstein's general theory of relativity. Few could understand relativity, but the basic postulates were so revolutionary and the scientific community was so obviously bedazzled that the physicist was acclaimed the greatest genius on Earth. Einstein himself was amazed at the reaction and apparently displeased, for he resented the consequent interruptions of his work. After his divorce he had, in the summer of 1919, married Elsa, the widowed daughter of his late father's cousin. He lived quietly with Elsa and her two daughters in Berlin, but, inevitably, his views as a foremost savant were sought on a variety of issues.

Despite the now deteriorating political situation in Germany, Einstein attacked nationalism and promoted pacifist ideals. With the rising tide of anti-Semitism in Berlin, Einstein was castigated for his "Bolshevism in physics," and the fury against him in right-wing circles grew when he began publicly to support the Zionist movement. Judaism had played little part in his life, but he insisted that, as a snail can shed his shell and still be a snail, so a Jew can shed his faith and still be a Jew.

Although Einstein was regarded warily in Berlin, such was the demand for him in other European cities that he travelled widely to lecture on relativity, usually arriving at each place by third-class rail carriage, with a violin tucked under his arm. So successful were his lectures that one enthusiastic impresario guaranteed him a three-week booking at the London Palladium. He ignored the offer, but, at the request of the Zionist leader Chaim Weizmann, toured the United States in the spring of 1921 to raise money for the Palestine Foundation Fund. Frequently treated like a circus freak and feted from morning to night, Einstein nevertheless was gratified by the standards of scientific research and the "idealistic attitudes" that he found prevailing in the United States.

During the next three years Einstein was constantly on the move, journeying not only to European capitals but also to the Orient, to the Middle East, and to South America. According to his diary notes, he found nobility among the Hindus of Ceylon, a pureness of soul among the Japanese, and a magnificent intellectual and moral calibre among the Jewish settlers in Palestine. His wife later wrote that, on steaming into one new harbour, Einstein had said to her, "Let us take it all in before we wake up."

In Shanghai a cable reached him announcing that he had been awarded the 1921 Nobel Prize for Physics "for your photoelectric law and your work in the field of theoretical physics." Relativity, still the centre of controversy, was not mentioned.

Though the 1920s were tumultuous times of wide acclaim, and some notoriety, Einstein did not waver from his new search--to find the mathematical relationship between electromagnetism and gravitation. This would be a first step, he felt, in discovering the common laws governing the behaviour of everything in the universe, from the electron to the planets. He sought to relate the universal properties of matter and energy in a single equation or formula, in what came to be called a unified field theory. This turned out to be a fruitless quest that occupied the rest of his life. Einstein's peers generally agreed quite early that his search was destined to fail because the rapidly developing quantum theory uncovered an uncertainty principle in all measurements of the motion of particles: the movement of a single particle simply could not be predicted because of a fundamental uncertainty in measuring simultaneously both its speed and its position, which means, in effect, that the future of any physical system at the subatomic level cannot be predicted. While fully recognizing the brilliance of quantum mechanics, Einstein rejected the idea that these theories were absolute and persevered with his theory of general relativity as the more satisfactory foundation to future discovery. He was widely quoted on his belief in an exactly engineered universe: "[God](#) is subtle but he is not malicious." On this point, he parted company with most theoretical physicists. The distinguished German quantum theorist Max Born, a close friend of Einstein, said at the time: "Many of us regard this as a tragedy, both for him, as he gropes his way in loneliness, and for us, who miss our leader and standard-bearer." This appraisal, and others pronouncing his work in later life as

largely wasted effort, will have to await the judgment of later generations.

The year of Einstein's 50th birthday, 1929, marked the beginning of the ebb flow of his life's work in a number of aspects. Early in the year the Prussian Academy published the first version of his unified-field theory, but, despite the sensation it caused, its very preliminary nature soon became apparent. The reception of the theory left him undaunted, but Einstein was dismayed by the preludes to certain disaster in the field of human affairs: Arabs launched savage attacks on Jewish colonists in Palestine; the Nazis gained strength in Germany; the League of Nations proved so impotent that Einstein resigned abruptly from its Committee on Intellectual Cooperation as a protest to its timidity; and the stock market crash in New York City heralded worldwide economic crisis.

Crushing Einstein's natural gaiety more than any of these events was the mental breakdown of his younger son, Edward. Edward had worshipped his father from a distance but now blamed him for deserting him and for ruining his life. Einstein's sorrow was eased only slightly by the amicable relationship he enjoyed with his older son, Hans Albert.

As visiting professor at Oxford University in 1931, Einstein spent as much time espousing pacifism as he did discussing science. He went so far as to authorize the establishment of the Einstein War Resisters' International Fund in order to bring massive public pressure to bear on the World Disarmament Conference, scheduled to meet in Geneva in February 1932. When these talks foundered, Einstein felt that his years of supporting world peace and human understanding had accomplished nothing. Bitterly disappointed, he visited Geneva to focus world attention on the "farce" of the disarmament conference. In a rare moment of fury, Einstein stated to a journalist,

They [the politicians and statesmen] have cheated us. They have fooled us. Hundreds of millions of people in Europe and in America, billions of men and women yet to be born, have been and are being cheated, traded and tricked out of their lives and health and well-being.

Shortly after this, in a famous exchange of letters with the Austrian psychiatrist Sigmund Freud, Einstein suggested that people must have an innate lust for hatred and destruction. Freud agreed, adding that war was biologically sound because of the love-hate instincts of man and that pacifism was an idiosyncrasy directly related to Einstein's high degree of cultural development. This exchange was only one of Einstein's many philosophic dialogues with renowned men of his age. With Rabindranath Tagore, Hindu poet and mystic, he discussed the nature of truth. While Tagore held that truth was realized through man, Einstein maintained that scientific truth must be conceived as a valid truth that is independent of humanity. "I cannot prove that I am right in this, but that is my religion," said Einstein. Firmly denying atheism, Einstein expressed a belief in "Spinoza's God who reveals himself in the harmony of what exists." The physicist's breadth of spirit and depth of enthusiasm were always most evident among truly intellectual men. He loved being with the physicists Paul Ehrenfest and Hendrick A. Lorentz at The Netherlands' Leiden University, and several times he visited the California Institute of Technology in Pasadena to attend seminars at the Mt. Wilson Observatory, which had become world renowned as a

centre for astrophysical research. At Mt. Wilson he heard the Belgian scientist Abbi Georges Lemantre detail his theory that the universe had been created by the explosion of a "primeval atom" and was still expanding. Gleefully, Einstein jumped to his feet, applauding. "This is the most beautiful and satisfactory explanation of creation to which I have ever listened," he said.

In 1933, soon after Adolf Hitler became chancellor of Germany, Einstein renounced his German citizenship and left the country. He later accepted a full-time position as a foundation member of the school of mathematics at the new Institute for Advanced Study in Princeton, New Jersey. In reprisal, Nazi storm troopers ransacked his beloved summer house at Caputh, near Berlin, and confiscated his sailboat. Einstein was so convinced that Nazi Germany was preparing for war that, to the horror of Romain Rolland and his other pacifist friends, he violated his pacifist ideals and urged free Europe to arm and recruit for defense.

Although his warnings about war were largely ignored, there were fears for Einstein's life. He was taken by private yacht from Belgium to England. By the time he arrived in Princeton in October 1933, he had noticeably aged. A friend wrote,

It was as if something had deadened in him. He sat in a chair at our place, twisting his white hair in his fingers and talking dreamily about everything under the sun. He was not laughing any more.

In Princeton Einstein set a pattern that was to vary little for more than 20 years. He lived with his wife in a simple, two-story frame house and most mornings walked a mile or so to the Institute, where he worked on his unified field theory and talked with colleagues. For relaxation he played his violin and sailed on a local lake. Only rarely did he travel, even to New York. In a letter to Queen Elisabeth of Belgium, he described his new refuge as a "wonderful little spot, . . . a quaint and ceremonious village of puny demigods on stilts." Eventually he acquired American citizenship, but he always continued to think of himself as a European. Pursuing his own line of theoretical research outside the mainstream of physics, he took on an air of fixed serenity. "Among my European friends, I am now called Der grosse Schweiger ("The Great Stone Face"), a title I well deserve," he said. Even his wife's death late in 1936 did not disturb his outward calm. "It seemed that the difference between life and death for Einstein consisted only in the difference between being able and not being able to do physics," wrote Leopold Infeld, the Polish physicist who arrived in Princeton at this time.

Niels Bohr, the great Danish atomic physicist, brought news to Einstein in 1939 that the German refugee physicist Lise Meitner had split the uranium atom, with a slight loss of total mass that had been converted into energy. Meitner's experiments, performed in Copenhagen, had been inspired by similar, though less precise, experiments done months earlier in Berlin by two German chemists, Otto Hahn and Fritz Strassmann. Bohr speculated that, if a controlled chain-reaction splitting of uranium atoms could be accomplished, a mammoth explosion would result. Einstein was skeptical, but laboratory experiments in the United States showed the feasibility of the idea. With a European war regarded as imminent and fears that Nazi scientists might build such a "bomb" first, Einstein was persuaded by colleagues to write a

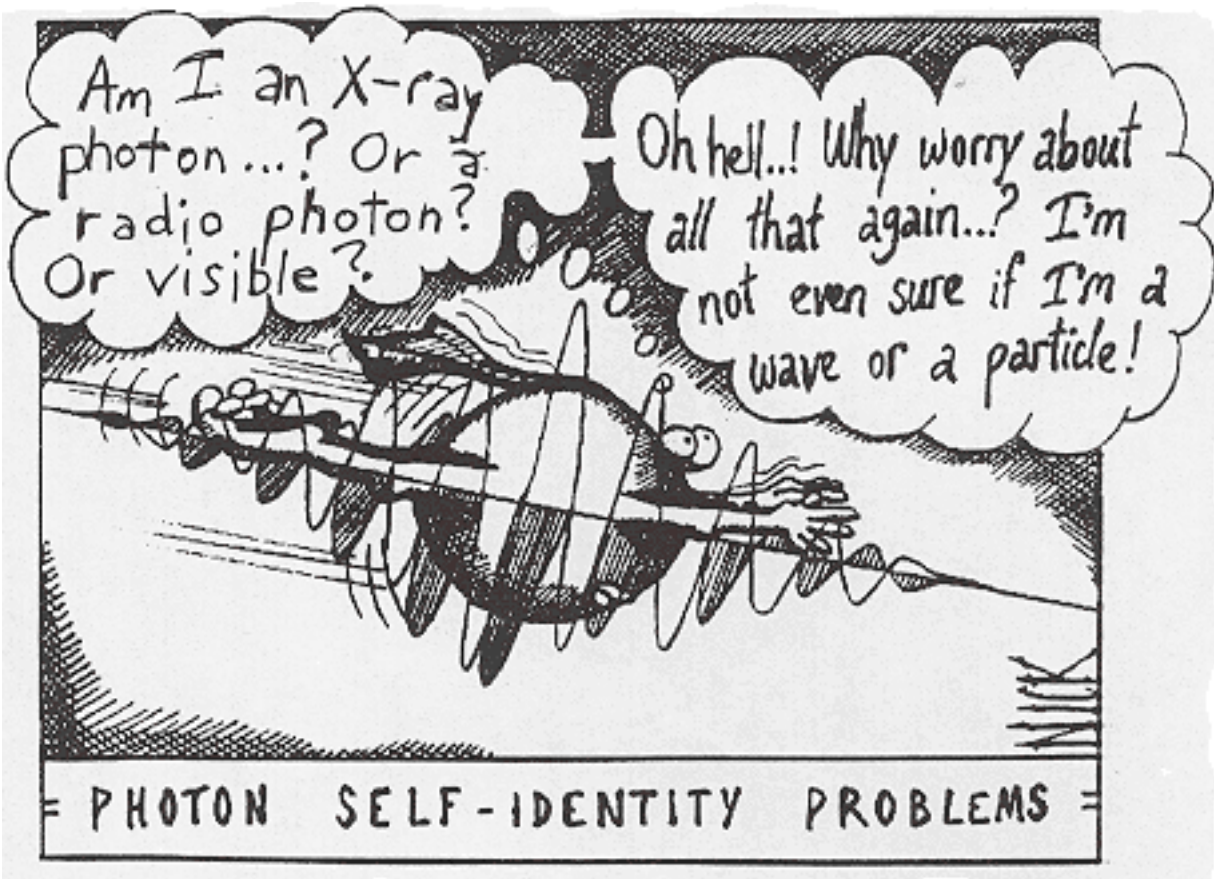
letter to President Franklin D. Roosevelt urging "watchfulness and, if necessary, quick action" on the part of the United States in atomic-bomb research. This recommendation marked the beginning of the Manhattan Project.

Although he took no part in the work at Los Alamos, New Mexico, and did not learn that a nuclear-fission bomb had been made until Hiroshima was razed in 1945, Einstein's name was emphatically associated with the advent of the atomic age. He readily joined those scientists seeking ways to prevent any future use of the bomb, his particular and urgent plea being the establishment of a world government under a constitution drafted by the United States, Britain, and Russia. With the spur of the atomic fear that haunted the world, he said "we must not be merely willing, but actively eager to submit ourselves to the binding authority necessary for world security." Once more, Einstein's name surged through the newspapers. Letters and statements tumbled out of his Princeton study, and in the public eye Einstein the physicist dissolved into Einstein the world citizen, a kind "grand old man" devoting his last years to bringing harmony to the world.

The rejection of his ideals by statesmen and politicians did not break him, because his prime obsession still remained with physics. "I cannot tear myself away from my work," he wrote at the time. "It has me inexorably in its clutches." In proof of this came his new version of the unified field in 1950, a most meticulous mathematical essay that was immediately but politely criticized by most physicists as untenable.

Compared with his renown of a generation earlier, Einstein was virtually neglected and said himself that he felt almost like a stranger in the world. His health deteriorated to the extent that he could no longer play the violin or sail his boat. Many years earlier, chronic abdominal pains had forced him to give up smoking his pipe and to watch his diet carefully.

On April 18, 1955, Einstein died in his sleep at Princeton Hospital. On his desk lay his last incomplete statement, written to honour Israeli Independence Day. It read in part: "What I seek to accomplish is simply to serve with my feeble capacity truth and justice at the risk of pleasing no one." His contribution to man's understanding of the universe was matchless, and he is established for all time as a giant of science. Broadly speaking, his crusades in human affairs seem to have had no lasting impact. Einstein perhaps anticipated such an assessment of his life when he said, "Politics are for the moment. An equation is for eternity."



Wave/Particle Duality :

Wave/particle duality is the possession by physical entities (such as light and electrons) of both wavelike and particle-like characteristics. On the basis of experimental evidence, the German physicist Albert Einstein first showed (1905) that light, which had been considered a form of electromagnetic waves, must also be thought of as particle-like, or localized in packets of discrete energy (see the [photoelectric effect](#)). The French physicist Louis de Broglie proposed (1924) that electrons and other discrete bits of matter, which until then had been conceived only as material particles, also have wave properties such as wavelength and frequency. Later (1927) the wave nature of electrons was experimentally established. An understanding of the complementary relation between the wave aspects and the particle aspects of the same phenomenon was announced in 1928.

Uncertainty Principle :

also called Heisenberg Uncertainty Principle, or Indeterminacy Principle, articulated (1927) by the German physicist Werner Heisenberg, that the position and the velocity of an object cannot both be measured exactly, at the same time, even in theory. The very concepts of exact position and exact velocity together, in fact, have no meaning in nature.

Ordinary experience provides no clue of this principle. It is easy to measure both the position and the velocity of, say, an automobile, because the uncertainties implied by this principle for ordinary objects are too small to be observed. The complete rule stipulates that the product of the uncertainties in position and velocity is equal to or greater than a tiny physical quantity, or constant (about 10^{-34} joule-second, the value of the quantity h (where h is Planck's constant)). Only for the exceedingly small masses of atoms and subatomic particles does the product of the uncertainties become significant.

Any attempt to measure precisely the velocity of a subatomic particle, such as an electron, will knock it about in an unpredictable way, so that a simultaneous measurement of its position has no validity. This result has nothing to do with inadequacies in the measuring instruments, the technique, or the observer; it arises out of the intimate connection in nature between particles and waves in the realm of subatomic dimensions.

Every particle has a wave associated with it; each particle actually exhibits wavelike behaviour. The particle is most likely to be found in those places where the undulations of the wave are greatest, or most intense. The more intense the undulations of the associated wave become, however, the more ill defined becomes the wavelength, which in turn determines the momentum of the particle. So a strictly localized wave has an indeterminate wavelength; its associated particle, while having a definite position, has no certain velocity. A particle wave having a well-defined wavelength, on the other hand, is spread out; the associated particle, while having a rather precise velocity, may be almost anywhere. A quite accurate measurement of one observable involves a relatively large uncertainty in the measurement of the other.

The uncertainty principle is alternatively expressed in terms of a particle's momentum and position. The momentum of a particle is equal to the product of its mass times its velocity. Thus, the product of the uncertainties in the momentum and the position of a particle equals $h/2$ or more. The principle applies to other related (conjugate) pairs of observables, such as energy and time: the product of the uncertainty in an energy measurement and the uncertainty in the time interval during which the measurement is made also equals $h/2$ or more. The same relation holds, for an unstable atom or nucleus, between the uncertainty in the quantity of energy radiated and the uncertainty in the lifetime of the unstable system as it makes a transition to a more stable state.

Complementarity :

A characteristic feature of quantum physics is the principle of complementarity, which "implies the impossibility of any sharp separation between the behaviour of atomic objects and the interaction with the measuring instruments which serve to define the conditions under which the phenomena appear." As a result, "evidence obtained under different experimental conditions cannot be comprehended within a single picture, but must be regarded as complementary in the sense that only the totality of the phenomena exhausts the possible information about the objects." This interpretation of the meaning of quantum physics, which implied an altered view of the meaning of physical explanation, gradually came to be accepted by the majority of physicists during the 1930's.

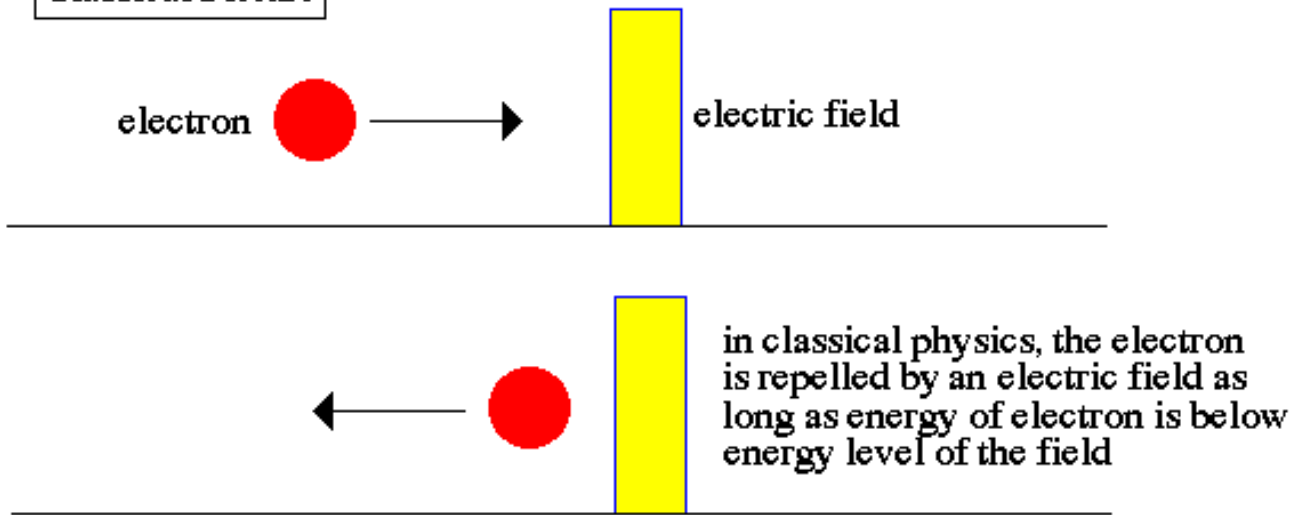
Quantum Wave Function :

The wave function, also called Schrodinger's Equation, is a mathematical description of all the possibilities for an object. For example, we could imagine the wave function as a deck of 52 cards where each card is a yet unobserved quantum state. The deck has 52 possibilities so the wave function has 52 humps.

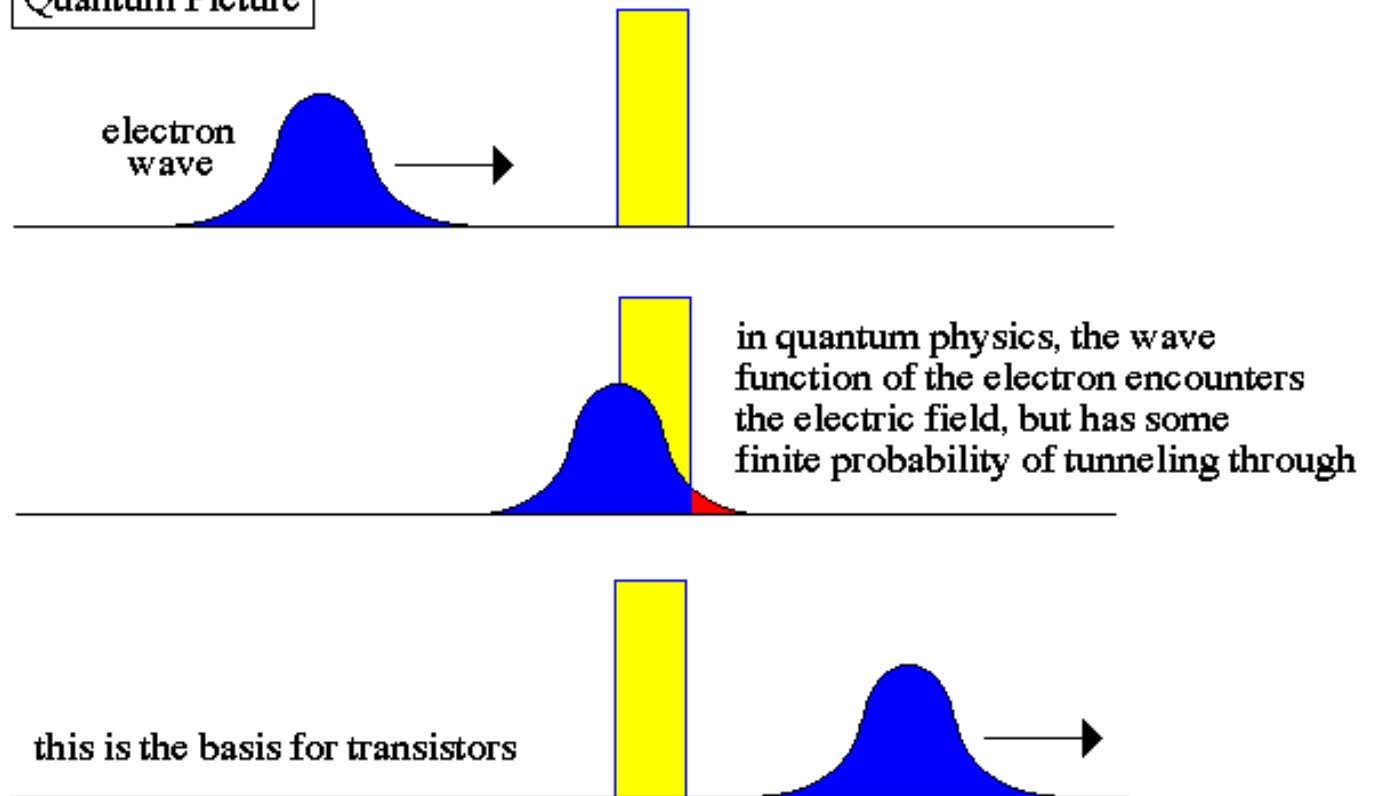
In quantum theory, all events are possible (because the initial state of the system is indeterminate), but some are more likely than others. While the quantum physicist can say very little about the likelihood of any single event's happening, quantum physics works as a science that can make predictions because patterns of probability emerge in large numbers of events. It is more likely that some events will happen than others, and over an average of many events, a given pattern of outcome is predictable. Thus, to make their science work for them, quantum physicists assign a probability to each of the possibilities represented in the wave function.

Quantum Tunneling

Classical Picture



Quantum Picture



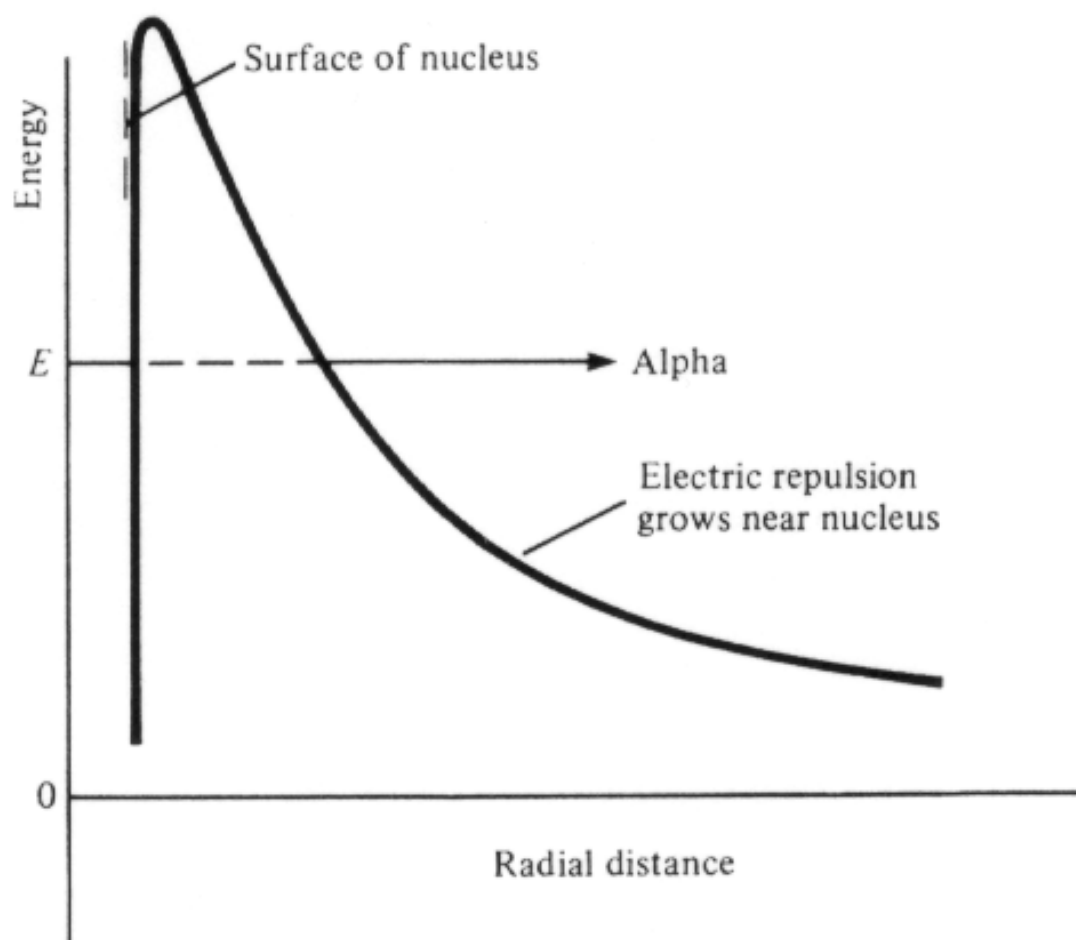
Quantum Tunneling :

The phenomenon of tunneling, which has no counterpart in classical physics, is an important consequence of quantum mechanics. Consider a particle with energy E in the inner region of a one-dimensional potential well $V(x)$. (A potential well

is a potential that has a lower value in a certain region of space than in the neighbouring regions.) In classical mechanics, if $E < V$ (the maximum height of the potential barrier), the particle remains in the well forever; if $E > V$, the particle escapes. In quantum mechanics, the situation is not so simple. The particle can escape even if its energy E is below the height of the barrier V , although the probability of escape is small unless E is close to V . In that case, the particle may tunnel through the potential barrier and emerge with the same energy E .

The phenomenon of tunneling has many important applications. For example, it describes a type of radioactive decay in which a nucleus emits an alpha particle (a helium nucleus). According to the quantum explanation given independently by George Gamow and by Ronald W. Gurney and Edward Condon in 1928, the alpha particle is confined before the decay by a potential. For a given nuclear species, it is possible to measure the energy E of the emitted alpha particle and the average lifetime of the nucleus before decay. The lifetime of the nucleus is a measure of the probability of tunneling through the barrier--the shorter the lifetime, the higher the probability.

Potential barrier around a uranium nucleus presented to an alpha particle. The central well is due to the average nuclear attraction of all the nucleons and the hill is due to the electric repulsion of the protons. Alpha particles with energy E trapped inside the nuclear well may still escape to become alpha rays, by quantum mechanically tunnelling through the barrier.



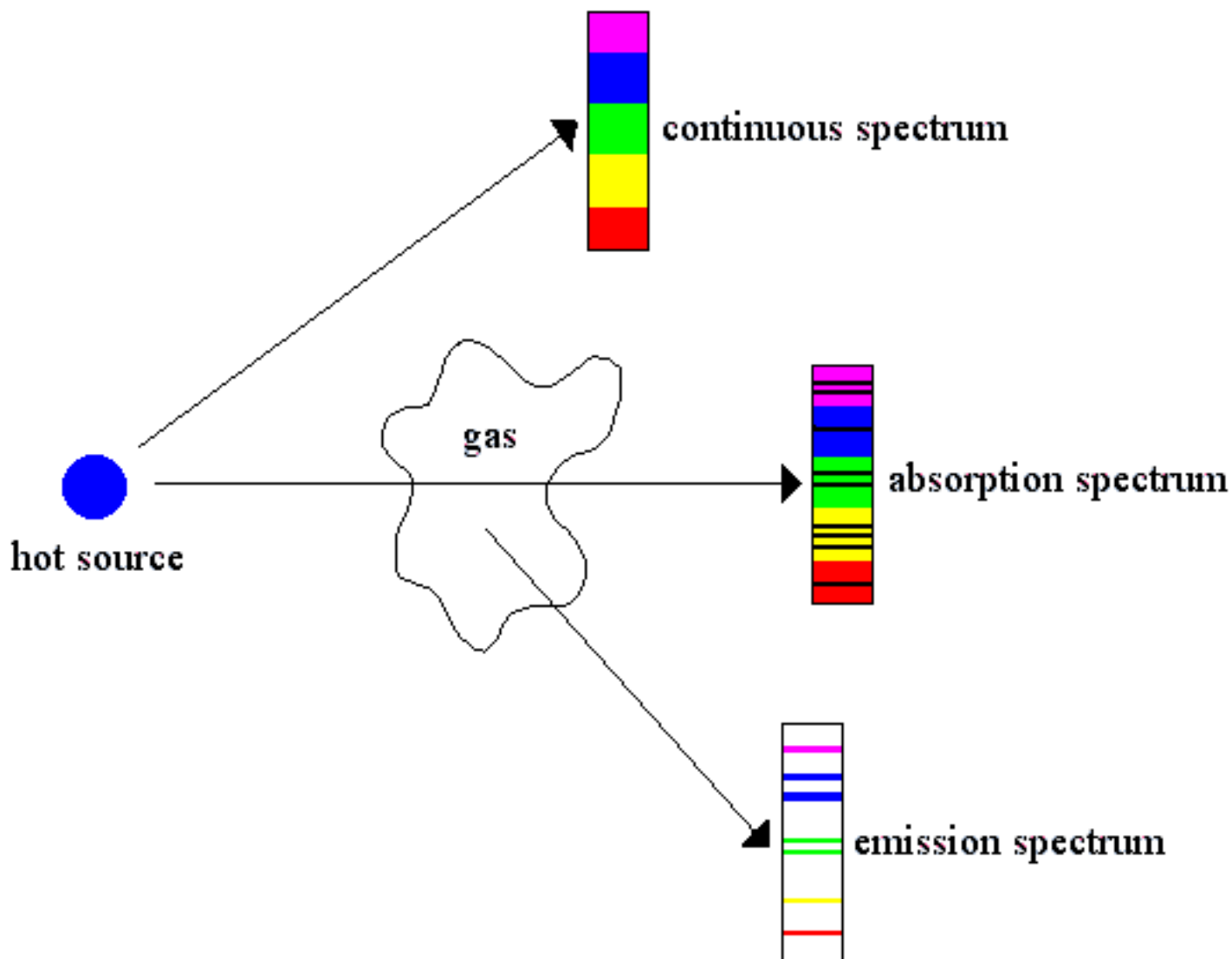
With plausible assumptions about the general form of the potential function, it is possible to calculate a relationship between E and the probability of tunneling that is applicable to all alpha emitters. This theory, which is borne out by experiment, shows that the probability of tunneling is extremely sensitive to the value of E . For all known alpha-particle emitters, the value of E varies from about 2 to 8 megaelectron volts, or MeV (1 MeV = 10^6 electron volts). Thus, the value of E varies only by a factor of 4, whereas the range of lifetime is from about 10^{11} years down to about 10^{-6} second, a factor of 10^{24} . It would be difficult to account for this sensitivity of lifetime to the value of E by any theory other than quantum mechanical tunneling.



Spectra of Light :

Spectrum, in optics, the arrangement according to wavelength of visible, ultraviolet, and infrared light. An instrument designed for visual observation of spectra is called a spectroscope; an instrument that photographs or maps spectra is a spectrograph. The typical spectroscope is a combination of a microscope and a prism. The prism breaks the light into its spectra components (by differential refraction) which is then magnified with a microscope.

Spectra may be classified according to the nature of their origin, i.e., emission or absorption. An emission spectrum consists of all the radiations emitted by atoms or molecules, whereas in an absorption spectrum, portions of a continuous spectrum (light containing all wavelengths) are missing because they have been absorbed by the medium through which the light has passed; the missing wavelengths appear as dark lines or gaps.



The spectrum of incandescent solids is said to be continuous because all wavelengths are present. The spectrum of incandescent gases, on the other hand, is called a line or emission spectrum because only a few wavelengths are emitted. These wavelengths appear to be a series of parallel lines because a slit is used as the light-imaging device. Line spectra are characteristic of the elements that emit the radiation. Line spectra are also called atomic spectra because the lines represent wavelengths radiated from atoms when electrons change from one energy level to another. Band spectra is the name given to groups of lines so closely spaced that each group appears to be a band, e.g., nitrogen spectrum. Band spectra, or molecular spectra, are produced by molecules radiating their rotational or vibrational energies, or both simultaneously.

Quantum Mechanics :

Quantum mechanics, the branch of mathematical physics that deals with atomic and subatomic systems and their interaction with radiation in terms of observable quantities. It is an outgrowth of the concept that all forms of energy are released in discrete units or bundles called quanta.

Quantum mechanics is concerned with phenomena that are so small-scale that they cannot be described in classical terms. Throughout the 1800s most physicists regarded Isaac Newton's dynamical laws as sacrosanct, but it became increasingly clear during the early years of the 20th century that many phenomena, especially those associated with radiation, defy explanation by Newtonian physics. It has come to be recognized that the principles of quantum mechanics rather than those of classical mechanics must be applied when dealing with the behaviour of electrons and nuclei within atoms and molecules. Although conventional quantum mechanics makes no pretense of describing completely what occurs inside the atomic nucleus, it has helped scientists to better understand many processes such as the emission of alpha particles and photodisintegration. Moreover, the field theory of quantum mechanics has provided insight into the properties of mesons and other subatomic particles associated with nuclear phenomena.

In the equations of quantum mechanics, Max Planck's constant of action $h = 6.626 \times 10^{-34}$ joule-second plays a central role. This constant, one of the most important in all of physics, has the dimensions energy time. The term "small-scale" used to delineate the domain of quantum mechanics should not be literally interpreted as necessarily relating to extent in space. A more precise criterion as to whether quantum modifications of Newtonian laws are important is whether or not the phenomenon in question is characterized by an "action" (i.e., time integral of kinetic energy) that is large compared to Planck's constant. Accordingly, if a great many quanta are involved, the notion that there is a discrete, indivisible quantum unit loses significance. This fact explains why ordinary physical processes appear to be so fully in accord with the laws of Newton. The laws of quantum mechanics, unlike Newton's deterministic laws, lead to a probabilistic description of nature. As a consequence, one of quantum mechanics' most important philosophical implications concerns the apparent breakdown, or at least a drastic reinterpretation, of the causality principle in atomic phenomena.

The history of quantum mechanics may be divided into three main periods. The first began with Planck's theory of black-body radiation in 1900; it may be described as the period in which the validity of Planck's constant was demonstrated but its real meaning was not fully understood. The second period began with the quantum theory of atomic structure and spectra proposed by Niels Bohr in 1913. Bohr's ideas gave an accurate formula for the frequency of spectral lines in many cases and were an enormous help in the codification and understanding of spectra. Nonetheless, they did not represent a consistent, unified theory, constituting as they did a sort of patchwork affair in which classical mechanics was subjected to a somewhat extraneous set of so-called quantum conditions that restrict the constants of integration to particular values. True quantum mechanics appeared in 1926, reaching fruition nearly simultaneously in a variety of forms--namely, the matrix theory of Max Born and Werner Heisenberg, the wave mechanics of Louis V. de Broglie and Erwin Schrödinger, and the transformation theory of P.A.M. Dirac and Pascual Jordan. These different formulations were in no sense alternative theories;

rather, they were different aspects of a consistent body of physical law.

Holism:

Holism as an idea or philosophical concept is diametrically opposed to atomism. Where the atomist believes that any whole can be broken down or analyzed into its separate parts and the relationships between them, the holist maintains that the whole is primary and often greater than the sum of its parts. The atomist divides things up in order to know them better; the holist looks at things or systems in aggregate and argues that we can know more about them viewed as such, and better understand their nature and their purpose.

The early Greek atomism of Leucippus and Democritus (fifth century B.C.) was a forerunner of classical physics. According to their view, everything in the universe consists of indivisible, indestructible atoms of various kinds. Change is a rearrangement of these atoms. This kind of thinking was a reaction to the still earlier holism of Parmenides, who argued that at some primary level the world is a changeless unity. According to him, "All is One. Nor is it divisible, wherefore it is wholly continuous.... It is complete on every side like the mass of a rounded sphere."

In the seventeenth century, at the same time that classical physics gave renewed emphasis to atomism and reductionism, Spinoza developed a holistic philosophy reminiscent of Parmenides. According to Spinoza, all the differences and apparent divisions we see in the world are really only aspects of an underlying single substance, which he called God or nature. Based on pantheistic religious experience, this emphasis on an underlying unity is reflected in the mystical thinking of most major spiritual traditions. It also reflects developments in modern quantum field theory, which describes all existence as an excitation of the underlying quantum vacuum, as though all existing things were like ripples on a universal pond.

Hegel, too, had mystical visions of the unity of all things, on which he based his own holistic philosophy of nature and the state. Nature consists of one timeless, unified, rational and spiritual reality. Hegel's state is a quasi-mystical collective, an "invisible and higher reality," from which participating individuals derive their authentic identity, and to which they owe their loyalty and obedience. All modern collectivist political thinkers - including, of course, Karl Marx - stress some higher collective reality, the unity, the whole, the group, though nearly always at the cost of minimizing the importance of difference, the part, the individual. Against individualism, all emphasize the social whole or social forces that somehow possess a character and have a will of their own, over and above the characters and wills of individual members.

The twentieth century has seen a tentative movement toward holism in such diverse areas as politics, social thinking, psychology, management theory, and medicine. These have included the practical application of Marx's thinking in Communist and Socialist states, experiments in collective living, the rise of Gestalt psychology, systems theory, and concern with the whole person in alternative medicine. All these have been reactions against excessive individualism with its attendant alienation and fragmentation, and exhibit a commonsense appreciation of human beings' interdependency with one another and with the environment.

Where atomism was apparently legitimized by the sweeping successes of classical physics, holism found no such foundation in the hard sciences. It remained a change of emphasis rather than a new philosophical position. There were attempts to found it on the idea of organism in biology - the emergence of biological form and the cooperative relation between biological and ecological systems - but these, too, were ultimately reducible to simpler parts, their properties, and the relation between them. Even systems theory, although it emphasizes the complexity of aggregates, does so in terms of causal feedback loops between various constituent parts. It is only with quantum theory and the dependence of the very being or identity of quantum entities upon their contexts and relationships that a genuinely new, "deep" holism emerges.

Relational Holism in Quantum Mechanics

Every quantum entity has both a wavelike and a particlelike aspect. The wavelike aspect is indeterminate, spread out all over space and time and the realm of possibility. The particlelike aspect is determinate, located at one place in space and time and limited to the domain of actuality. The particlelike aspect is fixed, but the wavelike aspect becomes fixed only in dialogue with its surroundings - in dialogue with an experimental context or in relationship to another entity in measurement or observation. It is the indeterminate, wavelike aspect - the set of potentialities associated with the entity - that unites quantum things or systems in a truly emergent, relational holism that cannot be reduced to any previously existing parts or their properties.

If two or more quantum entities are "introduced" - that is, issue from the same source - their potentialities are entangled. Their indeterminate wave aspects are literally interwoven, to the extent that a change in potentiality in one brings about a correlated change in the same potentiality of the other. In the nonlocality experiments, measuring the previously indeterminate polarization of a photon on one side of a room effects an instantaneous fixing of the polarization of a paired photon shot off to the other side of the room. The polarizations are said to be correlated; they are always determined simultaneously and always found to be opposite. This paired-but-opposite polarization is described as an emergent property of the photons' "relational holism" - a property that comes into being only through the entanglement of their potentialities. It is not based on individual polarizations, which are not present until the photons are observed. They literally do not previously exist, although their oppositeness was a fixed characteristic of their combined system when it was formed.

In the coming together or simultaneous measurement of any two entangled quantum entities, their relationship brings about a "further fact." Quantum relationship evokes a new reality that could not have been predicted by breaking down the two relational entities into their individual properties.

The emergence of a quantum entity's previously indeterminate properties in the context of a given experimental situation is another example of relational holism. We cannot say that a photon is a wave or a particle until it is measured, and how we measure it determines what we will see. The quantum entity acquires a certain new property - position, momentum, polarization - only in relation to its measuring apparatus. The property did not exist prior to this relationship. It was indeterminate.

Quantum relational holism, resting on the nonlocal entanglement of potentialities, is a kind of holism not previously defined. Because each related entity has some characteristics - mass, charge, spin - before its emergent properties are evoked, each can be reduced to some extent to atomistic parts, as in classical physics. The holism is not the extreme holism of Parmenides or Spinoza, where everything is an aspect of the One. Yet because some of their properties emerge only through relationship, quantum entities are not wholly subject to reduction either. The truth is somewhere between Newton and Spinoza. A quantum system may also vary between being more atomistic at some times and more holistic at others; the degree of entanglement vary.

Parmenides:

Parmenides was a Greek philosopher and poet, born of an illustrious family about 510 B.C., at Elea in Lower Italy, and is the chief representative of the Eleatic philosophy. He was held in high esteem by his fellow-citizens for his excellent legislation, to which they ascribed the prosperity and wealth of the town. He was also admired for his exemplary life. A "Parmenidean life" was proverbial among the Greeks. He is commonly represented as a disciple of Xenophanes. Parmenides wrote after Heraclitus, and in conscious opposition to him, given the evident allusion to Heraclitus: "for whom it is and is not, the same and not the same, and all things travel in opposite directions". Little more is known of his biography than that he stopped at Athens on a journey in his sixty-fifth year, and there became acquainted with the youthful Socrates. That must have been in the middle of the fifth century B.C., or shortly after it.

Parmenides broke with the older Ionic prose tradition by writing in hexameter verse. His didactic poem, called *On Nature*, survives in fragments, although the Proem (or introductory discourse) of the work has been preserved. Parmenides was a young man when he wrote it, for the goddess who reveals the truth to him addresses him as 'youth'. The work is considered inartistic. Its Hesiodic style was appropriate for the cosmogony he describes in the second part, but is unsuited to the arid dialectic of the first. Parmenides was no born poet, and we much ask what led him to take this new departure. The example of Xenophanes' poetic writings is not a complete explanation; for the poetry of Parmenides is as unlike that of Xenophanes as it well can be, and his style is more like Hesiod and the Orphics. In the Proem Parmenides describes his ascent to the home of the goddess who is supposed to speak the remainder of the verses; this is a reflexion of the conventional ascents into heaven which were almost as common as descents into hell in the apocalyptic literature of those days.

The Poem opens with Parmenides representing himself as borne on a chariot and attended by the Sunmaidens who have quitted the Halls of Night to guide him on his journey. They pass along the highway till they come to the Gate of Night and Day, which is locked and barred. The key is in the keeping of Dike (Right), the Avenger, who is persuaded to unlock it by the Sunmaidens. They pass in through the gate and are now, of course, in the realms of Day. The goal of the journey is the palace of a goddess who welcomes Parmenides and instructs him in the two ways, that of Truth and the deceptive way of Belief, in which is no truth at all. All this is described without inspiration and in a purely conventional manner, so it must be interpreted by the canons of the apocalyptic style. It is clearly meant to indicate that Parmenides had been converted, that he had passed from error (night) to truth (day), and the Two Ways must represent his former error and the truth which is now revealed to him.

There is reason to believe that the Way of Belief is an account of Pythagorean cosmology. In any case, it is surely impossible to regard it as anything else than a description of some error. The goddess says so in words that cannot be explained away. Further, this erroneous belief is not the ordinary man's view of the world, but an elaborate system, which seems to be a natural development the Ionian cosmology on certain lines, and there is no other system but the Pythagorean that fulfils this requirement. To this it has been objected that Parmenides would not have taken the trouble to expound in detail a system he had altogether rejected, but that is to mistake the character of the apocalyptic convention. It is not

Parmenides, but the goddess, that expounds the system, and it is for this reason that the beliefs described are said to be those of 'mortals'. Now a description of the ascent of the soul would be quite incomplete without a picture of the region from which it had escaped. The goddess must reveal the two ways at the parting of which Parmenides stands, and bid him choose the better. The rise of mathematics in the Pythagorean school had revealed for the first time the power of thought. To the mathematician of all men it is the same thing that can be thought and that can be, and this is the principle from which Parmenides starts. It is impossible to think what is not, and it is impossible for what cannot be thought to be. The great question, Is it or is it not? is therefore equivalent to the question, Can it be thought or not?

In any case, the work thus has two divisions. The first discusses the truth, and the second the world of illusion -- that is, the world of the senses and the erroneous opinions of mankind founded upon them. In his opinion truth lies in the perception that existence is, and error in the idea that non-existence also can be. Nothing can have real existence but what is conceivable; therefore to be imagined and to be able to exist are the same thing, and there is no development. The essence of what is conceivable is incapable of development, imperishable, immutable, unbounded, and indivisible. What is various and mutable, all development, is a delusive phantom. Perception is thought directed to the pure essence of being; the phenomenal world is a delusion, and the opinions formed concerning it can only be improbable.

Parmenides goes on to consider in the light of this principle the consequences of saying that anything is. In the first place, it cannot have come into being. If it had, it must have arisen from nothing or from something. It cannot have arisen from nothing; for there is no nothing. It cannot have arisen from something; for here is nothing else than what is. Nor can anything else besides itself come into being; for there can be no empty space in which it could do so. Is it or is it not? If it is, then it is now, all at once. In this way Parmenides refutes all accounts of the origin of the world. *Ex nihilo nihil fit*.

Further, if it is, it simply is, and it cannot be more or less. There is, therefore, as much of it in one place as in another. (That makes rarefaction and condensation impossible.) it is continuous and indivisible; for there is nothing but itself which could prevent its parts being in contact with one another. It is therefore full, a continuous indivisible plenum. (That is directed against the Pythagorean theory of a discontinuous reality.) Further, it is immovable. If it moved, it must move into empty space, and empty space is nothing, and there is no nothing. Also it is finite and spherical; for it cannot be in one direction any more than in another, and the sphere is the only figure of which this can be said. What is is, therefore a finite, spherical, motionless, continuous plenum, and there is nothing beyond it. Coming into being and ceasing to be are mere 'names', and so is motion, and still more color and the like. They are not even thoughts; for a thought must be a thought of something that is, and none of these can be.

Such is the conclusion to which the view of the real as a single body inevitably leads, and there is no escape from it. The 'matter' of our physical text-books is just the real of Parmenides; and, unless we can find room for something else than matter, we are shut up into his account of reality. No subsequent system could afford to ignore this, but of course it was impossible to acquiesce permanently in a doctrine like that of Parmenides. It deprives the world we know of all claim to existence, and reduces it to something which is hardly even an illusion. If we are to give an intelligible account of the world, we must certainly introduce motion again somehow. That can never be taken for granted any more, as it was

by the early cosmologists; we must attempt to explain it if we are to escape from the conclusions of Parmenides.

Parmenides was most famous for the following statements:

What is, is.

What is not, is not.

Although these statements appear simple, in fact they make a profound statement on the existence in the material plane. What is not, is not basically says there can be no vacuum, that the concept of void is a logical error. The substance that is the One fills all the Universe. Classical Physics would deny Parmenides claim, however, modern physics has found that the quantum vacuum in fact fills the Universe just as Parmenides predicted over 2000 years ago.

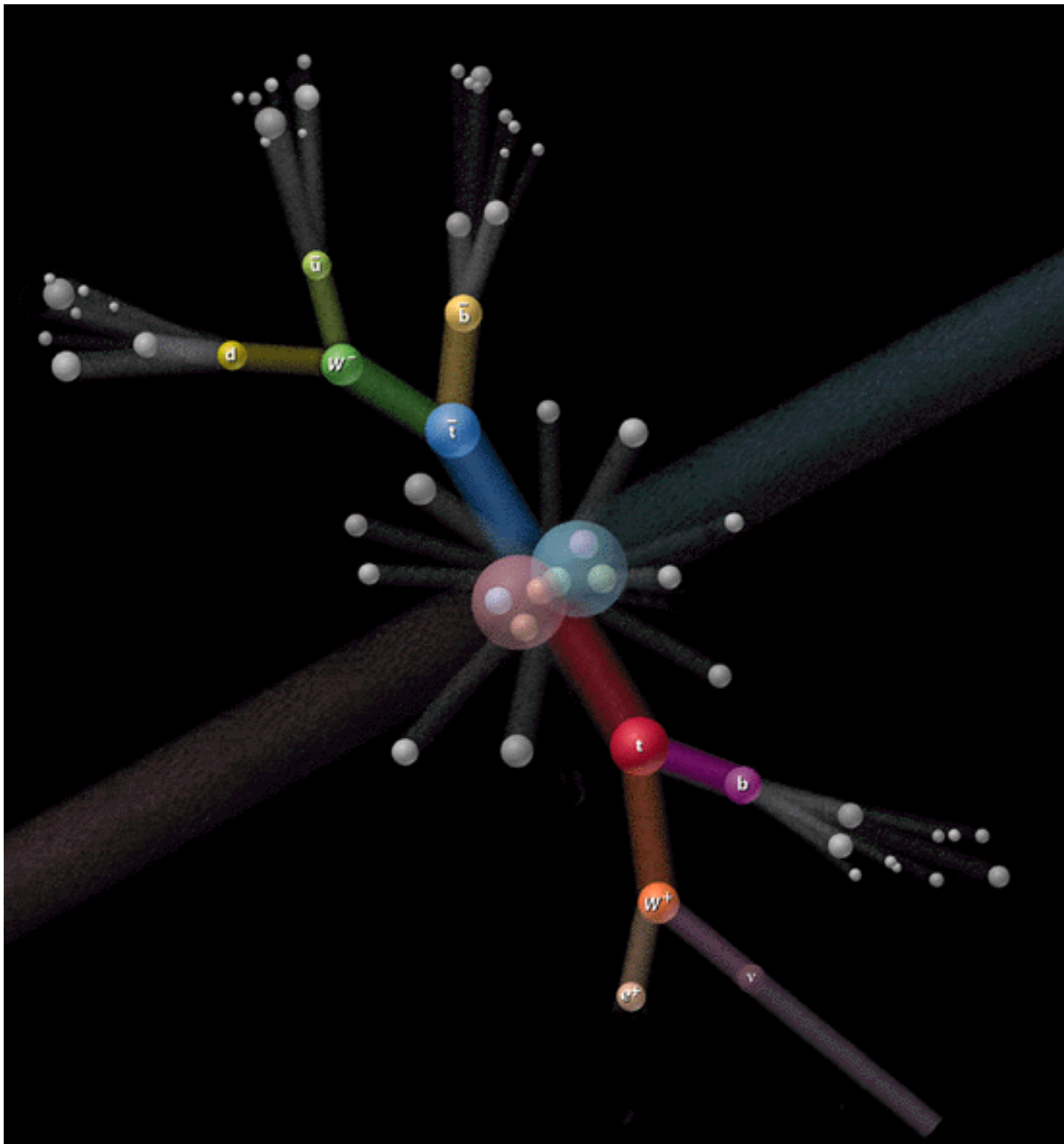
Quantum Vacuum :

The words "nothing," "void," and "vacuum" usually suggest uninteresting empty space. To modern quantum physicists, however, the vacuum has turned out to be rich with complex and unexpected behaviour. They envisage it as a state of minimum energy where quantum fluctuations, consistent with the uncertainty principle of the German physicist Werner Heisenberg, can lead to the temporary formation of particle-antiparticle pairs.

Elementary Particles :

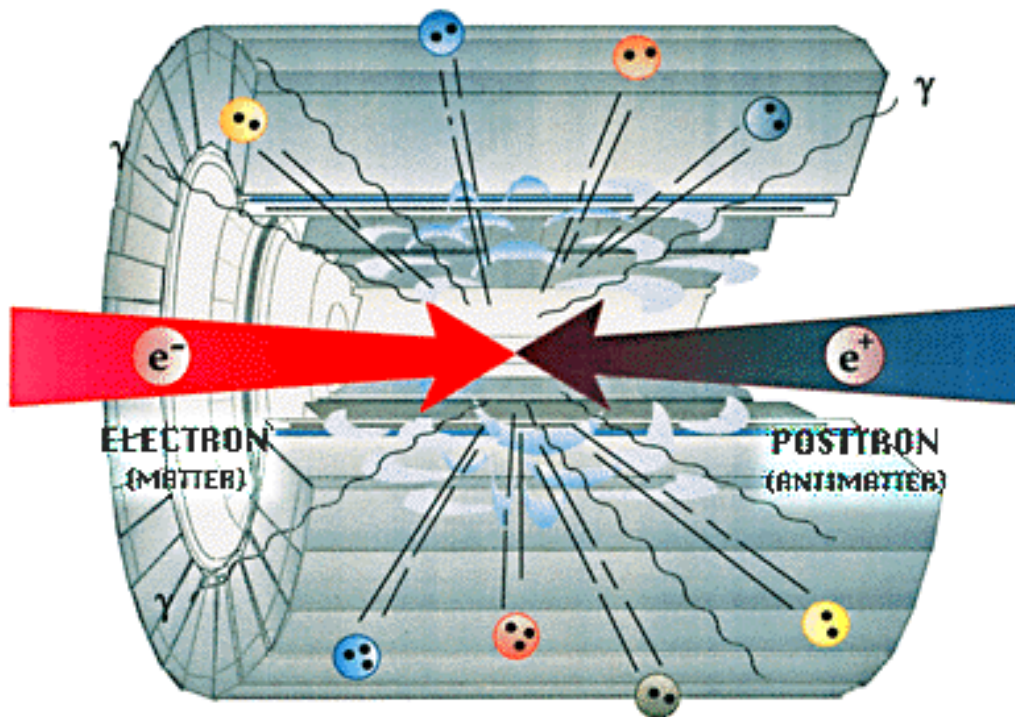
Until its development in the third decade of the 20th century, the scientific atomic theory did not differ philosophically very much from that of Dalton, although at first sight the difference may appear large. Dalton's atoms were no longer considered to be immutable and indivisible; new elementary particles sometimes appeared on the scene; and molecules were no longer seen as a mere juxtaposition of atoms--when entering into a compound atoms became ions. Yet, these differences were only accidental; the atoms revealed themselves as composed of more elementary particles--protons, neutrons, and electrons--but these particles themselves were considered then as immutable. Thus the general picture remained the same. The material world was still thought to be composed of smallest particles, which differed in nature and which in certain definite ways could form relatively stable structures (atoms). These structures were able to form new combinations (molecules) by exchanging certain component parts (electrons). The whole process was ruled by well-known mechanical and electrodynamic laws.

In contemporary atomic theory the differences from Dalton are much more fundamental. The hypothesis of the existence of immutable elementary particles has been abandoned: elementary particles can be transformed into radiation and vice versa. And when they combine into greater units, the particles do not necessarily preserve their identity; they can be absorbed into a greater whole.



Particle Physics :

One of the most significant branches of contemporary physics is the study of the fundamental subatomic constituents of matter, the elementary particles. This field, also called high-energy physics, emerged in the 1930s out of the developing experimental areas of nuclear and cosmic-ray physics. Initially investigators studied cosmic rays, the very-high-energy extraterrestrial radiations that fall upon the Earth and interact in the atmosphere (see below The methodology of physics). However, after World War II, scientists gradually began using high-energy particle accelerators to provide subatomic particles for study. Quantum field theory, a generalization of QED to other types of force fields, is essential for the analysis of high-energy physics.



During recent decades a coherent picture has evolved of the underlying strata of matter involving three types of particles called leptons, quarks, and field quanta, for whose existence evidence is good. (Other types of particles have been hypothesized but have not yet been detected.) Subatomic particles cannot be visualized as tiny analogues of ordinary material objects such as billiard balls, for they have properties that appear contradictory from the classical viewpoint. That is to say, while they possess charge, spin, mass, magnetism, and other complex characteristics, they are nonetheless regarded as pointlike. Leptons and quarks occur in pairs (e.g., one lepton pair consists of the electron and the neutrino). Each quark and each lepton have an antiparticle with properties that mirror those of its partner (the antiparticle of the negatively charged electron is the positive electron, or positron; that of the neutrino is the antineutrino). In addition to their electric and magnetic properties, quarks have very strong nuclear forces and also participate in the weak nuclear interaction, while leptons take part in only the weak interaction.

Ordinary matter consists of electrons surrounding the nucleus, which is composed of neutrons and protons, each of which is believed to contain three quarks. Quarks have charges that are either positive two-thirds

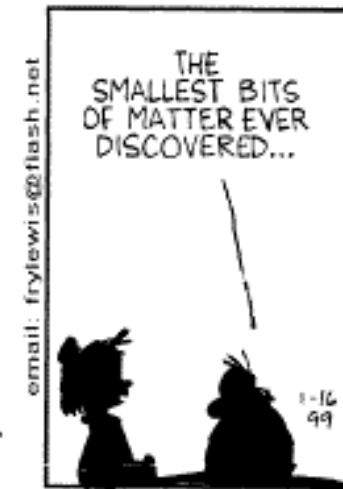
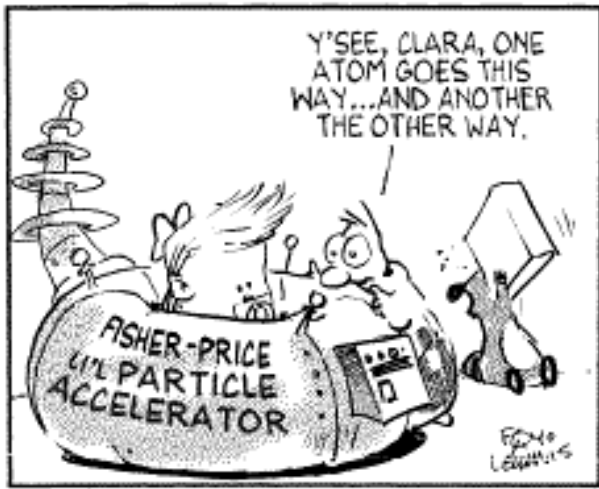
or negative one-third of the electron's charge, while antiquarks have the opposite charges. Mesons, responsible for the nuclear binding force, are composed of one quark and one antiquark. In addition to the particles in ordinary matter and their antiparticles, which are referred to as first-generation, there are probably two or more additional generations of quarks and leptons, more massive than the first. Evidence exists at present for the second generation and all but one quark of the third, namely the t (or top) quark, which may be so massive that a new higher-energy accelerator may be needed to produce it.

The quantum fields through which quarks and leptons interact with each other and with themselves consist of particle-like objects called quanta (from which quantum mechanics derives its name). The first known quanta were those of the electromagnetic field; they are also called photons because light consists of them. A modern unified theory of weak and electromagnetic interactions, known as the electroweak theory, proposes that the weak nuclear interaction involves the exchange of particles about 100 times as massive as protons. These massive quanta have been observed--namely, two charged particles, W_+ and W_- , and a neutral one, Z_0 .

In the theory of strong nuclear interactions known as quantum chromodynamics (QCD), eight quanta, called gluons, bind quarks to form protons and neutrons and also bind quarks to antiquarks to form mesons, the force itself being dubbed the "color force." (This unusual use of the term color is a somewhat forced analogue of ordinary color mixing.) Quarks are said to come in three colors--red, blue, and green. (The opposites of these imaginary colors, minus-red, minus-blue, and minus-green, are ascribed to antiquarks.) Only certain color combinations, namely color-neutral, or "white" (i.e., equal mixtures of the above colors cancel out one another, resulting in no net color), are conjectured to exist in nature in an observable form. The gluons and quarks themselves, being colored, are permanently confined (deeply bound within the particles of which they are a part), while the color-neutral composites such as protons can be directly observed. One consequence of color confinement is that the observable particles are either electrically neutral or have charges that are integral multiples of the charge of the electron. A number of specific predictions of QCD have been experimentally tested and found correct.

To see how a particle accelerator works, [click here](#)

To see what a particle collision looks like, [click here](#)



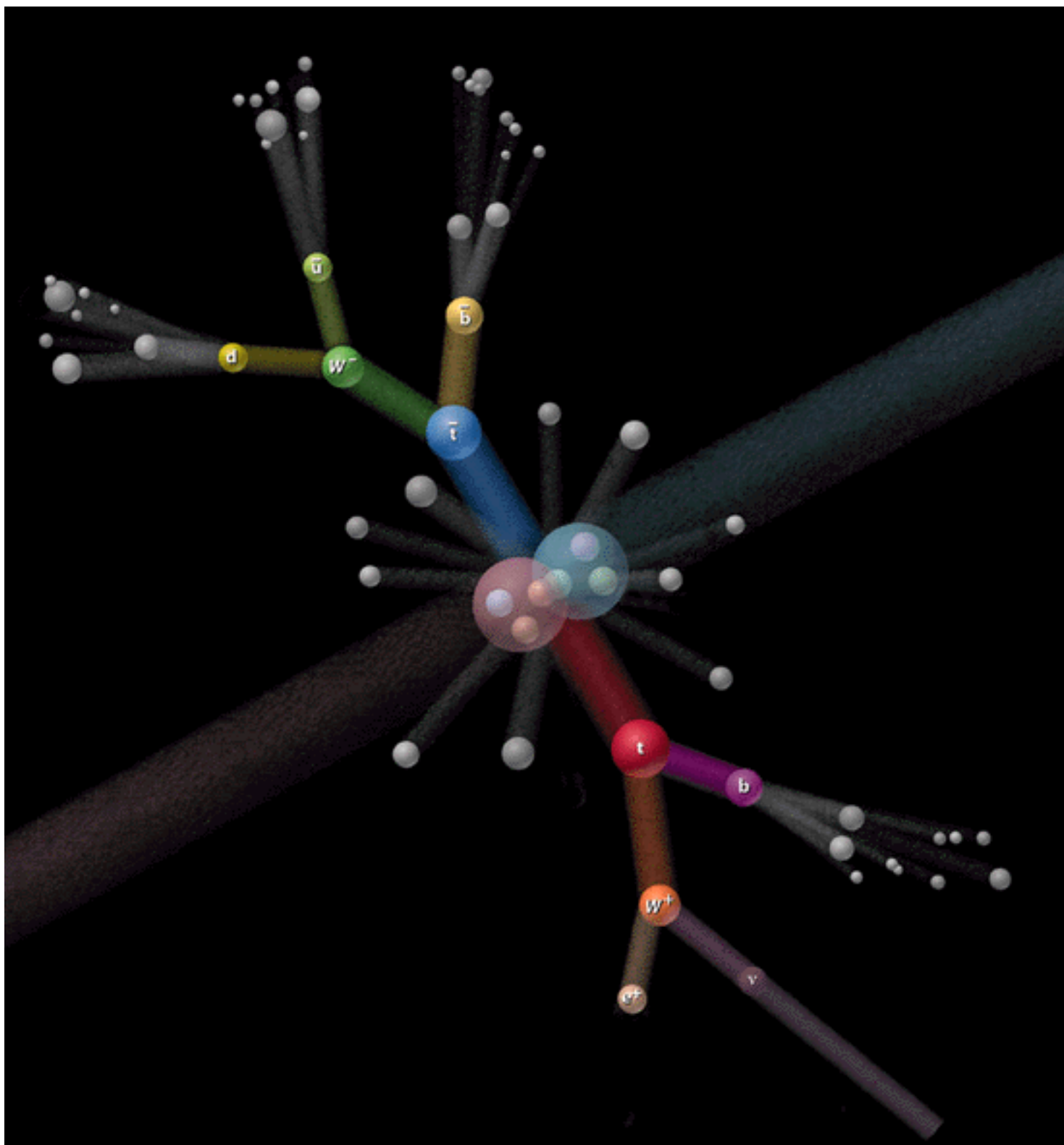
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Quark :

A quark is any of a group of subatomic particles believed to be among the fundamental constituents of matter. In much the same way that protons and neutrons make up atomic nuclei, these particles themselves are thought to consist of quarks. Quarks constitute all hadrons (baryons and mesons)--i.e., all particles that interact by means of the strong force, the force that binds the components of the nucleus.

According to prevailing theory, quarks have mass and exhibit a spin (i.e., type of intrinsic angular momentum corresponding to a rotation around an axis through the particle). Quarks appear to be truly fundamental. They have no apparent structure; that is, they cannot be resolved into something smaller. Quarks always seem to occur in combination with other quarks or antiquarks, never alone. For years physicists have attempted to knock a quark out of a baryon in experiments with particle accelerators to observe it in a free state but have not yet succeeded in doing so.

Throughout the 1960s theoretical physicists, trying to account for the ever-growing number of subatomic particles observed in experiments, considered the possibility that protons and neutrons were composed of smaller units of matter. In 1961 two physicists, Murray Gell-Mann of the United States and Yuval Ne'eman of Israel, proposed a particle classification scheme called the Eightfold Way, based on the mathematical symmetry group $SU(3)$, that described strongly interacting particles in terms of building blocks. In 1964 Gell-Mann introduced the concept of quarks as a physical basis for the scheme, adopting the fanciful term from a passage in James Joyce's novel *Finnegans Wake*. (The American physicist George Zweig developed a similar theory independently that same year and called his fundamental particles "aces.") Gell-Mann's model provided a simple picture in which all mesons are shown as consisting of a quark and an antiquark and all baryons as composed of three quarks. It postulated the existence of three types of quarks, distinguished by distinctive "flavours." These three quark types are now commonly designated as "up" (u), "down" (d), and "strange" (s). Each carries a fractional electric charge (i.e., a charge less than that of the electron). The up and down quarks are thought to make up protons and neutrons and are thus the ones observed in ordinary matter. Strange quarks occur as components of K mesons and various other extremely short-lived subatomic particles that were first observed in cosmic rays but that play no part in ordinary matter.



Most problems with quarks were resolved by the introduction of the concept of color, as formulated in quantum chromodynamics (QCD). In this theory of strong interactions, developed in 1977, the term color has nothing to do with the colors of the everyday world but rather represents a special quantum property of quarks. The colors red, green, and blue are ascribed to quarks, and their opposites, minus-red, minus-green, and minus-blue, to antiquarks. According to QCD, all combinations of quarks must contain equal mixtures of these imaginary colors so that they will cancel out one another, with the resulting particle having no net color. A baryon, for example, always consists of a combination of one red, one green, and

one blue quark. The property of color in strong interactions plays a role analogous to an electric charge in electromagnetic interactions. Charge implies the exchange of photons between charged particles. Similarly, color involves the exchange of massless particles called gluons among quarks. Just as photons carry electromagnetic force, gluons transmit the forces that bind quarks together. Quarks change their color as they emit and absorb gluons, and the exchange of gluons maintains proper quark color distribution.

Leptons :

Leptons are any member of a class of fermions that respond only to electromagnetic, weak, and gravitational forces and do not take part in strong interactions. Like all fermions, leptons have a half-integral spin. (In quantum-mechanical terms, spin constitutes the property of intrinsic angular momentum.) Leptons obey the Pauli exclusion principle, which prohibits any two identical fermions in a given population from occupying the same quantum state. Leptons are said to be fundamental particles; that is, they do not appear to be made up of smaller units of matter.

Leptons can either carry one unit of electric charge or be neutral. The charged leptons are the electrons, muons, and taus. Each of these types has a negative charge and a distinct mass. Electrons, the lightest leptons, have a mass only 0.0005 that of a proton. Muons are heavier, having more than 200 times as much mass as electrons. Taus, in turn, are approximately 3,700 times more massive than electrons. Each charged lepton has an associated neutral partner, or neutrino (i.e., electron-, muon-, and tau-neutrino), that has no electric charge and no significant mass. Moreover, all leptons, including the neutrinos, have antiparticles called antileptons. The mass of the antileptons is identical to that of the leptons, but all of the other properties are reversed.

Neutrino:

The neutrino is a type of fundamental particle with no electric charge, a very small mass, and one-half unit of spin. Neutrinos belong to the family of particles called leptons, which are not subject to the strong nuclear force. There are three types of neutrino, each associated with a charged lepton--i.e., the electron, muon, and tau.



Wolfgang Pauli (1900-1958), Austrian physicist who won the Physics Nobel price for his idea of the exclusion principle: two electrons, and more generally two fermions, cannot have the same quantum state (position, momentum, mass, spin)

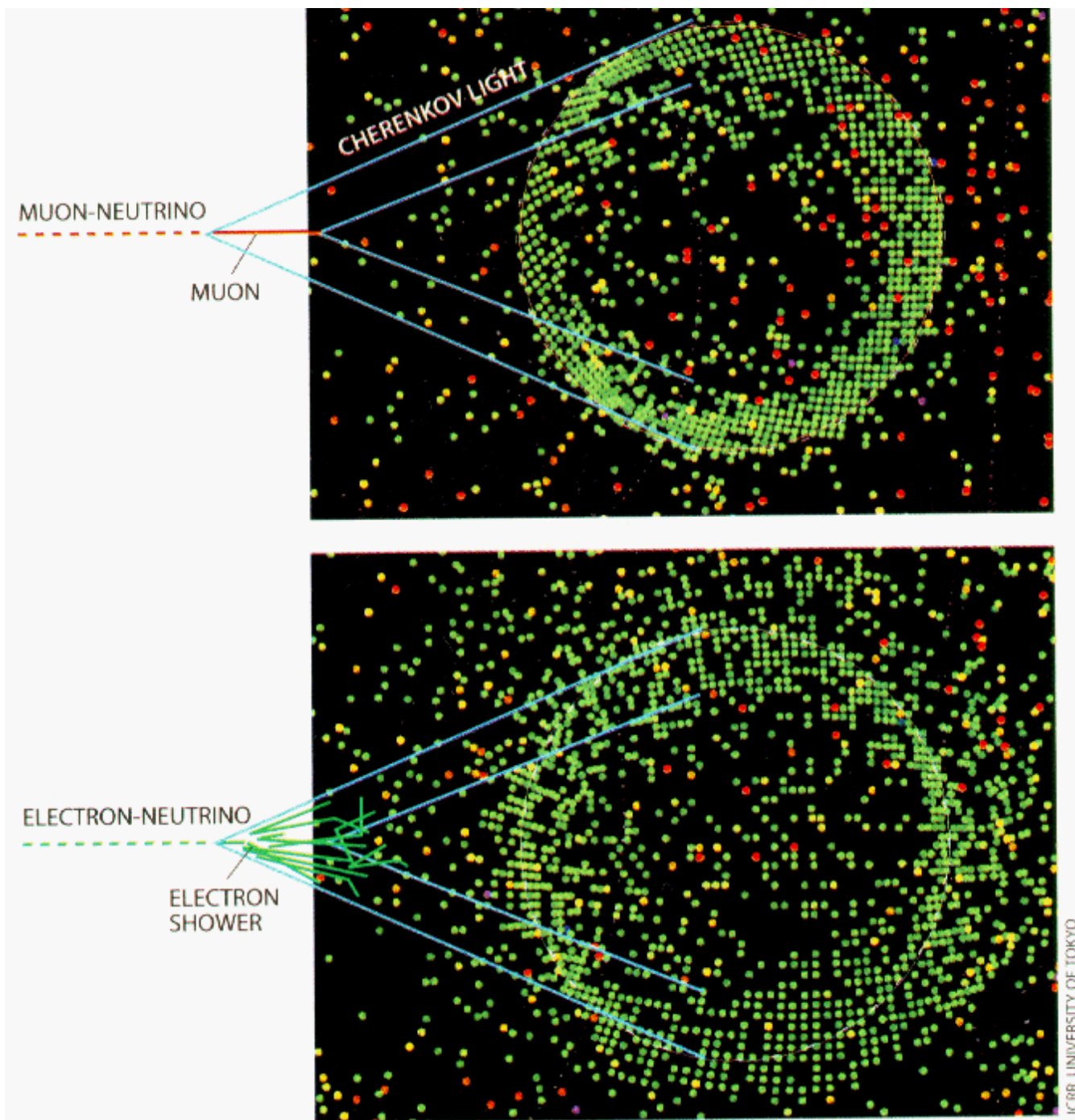
The electron-neutrino was proposed in 1930 by the Austrian physicist Wolfgang Pauli to explain the apparent loss of energy in the process of beta decay, a form of radioactivity. It seemed that examination of the reaction products always indicated that some variable ammount of energy was missing. Pauli concluded that the products must include a third particle, but one which didn't interact strongly enough for it to be detected.

The Italian-born physicist Enrico Fermi further elaborated (1934) the proposal and gave the particle its name, the neutrino which meant "little neutral one". An electron-neutrino is emitted along with a positron in positive beta decay, while an electron-antineutrino is emitted with an electron in negative beta decay.

Neutrinos are the most penetrating of subatomic particles because they react with matter only through the weak interaction. Neutrinos do not cause ionization, because they are not electrically charged. Only 1 in 10 billion, traveling through matter a distance equal to the Earth's diameter, reacts with a proton or neutron. Electron-neutrinos were first experimentally observed in 1956 by monitoring a volume of cadmium chloride with scintillating liquid near to a nuclear reactor. A beam of antineutrinos from a nuclear reactor produced neutrons and positrons by reacting with protons.

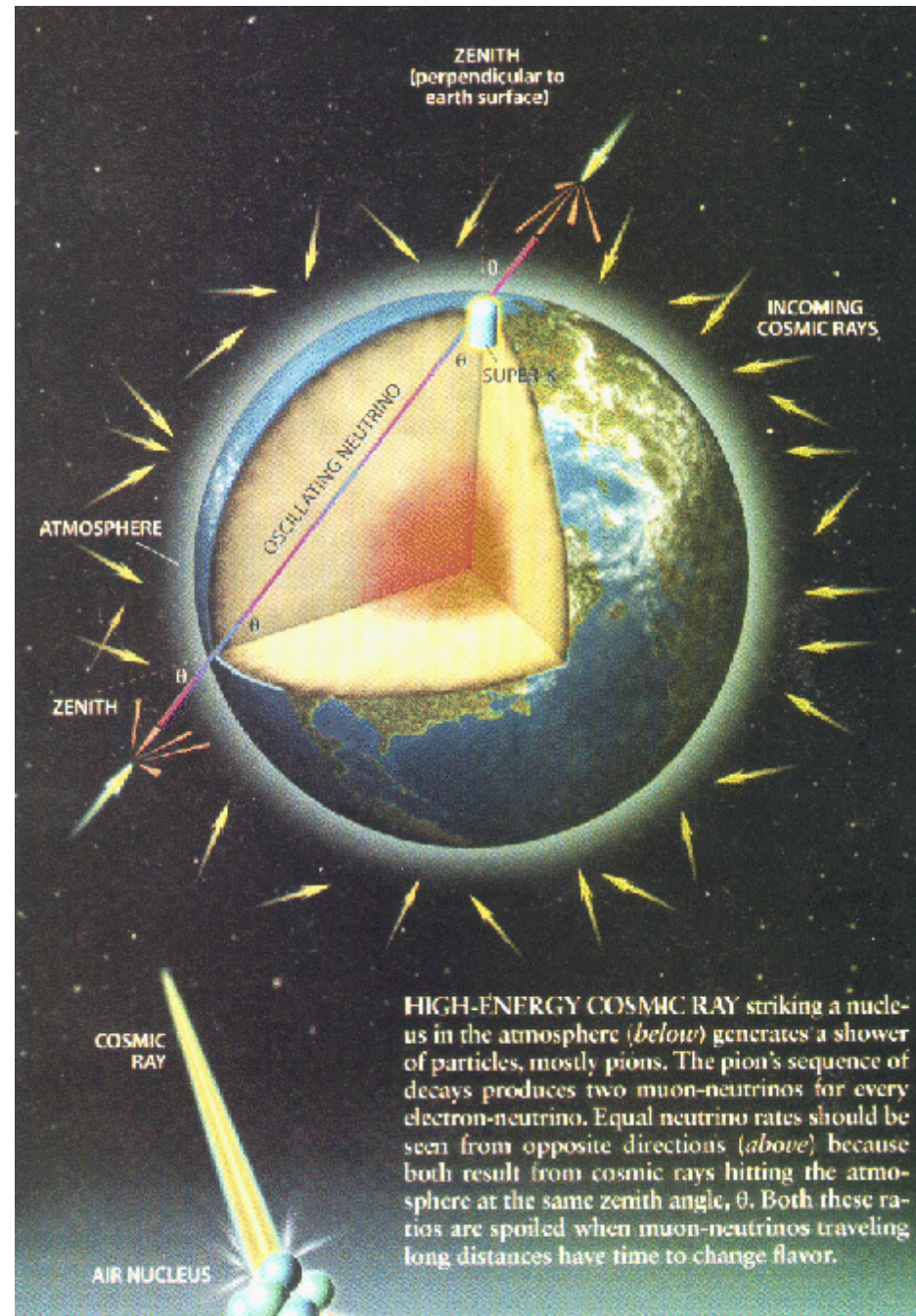
All types of neutrino have masses much smaller than those of their charged partners. For example, experiments show that the mass of the electron-neutrino must be less than 0.0004 that of the electron.

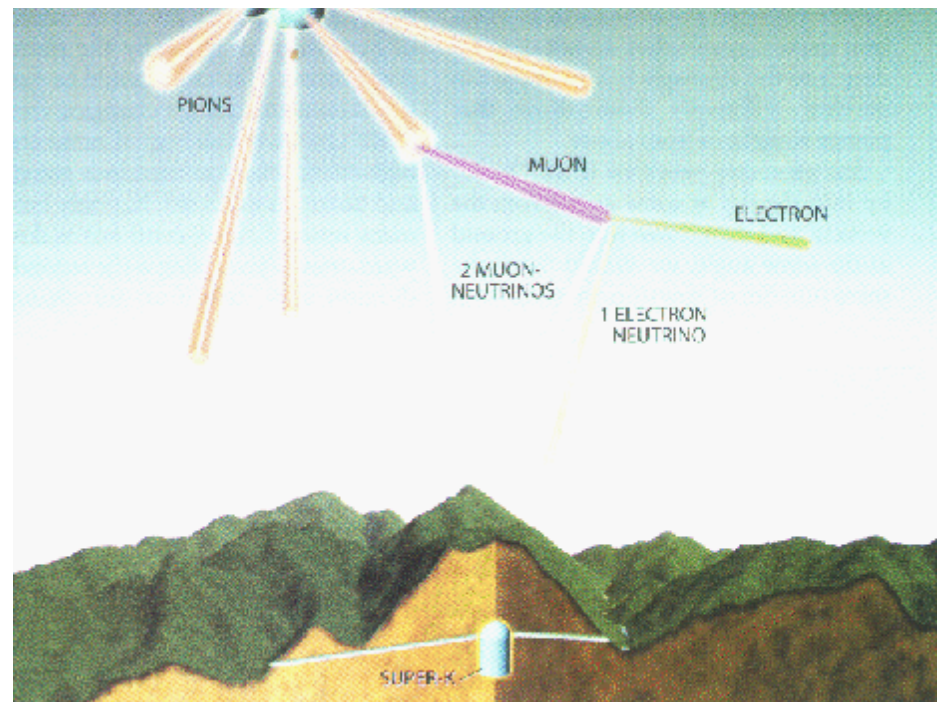


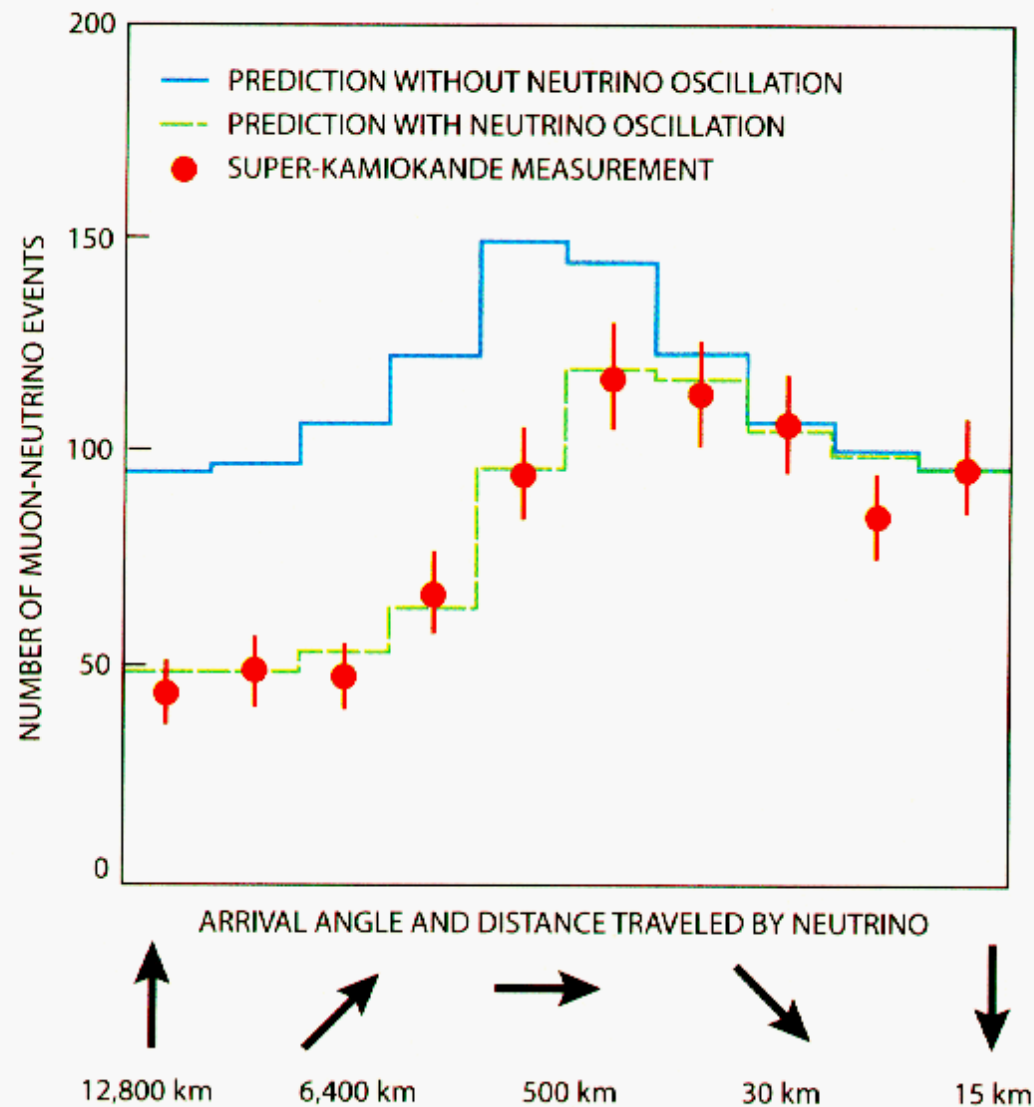


CONES OF CHERENKOV LIGHT are emitted when high-energy neutrinos hit a nucleus and produce a charged particle. A muon-neutrino (*top*) creates a muon, which travels perhaps one meter and projects a sharp ring of light onto the detec-

tors. An electron, produced by an electron-neutrino (*bottom*), generates a small shower of electrons and positrons, each with its own Cherenkov cone, resulting in a fuzzy ring of light. Green dots indicate light detected in the same narrow time interval.



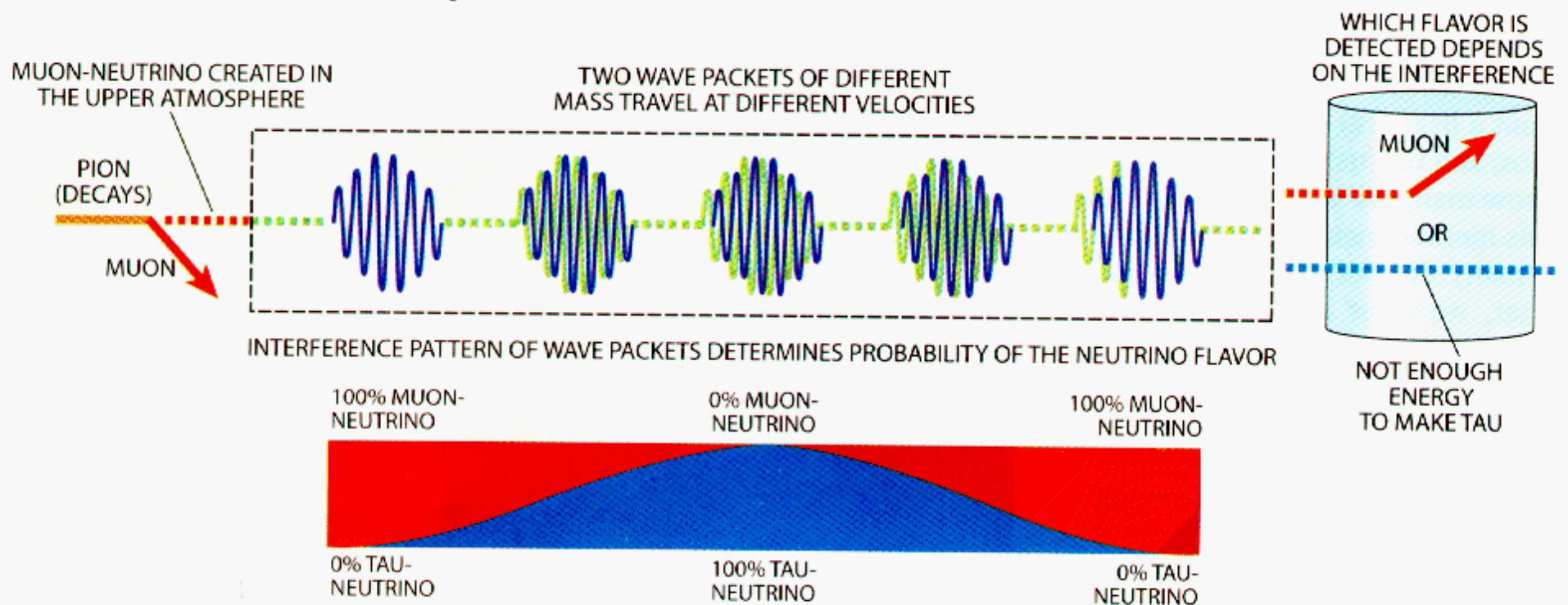




LAURIE GRACE

NUMBER OF HIGH-ENERGY MUON-NEUTRINOS seen arriving on different trajectories at Super-K clearly matches a prediction incorporating neutrino oscillations (*green*) and does not match the no-oscillation prediction (*blue*). Upward-going neutrinos (*plotted toward left of graph*) have traveled far enough for half of them to change flavor and escape detection.

How Quantum Waves Make a Neutrino Oscillate



When a pion decays (*top left*), it produces a neutrino. Described quantum-mechanically, the neutrino is apparently a superposition of two wave packets of different mass (*purple and green; top middle*). The wave packets propagate at different speeds, with the lighter wave packet getting ahead of the heavier one. As this proceeds, the waves interfere, and the interference pattern controls what flavor neutrino—muon (*red*) or tau (*blue*)—one is most likely to detect at any point along the flight path (*bottom*). Like all quantum effects, this is a

game of chance, with the chances heavily favoring a muon-neutrino close to where it was produced. But the probabilities oscillate back and forth, favoring the tau-neutrino at just the right distance and returning to favor the muon-neutrino farther on. When the neutrino finally interacts in the detector (*top right*), the quantum dice are rolled. If the outcome is muon-neutrino, a muon is produced. If chance favors the tau-neutrino, and the neutrino does not have enough energy to create a tau particle, Super-K detects nothing.

—E.K., T.K. and Y.T.

Electron :

The electron is the lightest stable subatomic particle known. It carries a negative charge which is considered the basic charge of electricity.

An electron is nearly massless. It has a rest mass of 9.1×10^{-28} gram, which is only 0.0005 the mass of a proton. The electron reacts only by the electromagnetic, weak, and gravitational forces; it does not respond to the short-range strong nuclear force that acts between quarks and binds protons and neutrons in the atomic nucleus. The electron has an antimatter counterpart called the positron. This antiparticle has precisely the same mass and spin, but it carries a positive charge. If it meets an electron, both are annihilated in a burst of energy. Positrons are rare on the Earth, being produced only in high-energy processes (e.g., by cosmic rays) and live only for brief intervals before annihilation by electrons that abound everywhere.

The electron was the first subatomic particle discovered. It was identified in 1897 by the British physicist J.J. Thomson during investigations of cathode rays. His discovery of electrons, which he initially called corpuscles, played a pivotal role in revolutionizing knowledge of atomic structure.

Under ordinary conditions, electrons are bound to the positively charged nuclei of atoms by the attraction between opposite electric charges. In a neutral atom the number of electrons is identical to the number of positive charges on the nucleus. Any atom, however, may have more or fewer electrons than positive charges and thus be negatively or positively charged as a whole; these charged atoms are known as ions. Not all electrons are associated with atoms. Some occur in a free state with ions in the form of matter known as plasma.

Quantum Electrodynamics :

Quantum electrodynamics, or QED, is a quantum theory of the interactions of charged particles with the electromagnetic field. It describes mathematically not only all interactions of light with matter but also those of charged particles with one another. QED is a relativistic theory in that Albert Einstein's theory of special relativity is built into each of its equations. Because the behaviour of atoms and molecules is primarily electromagnetic in nature, all of atomic physics can be considered a test laboratory for the theory. Agreement of such high accuracy makes QED one of the most successful physical theories so far devised.

In 1926 the British physicist P.A.M. Dirac laid the foundations for QED with his discovery of an equation describing the motion and spin of electrons that incorporated both the quantum theory and the theory of special relativity. The QED theory was refined and fully developed in the late 1940s by Richard P. Feynman, Julian S. Schwinger, and Shin'ichiro Tomonaga, independently of one another. QED rests on the idea that charged particles (e.g., electrons and positrons) interact by emitting and absorbing photons, the particles of light that transmit electromagnetic forces. These photons are virtual; that is, they cannot be seen or detected in any way because their existence violates the conservation of energy and momentum. The particle exchange is merely the "force" of the interaction, because the interacting particles change their speed and direction of travel as they release or absorb the energy of a photon. Photons also can be emitted in a free state, in which case they may be observed. The interaction of two charged particles occurs in a series of processes of increasing complexity. In the simplest, only one virtual photon is involved; in a second-order process, there are two; and so forth. The processes correspond to all the possible ways in which the particles can interact by the exchange of virtual photons, and each of them can be represented graphically by means of the diagrams developed by Feynman. Besides furnishing an intuitive picture of the process being considered, this type of diagram prescribes precisely how to calculate the variable involved.

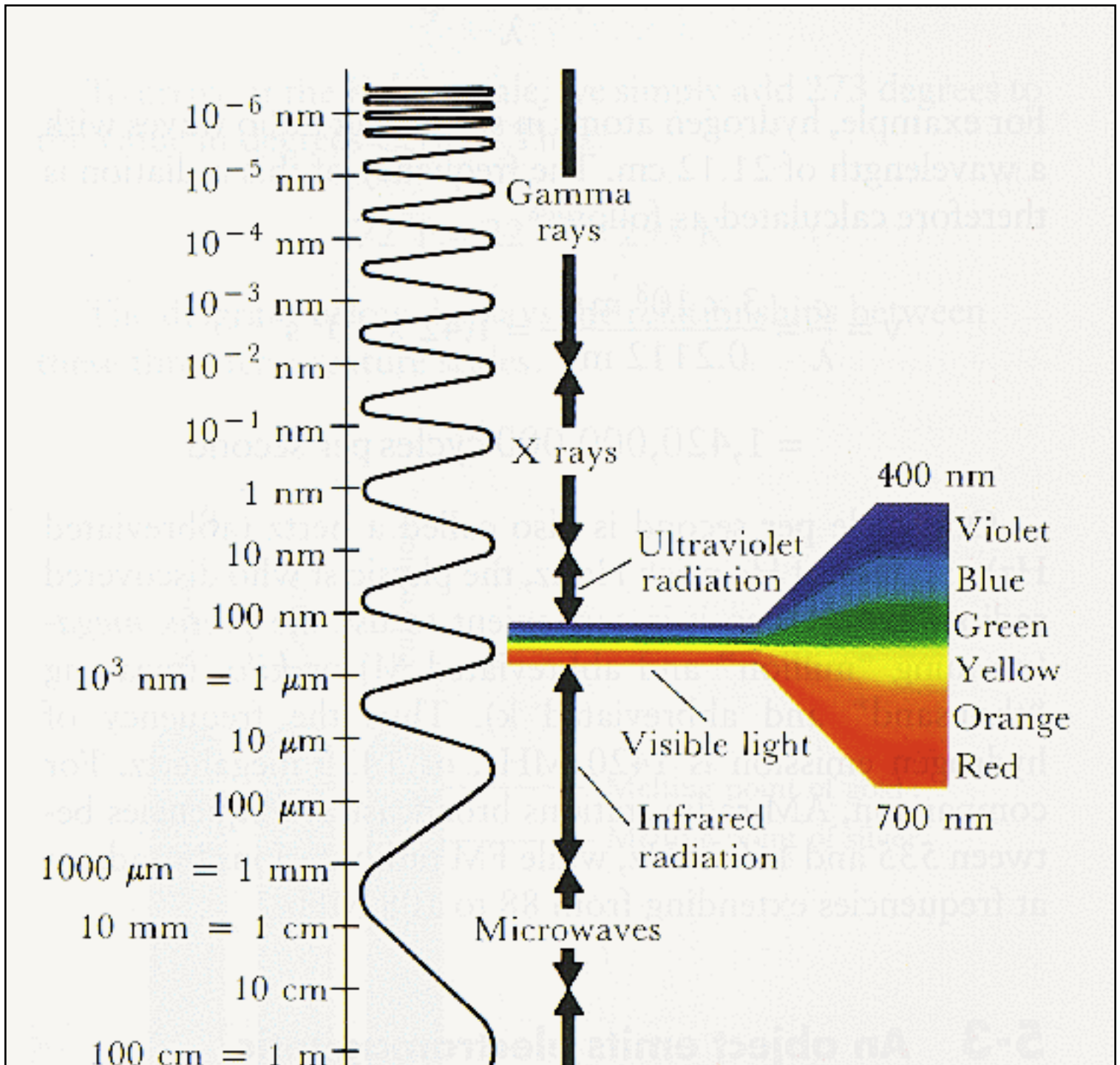
Action at a Distance :

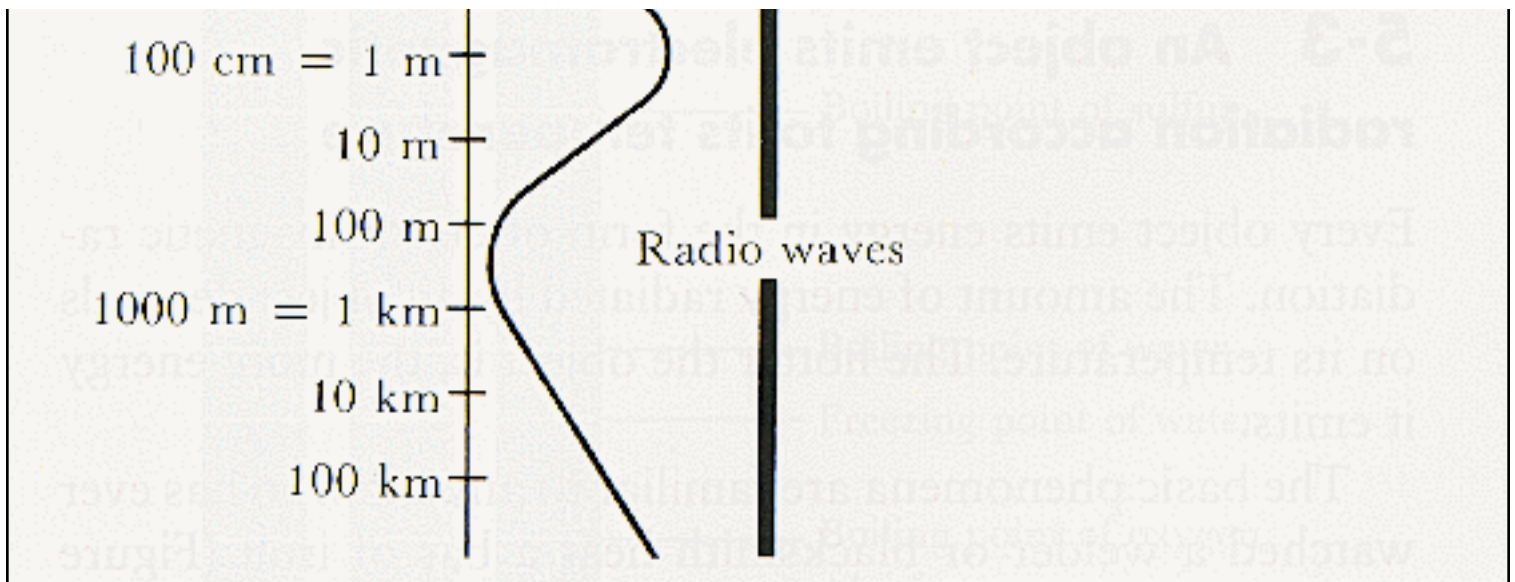
The Newtonian view of the universe may be described as a mechanistic interpretation. All components of the universe, small or large, obey the laws of mechanics, and all phenomena are in the last analysis based on matter in motion. A conceptual difficulty in Newtonian mechanics, however, is the way in which the gravitational force between two massive objects acts over a distance across empty space or in electromagnetism how a magnetic force operates between two charged particles. Newton did not address this question, but many of his contemporaries hypothesized that the forces were mediated through an invisible and frictionless medium which Aristotle had called the ether. The problem is that everyday experience of natural phenomena shows mechanical things to be moved by forces which make contact. Any cause and effect without a discernible contact, or *action at a distance*, contradicts common sense and has been an unacceptable notion since antiquity. Whenever the nature of the transmission of certain actions and effects over a distance was not yet understood, the ether was resorted to as a conceptual solution of the transmitting medium. By necessity, any description of how the ether functioned remained vague, but its existence was required by common sense and thus not questioned.

After 1916 Einstein strove to produce what is now called the theory of relativity into a formulation that includes gravitation, which was still being expressed in the form imparted to it by Newton; i.e., that of a theory of action at a distance. Einstein did succeed in the case of gravitation in reducing it to a *local-action* theory, but, in so doing, he increased the mathematical complexity considerably, as Maxwell, too, had done when he transformed electrodynamics from a theory of action at a distance to a local-action theory.

Photons :

Photons, also called light quantum, are minute energy packets of electromagnetic radiation. The concept originated in Einstein's explanation of the [photoelectric effect](#), in which he proposed the existence of discrete energy packets during the transmission of light. The concept came into general use after the U.S. physicist Arthur H. Compton demonstrated (1923) the corpuscular nature of X-rays. The term photon (from Greek phos, photos, "light"), however, was not used until 1926. The energy of a photon depends on radiation frequency; there are photons of all energies from high-energy gamma- and X-rays, through visible light, to low-energy infrared and radio waves. All photons travel at the speed of light. Considered among the subatomic particles, photons are bosons, having no electric charge or rest mass; they are field particles that are thought to be the carriers of the electromagnetic field.





Standard Model :

The Standard Model is the combination of two theories of particle physics into a single framework to describe all interactions of subatomic particles, except those due to gravity. The two components of the standard model are electroweak theory, which describes interactions via the electromagnetic and weak forces, and quantum chromodynamics, the theory of the strong nuclear force. Both these theories are gauge field theories, which describe the interactions between particles in terms of the exchange of intermediary "messenger" particles that have one unit of intrinsic angular momentum, or spin.

In addition to these force-carrying particles, the standard model encompasses two families of subatomic particles that build up matter and that have spins of one-half unit. These particles are the quarks and the leptons, and there are six varieties, or "flavours," of each, related in pairs in three "generations" of increasing mass. Everyday matter is built from the members of the lightest generation: the "up" and "down" quarks that make up the protons and neutrons of atomic nuclei; the electron that orbits within atoms and participates in binding atoms together to make molecules and more complex structures; and the electron-neutrino that plays a role in radioactivity and so influences the stability of matter. Heavier types of quark and lepton have been discovered in studies of high-energy particle interactions, both at scientific laboratories with particle accelerators and in the natural reactions of high-energy cosmic-ray particles in the atmosphere.

The standard model has proved a highly successful framework for predicting the interactions of quarks and leptons with great accuracy. Yet it has a number of weaknesses that lead physicists to search for a more complete theory of subatomic particles and their interactions. The present standard model, for example, cannot explain why there are three generations of quarks and leptons. It makes no predictions of the masses of the quarks and the leptons nor of the strengths of the various interactions. Physicists hope that, by probing the standard model in detail and making highly accurate measurements, they will discover some way in which the model begins to break down and thereby find a more complete theory. This may prove to be what is known as a grand unified theory, which uses a single theoretical structure to describe the strong, weak, and electromagnetic forces.

Unified Field Theory :

Unified field theory, in particle physics, is an attempt to describe all fundamental forces and the relationships between elementary particles in terms of a single theoretical framework. Forces can be described by fields that mediate interactions between separate objects. In the mid-19th century James Clerk Maxwell formulated the first field theory in his theory of electromagnetism. Then, in the early part of the 20th century, Albert Einstein developed general relativity, a field theory of gravitation. Later, Einstein and others attempted to construct a unified field theory in which electromagnetism and gravity would emerge as different aspects of a single fundamental field. They failed, and to this day gravity remains beyond attempts at a unified field theory.

At subatomic distances, fields are described by quantum field theories, which apply the ideas of quantum mechanics to the fundamental field. In the 1940s quantum electrodynamics (QED), the quantum field theory of electromagnetism, became fully developed. In QED, charged particles interact as they emit and absorb photons (minute packets of electromagnetic radiation), in effect exchanging the photons in a game of subatomic "catch." This theory works so well that it has become the prototype for theories of the other forces.

During the 1960s and '70s particle physicists discovered that matter is composed of two types of basic building block--the fundamental particles known as quarks and leptons. The quarks are always bound together within larger observable particles, such as protons and neutrons. They are bound by the short-range strong force, which overwhelms electromagnetism at subnuclear distances. The leptons, which include the electron, do not "feel" the strong force. However, quarks and leptons both experience a second nuclear force, the weak force. This force, which is responsible for certain types of radioactivity classed together as beta decay, is feeble in comparison with electromagnetism.

At the same time that the picture of quarks and leptons began to crystallize, major advances led to the possibility of developing a unified theory. Theorists began to invoke the concept of local gauge invariance, which postulates symmetries of the basic field equations at each point in space and time. Both electromagnetism and general relativity already involved such symmetries, but the important step was the discovery that a gauge-invariant quantum field theory of the weak force had to include an additional interaction--namely, the electromagnetic interaction. Sheldon Glashow, Abdus Salam, and Steven Weinberg independently proposed a unified "electroweak" theory of these forces based on the exchange of four particles: the photon for electromagnetic interactions, and two charged W particles and a neutral Z particle for weak interactions.

During the 1970s there was developed a similar quantum field theory for the strong force, called quantum chromodynamics (QCD). In QCD, quarks interact through the exchange of particles called gluons. The aim of researchers now is to discover whether the strong force can be unified with the electroweak force in a grand unified theory (GUT). There is evidence that the strengths of the different forces vary with energy in such a way that they converge at high energies. However, the energies involved are extremely high, more than a million million times as great as the energy scale of

electroweak unification, which has already been verified by many experiments.

Supergravity :

Supergravity is a type of quantum theory of elementary particles and their interactions that is based on the particle symmetry known as supersymmetry and that naturally includes gravity along with the other fundamental forces (the electromagnetic force, the weak nuclear force, and the strong nuclear force).

The electromagnetic and the weak forces are now understood to be different facets of a single underlying force that is described by the electroweak theory. Further unification of all four fundamental forces in a single quantum theory is a major goal of theoretical physics. Gravity, however, has proved difficult to treat with any quantum theory that describes the other forces in terms of messenger particles that are exchanged between interacting particles of matter. General relativity, which relates the gravitational force to the curvature of space-time, provides a respectable theory of gravity on a larger scale. To be consistent with general relativity, gravity at the quantum level must be carried by a particle, called the graviton, with an intrinsic angular momentum (spin) of 2 units, unlike the other fundamental forces, whose carriers (e.g., the photon and the gluon) have a spin of 1.

A particle with the properties of the graviton appears naturally in certain theories based on supersymmetry--a symmetry that relates fermions (particles with half-integral values of spin) and bosons (particles with integral values of spin). In these theories supersymmetry is treated as a "local" symmetry; in other words, its transformations vary over space-time, unlike a "global" symmetry, which transforms uniformly over space-time. Treating supersymmetry in this way relates it to general relativity, and so gravity is automatically included. Moreover, these supergravity theories seem to be free from various infinite quantities that usually arise in quantum theories of gravity. This is due to the effects of the additional particles that supersymmetry predicts (every particle must have a supersymmetric partner with the other type of spin). In the simplest form of supergravity, the only particles that exist are the graviton with spin 2 and its fermionic partner, the gravitino, with spin 3/2. (Neither has yet been observed.) More complicated variants also include particles with spin 1, spin 1/2, and spin 0, all of which are needed to account for the known particles. These variants, however, also predict many more particles than are known at present, and it is difficult to know how to relate the particles in the theory to those that do exist.